

JEE MAIN 2027 EDITION · SESSION 1: 22-30 JAN 2027 (TENTATIVE)

JEE MAIN

99+

THE COMPLETE PERSONALIZED PREPARATION
SYSTEM

*A Personalized Strategy, Formula, PYQ, Practice, Revision &
Mock-Test Handbook*

● Physics

● Chemistry

● Mathematics

● Strategy · PYQs · Mocks

PREPARED FOR

Gayathri Devu

MASTER BOOK

Strategy · Physics · Chemistry · Maths · Formulas
· PYQs · Mocks · Plans · Trackers

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PART I

Start Here



Welcome, verified exam facts, the 99+ roadmap and percentile strategy, your personal profile, the diagnostic test and the plan selector.

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Welcome, Gayathri

This book is a working system, not a reading book. Every page is here for one job: moving your JEE Main 2027 score up by the next few marks, again and again, until you are at the level where 99+ percentile becomes a realistic outcome.

It combines six things that are usually sold separately:

1 · A coaching module set

Concept summaries, formulas, traps, shortcuts and original practice questions for every chapter in the current NTA syllabus: 20 Physics units, 20 Chemistry units and 14 Mathematics units.

2 · A formula & fact handbook

Every formula, constant, reaction, reagent and NCERT fact you need, compressed for 1-day, 3-day, 7-day and final-24-hour revision.

3 · A strategy manual

Question selection, answer-choice techniques, speed methods, exam-hall timing and negative-marking control, built from verified exam rules.

4 · A mock-test companion

A full-length original mock test, a 60-question bank in six skill sets, a chapter test in every module, a mock-analysis sheet and a percentile-planning section based on historical data.

5 · A personal planner

30- to 180-day plans at 6, 8 and 10 hours a day, final 30/14/7/3-day plans, timetables, weekly scorecards and printable trackers.

6 · A mistake-elimination system

The Mistake Book: eight mistake types, how to diagnose each one, and a fix routine that stops you making the same mistake twice.

Your timeline, in one line

✓ VERIFIED FACT Verified: JEE Main 2027 Session 1 dates

NTA's tentative examination calendar (public notice dated 16 September 2026) schedules **JEE Main 2027 Session 1 on 22, 23, 24, 28, 29 and 30 January 2027**, with 31 January as a buffer day. NTA has said these dates are tentative. The detailed Information Bulletin, with eligibility, pattern and syllabus confirmation, had **not** been released when this book was compiled (28 September 2026). Session 2 is expected in April 2027 but had not been announced. Sources: [NTA-CAL27], [NTA-WEB].

Counting from 28 September 2026, you have about **116 days to Session 1** and about **185–190 days to an early-April Session 2**. That is why this book's **120-day plan** is built to peak in late January and its **180-day plan** carries you through to April. If you are reading this later, the Study-Plan Selector (page **30**) picks the right plan for the days you have left.

Three promises this book makes, and one it doesn't

What this book promises

- **Accuracy over hype.** Exam rules are taken from NTA documents and labelled *Verified*. Trend numbers from coaching analyses are labelled *Third-party analysis*. Our own advice is labelled *Recommendation*.
- **Complete coverage.** Every unit of the official syllabus has a module. Topics that NTA removed in 2024 are flagged so you don't waste time on them.
- **Original material.** Every practice question here was written for this book. PYQs are referenced and classified, not copied.

What it will not promise

No book, coach or strategy can *guarantee* a percentile, rank or college. Percentile depends on your raw score **and** on how everyone else in your shift performs. What we can do is show you, with historical ranges, what score band usually corresponds to 99 percentile, and then give you a system for reaching the top of that band with a safety margin.

How to start today (40 minutes)

1. Read **How to Use This Book** (page [7](#)): 5 minutes.
2. Fill in your **Personalization Profile** (page [20](#)): 10 minutes. Leave blanks where you're unsure.
3. Take the **Current-Level Diagnostic Test** (page [22](#)): 60 minutes, today or tomorrow.
4. Use the **Study-Plan Selector** (page [30](#)) to choose your plan, print that plan's pages, and start Day 1.

★ **IMPORTANT** The one habit that matters most

Every question you get wrong goes into **The Mistake Book** (page [449](#)) *the same day*, with its mistake type. Students who keep this habit reliably get more out of the same number of study hours, because they stop losing marks to the same errors.

Prepared with care for **Gayathri Devu**, target: **JEE Main 2027, 99+ percentile**.

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How to Use This Book

The visual code

Every box in this book has one job, and its colour and label tell you what that job is. After a week you will scan pages by colour.

FORMULA Formula

Formulas and standard results you must be able to write from memory. Example: $v^2 = u^2 + 2as$.

SHORTCUT Shortcut

A legitimate faster method, with *when to use it* and *when not to use it*.

IMPORTANT Important

A concept that questions are built on. If you skip this, later questions in the chapter stop making sense.

MISTAKE ALERT Mistake Alert

A trap that has cost students marks: sign errors, unit slips, misread conditions, NCERT exceptions.

PYQ PATTERN PYQ Pattern

How this idea is typically asked in JEE Main: the recurring question *structure*, not a copied question.

REVISION Revision

What to re-read on Day 2, Day 7 and Day 30. Tick the boxes as you go.

PRACTICE Practice

Original questions with answer, full solution, shortcut, concept tested, difficulty, target time and common trap.

MOCK TEST Mock Test

Timed test material and instructions: sit it like the real exam.

WARNING Warning

Rules where a mistake costs you directly, such as negative marking or exam-day regulations.

VERIFIED FACT Verified Fact

Information taken from an official document (NTA, NCERT, JoSAA). Re-check it against the 2027 bulletin.

REFERENCE Reference

Textbook reference data, such as reagent tables, trends and tests, collected for quick lookup.

THIRD-PARTY DATA Third-Party Data

Numbers from coaching or portal analyses, such as marks-vs-percentile ranges. Approximate by nature.

STRATEGY Strategy

A method or routine this book recommends: how to study, attempt or review.

Three kinds of statement, three labels

Label	Meaning	Example
 VERIFIED	Taken from an official NTA/NCERT/JoSAA document or an official announcement reported by a reputable outlet	"Each wrong answer carries –1 mark."
 THIRD-PARTY ANALYSIS	From published analyses by coaching institutes or education portals; useful but not official	"Current Electricity usually has 2–3 questions per shift."
 RECOMMENDATION	This book's strategy advice, derived from the rules, the trend data and learning research	"Attempt Chemistry first if it is your fastest subject."

Difficulty and question tags

Tag	Meaning
 Easy	Easy: direct formula or fact. Target: under 1.5 min.
 Medium	Medium: two steps, or one idea plus careful calculation. Target: 2–3 min.
 Hard	Hard: multi-concept, long algebra, or an unusual twist. Target: 3–5 min, attempt only in Round 2.
NV	Numerical-value answer (Section B style). There are no options to check against, so be careful: a wrong answer here also costs –1.
Speed	Time-pressure item: solve it in the target time or skip it.
Tricky	Contains a deliberate trap. Read the <i>Common trap</i> line after solving.

Navigating the PDF

On a laptop or tablet

- The **Contents** pages are clickable: tap any line to jump there.
- Every page has **quick-jump tabs** in the footer: *Contents · Physics · Chemistry · Maths · Formulas · Plans · Trackers*.
- Open your PDF reader's **bookmarks/outline panel** for a collapsible map of the whole book.
- "Page 123"-style references in the text are clickable too.
- The **progress bar** at the top of each page shows how far through the book you are, coloured by subject.

On paper

- Print the **Study Planner** and **Mock Analysis Template** PDFs separately; they're designed for A4 printing.
- Print the **Quick Revision Book** double-sided and keep it in your bag.
- The **Exam Day Command Sheet** is a single page. Print two copies: one for your desk, one for exam morning.
- Chapter modules print well in grayscale: every box also has a text label.

NOTE About interactivity, honestly

A standard PDF cannot play real animations reliably across devices, so this book doesn't try to. What it does have: clickable contents, internal links, bookmarks, quick-jump tabs, QR codes to official resources, and **fillable checkboxes and text fields**. These work in Adobe Acrobat Reader, Xodo, Foxit and most desktop browsers. On some phone viewers, fields may show but not save. In that case, print the trackers.

The daily loop this book is designed around

- 1 Learn**
Read the module's concept summary and formula box (40–60 min per sub-topic)
- 2 Practise**
Solve the module's practice set, timed, then check solutions
- 3 Log**
Every error goes into The Mistake Book with its type
- 4 Revise**
Same-day 10-min recap → Day 2 → Day 7 → Day 14 → Day 30
- 5 Test**
Chapter test → mixed tests → full mocks, analysed with the Mock Analysis Sheet

What to read when

Situation	Go to
Starting preparation	Roadmap (page 15) → Profile (page 20) → Diagnostic (page 22) → Plan Selector (page 30)
Learning a chapter	Its chapter module in Part II, III or IV
A mock went badly	Mock Analysis Sheet (page 446) → The Mistake Book (page 449) → Mental game (page 455)
Final month	Final 30/14/7/3-day plans (page 467) → Formula Handbook (page 367)
Night before and exam morning	Exam-Eve Plan (page 478) → Exam Day Command Sheet (page 483) → Last-Minute Revision Book (page 485)

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Know Your Exam: JEE Main 2027

WARNING Status as of 28 September 2026

The **JEE (Main) 2027 Information Bulletin has not yet been released**. The only official 2027-specific announcement so far is NTA's tentative calendar for Session 1. Everything else below comes from the **JEE (Main) 2026 Information Bulletin** and NTA notices, and is marked as such. Pattern and syllabus have been stable since 2025 and 2024 respectively, but **re-check every row against the 2027 bulletin when it is published**. The checklist at the end of this chapter is there for exactly that.

2027 dates

Item	Detail	Status
Session 1 exam days	22, 23, 24, 28, 29, 30 January 2027 (31 January buffer)	✓ VERIFIED tentative, NTA calendar dated 16 Sep 2026 [NTA-CAL27]
Information Bulletin & registration	Not released as of 28 Sep 2026. Portals expect late October 2026	 THIRD-PARTY ANALYSIS expected, not official
Session 2	Not announced. In 2026 it ran 2–8 April; 2025: 2–8 April	✓ VERIFIED past years only [R-DATES26], [R-DATES25]
Official websites	jeemain.nta.nic.in · nta.ac.in	✓ VERIFIED



JEE Main official portal
jeemain.nta.nic.in



NTA (public notices, calendar)
nta.ac.in



NCERT textbooks (free PDFs)
ncert.nic.in/textbook.php



JoSAA (counselling, seat matrix)
josaa.nic.in

Paper 1 (B.E./B.Tech.) pattern

VERIFIED FACT Verified from the JEE (Main) 2026 Information Bulletin [NTA-IB26]

	Physics	Chemistry	Mathematics	Total
Section A: MCQs (4 options, one correct)	20	20	20	60
Section B: Numerical-value questions	5	5	5	15
Questions to attempt	25 (all compulsory)	25	25	75
Marks	100	100	100	300

- **Mode:** Computer Based Test (CBT). **Duration:** 3 hours. In 2026 the shifts ran 9:00–12:00 and 15:00–18:00.
- **Section B has no internal choice.** Since 2025 all five numerical questions are compulsory. The old 10-given-attempt-5 format is gone.
- **Languages:** 13 (English, Hindi, Assamese, Bengali, Gujarati, Kannada, Malayalam, Marathi, Odia, Punjabi, Tamil, Telugu, Urdu).

Marking scheme

WARNING Verified: the numbers that decide your strategy [NTA-IB26]

Response	Section A (MCQ)	Section B (Numerical)
Correct	+4	+4
Incorrect	−1	−1
Unanswered / Marked for Review without an answer	0	0

Negative marking applies to numerical questions too. Older books that say "no negative marking in Section B" describe a pattern that no longer exists.

☆ IMPORTANT What −1 really means

A random guess on a 4-option MCQ has expected value $\frac{1}{4}(+4) + \frac{3}{4}(-1) = +0.25$ marks. That looks positive, but it's tiny, and the variance is large. If you can **eliminate two options with certainty**, the expected value rises to $\frac{1}{2}(4) + \frac{1}{2}(-1) = +1.5$. The honest rule, used throughout this book: **attempt when you have eliminated at least two options through real reasoning. Never attempt a pure blind guess.** On numerical questions a guess is almost always wrong, so answer only after solving.

Calculators, tables, rough work

- **No calculator of any kind.** NTA's November 2025 clarification states that the on-screen calculator mentioned in the first 2026 bulletin was a typographical error, and that calculators are not allowed in JEE Main. **✓ VERIFIED** [NTA-CALC], [R-CALC]
- **Numerical answers:** entered with the on-screen numeric keypad. In recent papers the Section B instructions have asked for answers rounded to the nearest integer. Read the instruction on your screen on the day.

||| **THIRD-PARTY ANALYSIS**

- Rough sheets are provided at the centre and must be returned. Don't carry your own paper. ✓ VERIFIED [NTA-IB26]

What this means for you: practise all mental arithmetic, approximation and long calculation **by hand from Day 1**. The Speed Manual (page **389**) has drills for this.

How your percentile is calculated

✓ VERIFIED FACT Verified: NTA normalisation [NTA-IB26], [R-NORM]

Each shift's raw scores are converted into **NTA percentile scores** separately:

$$\text{Percentile} = \frac{100 \times \text{number of candidates in that shift with raw score} \leq \text{yours}}{\text{total candidates who appeared in that shift}}$$

- The topper of every shift gets 100 percentile. Percentiles are reported to 7 decimal places.
- Percentiles are also computed **subject-wise** (Physics, Chemistry, Maths). Your total percentile is computed from your total raw score. It is **not** the average of the subject percentiles.
- If you take both sessions, **the better of your two NTA scores** is used for the merit list/rank.

Why the same marks give different percentiles in different shifts: in a harder shift, fewer candidates score high, so fewer marks are needed to beat 99% of that shift. The 99+ Strategy chapter (page **16**) shows the historical ranges.

Tie-breaking (for rank, when percentiles are equal)

✓ VERIFIED as per the 2026 bulletin, reported in [R-TIE]. Ties are resolved in this order:

1. Higher NTA score in **Mathematics** → 2. **Physics** → 3. **Chemistry** → 4. Lower ratio of incorrect to correct answers (all subjects) → 5. same ratio in Maths → 6. Physics → 7. Chemistry → 8. Otherwise the same rank.

Strategy note: Maths is the first tie-breaker, and a low wrong-to-right ratio is the next. That is two more reasons, beyond the marks themselves, to protect accuracy.

Eligibility essentials

Rule	Detail	Status
Age limit	None for JEE Main (institutes may have their own)	✓ VERIFIED 2026 bulletin
Year of passing Class 12	2026 exam: passed in 2024 or 2025, or appearing in 2026. For 2027 the same pattern would mean 2025, 2026 or appearing in 2027. Confirm in the 2027 bulletin.	✓ VERIFIED (2026); 2027 pending
75% criterion (for NIT/IIIT/CFTI admission via JoSAA)	75% aggregate in Class 12 (65% for SC/ST), or top-20 percentile of your board	✓ VERIFIED 2026 bulletin / JoSAA
JEE Advanced qualifying cut-off (General)	2026: 93.4123549 · 2025: 93.1023262 · 2024: 93.2362181 · 2023: 90.7788642	✓ VERIFIED as reported [R-CUTOFF]

Facts that matter specifically for you

Female-only seat pool (JoSAA) ✓ VERIFIED

In JoSAA counselling, IITs and the NIT+ system institutes run a *Female-only (including supernumerary)* pool alongside the *Gender-neutral* pool. The supernumerary seats exist to raise female enrolment to at least about 20%. A female candidate is considered for **both** pools, so the ranks you need can be meaningfully different from gender-neutral closing ranks. Check the official JoSAA past opening/closing ranks for your target branches. [JOSAA]

Application fee ✓ VERIFIED (2026 bulletin, reported [R-FEE])

For one paper at an Indian centre, the fee was **₹800 for female candidates of all categories**, against ₹1,000 for General-category male candidates. Check the 2027 bulletin for the current figure.

Exam-day rules (from 2026 guidelines) ✓ VERIFIED

Report early: in 2026, gates closed 30 minutes before the exam started. Carry your admit card, a valid photo ID and the photographs the admit card asks for. Metallic items and jewellery are restricted, and face authentication/biometric verification happens at entry. Candidates in customary or religious attire should report earlier for frisking. **The admit card's instructions override anything written here.**

Practical planning ☉ RECOMMENDATION

- Visit or map your centre route 1–2 days before. Plan to travel with a family member for early-morning shifts.
- If your exam dates may coincide with your period, plan comfort and medication with your doctor in advance, and take at least two full mocks at the same time of day to rehearse. Check the admit card for permitted items.
- Keep the exam-morning routine (page [479](#)) identical to your mock-day routine.

Recent official changes worth knowing

Year	Change	Status
2024	Syllabus reduced and aligned to the rationalised NCERT (e.g., Communication Systems, States of Matter, Surface Chemistry, s-Block, Hydrogen, Polymers, Mathematical Reasoning and Induction removed)	✓ VERIFIED [NTA-SYL]
2025	Section B optional questions scrapped: 5 compulsory numerical questions per subject	✓ VERIFIED
2026	Aadhaar/DigiLocker-based identity details at registration; face authentication at centres; calculator clarification	✓ VERIFIED [NTA-IB26], [NTA-CALC]
2026–27	Ministry-announced multi-level (four-tier) scrutiny of question papers to reduce errors. No change to difficulty, pattern or syllabus announced	THIRD-PARTY ANALYSIS reported [R-REFORM]

Verify-on-release checklist (when the 2027 bulletin comes out)

- ☐ Number of questions per section (20 MCQ + 5 NV?) and whether all are compulsory
- ☐ Marking: +4 / –1 in both sections?
- ☐ Syllabus PDF: any unit or topic change from the 2024–2026 syllabus?
- ☐ Eligibility: permitted years of passing Class 12
- ☐ Session 2 dates and registration window
- ☐ Shift timings, reporting time, gate-closing time
- ☐ Calculator/rough-sheet rules
- ☐ Tie-breaking order
- ☐ Fee for female candidates
- ☐ Exam-city choices near you (choose four with care)

- ☐ Photo/signature and document specifications
- ☐ Update page **10** of this book by hand where anything changed

PART I · START HERE

The 99+ Percentile Roadmap

The roadmap has five phases. Each phase ends with a **gate**: a measurable condition you must meet before the next phase gets most of your time. The phase lengths below assume the 180-day plan. Shorter plans compress them (see the Plan Selector, page 30).

1 Phase 0 · Diagnose

Days 1–3: profile, diagnostic test, pick plan, set up Mistake Book

2 Phase 1 · Build

Concepts + chapter practice for every Must-do and High chapter; weekly mixed tests

3 Phase 2 · Strengthen

Chapter-wise PYQs, second pass on weak chapters, formula mastery

4 Phase 3 · Mock & Fix

2–3 full mocks a week, each followed by analysis and targeted repair

5 Phase 4 · Peak

Final 14 days: revision-heavy, mocks in the exam time slot, sleep locked

Phase	Share of time (180-day plan)	Gate to pass before moving on	Status
0 · Diagnose	3 days	Profile filled, diagnostic scored, plan chosen, Mistake Book started	RECOMMENDATION
1 · Build	~45% (Days 4–85)	Every Must-do chapter: module read, practice set $\geq 75\%$ correct, chapter test taken	RECOMMENDATION
2 · Strengthen	~25% (Days 86–130)	PYQs for all Must-do + High chapters done; formula recall $\geq 90\%$ on a blank-sheet test	RECOMMENDATION
3 · Mock & Fix	~20% (Days 131–166)	Last 5 full mocks averaging your target score band with ≤ 8 negative marks	RECOMMENDATION
4 · Peak	Final 14 days	Revision cycle complete; exam-slot mocks stable; sleep 8 h	RECOMMENDATION

☆ IMPORTANT The central idea

99 percentile isn't reached by knowing "more chapters". It's reached by **converting what you know into marks at high accuracy within 180 minutes**. That's why this book puts as much weight on selection, speed, accuracy and mistake elimination as on content.

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The 99+ Percentile Strategy

What score has 99 percentile needed? (historical, approximate)

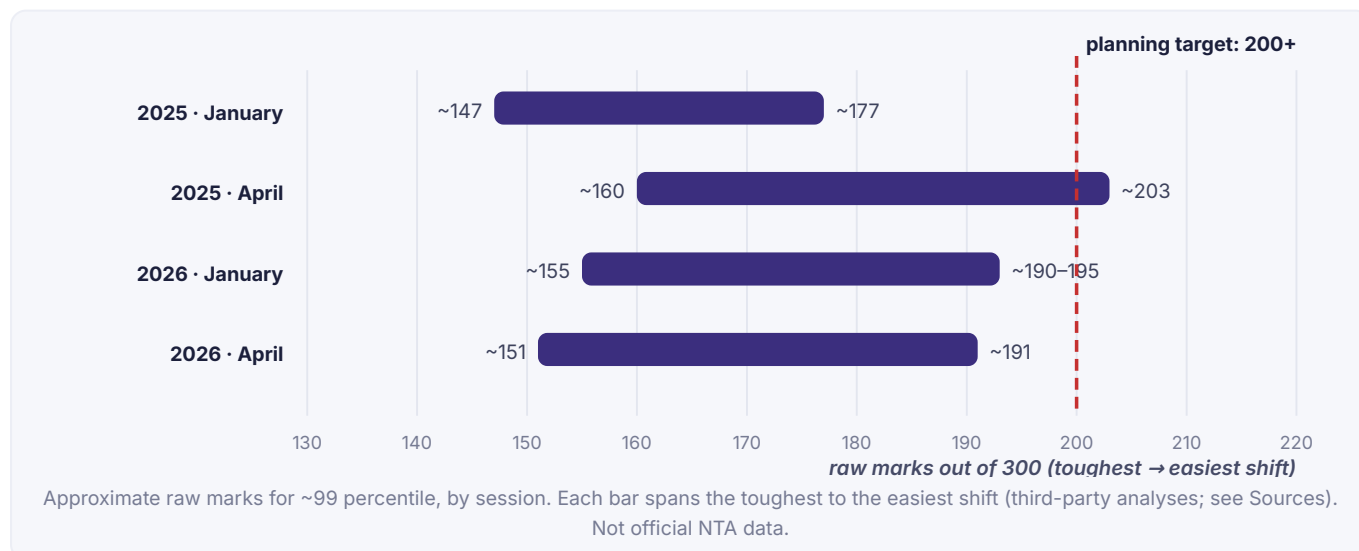


THIRD-PARTY DATA Third-party analyses of past results, labelled approximate

Marks needed for about the 99th percentile varied widely **by shift**. Ranges compiled from published shift-wise marks-vs-percentile analyses:

Session	Approx. raw marks for ~99 percentile (toughest → easiest shift)	Sources
2025 · January	~147 → ~177	[T-VED-25]
2025 · April	~160 → ~203	[T-VED-25]
2026 · January	~155 → ~190–195	[T-VED-26], [T-COMP]
2026 · April	~151 → ~191	[T-PW-S2]

Published tables for other percentile bands (95, 99.5, 99.9) **disagree with one another by large margins**, so this book doesn't reproduce them. Use the ranges above for 99 percentile only, and treat them as approximate.



Reading the chart: in a tough shift, 150-odd marks reached 99 percentile. In an easy shift, even 190–200 wasn't always enough. You won't know in advance which kind of shift you'll get.



STRATEGY The planning target this book uses

Plan for the **easy-shift case**. That means a target of **200+ raw marks in the actual exam**, which in turn means consistently scoring **200–215 in full-length mocks of JEE-Main standard** in the final month, with **≤ 8 negative marks**. If you hit this, you're in the 99+ zone in almost every historical shift. If your shift turns out tough, the same preparation lands you well above 99. **This is a target band, not a guarantee.**

Percentile → approximate rank

$$\text{Approx. rank} \approx \frac{(100 - \text{percentile}) \times N}{100}$$

where N is the number of unique candidates. In 2026, $N \approx 15.38$ lakh appeared ✓ VERIFIED [R-2026STAT]. So 99 percentile $\approx$ rank 15,000 and 99.5 percentile $\approx$ rank 7,700 (common rank list, before category and female-pool effects). This is a rough approximation: NTA's actual rank list merges sessions and applies tie-breaks.

Why percentile varies by shift

1. **Paper difficulty differs.** NTA sets a different paper for each shift. A lengthy Maths section in one shift pulls that shift's whole score distribution down.
2. **Normalisation is within the shift.** Your percentile compares you only with candidates in your shift (page 10).
3. **Candidate pool differs slightly by shift and session.** More repeaters and stronger candidates often appear in Session 2 (April), when many are attempting a second time. This is one reason April cut-offs have often been higher. THIRD-PARTY ANALYSIS

What you control: your raw score's safety margin above the band. Build the margin through accuracy and selection, not through more attempts.

Accuracy targets

Metric	99+ zone target	Why
Overall accuracy (correct ÷ attempted)	≥ 90%	At 90% accuracy on 58 attempts: $52 \times 4 - 6 = 202$
Negative marks per paper	≤ 8	Each wrong answer costs 5 marks of swing versus leaving it: the +4 you didn't get and the -1 you did
Section B (NV) accuracy	≥ 90%	No options to verify against, so every NV answer must be checked once
Silly-mistake losses	≤ 2 questions	Measured in the Mock Analysis Sheet
Unattempted "known" questions	0	Skipping a question you could solve is a selection mistake

Minimum safe preparation milestones RECOMMENDATION

- ☐ **Coverage:** 100% of Must-do and High-priority chapters (PYQ Priority Matrix, page 402); at least 90% of the full syllabus
- ☐ **Formulas:** blank-sheet recall of each subject's formula list at ≥ 90%
- ☐ **PYQs:** all 2024–2026 PYQs (post-syllabus-change papers) for Must-do chapters; High-priority chapters from 2022 onward
- ☐ **Practice volume:** 60–80 timed questions a day on normal days (6–8 h plan)
- ☐ **Mocks:** at least 20 full-length mocks before Session 1 (at least 30 before Session 2), each analysed
- ☐ **Stability:** last 5 mocks within ±15 marks of each other, averaging 200+
- ☐ **Mistake Book:** no mistake type accounts for more than 30% of lost marks in the last 3 mocks

Attempt strategy: six scenarios

All six aim at roughly 200+. Pick the one closest to your profile after 3–4 mocks, then refine it. **Time allocations are for the 180-minute paper**, with the subject order you choose (page 480).

A · High-Accuracy Strategy (for careful solvers, slower speed)

	Phy	Chem	Maths	Total
Attempt	19	21	16	56
Correct	18	20	15	53
Wrong	1	1	1	3
Marks	71	79	59	209
Time (min)	60	45	70	+5 review

Leave 19 questions. Accept only questions you can finish in time. Accuracy $\approx 95\%$.

B · Balanced Attempt Strategy (the default)

	Phy	Chem	Maths	Total
Attempt	20	22	18	60
Correct	18	20	16	54
Wrong	2	2	2	6
Marks	70	78	62	210
Time (min)	60	45	65	+10 review

Accuracy 90%. The most robust plan for most students.

C · Speed Strategy (fast and accurate students only)

	Phy	Chem	Maths	Total
Attempt	22	23	21	66
Correct	20	21	18	59
Wrong	2	2	3	7
Marks	78	82	69	229
Time (min)	55	40	75	+10 review

Use only if your mocks show $\geq 88\%$ accuracy at this attempt level. More attempts at lower accuracy lose marks.

D · Weak-Maths Strategy

	Phy	Chem	Maths	Total
Attempt	21	23	13	57
Correct	20	22	12	54
Wrong	1	1	1	3
Marks	79	87	47	213
Time (min)	60	45	65	+10 review

Maths: attempt only Must-do models (Matrices, Vectors & 3D, Sequences, Binomial, P&C, definite integrals, the conic basics). Every mark you gain here comes from Physics and Chemistry accuracy.

E · Weak-Physics Strategy

	Phy	Chem	Maths	Total
Attempt	15	23	20	58
Correct	14	22	18	54
Wrong	1	1	2	4
Marks	55	87	70	212
Time (min)	55	45	70	+10 review

Physics: bank Modern Physics, Semiconductors, Units & Errors, EM Waves, Ray and Wave Optics, KTG, and the experiment-based questions: formula-direct questions first.

F · Weak-Chemistry Strategy

	Phy	Chem	Maths	Total
Attempt	21	17	20	58
Correct	20	15	18	53
Wrong	1	2	2	5
Marks	79	58	70	207
Time (min)	60	45	65	+10 review

Chemistry is the most *fixable* weak subject: NCERT-line Inorganic, Biomolecules, Practical Chemistry and formula-based Physical Chemistry can add 20+ marks in 6 weeks (Chemistry strategy, page [161](#)).

⚠ MISTAKE ALERT The most common strategy error

Students raise the number of attempts when a mock goes badly. The data from your own Mock Analysis Sheets will usually show the opposite: **scores fall when attempts rise above your accuracy-safe level**. Adjust attempts only from evidence, and in steps of about 2 questions per subject.

Section timing checkpoints (default Balanced plan, order Chem → Phy → Maths)

Clock (from start)	You should be...
0:00–0:05	Reading instructions, calm breathing, first scan of Chemistry
0:45	Chemistry Round 1 done (≈ 20 attempted); move to Physics
1:45	Physics Round 1 done; move to Maths
2:40	Maths Round 1 done; start Round 2 (marked questions, all subjects)
2:50	Stop new questions. Review marked answers and numerical entries
3:00	Submit

Full exam-hall protocol: page [480](#).

PART I · START HERE

Student Personalization Profile

Fill this in on Day 1, in pen or directly in the PDF (the fields are fillable). Update it after every 5th mock. The rest of the book refers back to these answers.

Field	Your details	If unknown, this book assumes...
Student name	Gayathri Devu	—
Current class / year		Class 12 in 2026–27, or a Class 12 pass-out taking a drop year
Current preparation level	(Beginner / Intermediate / Advanced)	Intermediate: Class 11 done once, Class 12 partially done
Target percentile	99+	—
Target score (raw, /300)		200–215 in mocks → 200+ in the exam (page 16)
Strongest subject		Unknown: decided by the Diagnostic Test
Weakest subject		Unknown: decided by the Diagnostic Test
Average daily study time	hours	8 hours on school days, 10 on holidays
Exam attempt	(1st / 2nd / 3rd)	First attempt in Session 1, Jan 2027; second in Session 2, Apr 2027
Target college / branch (optional)		—
School / college schedule		School 8:00–14:00 on weekdays (Class 12 student)
Current mock score	/300	Not yet measured: take a baseline mock in Week 1
Exam date		Session 1: 22–30 Jan 2027 (tentative, NTA calendar)

NOTE About the assumptions column

Where you leave a field blank, the plans in this book use the assumption shown. Every assumption is marked; none is presented as fact about you. When you fill in the real detail, use the adjustment table below.

Personal adjustment table

If your profile says...	Then adjust like this	Go to
Level = Beginner	Use the next-longer plan if possible. Phase 1 is +25% time; start every chapter with its "Core concepts" and Basic questions only	Plan Selector, page 30
Level = Advanced	Skim Core concepts; start at the PYQ Pattern box; do only M/H practice questions; mocks begin 4 weeks earlier	Mock system, page 444
Weakest = Maths	Scenario D; 40% of daily problem-solving time to Maths; drill the Must-do models first	page 16, 285
Weakest = Physics	Scenario E; formula-first approach; daily 20-min "direct formula" drill	page 33
Weakest = Chemistry	Scenario F; NCERT reading block every day; Inorganic flash-review nightly	page 161
Study time $\leq$ 6 h/day (school days)	Use the 6-hour variants; move mocks to Sunday mornings; skip Low-frequency topics until Phase 3	Plans, page 459

If your profile says...	Then adjust like this	Go to
Mock score < 120	The priority is concept gaps, not speed. Phase 1 continues until chapter tests reach $\geq 70\%$	Roadmap, page <u>15</u>
Mock score 120–170	The priority is accuracy and selection. Mistake Book twice daily; attempt-limit drills	The Mistake Book, page <u>449</u>
Mock score 170–200	The priority is speed and Section B accuracy. Speed Manual drills; Round-1/Round-2 discipline	Speed Manual, page <u>389</u>

My why, in one sentence (optional)

Baseline numbers (fill after the Diagnostic Test and the first mock)

	Physics	Chemistry	Maths	Total
Diagnostic score (/40 each)				
Baseline mock (/100 each)				
Accuracy %				
Target by Session 1				

PART I · START HERE

Current-Level Diagnostic Test

⌚ MOCK TEST Instructions

30 questions · 60 minutes · +4 / -1 (same marking as JEE Main, NV questions included). Use a timer, no calculator, no notes. Mark the time at which you finish each subject. The purpose is **measurement**, not a good score. Don't prepare for it, and don't look at the solutions until you've finished.

🔗 Diagnostic · Physics

1 Easy ⌚ 1 min Basic

DP-01

The dimensional formula of Planck's constant h is:

- (A) $[ML^2T^{-1}]$ (B) $[ML^2T^{-2}]$ (C) $[MLT^{-1}]$ (D) $[ML^{-1}T^{-2}]$

2 Easy ⌚ 1 min Basic

DP-02

A ball is thrown vertically upward at 20 m s^{-1} ($g = 10 \text{ m s}^{-2}$). It returns to the thrower's hand after:

- (A) 2 s (B) 4 s (C) 6 s (D) 8 s

3 Easy ⌚ 1.5 min Concept

DP-03

A 2 kg block **sliding** on a rough horizontal floor ($\mu_k = 0.3$, $g = 10 \text{ m s}^{-2}$) is pulled by a horizontal force of 10 N in the direction of motion. Its acceleration is:

- (A) 1 m s^{-2} (B) 5 m s^{-2} (C) 3 m s^{-2} (D) 2 m s^{-2}

4 Medium ⌚ 1.5 min Concept

DP-04

A solid sphere rolls without slipping. The fraction of its total kinetic energy that is rotational is:

- (A) $2/5$ (B) $5/7$ (C) $2/7$ (D) $1/2$

5 Medium ⌚ 1.5 min Concept

DP-05

The escape speed from Earth is 11.2 km s^{-1} . For a planet of the same density but twice Earth's radius it is:

- (A) 11.2 km s^{-1} (B) 15.8 km s^{-1} (C) 44.8 km s^{-1} (D) 22.4 km s^{-1}

6 Medium ⌚ 1.5 min Concept

DP-06

An ideal monatomic gas is heated at constant pressure. The fraction of the heat supplied that is converted into work is:

- (A) $2/3$ (B) $3/5$ (C) $2/5$ (D) $1/3$

7 Easy ⌚ 1.5 min Basic

DP-07

Capacitors of $2 \mu\text{F}$ and $3 \mu\text{F}$ are connected in series across a 10 V battery. The charge on each capacitor is:

- (A) $20 \mu\text{C}$ (B) $12 \mu\text{C}$ (C) $30 \mu\text{C}$ (D) $50 \mu\text{C}$

8 Easy 1.5 min NV

DP-08

A uniform wire of resistance $32\ \Omega$ is bent into a circle. Find the resistance (in Ω) between two diametrically opposite points.

Your answer (numerical value):

9 Easy 1.5 min Basic

DP-09

An object is placed 30 cm in front of a convex lens of focal length 20 cm. The image is:

- (A) 12 cm behind the lens, virtual, erect
 (B) 60 cm behind the lens, real, inverted, twice the size
 (C) 60 cm in front of the lens, virtual, erect
 (D) 12 cm behind the lens, real, diminished

10 Easy 1 min NV

DP-10

Photons of energy 6.0 eV fall on a metal of work function 2.0 eV. Find the stopping potential (in volts).

Your answer (numerical value):

Diagnostic · Chemistry

1 Easy 1 min Basic

DC-01

The number of moles of oxygen **atoms** in 88 g of CO_2 is:

- (A) 2 (B) 6 (C) 4 (D) 8

2 Easy 1 min Basic

DC-02

The number of radial nodes in a 3p orbital is:

- (A) 1 (B) 0 (C) 2 (D) 3

3 Easy 1 min Basic

DC-03

Which molecule has zero dipole moment?

- (A) BF_3 (B) H_2O (C) NH_3 (D) CHCl_3

4 Medium 1.5 min Concept

DC-04

For a reaction, $\Delta H = -100\ \text{kJ mol}^{-1}$ and $\Delta S = -200\ \text{J K}^{-1}\text{mol}^{-1}$ (assume both are temperature independent). The reaction is spontaneous:

- (A) below 500 K (B) above 500 K
 (C) at all temperatures (D) at no temperature

5 Easy 1 min Basic

DC-05

The pH of $10^{-3}\ \text{M}$ NaOH at $25\ ^\circ\text{C}$ is:

- (A) 3 (B) 10 (C) 12 (D) 11

6 Easy 1 min NV

DC-06

A first-order reaction has a half-life of 10 min. Find the time (in minutes) needed for 87.5% completion.

Your answer (numerical value):

7 Medium 1.5 min Concept

DC-07

The correct order of first ionisation enthalpy is:

- (A) $B < C < N < O$ (B) $B < C < O < N$ (C) $C < B < O < N$ (D) $O < N < C < B$

8 Medium 1.5 min Concept

DC-08

The spin-only magnetic moment of $[\text{Fe}(\text{CN})_6]^{3-}$ is closest to:

- (A) 5.92 BM (B) 3.87 BM (C) 1.73 BM (D) 0 BM

9 Easy 1 min Basic

DC-09

The most stable carbocation among the following is:

- (A) CH_3^+ (B) CH_3CH_2^+ (C) $(\text{CH}_3)_2\text{CH}^+$ (D) $(\text{CH}_3)_3\text{C}^+$

10 Medium 1.5 min NV

DC-10

How many structurally isomeric **alcohols** have the formula $\text{C}_4\text{H}_{10}\text{O}$?Your answer (numerical value):

Diagnostic · Mathematics

1 Easy 1 min Basic

DM-01

If $f(x) = 2x + 3$ and $g(x) = x^2$, then $(f \circ g)(2)$ equals:

- (A) 7 (B) 49 (C) 14 (D) 11

2 Easy 1 min Basic

DM-02

 $\left| \frac{3+4i}{1-i} \right|$ equals:

- (A) $5\sqrt{2}$ (B) $5/\sqrt{2}$ (C) $5/2$ (D) $7/\sqrt{2}$

3 Easy 1 min Basic

DM-03

If α, β are the roots of $x^2 - 5x + 6 = 0$, then $\alpha^2 + \beta^2$ is:

- (A) 13 (B) 25 (C) 37 (D) 11

4 Medium 1 min Concept

DM-04

If A is a 3×3 matrix with $|A| = 4$, then $|\text{adj } A|$ is:

- (A) 4 (B) 64 (C) 16 (D) 8

5 Easy 1.5 min NV

DM-05

Find the number of distinct arrangements of the letters of the word LEVEL.

Your answer (numerical value):

6 Easy 1 min Basic

DM-06

The sum of the first 20 terms of the AP 3, 7, 11, ... is:

- (A) 840 (B) 800 (C) 780 (D) 820

7 Easy 1 min Basic

DM-07

 $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$ equals:

- (A) 1 (B) 2 (C) 4 (D) 1/2

8 Easy 1 min Basic

DM-08

 $\int_0^{\pi/2} \sin^2 x \, dx$ equals:

- (A) $\pi/4$ (B) $\pi/2$ (C) 1 (D) $\pi/8$

9 Easy 1 min NV

DM-09

Find the perpendicular distance of the point (1, 2) from the line $3x + 4y - 1 = 0$.Your answer (numerical value):

10 Easy 1 min Basic

DM-10

Two fair dice are thrown. The probability that the sum is 7 is:

- (A) 1/12 (B) 1/6 (C) 5/36 (D) 7/36

Diagnostic — Answers & Solutions

Diagnostic · Physics

1 (A) 2 (B) 3 (D) 4 (C) 5 (D) 6 (C) 7 (B) 8 8 9 (B) 10 4

Diagnostic · Chemistry

1 (C) 2 (A) 3 (A) 4 (A) 5 (D) 6 30 7 (B) 8 (C) 9 (D) 10 4

Diagnostic · Mathematics

1 (D) 2 (B) 3 (A) 4 (C) 5 30 6 (D) 7 (B) 8 (A) 9 2 10 (B)

DIAGNOSTIC · PHYSICS

1 Answer: (A) Dimensions of Planck's constant Easy 1 min Basic

DP-01

SOLUTION $E = h\nu \Rightarrow h = E/\nu = [ML^2T^{-2}]/[T^{-1}] = [ML^2T^{-1}]$.SHORTCUT Same dimensions as angular momentum ($L = nh/2\pi$ in Bohr's model).

COMMON TRAP Confusing with the dimensions of energy, which is option (B).

2 Answer: (B) Vertical projectile time of flight Easy 1 min Basic

DP-02

SOLUTION Time to top $= u/g = 2$ s; the descent takes the same time. Total $= 2u/g = 4$ s.SHORTCUT $T = 2u/g$ for return to the launch level.

COMMON TRAP Answering the time to the top (2 s).

3 Answer: (D) Kinetic friction Easy 1.5 min Concept

DP-03

SOLUTION Friction $f = \mu_k mg = 0.3 \times 20 = 6$ N opposing motion. $a = (10 - 6)/2 = 2 \text{ m s}^{-2}$.SHORTCUT Net force $= F - \mu mg$.

COMMON TRAP Ignoring friction gives 5.

4 Answer: (C) Rolling kinetic energy split Medium 1.5 min Concept

DP-04

SOLUTION $K_{\text{rot}}/K_{\text{tot}} = \frac{\frac{1}{2}I\omega^2}{\frac{1}{2}mv^2(1 + k^2/R^2)} = \frac{k^2/R^2}{1 + k^2/R^2} = \frac{2/5}{7/5} = \frac{2}{7}$.SHORTCUT Fraction $= \frac{k^2/R^2}{1 + k^2/R^2}$; for a solid sphere $k^2/R^2 = 2/5$.COMMON TRAP Answering $k^2/R^2 = 2/5$ itself.

5 Answer: (D) *Escape velocity scaling* **Medium** ⌚ 1.5 min **Concept**

DP-05

SOLUTION $v_e = \sqrt{2GM/R}$ and $M = \frac{4}{3}\pi R^3 \rho$, so $v_e = R\sqrt{\frac{8}{3}\pi G \rho} \propto R$ at fixed ρ . Doubling R doubles v_e .

SHORTCUT Fixed density: $v_e \propto R$. Fixed mass: $v_e \propto 1/\sqrt{R}$.

COMMON TRAP Using $v_e \propto 1/\sqrt{R}$ (fixed-mass case) gives the wrong direction.

6 Answer: (C) *First law, isobaric process* **Medium** ⌚ 1.5 min **Concept**

DP-06

SOLUTION $W = nR\Delta T$, $Q = nC_p\Delta T = \frac{5}{2}nR\Delta T$, so $W/Q = 2/5$.

SHORTCUT $W/Q = R/C_p = 1 - 1/\gamma$.

COMMON TRAP Using C_v instead of C_p gives $2/3$.

7 Answer: (B) *Series capacitors* **Easy** ⌚ 1.5 min **Basic**

DP-07

SOLUTION $C_{eq} = \frac{2 \times 3}{2+3} = 1.2 \mu\text{F}$; $Q = C_{eq}V = 12 \mu\text{C}$ (the same on both in series).

SHORTCUT Series: same Q ; $C_{eq} = C_1C_2/(C_1 + C_2)$.

COMMON TRAP Adding capacitances as if in parallel gives 50.

8 Answer: 8 *Resistance of a ring* **Easy** ⌚ 1.5 min **NV**

DP-08

SOLUTION Each semicircle is 16Ω ; they're in parallel: $16 \parallel 16 = 8 \Omega$.

SHORTCUT Two equal halves in parallel: $R/4$.

COMMON TRAP Answering 16 (one half only).

9 Answer: (B) *Thin lens formula* **Easy** ⌚ 1.5 min **Basic**

DP-09

SOLUTION $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ with $u = -30$: $\frac{1}{v} = \frac{1}{20} - \frac{1}{30} = \frac{1}{60}$, so $v = +60$ cm. $m = v/u = -2$: real, inverted, magnified.

SHORTCUT Object between f and $2f \rightarrow$ image beyond $2f$, real, magnified.

COMMON TRAP Sign error with u gives 12 cm.

10 Answer: 4 *Photoelectric equation* **Easy** ⌚ 1 min **NV**

DP-10

SOLUTION $K_{max} = h\nu - \phi = 4.0$ eV, so $eV_0 = 4.0$ eV and $V_0 = 4$ V.

SHORTCUT In eV units, V_0 (volts) = $E_{photon} - \phi$ numerically.

COMMON TRAP Adding instead of subtracting the work function.

DIAGNOSTIC · CHEMISTRY**1 Answer: (C)** *Mole concept* **Easy** ⌚ 1 min **Basic**

DC-01

SOLUTION $88/44 = 2$ mol CO_2 ; each has 2 O atoms, so 4 mol O atoms.

SHORTCUT Moles of atoms = moles of molecules $\times$ atomicity.

COMMON TRAP Stopping at 2 (moles of molecules).

2 Answer: (A) *Nodes of orbitals* **Easy** ⌚ 1 min **Basic**

DC-02

SOLUTION Radial nodes = $n - l - 1 = 3 - 1 - 1 = 1$. (Angular nodes = $l = 1$.)

SHORTCUT Radial = $n - l - 1$; angular = l ; total = $n - 1$.

COMMON TRAP Giving the total nodes (2).

3 Answer: (A) *Dipole moment & shape* **Easy** ⌚ 1 min **Basic**

DC-03

SOLUTION BF_3 is trigonal planar and symmetric, so the bond dipoles cancel. The others are pyramidal, bent or tetrahedral with unequal substituents.

SHORTCUT Symmetric shapes (linear AX_2 , trigonal planar AX_3 , tetrahedral AX_4) with identical atoms $\rightarrow \mu = 0$.

COMMON TRAP Polar bonds don't imply a polar molecule.

4 Answer: (A) *Gibbs energy & spontaneity* **Medium** ⌚ 1.5 min **Concept**

DC-04

SOLUTION $\Delta G = \Delta H - T\Delta S = -100 + 0.2T$ (kJ). $\Delta G < 0$ when $T < 500$ K.

SHORTCUT Both negative $\rightarrow$ enthalpy-driven $\rightarrow$ spontaneous at low T , below $T = \Delta H/\Delta S$.

COMMON TRAP Unit mismatch: J vs kJ gives 0.5 K or 500 000 K.

5 Answer: (D) *pH of strong base* Easy 1 min Basic

DC-05

SOLUTION $[\text{OH}^-] = 10^{-3}$, so $\text{pOH} = 3$ and $\text{pH} = 14 - 3 = 11$.**SHORTCUT** $\text{pH} + \text{pOH} = 14$ at 25°C .**COMMON TRAP** Reporting pOH as pH .**6 Answer: 30** *First-order half-life* Easy 1 min NV

DC-06

SOLUTION 87.5% complete means 12.5% = $1/8$ remains, which is 3 half-lives: $3 \times 10 = 30$ min.**SHORTCUT** Remaining fraction = $(1/2)^n$ for first order.**COMMON TRAP** Using linear (zero-order) thinking.**7 Answer: (B)** *Ionisation enthalpy trend* Medium 1.5 min Concept

DC-07

SOLUTION Increases across the period, with one exception: N ($2p^3$, half-filled) $>$ O ($2p^4$). So $\text{B} < \text{C} < \text{O} < \text{N}$.**SHORTCUT** Period-2 exceptions: $\text{Be} > \text{B}$ and $\text{N} > \text{O}$.**COMMON TRAP** Forgetting the $\text{N} > \text{O}$ exception gives (A).**8 Answer: (C)** *Low-spin complexes* Medium 1.5 min Concept

DC-08

SOLUTION Fe^{3+} is d^5 . CN^- is a strong-field ligand, so the complex is low spin: $t_{2g}^5 e_g^0$, one unpaired electron. $\mu = \sqrt{n(n+2)} = \sqrt{3} \approx 1.73$ BM.**SHORTCUT** Strong-field ligands (CN^- , CO) pair up d^4 – d^7 in octahedral complexes.**COMMON TRAP** Treating it as high spin gives 5.92 BM.**9 Answer: (D)** *Carbocation stability* Easy 1 min Basic

DC-09

SOLUTION Stability grows with hyperconjugation and +I donation: $3^\circ > 2^\circ > 1^\circ >$ methyl.**SHORTCUT** More α -hydrogens (hyperconjugation) $\rightarrow$ more stable alkyl carbocation.**COMMON TRAP** None here; this is a speed item.**10 Answer: 4** *Structural isomers of alcohols* Medium 1.5 min NV

DC-10

SOLUTION Butan-1-ol, butan-2-ol, 2-methylpropan-1-ol, 2-methylpropan-2-ol. (The 3 ethers are not alcohols. Counting stereoisomers is not asked.)**SHORTCUT** Place $-\text{OH}$ on each distinct carbon of each C_4 skeleton: n-butane gives 2, isobutane gives 2.**COMMON TRAP** Including ethers gives 7.**DIAGNOSTIC · MATHEMATICS****1 Answer: (D)** *Composition of functions* Easy 1 min Basic

DM-01

SOLUTION $(f \circ g)(2) = f(g(2)) = f(4) = 11$.**SHORTCUT** Apply the inner function first.**COMMON TRAP** Computing $g(f(2)) = 49$.**2 Answer: (B)** *Modulus of a quotient* Easy 1 min Basic

DM-02

SOLUTION $|z_1/z_2| = |z_1|/|z_2| = 5/\sqrt{2}$.**SHORTCUT** Never rationalise when only the modulus is asked.**COMMON TRAP** Rationalising wastes time and invites arithmetic slips.**3 Answer: (A)** *Symmetric functions of roots* Easy 1 min Basic

DM-03

SOLUTION $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 25 - 12 = 13$.**SHORTCUT** Use Vieta's relations; don't solve for the roots.**COMMON TRAP** Forgetting the $-2\alpha\beta$ term.**4 Answer: (C)** *Determinant of adjoint* Medium 1 min Concept

DM-04

SOLUTION $|\text{adj } A| = |A|^{n-1} = 4^2 = 16$ for $n = 3$.**SHORTCUT** $|\text{adj } A| = |A|^{n-1}$ and $|\text{adj}(\text{adj } A)| = |A|^{(n-1)^2}$.**COMMON TRAP** Using $|A|^n = 64$.

5 Answer: 30 *Permutations with repetition* **Easy** ⌚ 1.5 min **NV**

DM-05

SOLUTION 5 letters with L repeated twice and E repeated twice: $\frac{5!}{2!2!} = 30$.

SHORTCUT Divide by the factorial of each repeated letter's count.

COMMON TRAP Dividing by 2! only once gives 60.

6 Answer: (D) *AP sum* **Easy** ⌚ 1 min **Basic**

DM-06

SOLUTION $S_{20} = \frac{20}{2}[2(3) + 19(4)] = 10 \times 82 = 820$.

SHORTCUT $S_n = \frac{n}{2}(\text{first} + \text{last})$; the last term is $3 + 76 = 79$, so 10×82 .

COMMON TRAP Using $20d$ instead of $19d$ gives 840.

7 Answer: (B) *Standard trigonometric limit* **Easy** ⌚ 1 min **Basic**

DM-07

SOLUTION $1 - \cos 2x = 2 \sin^2 x$, so the limit is $2 \left(\frac{\sin x}{x} \right)^2 \rightarrow 2$.

SHORTCUT $1 - \cos kx \approx \frac{k^2 x^2}{2}$ near 0.

COMMON TRAP Forgetting to square k (giving 1) or dropping the $\frac{1}{2}$ (giving 4).

8 Answer: (A) *Definite integral* **Easy** ⌚ 1 min **Basic**

DM-08

SOLUTION By King's property, $\int_0^{\pi/2} \sin^2 x \, dx = \int_0^{\pi/2} \cos^2 x \, dx$. Their sum is $\int_0^{\pi/2} 1 \, dx = \pi/2$, so each is $\pi/4$.

SHORTCUT Average value of $\sin^2$ over a quarter period is $\frac{1}{2}$: $\frac{1}{2} \cdot \frac{\pi}{2}$.

COMMON TRAP Answering $\pi/2$ by forgetting the halving.

9 Answer: 2 *Distance of point from line* **Easy** ⌚ 1 min **NV**

DM-09

SOLUTION $d = \frac{|3(1) + 4(2) - 1|}{\sqrt{3^2 + 4^2}} = \frac{10}{5} = 2$.

SHORTCUT Substitute the point, take the modulus, divide by $\sqrt{a^2 + b^2}$.

COMMON TRAP Forgetting the constant term.

10 Answer: (B) *Classical probability* **Easy** ⌚ 1 min **Basic**

DM-10

SOLUTION Favourable pairs: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1), which is 6 of 36 outcomes. $P = 1/6$.

SHORTCUT Sum 7 is the most likely sum, with 6 outcomes.

COMMON TRAP Counting unordered pairs gives 3.

Interpreting your score

Score each subject out of 40 (+4 correct, -1 wrong, 0 blank).

Subject score (/40)	Starting level for that subject	What it tells you
32–40	Advanced start	Foundations are solid. Your gains will come from speed, PYQs and hard questions
20–31	Intermediate start	The core is there. Gaps are chapter-specific: use the self-audit below
below 20	Beginner start	Rebuild concepts first. Don't start mocks until the chapter tests reach 70%

Total (/120)	Overall level to enter in your profile
90+	Advanced
55–89	Intermediate
below 55	Beginner

Also note:

- Your **strongest subject** is the one with the highest score *per minute*, not just the highest score.
- If you didn't finish in 60 minutes, you have a **speed gap**: add the daily calculation drill (page [394](#)).
- Two or more wrong answers in a subject *where you felt confident* means an **accuracy gap**: start the Mistake Book routine immediately.

Syllabus self-audit

Rate each unit honestly: **0** = not studied · **1** = read, weak · **2** = can solve standard questions · **3** = can solve JEE-level questions under time.

Physics

Unit	0	1	2	3
Units & Measurements	☐	☐	☐	☐
Kinematics	☐	☐	☐	☐
Laws of Motion	☐	☐	☐	☐
Work, Energy, Power	☐	☐	☐	☐
Rotational Motion	☐	☐	☐	☐
Gravitation	☐	☐	☐	☐
Solids & Liquids	☐	☐	☐	☐
Thermodynamics	☐	☐	☐	☐
Kinetic Theory	☐	☐	☐	☐
Oscillations & Waves	☐	☐	☐	☐
Electrostatics	☐	☐	☐	☐
Current Electricity	☐	☐	☐	☐
Magnetism	☐	☐	☐	☐
EMI & AC	☐	☐	☐	☐
EM Waves	☐	☐	☐	☐
Optics	☐	☐	☐	☐
Dual Nature	☐	☐	☐	☐
Atoms & Nuclei	☐	☐	☐	☐
Electronic Devices	☐	☐	☐	☐
Experimental Skills	☐	☐	☐	☐

Chemistry

Unit	0	1	2	3
Basic Concepts	☐	☐	☐	☐
Atomic Structure	☐	☐	☐	☐
Chemical Bonding	☐	☐	☐	☐
Thermodynamics	☐	☐	☐	☐
Solutions	☐	☐	☐	☐
Equilibrium	☐	☐	☐	☐
Redox & Electrochem	☐	☐	☐	☐
Chemical Kinetics	☐	☐	☐	☐
Periodicity	☐	☐	☐	☐
p-Block	☐	☐	☐	☐
d- & f-Block	☐	☐	☐	☐
Coordination Cpds	☐	☐	☐	☐
Purification & Analysis	☐	☐	☐	☐
GOC	☐	☐	☐	☐
Hydrocarbons	☐	☐	☐	☐
Haloalkanes/arenes	☐	☐	☐	☐
Oxygen compounds	☐	☐	☐	☐
Nitrogen compounds	☐	☐	☐	☐
Biomolecules	☐	☐	☐	☐
Practical Chemistry	☐	☐	☐	☐

Mathematics

Unit	0	1	2	3
Sets, Relations, Functions	☐	☐	☐	☐
Complex & Quadratics	☐	☐	☐	☐
Matrices & Determinants	☐	☐	☐	☐
Permutations & Comb.	☐	☐	☐	☐
Binomial Theorem	☐	☐	☐	☐
Sequences & Series	☐	☐	☐	☐
Limits, Cont., Diff.	☐	☐	☐	☐
Integral Calculus	☐	☐	☐	☐
Differential Equations	☐	☐	☐	☐
Coordinate Geometry	☐	☐	☐	☐
3D Geometry	☐	☐	☐	☐
Vector Algebra	☐	☐	☐	☐
Statistics & Probability	☐	☐	☐	☐
Trigonometry	☐	☐	☐	☐

PART I · START HERE

Study-Plan Selector

Answer three questions and you'll have your plan. Each plan in Part X comes in **6 h, 8 h and 10 h per day** variants.

- 1 Q1 · How many days to your exam?**
Count from today to the first day of your session
- 2 Q2 · How many hours can you study per day?**
Honest average, including school days
- 3 Q3 · What is your level?**
From the Diagnostic Test (page [22](#))
- 4 Your plan**
Read it off the tables below and print those pages

Step 1: days → plan

Days left	Plan	Where
170 or more	180-Day Plan	page 461
110–169	120-Day Plan	page 463
80–109	90-Day Plan	page 464
50–79	60-Day Plan	page 465
25–49	30-Day Plan	page 466
14–24	Final 14 Days	page 469
fewer than 14	Final 7 / 3 Days + Exam-Eve	page 470

NOTE For Gayathri, starting on 28 September 2026 (calculation)

- **Session 1 (from 22 Jan 2027):** 116 days → **120-Day Plan**, starting from its Day 5. Days 1–4 are the diagnostic and set-up days, done in the first weekend.
- **Session 2 (expected early April 2027, not yet announced):** about 185 days → **180-Day Plan**, with Session 1 used as a full-length "real mock" at its Day ~116.
- Recommended: **follow the 120-Day Plan to Session 1, then the "Session 1 → Session 2 bridge"** (the last block of the 180-Day Plan) up to April.

Step 2: hours → variant

Honest daily average	Variant	Notes
up to 6.5 h	6 h	School-going Class 12 on weekdays; mocks on Sundays
6.5 – 9 h	8 h	Most common; school + evening study, or dropper with coaching
9 h or more	10 h	Dropper or holiday period. Not more than 10 h: beyond that, retention and sleep suffer
School weekdays + free weekends	Hybrid: 6 h Mon–Fri, 10 h Sat–Sun	Averages about 7.1 h/day. The plans mark which tasks move to the weekend

⚠ WARNING Sleep isn't negotiable

Every variant keeps **8 hours of sleep** (7.5 minimum on the occasional late night). For 13–18-year-olds, the American Academy of Sleep Medicine recommends 8–10 hours [LS-AASM]. Memory consolidation happens during sleep: cutting sleep to add study hours reliably *reduces* what you retain.

Step 3: level → modifications

Level	Modification
Beginner	Choose the <i>next longer</i> plan if your days allow. Otherwise drop Low-frequency topics (page 402) until Phase 3 and double the Basic-question practice in Phase 1
Intermediate	Use the plan as written
Advanced	Compress Phase 1 by 25%. Use the saved days for extra full mocks and Hard-question sets

Step 4: weak-subject weighting

Default split of *problem-solving time* is Physics 34% · Chemistry 30% · Maths 36%. Adjust by 10 percentage points towards your weakest subject:

Weakest subject	Physics	Chemistry	Maths
none / balanced	34%	30%	36%
Maths	30%	25%	45%
Physics	44%	25%	31%
Chemistry	30%	40%	30%

My selection

Days left today	
Plan chosen	
Hours variant	
Level modification	
Weak-subject weighting	
Start date	
Re-check date (after 5th mock)	

Physics



Strategy for maximising Physics marks, then 20 unit modules: concepts, formulas, graphs, PYQ patterns, traps, shortcuts, practice sets and chapter tests.

•	Physics: The Complete Strategy	33
01	Units & Measurements	36
02	Kinematics	43
03	Laws of Motion	50
04	Work, Energy & Power	57
05	Rotational Motion & Centre of Mass	63
06	Gravitation	70
07	Properties of Solids & Liquids (incl. Thermal Properties)	76
08	Thermodynamics	84
09	Kinetic Theory of Gases	90
10	Oscillations & Waves	95

11	Electrostatics & Capacitors	101
12	Current Electricity	108
13	Magnetic Effects of Current & Magnetism	114
14	Electromagnetic Induction & Alternating Current	121
15	Electromagnetic Waves	128
16	Optics: Ray & Wave	132
17	Dual Nature of Matter & Radiation	139
18	Atoms & Nuclei	144
19	Electronic Devices	150
20	Experimental Skills	155

PART II · PHYSICS

Physics: The Complete Strategy

What Physics in JEE Main actually tests

JEE Main Physics is mostly "**one idea + clean calculation**". A typical question identifies one principle (conservation of energy, Gauss's law, lens formula, Bohr's model), sets up a standard situation, and checks whether you can apply it quickly and without sign or unit errors. Long multi-concept problems do appear, but a minority of each paper is truly hard. The rest rewards **formula fluency, standard-model recognition and calculation discipline**.

||| **THIRD-PARTY ANALYSIS** [T-ANALYSIS26]

✓ VERIFIED FACT Syllabus structure (official)

The official Physics syllabus has **20 units**. Units 1–19 are theory. **Unit 20, Experimental Skills**, lists 18 standard experiments (Vernier callipers, screw gauge, metre bridge, resonance tube, and others). The syllabus document states that the theory part carries 80% weightage and the experimental part 20% [NTA-SYL]. In practice, directly lab-based questions have been fewer than this per shift, but the experiments' underlying physics (errors, least count, graphs) is tested often. ||| **THIRD-PARTY ANALYSIS**

Unit-wise weight and priority

"Typical questions per shift" is a range compiled from published PYQ analyses of 2024–2026 papers

||| **THIRD-PARTY ANALYSIS** [T-WT-VED], [T-WT-CD], [T-PHY26]. **Priority** is this book's recommendation

🕒 **RECOMMENDATION**, balancing frequency, difficulty and the time needed to master the unit.

#	Unit	Typical Q / shift	Difficulty	Priority	Study hours (Intermediate)
1	Units & Measurements (+ errors)	1–2	Easy	Must-do	6
2	Kinematics	1	Medium	High	10
3	Laws of Motion	1	Medium	High	10
4	Work, Energy & Power (+ collisions)	1–2	Medium	High	8
5	Rotational Motion (+ centre of mass)	1–2	Hard	Must-do	16
6	Gravitation	1–2	Easy–Med	Must-do	7
7	Properties of Solids & Liquids (+ thermal properties)	2–3	Medium	Must-do	14
8	Thermodynamics	1–2	Medium	Must-do	9
9	Kinetic Theory of Gases	1	Easy	Must-do	5
10	Oscillations & Waves	1–2	Medium	High	12
11	Electrostatics (+ capacitors)	2–3	Medium	Must-do	16
12	Current Electricity	2–3	Easy–Med	Must-do	12
13	Magnetic Effects & Magnetism	2–3	Medium	Must-do	14
14	EMI & AC	1–2	Medium	High	12

#	Unit	Typical Q / shift	Difficulty	Priority	Study hours (Intermediate)
15	Electromagnetic Waves	1	Easy	Must-do	3
16	Optics (ray + wave)	2–3	Medium	Must-do	16
17	Dual Nature of Matter & Radiation	1	Easy	Must-do	4
18	Atoms & Nuclei	1–2	Easy	Must-do	6
19	Electronic Devices	1–2	Easy	Must-do	5
20	Experimental Skills	0–2 direct	Easy	High	6

☆ **IMPORTANT** The "easy block"

Units **1, 9, 15, 17, 18, 19** (plus the direct parts of 6 and 12) together usually give **5–8 questions a shift** that are formula-direct and fast. This is the highest-return study time in Physics: about 30 hours of work for roughly 20–30 marks. Master it first if Physics is your weak subject.

How to study a Physics chapter (the 5-pass method) 📌 RECOMMENDATION

- Pass 1 · Understand (NCERT + module core theory).** Read NCERT for the chapter, then this module's *Concept summary*. Write the formula box by hand once.
- Pass 2 · Model bank.** List the chapter's standard models (block on incline, rolling body, charged sphere, LCR circuit). For each, write *the one equation you start from*.
- Pass 3 · Practice.** Do the module's practice set timed, then 30–60 questions from a standard question bank, then PYQs.
- Pass 4 · Error harvest.** Every wrong question goes into the Mistake Book with its type.
- Pass 5 · Compress.** Reduce the chapter to its half-page Formula Handbook entry and its trap list. Revise from those only.

Using NCERT for Physics

NCERT Physics isn't sufficient on its own for JEE-level problem solving, but it's **necessary** for four things:

- **Definitions and statements** (Lenz's law, Huygens' principle, Brewster's law, Kepler's laws): statement-type questions come from here.
- **Semiconductors, EM waves, Dual Nature, Atoms, Nuclei:** NCERT depth is close to JEE Main depth in these chapters.
- **Units & Measurements** (significant figures, error rules) and the **Lab Manual**, which covers the Unit 20 experiments.
- **Graphs and tables** printed in NCERT (binding-energy curve, stress–strain curve, I–V characteristics, EM spectrum table).

Physics: how to maximise marks

Strong student (Physics $\geq$ 75/100 in mocks)

- Aim for 22–23 attempts at 92%+ accuracy.
- Your losses are silly: signs, unit conversions, reading "radius" as "diameter". Keep a 10-item personal trap list and read it before every mock.
- Practise hard Rotation, EMI and Electrostatics *mixed* problems. These separate 90 from 75.
- Do Section B (NV) Physics first. NV questions are often direct, and strong students can bank them.

Average student (50–75)

- Master the easy block first (above): it's worth 20–30 marks.
- For every medium chapter, learn the **three most-asked models** only. Don't collect tricks.
- In the exam, do Round 1 on "formula $\rightarrow$ numbers" questions only. Leave all multi-concept questions for Round 2.
- Weekly target: 150 Physics questions, with 40 of them PYQs.

Weak student (< 50)

- Physics is where the formula handbook pays off most. Daily 20-minute blank-sheet formula recall.
- Chapter order: Units $\rightarrow$ Modern (17, 18, 19) $\rightarrow$ EM Waves $\rightarrow$ KTG $\rightarrow$ Current $\rightarrow$ Electrostatics $\rightarrow$ Optics $\rightarrow$ Thermodynamics $\rightarrow$ Gravitation, then Mechanics.
- Attempt 15–17 questions, all Easy/Medium, at 90%+ accuracy. That alone is 55–65 marks.
- Use dimensional analysis and limiting cases (Speed Manual) to verify answers.

Physics exam-hall rules RECOMMENDATION

- **Read the last line first.** Is it asking for speed or velocity, total or net, radius or diameter, in cm or m?
- **Write the "given" in SI units** before any algebra.
- **Estimate before you compute.** A rough order of magnitude eliminates 1–2 options in most numerical MCQs.
- **Two-minute rule.** No progress in 2 minutes means mark for review and move on.
- **Section B:** recompute once, using a different path where possible (energy vs force, for example), before entering the number.

PART II · PHYSICS · UNIT 01

Units & Measurements

01

1-2 per shift

TYPICAL QUESTIONS

Easy

DIFFICULTY

Must-do

PRIORITY

Class 11 · Ch 1

NCERT

~6 hours

STUDY TIME

Concept summary

- Every physical quantity = **numerical value** × **unit**. The SI system has **7 base units**. Everything else is derived.
- Dimensions** express a quantity in terms of $[M]$, $[L]$, $[T]$, $[A]$, $[K]$, $[\text{mol}]$, $[\text{cd}]$. They are used to check equations, derive relations and convert units.
- Significant figures** record how precisely something was measured. **Errors** measure how far a measurement may be from the true value.
- JEE Main asks almost only four things here: *dimensions of an unfamiliar constant*, *error propagation*, *significant figures / rounding*, and *instrument readings* (Vernier, screw gauge; see Unit 20).

Core theory

SI base units

Quantity	Unit	Symbol	Dimension
Length	metre	m	$[L]$
Mass	kilogram	kg	$[M]$
Time	second	s	$[T]$
Electric current	ampere	A	$[A]$
Thermodynamic temperature	kelvin	K	$[K]$
Amount of substance	mole	mol	$[\text{mol}]$
Luminous intensity	candela	cd	$[\text{cd}]$

Plane angle (radian) and solid angle (steradian) are **dimensionless**, but they do have units. Since the 2019 SI revision, base units are defined by fixing exact values of constants: h , e , k_B , N_A , c and others.

Dimensions you must know by heart

Quantity	Dimension	Quantity	Dimension
Force	$[MLT^{-2}]$	Pressure, stress, modulus, energy density	$[ML^{-1}T^{-2}]$
Work, energy, torque	$[ML^2T^{-2}]$	Power	$[ML^2T^{-3}]$
Momentum, impulse	$[MLT^{-1}]$	Angular momentum, h	$[ML^2T^{-1}]$
Gravitational constant G	$[M^{-1}L^3T^{-2}]$	Surface tension, spring constant	$[MT^{-2}]$
Coefficient of viscosity η	$[ML^{-1}T^{-1}]$	Moment of inertia	$[ML^2]$
Charge	$[AT]$	Potential (voltage)	$[ML^2T^{-3}A^{-1}]$
Resistance	$[ML^2T^{-3}A^{-2}]$	Capacitance	$[M^{-1}L^{-2}T^4A^2]$
Inductance	$[ML^2T^{-2}A^{-2}]$	Magnetic field B	$[MT^{-2}A^{-1}]$
Permittivity ϵ_0	$[M^{-1}L^{-3}T^4A^2]$	Permeability μ_0	$[MLT^{-2}A^{-2}]$
Electric field	$[MLT^{-3}A^{-1}]$	Magnetic flux	$[ML^2T^{-2}A^{-1}]$
Specific heat	$[L^2T^{-2}K^{-1}]$	Latent heat	$[L^2T^{-2}]$
Thermal conductivity	$[MLT^{-3}K^{-1}]$	Boltzmann constant	$[ML^2T^{-2}K^{-1}]$

⚡ **SHORTCUT** Build dimensions from a defining equation, never from memory alone

Unknown constant? Write the simplest equation it appears in and solve for it.

$$\epsilon_0: \text{from } F = \frac{q^2}{4\pi\epsilon_0 r^2} \text{ we get } [\epsilon_0] = \frac{[A^2 T^2]}{[MLT^{-2}][L^2]} = [M^{-1}L^{-3}T^4 A^2].$$

$$B: \text{from } F = qvB. L: \text{from } \epsilon = L dI/dt. \eta: \text{from } F = 6\pi\eta rv.$$

Time saved: 30–60 s per question, with far fewer memory errors.

Uses of dimensional analysis

1. **Checking** an equation: every added or subtracted term must have the same dimensions (principle of homogeneity).
2. **Deriving** a relation when the quantity depends on 3 others as a product of powers.
3. **Converting units:** $n_2 = n_1 \left[\frac{M_1}{M_2} \right]^a \left[\frac{L_1}{L_2} \right]^b \left[\frac{T_1}{T_2} \right]^c$ for a quantity of dimension $[M^a L^b T^c]$.

⚠ **MISTAKE ALERT** Limitations examiners test

Dimensional analysis **cannot**: find dimensionless constants ($2\pi, \frac{1}{2}$); derive relations that involve sums ($s = ut + \frac{1}{2}at^2$ as a whole) or trig/log/exp functions; tell apart quantities with the same dimensions (work vs torque). Arguments of sin, log, e^x **must be dimensionless**. That's often the key to the whole question.

Significant figures

- All non-zero digits count. Zeros **between** non-zeros count. **Leading** zeros never count (0.00520 has 3). **Trailing zeros after a decimal point** count (2.50 has 3).
- Trailing zeros in an integer without a decimal point are ambiguous. Use scientific notation: 4.50×10^3 has 3.
- **Multiplication/division:** the result has as many s.f. as the *least* precise factor.
- **Addition/subtraction:** the result has as many *decimal places* as the term with the fewest decimal places.
- **Rounding:** if the digit to drop is > 5 , round up; if < 5 , leave it; if exactly 5 (followed by nothing), make the preceding digit **even**.

Errors

Σ **FORMULA** Error formulas

Operation	Maximum error
$Z = A \pm B$	$\Delta Z = \Delta A + \Delta B$
$Z = AB$ or A/B	$\frac{\Delta Z}{Z} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$
$Z = \frac{A^p B^q}{C^r}$	$\frac{\Delta Z}{Z} = p \frac{\Delta A}{A} + q \frac{\Delta B}{B} + r \frac{\Delta C}{C}$
n readings	mean $\bar{a} = \frac{\sum a_i}{n}$, mean absolute error $\overline{\Delta a} = \frac{\sum a_i - \bar{a} }{n}$
Relative / percentage error	$\delta a = \frac{\overline{\Delta a}}{\bar{a}}$, percentage = $\delta a \times 100\%$

Errors always **add**, even in subtraction and division. That's the maximum-error convention used in JEE.

⚡ **SHORTCUT** Percentage error of a formula in one line

$g = 4\pi^2 L/T^2$: %error in g = % L + $2 \times$ % T . Multiply each percentage by the **absolute value of its power** and add. Constants ($4\pi^2$) contribute nothing.

When NOT to use: when a quantity appears in a sum, e.g. $Z = \frac{A}{A+B}$. There you must differentiate properly.

Instruments (details in Unit 20, page 155)

- **Vernier callipers:** least count = 1 MSD – 1 VSD. If N VSD = $(N - 1)$ MSD, then LC = MSD/ N .

- **Screw gauge:** least count = $\frac{\text{pitch}}{\text{number of circular-scale divisions}}$.
- **Zero error:** true reading = observed reading – (zero error), with the zero error's sign included.

Frequently tested concepts & question patterns

PYQ PATTERN Recurring structures (2024–2026 papers)

1. "Find the dimensions of a, b in $X = \dots$ " Van der Waals-type equations, $y = A \sin(\omega t - kx)$, $P = \frac{a - t^2}{bx}$. Use homogeneity term by term.
2. **Combination checks:** which of these has the dimensions of speed / time / energy? ($1/\sqrt{\mu_0 \epsilon_0}$, RC , L/R , $\sqrt{LC}$, hc/λ , $\epsilon_0 E^2$, B^2/μ_0)
3. **Error in a derived quantity** (density of a sphere, g from a pendulum, Young's modulus): the power rule.
4. **Significant-figure arithmetic** and rounding.
5. **Vernier/screw-gauge reading with zero error**, sometimes combined with an error calculation.
6. **New unit system:** "If the units of mass, length and time are changed to..., find the new value of 1 J."

Question classification

Type	Difficulty	Approach	Target time
Dimensions of a named quantity	E	Defining equation	45 s
Dimensions of constants in an unfamiliar formula	E–M	Homogeneity, term by term	1.5 min
Percentage error, power rule	E	One line	45 s
Error with a sum/difference inside	M	Differentiate / add absolute errors	2 min
Instrument reading + zero error	E–M	LC first, then sign of zero error	1.5 min
Dimensional derivation (3 variables)	M	Solve 3 equations in M, L, T	2 min

Selection rule: almost always Round-1 questions. Skip only if the unfamiliar formula is very long.

Common mistakes

MISTAKE ALERT Mistake alerts

- Treating an angle as having a dimension. It's dimensionless, even though it has a unit (rad).
- Subtracting errors in $A - B$. **Errors add.**
- Forgetting to multiply by the power: %error in r^3 is $3 \times \%r$.
- Getting the zero-error sign backwards. With the jaws closed, a reading *above* zero is a **positive** zero error, and you subtract it.
- Counting leading zeros as significant.
- Rounding intermediate results. Keep one extra digit and round at the end.

Practice questions

P01 · Practice

1 Easy 1 min Basic

P01-01

The dimensions of $\frac{1}{2}\epsilon_0 E^2$ (ϵ_0 = permittivity of free space, E = electric field) are:

- (A) $[MLT^{-2}]$ (B) $[ML^{-1}T^{-2}]$ (C) $[ML^2T^{-2}]$ (D) $[ML^{-2}T^{-1}]$

2 Medium 1.5 min Concept

P01-02

In $\left(P + \frac{a}{V^2}\right)(V - b) = RT$, the dimensions of a/b are those of:

- (A) energy (B) pressure (C) force (D) momentum

3 Easy 0.75 min Speed

P01-03

In $g = 4\pi^2 L/T^2$, the percentage errors in L and T are 1% and 2%. The maximum percentage error in g is:

- (A) 3% (B) 4% (C) 5% (D) 1%

4 Easy 1 min Basic

P01-04

The product 12.5×1.25 , reported to the correct number of significant figures, is:

- (A) 15.625 (B) 15.63 (C) 16 (D) 15.6

5 Medium 2 min JEE

P01-05

A Vernier calliper has 1 MSD = 1 mm and 10 VSD = 9 MSD. With the jaws closed, the Vernier zero lies to the **right** of the main-scale zero and the 3rd Vernier division coincides with a main-scale division. When a rod is measured, the main-scale reading is 32 mm and the 6th Vernier division coincides. The corrected diameter is:

- (A) 3.29 cm (B) 3.26 cm (C) 3.20 cm (D) 3.23 cm

6 Medium 2 min JEE

P01-06

A screw gauge has pitch 0.5 mm and 50 circular-scale divisions. With the studs closed, the reference line reads +4 divisions on the circular scale. Measuring a wire, the main-scale reading is 2.5 mm and the circular-scale reading is 36. The diameter of the wire is:

- (A) 2.86 mm (B) 2.90 mm (C) 2.82 mm (D) 2.54 mm

7 Medium 1.5 min NV

P01-07

In a new system, the unit of mass is 5 kg, the unit of length is 2 m and the unit of time is 2 s. What is the numerical value of 100 J in this system?

Your answer (numerical value):

8 Medium 2 min Concept

P01-08

The time period T of oscillation of a small liquid drop depends on its density ρ , radius r and surface tension σ . Dimensional analysis gives $T \propto$:

- (A) $\sqrt{\sigma/(\rho r^3)}$ (B) $\sqrt{\rho r^3/\sigma}$ (C) $\sqrt{\rho r/\sigma}$ (D) $\rho r^3/\sigma$

9 Medium 2 min Calc

P01-09

Five measurements of a time period are 1.52 s, 1.48 s, 1.55 s, 1.45 s and 1.50 s. The period, with its mean absolute error, is:

- (A) (1.50 ± 0.03) s (B) (1.50 ± 0.05) s (C) (1.51 ± 0.02) s (D) (1.50 ± 0.10) s

P01 · Chapter Test

1 Easy 0.75 min Speed

P01-T1

The dimensions of $\mu_0 \epsilon_0$ are:

- (A) $[L^2 T^{-2}]$ (B) $[L T^{-1}]$ (C) $[L^{-2} T^2]$ (D) dimensionless

2 Easy 0.5 min Basic

P01-T2

Which pair has the same dimensions?

- (A) power and energy (B) momentum and impulse per unit time
(C) stress and strain (D) torque and work

3 Easy 1 min Basic

P01-T3

 $V = (100 \pm 5) \text{ V}$ and $I = (10 \pm 0.2) \text{ A}$. The percentage error in $R = V/I$ is:

- (A) 7% (B) 5% (C) 3% (D) 2.5%

4 Easy 0.5 min NV

P01-T4

How many significant figures are there in 0.007020?

Your answer (numerical value):

5 Medium 1.5 min Concept

P01-T5

E = energy, J = angular momentum, M = mass, G = gravitational constant. The quantity $\frac{EJ^2}{M^5G^2}$ has the dimensions of:

- (A) length (B) an angle (dimensionless)
(C) mass (D) time

Answers & Solutions

P01 · Practice

1 (B) 2 (A) 3 (C) 4 (D) 5 (D) 6 (C) 7 20 8 (B) 9 (A)

P01 · Chapter Test

1 (C) 2 (D) 3 (A) 4 4 5 (B)

P01 · PRACTICE

1 Answer: (B) Dimensions of energy density Easy 1 min Basic

P01-01

SOLUTION $\frac{1}{2}\epsilon_0 E^2$ is the **energy per unit volume** of an electric field: $[ML^2T^{-2}]/[L^3] = [ML^{-1}T^{-2}]$.

SHORTCUT Energy density has the dimensions of pressure. Same for $B^2/2\mu_0$.

COMMON TRAP Answering the dimension of energy.

2 Answer: (A) Homogeneity in van der Waals equation Medium 1.5 min Concept

P01-02

SOLUTION a/V^2 must be a pressure, so $[a] = [PV^2]$. b must be a volume, so $[b] = [V]$. Then $[a/b] = [PV]$, which is energy (PV has units $\text{N m}^{-2} \times \text{m}^3 = \text{J}$).

SHORTCUT $a/b \sim PV \rightarrow \text{energy}$.

COMMON TRAP Stopping at " a has dimensions of PV^2 " and choosing pressure.

3 Answer: (C) Percentage error with powers Easy 0.75 min Speed

P01-03

SOLUTION $\frac{\Delta g}{g} = \frac{\Delta L}{L} + 2\frac{\Delta T}{T} = 1\% + 4\% = 5\%$.

SHORTCUT Power rule: add |power| $\times$ %error.

COMMON TRAP Using -2 and subtracting gives 3%.

4 Answer: (D) Significant figures in multiplication Easy 1 min Basic

P01-04

SOLUTION Both factors have 3 s.f., so the result keeps 3 s.f.: $15.625 \rightarrow 15.6$.

SHORTCUT Multiplication: keep the least number of s.f.

COMMON TRAP Applying the decimal-place rule, which is for addition only.

5 Answer: (D) Vernier reading with zero error **Medium** ⌚ 2 min JEE P01-05

SOLUTION $LC = 1 \text{ MSD} - 1 \text{ VSD} = 1 - 0.9 = 0.1 \text{ mm}$. Zero error = $+3 \times 0.1 = +0.3 \text{ mm}$ (zero to the right means positive).
Observed = $32 + 6(0.1) = 32.6 \text{ mm}$. Corrected = $32.6 - 0.3 = 32.3 \text{ mm} = 3.23 \text{ cm}$.

SHORTCUT Corrected = observed – (signed zero error).

COMMON TRAP Adding the positive zero error gives 3.29 cm.

6 Answer: (C) Screw gauge with zero error **Medium** ⌚ 2 min JEE P01-06

SOLUTION $LC = 0.5/50 = 0.01 \text{ mm}$. Zero error = $+4 \times 0.01 = +0.04 \text{ mm}$. Observed = $2.5 + 36(0.01) = 2.86 \text{ mm}$. Corrected = $2.86 - 0.04 = 2.82 \text{ mm}$.

SHORTCUT Observed – zero error.

COMMON TRAP Forgetting the correction gives 2.86.

7 Answer: 20 Conversion to a new unit system **Medium** ⌚ 1.5 min NV P01-07

SOLUTION New energy unit = $(5 \text{ kg})(2 \text{ m})^2/(2 \text{ s})^2 = 5 \text{ J}$. So $100 \text{ J} = 100/5 = 20$ new units.

SHORTCUT Compute the size of the new unit, then divide.

COMMON TRAP Inverting the ratio gives 500.

8 Answer: (B) Dimensional derivation **Medium** ⌚ 2 min Concept P01-08

SOLUTION Let $T = k\rho^a r^b \sigma^c$. $[\sigma] = [MT^{-2}]$. M: $a + c = 0$. L: $-3a + b = 0$. T: $-2c = 1$. So $c = -\frac{1}{2}$, $a = \frac{1}{2}$, $b = \frac{3}{2}$, giving $T \propto \sqrt{\rho r^3/\sigma}$.

SHORTCUT The time dimension comes only from σ , so σ must sit under a square root in the denominator. That leaves (B) or (C), and only (B) balances M and L .

COMMON TRAP Taking $[\sigma] = [MLT^{-2}]$ (the dimensions of force).

9 Answer: (A) Mean absolute error **Medium** ⌚ 2 min Calc P01-09

SOLUTION Mean = $7.50/5 = 1.50 \text{ s}$. Deviations: 0.02, 0.02, 0.05, 0.05, 0.00. Mean absolute error = $0.14/5 = 0.028 \approx 0.03 \text{ s}$.

SHORTCUT Round the error to the same decimal place as the mean.

COMMON TRAP Using the largest deviation (0.05) instead of the mean deviation.

P01 · CHAPTER TEST

1 Answer: (C) Speed of light from constants **Easy** ⌚ 0.75 min Speed P01-T1

SOLUTION $c = 1/\sqrt{\mu_0 \epsilon_0}$, so $\mu_0 \epsilon_0 = 1/c^2$, which has dimensions $[L^{-2}T^2]$.

SHORTCUT Memorise: $1/\sqrt{\mu_0 \epsilon_0} = \text{speed}$.

COMMON TRAP Answering the dimension of c^2 .

2 Answer: (D) Same dimensions **Easy** ⌚ 0.5 min Basic P01-T2

SOLUTION Torque ($r \times F$) and work ($F \cdot d$) are both $[ML^2T^{-2}]$. Impulse per unit time is force. Strain is dimensionless. Power is energy per unit time.

SHORTCUT Same dimensions don't mean the same physical quantity: torque is a vector, work a scalar.

COMMON TRAP Thinking momentum = impulse per unit time. Momentum = impulse.

3 Answer: (A) Error in a quotient **Easy** ⌚ 1 min Basic P01-T3

SOLUTION $\frac{\Delta R}{R} = \frac{5}{100} + \frac{0.2}{10} = 5\% + 2\% = 7\%$.

SHORTCUT Quotient: add the relative errors.

COMMON TRAP Subtracting (3%).

4 Answer: 4 Counting significant figures **Easy** ⌚ 0.5 min NV P01-T4

SOLUTION Leading zeros don't count. 7, 0 (between non-zeros), 2 and the trailing 0 after the decimal point do: 4 s.f.

SHORTCUT Start counting from the first non-zero digit.

COMMON TRAP Excluding the final zero gives 3.

5 Answer: (B) Dimensionless combination **Medium** ⌚ 1.5 min Concept P01-T5

SOLUTION $[EJ^2] = [ML^2T^{-2}][M^2L^4T^{-2}] = [M^3L^6T^{-4}]$. $[M^5G^2] = [M^5][M^{-2}L^6T^{-4}] = [M^3L^6T^{-4}]$. The ratio is dimensionless.

SHORTCUT Count powers of T first: numerator T^{-4} , denominator T^{-4} .

COMMON TRAP Arithmetic slip in the powers of G .

**REVISION** Revision tracker · Units & Measurements

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART II · PHYSICS · UNIT 02

Kinematics

02

~1 per shift
TYPICAL QUESTIONSMedium
DIFFICULTYHigh
PRIORITYClass 11 · Ch 2, 3
NCERT~10 hours
STUDY TIME

Concept summary

- **Kinematics** describes motion (position, velocity, acceleration) without asking what causes it.
- Everything comes from three definitions: $v = \frac{dx}{dt}$, $a = \frac{dv}{dt} = v \frac{dv}{dx}$, and the **slope/area** rules for graphs.
- For constant acceleration, the equations of motion are exact. For variable acceleration, **integrate**.
- 2D motion (projectiles, relative velocity, circular motion) = two independent 1D motions plus vector addition.

Core theory & definitions

Term	Meaning	Key point
Distance	Total path length (scalar)	Never decreases; $\geq$ displacement
Displacement	Change in position vector	Can be zero for non-zero distance
Average velocity	$\Delta \vec{r} / \Delta t$	Average speed = distance/time $\geq$ average velocity
Instantaneous velocity	$d\vec{r}/dt$	Always tangent to the path
Acceleration	$d\vec{v}/dt$	Can be non-zero at an instant where $v = 0$ (top of a vertical throw)

Formulas

Σ FORMULA Uniformly accelerated motion (1D)

$$v = u + at \quad s = ut + \frac{1}{2}at^2 \quad v^2 = u^2 + 2as \quad s = \frac{(u+v)}{2}t$$

$$s_{n^{\text{th}}} = u + \frac{a}{2}(2n - 1) \quad (\text{displacement in the } n^{\text{th}} \text{ second})$$

Free fall from rest: $h = \frac{1}{2}gt^2$, $v = \sqrt{2gh}$. **Thrown up with u :** $H_{\text{max}} = \frac{u^2}{2g}$, time to top = u/g , return time = $2u/g$.

From rest, distances in successive equal intervals are in the ratio 1 : 3 : 5 : 7 : ...

Σ FORMULA Variable acceleration

$$v = \int a \, dt \quad x = \int v \, dt \quad a = v \frac{dv}{dx} \Rightarrow \int v \, dv = \int a \, dx$$

Use $a = v \, dv/dx$ whenever a or v is given as a function of position.

Σ FORMULA Projectile motion (ground to ground, launch speed u at angle θ)

$$T = \frac{2u \sin \theta}{g} \quad H = \frac{u^2 \sin^2 \theta}{2g} \quad R = \frac{u^2 \sin 2\theta}{g} \quad R_{max} = \frac{u^2}{g} \quad (\theta = 45^\circ)$$

$$y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta} = x \tan \theta \left(1 - \frac{x}{R}\right)$$

- Complementary angles (θ and $90^\circ - \theta$) give the **same range**. $H_1/H_2 = \tan^2 \theta$ for the pair, and $H_1 H_2 = R^2/16$.
- $\frac{H}{R} = \frac{\tan \theta}{4}$ and $\frac{H}{T^2} = \frac{g}{8}$.
- Horizontal launch** from height h at speed u : $t = \sqrt{2h/g}$, range $= u\sqrt{2h/g}$.
- At the highest point: speed $= u \cos \theta$, acceleration $= g$ (downward), and velocity $\perp$ acceleration.

Σ FORMULA Relative velocity

$$\vec{v}_{AB} = \vec{v}_A - \vec{v}_B$$

River crossing (width d , boat speed v relative to water, river speed u):

- Minimum time: head straight across. $t_{min} = d/v$, drift $= ud/v$.
- Zero drift (shortest path, needs $v > u$): head upstream at $\sin \alpha = u/v$ from the perpendicular. Then $t = \frac{d}{\sqrt{v^2 - u^2}}$.

Rain-person problems: $\vec{v}_{\text{rain, rel. person}} = \vec{v}_{\text{rain}} - \vec{v}_{\text{person}}$. Hold the umbrella along the **relative** velocity.

Σ FORMULA Circular motion

$$v = \omega r \quad a_c = \frac{v^2}{r} = \omega^2 r \quad a_t = \frac{dv}{dt} = \alpha r \quad a_{net} = \sqrt{a_c^2 + a_t^2}$$

$\omega = 2\pi/T = 2\pi f$. In uniform circular motion, speed is constant but velocity is not, so the particle accelerates towards the centre.

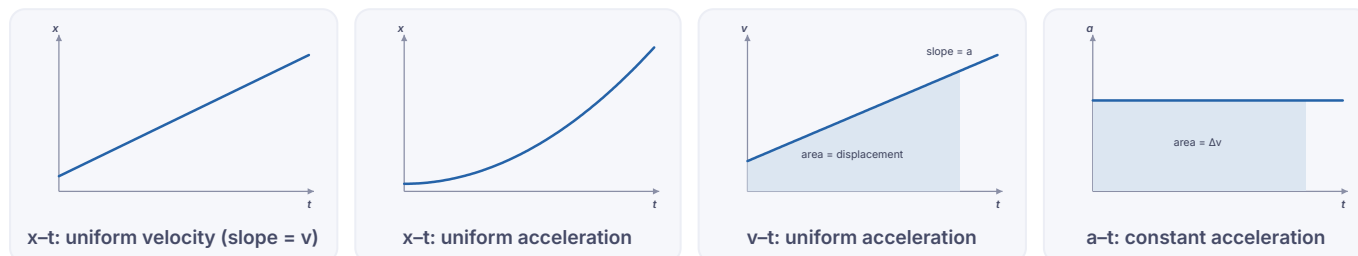
Σ FORMULA Vectors (used everywhere)

$$|\vec{A} + \vec{B}| = \sqrt{A^2 + B^2 + 2AB \cos \theta}; \text{ direction } \tan \alpha = \frac{B \sin \theta}{A + B \cos \theta} \text{ from } \vec{A}.$$

$$\vec{A} \cdot \vec{B} = AB \cos \theta; |\vec{A} \times \vec{B}| = AB \sin \theta; \text{ component of } \vec{A} \text{ along } \vec{B} = \vec{A} \cdot \hat{B}.$$

$$|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}| \iff \vec{A} \perp \vec{B}.$$

Graphs and how to read them



Graph	Slope gives	Area gives
$x-t$	velocity	—
$v-t$	acceleration	displacement (signed); distance = total unsigned area
$a-t$	jerk (not tested)	change in velocity

Graph	Slope gives	Area gives
$v-t$	dv/dx , so $a = v \times \text{slope}$	—

MISTAKE ALERT Graph traps

- A straight $x-t$ line means **constant velocity**, not constant acceleration.
- A curved $x-t$ graph bending upward means positive acceleration.
- On a $v-t$ graph, the area **below** the time axis is negative displacement. Distance counts it as positive.
- A $v-t$ graph can never be vertical (that would need infinite acceleration). An $x-t$ graph can never go backwards in time.

Standard models & assumptions

- Air resistance neglected; g constant (take 10 m s^{-2} unless told otherwise).
- Projectiles: horizontal velocity constant, vertical motion is free fall.
- "Particle" means size is ignored and there is no rotation.

Frequently tested concepts & question patterns

PYQ PATTERN Recurring structures

1. **Variable acceleration:** $x(t)$ or $v(t)$ given as a polynomial, asked for the distance (not displacement) in an interval. Find where $v = 0$ first.
2. **Projectile relations:** H vs R , complementary angles, equation of trajectory $\rightarrow u$ and θ .
3. **River-boat / rain-umbrella** relative velocity.
4. **Graph conversion:** given an $a-t$ or $v-t$ graph, find displacement or the shape of another graph.
5. n^{th} -**second distance** and ratio questions.
6. **Circular motion:** net acceleration with a tangential component.

Shortcuts

SHORTCUT Distance vs displacement for polynomial motion

If $v(t)$ changes sign inside the interval, split the interval at the root of $v = 0$. Compute $|x(t_1) - x(0)| + |x(t_2) - x(t_1)|$.

Time saved: avoids integrating $|v|$. **Common mistake:** forgetting that the particle may reverse direction.

SHORTCUT Trajectory equation $\rightarrow$ everything

Write $y = ax - bx^2$. Then $\tan \theta = a$, $R = a/b$, $H = a^2/(4b)$, and $u^2 = \frac{g}{2b \cos^2 \theta}$.

When NOT to use: when the launch point isn't at the origin.

Question classification

Type	Difficulty	Target time
Equations of motion, free fall	E	1 min
Projectile formulas / ratios	E-M	1.5 min
Variable acceleration with calculus	M	2–3 min
Relative motion (river, rain, two bodies)	M	2 min
Graph interpretation	E-M	1.5 min
Projectile on an incline, or two-body chase	H	3–4 min (Round 2)

Common mistakes

MISTAKE ALERT Mistake alerts

- Using equations of motion when acceleration is **not** constant.
- Sign convention: pick "up = +" and **stay with it**. With up positive, g enters as -10 .
- In river problems, confusing "shortest time" (head straight across) with "shortest path" (head upstream).
- Treating speed at the top of a projectile's flight as zero. It's $u \cos \theta$.

Practice questions

P02 · Practice

1 Easy 1 min Speed P02-01

A particle starts from rest with uniform acceleration and covers 15 m in the 3rd second. The distance it covers in the 5th second is:

(A) 21 m (B) 25 m (C) 27 m (D) 30 m

2 Medium 1.5 min Concept P02-02

A projectile's maximum height equals its horizontal range. The angle of projection is:

(A) $\tan^{-1} 4$ (B) $\tan^{-1} 2$ (C) 45° (D) $\tan^{-1}(1/4)$

3 Medium 1.5 min Concept P02-03

Rain falls vertically at 12 m s^{-1} . A woman walks east at 5 m s^{-1} . To protect herself she should hold her umbrella:

(A) vertically
 (B) tilted towards the east at $\tan^{-1}(12/5)$ from the vertical
 (C) tilted towards the west at $\tan^{-1}(5/12)$ from the vertical
 (D) tilted towards the east at $\tan^{-1}(5/12)$ from the vertical

4 Medium 1.5 min NV P02-04

A river 100 m wide flows at 3 m s^{-1} . A boat moves at 5 m s^{-1} relative to the water. Find the time (in s) to cross along the shortest path, landing directly opposite.

Your answer (numerical value):

5 Medium 2 min Tricky P02-05

A particle moves along the x -axis with $v = 3t^2 - 6t$ (SI units), starting at $x = 0$. The distance it travels in the first 3 s is:

(A) 0 m (B) 4 m (C) 8 m (D) 12 m

6 Easy 1 min Concept P02-06

A particle moves so that $v = 2\sqrt{x}$ (SI units). Its acceleration is:

(A) 1 m s^{-2} , constant (B) 2 m s^{-2} , constant
 (C) 4 m s^{-2} , constant (D) proportional to $\sqrt{x}$

7 Easy 1.5 min NV

P02-07

A ball is thrown horizontally at 20 m s^{-1} from the top of a 45 m tower ($g = 10 \text{ m s}^{-2}$). Find the horizontal distance (in m) from the foot of the tower where it lands.

Your answer (numerical value):

8 Medium 1.5 min JEE

P02-08

A particle moves on a circle of radius 2 m . Its speed increases at a constant 3 m s^{-2} . When its speed is 4 m s^{-1} , the magnitude of its acceleration is:

- (A) 11 m s^{-2} (B) $\sqrt{73} \text{ m s}^{-2}$ (C) 5 m s^{-2} (D) 8 m s^{-2}

9 Easy 0.5 min Speed

P02-09

If $|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$ for non-zero vectors, the angle between them is:

- (A) 90° (B) 45° (C) 0° (D) 180°

P02 · Chapter Test

1 Easy 1 min Basic

P02-T1

A stone dropped from height h reaches the ground in time T . At time $T/2$, its height above the ground is:

- (A) $h/2$ (B) $3h/4$ (C) $h/4$ (D) $h/8$

2 Easy 1 min Basic

P02-T2

Two projectiles are launched at the same speed at 30° and 60° . The ratio of their maximum heights $H_{30} : H_{60}$ is:

- (A) $1 : 1$ (B) $3 : 1$ (C) $1 : 3$ (D) $1 : \sqrt{3}$

3 Medium 1.5 min JEE

P02-T3

A projectile's trajectory is $y = \sqrt{3}x - \frac{x^2}{20}$ (SI units, $g = 10 \text{ m s}^{-2}$). Its speed of projection is:

- (A) 20 m s^{-1} (B) 10 m s^{-1} (C) $20\sqrt{3} \text{ m s}^{-1}$ (D) 40 m s^{-1}

4 Easy 1 min NV

P02-T4

A car moving at 20 m s^{-1} brakes uniformly and stops in 40 m . Find the magnitude of its deceleration (in m s^{-2}).

Your answer (numerical value):

5 Easy 0.5 min Speed

P02-T5

The angular speed of the minute hand of a clock is:

- (A) $\pi/30 \text{ rad s}^{-1}$ (B) $\pi/3600 \text{ rad s}^{-1}$ (C) $\pi/1800 \text{ rad s}^{-1}$ (D) $\pi/21600 \text{ rad s}^{-1}$

Answers & Solutions

P02 · Practice

1 (C) 2 (A) 3 (D) 4 25 5 (C) 6 (B) 7 60 8 (B) 9 (A)

P02 · Chapter Test

1 (B) 2 (C) 3 (A) 4 5 5 (C)

P02 · PRACTICE

1 Answer: (C) Distance in n th second Easy ⌚ 1 min Speed P02-01

SOLUTION $s_n = \frac{a}{2}(2n-1)$ for $u = 0$. $s_3 = \frac{5a}{2} = 15 \Rightarrow a = 6 \text{ m s}^{-2}$. $s_5 = \frac{9a}{2} = 27 \text{ m}$.

SHORTCUT $s_5 : s_3 = 9 : 5$, so $15 \times 9/5 = 27$.

COMMON TRAP Using $s = \frac{1}{2}at^2$ for the whole 5 s.

2 Answer: (A) $H = R$ condition Medium ⌚ 1.5 min Concept P02-02

SOLUTION $\frac{H}{R} = \frac{\tan \theta}{4} = 1 \Rightarrow \tan \theta = 4$.

SHORTCUT Memorise $H/R = \tan \theta/4$.

COMMON TRAP Guessing 45° (which gives the maximum range, not $H = R$).

3 Answer: (D) Rain-person relative velocity Medium ⌚ 1.5 min Concept P02-03

SOLUTION $\vec{v}_{\text{rain, woman}} = \vec{v}_{\text{rain}} - \vec{v}_{\text{woman}} = (-5\hat{i} - 12\hat{j}) \text{ m s}^{-1}$. The rain appears to come from the east, falling towards the west. The umbrella must face the incoming rain, so tilt it **forward (east)** at $\tan^{-1}(5/12)$ from the vertical. Relative speed = 13 m s^{-1} .

SHORTCUT Tilt the umbrella in the direction you are walking.

COMMON TRAP Measuring the angle from the horizontal gives (B).

4 Answer: 25 River crossing: zero drift Medium ⌚ 1.5 min NV P02-04

SOLUTION For zero drift the boat heads upstream so that its upstream component cancels the river: $5 \sin \alpha = 3$. The across-component is $\sqrt{25-9} = 4 \text{ m s}^{-1}$, so $t = 100/4 = 25 \text{ s}$.

SHORTCUT $t = d/\sqrt{v^2 - u^2}$.

COMMON TRAP Answering $100/5 = 20 \text{ s}$ (that's the minimum-time case, which has drift).

5 Answer: (C) Distance for velocity with sign change Medium ⌚ 2 min Tricky P02-05

SOLUTION $x = t^3 - 3t^2$. $v = 0$ at $t = 0$ and $t = 2 \text{ s}$. $x(2) = -4$, $x(3) = 0$. Distance = $|-4 - 0| + |0 - (-4)| = 8 \text{ m}$. (Displacement = 0.)

SHORTCUT Split at the root of $v = 0$.

COMMON TRAP Answering the displacement (0 m).

6 Answer: (B) Acceleration from $v(x)$ Easy ⌚ 1 min Concept P02-06

SOLUTION $a = v \frac{dv}{dx} = 2\sqrt{x} \cdot \frac{1}{\sqrt{x}} = 2 \text{ m s}^{-2}$.

SHORTCUT $v^2 = 4x$; compare with $v^2 = 2ax$, giving $a = 2$.

COMMON TRAP Differentiating v with respect to x and stopping there.

7 Answer: 60 Horizontal projection Easy ⌚ 1.5 min NV P02-07

SOLUTION Fall time $t = \sqrt{2h/g} = \sqrt{9} = 3 \text{ s}$. $x = ut = 60 \text{ m}$.

SHORTCUT Fall time depends only on height.

COMMON TRAP Using the projectile range formula $u^2 \sin 2\theta/g$ (which is for ground-to-ground launches).

8 Answer: (B) Net acceleration in non-uniform circular motion Medium ⌚ 1.5 min JEE P02-08

SOLUTION $a_c = v^2/r = 16/2 = 8$; $a_t = 3$. $a = \sqrt{64+9} = \sqrt{73} \approx 8.5 \text{ m s}^{-2}$.

SHORTCUT The two components are perpendicular: add in quadrature.

COMMON TRAP Adding them directly (11).

9 Answer: (A) Equal sum and difference Easy ⌚ 0.5 min Speed P02-09

SOLUTION Squaring: $A^2 + B^2 + 2\vec{A} \cdot \vec{B} = A^2 + B^2 - 2\vec{A} \cdot \vec{B} \Rightarrow \vec{A} \cdot \vec{B} = 0$.

SHORTCUT Diagonals of a parallelogram are equal only for a rectangle.

COMMON TRAP None; this is a pure speed item.

P02 · CHAPTER TEST

1 Answer: (B) Free fall at half time Easy ⌚ 1 min Basic P02-T1

SOLUTION Distance fallen $\propto t^2$: in $T/2$ it falls $h/4$. Height above ground = $3h/4$.

SHORTCUT Ratio 1 : 3 in two equal halves of the time.

COMMON TRAP Answering the distance fallen ($h/4$).

2 Answer: (C) Complementary angles **Easy** ⌚ 1 min **Basic**

P02-T2

SOLUTION $H \propto \sin^2 \theta$: $\sin^2 30^\circ : \sin^2 60^\circ = \frac{1}{4} : \frac{3}{4} = 1 : 3$. (The ranges are equal.)

SHORTCUT $H_1/H_2 = \tan^2 \theta_1$ for complementary angles: $\tan^2 30^\circ = 1/3$.

COMMON TRAP Confusing heights with ranges (1:1).

3 Answer: (A) Speed from trajectory equation **Medium** ⌚ 1.5 min **JEE**

P02-T3

SOLUTION $\tan \theta = \sqrt{3} \Rightarrow \theta = 60^\circ$, $\cos^2 \theta = \frac{1}{4}$. $\frac{g}{2u^2 \cos^2 \theta} = \frac{1}{20} \Rightarrow \frac{10}{u^2/2} = \frac{1}{20} \Rightarrow u^2 = 400$, so $u = 20 \text{ m s}^{-1}$.

SHORTCUT $u^2 = \frac{g}{2b \cos^2 \theta}$ with $b = 1/20$.

COMMON TRAP Forgetting $\cos^2 \theta$.

4 Answer: 5 Braking distance **Easy** ⌚ 1 min **NV**

P02-T4

SOLUTION $0 = u^2 - 2as \Rightarrow a = \frac{400}{80} = 5 \text{ m s}^{-2}$.

SHORTCUT Stopping distance $= u^2/2a$.

COMMON TRAP Forgetting the factor 2.

5 Answer: (C) Angular speed of a clock hand **Easy** ⌚ 0.5 min **Speed**

P02-T5

SOLUTION $\omega = 2\pi/T = 2\pi/3600 = \pi/1800 \text{ rad s}^{-1}$.

SHORTCUT Minute hand: $T = 1 \text{ h}$. Hour hand: $T = 12 \text{ h}$.

COMMON TRAP Using 60 s (the second hand).

REVISION Revision tracker · Kinematics

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART II · PHYSICS · UNIT 03

Laws of Motion

03

~1 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

High

PRIORITY

Class 11 · Ch 4

NCERT

~10 hours

STUDY TIME

Concept summary

- **Newton I:** a body stays at rest or in uniform motion unless a net external force acts. It defines inertial frames.
- **Newton II:** $\vec{F}_{net} = \frac{d\vec{p}}{dt}$, which is $m\vec{a}$ for constant mass. **Newton III:** forces come in equal and opposite pairs acting on *different* bodies.
- **Method for every problem:** isolate each body → draw a **free-body diagram (FBD)** → choose axes → write $\sum F = ma$ per axis → add the **constraint** relations (string length, contact) → solve.
- **Friction** adjusts itself up to $\mu_s N$ (static) and is $\mu_k N$ once sliding. **Circular motion** needs a net inward force mv^2/r supplied by real forces.

Formulas

Σ FORMULA Momentum & impulse

$$\vec{p} = m\vec{v} \quad \vec{J} = \int \vec{F} dt = \Delta\vec{p} \quad \text{No external force} \Rightarrow \vec{p}_{total} = \text{constant}$$

Recoil: $m_{bullet}v_{bullet} = M_{gun}V_{gun}$. For a rebound from a wall: $|\Delta p| = m(v_1 + v_2)$.

Σ FORMULA Standard systems

System	Acceleration	Tension / normal force
Atwood machine ($m_2 > m_1$)	$a = \frac{(m_2 - m_1)g}{m_1 + m_2}$	$T = \frac{2m_1m_2g}{m_1 + m_2}$
Block m_1 on smooth table pulled by hanging m_2	$a = \frac{m_2g}{m_1 + m_2}$	$T = \frac{m_1m_2g}{m_1 + m_2}$
Lift accelerating up at a	—	apparent weight $N = m(g + a)$
Lift accelerating down at a	—	$N = m(g - a)$; free fall: $N = 0$
Smooth incline	$a = g \sin \theta$	$N = mg \cos \theta$
Rough incline, sliding down	$a = g(\sin \theta - \mu_k \cos \theta)$	—
Pendulum in a car accelerating at a	string at $\tan \theta = a/g$ from the vertical	$T = m\sqrt{g^2 + a^2}$

Σ FORMULA Friction

$$f_s \leq \mu_s N \quad f_k = \mu_k N \quad \mu_k < \mu_s \quad \text{angle of repose: } \tan \theta_r = \mu_s \quad \text{angle of friction: } \tan \lambda = \mu_s$$

- Minimum force to move a block on a horizontal surface (pull at angle ϕ above the horizontal): $F_{min} = \frac{\mu mg}{\sqrt{1 + \mu^2}}$ at $\tan \phi = \mu$.
- Two blocks (top m on bottom M , force on the bottom block): they move together as long as $F \leq \mu_s(m + M)g$.

Σ FORMULA Circular dynamics

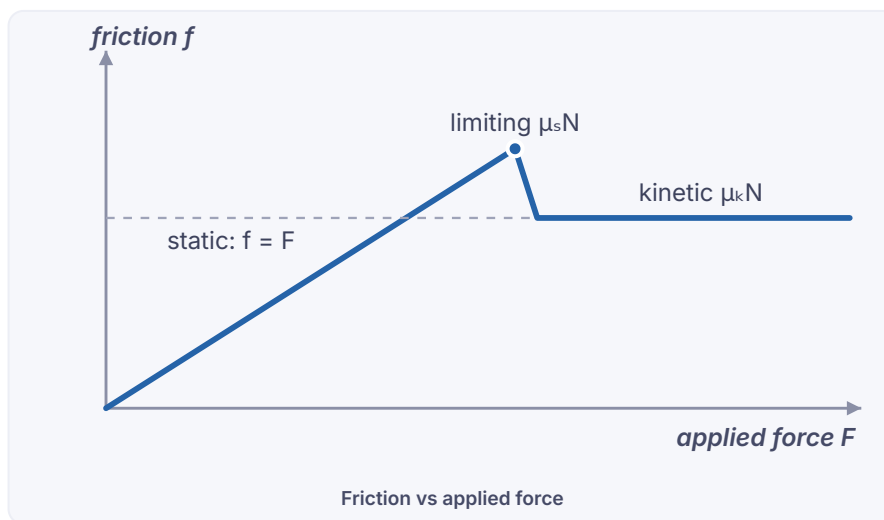
Level road: $v_{max} = \sqrt{\mu rg}$ Banked road, no friction: $v = \sqrt{rg \tan \theta}$

Banked road with friction: $v_{max} = \sqrt{rg \frac{\mu + \tan \theta}{1 - \mu \tan \theta}}$ $v_{min} = \sqrt{rg \frac{\tan \theta - \mu}{1 + \mu \tan \theta}}$

Conical pendulum (string length L , angle θ): $\tan \theta = \frac{v^2}{rg}$, $T = 2\pi \sqrt{\frac{L \cos \theta}{g}}$.

Σ FORMULA Equilibrium of concurrent forces

$\sum \vec{F} = 0$. **Lami's theorem** for three forces: $\frac{F_1}{\sin \alpha} = \frac{F_2}{\sin \beta} = \frac{F_3}{\sin \gamma}$, where each angle is the one **opposite** that force (between the other two).

**☆ IMPORTANT Friction is a self-adjusting force**

Below the limit, static friction equals exactly what's needed to prevent slipping. It is **not** $\mu_s N$. The value $\mu_s N$ is only the maximum. Always first check whether the body slips: compare the applied force (or the force needed for common motion) with $\mu_s N$.

Standard models & assumptions

- Strings are massless and inextensible (same tension throughout, same acceleration magnitude at both ends). Pulleys are massless and frictionless unless stated otherwise.
- "Smooth" means no friction. "Rough" means friction, with μ given.
- **Pseudo force** $-m\vec{a}_0$ acts on every body when you analyse from a frame accelerating at $\vec{a}_0$.

Frequently tested concepts & question patterns



PYQ PATTERN Recurring structures

1. **Two-block systems** with friction between blocks: find the maximum force for common motion, or the accelerations when slipping occurs.
2. **Incline + friction**: acceleration, minimum/maximum force to hold a block, time to slide down.
3. **Banked roads** and conical pendulums.
4. **Connected bodies over pulleys**: tension and acceleration, sometimes with a movable pulley ($a_1 = 2a_2$ constraint).
5. **Impulse from a force–time graph**: area under the $F-t$ graph.
6. **Equilibrium of a mass hung by two strings**: Lami's theorem or components.

Shortcuts



SHORTCUT System approach for acceleration

For connected bodies moving with the same acceleration magnitude: $a = \frac{(\text{net driving force})}{(\text{total mass})}$. Then find any tension from **one** body's FBD.

When NOT to use: when bodies can slip relative to each other. Check the friction limit first.

Time saved: about 1 minute per pulley problem.



SHORTCUT Pulley constraint in 5 seconds

For a movable pulley carrying block B, with the string's free end attached to A: $x_A + 2x_B = \text{const} \Rightarrow a_A = 2a_B$. Count the strings supporting the pulley.

Question classification

Type	Difficulty	Target time
Lift, recoil, impulse	E	1 min
Atwood / pulley system	E–M	1.5 min
Incline with friction	M	2 min
Banked road / conical pendulum	M	2 min
Two-block friction (slip check)	M–H	3 min
Variable force + pseudo force combined	H	3–4 min (Round 2)

Common mistakes



MISTAKE ALERT Mistake alerts

- Writing friction $= \mu_s N$ when the body doesn't move.
- Putting a Newton-III pair on the **same** FBD.
- Forgetting that $N \neq mg$ on an incline, or when a force is applied at an angle.
- Using the pseudo force in an inertial frame too, which counts it twice.
- Impulse on a rebound: $m(v_1 + v_2)$, **not** $m(v_1 - v_2)$.

Practice questions

P03 · Practice

1 Easy 1 min Basic P03-01

Masses of 3 kg and 5 kg hang from the two ends of a light string over a frictionless pulley ($g = 10 \text{ m s}^{-2}$). The tension in the string is:

- (A) 37.5 N (B) 30 N (C) 40 N (D) 50 N

2 Easy 0.75 min Speed P03-02

A 60 kg person stands on a scale in a lift that accelerates **downward** at 2 m s^{-2} ($g = 10$). The scale reads:

- (A) 720 N (B) 600 N (C) 120 N (D) 480 N

3 Medium 1.5 min Concept P03-03

A block slides down a 30° incline with $\mu_k = \frac{1}{2\sqrt{3}}$ ($g = 10 \text{ m s}^{-2}$). Its acceleration is:

- (A) 5 m s^{-2} (B) 2.5 m s^{-2} (C) 7.5 m s^{-2} (D) 1.25 m s^{-2}

4 Easy 1 min NV P03-04

A road of radius 50 m is banked at an angle θ with $\tan \theta = 0.2$. Find the speed (in m s^{-1}) at which a car needs no friction ($g = 10 \text{ m s}^{-2}$).

Your answer (numerical value):

5 Medium 2 min NV P03-05

A 2 kg block rests on a 4 kg block that lies on a frictionless floor. The coefficient of static friction between the blocks is 0.3. Find the maximum horizontal force (in N) that can be applied to the **lower** block so that the blocks move together ($g = 10 \text{ m s}^{-2}$).

Your answer (numerical value):

6 Easy 1 min Basic P03-06

A 0.2 kg ball hits a wall perpendicularly at 10 m s^{-1} and rebounds at 8 m s^{-1} . The magnitude of the impulse on the ball is:

- (A) 0.4 N s (B) 2.0 N s (C) 1.8 N s (D) 3.6 N s

7 Medium 1 min Concept P03-07

A small bob hangs from the roof of a car accelerating horizontally at $g/\sqrt{3}$. In equilibrium relative to the car, the string makes an angle with the vertical of:

- (A) 60° , towards the back of the car (B) 30° , towards the front of the car
(C) 30° , towards the back of the car (D) 45° , towards the back of the car

8 Medium 2 min JEE P03-08

A 10 kg mass hangs from two strings making angles of 30° and 60° with the horizontal ceiling ($g = 10 \text{ m s}^{-2}$). The tension in the string at 60° to the horizontal is:

- (A) $50\sqrt{3} \text{ N}$ (B) 50 N (C) 100 N (D) $100/\sqrt{3} \text{ N}$

9 Easy 0.75 min Speed

P03-09

A 5 kg gun fires a 50 g bullet at 400 m s^{-1} . The recoil speed of the gun is:

- (A) 0.4 m s^{-1} (B) 40 m s^{-1} (C) 4 m s^{-1} (D) 2 m s^{-1}

P03 · Chapter Test

1 Medium 1 min Tricky

P03-T1

A 5 kg block rests on a horizontal floor with $\mu_s = 0.5$ ($g = 10$). A horizontal force of 20 N is applied. The friction force on the block is:

- (A) 25 N (B) 20 N (C) 5 N (D) 0 N

2 Easy 0.5 min Speed

P03-T2

If $\mu_s = 1/\sqrt{3}$ between a block and an incline, the angle of repose is:

- (A) 15° (B) 45° (C) 60° (D) 30°

3 Easy 1 min Basic

P03-T3

The maximum safe speed on a level circular road of radius 40 m with $\mu = 0.4$ ($g = 10$) is:

- (A) 8 m s^{-1} (B) 16 m s^{-1} (C) $4\sqrt{10} \text{ m s}^{-1}$ (D) 40 m s^{-1}

4 Medium 1.5 min NV

P03-T4

A force $F = 6t \text{ N}$ acts on a 2 kg body initially at rest. Find its speed (in m s^{-1}) at $t = 2 \text{ s}$.

Your answer (numerical value):

5 Easy 1 min Basic

P03-T5

A 2 kg block on a smooth table is connected by a string over a pulley to a hanging 3 kg block ($g = 10$). The tension in the string is:

- (A) 12 N (B) 20 N (C) 30 N (D) 18 N

Answers & Solutions

P03 · Practice

1 (A) 2 (D) 3 (B) 4 10 5 18 6 (D) 7 (C) 8 (A) 9 (C)

P03 · Chapter Test

1 (B) 2 (D) 3 (C) 4 6 5 (A)

P03 · PRACTICE

1 Answer: (A) Atwood machine tension Easy 1 min Basic

P03-01

SOLUTION $T = \frac{2m_1m_2g}{m_1 + m_2} = \frac{2(3)(5)(10)}{8} = 37.5 \text{ N}$.

SHORTCUT T lies between the two weights (30 N and 50 N). It's their harmonic mean.

COMMON TRAP Taking T equal to the lighter weight (30 N).

2 Answer: (D) Apparent weight Easy 0.75 min Speed

P03-02

SOLUTION $N = m(g - a) = 60 \times 8 = 480 \text{ N}$.

SHORTCUT Accelerating down means you feel lighter.

COMMON TRAP Adding a gives 720 N.

3 Answer: (B) Rough incline acceleration **Medium** ⌚ 1.5 min **Concept**

P03-03

SOLUTION $a = g(\sin \theta - \mu_k \cos \theta) = 10 \left(\frac{1}{2} - \frac{1}{2\sqrt{3}} \cdot \frac{\sqrt{3}}{2} \right) = 10(0.5 - 0.25) = 2.5 \text{ m s}^{-2}$.

SHORTCUT Friction removes $\mu \cot \theta$ of the smooth-incline value: $\mu \cot 30^\circ = 0.5$, so half of $g \sin \theta = 5$.

COMMON TRAP Using $\sin \theta$ in the friction term.

4 Answer: 10 Banking speed **Easy** ⌚ 1 min **NV**

P03-04

SOLUTION $v = \sqrt{rg \tan \theta} = \sqrt{50 \times 10 \times 0.2} = \sqrt{100} = 10 \text{ m s}^{-1}$.

SHORTCUT Design speed of a banked road: $v^2 = rg \tan \theta$.

COMMON TRAP Using $\sin \theta$ instead of $\tan \theta$.

5 Answer: 18 Two-block common motion **Medium** ⌚ 2 min **NV**

P03-05

SOLUTION The upper block is accelerated only by friction, so $a_{\max} = \mu_s g = 3 \text{ m s}^{-2}$. For common motion $F_{\max} = (2 + 4)(3) = 18 \text{ N}$.

SHORTCUT $F_{\max} = \mu_s g (m + M)$ when pushing the lower block.

COMMON TRAP Using the lower block's mass alone (12 N), or μ times the total weight used differently.

6 Answer: (D) Impulse on rebound **Easy** ⌚ 1 min **Basic**

P03-06

SOLUTION $|\Delta p| = m(v_1 + v_2) = 0.2 \times 18 = 3.6 \text{ N s}$ (the velocity reverses).

SHORTCUT Rebound: add the speeds.

COMMON TRAP Subtracting the speeds gives 0.4 N s.

7 Answer: (C) Pendulum in accelerating car **Medium** ⌚ 1 min **Concept**

P03-07

SOLUTION In the car's frame, a pseudo force ma acts backward. $\tan \theta = a/g = 1/\sqrt{3} \Rightarrow \theta = 30^\circ$, with the bob hanging towards the **back**.

SHORTCUT The bob lags behind the acceleration.

COMMON TRAP Choosing "front".

8 Answer: (A) Equilibrium with two strings **Medium** ⌚ 2 min **JEE**

P03-08

SOLUTION Horizontal: $T_1 \cos 30^\circ = T_2 \cos 60^\circ \Rightarrow T_2 = \sqrt{3} T_1$. Vertical: $T_1 \sin 30^\circ + T_2 \sin 60^\circ = 100 \Rightarrow \frac{T_1}{2} + \frac{3T_1}{2} = 100 \Rightarrow T_1 = 50 \text{ N}$, $T_2 = 50\sqrt{3} \text{ N}$.

SHORTCUT The strings are perpendicular to each other ($30^\circ + 60^\circ = 90^\circ$), so $T_2 = W \cos 30^\circ$ and $T_1 = W \sin 30^\circ$.

COMMON TRAP Swapping which string carries more tension. The steeper string carries more.

9 Answer: (C) Recoil of a gun **Easy** ⌚ 0.75 min **Speed**

P03-09

SOLUTION $V = \frac{mv}{M} = \frac{0.05 \times 400}{5} = 4 \text{ m s}^{-1}$.

SHORTCUT Momentum conservation from rest.

COMMON TRAP Using 50 kg for 50 g.

P03 · CHAPTER TEST**1 Answer: (B)** Static friction below the limit **Medium** ⌚ 1 min **Tricky**

P03-T1

SOLUTION Limiting friction $= \mu_s N = 25 \text{ N} > 20 \text{ N}$, so the block doesn't move. Static friction balances the applied force: 20 N.

SHORTCUT Friction = applied force whenever there is no slipping.

COMMON TRAP Answering $\mu_s N$ (25 N).

2 Answer: (D) Angle of repose **Easy** ⌚ 0.5 min **Speed**

P03-T2

SOLUTION $\tan \theta_r = \mu_s = 1/\sqrt{3} \Rightarrow \theta_r = 30^\circ$.

SHORTCUT Angle of repose = angle of friction.

COMMON TRAP None.

3 Answer: (C) Level-road maximum speed **Easy** ⌚ 1 min **Basic**

P03-T3

SOLUTION $v = \sqrt{\mu r g} = \sqrt{0.4 \times 40 \times 10} = \sqrt{160} = 4\sqrt{10} \approx 12.6 \text{ m s}^{-1}$.

SHORTCUT $\sqrt{160}$ lies between 12 and 13.

COMMON TRAP Forgetting the square root.

4 Answer: 6 Impulse of variable force **Medium** ⌚ 1.5 min **NV**

P03-T4

SOLUTION $J = \int_0^2 6t \, dt = 3t^2 \Big|_0^2 = 12 \, \text{N s}$. $v = J/m = 6 \, \text{m s}^{-1}$.

SHORTCUT Impulse = area under the F - t graph (a triangle: $\frac{1}{2} \times 2 \times 12$).

COMMON TRAP Using F at $t = 2$ with $v = at$ (constant acceleration assumed): $12 \, \text{m s}^{-1}$.

5 Answer: (A) Table-pulley system **Easy** ⌚ 1 min **Basic**

P03-T5

SOLUTION $a = \frac{m_2 g}{m_1 + m_2} = \frac{30}{5} = 6 \, \text{m s}^{-2}$. $T = m_1 a = 12 \, \text{N}$.

SHORTCUT $T = \frac{m_1 m_2 g}{m_1 + m_2} = \frac{60}{5}$.

COMMON TRAP Taking T = the hanging weight (30 N).

REVISION Revision tracker · Laws of Motion

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART II · PHYSICS · UNIT 04

Work, Energy & Power

04

1-2 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

High

PRIORITY

Class 11 · Ch 5

NCERT

~8 hours

STUDY TIME

Concept summary

- **Work-energy theorem:** $W_{net} = \Delta K$. This is the single most powerful problem-solving tool in mechanics. It skips accelerations and time entirely.
- **Conservative forces** (gravity, spring, electrostatic) store potential energy: $F = -\frac{dU}{dx}$. If only conservative forces do work, $K + U$ is constant.
- **Power** is the rate of doing work: $P = \vec{F} \cdot \vec{v}$.
- **Collisions:** momentum is always conserved (no external impulse). Kinetic energy is conserved only in elastic collisions.

Formulas

Σ FORMULA Work & energy

$$W = \vec{F} \cdot \vec{s} = F s \cos \theta \quad W = \int \vec{F} \cdot d\vec{r} \text{ (area under } F-x \text{)} \quad K = \frac{1}{2}mv^2 = \frac{p^2}{2m}$$

$$U_{gravity} = mgh \quad U_{spring} = \frac{1}{2}kx^2 \quad W_{spring} = \frac{1}{2}k(x_1^2 - x_2^2)$$

$$W_{all \text{ forces}} = \Delta K \quad W_{non-conservative} = \Delta(K + U) \quad P = \frac{dW}{dt} = \vec{F} \cdot \vec{v}$$

Units: 1 kWh = 3.6×10^6 J; 1 hp = 746 W; 1 eV = 1.6×10^{-19} J.

Σ FORMULA Potential energy curves

$F = -dU/dx$. Equilibrium where $dU/dx = 0$: **stable** if $d^2U/dx^2 > 0$ (minimum of U), **unstable** if < 0 (maximum), **neutral** if U is flat. A particle with total energy E is confined to where $U(x) \leq E$ (turning points at $U = E$).

Σ FORMULA Vertical circle (string / light rod, radius R)

Case	Minimum speed at the lowest point	Minimum speed at the top
String (or inside a track)	$\sqrt{5gR}$	$\sqrt{gR}$
Light rod (or tube)	$\sqrt{4gR}$	0

$T_{bottom} - T_{top} = 6mg$ (string, full circle). If $\sqrt{2gR} < u < \sqrt{5gR}$ at the bottom, the string goes slack somewhere above the horizontal diameter. If $u \leq \sqrt{2gR}$, the bob only oscillates below the horizontal diameter.

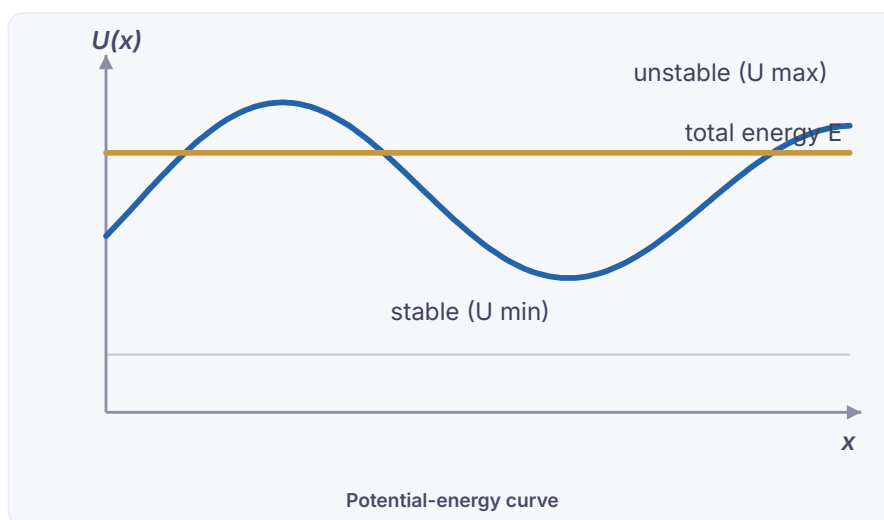
Σ FORMULA Collisions

Coefficient of restitution: $e = \frac{\text{velocity of separation}}{\text{velocity of approach}}$ (along the line of impact). Elastic: $e = 1$. Perfectly inelastic: $e = 0$.

1D elastic, m_2 initially at rest:

$$v_1 = \frac{m_1 - m_2}{m_1 + m_2} u_1 \quad v_2 = \frac{2m_1}{m_1 + m_2} u_1$$

- Equal masses exchange velocities. A heavy body hitting a light one at rest: the light one leaves at about $2u_1$.
- **KE lost in any 1D collision:** $\Delta K = \frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} (u_1 - u_2)^2 (1 - e^2)$.
- **Perfectly inelastic, m_2 at rest:** fraction of KE lost = $\frac{m_2}{m_1 + m_2}$.
- **Bouncing ball:** $h_n = e^{2n} h_0$, $e = \sqrt{h_1/h_0}$.
- **Oblique elastic collision of equal masses (one at rest):** they move off at 90° to each other.



Standard models & assumptions

- Springs are massless and obey Hooke's law. "Smooth" surfaces do no work (the normal force $\perp$ displacement).
- The normal force and tension in a vertical circle do no work. Only gravity changes the speed.
- In collisions, impulsive forces dominate: ignore gravity and friction *during* the brief impact.

Frequently tested concepts & question patterns

▮ PYQ PATTERN Recurring structures

1. **Work by a variable force** $F(x)$ or from an F - x graph (area).
2. **Spring-block** energy: maximum compression, speed at a given compression.
3. **Vertical circle:** minimum speed, tension at a point, where the string slackens.
4. **Collisions:** final velocities, KE loss, coefficient of restitution from bounce heights.
5. **Power:** vehicle on an incline at constant speed; pumping water; $P = Fv$ with variable force.
6. $U(x) \rightarrow$ **force and equilibrium type**.
7. **Kinetic energy-momentum relation:** percentage changes, $p = \sqrt{2mK}$.

Shortcuts

SHORTCUT Energy method first

If the question asks for a **speed, height or compression** (not a time), use energy conservation or the work–energy theorem straight away.

Time saved: 1–2 minutes compared with a force/kinematics approach. **When NOT to use:** when the question asks for time or acceleration, or when the path matters for a non-conservative force whose work you can't easily compute.

SHORTCUT Percentage changes

$K \propto p^2$. If p rises by $x\%$ (small), K rises by about $2x\%$. For large changes use exact ratios: $K \times 4 \Rightarrow p \times 2$ (a 300% rise in K is a 100% rise in p).

Question classification

Type	Difficulty	Target time
Work from $F(x)$ or graph area	E	1 min
Spring–block energy	E	1 min
Power on an incline	E–M	1.5 min
1D collision velocities / KE loss	M	2 min
Vertical circle tension/slack point	M–H	2.5 min
2D collision, or collision + spring combined	H	3–4 min (Round 2)

Common mistakes

MISTAKE ALERT Mistake alerts

- Taking work by friction as $\mu mg \times d$ on an incline. Use $\mu mg \cos \theta \times d$.
- Sign errors in W_{spring} when the spring is stretched from x_1 to x_2 .
- Conserving KE in an inelastic collision.
- For a rod in a vertical circle, using the string condition $\sqrt{5gR}$.
- Forgetting that $P = Fv$ uses the **force exerted by the engine**, not the net force.

Practice questions

P04 · Practice

1 Easy 1 min NV

P04-01

A force $F = 3x^2$ N acts along the x -axis. Find the work done (in J) as a body moves from $x = 0$ to $x = 2$ m.

Your answer (numerical value):

2 Easy 1 min Basic

P04-02

A 2 kg block moving at 4 m s^{-1} on a smooth floor hits a spring of constant 200 N m^{-1} . The maximum compression is:

- (A) 0.4 m (B) 0.2 m (C) 0.8 m (D) 0.04 m

3 Medium 1.5 min JEE

P04-03

A 1000 kg car climbs an incline with $\sin \theta = 1/20$ at a constant 10 m s^{-1} against a resistive force of 500 N ($g = 10$). The engine's power is:

- (A) 10 kW (B) 5 kW (C) 15 kW (D) 20 kW

4 Medium 1 min Basic

P04-04

A bob on a light string of length 0.9 m is to complete a vertical circle ($g = 10$). The minimum speed needed at the lowest point is:

- (A) 3 m s^{-1} (B) 6 m s^{-1} (C) $3\sqrt{2} \text{ m s}^{-1}$ (D) $3\sqrt{5} \text{ m s}^{-1}$

5 Easy 1 min Basic

P04-05

A 2 kg ball moving at 6 m s^{-1} collides elastically head-on with a stationary 4 kg ball. The velocity of the 4 kg ball afterwards is:

- (A) 2 m s^{-1} (B) 3 m s^{-1} (C) 4 m s^{-1} (D) 6 m s^{-1}

6 Medium 1.5 min NV

P04-06

A 1 kg block moving at 10 m s^{-1} collides with and sticks to a stationary 4 kg block. Find the percentage of the initial kinetic energy lost.

Your answer (numerical value):

7 Easy 1 min Concept

P04-07

A particle has potential energy $U(x) = x^2 - 4x$ (SI units). Its equilibrium position and type are:

- (A) $x = 2 \text{ m}$, unstable (B) $x = 2 \text{ m}$, stable (C) $x = 4 \text{ m}$, stable (D) $x = 0$, neutral

8 Easy 1 min Basic

P04-08

A ball dropped from 10 m rebounds to 6.4 m. The coefficient of restitution is:

- (A) 0.64 (B) 0.36 (C) 0.8 (D) 0.6

9 Easy 1 min Basic

P04-09

A 1 kg block slides on a floor with $\mu_k = 0.2$ from an initial speed of 6 m s^{-1} ($g = 10$). It stops after:

- (A) 3 m (B) 9 m (C) 6 m (D) 18 m

P04 · Chapter Test

1 Easy 0.5 min Speed

P04-T1

The work done by Earth's gravity on a satellite in a circular orbit during one full revolution is:

- (A) positive (B) negative (C) $2\pi r \times mg$ (D) zero

2 Easy 1 min Basic

P04-T2

If the kinetic energy of a body increases by 300%, its momentum increases by:

- (A) 100% (B) 200% (C) 300% (D) 50%

3 Medium 1 min Calc

P04-T3

A machine gun fires 10 bullets per second, each of mass 50 g, at 1000 m s^{-1} . The power delivered to the bullets is:

- (A) 250 kW (B) 25 kW (C) 500 kW (D) 2.5 kW

4 Easy 0.75 min NV

P04-T4

The potential energy of a particle is $U = 5x^3 \text{ J}$ (x in m). Find the magnitude of the force (in N) on it at $x = 1 \text{ m}$.

Your answer (numerical value):

5 Easy 0.5 min Tricky

P04-T5

A 2 kg body is lifted 5 m vertically at **constant speed** ($g = 10$). The net work done on the body is:

- (A) 100 J (B) -100 J (C) 200 J (D) 0

Answers & Solutions

P04 · Practice

1 8 2 (A) 3 (A) 4 (D) 5 (C) 6 80 7 (B) 8 (C) 9 (B)

P04 · Chapter Test

1 (D) 2 (A) 3 (A) 4 15 5 (D)

P04 · PRACTICE

1 Answer: 8 Work by variable force Easy 1 min NV

P04-01

SOLUTION $W = \int_0^2 3x^2 dx = x^3 \Big|_0^2 = 8 \text{ J}$.

SHORTCUT Integrate: the antiderivative of $3x^2$ is x^3 .

COMMON TRAP Using $F(2) \times 2 = 24 \text{ J}$.

2 Answer: (A) Spring maximum compression Easy 1 min Basic

P04-02

SOLUTION $\frac{1}{2}mv^2 = \frac{1}{2}kx^2 \Rightarrow x = v\sqrt{m/k} = 4\sqrt{0.01} = 0.4 \text{ m}$.

SHORTCUT $x = v\sqrt{m/k}$.

COMMON TRAP Forgetting the square root gives 0.04.

3 Answer: (A) Power on an incline Medium 1.5 min JEE

P04-03

SOLUTION At constant speed, the engine force $= mg \sin \theta + f = 500 + 500 = 1000 \text{ N}$. $P = Fv = 10 \text{ kW}$.

SHORTCUT Constant speed means driving force = sum of resisting forces.

COMMON TRAP Omitting the gravity component (5 kW).

4 Answer: (D) Vertical circle, minimum speed Medium 1 min Basic

P04-04

SOLUTION $u_{\min} = \sqrt{5gR} = \sqrt{45} = 3\sqrt{5} \approx 6.7 \text{ m s}^{-1}$.

SHORTCUT String: $\sqrt{5gR}$ at the bottom. Rod: $\sqrt{4gR}$.

COMMON TRAP Using the rod condition gives 6.

5 Answer: (C) 1D elastic collision Easy 1 min Basic

P04-05

SOLUTION $v_2 = \frac{2m_1}{m_1 + m_2} u_1 = \frac{4}{6} \times 6 = 4 \text{ m s}^{-1}$. (The 2 kg ball rebounds at 2 m s^{-1} .)

SHORTCUT Check: relative speed of separation = relative speed of approach: $4 - (-2) = 6$.

COMMON TRAP Assuming the balls exchange velocities (true only for equal masses).

- 6 Answer: 80** KE loss in perfectly inelastic collision **Medium** ⌚ 1.5 min **NV** P04-06
- SOLUTION** $v = \frac{1 \times 10}{5} = 2 \text{ m s}^{-1}$. $K_i = 50 \text{ J}$, $K_f = \frac{1}{2}(5)(4) = 10 \text{ J}$. Loss = $40/50 = 80\%$.
- SHORTCUT** Fraction lost = $\frac{m_2}{m_1 + m_2} = \frac{4}{5}$.
- COMMON TRAP** Reporting the percentage retained (20).

- 7 Answer: (B)** Equilibrium from $U(x)$ **Easy** ⌚ 1 min **Concept** P04-07
- SOLUTION** $dU/dx = 2x - 4 = 0 \Rightarrow x = 2$. $d^2U/dx^2 = 2 > 0$, a minimum, so stable.
- SHORTCUT** U is an upward parabola, so its vertex is a stable equilibrium.
- COMMON TRAP** Taking the root of $U = 0$ ($x = 4$) as the equilibrium.

- 8 Answer: (C)** Restitution from bounce height **Easy** ⌚ 1 min **Basic** P04-08
- SOLUTION** $e = \sqrt{h_1/h_0} = \sqrt{0.64} = 0.8$.
- SHORTCUT** Speed $\propto \sqrt{h}$.
- COMMON TRAP** Using h_1/h_0 directly (0.64).

- 9 Answer: (B)** Stopping distance with friction **Easy** ⌚ 1 min **Basic** P04-09
- SOLUTION** $\mu mg d = \frac{1}{2}mv^2 \Rightarrow d = \frac{v^2}{2\mu g} = \frac{36}{4} = 9 \text{ m}$.
- SHORTCUT** Mass cancels.
- COMMON TRAP** Forgetting the 2 gives 18 m.

P04 · CHAPTER TEST

- 1 Answer: (D)** Work by centripetal force **Easy** ⌚ 0.5 min **Speed** P04-T1
- SOLUTION** Gravity is always perpendicular to the velocity in a circular orbit, so it does no work at any instant.
- SHORTCUT** Force $\perp$ velocity $\rightarrow$ no work.
- COMMON TRAP** Multiplying force by the circumference.

- 2 Answer: (A)** KE-momentum relation **Easy** ⌚ 1 min **Basic** P04-T2
- SOLUTION** $K' = 4K \Rightarrow p' = \sqrt{2mK'} = 2p$, an increase of 100%.
- SHORTCUT** $p \propto \sqrt{K}$.
- COMMON TRAP** Reading "+300%" as $\times 3$ instead of $\times 4$.

- 3 Answer: (A)** Power of a machine gun **Medium** ⌚ 1 min **Calc** P04-T3
- SOLUTION** KE per bullet = $\frac{1}{2}(0.05)(10^6) = 2.5 \times 10^4 \text{ J}$. Power = $10 \times 2.5 \times 10^4 = 2.5 \times 10^5 \text{ W} = 250 \text{ kW}$.
- SHORTCUT** $P = \frac{1}{2}nmv^2$.
- COMMON TRAP** Unit slip: 50 g as 0.5 kg.

- 4 Answer: 15** Force from potential energy **Easy** ⌚ 0.75 min **NV** P04-T4
- SOLUTION** $F = -dU/dx = -15x^2$; at $x = 1$, $|F| = 15 \text{ N}$.
- SHORTCUT** Force = $-\text{slope of } U$.
- COMMON TRAP** Reporting $U(1) = 5$.

- 5 Answer: (D)** Net work at constant speed **Easy** ⌚ 0.5 min **Tricky** P04-T5
- SOLUTION** $W_{\text{net}} = \Delta K = 0$ (constant speed). The applied force does +100 J, and gravity does -100 J .
- SHORTCUT** Constant speed means zero net work.
- COMMON TRAP** Answering the work done by the lifting force alone.

 REVISION Revision tracker · Work, Energy & Power

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART II · PHYSICS · UNIT 05

Rotational Motion & Centre of Mass

05

1–2 per shift
TYPICAL QUESTIONSMedium–Hard
DIFFICULTYMust-do
PRIORITYClass 11 · Ch 6
NCERT~16 hours
STUDY TIME

Concept summary

- The **centre of mass (COM)** moves as if all the mass and all the external forces were concentrated there: $M\vec{a}_{cm} = \vec{F}_{ext}$.
- Rotation** has an exact analogue for every linear quantity: $\theta \leftrightarrow x, \omega \leftrightarrow v, \alpha \leftrightarrow a, I \leftrightarrow m, \tau \leftrightarrow F, L \leftrightarrow p$.
- Moment of inertia** depends on the mass *and* how far it lies from the axis. The parallel and perpendicular axis theorems give I about new axes.
- Rolling without slipping:** $v_{cm} = \omega R$. Rolling KE splits into translational and rotational parts in a fixed ratio.
- Angular momentum is conserved** when there is no external torque: $I_1\omega_1 = I_2\omega_2$.

Formulas

Σ FORMULA Centre of mass

$$\vec{r}_{cm} = \frac{\sum m_i \vec{r}_i}{\sum m_i} \quad x_{cm} = \frac{\int x dm}{\int dm} \quad \vec{v}_{cm} = \frac{\sum m_i \vec{v}_i}{M}$$

Body	COM location
Semicircular ring (radius R)	$\frac{2R}{\pi}$ from the centre
Semicircular disc	$\frac{4R}{3\pi}$ from the centre
Solid hemisphere	$\frac{3R}{8}$ from the centre of the flat face
Hollow hemisphere	$\frac{R}{2}$ from the centre
Solid cone (height h)	$\frac{h}{4}$ from the base
Triangle (lamina)	centroid

Removed-part trick: $x_{cm} = \frac{m_1 x_1 - m_2 x_2}{m_1 - m_2}$ (treat the hole as negative mass).

Σ FORMULA Rotational kinematics & dynamics

$$\omega = \omega_0 + \alpha t \quad \theta = \omega_0 t + \frac{1}{2}\alpha t^2 \quad \omega^2 = \omega_0^2 + 2\alpha\theta$$

$$\vec{\tau} = \vec{r} \times \vec{F} \quad \tau = I\alpha \quad \vec{L} = \vec{r} \times \vec{p} = I\vec{\omega} \quad \vec{\tau}_{ext} = \frac{d\vec{L}}{dt}$$

$$K_{rot} = \frac{1}{2}I\omega^2 = \frac{L^2}{2I} \quad W = \int \tau d\theta \quad P = \tau\omega \quad \text{angular impulse} = \int \tau dt = \Delta L$$

Σ FORMULA Moments of inertia

Body (mass M)	Axis	I
Thin ring, radius R	central, $\perp$ plane	MR^2
Thin ring	diameter	$\frac{1}{2}MR^2$
Disc / solid cylinder	central axis	$\frac{1}{2}MR^2$
Disc	diameter	$\frac{1}{4}MR^2$
Hollow cylinder (thin)	central axis	MR^2
Solid sphere	diameter	$\frac{2}{5}MR^2$
Hollow sphere (thin shell)	diameter	$\frac{2}{3}MR^2$
Thin rod, length L	$\perp$ through centre	$\frac{1}{12}ML^2$
Thin rod	$\perp$ through one end	$\frac{1}{3}ML^2$
Rectangular plate $a \times b$	$\perp$ plane through centre	$\frac{1}{12}M(a^2 + b^2)$
Solid cylinder (length l)	$\perp$ axis through centre	$M \left(\frac{R^2}{4} + \frac{l^2}{12} \right)$

Parallel axis: $I = I_{cm} + Md^2$ (any body). **Perpendicular axis:** $I_z = I_x + I_y$ (planar bodies only). **Radius of gyration:** $k = \sqrt{I/M}$.

Σ FORMULA Rolling without slipping

$$v_{cm} = \omega R \quad K = \frac{1}{2}mv^2 \left(1 + \frac{k^2}{R^2} \right) \quad a_{incline} = \frac{g \sin \theta}{1 + k^2/R^2} \quad \mu_{min} = \frac{\tan \theta}{1 + R^2/k^2}$$

Body	k^2/R^2	K_{rot}/K_{total}	a on incline
Ring / hollow cylinder	1	1/2	$\frac{1}{2}g \sin \theta$
Disc / solid cylinder	1/2	1/3	$\frac{2}{3}g \sin \theta$
Hollow sphere	2/3	2/5	$\frac{3}{5}g \sin \theta$
Solid sphere	2/5	2/7	$\frac{5}{7}g \sin \theta$

Race down an incline (smallest k^2/R^2 wins): solid sphere > disc > hollow sphere > ring. The race result is independent of mass and radius.

Sliding $\rightarrow$ rolling on a rough floor (initial speed v_0 , $\omega_0 = 0$): the final rolling speed is $\frac{v_0}{1 + k^2/R^2}$. For a solid sphere, $\frac{5}{7}v_0$.

Σ FORMULA Equilibrium of a rigid body

$\sum \vec{F} = 0$ and $\sum \vec{\tau} = 0$ about *any* point. Take torques about the point where the most unknown forces act.

Ladder (uniform, smooth wall, rough floor, at angle θ with the floor): on the verge of slipping, $\mu = \frac{1}{2 \tan \theta}$.

Linear $\leftrightarrow$ rotational comparison

Linear	Rotational
x, v, a	θ, ω, α
m	I
$F = ma$	$\tau = I\alpha$
$p = mv$	$L = I\omega$
$K = \frac{1}{2}mv^2$	$K = \frac{1}{2}I\omega^2$

Linear	Rotational
$P = Fv$	$P = \tau\omega$

Frequently tested concepts & question patterns

PYQ PATTERN Recurring structures

1. **MI about a shifted axis:** parallel/perpendicular axis on a disc, ring, rod or plate; MI of composite bodies (rod + spheres, square of rods).
2. **Rolling on an incline:** acceleration, speed at the bottom, which body reaches first, minimum friction.
3. **Angular momentum conservation:** a person walking on a turntable, a skater, two discs coupled (KE lost).
4. **COM of composite or cut-out bodies**, and COM motion after an explosion.
5. **Torque and angular momentum as cross products** in vector form.
6. **Rolling KE ratios**, and the fraction of KE that is rotational.

Shortcuts

SHORTCUT The k^2/R^2 key

Memorise one number per body: ring 1, disc 1/2, hollow sphere 2/3, solid sphere 2/5. Every rolling formula in the table above is generated from it.

Time saved: 1–2 min per rolling question. **Common mistake:** using $I/(MR^2)$ about the wrong axis, such as a disc about its diameter (1/4) when it rolls about its central axis (1/2).

SHORTCUT Coupled rotating bodies

Two coaxial bodies stick together: $\omega_f = \frac{I_1\omega_1 + I_2\omega_2}{I_1 + I_2}$ and $\Delta K = \frac{I_1 I_2 (\omega_1 - \omega_2)^2}{2(I_1 + I_2)}$. This is the rotational version of a perfectly inelastic collision.

Question classification

Type	Difficulty	Target time
MI from the table + one theorem	E	1 min
Rolling acceleration / KE ratios	E–M	1.5 min
Angular momentum conservation	M	2 min
COM of a cut-out body	M	2 min
Torque/L as a cross product	E	1 min
Rolling + slipping, or impulse on a rod	H	3–4 min (Round 2)

Common mistakes

MISTAKE ALERT Mistake alerts

- Applying the perpendicular axis theorem to a 3D body (a sphere or cylinder). It works for **planar** bodies only.
- Using the parallel axis theorem between two axes, neither of which passes through the COM. Always go through I_{cm} .
- Assuming friction does work in pure rolling. Static friction at the contact point does **no** work.
- Using $v = \omega R$ when the body is still slipping.
- Conserving KE when two rotating bodies couple. Only angular momentum is conserved.

Practice questions

P05 · Practice

1 Medium 1.5 min Concept

P05-01

The moment of inertia of a uniform disc (mass M , radius R) about a **tangent lying in its plane** is:

- (A) $\frac{5}{4}MR^2$ (B) $\frac{3}{2}MR^2$ (C) $\frac{1}{2}MR^2$ (D) $\frac{3}{4}MR^2$

2 Easy 1 min Basic

P05-02

A solid cylinder rolls without slipping down a 30° incline ($g = 10$). Its acceleration is:

- (A) 5 m s^{-2} (B) 2.5 m s^{-2} (C) $10/3 \text{ m s}^{-2}$ (D) $25/7 \text{ m s}^{-2}$

3 Easy 1 min Concept

P05-03

A skater spins with MI 4 kg m^2 at 2 rev/s. She pulls in her arms, reducing her MI to 2 kg m^2 . The ratio (new KE)/(old KE) is:

- (A) $1/2$ (B) 2 (C) 1 (D) 4

4 Easy 0.75 min Speed

P05-04

The centre of mass of a thin uniform wire bent into a semicircle of radius R is at a distance from the centre of:

- (A) $\pi R/4$ (B) $4R/3\pi$ (C) $R/2$ (D) $2R/\pi$

5 Easy 1 min NV

P05-05

A force $\vec{F} = (2\hat{i} + 3\hat{j}) \text{ N}$ acts at the point $\vec{r} = (\hat{i} - \hat{j}) \text{ m}$. Find the magnitude of the torque (in N m) about the origin.

Your answer (numerical value):

6 Easy 0.75 min Speed

P05-06

A solid sphere, a disc and a ring (all of different masses and radii) are released together from rest at the top of the same incline and roll without slipping. The order in which they reach the bottom is:

- (A) ring, disc, sphere (B) sphere, disc, ring
(C) disc, sphere, ring (D) all together

7 Easy 1 min NV

P05-07

A 2 kg disc rolls without slipping at 2 m s^{-1} . Find its total kinetic energy (in J).

Your answer (numerical value):

8 Hard 2.5 min JEE

P05-08

A uniform ladder rests against a smooth vertical wall, making 60° with a rough horizontal floor. The minimum coefficient of friction at the floor for equilibrium is:

- (A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{3}}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1}{2\sqrt{3}}$

9 Medium 1.5 min Concept

P05-09

A solid sphere is set sliding (not rotating) at speed v_0 on a rough horizontal floor. When it starts rolling without slipping, its speed is:

- (A) v_0 (B) $\frac{2}{3}v_0$ (C) $\frac{2}{7}v_0$ (D) $\frac{5}{7}v_0$

10 Easy 0.75 min Speed

P05-10

A wheel starts from rest with a constant angular acceleration of 2 rad s^{-2} . The angle it turns through in the first 5 s is:

- (A) 25 rad (B) 10 rad (C) 50 rad (D) 5 rad

P05 · Chapter Test

1 Easy 0.5 min Speed

P05-T1

The radius of gyration of a solid sphere of radius R about a diameter is:

- (A) $\sqrt{2/3} R$ (B) $\frac{2}{5} R$ (C) $\sqrt{2/5} R$ (D) $R/\sqrt{2}$

2 Easy 0.75 min Basic

P05-T2

The moment of inertia of a thin ring (mass M , radius R) about a diameter is:

- (A) $\frac{1}{2} MR^2$ (B) MR^2 (C) $\frac{1}{4} MR^2$ (D) $2MR^2$

3 Easy 1 min NV

P05-T3

A constant torque of 10 N m acts on a flywheel of MI 2 kg m^2 that starts from rest. Find its angular speed (in rad/s) after 4 s.

Your answer (numerical value):

4 Medium 1 min Concept

P05-T4

A disc rotates freely at ω_0 . An identical disc, initially at rest, is gently placed on it coaxially, and they rotate together. The final angular speed is:

- (A) ω_0 (B) $2\omega_0$ (C) $\omega_0/\sqrt{2}$ (D) $\omega_0/2$

5 Medium 1 min Concept

P05-T5

A hollow sphere and a solid sphere of equal mass and radius roll without slipping at the same speed. The ratio of their kinetic energies (hollow : solid) is:

- (A) 21 : 25 (B) 25 : 21 (C) 5 : 3 (D) 1 : 1

Answers & Solutions

P05 · Practice

1 (A) 2 (C) 3 (B) 4 (D) 5 5 6 (B) 7 6 8 (D) 9 (D) 10 (A)

P05 · Chapter Test

1 (C) 2 (A) 3 20 4 (D) 5 (B)

P05 · PRACTICE

1 Answer: (A) Parallel + perpendicular axis Medium 1.5 min Concept

P05-01

SOLUTION About a diameter: $\frac{1}{4} MR^2$ (perpendicular axis theorem: $I_x + I_y = \frac{1}{2} MR^2$). A tangent in the plane is parallel to a diameter at distance R : $\frac{1}{4} MR^2 + MR^2 = \frac{5}{4} MR^2$.

SHORTCUT In-plane tangent: $\frac{5}{4}$. Tangent $\perp$ plane: $\frac{3}{2}$.

COMMON TRAP Starting from the central-axis value gives $\frac{3}{2} MR^2$ (the tangent perpendicular to the plane).

2 Answer: (C) Rolling acceleration Easy 1 min Basic

P05-02

SOLUTION $a = \frac{g \sin \theta}{1 + k^2/R^2} = \frac{5}{3/2} = \frac{10}{3} \text{ m s}^{-2}.$

SHORTCUT Disc/cylinder: $\frac{2}{3}g \sin \theta.$

COMMON TRAP Using the solid sphere's k^2/R^2 gives 25/7.

3 Answer: (B) Angular momentum conservation Easy 1 min Concept

P05-03

SOLUTION L is conserved: $\omega_2 = 4 \text{ rev/s}$. $K = L^2/2I$ with L fixed, so $K \propto 1/I$ and the ratio is 2. The extra energy comes from the work she does pulling her arms in.

SHORTCUT Constant L : $K \propto 1/I$.

COMMON TRAP Assuming KE is conserved (ratio 1).

4 Answer: (D) COM of a semicircular wire Easy 0.75 min Speed

P05-04

SOLUTION For a semicircular ring, $y_{cm} = \frac{\int R \sin \theta \lambda R d\theta}{\lambda \pi R} = \frac{2R}{\pi}.$

SHORTCUT Ring: $2R/\pi$. Disc (lamina): $4R/3\pi$.

COMMON TRAP Mixing up the wire and lamina results.

5 Answer: 5 Torque as a cross product Easy 1 min NV

P05-05

SOLUTION $\vec{\tau} = \vec{r} \times \vec{F} = (1)(3) - (-1)(2) = 5$ along $\hat{k}$. Magnitude 5 N m.

SHORTCUT In 2D, $\tau_z = xF_y - yF_x$.

COMMON TRAP Computing $\vec{F} \times \vec{r}$ changes the sign only, not the magnitude. The real trap is arithmetic with the negative y .

6 Answer: (B) Rolling race Easy 0.75 min Speed

P05-06

SOLUTION $a \propto 1/(1 + k^2/R^2)$: sphere (2/5) > disc (1/2) > ring (1). Mass and radius don't matter.

SHORTCUT Smaller k^2/R^2 reaches first.

COMMON TRAP Thinking the heavier body wins.

7 Answer: 6 Rolling kinetic energy Easy 1 min NV

P05-07

SOLUTION $K = \frac{1}{2}mv^2(1 + \frac{1}{2}) = \frac{1}{2}(2)(4)(1.5) = 6 \text{ J}.$

SHORTCUT Disc: $K = \frac{3}{4}mv^2$.

COMMON TRAP Counting only the translational KE (4 J).

8 Answer: (D) Ladder friction Hard 2.5 min JEE

P05-08

SOLUTION Forces: weight W at the middle, floor normal $N_1 = W$, friction f , wall normal $N_2 = f$. Torques about the foot:

$$N_2 L \sin 60^\circ = W \frac{L}{2} \cos 60^\circ \Rightarrow N_2 = \frac{W}{2 \tan 60^\circ}. \text{ Then } \mu \geq f/N_1 = \frac{1}{2 \tan 60^\circ} = \frac{1}{2\sqrt{3}}.$$

SHORTCUT $\mu_{min} = \frac{1}{2 \tan \theta}$ (θ measured from the floor).

COMMON TRAP Measuring θ from the wall, which gives $\frac{\tan \theta}{2}$ with the wrong angle.

9 Answer: (D) Sliding to rolling Medium 1.5 min Concept

P05-09

SOLUTION Angular momentum about the contact point is conserved (friction acts through that point): $mv_0 R = mvR +$

$$\frac{2}{5}mR^2(v/R) \Rightarrow v = \frac{5}{7}v_0.$$

SHORTCUT $v = \frac{v_0}{1 + k^2/R^2}.$

COMMON TRAP Conserving kinetic energy, which friction dissipates during slipping.

10 Answer: (A) Rotational kinematics Easy 0.75 min Speed

P05-10

SOLUTION $\theta = \frac{1}{2}\alpha t^2 = \frac{1}{2}(2)(25) = 25 \text{ rad}.$

SHORTCUT Same form as $s = \frac{1}{2}at^2$.

COMMON TRAP Using $\theta = \alpha t$ (10).

P05 · CHAPTER TEST

1 Answer: (C) Radius of gyration **Easy** ⌚ 0.5 min **Speed**

P05-T1

SOLUTION $k = \sqrt{I/M} = \sqrt{\frac{2}{5}}R$.

SHORTCUT $k^2/R^2 =$ the fraction in $I = (\cdot)MR^2$.

COMMON TRAP Forgetting the square root.

2 Answer: (A) Perpendicular axis for a ring **Easy** ⌚ 0.75 min **Basic**

P05-T2

SOLUTION $I_z = I_x + I_y = 2I_{dia} = MR^2 \Rightarrow I_{dia} = \frac{1}{2}MR^2$.

SHORTCUT Diameter MI = half the central-axis MI (for any symmetric planar body).

COMMON TRAP Using the disc value (1/4).

3 Answer: 20 Torque and angular speed **Easy** ⌚ 1 min **NV**

P05-T3

SOLUTION $\alpha = \tau/I = 5 \text{ rad s}^{-2}$; $\omega = \alpha t = 20 \text{ rad/s}$.

SHORTCUT Angular impulse $\tau t = I\omega$: $40 = 2\omega$.

COMMON TRAP None; speed item.

4 Answer: (D) Coupling identical discs **Medium** ⌚ 1 min **Concept**

P05-T4

SOLUTION $I\omega_0 = 2I\omega \Rightarrow \omega = \omega_0/2$. (Half of the KE is lost.)

SHORTCUT Rotational perfectly inelastic collision.

COMMON TRAP Conserving KE gives $\omega_0/\sqrt{2}$.

5 Answer: (B) Rolling KE ratio **Medium** ⌚ 1 min **Concept**

P05-T5

SOLUTION $K \propto 1 + k^2/R^2$: $(1 + \frac{2}{3}) : (1 + \frac{2}{5}) = \frac{5}{3} : \frac{7}{5} = 25 : 21$.

SHORTCUT Compare the $(1 + k^2/R^2)$ factors.

COMMON TRAP Inverting the ratio.

 REVISION Revision tracker · Rotational Motion

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

Gravitation

06

1-2 per shift

TYPICAL QUESTIONS

Easy-Medium

DIFFICULTY

Must-do

PRIORITY

Class 11 · Ch 7

NCERT

~7 hours

STUDY TIME

Concept summary

- **Newton's law:** every pair of masses attracts with $F = Gm_1m_2/r^2$ along the line joining them. Spherically symmetric bodies act as point masses for external points.
- g falls with height and with depth. It is maximum at the surface.
- **Potential energy** is negative, $U = -GMm/r$, with zero at infinity. Bound orbits have negative total energy.
- **Satellites:** gravity provides the centripetal force. Everything (speed, period, energy) follows from $\frac{GMm}{r^2} = \frac{mv^2}{r}$.
- **Kepler's laws** summarise planetary motion. The second law is angular momentum conservation.

Formulas

Σ FORMULA Acceleration due to gravity

$$g = \frac{GM}{R^2} \quad g_h = \frac{g}{(1 + h/R)^2} \approx g \left(1 - \frac{2h}{R}\right) \quad (h \ll R) \quad g_d = g \left(1 - \frac{d}{R}\right)$$

- g is zero at the centre of the Earth.
- For the same small fractional decrease in g : $d = 2h$.
- Effect of rotation at latitude λ : $g' = g - \omega^2 R \cos^2 \lambda$. g is minimum at the equator.

Σ FORMULA Field and potential

Source	Field E_g	Potential V
Point mass / outside a sphere ($r \geq R$)	$\frac{GM}{r^2}$	$-\frac{GM}{r}$
Inside a thin shell ($r < R$)	0	$-\frac{GM}{R}$ (constant)
Inside a solid sphere ($r < R$)	$\frac{GMr}{R^3}$	$-\frac{GM(3R^2 - r^2)}{2R^3}$

At the centre of a solid sphere, $V = -\frac{3}{2} \frac{GM}{R}$. $E_g = -dV/dr$.

Σ FORMULA Energy, escape and orbits

$$U = -\frac{GMm}{r} \quad v_{esc} = \sqrt{\frac{2GM}{R}} = \sqrt{2gR} \approx 11.2 \text{ km s}^{-1}$$

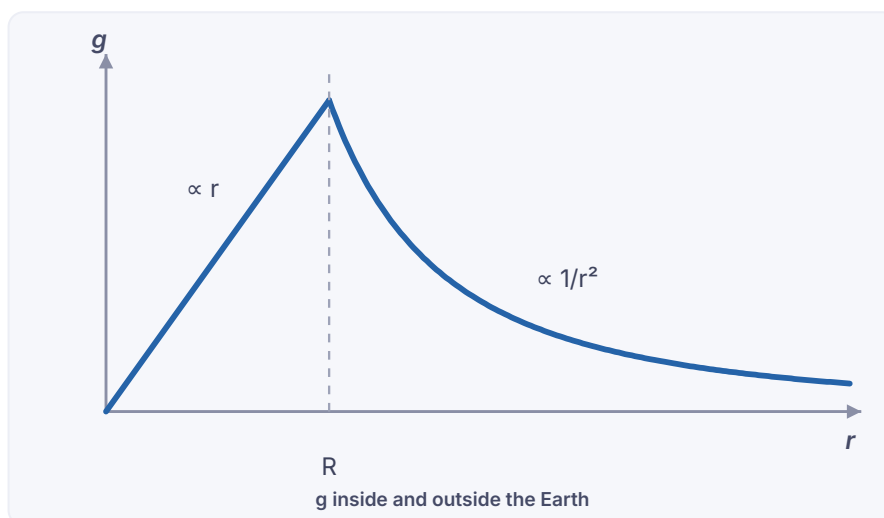
$$v_{orb} = \sqrt{\frac{GM}{r}} \quad T = 2\pi\sqrt{\frac{r^3}{GM}} \quad \text{near the surface: } v_o = \sqrt{gR} \approx 7.9 \text{ km s}^{-1}, T \approx 84 \text{ min}$$

$$K = \frac{GMm}{2r} \quad U = -\frac{GMm}{r} \quad E = -\frac{GMm}{2r} \quad K = -E = -\frac{1}{2}U$$

- $v_{esc} = \sqrt{2} v_{orb}$ at the same point. Binding energy $= +\frac{GMm}{2r}$.
- Energy to move a satellite from r_1 to r_2 : $\Delta E = \frac{GMm}{2} \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$.
- **Geostationary orbit:** $T = 24 \text{ h}$, equatorial, same sense as Earth's rotation, $r \approx 42\,000 \text{ km}$ from the centre ($h \approx 36\,000 \text{ km}$).
- Escape speed does **not** depend on the mass of the body or the direction of launch.

Σ FORMULA Kepler's laws

1. Orbits are ellipses with the Sun at one focus.
2. The radius vector sweeps equal areas in equal times: $\frac{dA}{dt} = \frac{L}{2m} = \text{constant}$, so speed is maximum at perihelion. $v_p r_p = v_a r_a$.
3. $T^2 \propto a^3$, where a is the semi-major axis: $T^2 = \frac{4\pi^2}{GM} a^3$.



Standard models & assumptions

- Earth is a uniform sphere, $R = 6400 \text{ km}$, $g = 9.8$ (or 10) m s^{-2} , $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{kg}^{-2}$.
- Satellites are in circular orbits unless an ellipse is stated. The satellite's own mass doesn't affect its orbit.

Frequently tested concepts & question patterns

PYQ PATTERN Recurring structures

1. **g vs height and depth:** ratios, "at what height/depth does g become $x\%$ ", comparisons.
2. **Satellite energies:** KE/PE/total relations, energy to shift orbits, binding energy.
3. **Escape speed** for planets with changed M , R or density; escape vs orbital speed.
4. **Kepler's third law:** period ratios; areal velocity.
5. **Field/potential inside shells and spheres** (graph-based).
6. **Null point** between two masses.

Shortcuts

SHORTCUT Scaling instead of computing

Most gravitation MCQs are ratio questions. Write the dependence and scale it:

$$g \propto M/R^2 \propto \rho R, v_{esc} \propto \sqrt{M/R} \propto R\sqrt{\rho}, T \propto r^{3/2}, v_{orb} \propto r^{-1/2}.$$

Time saved: 1 minute. **Common mistake:** holding the wrong variable fixed (mass vs density).

SHORTCUT Null point between masses m_1, m_2 a distance d apart

$$\text{From } m_1: x = \frac{d\sqrt{m_1}}{\sqrt{m_1} + \sqrt{m_2}}.$$

Question classification

Type	Difficulty	Target time
g at height/depth	E	1 min
Kepler III ratio	E	45 s
Satellite energy relations	E-M	1 min
Escape/orbital speed scaling	E	1 min
Field/potential inside bodies	M	2 min
Elliptical orbits (angular momentum + energy)	M-H	3 min

Common mistakes

MISTAKE ALERT Mistake alerts

- Using h (height above the surface) where $r = R + h$ is needed.
- Applying $g(1 - 2h/R)$ when h is not small compared with R .
- Treating PE as positive, or forgetting that total energy is negative.
- Claiming g inside a **shell** is non-zero. The field is zero, but the potential is not.
- Taking $T^2 \propto r^3$ with r measured from the surface.

Practice questions

P06 · Practice

1 Easy 0.75 min Speed

P06-01

The acceleration due to gravity at a height equal to Earth's radius, above the surface, is:

- (A) $g/2$ (B) $g/\sqrt{2}$ (C) $g/4$ (D) $g/8$

2 Easy 1 min Basic

P06-02

At what height above Earth's surface does g decrease by 1%? ($R = 6400$ km; assume $h \ll R$.)

- (A) 64 km (B) 32 km (C) 16 km (D) 128 km

3 Easy 0.75 min Speed

P06-03

A planet's orbital radius is 4 times Earth's. Its orbital period is:

- (A) 2 years (B) 4 years (C) 8 years (D) 16 years

4 Easy 0.75 min Basic

P06-04

A satellite in a circular orbit has kinetic energy K . Its total mechanical energy is:

- (A) K (B) $2K$ (C) $-2K$ (D) $-K$

5 Medium 1 min Concept

P06-05

A satellite orbits just above Earth's surface ($v_o \approx 7.9$ km s⁻¹). The minimum *additional* speed it needs to escape is about:

- (A) 3.3 km/s (B) 11.2 km/s (C) 7.9 km/s (D) 1.4 km/s

6 Medium 1.5 min JEE

P06-06

The energy needed to move a satellite of mass m from a circular orbit of radius $2R$ to one of radius $3R$ ($R =$ Earth's radius) is:

- (A) $mgR/2$ (B) $mgR/6$ (C) $mgR/3$ (D) $mgR/12$

7 Easy 0.75 min Concept

P06-07

Inside a uniform thin spherical shell of mass M and radius R :

- (A) field and potential are both zero
 (B) the field is zero and the potential is $-GM/R$ everywhere
 (C) the field is GM/R^2 and the potential is zero
 (D) the field increases linearly with r

8 Easy 1 min NV

P06-08

Masses of 1 kg and 4 kg are 3 m apart. Find the distance (in m) from the 1 kg mass of the point between them where the net gravitational field is zero.

Your answer (numerical value):

9 Easy 0.75 min Speed

P06-09

A planet has twice Earth's mass and twice its radius. Its surface gravity is:

- (A) $g/2$ (B) $2g$ (C) g (D) $g/4$

P06 · Chapter Test

1 Easy 0.5 min Concept

P06-T1

The escape speed from a planet depends on:

- (A) the mass of the object being launched (B) the angle of launch
 (C) the mass and radius of the planet (D) all of the above

2 Easy 0.5 min Speed

P06-T2

The weight of a body at the centre of the Earth is:

- (A) the same as at the surface (B) zero
(C) half of that at the surface (D) infinite

3 Easy 1 min Basic

P06-T3

If the orbital speed just above Earth's surface is 7.9 km/s, the orbital speed at a height R above the surface is about:

- (A) 3.95 km/s (B) 11.2 km/s (C) 7.9 km/s (D) 5.6 km/s

4 Easy 0.5 min NV

P06-T4

The orbital radius of a satellite is made 4 times larger. By what factor does its period increase?

Your answer (numerical value):

5 Easy 0.5 min Basic

P06-T5

The binding energy of a satellite of mass m in a circular orbit of radius r around a planet of mass M is:

- (A) GMm/r (B) $-GMm/2r$ (C) $GMm/2r$ (D) $2GMm/r$

Answers & Solutions

P06 · Practice

1 (C) 2 (B) 3 (C) 4 (D) 5 (A) 6 (D) 7 (B) 8 1 9 (A)

P06 · Chapter Test

1 (C) 2 (B) 3 (D) 4 8 5 (C)

P06 · PRACTICE

1 Answer: (C) g at height R Easy 0.75 min Speed

P06-01

SOLUTION $g_h = g \left(\frac{R}{R+h} \right)^2 = g \left(\frac{1}{2} \right)^2 = g/4$.

SHORTCUT r doubles, so g falls by a factor of 4.

COMMON TRAP Using $g(1 - 2h/R)$, which gives $-g$ and is invalid for large h .

2 Answer: (B) Small-height decrease Easy 1 min Basic

P06-02

SOLUTION $\frac{\Delta g}{g} = \frac{2h}{R} = 0.01 \Rightarrow h = \frac{0.01 \times 6400}{2} = 32$ km.

SHORTCUT Depth for the same 1% decrease: 64 km, which is $2h$.

COMMON TRAP Forgetting the 2 gives 64 km.

3 Answer: (C) Kepler's third law Easy 0.75 min Speed

P06-03

SOLUTION $T \propto r^{3/2}$: $4^{3/2} = 8$, so 8 years.

SHORTCUT $\sqrt{4^3} = 8$.

COMMON TRAP Using $T \propto r^2$ gives 16.

4 Answer: (D) Satellite energy relations Easy 0.75 min Basic

P06-04

SOLUTION $E = -\frac{GMm}{2r} = -K$ (and $U = -2K$).

SHORTCUT $K : U : E = 1 : -2 : -1$.

COMMON TRAP Confusing U with E .

5 Answer: (A) Extra speed to escape **Medium** ⌚ 1 min **Concept**

P06-05

SOLUTION $v_e = \sqrt{2}v_o \approx 11.2 \text{ km/s}$. Additional $= (\sqrt{2} - 1)v_o \approx 0.414 \times 7.9 \approx 3.3 \text{ km/s}$.**SHORTCUT** $\Delta v = (\sqrt{2} - 1)v_o$.**COMMON TRAP** Answering v_e itself.**6 Answer: (D)** Energy to shift orbits **Medium** ⌚ 1.5 min **JEE**

P06-06

SOLUTION $\Delta E = \frac{GMm}{2} \left(\frac{1}{2R} - \frac{1}{3R} \right) = \frac{GMm}{12R} = \frac{mgR}{12}$, using $GM = gR^2$.**SHORTCUT** Take the difference of total energies $-GMm/2r$.**COMMON TRAP** Using the difference of potential energies gives $mgR/6$.**7 Answer: (B)** Field inside a shell **Easy** ⌚ 0.75 min **Concept**

P06-07

SOLUTION By the shell theorem the field inside is zero, so the potential is constant and equal to its surface value $-GM/R$.**SHORTCUT** Zero field $\Rightarrow$ constant (not zero) potential.**COMMON TRAP** Assuming zero potential.**8 Answer: 1** Null point between two masses **Easy** ⌚ 1 min **NV**

P06-08

SOLUTION $\frac{G(1)}{x^2} = \frac{G(4)}{(3-x)^2} \Rightarrow 3-x = 2x \Rightarrow x = 1 \text{ m}$.**SHORTCUT** $x = \frac{d\sqrt{m_1}}{\sqrt{m_1} + \sqrt{m_2}} = \frac{3}{3}$.**COMMON TRAP** Measuring from the heavier mass (2 m).**9 Answer: (A)** Surface gravity of another planet **Easy** ⌚ 0.75 min **Speed**

P06-09

SOLUTION $g \propto M/R^2 = 2/4 = 1/2$.**SHORTCUT** Ratio scaling.**COMMON TRAP** Using M/R gives g .**P06 · CHAPTER TEST****1 Answer: (C)** What escape speed depends on **Easy** ⌚ 0.5 min **Concept**

P06-T1

SOLUTION $v_e = \sqrt{2GM/R}$ is independent of the object's mass and of the launch direction (energy is a scalar).**SHORTCUT** Energy is a scalar, so direction doesn't matter.**COMMON TRAP** Choosing (B).**2 Answer: (B)** Weight at Earth's centre **Easy** ⌚ 0.5 min **Speed**

P06-T2

SOLUTION $g_d = g(1 - d/R) = 0$ at $d = R$.**SHORTCUT** g grows linearly from 0 at the centre to the surface value.**COMMON TRAP** Thinking gravity is strongest at the centre.**3 Answer: (D)** Orbital speed at height R **Easy** ⌚ 1 min **Basic**

P06-T3

SOLUTION $v \propto 1/\sqrt{r}$; r doubles, so $v = 7.9/\sqrt{2} \approx 5.6 \text{ km/s}$.**SHORTCUT** Divide by $\sqrt{2}$.**COMMON TRAP** Halving the speed (3.95).**4 Answer: 8** Period scaling **Easy** ⌚ 0.5 min **NV**

P06-T4

SOLUTION $T \propto r^{3/2} = 4^{3/2} = 8$.**COMMON TRAP** Answering 16 (r^2).**5 Answer: (C)** Binding energy **Easy** ⌚ 0.5 min **Basic**

P06-T5

SOLUTION Binding energy $= -E = +\frac{GMm}{2r}$: the energy needed to free the satellite completely.**SHORTCUT** Binding energy is positive.**COMMON TRAP** Choosing the (negative) total energy itself.**REVISION Revision tracker · Gravitation**☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

Properties of Solids & Liquids (incl. Thermal Properties)

07

2–3 per shift
TYPICAL QUESTIONS

Medium
DIFFICULTY

Must-do
PRIORITY

Class 11 · Ch 8, 9, 10
NCERT

~14 hours
STUDY TIME

Concept summary

This unit bundles four mini-chapters, and each regularly gets a question:

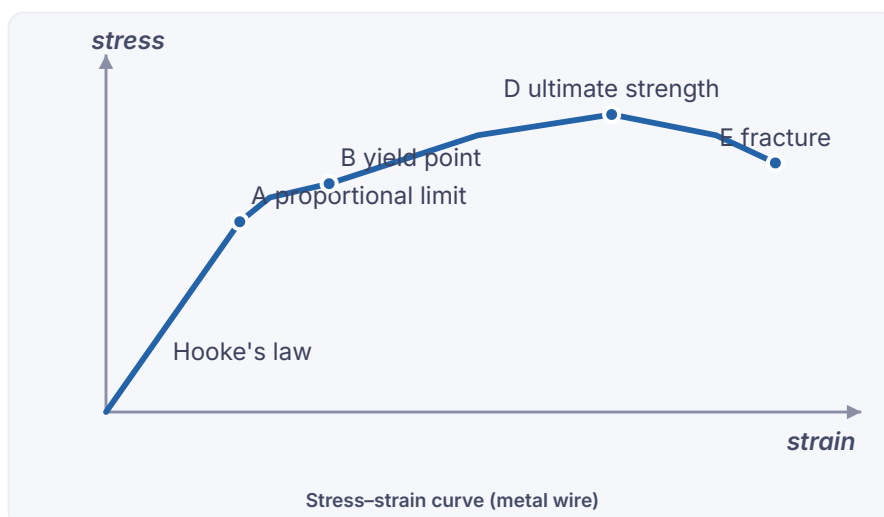
1. **Elasticity:** stress, strain, the three moduli, stress–strain curve, elastic energy.
2. **Fluids:** pressure, Pascal's law, buoyancy, continuity, Bernoulli, viscosity (Stokes, terminal velocity).
3. **Surface tension:** surface energy, excess pressure, capillary rise, angle of contact.
4. **Thermal properties:** expansion, calorimetry, latent heat, conduction, radiation, Newton's law of cooling.

Formulas: Elasticity

Σ FORMULA Moduli & energy

$$\text{Stress} = \frac{F}{A} \quad \text{Strain} = \frac{\Delta L}{L} \quad Y = \frac{FL}{A \Delta L} \quad B = -\frac{\Delta P}{\Delta V/V} \quad \eta_{\text{rigidity}} = \frac{F/A}{\theta}$$

- **Compressibility** $= 1/B$. **Poisson's ratio** $\sigma = -\frac{\text{lateral strain}}{\text{longitudinal strain}}$ (between 0 and 0.5 for real materials).
- **Elastic energy:** $U = \frac{1}{2} F \Delta L$; energy density $u = \frac{1}{2} \text{stress} \times \text{strain} = \frac{\text{stress}^2}{2Y}$.
- **Thermal stress** in a rod clamped at both ends: $Y \propto \Delta T$.
- **Wire under its own weight** (mass M , hanging): $\Delta L = \frac{MgL}{2AY}$.
- For the same material and load: $\Delta L \propto \frac{L}{r^2}$.



☆ IMPORTANT Stress–strain curve vocabulary

Proportional limit (Hooke's law holds up to here) → elastic limit (full recovery up to here) → yield point (plastic flow begins) → ultimate tensile strength → fracture. **Ductile** materials (copper) have a long plastic region. **Brittle** materials (glass) fracture soon after the elastic limit. **Elastomers** (rubber) have large elastic strain but no Hooke's-law region.

Formulas: Fluids

Σ FORMULA Statics

$$P = P_0 + \rho gh \quad \text{Pascal (hydraulic lift): } \frac{F_1}{A_1} = \frac{F_2}{A_2} \quad \text{Buoyancy: } F_B = \rho_{fluid} V_{sub} g$$

Floating body: fraction submerged = $\rho_{body} / \rho_{fluid}$. Gauge pressure = $P - P_0$.

Σ FORMULA Dynamics

$$A_1 v_1 = A_2 v_2 \quad P + \frac{1}{2} \rho v^2 + \rho gh = \text{constant} \quad \text{Torricelli: } v = \sqrt{2gh}$$

- Hole at depth h below the surface in a tank of height H (on the ground): horizontal range = $2\sqrt{h(H-h)}$, maximum at $h = H/2$.
- Venturi meter:** $v_1 = A_2 \sqrt{\frac{2\Delta P}{\rho(A_1^2 - A_2^2)}}$.
- Lift on a wing / roof blown off: faster flow means lower pressure (Bernoulli).

Σ FORMULA Viscosity

$$F = -\eta A \frac{dv}{dx} \quad \text{Stokes: } F = 6\pi\eta r v \quad \text{Terminal velocity: } v_t = \frac{2r^2(\rho - \sigma)g}{9\eta}$$

ρ = density of the sphere, σ = density of the fluid. **Reynolds number** $R_e = \frac{\rho v D}{\eta}$: flow is streamline for $R_e \lesssim 1000$ and turbulent for $R_e \gtrsim 2000$. Critical velocity $v_c = \frac{R_e \eta}{\rho D}$. The viscosity of liquids falls with temperature, while that of gases rises.

Formulas: Surface tension

Σ FORMULA Surface tension

$$T = \frac{F}{L} \quad W = T \Delta A \quad \text{Excess pressure: drop } \frac{2T}{R}, \text{ soap bubble } \frac{4T}{R}, \text{ air bubble in liquid } \frac{2T}{R}$$

$$\text{Capillary rise: } h = \frac{2T \cos \theta}{\rho g r}$$

- $\theta < 90^\circ$ (water-glass): the liquid rises and the meniscus is concave. $\theta > 90^\circ$ (mercury-glass): the liquid is depressed and the meniscus is convex.
- Combining n drops** of radius r into one: $R = n^{1/3} r$; energy released = $4\pi T(nr^2 - R^2)$.
- Detergents **lower** surface tension. Surface tension falls as temperature rises.
- If a capillary tube is shorter than h , the liquid doesn't overflow. The radius of curvature of the meniscus adjusts instead.

Formulas: Thermal properties

Σ FORMULA Expansion & calorimetry

$$\Delta L = L \alpha \Delta T \quad \beta = 2\alpha \quad \gamma = 3\alpha \quad Q = mc \Delta T \quad Q = mL$$

- Principle of calorimetry:** heat lost = heat gained (no losses).
- Water: $c = 4186 \text{ J kg}^{-1} \text{ K}^{-1}$ ($1 \text{ cal g}^{-1} \text{ }^\circ\text{C}^{-1}$). $L_f = 80 \text{ cal/g}$ ($3.34 \times 10^5 \text{ J/kg}$). $L_v = 540 \text{ cal/g}$ ($2.26 \times 10^6 \text{ J/kg}$).
- Anomalous expansion of water:** it contracts from 0°C to 4°C and has maximum density at 4°C .

Σ FORMULA Heat transfer

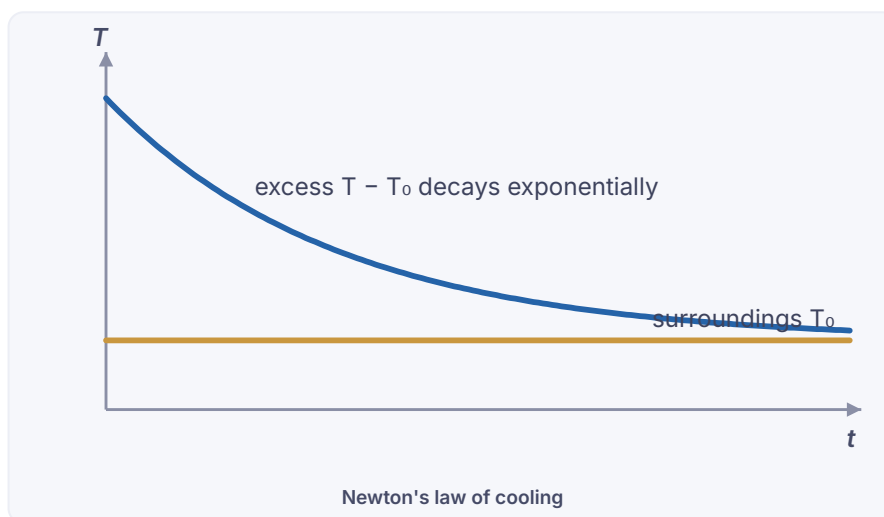
$$\frac{dQ}{dt} = \frac{kA \Delta T}{L} \quad R_{th} = \frac{L}{kA} \quad \text{series: } R = R_1 + R_2 \quad \text{parallel: } \frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

Two slabs of equal length in series: $k_{eq} = \frac{2k_1k_2}{k_1 + k_2}$. Equal area in parallel: $k_{eq} = \frac{k_1 + k_2}{2}$.

Radiation: Stefan–Boltzmann $P = e\sigma AT^4$; net $P = e\sigma A(T^4 - T_0^4)$; $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2}\text{K}^{-4}$.

Wien's law: $\lambda_m T = b = 2.9 \times 10^{-3} \text{ m K}$.

Newton's law of cooling: $\frac{dT}{dt} = -k(T - T_0)$, approximately $\frac{T_1 - T_2}{t} = k \left(\frac{T_1 + T_2}{2} - T_0 \right)$.



Frequently tested concepts & question patterns

▮ PYQ PATTERN Recurring structures

1. **Young's modulus / extension** comparisons between wires (same material, different L and r); elastic energy.
2. **Bernoulli + continuity:** pressure difference in a pipe, efflux speed, range of a jet.
3. **Terminal velocity** scaling (drops merging), Stokes force.
4. **Capillary rise** changes when r or θ changes; excess pressure in bubbles; energy when drops merge.
5. **Calorimetry with phase change:** ice + water mixing. Check whether all the ice melts.
6. **Conduction through composite slabs** (junction temperature), and **Newton's law of cooling** (time for successive intervals).
7. **Stefan/Wien:** power ratios for temperature changes; peak wavelength.

Shortcuts

⚡ SHORTCUT Junction temperature of two slabs in series

$\theta_{\text{junction}} = \frac{\theta_1/R_1 + \theta_2/R_2}{1/R_1 + 1/R_2}$, the weighted mean with weights kA/L . This is the "Kirchhoff" approach used for resistors, with heat current in place of electric current.

⚡ SHORTCUT Is all the ice melted?

First compute the heat the warm water can give up by cooling to 0°C , Q_{avail} . Compare it with $m_{\text{ice}}L_f$ (plus any heat needed to warm the ice to 0°C). If Q_{avail} is smaller, the final temperature is 0°C with some ice left. **Saves** you from impossible negative answers.

Question classification

Type	Difficulty	Target time
Moduli / extension ratios	E	1 min
Bernoulli / continuity	M	2 min
Terminal velocity scaling	E	1 min
Capillarity / excess pressure	E-M	1.5 min
Calorimetry with phase change	M	2.5 min
Composite conduction / Newton cooling	M	2 min
Stefan / Wien ratios	E	1 min

Common mistakes



MISTAKE ALERT Mistake alerts

- Soap bubbles have **two** surfaces: $4T/R$, not $2T/R$.
- Stokes' law uses the **radius**, not the diameter.
- Using $^{\circ}\text{C}$ in Stefan's law. T must be in **kelvin**.
- Assuming the final temperature is above 0°C without checking that all the ice has melted.
- Forgetting that $\gamma = 3\alpha$ (volume) when a question switches from length to volume.
- In Bernoulli, mixing gauge and absolute pressure.

Practice questions

P07 · Practice

1

Easy

⌚ 1 min

Calc

P07-01

A 2 m wire of cross-section 1 mm^2 stretches by 1 mm under a 10 kg load ($g = 10$). Young's modulus is:

- (A) $2 \times 10^{10} \text{ N m}^{-2}$ (B) $2 \times 10^9 \text{ N m}^{-2}$ (C) $1 \times 10^{11} \text{ N m}^{-2}$ (D) $2 \times 10^{11} \text{ N m}^{-2}$

2

Easy

⌚ 1 min

Basic

P07-02

A wire stretched by 2 mm under a load of 50 N stores elastic energy of:

- (A) 0.1 J (B) 0.05 J (C) 0.025 J (D) 100 J

3

Easy

⌚ 0.75 min

NV

P07-03

Water flows out of a small hole 5 m below the free surface of a large open tank. Find the efflux speed (in m/s, $g = 10$).

Your answer (numerical value):

4

Easy

⌚ 0.5 min

Speed

P07-04

If a pipe's diameter halves, the speed of an incompressible fluid in it becomes:

- (A) twice (B) half (C) four times (D) one-fourth

5

Medium

⌚ 1 min

Concept

P07-05

Eight identical rain drops, each falling at terminal velocity v , merge into one drop. Its terminal velocity is:

- (A) $4v$ (B) $2v$ (C) $8v$ (D) $16v$

6 Medium 1.5 min NV

P07-06

Find the height (in cm) to which water rises in a clean glass capillary of radius 0.07 mm. Take surface tension 0.07 N/m, contact angle 0° , $\rho = 1000 \text{ kg m}^{-3}$, $g = 10$.

Your answer (numerical value):

7 Easy 0.75 min Basic

P07-07

The excess pressure inside a soap bubble of radius 1 cm (surface tension 0.03 N m^{-1}) is:

- (A) 6 Pa (B) 24 Pa (C) 3 Pa (D) 12 Pa

8 Easy 1 min NV

P07-08

100 g of water at 80°C is mixed with 200 g of water at 20°C . Find the final temperature (in $^\circ\text{C}$), neglecting heat losses.

Your answer (numerical value):

9 Easy 1 min Basic

P07-09

The heat needed to convert 10 g of ice at 0°C into water at 20°C is ($L_f = 80 \text{ cal/g}$):

- (A) 1000 cal (B) 800 cal (C) 200 cal (D) 1600 cal

10 Medium 1 min Concept

P07-10

Two rods of equal length and area, with conductivities k and $2k$, are joined end to end. The equivalent conductivity of the combination is:

- (A) $3k/2$ (B) $4k/3$ (C) $3k$ (D) $2k/3$

11 Easy 0.5 min Speed

P07-11

If the absolute temperature of a black body doubles, the power it radiates becomes:

- (A) 2 times (B) 16 times (C) 8 times (D) 4 times

12 Medium 2 min NV

P07-12

A body cools from 80°C to 70°C in 5 minutes in surroundings at 15°C . Using the average-temperature form of Newton's law of cooling, find the time (in minutes) it takes to cool from 70°C to 60°C .

Your answer (numerical value):

P07 · Chapter Test

1 Easy 0.5 min Basic

P07-T1

A wire's longitudinal strain is 2×10^{-3} and its lateral strain is -6×10^{-4} . Poisson's ratio is:

- (A) 3.3 (B) 0.3 (C) 0.03 (D) 0.12

2 Easy 0.5 min Speed

P07-T2

The pistons of a hydraulic lift have areas in the ratio 1 : 50. A 200 N force on the small piston can support a load of weight:

- (A) 10 000 N (B) 250 N (C) 4 N (D) 2 000 N

3 Easy 0.75 min Basic

P07-T3

A star's spectrum peaks at 500 nm. Its surface temperature is about ($b = 2.9 \times 10^{-3} \text{ m K}$):

- (A) 580 K (B) 58 000 K (C) 5800 K (D) 1450 K

4 Easy 0.5 min NV

P07-T4

A solid has $\alpha = 2 \times 10^{-5} \text{ K}^{-1}$. Its volume expansion coefficient is $x \times 10^{-5} \text{ K}^{-1}$. Find x .

Your answer (numerical value):

5 Easy 1 min Calc

P07-T5

A sphere of radius 1 mm moves at 0.1 m s^{-1} through a liquid of viscosity 1 Pa s. The viscous drag is about:

- (A) $1.9 \times 10^{-2} \text{ N}$ (B) $6.0 \times 10^{-4} \text{ N}$ (C) $1.9 \times 10^{-3} \text{ N}$ (D) $3.8 \times 10^{-3} \text{ N}$

Answers & Solutions

P07 · Practice

- 1 (D) 2 (B) 3 10 4 (C) 5 (A) 6 20 7 (D) 8 40 9 (A) 10 (B)
11 (B) 12 6

P07 · Chapter Test

- 1 (B) 2 (A) 3 (C) 4 6 5 (C)

P07 · PRACTICE

1 Answer: (D) Young's modulus Easy 1 min Calc

P07-01

SOLUTION $Y = \frac{FL}{A\Delta L} = \frac{100 \times 2}{10^{-6} \times 10^{-3}} = 2 \times 10^{11} \text{ N m}^{-2}$.

SHORTCUT Convert $\text{mm}^2 \rightarrow 10^{-6} \text{ m}^2$ and $\text{mm} \rightarrow 10^{-3} \text{ m}$ first.

COMMON TRAP Converting 1 mm^2 as 10^{-3} m^2 .

2 Answer: (B) Elastic energy Easy 1 min Basic

P07-02

SOLUTION $U = \frac{1}{2} F \Delta L = \frac{1}{2} (50) (2 \times 10^{-3}) = 0.05 \text{ J}$.

SHORTCUT The factor $\frac{1}{2}$ is there because the force rises linearly from 0 to F .

COMMON TRAP Forgetting the $\frac{1}{2}$ (0.1 J).

3 Answer: 10 Torricelli's theorem Easy 0.75 min NV

P07-03

SOLUTION $v = \sqrt{2gh} = \sqrt{100} = 10 \text{ m s}^{-1}$.

SHORTCUT Same as the speed of free fall through h .

COMMON TRAP Measuring h from the bottom instead of from the free surface.

4 Answer: (C) Continuity equation Easy 0.5 min Speed

P07-04

SOLUTION Av is constant and $A \propto d^2$: the area becomes $1/4$, so the speed is $\times 4$.

SHORTCUT Speed $\propto 1/d^2$.

COMMON TRAP Taking speed $\propto 1/d$.

5 Answer: (A) Terminal velocity of merged drops Medium 1 min Concept

P07-05

SOLUTION Volume $\times 8$ means $R = 2r$. $v_t \propto r^2$, so $v' = 4v$.

SHORTCUT $v' = n^{2/3}v$.

COMMON TRAP Assuming $v_t \propto$ volume (8v).

6 Answer: 20 *Capillary rise* **Medium** ⌚ 1.5 min **NV**

P07-06

SOLUTION $h = \frac{2T}{\rho g r} = \frac{0.14}{1000 \times 10 \times 7 \times 10^{-5}} = \frac{0.14}{0.7} = 0.2 \text{ m} = 20 \text{ cm}.$

SHORTCUT hr is constant for a given liquid: here $hr = 1.4 \times 10^{-5} \text{ m}^2$.

COMMON TRAP Using the diameter for r gives 10 cm.

7 Answer: (D) *Excess pressure in a soap bubble* **Easy** ⌚ 0.75 min **Basic**

P07-07

SOLUTION $\Delta P = 4T/R = 0.12/0.01 = 12 \text{ Pa}.$

SHORTCUT Two surfaces, so $4T/R$.

COMMON TRAP Using $2T/R$ gives 6 Pa.

8 Answer: 40 *Method of mixtures* **Easy** ⌚ 1 min **NV**

P07-08

SOLUTION $100(80 - T) = 200(T - 20) \Rightarrow 8000 + 4000 = 300T \Rightarrow T = 40^\circ\text{C}.$

SHORTCUT Mass-weighted mean: $\frac{100 \times 80 + 200 \times 20}{300}.$

COMMON TRAP Taking a simple average (50).

9 Answer: (A) *Latent heat + sensible heat* **Easy** ⌚ 1 min **Basic**

P07-09

SOLUTION Melt: $10 \times 80 = 800 \text{ cal}.$ Warm: $10 \times 1 \times 20 = 200 \text{ cal}.$ Total 1000 cal.

SHORTCUT Always add the phase-change step.

COMMON TRAP Stopping after melting (800).

10 Answer: (B) *Series conduction* **Medium** ⌚ 1 min **Concept**

P07-10

SOLUTION Series, equal lengths: $k_{eq} = \frac{2k_1k_2}{k_1 + k_2} = \frac{2(k)(2k)}{3k} = \frac{4k}{3}.$

SHORTCUT The harmonic mean is closer to the smaller value.

COMMON TRAP Using the arithmetic mean (the parallel formula).

11 Answer: (B) *Stefan's law* **Easy** ⌚ 0.5 min **Speed**

P07-11

SOLUTION $P \propto T^4 = 2^4 = 16.$

COMMON TRAP Doubling a Celsius temperature is **not** doubling T .

12 Answer: 6 *Newton's law of cooling* **Medium** ⌚ 2 min **NV**

P07-12

SOLUTION $\frac{10}{5} = k(75 - 15) \Rightarrow k = \frac{1}{30}.$ Next: $\frac{10}{t} = k(65 - 15) = \frac{50}{30} \Rightarrow t = 6 \text{ min}.$

SHORTCUT Equal temperature drops: $t_2/t_1 = \frac{\text{mean excess}_1}{\text{mean excess}_2} = \frac{60}{50}.$

COMMON TRAP Assuming the same 5 min, i.e. constant cooling rate.

P07 · CHAPTER TEST**1 Answer: (B)** *Poisson's ratio* **Easy** ⌚ 0.5 min **Basic**

P07-T1

SOLUTION $\sigma = -\frac{-6 \times 10^{-4}}{2 \times 10^{-3}} = 0.3.$

SHORTCUT Must lie between 0 and 0.5.

COMMON TRAP Inverting the ratio gives 3.3.

2 Answer: (A) *Hydraulic lift* **Easy** ⌚ 0.5 min **Speed**

P07-T2

SOLUTION $F_2 = F_1 \frac{A_2}{A_1} = 200 \times 50 = 10\,000 \text{ N}.$

SHORTCUT Pascal's law multiplies the force by the area ratio.

COMMON TRAP Dividing by the ratio.

3 Answer: (C) *Wien's displacement law* **Easy** ⌚ 0.75 min **Basic**

P07-T3

SOLUTION $T = b/\lambda_m = \frac{2.9 \times 10^{-3}}{5 \times 10^{-7}} = 5800 \text{ K}.$

SHORTCUT Remember: the Sun $\approx 5800 \text{ K} \approx 500 \text{ nm}.$

COMMON TRAP Using $500 \times 10^{-6} \text{ m}$ for nm.

4 Answer: 6 Volume expansion coefficient **Easy** ⌚ 0.5 min **NV**

P07-T4

SOLUTION $\gamma = 3\alpha = 6 \times 10^{-5} \text{ K}^{-1}$.

SHORTCUT $\alpha : \beta : \gamma = 1 : 2 : 3$.

COMMON TRAP Using $\gamma = \alpha^3$.

5 Answer: (C) Stokes' law **Easy** ⌚ 1 min **Calc**

P07-T5

SOLUTION $F = 6\pi\eta rv = 6\pi(1)(10^{-3})(0.1) = 6\pi \times 10^{-4} \approx 1.88 \times 10^{-3} \text{ N}$.

SHORTCUT $6\pi \approx 18.85$.

COMMON TRAP Omitting the factor π gives (B).

**REVISION** Revision tracker · Solids & Liquids

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

Thermodynamics

08

1-2 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

Must-do

PRIORITY

Class 11 · Ch 11

NCERT

~9 hours

STUDY TIME

Concept summary

- **Zeroth law:** thermal equilibrium is transitive, which defines temperature.
- **First law:** energy conservation for heat and work: $Q = \Delta U + W$. Here W is the work done **by** the gas (NCERT convention).
- **Internal energy** of an ideal gas depends **only on temperature**: $\Delta U = nC_v\Delta T$ for any process.
- **Work** is the area under the P - V curve. It depends on the path. Heat also depends on the path.
- **Second law:** no engine converts heat entirely into work in a cycle (Kelvin-Planck), and heat doesn't flow by itself from cold to hot (Clausius). The Carnot engine sets the maximum efficiency.

Formulas

FORMULA First law & process table (ideal gas, n moles)

Process	Condition	W (by gas)	ΔU	Q
Isochoric	V const	0	$nC_v\Delta T$	$nC_v\Delta T$
Isobaric	P const	$P\Delta V = nR\Delta T$	$nC_v\Delta T$	$nC_p\Delta T$
Isothermal	T const	$nRT \ln \frac{V_2}{V_1}$	0	$= W$
Adiabatic	$Q = 0, PV^\gamma$ const	$\frac{P_1V_1 - P_2V_2}{\gamma - 1} = \frac{nR(T_1 - T_2)}{\gamma - 1}$	$-W$	0
Cyclic	returns to start	area enclosed	0	$= W$
Free expansion	into vacuum, insulated	0	0	0

Adiabatic relations: $PV^\gamma = \text{const}$, $TV^{\gamma-1} = \text{const}$, $P^{1-\gamma}T^\gamma = \text{const}$.

FORMULA Heat capacities

$$C_p - C_v = R \quad \gamma = \frac{C_p}{C_v}$$

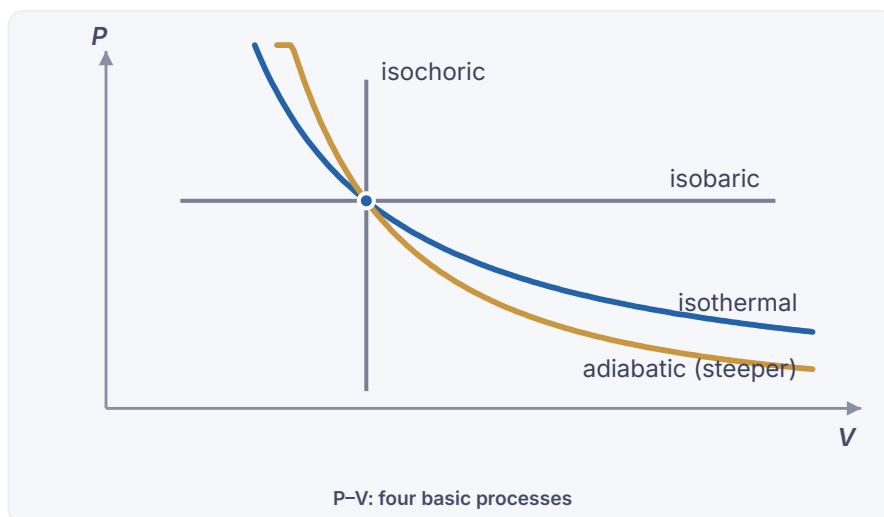
Gas	C_v	C_p	γ
Monatomic	$\frac{3}{2}R$	$\frac{5}{2}R$	$5/3 \approx 1.67$
Diatomic (rigid, room temperature)	$\frac{5}{2}R$	$\frac{7}{2}R$	$7/5 = 1.4$
Polyatomic (nonlinear, rigid)	$3R$	$4R$	$4/3 \approx 1.33$

Mixture: $C_{v,mix} = \frac{n_1C_{v1} + n_2C_{v2}}{n_1 + n_2}$. In an isobaric process, the fractions of Q are $\Delta U/Q = 1/\gamma$ and $W/Q = 1 - 1/\gamma$.

FORMULA Engines & refrigerators

$$\eta = \frac{W}{Q_1} = 1 - \frac{Q_2}{Q_1} \quad \eta_{\text{Carnot}} = 1 - \frac{T_2}{T_1} \quad \text{COP}_{\text{ref}} = \frac{Q_2}{W} = \frac{T_2}{T_1 - T_2} \quad \text{COP} = \frac{1 - \eta}{\eta}$$

Temperatures are in **kelvin**. Carnot cycle: two isothermals + two adiabatics. It is reversible, and no engine working between the same two temperatures is more efficient.



☆ IMPORTANT Reading P-V diagrams

- The **adiabatic** curve is steeper than the isothermal through the same point: slope ratio = γ .
- A clockwise cycle on a P - V diagram means net work done **by** the gas ($W > 0$), which is an engine. Anticlockwise means work is done on the gas (a refrigerator).
- On a P - T or V - T diagram, convert to P - V before finding work. Straight lines through the origin on a V - T diagram are isobaric.

Standard models & assumptions

- Ideal gas: $PV = nRT$, $R = 8.314 \text{ J mol}^{-1}\text{K}^{-1}$.
- Quasi-static (slow) processes for the formulas above. Free expansion is not quasi-static.
- "Insulated/thermally isolated/sudden" suggests adiabatic. "Slow in contact with a reservoir" means isothermal.

Frequently tested concepts & question patterns

▮ PYQ PATTERN Recurring structures

1. **Process identification** from graphs (P - V , V - T , P - T) and signs of Q , W , ΔU for each leg.
2. **Adiabatic compression/expansion**: final T or P with γ .
3. **Work in a cycle** as an area; efficiency of a given cycle.
4. **Carnot efficiency / refrigerator COP**, including "what T_1 increase raises η to ...".
5. **Heat supplied in isobaric process** and the split between ΔU and W .
6. **Mixtures of gases**: γ_{mix} .

Shortcuts

⚡ SHORTCUT ΔU first, always

For any process of an ideal gas, write $\Delta U = nC_v\Delta T$ first. Then get whichever of Q or W is easier, and the other from the first law.

Time saved: avoids integrating heat along a path.

⚡ SHORTCUT Temperature ratio in adiabatics

$T_2/T_1 = (V_1/V_2)^{\gamma-1}$. Monatomic: $\gamma - 1 = 2/3$, so compressing to $1/8$ of the volume gives $8^{2/3} = 4$, i.e. $T \times 4$.

Question classification

Type	Difficulty	Target time
First law sign bookkeeping	E	45 s
Carnot / COP	E	1 min
Adiabatic T , P , V relations	E-M	1.5 min
Cyclic process work/efficiency from graph	M	2–3 min
Isothermal work with logs	M	1.5 min
γ of mixtures, polytropic processes	M-H	2.5 min

Common mistakes

MISTAKE ALERT Mistake alerts

- Sign convention: in this book (NCERT), W is work done **by** the system. Some books use work **on** the system, which flips the sign.
- Using $^{\circ}\text{C}$ in Carnot efficiency.
- Assuming $\Delta U = 0$ in an adiabatic process. It's Q that is zero there. $\Delta U = 0$ in isothermal and cyclic processes.
- Mixing $\ln$ and $\log_{10}$: $\ln 2 = 0.693$, $\ln 10 = 2.303$.
- Using $\gamma = 5/3$ for air (diatomic, so $7/5$).

Practice questions

P08 · Practice

1 **Medium** ⌚ 1.5 min **Calc** P08-01

Two moles of an ideal gas expand isothermally at 300 K from 10 L to 20 L. The work done by the gas is about ($R = 8.314 \text{ J mol}^{-1}\text{K}^{-1}$, $\ln 2 = 0.693$):

- (A) 3.46 kJ (B) 1.73 kJ (C) 6.92 kJ (D) 4.99 kJ

2 **Medium** ⌚ 1 min **Concept** P08-02

A monatomic ideal gas at temperature T is compressed adiabatically to one-eighth of its volume. Its new temperature is:

- (A) $2T$ (B) $16T$ (C) $8T$ (D) $4T$

3 **Easy** ⌚ 1 min **NV** P08-03

An ideal gas goes round a rectangular cycle on a P - V diagram with corners at $V = 1 \text{ L}$ and 3 L and $P = 1 \times 10^5 \text{ Pa}$ and $3 \times 10^5 \text{ Pa}$. Find the magnitude of the net work per cycle (in J).

Your answer (numerical value):

4 **Easy** ⌚ 0.5 min **Speed** P08-04

A gas absorbs 500 J of heat and does 200 J of work. The change in its internal energy is:

- (A) 700 J (B) 200 J (C) -300 J (D) 300 J

5 Easy 0.5 min Speed

P08-05

A Carnot engine works between 500 K and 300 K. Its efficiency is:

- (A) 40% (B) 60% (C) 20% (D) 66.7%

6 Easy 0.75 min NV

P08-06

An ideal (Carnot) refrigerator keeps its interior at 250 K in a room at 300 K. Find its coefficient of performance.

Your answer (numerical value):

7 Medium 1 min Concept

P08-07

Heat is supplied to a rigid diatomic ideal gas at constant pressure. The fraction of the heat that increases internal energy is:

- (A) $2/7$ (B) $3/5$ (C) $5/7$ (D) $2/5$

8 Easy 0.75 min Concept

P08-08

At the same point on a P - V diagram, the ratio (slope of adiabatic)/(slope of isothermal) is:

- (A) 1 (B) γ (C) $1/\gamma$ (D) $\gamma - 1$

9 Easy 0.5 min Tricky

P08-09

An ideal gas expands freely into a vacuum inside an insulated container. Its temperature:

- (A) rises (B) falls
(C) falls to absolute zero (D) stays the same

P08 · Chapter Test

1 Easy 0.5 min Basic

P08-T1

Which of the following is a state function?

- (A) internal energy (B) work
(C) heat (D) the heat rejected in a cycle

2 Easy 0.5 min Speed

P08-T2

For a rigid diatomic ideal gas, γ is:

- (A) $5/3$ (B) $4/3$ (C) $7/5$ (D) $9/7$

3 Easy 0.75 min Basic

P08-T3

One mole of an ideal gas is heated by 10 K at constant pressure. The work done by the gas is ($R = 8.314$):

- (A) 8.3 J (B) 207.9 J (C) 83.1 J (D) 124.7 J

4 Easy 0.5 min NV

P08-T4

An engine absorbs 1000 J per cycle and rejects 750 J. Find its efficiency (in %).

Your answer (numerical value):

P08-T5

5 Easy 0.5 min Concept

A cyclic process on a P - V diagram is traversed anticlockwise. The net work done by the gas is:

- (A) positive
(B) negative
(C) zero
(D) equal to the heat absorbed, which is positive

Answers & Solutions

P08 · Practice

1 (A) 2 (D) 3 400 4 (D) 5 (A) 6 5 7 (C) 8 (B) 9 (D)

P08 · Chapter Test

1 (A) 2 (C) 3 (C) 4 25 5 (B)

P08 · PRACTICE

1 Answer: (A) Isothermal work Medium 1.5 min Calc

P08-01

SOLUTION $W = nRT \ln(V_2/V_1) = 2 \times 8.314 \times 300 \times 0.693 \approx 3457 \text{ J} \approx 3.46 \text{ kJ}$.

SHORTCUT $2 \times 8.314 \times 300 \approx 4988$; $\times 0.7 \approx 3.49$. Closest option: 3.46.

COMMON TRAP Using $\log_{10} 2 = 0.301$ gives 1.5 kJ.

2 Answer: (D) Adiabatic temperature change Medium 1 min Concept

P08-02

SOLUTION $TV^{\gamma-1} = \text{const}$ with $\gamma - 1 = 2/3$: $T_2 = T \cdot 8^{2/3} = 4T$.

SHORTCUT $8^{2/3} = (8^{1/3})^2 = 4$.

COMMON TRAP Using the isothermal relation (unchanged T), or γ in place of $\gamma - 1$.

3 Answer: 400 Work in a rectangular cycle Easy 1 min NV

P08-03

SOLUTION Area = $\Delta P \Delta V = (2 \times 10^5)(2 \times 10^{-3}) = 400 \text{ J}$.

SHORTCUT Work = enclosed area. $1 \text{ L} = 10^{-3} \text{ m}^3$.

COMMON TRAP Leaving litres unconverted.

4 Answer: (D) First law bookkeeping Easy 0.5 min Speed

P08-04

SOLUTION $\Delta U = Q - W = 500 - 200 = 300 \text{ J}$.

SHORTCUT $Q = \Delta U + W$.

COMMON TRAP Adding the two.

5 Answer: (A) Carnot efficiency Easy 0.5 min Speed

P08-05

SOLUTION $\eta = 1 - 300/500 = 0.4$.

COMMON TRAP Using T_2/T_1 (60%).

6 Answer: 5 Refrigerator COP Easy 0.75 min NV

P08-06

SOLUTION $\text{COP} = \frac{T_2}{T_1 - T_2} = \frac{250}{50} = 5$.

COMMON TRAP Using $T_1/(T_1 - T_2) = 6$ (the heat-pump COP).

7 Answer: (C) Isobaric heat split Medium 1 min Concept

P08-07

SOLUTION $\frac{\Delta U}{Q} = \frac{C_v}{C_p} = \frac{5/2}{7/2} = \frac{5}{7}$.

SHORTCUT $\Delta U/Q = 1/\gamma$.

COMMON TRAP Giving the work fraction (2/7).

8 Answer: (B) Slopes of adiabatic vs isothermal **Easy** ⌚ 0.75 min **Concept**

P08-08

SOLUTION Isothermal: $dP/dV = -P/V$. Adiabatic: $dP/dV = -\gamma P/V$. Ratio = γ .

SHORTCUT Adiabatic is steeper by a factor $\gamma > 1$.

COMMON TRAP Choosing $1/\gamma$.

9 Answer: (D) Free expansion **Easy** ⌚ 0.5 min **Tricky**

P08-09

SOLUTION $Q = 0$ (insulated) and $W = 0$ (no external pressure), so $\Delta U = 0$. For an ideal gas, U depends only on T , so T is unchanged.

SHORTCUT Free expansion of an ideal gas: $\Delta T = 0$, though it's **not** an isothermal quasi-static process.

COMMON TRAP Treating it as an adiabatic *reversible* expansion, which cools the gas.

P08 · CHAPTER TEST

1 Answer: (A) State functions **Easy** ⌚ 0.5 min **Basic**

P08-T1

SOLUTION Internal energy depends only on the state. Heat and work depend on the path. (Their difference $Q - W = \Delta U$ is path-independent, but each alone is not.)

SHORTCUT U, H, P, V, T are state functions; Q and W are path functions.

COMMON TRAP Treating heat as a property the system "has".

2 Answer: (C) γ for a diatomic gas **Easy** ⌚ 0.5 min **Speed**

P08-T2

SOLUTION $C_v = \frac{5}{2}R$ and $C_p = \frac{7}{2}R$, so $\gamma = 7/5$.

SHORTCUT $\gamma = 1 + \frac{2}{f}$ with $f = 5$.

COMMON TRAP Using the monatomic value.

3 Answer: (C) Isobaric work **Easy** ⌚ 0.75 min **Basic**

P08-T3

SOLUTION $W = nR\Delta T = 8.314 \times 10 \approx 83.1$ J.

SHORTCUT $P\Delta V = nR\Delta T$.

COMMON TRAP Computing the heat $nC_p\Delta T$ instead.

4 Answer: 25 Engine efficiency **Easy** ⌚ 0.5 min **NV**

P08-T4

SOLUTION $\eta = 1 - 750/1000 = 0.25 = 25\%$.

SHORTCUT $W = 250$ J.

COMMON TRAP Answering 75.

5 Answer: (B) Anticlockwise cycle **Easy** ⌚ 0.5 min **Concept**

P08-T5

SOLUTION Anticlockwise means net work is done **on** the gas, so $W_{by} < 0$ and the gas rejects net heat ($Q = W < 0$).

SHORTCUT Clockwise = engine; anticlockwise = refrigerator.

COMMON TRAP Taking work as always positive.

REVISION Revision tracker · Thermodynamics

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

Kinetic Theory of Gases

09

~1 per shift

TYPICAL QUESTIONS

Easy

DIFFICULTY

Must-do

PRIORITY

Class 11 · Ch 12

NCERT

~5 hours

STUDY TIME

Concept summary

- A gas is a huge number of point molecules in random motion. They collide elastically, and their pressure comes from momentum transfer to the walls.
- Temperature is a measure of average translational KE:** $\langle \frac{1}{2}mv^2 \rangle = \frac{3}{2}k_B T$, independent of the gas.
- Equipartition:** each quadratic degree of freedom carries $\frac{1}{2}k_B T$ on average, and that fixes C_v and γ .
- The **mean free path** is the average distance between collisions.

Formulas

Σ FORMULA Ideal gas & pressure

$$PV = nRT = Nk_B T \quad P = \frac{1}{3}\rho v_{rms}^2 = \frac{1}{3}\frac{N}{V}m v_{rms}^2 \quad P = \frac{2}{3}E_{tr} \text{ (per unit volume)}$$

$R = 8.314 \text{ J mol}^{-1}\text{K}^{-1}$, $k_B = 1.38 \times 10^{-23} \text{ J K}^{-1}$, $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$. Total translational KE of a gas = $\frac{3}{2}PV = \frac{3}{2}nRT$.

Σ FORMULA Molecular speeds (M = molar mass in kg/mol)

$$v_{rms} = \sqrt{\frac{3RT}{M}} \quad v_{avg} = \sqrt{\frac{8RT}{\pi M}} \quad v_{mp} = \sqrt{\frac{2RT}{M}}$$

$v_{rms} : v_{avg} : v_{mp} = \sqrt{3} : \sqrt{8/\pi} : \sqrt{2} \approx 1.73 : 1.60 : 1.41$. At the same T , speed $\propto 1/\sqrt{M}$. For one gas, speed $\propto \sqrt{T}$. Average translational KE per molecule: $\frac{3}{2}k_B T$ (about $6.2 \times 10^{-21} \text{ J}$ at 300 K).

Σ FORMULA Degrees of freedom & heat capacities

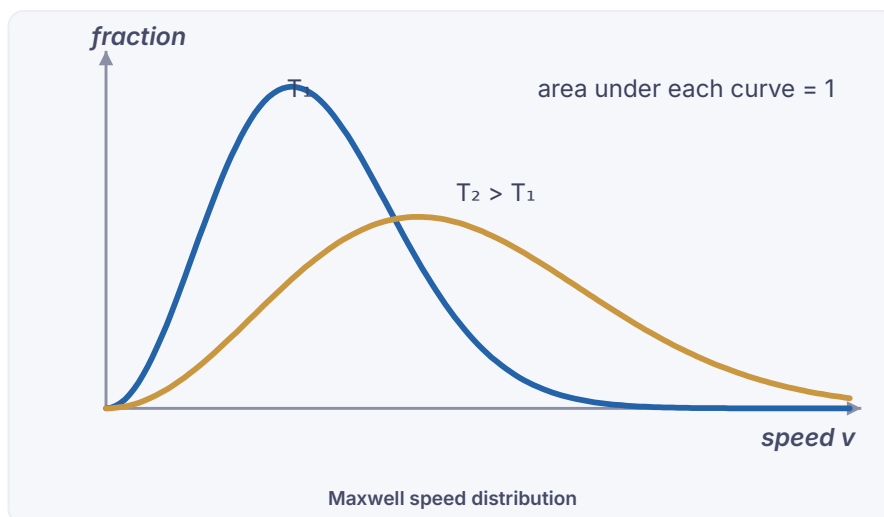
Molecule	Translational + rotational f	U per mole	C_v	$\gamma = 1 + 2/f$
Monatomic (He, Ar)	3	$\frac{3}{2}RT$	$\frac{3}{2}R$	5/3
Diatomic, rigid (O_2 , N_2 at room T)	5	$\frac{5}{2}RT$	$\frac{5}{2}R$	7/5
Diatomic with vibration (high T)	7	$\frac{7}{2}RT$	$\frac{7}{2}R$	9/7
Nonlinear polyatomic, rigid	6	$3RT$	$3R$	4/3

Each vibrational mode contributes 2 quadratic terms (kinetic + potential), i.e. $k_B T$ per molecule.

Σ FORMULA Mean free path

$$\lambda = \frac{1}{\sqrt{2}\pi d^2 n} = \frac{k_B T}{\sqrt{2}\pi d^2 P}$$

n = number density, d = molecular diameter. At constant T , $\lambda \propto 1/P$. At constant n , λ doesn't depend on T .



Standard models & assumptions

- Molecules are point masses (negligible volume), with no intermolecular forces except during collisions, and elastic collisions.
- Large number of molecules, random motion, equally likely in all directions (so $\overline{v_x^2} = \frac{1}{3}\overline{v^2}$).

Frequently tested concepts & question patterns

PYQ PATTERN Recurring structures

- rms speed at a temperature; temperature at which two gases have equal speeds.
- Speed ratios (rms/avg/most probable) and scaling with T , M .
- Degrees of freedom $\rightarrow C_v, C_p, \gamma$, including mixtures.
- Pressure-KE relation: $E = \frac{3}{2}PV$.
- Mean free path dependence on P , T , d .
- Maxwell distribution curve: how the peak shifts with T (the area stays constant).

Shortcuts

SHORTCUT Equal speeds of two gases

Speeds are equal when T/M is equal: $\frac{T_1}{M_1} = \frac{T_2}{M_2}$. No square roots needed.

SHORTCUT γ of a mixture

$\gamma_{\text{mix}} = \frac{\sum n_i C_{p,i}}{\sum n_i C_{v,i}}$. Never average the γ values directly.

Question classification

Type	Difficulty	Target time
rms speed / temperature scaling	E	1 min
$f \rightarrow \gamma, C_v$	E	45 s
Mixture γ	M	1.5 min
Mean free path	E	1 min
KE-pressure relation	E	1 min

Common mistakes

MISTAKE ALERT Mistake alerts

- Using M in g/mol inside $\sqrt{3RT/M}$. Convert to **kg/mol**.
- Using $^{\circ}\text{C}$ instead of K.
- Averaging γ values for mixtures.
- Forgetting that vibration adds 2 (not 1) to f per mode.
- Thinking $\frac{3}{2}k_B T$ depends on the gas's molar mass. It doesn't.

Practice questions

P09 · Practice

1 Easy ⌚ 1 min Calc P09-01

The rms speed of oxygen molecules ($M = 32$ g/mol) at 27°C is about ($R = 8.314$):

- (A) 484 m s^{-1} (B) 342 m s^{-1} (C) 1600 m s^{-1} (D) 250 m s^{-1}

2 Easy ⌚ 1 min NV P09-02

Find the temperature (in K) at which the rms speed of hydrogen equals that of oxygen at 47°C .

Your answer (numerical value):

3 Easy ⌚ 0.5 min Speed P09-03

To double the rms speed of a gas's molecules, its absolute temperature must be:

- (A) doubled (B) increased by $\sqrt{2}$ (C) halved (D) quadrupled

4 Easy ⌚ 0.5 min Speed P09-04

A gas has molecules with 6 degrees of freedom. Its γ is:

- (A) $5/3$ (B) $4/3$ (C) $7/5$ (D) $9/7$

5 Medium ⌚ 1.5 min Concept P09-05

One mole of a monatomic gas is mixed with one mole of a rigid diatomic gas. γ of the mixture is:

- (A) 1.5 (B) 1.4 (C) 1.53 (D) 1.67

6 Easy ⌚ 1 min Basic P09-06

The total translational kinetic energy of the molecules of an ideal gas filling 1 m^3 at a pressure of 10^5 Pa is:

- (A) 10^5 J (B) $3 \times 10^5 \text{ J}$ (C) $1.5 \times 10^5 \text{ J}$ (D) $0.67 \times 10^5 \text{ J}$

7 Easy ⌚ 0.5 min Concept P09-07

At constant temperature, if the pressure of a gas is doubled, its mean free path:

- (A) doubles (B) is unchanged (C) halves (D) becomes 4 times

8 Easy ⌚ 1 min NV P09-08

The average translational kinetic energy of a gas molecule at 300 K is $x \times 10^{-21} \text{ J}$. Find x to the nearest integer ($k_B = 1.38 \times 10^{-23} \text{ J K}^{-1}$).

Your answer (numerical value):

P09 · Chapter Test

1 Easy 0.5 min Speed

P09-T1

The ratio $v_{rms} : v_{avg} : v_{mp}$ is:

- (A) $\sqrt{2} : \sqrt{8/\pi} : \sqrt{3}$ (B) $1 : 1 : 1$ (C) $\sqrt{3} : \sqrt{2} : \sqrt{8/\pi}$ (D) $\sqrt{3} : \sqrt{8/\pi} : \sqrt{2}$

2 Easy 0.5 min Basic

P09-T2

The internal energy of 2 mol of a rigid diatomic ideal gas at temperature T is:

- (A) $3RT$ (B) $\frac{5}{2}RT$ (C) $7RT$ (D) $5RT$

3 Easy 0.75 min Basic

P09-T3

At the same temperature, the ratio of rms speeds of H_2 and O_2 molecules is:

- (A) $1 : 4$ (B) $4 : 1$ (C) $16 : 1$ (D) $1 : 16$

4 Easy 0.5 min NV

P09-T4

How many degrees of freedom (translational + rotational) does a rigid **nonlinear** triatomic molecule have?

Your answer (numerical value):

5 Easy 0.75 min Basic

P09-T5

The pressure of a gas at constant volume doubles when it is heated from 27°C to:

- (A) 54°C (B) 327°C (C) 600°C (D) 273°C

Answers & Solutions

P09 · Practice

1 (A) 2 20 3 (D) 4 (B) 5 (A) 6 (C) 7 (C) 8 6

P09 · Chapter Test

1 (D) 2 (D) 3 (B) 4 6 5 (B)

P09 · PRACTICE

1 Answer: (A) rms speed Easy 1 min Calc

P09-01

SOLUTION $v_{rms} = \sqrt{\frac{3 \times 8.314 \times 300}{0.032}} = \sqrt{2.34 \times 10^5} \approx 484 \text{ m s}^{-1}$.

SHORTCUT $3RT \approx 7483$; $7483/0.032 \approx 233\,800$; $\sqrt{} \approx 484$.

COMMON TRAP Using $M = 32$ (g) gives about 15 m/s, which isn't among the options, so re-check the units.

2 Answer: 20 Equal rms speeds Easy 1 min NV

P09-02

SOLUTION Equal speeds need T/M equal: $\frac{T_{H_2}}{2} = \frac{320}{32} \Rightarrow T_{H_2} = 20 \text{ K}$.

SHORTCUT $T \propto M$ for equal speeds.

COMMON TRAP Using 47 instead of 320 K.

3 Answer: (D) Speed-temperature scaling Easy 0.5 min Speed

P09-03

SOLUTION $v_{rms} \propto \sqrt{T}$, so $T \rightarrow 4T$.

COMMON TRAP Doubling it.

4 Answer: (B) γ from degrees of freedom Easy 0.5 min Speed

P09-04

SOLUTION $\gamma = 1 + 2/f = 1 + 1/3 = 4/3$.

5 Answer: (A) γ of a mixture **Medium** ⌚ 1.5 min **Concept**

P09-05

SOLUTION $C_v = \frac{1.5R + 2.5R}{2} = 2R$, $C_p = C_v + R = 3R$, so $\gamma = 1.5$.

SHORTCUT Average C_v , then add R .

COMMON TRAP Averaging the γ values: $(1.67 + 1.4)/2 \approx 1.53$.

6 Answer: (C) Translational KE from pressure **Easy** ⌚ 1 min **Basic**

P09-06

SOLUTION $E = \frac{3}{2}PV = 1.5 \times 10^5 \text{ J}$.

SHORTCUT $P = \frac{2}{3}(E/V)$.

COMMON TRAP Using $P = \frac{1}{3}(E/V)$.

7 Answer: (C) Mean free path vs pressure **Easy** ⌚ 0.5 min **Concept**

P09-07

SOLUTION $\lambda = \frac{k_B T}{\sqrt{2} \pi d^2 P} \propto 1/P$.

SHORTCUT More molecules per volume means more frequent collisions.

8 Answer: 6 Average KE per molecule **Easy** ⌚ 1 min **NV**

P09-08

SOLUTION $\frac{3}{2}k_B T = 1.5 \times 1.38 \times 10^{-23} \times 300 = 6.21 \times 10^{-21} \text{ J}$, so $x \approx 6$.

SHORTCUT Worth memorising: about $6.2 \times 10^{-21} \text{ J}$ at room temperature ($\approx 0.04 \text{ eV}$).

COMMON TRAP Using $\frac{1}{2}k_B T$ per molecule.

P09 · CHAPTER TEST

1 Answer: (D) Speed ratio **Easy** ⌚ 0.5 min **Speed**

P09-T1

SOLUTION From the formulas: $\sqrt{3}$, $\sqrt{8/\pi} \approx 1.60$, $\sqrt{2}$.

SHORTCUT $rms > avg > mp$.

2 Answer: (D) Internal energy **Easy** ⌚ 0.5 min **Basic**

P09-T2

SOLUTION $U = n \frac{f}{2} RT = 2 \times \frac{5}{2} RT = 5RT$.

COMMON TRAP Forgetting $n = 2$.

3 Answer: (B) rms speed ratio of two gases **Easy** ⌚ 0.75 min **Basic**

P09-T3

SOLUTION $v \propto 1/\sqrt{M}$: $\sqrt{32/2} = 4$.

COMMON TRAP Inverting the ratio.

4 Answer: 6 Degrees of freedom **Easy** ⌚ 0.5 min **NV**

P09-T4

SOLUTION 3 translational + 3 rotational = 6.

SHORTCUT Linear molecules have 2 rotational degrees of freedom; nonlinear ones have 3.

COMMON TRAP Answering 5 (as for a linear molecule).

5 Answer: (B) Gay-Lussac's law **Easy** ⌚ 0.75 min **Basic**

P09-T5

SOLUTION $P \propto T$: $T_2 = 2 \times 300 = 600 \text{ K} = 327^\circ \text{C}$.

SHORTCUT Double the kelvin temperature, then convert back.

COMMON TRAP Doubling the Celsius value.

REVISION Revision tracker · Kinetic Theory

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART II · PHYSICS · UNIT 10

Oscillations & Waves

10

1-2 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

High

PRIORITY

Class 11 · Ch 13,
14

NCERT

~12 hours

STUDY TIME

Concept summary

- **SHM** is motion under a restoring force proportional to displacement: $a = -\omega^2 x$. Its solution is a sine or cosine, and its energy oscillates between KE and PE.
- To find the period of **any** SHM system, write the restoring force or torque as $-(\text{const}) \times \text{displacement}$ and read off ω .
- A **wave** carries energy, not matter. Its speed depends on the medium: tension/density for strings, elasticity/density for sound.
- **Superposition** gives interference, standing waves (strings, pipes) and beats.

 WARNING Syllabus note

The official Unit 10 text (2024–2026) lists waves up to "**Standing waves in strings and organ pipes, fundamental mode and harmonics. Beats.**" The **Doppler effect is not listed**, so treat it as low priority [NTA-SYL].

Formulas: Oscillations

 FORMULA SHM kinematics & energy

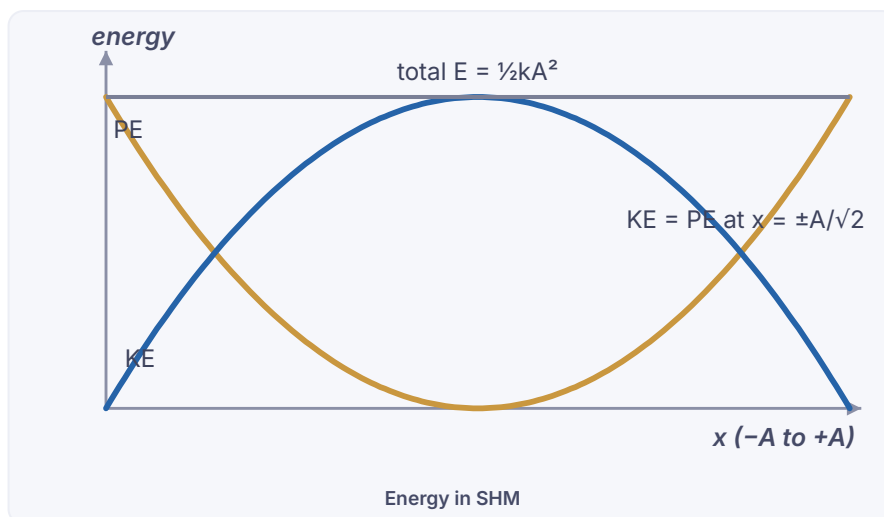
$$x = A \sin(\omega t + \phi) \quad v = \omega \sqrt{A^2 - x^2} \quad a = -\omega^2 x \quad v_{\max} = A\omega \quad a_{\max} = A\omega^2$$

$$K = \frac{1}{2}m\omega^2(A^2 - x^2) \quad U = \frac{1}{2}m\omega^2 x^2 \quad E = \frac{1}{2}m\omega^2 A^2 = \frac{1}{2}kA^2$$

- $K = U$ at $x = A/\sqrt{2}$. KE and PE each oscillate at frequency $2f$.
- Phase difference: v leads x by $\pi/2$; a leads x by π .

 FORMULA Time periods of standard systems

System	Period
Spring-mass	$T = 2\pi\sqrt{m/k}$ (horizontal or vertical; the same)
Springs in series / parallel	$\frac{1}{k_s} = \frac{1}{k_1} + \frac{1}{k_2}$; $k_p = k_1 + k_2$
Spring cut into n equal parts	each part has nk
Simple pendulum (small angle)	$T = 2\pi\sqrt{L/g}$
Pendulum in a lift accelerating up / down at a	$2\pi\sqrt{L/(g \pm a)}$
Pendulum in a car accelerating horizontally at a	$2\pi\sqrt{L/\sqrt{g^2 + a^2}}$
Physical (compound) pendulum	$2\pi\sqrt{I/(mgd)}$, d = pivot-to-COM distance
Liquid in a U-tube (total column length L)	$2\pi\sqrt{L/2g}$
Floating cylinder (submerged length h)	$2\pi\sqrt{h/g}$



Formulas: Waves

Σ FORMULA Wave speed & equation

$$v = f\lambda = \frac{\omega}{k} \quad \text{string: } v = \sqrt{T/\mu} \quad \text{sound: } v = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma RT}{M}} \text{ (Laplace)}$$

$$y = A \sin(kx - \omega t) \text{ (travelling in } +x) \quad k = \frac{2\pi}{\lambda} \quad \omega = 2\pi f \quad v_{\text{particle}} = \frac{\partial y}{\partial t} = -v \frac{\partial y}{\partial x}$$

- Speed of sound $\propto \sqrt{T}$ (kelvin), independent of pressure at constant T . Newton's formula $\sqrt{P/\rho}$ was 15% too low.
- Intensity $\propto A^2 \omega^2$. Phase difference $\Delta\phi = \frac{2\pi}{\lambda} \Delta x$.
- Reflection from a fixed end: phase change π . From a free end: no phase change.

Σ FORMULA Superposition, standing waves, beats

Interference: $A_R = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \phi}$; $I_R = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$.

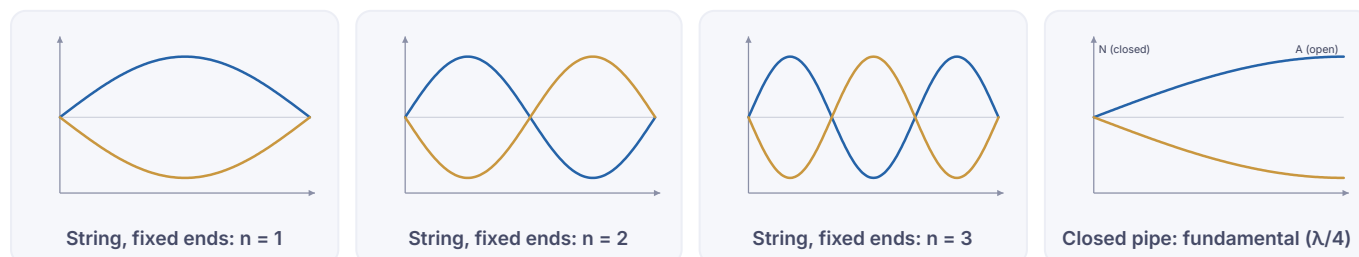
Standing wave $y = 2A \sin kx \cos \omega t$: nodes $\lambda/2$ apart, node-to-antinode $\lambda/4$.

System	Allowed frequencies	Harmonics present
String fixed at both ends	$f_n = \frac{nv}{2L}$	all ($n = 1, 2, 3, \dots$)
Open pipe (both ends open)	$f_n = \frac{nv}{2L}$	all
Closed pipe (one end closed)	$f_n = \frac{(2n-1)v}{4L}$	odd only (1, 3, 5, ...)

End correction $e \approx 0.6r$: closed pipe $L \rightarrow L + e$; open pipe $L \rightarrow L + 2e$.

Resonance tube: $\lambda = 2(L_2 - L_1)$, end correction $e = \frac{L_2 - 3L_1}{2}$.

Beats: $f_{\text{beat}} = |f_1 - f_2|$. Loading a fork with wax **lowers** its frequency; filing its prongs **raises** it.



Nodes (N) where the envelope pinches; antinodes (A) at maximum amplitude.

Frequently tested concepts & question patterns

PYQ PATTERN Recurring structures

1. **Period of a system:** springs in combination, pendulum in an accelerating frame, liquid columns.
2. **SHM energy:** where $K = U$; speed at a given displacement; phase from given x and v .
3. **Wave equation reading:** find v , λ , f , direction and particle velocity from $y(x, t)$.
4. **Organ pipes and strings:** harmonics, overtones, open vs closed comparisons, resonance tube.
5. **Beats with wax/filing** to determine an unknown frequency.
6. **Superposition of two SHMs** in the same direction (resultant amplitude and phase).

Shortcuts

SHORTCUT Overtone vs harmonic

For a closed pipe, the **first overtone is the 3rd harmonic**. For open pipes and strings, the first overtone is the 2nd harmonic. Translate the question's word before computing.

SHORTCUT Period by "restoring constant"

Write $F = -k_{eff}x$ (or $\tau = -\kappa\theta$). Then $T = 2\pi\sqrt{m/k_{eff}}$ (or $2\pi\sqrt{I/\kappa}$). Works for liquids, floating bodies and springs with pulleys.

Question classification

Type	Difficulty	Target time
SHM formulas (v_{max} , a_{max} , energy)	E	1 min
Spring combinations / pendulum variants	E-M	1.5 min
Wave-equation reading	E	1 min
Pipes and strings harmonics	M	2 min
Beats with loading	M	1.5 min
Composite SHM (pulley + spring, rolling + spring)	H	3 min (Round 2)

Common mistakes

MISTAKE ALERT Mistake alerts

- Closed pipe: only **odd** harmonics. Asked for the "second harmonic of a closed pipe"? It doesn't exist.
- Confusing μ (mass per unit **length**) with density.
- Using $^{\circ}\text{C}$ in $v \propto \sqrt{T}$.
- Wax lowers frequency. After deciding the sign of the change, check which way the beats moved.
- The spring-mass period is independent of g . The pendulum period is independent of mass.

Practice questions

P10 · Practice

1 Easy 0.75 min Basic

P10-01

A particle performs SHM with amplitude 10 cm and period 2 s. Its maximum speed is:

- (A) $\pi \text{ m s}^{-1}$ (B) $0.2\pi \text{ m s}^{-1}$ (C) $0.05\pi \text{ m s}^{-1}$ (D) $0.1\pi \text{ m s}^{-1}$

2 Easy 0.5 min Speed

P10-02

In SHM of amplitude A , kinetic energy equals potential energy at a displacement of:

- (A) $A/2$ (B) $A/4$ (C) $A/\sqrt{2}$ (D) $\sqrt{3}A/2$

3 Medium 1 min Basic

P10-03

A mass m hangs from two springs of constants k and $2k$ joined in series. The period of oscillation is:

- (A) $2\pi\sqrt{m/3k}$ (B) $2\pi\sqrt{3m/2k}$ (C) $2\pi\sqrt{2m/3k}$ (D) $2\pi\sqrt{3m/k}$

4 Medium 1 min Concept

P10-04

A simple pendulum has period T . In a lift accelerating **upward** at $g/3$, its period becomes:

- (A) $\frac{\sqrt{3}}{2}T$ (B) $\frac{2}{\sqrt{3}}T$ (C) $\frac{3}{4}T$ (D) T

5 Easy 0.75 min NV

P10-05

A string with linear mass density 0.01 kg m^{-1} is under 100 N tension. Find the transverse wave speed (in m/s).

Your answer (numerical value):

6 Medium 1.5 min Concept

P10-06

A pipe closed at one end is 25 cm long, and the speed of sound is 340 m/s (ignore end correction). The frequency of its **first overtone** is:

- (A) 680 Hz (B) 1020 Hz (C) 340 Hz (D) 1360 Hz

7 Medium 1.5 min Tricky

P10-07

Fork A (256 Hz) and fork B give 4 beats per second. When B is loaded with a little wax, the beat frequency **decreases**. The original frequency of B was:

- (A) 260 Hz (B) 252 Hz (C) 256 Hz (D) 248 Hz

8 Easy 1 min NV

P10-08

A wave is described by $y = 0.02 \sin[2\pi(10t - 0.5x)]$ (SI units). Find its speed (in m/s).

Your answer (numerical value):

9 Medium 1.5 min JEE

P10-09

In a resonance-tube experiment with a 500 Hz fork, the first and second resonances occur at 16.0 cm and 50.0 cm. The speed of sound is:

- (A) 340 m/s (B) 330 m/s (C) 350 m/s (D) 320 m/s

10 Easy 1 min Basic

P10-10

A 1 m string fixed at both ends vibrates in 3 loops at 300 Hz. The wave speed on the string is:

- (A) 100 m/s (B) 600 m/s (C) 300 m/s (D) 200 m/s

P10 · Chapter Test

1 Easy 0.5 min Speed

P10-T1

The maximum acceleration of a particle in SHM with $A = 5$ cm and $\omega = 10$ rad/s is:

- (A) 0.5 m s^{-2} (B) 50 m s^{-2} (C) 5 m s^{-2} (D) 0.05 m s^{-2}

2 Easy 0.5 min Speed

P10-T2

If the absolute temperature of air is made 4 times larger, the speed of sound in it becomes:

- (A) 4 times (B) unchanged (C) 16 times (D) 2 times

3 Medium 1 min Concept

P10-T3

A mass on a spring has period T . The spring is cut into two equal halves and the same mass is hung from one half. The new period is:

- (A) $T/\sqrt{2}$ (B) $T/2$ (C) $\sqrt{2}T$ (D) $2T$

4 Easy 0.5 min NV

P10-T4

Two sources of 510 Hz and 514 Hz sound together. Find the number of beats heard per second.

Your answer (numerical value):

5 Easy 0.75 min Basic

P10-T5

An open pipe and a closed pipe have the same length. The ratio of their fundamental frequencies (open : closed) is:

- (A) 1 : 2 (B) 1 : 1 (C) 2 : 1 (D) 4 : 1

Answers & Solutions

P10 · Practice

1 (D) 2 (C) 3 (B) 4 (A) 5 100 6 (B) 7 (A) 8 20 9 (A) 10 (D)

P10 · Chapter Test

1 (C) 2 (D) 3 (A) 4 4 5 (C)

P10 · PRACTICE

1 Answer: (D) Maximum speed in SHM Easy 0.75 min Basic

P10-01

SOLUTION $\omega = 2\pi/T = \pi \text{ rad/s}$; $v_{\max} = A\omega = 0.1\pi \approx 0.314 \text{ m s}^{-1}$.

COMMON TRAP Forgetting to convert cm to m.

2 Answer: (C) Equal kinetic and potential energy Easy 0.5 min Speed

P10-02

SOLUTION $\frac{1}{2}k(A^2 - x^2) = \frac{1}{2}kx^2 \Rightarrow x = A/\sqrt{2}$.

COMMON TRAP Answering $A/2$.

3 Answer: (B) Springs in series Medium 1 min Basic

P10-03

SOLUTION $k_s = \frac{k \cdot 2k}{3k} = \frac{2k}{3}$; $T = 2\pi\sqrt{m/k_s} = 2\pi\sqrt{3m/2k}$.

SHORTCUT Series springs behave like parallel resistors.

COMMON TRAP Adding them ($3k$), which is the parallel case.

4 Answer: (A) *Pendulum in accelerating lift* **Medium** ⌚ 1 min **Concept**

P10-04

SOLUTION $g_{eff} = g + g/3 = \frac{4}{3}g$. $T' = T\sqrt{g/g_{eff}} = T\sqrt{3/4} = \frac{\sqrt{3}}{2}T$.

SHORTCUT Upward acceleration raises g_{eff} , so the period gets shorter.

COMMON TRAP Using $g - a$.

5 Answer: 100 *Wave speed on a string* **Easy** ⌚ 0.75 min **NV**

P10-05

SOLUTION $v = \sqrt{T/\mu} = \sqrt{10^4} = 100 \text{ m s}^{-1}$.

6 Answer: (B) *Overtone of a closed pipe* **Medium** ⌚ 1.5 min **Concept**

P10-06

SOLUTION Fundamental = $\frac{v}{4L} = \frac{340}{1} = 340 \text{ Hz}$. The first overtone of a closed pipe is the 3rd harmonic: 1020 Hz.

SHORTCUT Closed pipe: 1, 3, 5, ... × fundamental.

COMMON TRAP Taking the 2nd harmonic (680 Hz).

7 Answer: (A) *Beats with wax loading* **Medium** ⌚ 1.5 min **Tricky**

P10-07

SOLUTION B is 252 or 260 Hz. Wax lowers B's frequency. If B were 252, lowering it would *increase* the beats. The beats decreased, so B = 260 Hz.

SHORTCUT Wax pulls f down. Fewer beats means f_B was above f_A .

COMMON TRAP Assuming wax raises the frequency.

8 Answer: 20 *Speed from wave equation* **Easy** ⌚ 1 min **NV**

P10-08

SOLUTION $\omega = 20\pi$, $k = \pi$, so $v = \omega/k = 20 \text{ m s}^{-1}$ (towards $+x$).

SHORTCUT $v = \text{coefficient of } t \div \text{coefficient of } x = 10/0.5$.

COMMON TRAP Reading $\lambda = 0.5 \text{ m}$. It is actually $1/0.5 = 2 \text{ m}$, so the wrong reading gives 5 m/s.

9 Answer: (A) *Resonance tube* **Medium** ⌚ 1.5 min **JEE**

P10-09

SOLUTION $\lambda/2 = L_2 - L_1 = 34 \text{ cm}$, so $\lambda = 0.68 \text{ m}$ and $v = f\lambda = 340 \text{ m s}^{-1}$. (End correction = $(L_2 - 3L_1)/2 = 1 \text{ cm}$.)

SHORTCUT The difference between successive resonances is always $\lambda/2$, so the end correction cancels.

COMMON TRAP Using $4L_1$ as λ gives 320 m/s.

10 Answer: (D) *Standing wave on a string* **Easy** ⌚ 1 min **Basic**

P10-10

SOLUTION 3 loops means $3\lambda/2 = 1 \text{ m}$, so $\lambda = 2/3 \text{ m}$. $v = 300 \times \frac{2}{3} = 200 \text{ m s}^{-1}$.

SHORTCUT $f_1 = 100 \text{ Hz} = v/2L$, so $v = 200$.

COMMON TRAP Taking $\lambda = L/3$.

P10 · CHAPTER TEST

1 Answer: (C) *Maximum acceleration* **Easy** ⌚ 0.5 min **Speed**

P10-T1

SOLUTION $a_{max} = \omega^2 A = 100 \times 0.05 = 5 \text{ m s}^{-2}$.

2 Answer: (D) *Sound speed vs temperature* **Easy** ⌚ 0.5 min **Speed**

P10-T2

SOLUTION $v \propto \sqrt{T}$.

3 Answer: (A) *Cutting a spring* **Medium** ⌚ 1 min **Concept**

P10-T3

SOLUTION Each half has spring constant $2k$, so $T' = 2\pi\sqrt{m/2k} = T/\sqrt{2}$.

SHORTCUT $k \propto 1/\text{length}$.

COMMON TRAP Thinking a shorter spring is "weaker".

4 Answer: 4 *Beat frequency* **Easy** ⌚ 0.5 min **NV**

P10-T4

SOLUTION $|514 - 510| = 4$.

5 Answer: (C) *Open vs closed pipe* **Easy** ⌚ 0.75 min **Basic**

P10-T5

SOLUTION Open: $v/2L$. Closed: $v/4L$. Ratio 2 : 1.



REVISION Revision tracker · Oscillations & Waves

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

Electrostatics & Capacitors

11

2–3 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

Must-do

PRIORITY

Class 12 · Ch 1, 2

NCERT

~16 hours

STUDY TIME

Concept summary

- Charge is **quantised** ($q = ne$) and **conserved**. **Coulomb's law** gives the force between point charges, and superposition handles many charges.
- The **electric field** $\vec{E} = \vec{F}/q_0$ is a vector. The **potential** V is a scalar, and $\vec{E} = -\nabla V$. Scalar potentials are usually easier to add.
- Gauss's law** gives the field of symmetric distributions (line, sheet, sphere) in one line.
- Conductors** in equilibrium: $E = 0$ inside, all charge on the surface, the surface is an equipotential, and the field just outside is σ/ϵ_0 ($\perp$ to the surface).
- Capacitors** store energy $\frac{1}{2}CV^2$. Dielectrics increase C by a factor K .

Formulas

Σ FORMULA Coulomb, field, potential

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \quad k = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2 \text{C}^{-2} \quad E = \frac{kq}{r^2} \quad V = \frac{kq}{r}$$

$$\vec{E} = -\frac{dV}{dr} \hat{r} \quad V_A - V_B = -\int_B^A \vec{E} \cdot d\vec{l} \quad U_{12} = \frac{kq_1 q_2}{r} \quad W_{ext} = q(V_f - V_i)$$

In a medium of dielectric constant K : F and E are reduced by a factor K .

Σ FORMULA Electric dipole ($\vec{p} = q\vec{d}$, from – to +)

Quantity	Axial point (distance $r \gg d$)	Equatorial point
Field	$\frac{2kp}{r^3}$ (along $\vec{p}$)	$\frac{kp}{r^3}$ (opposite to $\vec{p}$)
Potential	$\frac{kp}{r^2}$	0

General point: $V = \frac{kp \cos \theta}{r^2}$, $E = \frac{kp}{r^3} \sqrt{1 + 3 \cos^2 \theta}$.

In a uniform field: torque $\vec{\tau} = \vec{p} \times \vec{E}$, energy $U = -\vec{p} \cdot \vec{E}$, and net force zero. Work to rotate from θ_1 to θ_2 is $pE(\cos \theta_1 - \cos \theta_2)$. Stable equilibrium at $\theta = 0$, unstable at $\theta = 180^\circ$.

Σ FORMULA Gauss's law results

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0}$$

Distribution	Field
Infinite line charge (λ)	$\frac{\lambda}{2\pi\epsilon_0 r}$
Infinite plane sheet (σ)	$\frac{\sigma}{2\epsilon_0}$ (independent of distance)
Near a charged conductor's surface	$\frac{\sigma}{\epsilon_0}$
Thin spherical shell (Q, R)	0 inside; $\frac{kQ}{r^2}$ outside
Uniformly charged solid sphere	$\frac{kQr}{R^3}$ inside; $\frac{kQ}{r^2}$ outside

Flux through a closed surface depends **only** on the enclosed charge. A charge q at the centre of a cube gives flux $q/6\epsilon_0$ through each face. At a corner of the cube, the flux through the whole cube is $q/8\epsilon_0$.

Σ FORMULA Capacitors

$$C = \frac{Q}{V} \quad \text{parallel plate: } C = \frac{\epsilon_0 A}{d} \quad \text{with dielectric filling the gap: } C = \frac{K\epsilon_0 A}{d}$$

$$\text{slab of thickness } t < d : C = \frac{\epsilon_0 A}{d - t + t/K} \quad \text{metal slab: } C = \frac{\epsilon_0 A}{d - t}$$

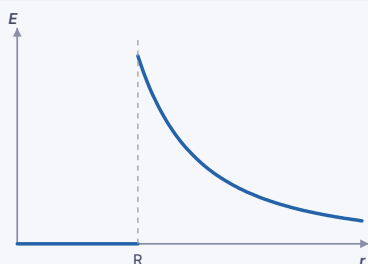
$$\text{isolated sphere: } C = 4\pi\epsilon_0 R \quad U = \frac{1}{2}CV^2 = \frac{Q^2}{2C} = \frac{1}{2}QV \quad u = \frac{1}{2}\epsilon_0 E^2$$

Series: $\frac{1}{C} = \sum \frac{1}{C_i}$ (same Q). **Parallel:** $C = \sum C_i$ (same V).

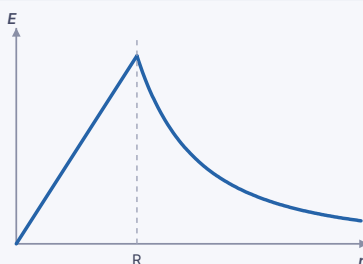
Sharing charge between two capacitors: common $V = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$; energy lost = $\frac{C_1 C_2 (V_1 - V_2)^2}{2(C_1 + C_2)}$.

Inserting a dielectric: battery connected $\rightarrow V$ fixed, so Q, C, U all rise by K . Battery disconnected $\rightarrow Q$ fixed, so V and U fall by K .

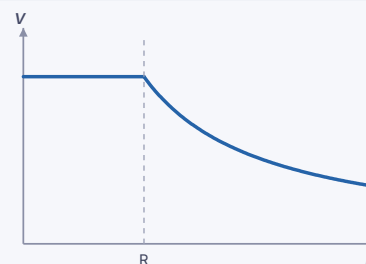
Force between the plates: $F = \frac{Q^2}{2\epsilon_0 A}$.



Conducting / hollow sphere



Uniformly charged solid sphere



Potential of a conducting sphere

☆ IMPORTANT Conductors vs insulators in one table

	Conductor (charged, isolated)	Insulator (uniform volume charge)
Charge location	outer surface only	throughout the volume
E inside	0	grows $\propto r$ (sphere)
V inside	constant = surface value	maximum at the centre, $\frac{3}{2}kQ/R$
Sharp points	high σ , high E (corona discharge)	—

Standard models & assumptions

- Point charges; infinite sheets and lines as idealisations; fringing ignored for parallel-plate capacitors.
- "Earthed" means $V = 0$, not $Q = 0$.
- The potential at infinity is zero.

Frequently tested concepts & question patterns

▮ PYQ PATTERN Recurring structures

1. **Superposition:** net force or field at a point due to charges at the corners of a triangle or square; null points on a line.
2. **Gauss's law / flux:** charge inside a cube, flux through a face; field of a sphere at $r < R$ vs $r > R$.
3. **Dipoles:** torque, work to rotate, axial vs equatorial ratios.
4. **Capacitor networks:** equivalent C , charge on one capacitor, energy stored.
5. **Dielectric insertion** with and without a battery; partially filled capacitors.
6. **Potential energy of a system of charges** (work to assemble).
7. **Motion of a charge in a uniform field** (projectile analogue: $a = qE/m$).

Shortcuts

⚡ SHORTCUT Symmetry kills vectors

Equal charges at the vertices of a regular polygon give $E = 0$ at the centre, but $V = nkq/r \neq 0$. Use symmetry to cancel field components before writing any vector sum.

Time saved: 1–2 minutes on "square of charges" questions.

⚡ SHORTCUT Dielectric slab as series capacitors

A slab of thickness t in a gap d acts as an air capacitor $(d - t)$ in series with a dielectric capacitor (t, K) . Slabs placed side by side (each covering part of the area) act in **parallel**.

Question classification

Type	Difficulty	Target time
Coulomb / field superposition	E–M	1.5 min
Gauss flux (cube/face)	E	45 s
Dipole torque/energy	E	1 min
Capacitor networks	M	2 min
Dielectric with/without battery	M	1.5 min
System energy / work to assemble	M	2 min
Non-uniform charge density + Gauss	H	3 min (Round 2)

Common mistakes

MISTAKE ALERT Mistake alerts

- Adding fields as scalars. Resolve into components first.
- Sheet field: $\sigma/2\epsilon_0$ for an isolated sheet but σ/ϵ_0 near a conductor's surface.
- Dielectric questions: always check whether the **battery stays connected**.
- Potential energy of a system: count **each pair once**, $\frac{n(n-1)}{2}$ pairs.
- Forgetting the sign of charges in potential sums.

Practice questions

P11 · Practice

1 Easy 1 min Basic P11-01

Two point charges repel with force F . If each charge is doubled and the distance between them is also doubled, the force becomes:

- (A) $4F$ (B) $2F$ (C) $F/2$ (D) F

2 Easy 0.75 min Speed P11-02

A charge q sits at the centre of a cube. The electric flux through **one face** of the cube is:

- (A) q/ϵ_0 (B) $q/8\epsilon_0$ (C) $q/6\epsilon_0$ (D) $6q/\epsilon_0$

3 Medium 1.5 min Concept P11-03

A dipole of moment p is aligned with a uniform field E . The work needed to rotate it by 180° is:

- (A) 0 (B) $2pE$ (C) pE (D) $-2pE$

4 Medium 2 min JEE P11-04

Three capacitors of $2\ \mu\text{F}$, $3\ \mu\text{F}$ and $6\ \mu\text{F}$ are in series across a 12 V battery. The charge on the $3\ \mu\text{F}$ capacitor and the energy it stores are:

- (A) $12\ \mu\text{C}$, $24\ \mu\text{J}$ (B) $12\ \mu\text{C}$, $48\ \mu\text{J}$ (C) $24\ \mu\text{C}$, $24\ \mu\text{J}$ (D) $6\ \mu\text{C}$, $6\ \mu\text{J}$

5 Medium 1.5 min Concept P11-05

A charged parallel-plate capacitor is **disconnected** from its battery, and a dielectric slab ($K = 4$) is then inserted to fill the gap. The stored energy:

- (A) becomes $1/4$ (B) becomes 4 times (C) stays the same (D) becomes 16 times

6 Medium 1.5 min Concept P11-06

A uniformly charged **non-conducting** sphere of radius R carries total charge Q . The field at $r = R/2$ is:

- (A) 0 (B) $\frac{kQ}{2R^2}$ (C) $\frac{4kQ}{R^2}$ (D) $\frac{kQ}{4R^2}$

7 Medium 2 min NV P11-07

Three charges of $+1\ \mu\text{C}$ each are placed at the corners of an equilateral triangle of side 0.3 m. Find the electrostatic potential energy of the system (in mJ). ($k = 9 \times 10^9\ \text{SI.}$)

Your answer (numerical value):

8 Easy 1 min NV

P11-08

A $2\ \mu\text{F}$ capacitor charged to 100 V is connected in parallel to an uncharged $3\ \mu\text{F}$ capacitor. Find the common potential (in V).

Your answer (numerical value):

9 Easy 1 min Basic

P11-09

Charges $+4q$ and $+q$ are 30 cm apart. The point where the net field is zero lies on the line joining them, at a distance from $+4q$ of:

- (A) 10 cm (B) 15 cm (C) 24 cm (D) 20 cm

P11 · Chapter Test

1 Easy 0.5 min Speed

P11-T1

Inside a charged hollow conducting sphere, the electric field is:

- (A) maximum at the centre (B) kQ/R^2
(C) zero (D) proportional to r

2 Easy 0.75 min Basic

P11-T2

If the plate separation of a parallel-plate capacitor is halved and its plate area is doubled, its capacitance becomes:

- (A) 4 times (B) 2 times (C) the same (D) 1/4 times

3 Medium 1 min Concept

P11-T3

At a point on the equatorial line of a short dipole:

- (A) $V = 0$ and $\vec{E}$ is parallel to $\vec{p}$ (B) $V = 0$ and $\vec{E}$ is antiparallel to $\vec{p}$
(C) $V \neq 0$ and $\vec{E} = 0$ (D) V and E are both zero

4 Easy 1 min NV

P11-T4

A $10\ \mu\text{F}$ capacitor is charged to 20 V. Find the stored energy (in mJ).

Your answer (numerical value):

5 Medium 1.5 min JEE

P11-T5

A parallel-plate capacitor with gap d has capacitance C_0 . A dielectric slab of $K = 2$ and thickness $d/2$ is inserted parallel to the plates. The new capacitance is:

- (A) $\frac{4}{3}C_0$ (B) $\frac{3}{2}C_0$ (C) $2C_0$ (D) $\frac{5}{4}C_0$

Answers & Solutions

P11 · Practice

1 (D) 2 (C) 3 (B) 4 (A) 5 (A) 6 (B) 7 90 8 40 9 (D)

P11 · Chapter Test

1 (C) 2 (A) 3 (B) 4 2 5 (A)

P11 · PRACTICE

1 Answer: (D) *Coulomb force change* **Easy** ⌚ 1 min **Basic** P11-01

SOLUTION $F \propto q_1 q_2 / r^2 = \frac{(2)(2)}{2^2} = 1$, so the force is unchanged.

SHORTCUT Numerator $\times 4$, denominator $\times 4$.

2 Answer: (C) *Flux through a cube face* **Easy** ⌚ 0.75 min **Speed** P11-02

SOLUTION The total flux q/ϵ_0 divides equally among 6 faces by symmetry.

COMMON TRAP Choosing the corner-charge value $q/8\epsilon_0$.

3 Answer: (B) *Work to rotate a dipole* **Medium** ⌚ 1.5 min **Concept** P11-03

SOLUTION $W = pE(\cos 0^\circ - \cos 180^\circ) = 2pE$.

SHORTCUT From stable ($U = -pE$) to unstable ($U = +pE$).

COMMON TRAP Using the torque $pE \sin \theta$, which is zero at both ends.

4 Answer: (A) *Capacitor network* **Medium** ⌚ 2 min **JEE** P11-04

SOLUTION $\frac{1}{C} = \frac{1}{2} + \frac{1}{3} + \frac{1}{6} = 1 \Rightarrow C = 1 \mu\text{F}$. $Q = 12 \mu\text{C}$ on each. $U_3 = \frac{Q^2}{2C_3} = \frac{144}{6} = 24 \mu\text{J}$.

SHORTCUT In series: same Q ; $V_3 = Q/C_3 = 4 \text{ V}$, so $U = \frac{1}{2}QV = 24 \mu\text{J}$.

COMMON TRAP Using $\frac{1}{2}CV^2$ with the full 12 V.

5 Answer: (A) *Dielectric with battery disconnected* **Medium** ⌚ 1.5 min **Concept** P11-05

SOLUTION Q is fixed and $C \rightarrow 4C$, so $U = \frac{Q^2}{2C}$ falls to $1/4$.

SHORTCUT Isolated: $U \propto 1/C$. Connected: $U \propto C$.

COMMON TRAP Assuming the battery stays connected ($\times 4$).

6 Answer: (B) *Field of a charged solid sphere* **Medium** ⌚ 1.5 min **Concept** P11-06

SOLUTION Inside: $E = \frac{kQr}{R^3} = \frac{kQ}{2R^2}$ at $r = R/2$.

SHORTCUT Inside $E \propto r$: at half the radius, half the surface field.

COMMON TRAP Treating it as a conductor ($E = 0$).

7 Answer: 90 *Potential energy of a charge system* **Medium** ⌚ 2 min **NV** P11-07

SOLUTION Each pair: $U = \frac{kq^2}{r} = \frac{9 \times 10^9 \times 10^{-12}}{0.3} = 0.03 \text{ J}$. Three pairs: $0.09 \text{ J} = 90 \text{ mJ}$.

SHORTCUT Count the 3 pairs once each.

COMMON TRAP Counting 6 ordered pairs.

8 Answer: 40 *Common potential after sharing* **Easy** ⌚ 1 min **NV** P11-08

SOLUTION $V = \frac{C_1 V_1}{C_1 + C_2} = \frac{200}{5} = 40 \text{ V}$.

SHORTCUT Charge is conserved; energy is not.

COMMON TRAP Conserving energy instead of charge.

9 Answer: (D) *Null point of two charges* **Easy** ⌚ 1 min **Basic** P11-09

SOLUTION $\frac{4q}{x^2} = \frac{q}{(30-x)^2} \Rightarrow 2(30-x) = x \Rightarrow x = 20 \text{ cm}$.

SHORTCUT For like charges, the null point lies between them, closer to the smaller charge.

COMMON TRAP Measuring from the smaller charge.

P11 · CHAPTER TEST

1 Answer: (C) *Field inside a conductor* **Easy** ⌚ 0.5 min **Speed** P11-T1

SOLUTION In electrostatic equilibrium, the field inside a conductor (and inside its cavity, if there's no charge in it) is zero.

2 Answer: (A) *Parallel-plate capacitance* **Easy** ⌚ 0.75 min **Basic** P11-T2

SOLUTION $C \propto A/d = 2/(1/2) = 4$.

3 Answer: (B) Dipole on its equatorial line **Medium** ⌚ 1 min **Concept**

P11-T3

SOLUTION On the equatorial line the potentials of $+q$ and $-q$ cancel, but their fields add to a resultant pointing opposite to $\vec{p}$.

SHORTCUT Zero potential doesn't mean zero field.

COMMON TRAP Choosing (D).

4 Answer: 2 Energy of a capacitor **Easy** ⌚ 1 min **NV**

P11-T4

SOLUTION $U = \frac{1}{2}CV^2 = \frac{1}{2} \times 10^{-5} \times 400 = 2 \times 10^{-3} \text{ J} = 2 \text{ mJ}$.

COMMON TRAP Forgetting the $\frac{1}{2}$ (4 mJ).

5 Answer: (A) Partially filled capacitor **Medium** ⌚ 1.5 min **JEE**

P11-T5

SOLUTION $C = \frac{\epsilon_0 A}{d - t + t/K} = \frac{\epsilon_0 A}{d/2 + d/4} = \frac{4\epsilon_0 A}{3d} = \frac{4}{3}C_0$.

SHORTCUT Effective gap: $d - t(1 - 1/K)$.

COMMON TRAP Treating the halves as parallel gives $\frac{3}{2}C_0$.

REVISION Revision tracker · Electrostatics

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART II · PHYSICS · UNIT 12

Current Electricity

12

2–3 per shift

TYPICAL QUESTIONS

Easy–Medium

DIFFICULTY

Must-do

PRIORITY

Class 12 · Ch 3

NCERT

~12 hours

STUDY TIME

Concept summary

- **Current** is the rate of flow of charge. In metals, free electrons **drift** slowly ($\sim 10^{-4}$ m/s) against the field, while the field itself sets up almost instantly.
- **Ohm's law** $V = IR$ holds for ohmic conductors at constant temperature. Diodes and electrolytes are non-ohmic.
- **Resistance** depends on geometry and material: $R = \rho L/A$. Resistivity rises with temperature for metals and falls for semiconductors.
- **Kirchhoff's laws** (junction = charge conservation, loop = energy conservation) solve every circuit. The Wheatstone bridge and metre bridge are special cases.

⚠ WARNING Syllabus note

The 2024–2026 syllabus text for this unit ends at "**Kirchhoff's laws and their applications. Wheatstone bridge. Metre Bridge.**" The **potentiometer** and **colour code** are not listed. Keep them to a quick read only [NTA-SYL].

Formulas

Σ FORMULA Microscopic view

$$I = \frac{dq}{dt} = neAv_d \quad v_d = \frac{eE\tau}{m} \quad \mu = \frac{v_d}{E} = \frac{e\tau}{m} \quad J = \frac{I}{A} = \sigma E \quad \rho = \frac{m}{ne^2\tau}$$

Σ FORMULA Resistance

$$R = \frac{\rho L}{A} \quad R_T = R_0(1 + \alpha \Delta T) \quad \text{Series: } R = \sum R_i \quad \text{Parallel: } \frac{1}{R} = \sum \frac{1}{R_i}$$

- **Stretching a wire** (volume constant) to n times its length: $R' = n^2 R$. If its radius becomes r/n : $R' = n^4 R$.
- n equal resistors: $R_{\text{series}}/R_{\text{parallel}} = n^2$.

Σ FORMULA Cells

$$V_{\text{terminal}} = \varepsilon - Ir \text{ (discharging)} \quad V = \varepsilon + Ir \text{ (charging)} \quad I = \frac{\varepsilon}{R + r}$$

- n identical cells in series: $n\varepsilon, nr$. In parallel: $\varepsilon, r/n$. Mixed (m rows of n cells): $I = \frac{n\varepsilon}{R + nr/m}$, maximum when $R = nr/m$.
- Two different cells in parallel: $\varepsilon_{eq} = \frac{\varepsilon_1/r_1 + \varepsilon_2/r_2}{1/r_1 + 1/r_2}$, $r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$.
- **Maximum power transfer:** $R = r$ gives $P_{\text{max}} = \frac{\varepsilon^2}{4r}$ (efficiency 50%).

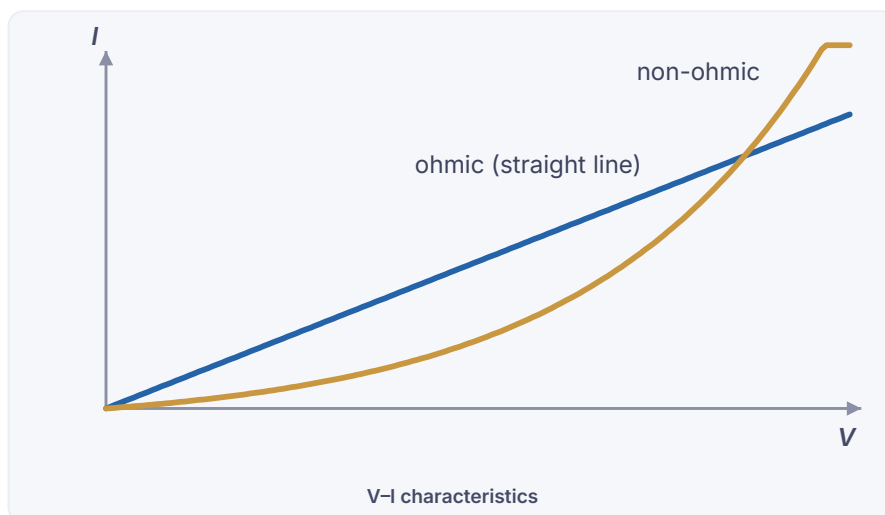
Σ FORMULA Power & bridges

$$P = VI = I^2 R = \frac{V^2}{R} \quad \text{rated bulb: } R = \frac{V_{\text{rated}}^2}{P_{\text{rated}}}$$

Wheatstone bridge balance: $\frac{P}{Q} = \frac{R}{S}$ (no current through the galvanometer; remove that arm).

Metre bridge: $\frac{R}{S} = \frac{l}{100-l}$ (l in cm from the end next to R). Most sensitive near the middle.

- Bulbs in **series**: the lower-wattage bulb (higher R) glows brighter. In **parallel**: the higher-wattage bulb glows brighter.
- Heater at a lower voltage: $P' = P(V'/V)^2$.



Kirchhoff's laws in practice

1. Mark currents in each branch with arbitrary directions.
2. **Junction rule:** $\sum I_{\text{in}} = \sum I_{\text{out}}$.
3. **Loop rule:** going around a loop, $\sum \Delta V = 0$. Across a resistor in the direction of current: $-IR$. Across a cell from $-$ to $+$: $+\mathcal{E}$.
4. Solve. A negative current just means the real direction is opposite to the one you assumed.

⚡ SHORTCUT Nodal (potential) method

Set one node at 0 V. Write the other node potentials as unknowns, and use $\sum \frac{V - V_j}{R_j} = 0$ at each node. This usually needs fewer equations than loops.

Time saved: 1–2 minutes on two-loop circuits.

⚡ SHORTCUT Symmetry & balanced bridges

If a network has a symmetry plane, points on it are at equal potential: join them or remove the resistors between them. Before solving, check $P/Q = R/S$.

Frequently tested concepts & question patterns

PYQ PATTERN Recurring structures

1. **Equivalent resistance** of networks: bridges, symmetric cubes/triangles, infinite ladders.
2. **Cells**: terminal voltage, internal resistance, combinations, maximum power.
3. **Kirchhoff two-loop circuits**: current in a branch.
4. **Drift velocity / current density / mobility** numericals.
5. **Stretched wire** and temperature dependence of resistance.
6. **Metre-bridge balance** and interchange of resistances (part of the experimental skills).
7. **Rated bulbs** in series/parallel: brightness and power.

Question classification

Type	Difficulty	Target time
R from geometry / stretching	E	45 s
Series-parallel networks	E	1 min
Cells, terminal voltage, max power	E	1 min
Kirchhoff two loops	M	2 min
Bridge (balanced/unbalanced)	M	1.5 min
Infinite ladder / cube networks	M-H	2.5 min

Common mistakes

MISTAKE ALERT Mistake alerts

- Forgetting the internal resistance when finding the current.
- Using $P = I^2 R$ for bulbs in parallel when V is the common quantity. Use V^2/R .
- In the metre bridge, measuring l from the wrong end.
- Assuming drift speed equals the speed at which the signal travels.
- Sign errors in the loop rule when you cross a cell from + to –.

Practice questions

P12 · Practice

1 Medium 1.5 min Calc P12-01

A copper wire of cross-section 1 mm^2 carries 1.6 A . With $n = 8 \times 10^{28} \text{ m}^{-3}$, the drift speed of the electrons is:

- (A) $1.25 \times 10^{-4} \text{ m s}^{-1}$ (B) $1.25 \times 10^{-3} \text{ m s}^{-1}$ (C) $2.5 \times 10^{-5} \text{ m s}^{-1}$ (D) $1.25 \times 10^{-2} \text{ m s}^{-1}$

2 Easy 0.5 min Speed P12-02

A wire is stretched uniformly to twice its original length. Its resistance becomes:

- (A) 2 times (B) 4 times (C) $1/2$ (D) 8 times

3 Medium 1.5 min NV P12-03

In a bridge network, $P = 1 \Omega$, $Q = 2 \Omega$ (in series on one side) and $R = 2 \Omega$, $S = 4 \Omega$ (in series on the other side). A 5Ω resistor connects the midpoints. Find the equivalent resistance (in Ω) between the two outer ends.

Your answer (numerical value):

4 Easy 1 min Basic

P12-04

A cell of emf 2 V and internal resistance $0.5\ \Omega$ drives current through a $3.5\ \Omega$ resistor. The terminal voltage is:

- (A) 2.0 V (B) 1.5 V (C) 1.75 V (D) 0.25 V

5 Easy 1 min NV

P12-05

A battery of emf 12 V and internal resistance $2\ \Omega$ is connected to a variable load. Find the maximum power (in W) the load can draw.

Your answer (numerical value):

6 Medium 2 min JEE

P12-06

Cells of 6 V and 4 V, each with internal resistance $1\ \Omega$, are connected in parallel (positive to positive) across a $2\ \Omega$ resistor. The current through the resistor is:

- (A) 1 A (B) 5 A (C) 2.5 A (D) 2 A

7 Easy 1 min Basic

P12-07

In a metre bridge, the balance point is at 40 cm from the end where a $20\ \Omega$ resistor is connected. The other resistance is:

- (A) $13.3\ \Omega$ (B) $30\ \Omega$ (C) $20\ \Omega$ (D) $50\ \Omega$

8 Medium 1 min Basic

P12-08

A wire's resistance is $10\ \Omega$ at 20°C and $12\ \Omega$ at 120°C . Taking the 20°C value as the reference, its temperature coefficient of resistance is:

- (A) $2 \times 10^{-2}\ ^\circ\text{C}^{-1}$ (B) $2 \times 10^{-3}\ ^\circ\text{C}^{-1}$ (C) $1.67 \times 10^{-3}\ ^\circ\text{C}^{-1}$ (D) $2 \times 10^{-4}\ ^\circ\text{C}^{-1}$

9 Easy 1 min Concept

P12-09

A 100 W and a 60 W bulb (both rated at 220 V) are connected in series across 220 V. Then:

- (A) the 100 W bulb glows brighter (B) neither glows
(C) both glow equally (D) the 60 W bulb glows brighter

P12 · Chapter Test

1 Easy 0.5 min Speed

P12-T1

Four identical resistors are connected first in series, then in parallel. The ratio $R_{\text{series}}/R_{\text{parallel}}$ is:

- (A) 4 (B) 8 (C) $1/16$ (D) 16

2 Easy 0.75 min Basic

P12-T2

A heater rated 1000 W at 220 V is operated at 110 V. Its power is:

- (A) 250 W (B) 500 W (C) 1000 W (D) 125 W

3 Easy 1 min NV

P12-T3

A current $I = (2 + 3t)$ A flows in a wire. Find the charge (in C) that passes in the first 2 s.

Your answer (numerical value):

4 Easy 1 min Basic

P12-T4

Five cells, each of emf 1.5 V and internal resistance 0.2Ω , are connected in series across a 4Ω resistor. The current is:

- (A) 0.375 A (B) 1.875 A (C) 1.5 A (D) 7.5 A

5 Easy 0.5 min Concept

P12-T5

Which of the following is a non-ohmic device?

- (A) a nichrome wire at constant temperature (B) a copper wire at constant temperature
(C) a semiconductor diode (D) a manganin resistor

Answers & Solutions

P12 · Practice

1 (A) 2 (B) 3 2 4 (C) 5 18 6 (D) 7 (B) 8 (B) 9 (D)

P12 · Chapter Test

1 (D) 2 (A) 3 10 4 (C) 5 (C)

P12 · PRACTICE

1 Answer: (A) Drift velocity Medium 1.5 min Calc

P12-01

SOLUTION $v_d = \frac{I}{neA} = \frac{1.6}{8 \times 10^{28} \times 1.6 \times 10^{-19} \times 10^{-6}} = \frac{1}{8 \times 10^3} = 1.25 \times 10^{-4} \text{ m s}^{-1}$.

SHORTCUT 1.6 cancels with e 's 1.6.

COMMON TRAP The $\text{mm}^2 \rightarrow \text{m}^2$ conversion.

2 Answer: (B) Stretching a wire Easy 0.5 min Speed

P12-02

SOLUTION Volume constant: A halves while L doubles, so $R \propto L/A$ rises $\times 4$.

SHORTCUT $R \propto L^2$ at constant volume.

COMMON TRAP Assuming A stays constant ($\times 2$).

3 Answer: 2 Balanced Wheatstone network Medium 1.5 min NV

P12-03

SOLUTION $P/Q = R/S = 1/2$: balanced, so the 5Ω carries no current. $R_{eq} = (1 + 2) \parallel (2 + 4) = 3 \parallel 6 = 2 \Omega$.

SHORTCUT Check the balance first.

COMMON TRAP Trying to include the middle resistor.

4 Answer: (C) Terminal voltage Easy 1 min Basic

P12-04

SOLUTION $I = 2/4 = 0.5 \text{ A}$; $V = \varepsilon - Ir = 2 - 0.25 = 1.75 \text{ V}$ (the same as IR).

SHORTCUT $V = \varepsilon \frac{R}{R+r}$.

COMMON TRAP Reporting the emf.

5 Answer: 18 Maximum power transfer Easy 1 min NV

P12-05

SOLUTION At $R = r$: $P_{max} = \frac{\varepsilon^2}{4r} = \frac{144}{8} = 18 \text{ W}$.

COMMON TRAP Using $\varepsilon^2/r = 72 \text{ W}$.

6 Answer: (D) Cells in parallel Medium 2 min JEE

P12-06

SOLUTION $\varepsilon_{eq} = \frac{6/1 + 4/1}{1+1} = 5 \text{ V}$; $r_{eq} = 0.5 \Omega$. $I = \frac{5}{2+0.5} = 2 \text{ A}$.

SHORTCUT Nodal method: $\frac{V-6}{1} + \frac{V-4}{1} + \frac{V}{2} = 0 \Rightarrow V = 4 \text{ V}$, so $I = 2 \text{ A}$.

COMMON TRAP Adding the emfs (10 V).

7 Answer: (B) Metre bridge balance **Easy** ⌚ 1 min **Basic**

P12-07

SOLUTION $\frac{20}{S} = \frac{40}{60} \Rightarrow S = 30 \Omega.$

COMMON TRAP Inverting the ratio.

8 Answer: (B) Temperature coefficient **Medium** ⌚ 1 min **Basic**

P12-08

SOLUTION $\alpha = \frac{\Delta R}{R_0 \Delta T} = \frac{2}{10 \times 100} = 2 \times 10^{-3} ^\circ\text{C}^{-1}.$

COMMON TRAP Dividing by 12 gives (C).

9 Answer: (D) Rated bulbs in series **Easy** ⌚ 1 min **Concept**

P12-09

SOLUTION $R = V^2/P$, so the 60 W bulb has the larger R . In series (same I), $P = I^2 R$ is larger for it.

SHORTCUT Series: the lower rating glows brighter.

COMMON TRAP Thinking "100 W is always brighter".

P12 · CHAPTER TEST

1 Answer: (D) Series vs parallel ratio **Easy** ⌚ 0.5 min **Speed**

P12-T1

SOLUTION $4R \div (R/4) = 16.$

SHORTCUT n^2 .

2 Answer: (A) Heater at lower voltage **Easy** ⌚ 0.75 min **Basic**

P12-T2

SOLUTION $P \propto V^2$ (fixed R): $1000 \times \frac{1}{4} = 250 \text{ W}.$

COMMON TRAP Using $P \propto V$.

3 Answer: 10 Charge from variable current **Easy** ⌚ 1 min **NV**

P12-T3

SOLUTION $q = \int_0^2 (2 + 3t) dt = 4 + 6 = 10 \text{ C}.$

SHORTCUT Area under the I - t graph (a trapezium: $\frac{2+8}{2} \times 2$).

COMMON TRAP Using $I(2) \times 2 = 16 \text{ C}.$

4 Answer: (C) Cells in series **Easy** ⌚ 1 min **Basic**

P12-T4

SOLUTION $\varepsilon = 7.5 \text{ V}, r = 1 \Omega; I = 7.5/(4 + 1) = 1.5 \text{ A}.$

COMMON TRAP Ignoring the internal resistance gives 1.875 A.

5 Answer: (C) Non-ohmic devices **Easy** ⌚ 0.5 min **Concept**

P12-T5

SOLUTION A diode's I - V curve is non-linear and depends on the direction of the voltage.

REVISION Revision tracker · Current Electricity

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART II · PHYSICS · UNIT 13

Magnetic Effects of Current & Magnetism

13

2–3 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

Must-do

PRIORITY

Class 12 · Ch 4, 5

NCERT

~14 hours

STUDY TIME

Concept summary

- **Moving charges create magnetic fields** (Biot–Savart, Ampère). **Magnetic fields exert forces on moving charges** (Lorentz) and on currents.
- The magnetic force $q\vec{v} \times \vec{B}$ is always $\perp$ to the velocity. It **does no work** and changes only direction, which is why charges move on circles or helices.
- A **current loop is a magnetic dipole**, $\vec{m} = NI\vec{A}$. It feels a torque $\vec{m} \times \vec{B}$ in a uniform field. That is the working principle of the moving-coil galvanometer.
- **Magnetic materials** are dia-, para- or ferromagnetic, depending on their atomic moments and how those moments respond to a field and to temperature.

⚠ WARNING Syllabus note

Earth's magnetism (elements, dip, declination) and the tangent galvanometer are **not** in the 2024–2026 syllabus text [NTA-SYL].

Formulas

Σ FORMULA Fields of standard currents

Source	B
Finite straight wire (angles α, β at the ends, distance d)	$\frac{\mu_0 I}{4\pi d} (\sin \alpha + \sin \beta)$
Infinite straight wire	$\frac{\mu_0 I}{2\pi d}$
Semi-infinite wire, at a point level with its end	$\frac{\mu_0 I}{4\pi d}$
Centre of a circular loop (N turns)	$\frac{\mu_0 NI}{2R}$
Arc subtending angle θ at the centre	$\frac{\mu_0 I \theta}{4\pi R}$
Axis of a loop, distance x	$\frac{\mu_0 NI R^2}{2(R^2 + x^2)^{3/2}}$
Long solenoid (n turns per metre)	$\mu_0 nI$ inside; $\frac{1}{2}\mu_0 nI$ at an end
Toroid (N turns, radius r)	$\frac{\mu_0 NI}{2\pi r}$
Thick wire (radius R), inside at r	$\frac{\mu_0 I r}{2\pi R^2}$

$\mu_0 = 4\pi \times 10^{-7}$ T m/A. **Ampère's law:** $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc}$.

Points lying on the line of a straight wire get **zero** field from it.

Σ FORMULA Forces

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B}) \quad r = \frac{mv}{qB} = \frac{p}{qB} = \frac{\sqrt{2mK}}{qB} = \frac{1}{B} \sqrt{\frac{2mV}{q}}$$

$$T = \frac{2\pi m}{qB} \text{ (independent of speed)} \quad \text{pitch of helix} = v \cos \theta \cdot T \quad \text{velocity selector: } v = \frac{E}{B}$$

$$\vec{F} = I\vec{L} \times \vec{B} \quad \frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d} \text{ (parallel currents attract, antiparallel repel)}$$

A closed current loop in a **uniform** field feels zero net force.

Σ FORMULA Torque, dipoles, galvanometer

$$\vec{m} = NI\vec{A} \quad \vec{\tau} = \vec{m} \times \vec{B} \quad U = -\vec{m} \cdot \vec{B}$$

Moving-coil galvanometer: $NIAB = k\phi$. Current sensitivity $\frac{\phi}{I} = \frac{NAB}{k}$; voltage sensitivity $\frac{\phi}{V} = \frac{NAB}{kR_G}$.

Ammeter: shunt $S = \frac{I_g G}{I - I_g}$ in parallel. **Voltmeter:** series $R = \frac{V}{I_g} - G$.

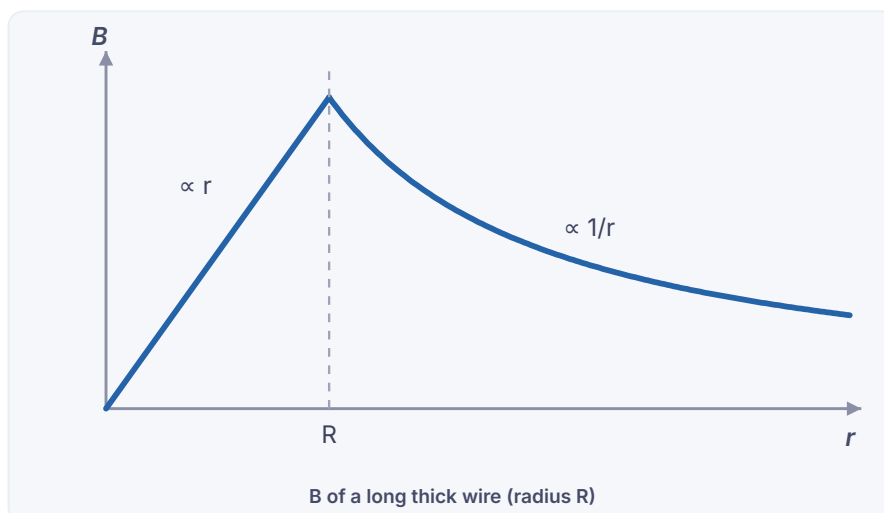
Orbiting electron: $\mu = \frac{evr}{2} = \frac{e}{2m_e} L$. Bohr magneton $\mu_B = \frac{eh}{4\pi m_e} \approx 9.27 \times 10^{-24} \text{ J T}^{-1}$.

Bar magnet (short dipole): axial $B = \frac{\mu_0}{4\pi} \frac{2m}{r^3}$, equatorial $B = \frac{\mu_0}{4\pi} \frac{m}{r^3}$.

Σ FORMULA Magnetic materials

Type	χ	μ_r	Behaviour	Temperature	Examples
Diamagnetic	small, negative	< 1	weakly repelled, moves to weaker field	independent	Bi, Cu, water, superconductors ($\chi = -1$)
Paramagnetic	small, positive	> 1	weakly attracted	$\chi = C/T$ (Curie law)	Al, Na, O ₂
Ferromagnetic	large, positive	$\gg 1$	strongly attracted; domains	above the Curie temperature it becomes paramagnetic	Fe, Co, Ni

$B = \mu_0(H + M)$, $M = \chi H$, $\mu_r = 1 + \chi$. **Soft iron** (low coercivity, narrow hysteresis loop) suits electromagnets and transformer cores. **Steel** (high retentivity and coercivity) suits permanent magnets.



Frequently tested concepts & question patterns



PYQ PATTERN Recurring structures

1. **Field at the centre of composite shapes:** arcs + straight segments, loops of several radii, semicircles with radial leads.
2. **Charged particle in B:** radius comparisons (proton, deuteron, α), period, helix pitch, crossed E and B .
3. **Force between parallel wires;** force on a wire loop near a long wire.
4. **Galvanometer conversion** to ammeter/voltmeter.
5. **Torque on a coil** and its potential energy.
6. **Magnetic materials:** identifying types from χ or behaviour; Curie law.
7. **Solenoid/toroid fields** with Ampère's law.

Shortcuts



SHORTCUT Radius ratios for particles

Same **KE**: $r \propto \sqrt{m/q}$. Same **accelerating voltage**: $r \propto \sqrt{m/q}$. Same **momentum**: $r \propto 1/q$.
Proton : deuteron : α with the same KE: $1 : \sqrt{2} : 1$.



SHORTCUT Composite-loop fields

Break the shape into arcs (use $\mu_0 I \theta / 4\pi R$) and straight pieces (radial pieces give **0**). Add with signs from the right-hand rule. That's a full answer in under a minute.

Question classification

Type	Difficulty	Target time
Standard-field formula	E	45 s
Composite loop field	M	1.5 min
Particle radius / period	E–M	1 min
Galvanometer conversion	E	1 min
Force between wires / on loops	M	1.5 min
Magnetic materials facts	E	30 s
Non-uniform B or combined E – B motion	H	3 min (Round 2)

Common mistakes



MISTAKE ALERT Mistake alerts

- Taking the magnetic force as doing work. It never changes the speed.
- For the shunt, using I instead of $(I - I_g)$ in the denominator.
- Diamagnetic susceptibility depends on temperature? **No**, it's independent.
- For a loop's field at its centre, forgetting the number of turns N .
- Using $B = \mu_0 I / 2\pi d$ for a finite wire.

Practice questions

P13 · Practice

1 Easy 0.75 min Calc P13-01

The magnetic field at the centre of a circular loop of radius 10 cm carrying 5 A is:

- (A) $2\pi \times 10^{-5}$ T (B) $\pi \times 10^{-5}$ T (C) 10^{-5} T (D) $\pi \times 10^{-6}$ T

2 Medium 1 min Concept P13-02

A proton and an α -particle with **equal kinetic energies** enter the same uniform magnetic field perpendicularly. The ratio of their radii $r_p : r_\alpha$ is:

- (A) 1 : 1 (B) 1 : 2 (C) 2 : 1 (D) $1 : \sqrt{2}$

3 Easy 1 min Basic P13-03

Two long parallel wires 10 cm apart carry 10 A and 20 A in the same direction. The force per metre between them is:

- (A) 8×10^{-4} N/m, attractive (B) 4×10^{-4} N/m, repulsive
(C) 2×10^{-4} N/m, attractive (D) 4×10^{-4} N/m, attractive

4 Easy 1 min Basic P13-04

A galvanometer of resistance 50Ω gives full-scale deflection at 1 mA. The shunt needed to convert it into an ammeter of range 1 A is about:

- (A) 50Ω in parallel (B) 0.05Ω in series
(C) 0.05Ω in parallel (D) 0.5Ω in parallel

5 Easy 1 min NV P13-05

The same galvanometer (50Ω , 1 mA full scale) is to be converted into a voltmeter of range 10 V. Find the series resistance needed (in Ω).

Your answer (numerical value):

6 Easy 1 min Calc P13-06

A 0.5 m long solenoid has 500 turns and carries 2 A. The field inside is about:

- (A) 5.0×10^{-3} T (B) 1.26×10^{-3} T (C) 2.5×10^{-3} T (D) 2.5×10^{-4} T

7 Medium 1.5 min JEE P13-07

A wire runs radially inward to a point on a circle, follows a semicircular arc of radius R to the diametrically opposite point, then runs radially outward, carrying current I throughout. The field at the centre is:

- (A) $\frac{\mu_0 I}{2R}$ (B) $\frac{\mu_0 I}{4R}$ (C) $\frac{\mu_0 I}{4\pi R}$ (D) 0

8 Easy 0.5 min Concept P13-08

The time period of a charged particle moving in a circle in a uniform magnetic field:

- (A) increases with its speed (B) is independent of its speed
(C) decreases with its speed (D) is proportional to the radius

9 Easy 0.5 min Speed

P13-09

The susceptibility of a paramagnetic substance varies with absolute temperature T as:

- (A) $1/T$ (B) T
(C) independent of T (D) T^2

10 Easy 1 min NV

P13-10

A 50-turn coil of area 0.02 m^2 carries 2 A in a uniform 1 T field, with the field **parallel to the plane** of the coil. Find the torque (in N m).

Your answer (numerical value):

P13 · Chapter Test

1 Easy 0.5 min Speed

P13-T1

A charge moves parallel to a uniform magnetic field. The magnetic force on it is:

- (A) qvB (B) $qvB/2$
(C) zero (D) along the velocity

2 Medium 1 min Concept

P13-T2

A long straight wire of radius R carries current I uniformly distributed over its cross-section. The field at $r = R/2$ is:

- (A) 0 (B) $\frac{\mu_0 I}{2\pi R}$ (C) $\frac{\mu_0 I}{\pi R}$ (D) $\frac{\mu_0 I}{4\pi R}$

3 Medium 1 min Concept

P13-T3

A charge q (mass m) enters a uniform field B with speed v at 60° to the field. The pitch of its helical path is:

- (A) $\frac{\pi mv}{2qB}$ (B) $\frac{2\pi mv}{qB}$ (C) $\frac{\sqrt{3}\pi mv}{qB}$ (D) $\frac{\pi mv}{qB}$

4 Easy 0.5 min NV

P13-T4

A 100-turn coil of area 0.01 m^2 carries 5 A. Find its magnetic moment (in A m^2).

Your answer (numerical value):

5 Easy 0.5 min Speed

P13-T5

Above its Curie temperature, a ferromagnetic material becomes:

- (A) paramagnetic (B) diamagnetic
(C) a stronger ferromagnet (D) superconducting

Answers & Solutions

P13 · Practice

1 (B) 2 (A) 3 (D) 4 (C) 5 9950 6 (C) 7 (B) 8 (B) 9 (A) 10 2

P13 · Chapter Test

1 (C) 2 (D) 3 (D) 4 5 5 (A)

P13 · PRACTICE

1 Answer: (B) Field at the centre of a loop **Easy** ⌚ 0.75 min **Calc** P13-01

SOLUTION $B = \frac{\mu_0 I}{2R} = \frac{4\pi \times 10^{-7} \times 5}{0.2} = \pi \times 10^{-5} \text{ T.}$

COMMON TRAP Using the diameter in place of R .

2 Answer: (A) Radii of proton and α particle **Medium** ⌚ 1 min **Concept** P13-02

SOLUTION $r = \frac{\sqrt{2mK}}{qB} \propto \frac{\sqrt{m}}{q}$: proton $\sqrt{1}/1 = 1$, α $\sqrt{4}/2 = 1$.

COMMON TRAP Using the same-speed rule ($r \propto m/q$), which gives 1 : 2.

3 Answer: (D) Force between parallel wires **Easy** ⌚ 1 min **Basic** P13-03

SOLUTION $\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d} = \frac{2 \times 10^{-7} \times 200}{0.1} = 4 \times 10^{-4} \text{ N/m.}$ Like currents attract.

SHORTCUT $\frac{\mu_0}{2\pi} = 2 \times 10^{-7}$.

COMMON TRAP Saying "same direction repels", which is the opposite of electrostatics intuition.

4 Answer: (C) Galvanometer $\rightarrow$ ammeter **Easy** ⌚ 1 min **Basic** P13-04

SOLUTION $S = \frac{I_g G}{I - I_g} = \frac{0.001 \times 50}{0.999} \approx 0.05 \Omega$, in parallel.

COMMON TRAP Putting the shunt in series.

5 Answer: 9950 Galvanometer $\rightarrow$ voltmeter **Easy** ⌚ 1 min **NV** P13-05

SOLUTION $R = \frac{V}{I_g} - G = 10000 - 50 = 9950 \Omega$.

COMMON TRAP Forgetting to subtract G .

6 Answer: (C) Solenoid field **Easy** ⌚ 1 min **Calc** P13-06

SOLUTION $n = 1000 \text{ m}^{-1}$; $B = \mu_0 n I = 4\pi \times 10^{-7} \times 1000 \times 2 = 8\pi \times 10^{-4} \approx 2.5 \times 10^{-3} \text{ T.}$

COMMON TRAP Using N in place of n .

7 Answer: (B) Semicircle with radial leads **Medium** ⌚ 1.5 min **JEE** P13-07

SOLUTION The radial segments point through the centre, so they contribute nothing. The semicircle gives $\frac{\mu_0 I \pi}{4\pi R} = \frac{\mu_0 I}{4R}$.

SHORTCUT Radial leads contribute zero.

COMMON TRAP Adding contributions from the leads.

8 Answer: (B) Cyclotron frequency **Easy** ⌚ 0.5 min **Concept** P13-08

SOLUTION $T = 2\pi m/(qB)$. A faster particle moves on a larger circle in the same time. This is the basis of the cyclotron.

9 Answer: (A) Paramagnetic susceptibility **Easy** ⌚ 0.5 min **Speed** P13-09

SOLUTION Curie's law: $\chi = C/T$. Thermal agitation disturbs the alignment of the atomic moments.

COMMON TRAP Confusing it with diamagnetism (independent of T).

10 Answer: 2 Torque on a coil **Easy** ⌚ 1 min **NV** P13-10

SOLUTION $\tau = NIAB \sin 90^\circ = 50 \times 2 \times 0.02 \times 1 = 2 \text{ N m.}$ The field lies in the plane, so it is perpendicular to $\vec{m}$.

SHORTCUT The torque is maximum when the field is parallel to the plane.

COMMON TRAP Using $\sin 0^\circ$ by confusing "plane" with "normal".

P13 · CHAPTER TEST

1 Answer: (C) Charge moving along B **Easy** ⌚ 0.5 min **Speed** P13-T1

SOLUTION $\vec{v} \times \vec{B} = 0$ when they are parallel.

2 Answer: (D) Field inside a thick wire **Medium** ⌚ 1 min **Concept** P13-T2

SOLUTION $B = \frac{\mu_0 I r}{2\pi R^2} = \frac{\mu_0 I}{4\pi R}$ at $r = R/2$.

SHORTCUT Half of the surface value $\mu_0 I / 2\pi R$.

COMMON TRAP Using the outside formula with $r = R/2$.

3 Answer: (D) Helix pitch **Medium** ⌚ 1 min Concept

P13-T3

SOLUTION Pitch = $v \cos 60^\circ \times \frac{2\pi m}{qB} = \frac{\pi mv}{qB}$.

SHORTCUT Parallel component $\times$ period.

COMMON TRAP Using $v \sin 60^\circ$.

4 Answer: 5 Magnetic moment of a coil **Easy** ⌚ 0.5 min NV

P13-T4

SOLUTION $m = NIA = 100 \times 5 \times 0.01 = 5 \text{ A m}^2$.

5 Answer: (A) Curie temperature **Easy** ⌚ 0.5 min Speed

P13-T5

SOLUTION Thermal agitation destroys the domain alignment, leaving paramagnetic behaviour.

 REVISION Revision tracker · Magnetism

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART II · PHYSICS · UNIT 14

Electromagnetic Induction & Alternating Current

14

1–2 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

High

PRIORITY

Class 12 · Ch 6, 7

NCERT

~12 hours

STUDY TIME

Concept summary

- **Faraday's law:** a changing magnetic flux induces an emf, $\varepsilon = -N \frac{d\Phi}{dt}$. The flux can change through B , the area, or the orientation.
- **Lenz's law:** the induced current opposes the *change* that produced it. It is a consequence of energy conservation.
- **Inductance** is the "inertia" of current: $L = N\Phi/I$. It stores energy $\frac{1}{2}LI^2$.
- **AC circuits:** resistors keep V and I in phase. In an inductor the current *lags* by 90° ; in a capacitor it *leads* by 90° . Impedance combines them like vectors (phasors).
- Only resistors dissipate average power. Pure L and C carry **wattless** current.

Formulas: Induction

Σ FORMULA Faraday–Lenz & motional emf

$$\Phi = \vec{B} \cdot \vec{A} = BA \cos \theta \quad \varepsilon = -N \frac{d\Phi}{dt} \quad \text{induced charge: } q = \frac{N \Delta\Phi}{R}$$

$$\text{rod moving } \perp B : \varepsilon = Blv \quad \text{rod rotating about one end: } \varepsilon = \frac{1}{2}B\omega l^2 \quad \text{coil rotating: } \varepsilon = NBA\omega \sin \omega t$$

The force needed to pull a rod at constant speed in a closed circuit of resistance R : $F = \frac{B^2 l^2 v}{R}$, with power $\frac{B^2 l^2 v^2}{R}$ (all dissipated as heat).

Σ FORMULA Inductance

$$L = \frac{N\Phi}{I} \quad \varepsilon = -L \frac{dI}{dt} \quad \text{solenoid: } L = \mu_0 n^2 Al = \frac{\mu_0 N^2 A}{l} \quad U = \frac{1}{2}LI^2$$

$$M = \mu_0 n_1 n_2 Al \quad M = k\sqrt{L_1 L_2} \quad (k \leq 1) \quad \text{series (no coupling): } L_1 + L_2 \quad \text{parallel: } \frac{L_1 L_2}{L_1 + L_2}$$

LR circuit: growth $I = I_0(1 - e^{-t/\tau})$, decay $I = I_0 e^{-t/\tau}$, with $\tau = L/R$. After one τ the current reaches 63% of its final value.

Energy density of a magnetic field: $u = \frac{B^2}{2\mu_0}$.

Formulas: Alternating current

Σ FORMULA AC basics

$$V = V_0 \sin \omega t \quad V_{rms} = \frac{V_0}{\sqrt{2}} \quad V_{avg, half\ cycle} = \frac{2V_0}{\pi} \quad V_{avg, full\ cycle} = 0$$

Household mains in India: $V_{rms} = 220\text{--}230\text{ V}$, $f = 50\text{ Hz}$. AC meters read **rms** values.

Σ FORMULA Series LCR circuit

$$X_L = \omega L \quad X_C = \frac{1}{\omega C} \quad Z = \sqrt{R^2 + (X_L - X_C)^2} \quad \tan \phi = \frac{X_L - X_C}{R}$$

$$\text{Resonance: } \omega_0 = \frac{1}{\sqrt{LC}}, \quad Z = R, \quad I = I_{max} \quad Q = \frac{\omega_0 L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}} \quad \text{bandwidth } \Delta\omega = \frac{\omega_0}{Q} = \frac{R}{L}$$

$$P_{avg} = V_{rms} I_{rms} \cos \phi = I_{rms}^2 R \quad \cos \phi = \frac{R}{Z} \quad \text{wattless component} = I_{rms} \sin \phi$$

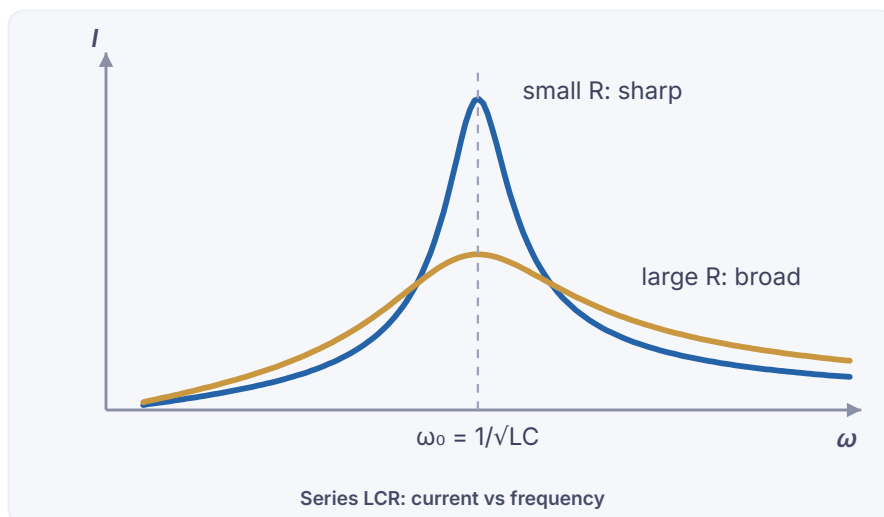
Element	Phase of I relative to V	Average power
R	in phase	$V_{rms} I_{rms}$
L	lags by 90°	0
C	leads by 90°	0

Σ FORMULA LC oscillations, generator, transformer

LC circuit: $\omega = \frac{1}{\sqrt{LC}}$; energy swaps between $\frac{q^2}{2C}$ and $\frac{1}{2}LI^2$ (analogue: $q \leftrightarrow x$, $L \leftrightarrow m$, $1/C \leftrightarrow k$).

AC generator: peak emf $\varepsilon_0 = NBA\omega$.

Ideal transformer: $\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$. Step-up raises V and lowers I . Losses: copper ($I^2 R$), eddy currents (reduced by laminating the core), hysteresis (reduced with soft iron), and flux leakage.



Frequently tested concepts & question patterns

PYQ PATTERN Recurring structures

1. **Motional emf:** rods on rails, rotating rods and discs, force/power needed.
2. **Flux-change emf and induced charge** in coils ($q = N\Delta\Phi/R$, independent of time).
3. **LCR series:** impedance, phase, power factor, average power, resonance frequency, Q factor.
4. **rms/average values** from a given $V(t)$.
5. **Transformer** turns/voltage/current ratios and efficiency.
6. **LR growth/decay** and energy stored in inductors.
7. **LC oscillation** frequency and energy exchange.

Shortcuts

SHORTCUT Phasor triangle

Draw R horizontally and $(X_L - X_C)$ vertically. The hypotenuse is Z , the angle is ϕ , and the power factor is the cosine. Use 3-4-5 triangles: numbers like $R = 30$, $X = 40$ are chosen deliberately.

Time saved: about 1 minute.

SHORTCUT Resonance in rad/s vs Hz

$\omega_0 = 1/\sqrt{LC}$ is in **rad/s**. $f_0 = \omega_0/2\pi$. Options often include both. Check the unit asked for.

Question classification

Type	Difficulty	Target time
Motional emf / rotating rod	E	1 min
Induced charge / emf from $\Delta\Phi$	E	1 min
LCR impedance/power	M	2 min
Resonance / Q factor	E-M	1 min
Transformer	E	45 s
LR growth / energy	M	1.5 min
Rod on rails with mass (terminal velocity)	H	3 min (Round 2)

Common mistakes

MISTAKE ALERT Mistake alerts

- Using V_0 instead of V_{rms} in the power formula.
- Taking the full-cycle average of a sinusoid as $2V_0/\pi$. It's 0 over a full cycle.
- Missing the $\frac{1}{2}$ in the rotating-rod emf $\frac{1}{2}B\omega l^2$.
- Mixing up "lead" and "lag": **ELI the ICE man**. In L, E (voltage) leads I; in C, I leads E.
- Forgetting N (the number of turns) in Faraday's law.

Practice questions

P14 · Practice

- 1** Easy 0.5 min Speed P14-01
 A 0.5 m rod moves at 4 m s^{-1} perpendicular to a 0.2 T field (and perpendicular to its own length). The emf across its ends is:
 (A) 0.1 V (B) 4 V (C) 0.8 V (D) 0.4 V
- 2** Easy 1 min NV P14-02
 A 100-turn coil of area 0.1 m^2 lies perpendicular to a field that falls uniformly from 0.5 T to zero in 0.1 s. Find the average induced emf (in V).
 Your answer (numerical value):
- 3** Easy 1 min Basic P14-03
 The flux through each turn of a 100-turn coil changes by 0.05 Wb. The coil's total resistance is 10Ω . The charge that flows is:
 (A) 0.005 C (B) 0.05 C (C) 0.5 C (D) 5 C
- 4** Easy 0.5 min NV P14-04
 Find the energy (in J) stored in a 2 H inductor carrying 3 A.
 Your answer (numerical value):
- 5** Medium 2 min JEE P14-05
 A series LCR circuit has $R = 30 \Omega$, $X_L = 80 \Omega$, $X_C = 40 \Omega$ and is connected to a 100 V (rms) source. The average power consumed is:
 (A) 120 W (B) 200 W (C) 160 W (D) 333 W
- 6** Medium 1 min Tricky P14-06
 In a series LCR circuit, $L = 0.1 \text{ H}$ and $C = 10 \mu\text{F}$. The resonant **angular** frequency is:
 (A) 159 rad/s (B) 1000 rad/s (C) 100 rad/s (D) 10 000 rad/s
- 7** Easy 0.5 min Speed P14-07
 For $V = 220\sqrt{2}\sin(100\pi t)$ volts, the rms voltage and frequency are:
 (A) $220\sqrt{2} \text{ V}$, 100 Hz (B) 220 V, 50 Hz (C) 220 V, 100 Hz (D) 311 V, 50 Hz
- 8** Easy 1 min Basic P14-08
 An ideal transformer has 200 primary and 1000 secondary turns. The primary is at 220 V, and the secondary delivers 0.2 A. The primary current is:
 (A) 0.04 A (B) 5 A (C) 0.2 A (D) 1 A
- 9** Easy 0.5 min Speed P14-09
 A 4 H inductor is in series with a 2Ω resistor. The time constant of the circuit is:
 (A) 8 s (B) 6 s (C) 0.5 s (D) 2 s

10 Easy 1 min Basic

P14-10

A 1 m metal rod rotates at 20 rad/s about one end, in a plane perpendicular to a uniform 0.5 T field. The emf between its ends is:

- (A) 5 V (B) 10 V (C) 20 V (D) 2.5 V

P14 · Chapter Test

1 Easy 0.5 min Concept

P14-T1

Lenz's law is a consequence of the conservation of:

- (A) energy (B) momentum (C) charge (D) mass

2 Easy 0.5 min Speed

P14-T2

In a purely inductive AC circuit, the current:

- (A) leads the voltage by $\pi/2$ (B) lags the voltage by $\pi/2$
(C) is in phase with the voltage (D) lags by π

3 Easy 0.5 min Speed

P14-T3

If the frequency of an AC source is doubled, the capacitive reactance:

- (A) doubles (B) stays the same (C) halves (D) becomes 4 times

4 Easy 1 min NV

P14-T4

A 100-turn coil of area 0.01 m^2 rotates at 60 rad/s in a 0.5 T field. Find the peak emf (in V).

Your answer (numerical value):

5 Easy 0.5 min Speed

P14-T5

At resonance in a series LCR circuit, the power factor is:

- (A) 0 (B) 1 (C) 0.5 (D) $1/\sqrt{2}$

Answers & Solutions

P14 · Practice

1 (D) 2 50 3 (C) 4 9 5 (A) 6 (B) 7 (B) 8 (D) 9 (D) 10 (A)

P14 · Chapter Test

1 (A) 2 (B) 3 (C) 4 30 5 (B)

P14 · PRACTICE

1 Answer: (D) Motional emf Easy 0.5 min Speed

P14-01

SOLUTION $\varepsilon = Blv = 0.2 \times 0.5 \times 4 = 0.4 \text{ V}$.

2 Answer: 50 Faraday's law for a coil Easy 1 min NV

P14-02

SOLUTION $\varepsilon = N \frac{\Delta(BA)}{\Delta t} = 100 \times \frac{0.5 \times 0.1}{0.1} = 50 \text{ V}$.

COMMON TRAP Omitting N (0.5 V).

3 Answer: (C) Induced charge **Easy** ⌚ 1 min Basic

P14-03

SOLUTION $q = \frac{N\Delta\Phi}{R} = \frac{100 \times 0.05}{10} = 0.5 \text{ C}$, regardless of how quickly the change happens.

SHORTCUT Charge depends on the change in flux, not its rate.

COMMON TRAP Thinking you need the time interval.

4 Answer: 9 Energy in an inductor **Easy** ⌚ 0.5 min NV

P14-04

SOLUTION $U = \frac{1}{2}LI^2 = \frac{1}{2} \times 2 \times 9 = 9 \text{ J}$.

5 Answer: (A) LCR average power **Medium** ⌚ 2 min JEE

P14-05

SOLUTION $Z = \sqrt{30^2 + 40^2} = 50 \Omega$; $I_{rms} = 2 \text{ A}$; $\cos \phi = 30/50 = 0.6$. $P = 100 \times 2 \times 0.6 = 120 \text{ W}$ (check: $I^2R = 4 \times 30$).

SHORTCUT $P = I_{rms}^2 R$ once you have I .

COMMON TRAP Using V^2/R (333 W).

6 Answer: (B) Resonance: rad/s vs Hz **Medium** ⌚ 1 min Tricky

P14-06

SOLUTION $\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10^{-6}}} = 1000 \text{ rad/s}$. (That's $f_0 \approx 159 \text{ Hz}$. The trap is option (A), with the wrong unit.)

COMMON TRAP Choosing 159 rad/s.

7 Answer: (B) Reading an AC expression **Easy** ⌚ 0.5 min Speed

P14-07

SOLUTION $V_{rms} = V_0/\sqrt{2} = 220 \text{ V}$; $\omega = 100\pi \Rightarrow f = 50 \text{ Hz}$.

COMMON TRAP Reading $f = 100$ from 100π .

8 Answer: (D) Ideal transformer **Easy** ⌚ 1 min Basic

P14-08

SOLUTION $\frac{I_p}{I_s} = \frac{N_s}{N_p} = 5 \Rightarrow I_p = 1 \text{ A}$. ($V_s = 1100 \text{ V}$; power $220 = 220 \text{ W}$.)

SHORTCUT Power in = power out.

COMMON TRAP Inverting the ratio (0.04 A).

9 Answer: (D) LR time constant **Easy** ⌚ 0.5 min Speed

P14-09

SOLUTION $\tau = L/R = 2 \text{ s}$.

COMMON TRAP R/L .

10 Answer: (A) Rotating rod **Easy** ⌚ 1 min Basic

P14-10

SOLUTION $\varepsilon = \frac{1}{2}B\omega l^2 = \frac{1}{2} \times 0.5 \times 20 \times 1 = 5 \text{ V}$.

COMMON TRAP Omitting the $\frac{1}{2}$.

P14 · CHAPTER TEST**1 Answer: (A)** Basis of Lenz's law **Easy** ⌚ 0.5 min Concept

P14-T1

SOLUTION If the induced current aided the change, it would create energy from nothing.

2 Answer: (B) Phase in a pure inductor **Easy** ⌚ 0.5 min Speed

P14-T2

SOLUTION The inductor opposes changes in current, so the current lags behind the voltage by 90° .

SHORTCUT ELI.

3 Answer: (C) Reactance vs frequency **Easy** ⌚ 0.5 min Speed

P14-T3

SOLUTION $X_C = 1/(\omega C) \propto 1/f$.

COMMON TRAP Applying the inductive behaviour instead.

4 Answer: 30 Generator peak emf **Easy** ⌚ 1 min NV

P14-T4

SOLUTION $\varepsilon_0 = NBA\omega = 100 \times 0.5 \times 0.01 \times 60 = 30 \text{ V}$.

5 Answer: (B) Power factor at resonance **Easy** ⌚ 0.5 min Speed

P14-T5

SOLUTION $X_L = X_C$, so $Z = R$, $\phi = 0$ and $\cos \phi = 1$.

**REVISION** Revision tracker · EMI & AC

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART II · PHYSICS · UNIT 15

Electromagnetic Waves

15

~1 per shift
TYPICAL QUESTIONS

Easy
DIFFICULTY

Must-do (easy
marks)
PRIORITY

Class 12 · Ch 8
NCERT

~3 hours
STUDY TIME

Concept summary

- Maxwell added the **displacement current** $I_d = \epsilon_0 d\Phi_E/dt$ to Ampère's law. A changing E creates B , just as a changing B creates E . Together they give self-sustaining **electromagnetic waves**.
- EM waves are **transverse**: $\vec{E} \perp \vec{B} \perp$ direction of travel ($\vec{E} \times \vec{B}$ points along the propagation). They need no medium, and in vacuum they travel at $c = 1/\sqrt{\mu_0\epsilon_0}$.
- They carry energy and momentum, so they exert radiation pressure.
- The **spectrum** runs from radio to gamma rays. Only the wavelength or frequency differs.

Formulas

Σ FORMULA EM wave relations

$$c = \frac{1}{\sqrt{\mu_0\epsilon_0}} = 3 \times 10^8 \text{ m s}^{-1} \quad \frac{E_0}{B_0} = c \quad v_{\text{medium}} = \frac{1}{\sqrt{\mu\epsilon}} = \frac{c}{\sqrt{\mu_r\epsilon_r}}$$

$$u = \frac{1}{2}\epsilon_0 E^2 + \frac{B^2}{2\mu_0} \text{ (equal shares)} \quad \langle u \rangle = \frac{1}{2}\epsilon_0 E_0^2 \quad I = \langle u \rangle c = \frac{1}{2}\epsilon_0 c E_0^2$$

$$\text{momentum } p = \frac{U}{c} \quad \text{radiation pressure: } \frac{I}{c} \text{ (absorbed), } \frac{2I}{c} \text{ (reflected)} \quad I_d = \epsilon_0 \frac{d\Phi_E}{dt}$$

Inside the gap of a charging capacitor, the displacement current equals the conduction current in the wires.

The electromagnetic spectrum (NCERT ranges)

Type	Wavelength range	Produced by	Main uses
Radio	> 0.1 m	oscillating charges in antennas	radio/TV communication
Microwaves	0.1 m – 1 mm	klystron, magnetron, Gunn diode	radar, microwave ovens, satellite links
Infrared	1 mm – 700 nm	hot bodies, molecular vibrations	remote controls, night vision, greenhouse effect, physiotherapy
Visible	700 nm – 400 nm	electron transitions in atoms	vision, photography
Ultraviolet	400 nm – 1 nm	very hot bodies (Sun), discharge lamps	sterilisation, water purification, LASIK; mostly absorbed by ozone
X-rays	1 nm – 10^{-3} nm	X-ray tubes (fast electrons hitting metal)	medical imaging, crystallography
Gamma rays	< 10^{-3} nm	radioactive nuclei	cancer therapy, nuclear physics

⚡ SHORTCUT Memory order

"**R**aman's **M**other **I**nited **V**ery **U**nusual **eX**tra **G**uests": increasing frequency, decreasing wavelength. Energy per photon increases along the same order.

PYQ PATTERN Recurring structures

1. $E_0 \leftrightarrow B_0$ conversion, direction of $\vec{B}$ from a given $\vec{E}$ and propagation direction.
2. Intensity / energy density / radiation pressure numericals.
3. **Spectrum ordering** and matching uses (UV–sterilisation, IR–remote, microwave–radar).
4. Displacement current in a charging capacitor.
5. Speed in a medium from ϵ_r, μ_r .

MISTAKE ALERT Mistake alerts

- Energy is shared **equally** between the electric and magnetic fields, even though $B_0 = E_0/c$ looks small.
- Radiation pressure doubles for a perfectly reflecting surface.
- Direction: $\vec{E} \times \vec{B}$ gives the propagation direction. If E is along $\hat{x}$ and the wave travels along $+\hat{z}$, then B is along $+\hat{y}$.

Practice questions

P15 · Practice

1 Easy 0.5 min Speed P15-01

An EM wave in vacuum has electric field amplitude 30 V/m. Its magnetic field amplitude is:

- (A) 10^{-8} T (B) 9×10^9 T (C) 10^{-7} T (D) 3×10^{-7} T

2 Easy 0.5 min Speed P15-02

Which of the following has the shortest wavelength?

- (A) γ -rays (B) microwaves (C) X-rays (D) ultraviolet

3 Easy 0.5 min NV P15-03

A capacitor is charged by a steady 2 A current. Find the displacement current (in A) between its plates.

Your answer (numerical value):

4 Medium 1 min Calc P15-04

An EM wave has $E_0 = 100$ V/m. Its intensity is about ($\epsilon_0 = 8.85 \times 10^{-12}$ SI):

- (A) 26.6 W/m² (B) 13.3 W/m² (C) 1.33 W/m² (D) 133 W/m²

5 Medium 1 min Concept P15-05

An EM wave travels along $+z$ with $\vec{E}$ along $+x$. At that instant, $\vec{B}$ points along:

- (A) $-x$ (B) $-y$ (C) $+z$ (D) $+y$

6 Easy 0.75 min Basic P15-06

An EM wave has a wavelength of 3 cm. Its frequency and type are:

- (A) 10 MHz, radio (B) 1 GHz, radio
(C) 100 GHz, infrared (D) 10 GHz, microwave

P15 · Chapter Test

1 Easy 0.5 min Concept

P15-T1

In an EM wave, the average energy density associated with the magnetic field is:

- (A) equal to that of the electric field (B) smaller than that of the electric field
(C) greater than that of the electric field (D) zero

2 Easy 0.5 min Speed

P15-T2

The EM radiation used to sterilise surgical instruments is:

- (A) infrared (B) ultraviolet (C) microwaves (D) radio waves

3 Easy 0.5 min Basic

P15-T3

The speed of EM waves in a non-magnetic medium with $\epsilon_r = 4$ is:

- (A) 3×10^8 m/s (B) 0.75×10^8 m/s (C) 1.5×10^8 m/s (D) 6×10^8 m/s

4 Easy 0.75 min NV

P15-T4

Radiation of intensity 900 W/m^2 falls normally on a perfectly absorbing surface. The pressure is $x \times 10^{-6} \text{ Pa}$. Find x .

Your answer (numerical value):

Answers & Solutions

P15 · Practice

1 (C) 2 (A) 3 2 4 (B) 5 (D) 6 (D)

P15 · Chapter Test

1 (A) 2 (B) 3 (C) 4 3

P15 · PRACTICE

1 Answer: (C) E - B amplitude relation Easy 0.5 min Speed

P15-01

SOLUTION $B_0 = E_0/c = 30/(3 \times 10^8) = 10^{-7} \text{ T}$.

COMMON TRAP Multiplying by c .

2 Answer: (A) Spectrum order Easy 0.5 min Speed

P15-02

SOLUTION Gamma rays lie at the high-frequency end of the spectrum.

COMMON TRAP Choosing X-rays.

3 Answer: 2 Displacement current Easy 0.5 min NV

P15-03

SOLUTION Continuity of current: $I_d = I_c = 2 \text{ A}$.

COMMON TRAP Answering zero because "no charge crosses the gap".

4 Answer: (B) Intensity from field amplitude Medium 1 min Calc

P15-04

SOLUTION $I = \frac{1}{2} \epsilon_0 c E_0^2 = 0.5 \times 8.85 \times 10^{-12} \times 3 \times 10^8 \times 10^4 \approx 13.3 \text{ W m}^{-2}$.

COMMON TRAP Forgetting the $\frac{1}{2}$ (26.6).

5 Answer: (D) Field directions Medium 1 min Concept

P15-05

SOLUTION $\vec{E} \times \vec{B}$ must point along $+\hat{z}$: $\hat{x} \times \hat{y} = \hat{z}$, so $\vec{B}$ is along $+\hat{y}$.

COMMON TRAP Using $\vec{B} \times \vec{E}$.

6 Answer: (D) Identify the band **Easy** ⌚ 0.75 min Basic

P15-06

SOLUTION $f = c/\lambda = 3 \times 10^8 / 0.03 = 10^{10}$ Hz; 3 cm lies in the microwave band (1 mm – 0.1 m).

P15 · CHAPTER TEST

1 Answer: (A) Energy sharing **Easy** ⌚ 0.5 min Concept

P15-T1

SOLUTION $\frac{1}{2}\epsilon_0 E^2 = \frac{B^2}{2\mu_0}$ since $E = cB$ and $c^2 = 1/\mu_0\epsilon_0$.

2 Answer: (B) Uses of UV **Easy** ⌚ 0.5 min Speed

P15-T2

SOLUTION UV kills bacteria by damaging their DNA.

3 Answer: (C) Speed in a medium **Easy** ⌚ 0.5 min Basic

P15-T3

SOLUTION $v = c/\sqrt{\epsilon_r} = c/2$.

COMMON TRAP Dividing by 4.

4 Answer: 3 Radiation pressure **Easy** ⌚ 0.75 min NV

P15-T4

SOLUTION $P = I/c = 900/(3 \times 10^8) = 3 \times 10^{-6}$ Pa.

COMMON TRAP Using $2I/c$ (for a reflecting surface).

REVISION Revision tracker · EM Waves

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

Optics: Ray & Wave

16

2–3 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

Must-do

PRIORITY

Class 12 · Ch 9, 10

NCERT

~16 hours

STUDY TIME

Concept summary

- **Ray optics** treats light as rays. It covers reflection (mirrors), refraction (Snell's law, lenses, prisms), total internal reflection and optical instruments.
- **Wave optics** treats light as a wave. It covers Huygens' principle, interference (YDSE), diffraction (single slit) and polarisation (Malus, Brewster).
- **Sign convention (Cartesian)**: distances are measured from the pole or optical centre. Positive means in the direction of the incident light, negative means against it. Heights are positive upward.

Formulas: Ray optics

Σ FORMULA Mirrors & refraction

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \quad f = \frac{R}{2} \quad m = -\frac{v}{u}$$

$$n_1 \sin \theta_1 = n_2 \sin \theta_2 \quad n = \frac{c}{v} \quad \text{apparent depth} = \frac{\text{real depth}}{n} \quad \text{shift} = t \left(1 - \frac{1}{n} \right)$$

Total internal reflection (dense → rare, $i > C$): $\sin C = \frac{n_{\text{rare}}}{n_{\text{dense}}}$. Applications: optical fibres, totally reflecting prisms, sparkle of diamond, mirage.

Plane mirror rotated by $\theta \rightarrow$ the reflected ray turns by 2θ . Two mirrors at angle θ form $\frac{360^\circ}{\theta} - 1$ images (when $360/\theta$ is even).

Σ FORMULA Spherical surfaces & lenses

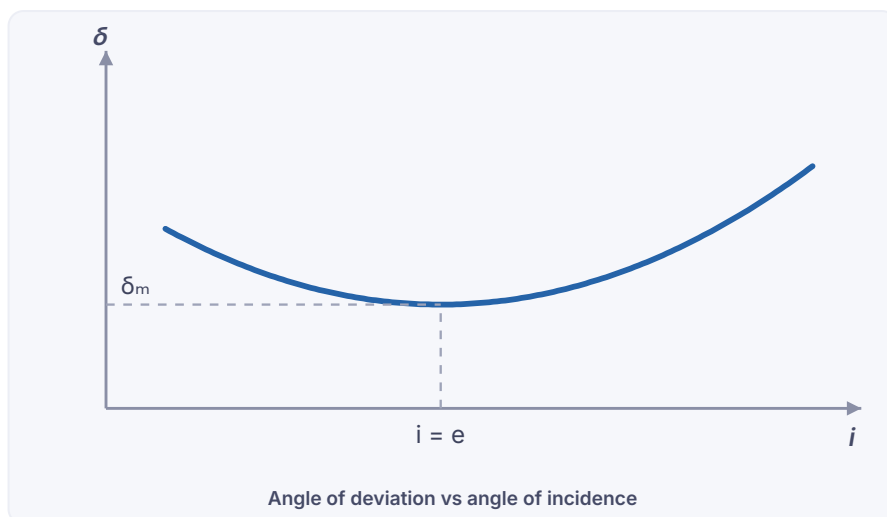
$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R} \quad \frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad \frac{1}{v} - \frac{1}{u} = \frac{1}{f} \quad m = \frac{v}{u}$$

- Lens in a medium: replace $(n - 1)$ with $\left(\frac{n_{\text{lens}}}{n_{\text{medium}}} - 1 \right)$. If $n_{\text{lens}} < n_{\text{medium}}$, a convex lens diverges.
- **Power** $P = \frac{1}{f \text{ (m)}}$ dioptre. **In contact**: $P = P_1 + P_2$. **Separated by d** : $P = P_1 + P_2 - dP_1P_2$.
- **Silvered lens** acts as a mirror with $P_{\text{eq}} = 2P_{\text{lens}} + P_{\text{mirror}}$. For a plano-convex lens (curved radius R): silvered on the **plane** face, $f = \frac{R}{2(n - 1)}$; silvered on the **curved** face, $f = \frac{R}{2n}$.
- **Displacement method**: $f = \frac{D^2 - x^2}{4D}$ (D = object–screen distance, x = distance between the two lens positions). Needs $D \geq 4f$.

Σ FORMULA Prism

$$A = r_1 + r_2 \quad \delta = i + e - A \quad n = \frac{\sin \frac{A + \delta_m}{2}}{\sin \frac{A}{2}} \quad \text{thin prism: } \delta = (n - 1)A$$

At minimum deviation: $i = e$, $r_1 = r_2 = A/2$, and the ray passes symmetrically. The δ – i graph is U-shaped (see below). Dispersion: $\delta_v > \delta_r$ because $n_v > n_r$.



Σ FORMULA Optical instruments ($D = 25\text{ cm}$)

Instrument	Image at infinity (normal adjustment)	Image at D
Simple microscope	$M = \frac{D}{f}$	$M = 1 + \frac{D}{f}$
Compound microscope	$M \approx \frac{L}{f_o} \cdot \frac{D}{f_e}$	$M = m_o \left(1 + \frac{D}{f_e}\right)$
Astronomical telescope	$M = \frac{f_o}{f_e}$, tube length $f_o + f_e$	$M = \frac{f_o}{f_e} \left(1 + \frac{f_e}{D}\right)$

Telescope: large-aperture objective (more light, better resolution), short- f eyepiece. **Reflecting telescopes** (Newtonian, Cassegrain) avoid chromatic aberration, and mirrors are easier to support than large lenses.

Formulas: Wave optics

Σ FORMULA Interference (YDSE)

$$\Delta x = d \sin \theta \approx \frac{yd}{D} \quad \beta = \frac{\lambda D}{d} \quad \text{bright: } \Delta x = n\lambda \quad \text{dark: } \Delta x = (2n-1)\frac{\lambda}{2}$$

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi \quad \text{equal slits: } I = 4I_0 \cos^2 \frac{\phi}{2} \quad \frac{I_{\max}}{I_{\min}} = \left(\frac{\sqrt{I_1} + \sqrt{I_2}}{\sqrt{I_1} - \sqrt{I_2}} \right)^2$$

- Apparatus immersed in a medium of index n : $\beta' = \beta/n$.
- A thin film (index μ , thickness t) over one slit shifts the pattern **towards that slit** by $\frac{(\mu-1)tD}{d}$. The number of fringes shifted is $\frac{(\mu-1)t}{\lambda}$.
- Sustained interference needs **coherent** sources (constant phase difference).

Σ FORMULA Diffraction (single slit, width a)

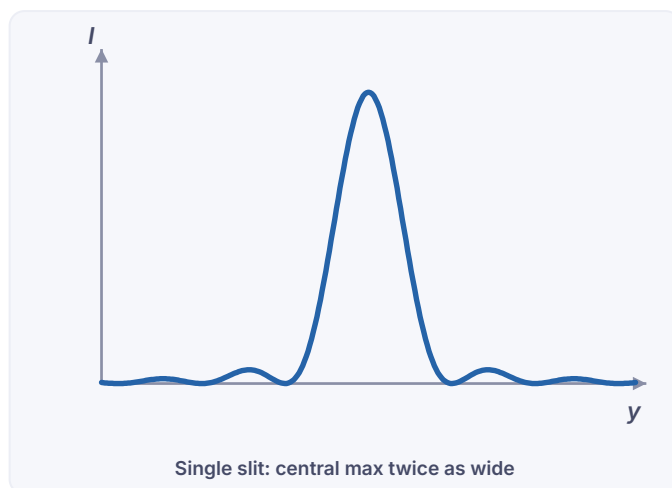
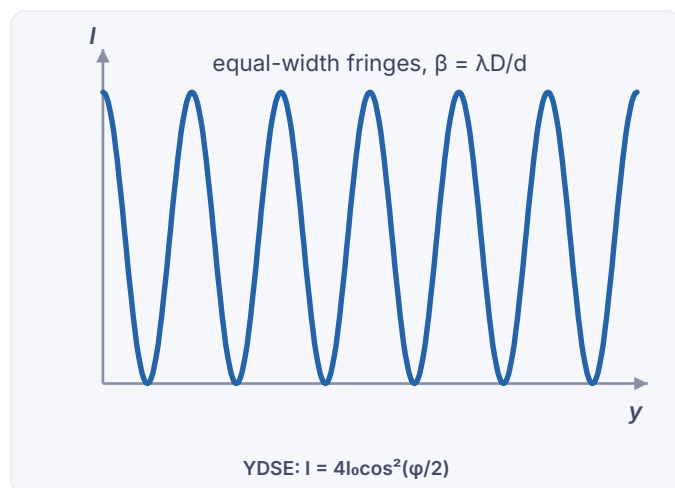
Minima: $a \sin \theta = n\lambda$ ($n = \pm 1, \pm 2, \dots$). Angular width of the central maximum = $\frac{2\lambda}{a}$; linear width = $\frac{2\lambda D}{a}$.
Secondary maxima get rapidly weaker. Narrower slit $\rightarrow$ wider central maximum.

Σ FORMULA Polarisation

Malus's law: $I = I_0 \cos^2 \theta$. Unpolarised light through one polaroid $\rightarrow I_0/2$.

Brewster's law: $\tan \theta_B = n$; at θ_B the reflected light is completely polarised, and the reflected and refracted rays are **perpendicular**.

Polarisation proves light is **transverse**.



Frequently tested concepts & question patterns

PYQ PATTERN Recurring structures

1. **Mirror/lens formula** with sign conventions: nature, position and size of the image.
2. **Lens maker's formula**, including a lens in a liquid, and combinations (contact/separated).
3. **TIR/critical angle**, apparent depth, glass slab shift.
4. **Prism minimum deviation**; thin-prism deviation.
5. **Microscope/telescope magnification** and tube length.
6. **YDSE**: fringe width changes (medium, D , d , λ), intensity at a point, thin-film shift, I_{max}/I_{min} .
7. **Single-slit width** of the central maximum; **Malus/Brewster** numericals.

Shortcuts

SHORTCUT Image-nature table (concave mirror / convex lens)

Object position	Image
beyond $2f$ (C)	between f and $2f$, real, inverted, diminished
at $2f$	at $2f$, real, inverted, same size
between f and $2f$	beyond $2f$, real, inverted, magnified
inside f	virtual, erect, magnified

Convex mirrors and concave lenses: **always virtual, erect, diminished**. This settles many MCQs without any calculation.

SHORTCUT Fringe-width checks

$\beta \propto \lambda D/d$. Any change $\rightarrow$ multiply the old fringe width by the ratios. For a medium: divide by n . Changing the slit **width** doesn't change β , only the intensity and visibility.

Question classification

Type	Difficulty	Target time
Mirror/lens formula	E	1 min
Lens maker / lens in liquid	M	1.5 min
Prism min deviation	E	1 min
TIR / apparent depth	E	1 min
Instruments	E	1 min

Type	Difficulty	Target time
YDSE fringe/intensity	E-M	1.5 min
Thin film shift / multiple wavelengths coinciding	M	2 min
Combined systems (lens + mirror, refraction at two surfaces)	H	3 min (Round 2)

Common mistakes

MISTAKE ALERT Mistake alerts

- Sign convention: for a concave mirror $f < 0$. Object distance u is **negative** for real objects.
- Lens formula has a **minus** ($1/v - 1/u$); mirror formula has a **plus**.
- Critical angle is defined only for dense $\rightarrow$ rare.
- The intensity after the first polaroid is $I_0/2$, **before** applying $\cos^2 \theta$ at the second.
- In a medium, the wavelength changes but the frequency doesn't.

Practice questions

P16 · Practice

1 Medium 1.5 min Concept

P16-01

An object is 10 cm in front of a concave mirror of focal length 15 cm. The image is:

- (A) 30 cm in front, real, inverted, $3\times$ magnified
 (B) 30 cm behind the mirror, virtual, erect, $3\times$ magnified
 (C) 6 cm behind, virtual, erect, diminished
 (D) at infinity

2 Easy 0.5 min Speed

P16-02

The critical angle for glass of refractive index $\sqrt{2}$ (against air) is:

- (A) 30° (B) 60° (C) 45° (D) 90°

3 Medium 1.5 min JEE

P16-03

A biconvex glass lens ($n = 1.5$) with both radii 20 cm has focal length 20 cm in air. Immersed in water ($n = 4/3$), its focal length becomes:

- (A) 80 cm (B) 40 cm (C) 60 cm (D) 20 cm

4 Easy 1 min NV

P16-04

A convex lens of focal length 20 cm is placed in contact with a concave lens of focal length 40 cm. Find the focal length (in cm) of the combination.

Your answer (numerical value):

5 Medium 1 min NV

P16-05

A prism with $A = 60^\circ$ is made of glass with $n = \sqrt{2}$. Find its angle of minimum deviation (in degrees).

Your answer (numerical value):

6 Easy 0.75 min Basic

P16-06

A coin lies at the bottom of a tank filled with water ($n = 4/3$) to a depth of 12 cm. Viewed from above, the coin appears raised by:

- (A) 9 cm (B) 16 cm (C) 4 cm (D) 3 cm

7 Easy 0.75 min Calc

P16-07

In YDSE, $d = 1$ mm, $D = 1$ m and $\lambda = 600$ nm. The fringe width is:

- (A) 1.2 mm (B) 6 mm (C) 0.06 mm (D) 0.6 mm

8 Medium 1 min NV

P16-08

In YDSE the intensities from the two slits are in the ratio 9 : 1. Find I_{max}/I_{min} .

Your answer (numerical value):

9 Easy 1 min Basic

P16-09

Light of 500 nm falls on a slit 0.2 mm wide, and the screen is 1 m away. The width of the central bright band is:

- (A) 2.5 mm (B) 5 mm (C) 10 mm (D) 1.25 mm

10 Medium 1 min Concept

P16-10

Unpolarised light of intensity I_0 passes through two polaroids whose axes are at 60° . The emerging intensity is:

- (A) $I_0/2$ (B) $I_0/4$ (C) $3I_0/8$ (D) $I_0/8$

11 Easy 0.5 min Speed

P16-11

The Brewster angle for glass of refractive index $\sqrt{3}$ is:

- (A) 30° (B) 60° (C) 45° (D) 90°

12 Easy 0.75 min Basic

P16-12

An astronomical telescope has $f_o = 100$ cm and $f_e = 5$ cm. In normal adjustment, its magnifying power and tube length are:

- (A) 500, 105 cm (B) 20, 95 cm (C) 20, 105 cm (D) 0.05, 105 cm

P16 · Chapter Test

1 Easy 0.5 min Speed

P16-T1

A plane mirror is rotated through θ while the incident ray stays fixed. The reflected ray turns through:

- (A) 2θ (B) θ (C) $\theta/2$ (D) 0

2 Easy 0.5 min Speed

P16-T2

The speed of light in glass of refractive index 1.5 is:

- (A) 2×10^8 m/s (B) 4.5×10^8 m/s (C) 3×10^8 m/s (D) 1.5×10^8 m/s

3 Easy 0.5 min Speed

P16-T3

A lens has focal length -25 cm. Its power is:

- (A) -4 D (B) +4 D (C) -0.04 D (D) -25 D

4 Easy 0.75 min Basic

P16-T4

A YDSE apparatus is immersed in water ($n = 4/3$). The fringe width becomes:

- (A) $4/3$ of the original (B) $9/16$ of the original
(C) unchanged (D) $3/4$ of the original

5 Easy 0.5 min NV

P16-T5

A simple microscope has focal length 5 cm. Find its magnifying power when the image forms at the near point ($D = 25$ cm).

Your answer (numerical value):

6 Medium 1.5 min JEE

P16-T6

A thin film of refractive index 1.5 placed over one slit of a YDSE shifts the central fringe by 5 fringe widths ($\lambda = 600$ nm). The film's thickness is:

- (A) $3 \mu\text{m}$ (B) $2 \mu\text{m}$ (C) $6 \mu\text{m}$ (D) $12 \mu\text{m}$

Answers & Solutions

P16 · Practice

- 1 (B) 2 (C) 3 (A) 4 40 5 30 6 (D) 7 (D) 8 4 9 (B) 10 (D)
11 (B) 12 (C)

P16 · Chapter Test

- 1 (A) 2 (A) 3 (A) 4 (D) 5 6 6 (C)

P16 · PRACTICE

1 Answer: (B) Concave mirror, object inside f Medium 1.5 min Concept

P16-01

SOLUTION $u = -10$, $f = -15$: $\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = -\frac{1}{15} + \frac{1}{10} = \frac{1}{30}$, so $v = +30$ cm (behind the mirror). $m = -v/u = 3$: virtual, erect.

SHORTCUT Object inside f of a concave mirror $\rightarrow$ virtual, erect, magnified. Only (B) fits.

COMMON TRAP Using $+15$ for f .

2 Answer: (C) Critical angle Easy 0.5 min Speed

P16-02

SOLUTION $\sin C = 1/\sqrt{2} \Rightarrow C = 45^\circ$.

3 Answer: (A) Lens in water Medium 1.5 min JEE

P16-03

SOLUTION In water: $\frac{1}{f'} = \left(\frac{1.5}{4/3} - 1\right) \frac{2}{20} = 0.125 \times 0.1 \Rightarrow f' = 80$ cm.

SHORTCUT $\frac{f'}{f} = \frac{n-1}{n/n_m-1} = \frac{0.5}{0.125} = 4$.

COMMON TRAP Assuming f is unchanged.

4 Answer: 40 Lenses in contact Easy 1 min NV

P16-04

SOLUTION $P = \frac{100}{20} - \frac{100}{40} = 5 - 2.5 = 2.5$ D, so $f = 40$ cm (converging).

COMMON TRAP Adding focal lengths.

5 Answer: 30 Minimum deviation Medium 1 min NV

P16-05

SOLUTION $\sqrt{2} = \frac{\sin \frac{60^\circ + \delta_m}{2}}{\sin 30^\circ} \Rightarrow \sin \frac{60^\circ + \delta_m}{2} = \frac{\sqrt{2}}{2} \Rightarrow \frac{60^\circ + \delta_m}{2} = 45^\circ \Rightarrow \delta_m = 30^\circ$.

COMMON TRAP Using the thin-prism formula: $(n-1)A \approx 24.9^\circ$.

6 Answer: (D) Apparent depth Easy 0.75 min Basic

P16-06

SOLUTION Apparent depth = $12/(4/3) = 9$ cm, so the rise is $12 - 9 = 3$ cm.**SHORTCUT** Shift = $t(1 - 1/n)$.**COMMON TRAP** Answering the apparent depth (9 cm).**7 Answer: (D)** YDSE fringe width Easy 0.75 min Calc

P16-07

SOLUTION $\beta = \frac{\lambda D}{d} = \frac{6 \times 10^{-7} \times 1}{10^{-3}} = 6 \times 10^{-4} \text{ m} = 0.6 \text{ mm}.$ **8 Answer: 4** Maximum to minimum intensity Medium 1 min NV

P16-08

SOLUTION Amplitude ratio 3 : 1. $\frac{I_{max}}{I_{min}} = \left(\frac{3+1}{3-1}\right)^2 = 4.$ **SHORTCUT** Work with amplitudes ($\sqrt{I}$).**COMMON TRAP** Using intensities directly: $(9+1)/(9-1).$ **9 Answer: (B)** Single-slit central maximum Easy 1 min Basic

P16-09

SOLUTION Width = $\frac{2\lambda D}{a} = \frac{2 \times 5 \times 10^{-7}}{2 \times 10^{-4}} = 5 \times 10^{-3} \text{ m}.$ **COMMON TRAP** Forgetting the 2 (half-width).**10 Answer: (D)** Two polaroids Medium 1 min Concept

P16-10

SOLUTION After the first: $I_0/2$. After the second: $\frac{I_0}{2} \cos^2 60^\circ = I_0/8.$ **COMMON TRAP** Skipping the first halving ($I_0/4$).**11 Answer: (B)** Brewster's angle Easy 0.5 min Speed

P16-11

SOLUTION $\tan \theta_B = \sqrt{3} \Rightarrow \theta_B = 60^\circ.$ **COMMON TRAP** Using sin (critical-angle thinking).**12 Answer: (C)** Astronomical telescope Easy 0.75 min Basic

P16-12

SOLUTION $M = f_o/f_e = 20$; $L = f_o + f_e = 105 \text{ cm}.$ **P16 · CHAPTER TEST****1 Answer: (A)** Rotating a plane mirror Easy 0.5 min Speed

P16-T1

SOLUTION Both the angle of incidence and the angle of reflection change by θ .**2 Answer: (A)** Speed of light in glass Easy 0.5 min Speed

P16-T2

SOLUTION $v = c/n = 2 \times 10^8 \text{ m/s}.$ **3 Answer: (A)** Power of a lens Easy 0.5 min Speed

P16-T3

SOLUTION $P = 1/f(\text{m}) = 1/(-0.25) = -4 \text{ D}$ (diverging).**COMMON TRAP** Using cm (-0.04).**4 Answer: (D)** YDSE in water Easy 0.75 min Basic

P16-T4

SOLUTION $\lambda' = \lambda/n$, so $\beta' = \beta/n = \frac{3}{4}\beta.$ **5 Answer: 6** Simple microscope Easy 0.5 min NV

P16-T5

SOLUTION $M = 1 + D/f = 1 + 5 = 6.$ **COMMON TRAP** Using $D/f = 5$ (the normal-adjustment case).**6 Answer: (C)** Film over one slit Medium 1.5 min JEE

P16-T6

SOLUTION Extra path $(\mu - 1)t = 5\lambda \Rightarrow t = \frac{5 \times 600 \times 10^{-9}}{0.5} = 6 \times 10^{-6} \text{ m}.$ **COMMON TRAP** Using μt instead of $(\mu - 1)t$ gives $2 \mu\text{m}.$ **REVISION Revision tracker · Optics**☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART II · PHYSICS · UNIT 17

Dual Nature of Matter & Radiation

17

~1 per shift
TYPICAL QUESTIONS

Easy
DIFFICULTY

Must-do (easy
marks)
PRIORITY

Class 12 · Ch 11
NCERT

~4 hours
STUDY TIME

Concept summary

- The **photoelectric effect** (observed by Hertz and Lenard, explained by Einstein) shows that light is absorbed in quanta: **photons** of energy $h\nu$.
- One photon ejects at most one electron. The electron's **maximum KE** depends on frequency, not intensity. Intensity controls the **number** of electrons, and so the photocurrent.
- Matter has wave nature** (de Broglie): $\lambda = h/p$. This was confirmed by electron diffraction in the Davisson–Germer experiment.

Formulas

Σ FORMULA Photons

$$E = h\nu = \frac{hc}{\lambda} \quad hc \approx 1240 \text{ eV nm} \quad p = \frac{h}{\lambda} = \frac{E}{c} \quad \text{photon flux } n = \frac{P}{h\nu} = \frac{P\lambda}{hc}$$

$$h = 6.63 \times 10^{-34} \text{ J s}; 1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}.$$

Σ FORMULA Photoelectric effect

$$K_{max} = h\nu - \phi = eV_0 \quad \nu_0 = \frac{\phi}{h} \quad \lambda_0 = \frac{hc}{\phi} \quad V_0 = \frac{h}{e}\nu - \frac{\phi}{e}$$

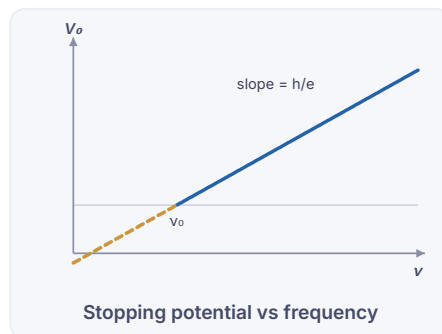
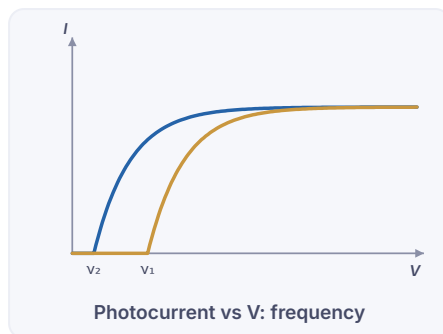
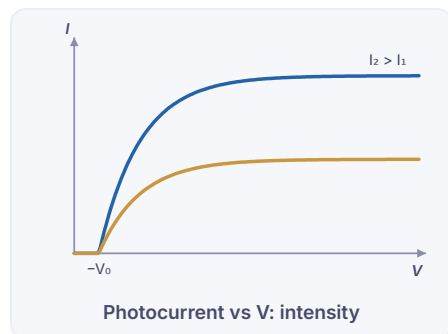
Change	Saturation current	Stopping potential V_0	K_{max}
Intensity $\uparrow$ (same ν)	$\uparrow$ (proportional)	unchanged	unchanged
Frequency $\uparrow$ (same intensity)	about the same (fewer photons per joule)	$\uparrow$	$\uparrow$
$\nu < \nu_0$	zero, whatever the intensity	—	—

- Emission is **instantaneous** (no time lag), even for very weak light.
- The V_0 vs ν graph is a straight line with slope h/e (**the same for every metal**) and ν -intercept ν_0 .

Σ FORMULA de Broglie waves

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{2mqV}}$$

- Electron through V volts: $\lambda = \frac{1.227}{\sqrt{V}} \text{ nm}$.
- Thermal particle at temperature T : $\lambda = \frac{h}{\sqrt{3mk_B T}}$.
- Davisson–Germer: 54 V electrons diffracted by a nickel crystal, $\lambda \approx 0.167 \text{ nm}$, matching de Broglie's prediction.



Frequently tested concepts & question patterns

PYQ PATTERN Recurring structures

1. K_{max} , V_0 , threshold wavelength for given λ and ϕ (use 1240 eV nm).
2. **Graph questions:** $I-V$ for different intensities/frequencies; $V_0-\nu$ slope and intercept.
3. **Two-wavelength problems:** V_1 , V_2 for λ_1 , $\lambda_2 \rightarrow$ find ϕ or h .
4. **de Broglie wavelength ratios** for particles accelerated through the same potential, or with the same KE/momentum.
5. **Photon flux** from a source of given power.

Shortcuts

SHORTCUT 1240 rule

$$E(\text{eV}) = \frac{1240}{\lambda(\text{nm})}. \quad 620 \text{ nm} \rightarrow 2 \text{ eV}, 400 \text{ nm} \rightarrow 3.1 \text{ eV}, 310 \text{ nm} \rightarrow 4 \text{ eV}, 248 \text{ nm} \rightarrow 5 \text{ eV}.$$

Time saved: avoids $6.63 \times 10^{-34} \times 3 \times 10^8$ arithmetic. **Common mistake:** using Å (divide by 10 first).

SHORTCUT de Broglie ratios

Same V : $\lambda \propto 1/\sqrt{mq}$. Same K : $\lambda \propto 1/\sqrt{m}$. Same p : equal λ .

Question classification

Type	Difficulty	Target time
Photon energy / flux	E	45 s
Einstein equation numericals	E	1 min
Graph interpretation	E	45 s
Two-wavelength elimination	M	1.5 min
de Broglie ratios	E	1 min

Common mistakes

MISTAKE ALERT Mistake alerts

- Thinking higher intensity gives faster electrons. It gives **more** electrons.
- Using λ in Å with 1240 (which needs nm).
- For de Broglie of a charged particle through V , forgetting the charge q (α -particle: $q = 2e$).
- Stopping potential in volts = K_{max} in eV, numerically. Don't multiply by e again.

Practice questions

P17 · Practice

1 Easy 0.5 min Speed

P17-01

The energy of a photon of wavelength 620 nm is about:

- (A) 1 eV (B) 3.1 eV (C) 2 eV (D) 0.5 eV

2 Easy 1 min Basic

P17-02

Light of 400 nm falls on a metal with work function 2.3 eV. The stopping potential is:

- (A) 3.1 V (B) 0.8 V (C) 2.3 V (D) 5.4 V

3 Easy 0.75 min NV

P17-03

A metal has work function 4.13 eV. Find its threshold wavelength (in nm). Take $hc = 1240$ eV nm and give the nearest integer.

Your answer (numerical value):

4 Easy 0.5 min Concept

P17-04

The intensity of light on a photo-cell is doubled at the same frequency. Then:

- (A) the saturation current doubles and the stopping potential stays the same
(B) the stopping potential doubles
(C) both double
(D) neither changes

5 Easy 0.75 min Basic

P17-05

An electron is accelerated from rest through 100 V. Its de Broglie wavelength is about:

- (A) 1.23 nm (B) 0.123 nm (C) 12.3 nm (D) 0.0123 nm

6 Medium 1 min Concept

P17-06

A proton and an α -particle are accelerated from rest through the same potential difference. The ratio $\lambda_p : \lambda_\alpha$ is:

- (A) 1 : 1 (B) $1 : 2\sqrt{2}$ (C) 2 : 1 (D) $2\sqrt{2} : 1$

7 Easy 0.5 min Concept

P17-07

The slope of the stopping potential vs frequency graph for a photoelectric metal is:

- (A) h/e (B) h (C) e/h (D) ϕ/e

8 Medium 1 min Calc

P17-08

A 10 W source emits light of wavelength 662 nm. The number of photons it emits per second is about ($h = 6.62 \times 10^{-34}$ J s):

- (A) 1.0×10^{20} (B) 3.3×10^{18} (C) 3.3×10^{19} (D) 6.6×10^{19}

P17 · Chapter Test

1 Easy 0.5 min Concept

P17-T1

In the photoelectric effect, the time lag between incidence of light and emission of electrons is:

- (A) about 1 s (B) inversely proportional to frequency
(C) proportional to intensity (D) negligible ($< 10^{-9}$ s)

2 Easy 0.5 min Speed

P17-T2

A photon of energy E has momentum:

- (A) E/c (B) Ec (C) E/c^2 (D) $\sqrt{2mE}$

3 Easy 0.5 min NV

P17-T3

An electron's de Broglie wavelength after acceleration through V volts is 0.1227 nm. Find V .

Your answer (numerical value):

4 Medium 1 min Concept

P17-T4

An electron and a photon have the same wavelength, 0.1 nm. Then:

- (A) they have equal energies
(B) the electron's KE is greater
(C) the photon has more energy than the electron's kinetic energy
(D) they have different momenta

Answers & Solutions

P17 · Practice

1 (C) 2 (B) 3 300 4 (A) 5 (B) 6 (D) 7 (A) 8 (C)

P17 · Chapter Test

1 (D) 2 (A) 3 100 4 (C)

P17 · PRACTICE

1 Answer: (C) Photon energy Easy 0.5 min Speed

P17-01

SOLUTION $E = 1240/620 = 2$ eV.

2 Answer: (B) Stopping potential Easy 1 min Basic

P17-02

SOLUTION $h\nu = 1240/400 = 3.1$ eV; $K_{max} = 0.8$ eV, so $V_0 = 0.8$ V.

COMMON TRAP Adding the work function.

3 Answer: 300 Threshold wavelength Easy 0.75 min NV

P17-03

SOLUTION $\lambda_0 = hc/\phi = 1240/4.13 \approx 300$ nm.

4 Answer: (A) Effect of intensity Easy 0.5 min Concept

P17-04

SOLUTION Twice as many photons eject twice as many electrons, but each electron's maximum energy is unchanged.

COMMON TRAP Choosing (B).

5 Answer: (B) Electron de Broglie wavelength Easy 0.75 min Basic

P17-05

SOLUTION $\lambda = 1.227/\sqrt{100}$ nm = 0.1227 nm ≈ 1.23 Å.

COMMON TRAP Forgetting the square root.

6 Answer: (D) de Broglie ratio: proton vs α **Medium** ⌚ 1 min **Concept**

P17-06

SOLUTION $\lambda \propto 1/\sqrt{mq}$: $\frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{m_\alpha q_\alpha}{m_p q_p}} = \sqrt{4 \times 2} = 2\sqrt{2}$.

COMMON TRAP Ignoring the charge ($\sqrt{4} = 2$).

7 Answer: (A) Slope of V_0 - v graph **Easy** ⌚ 0.5 min **Concept**

P17-07

SOLUTION $V_0 = \frac{h}{e}\nu - \frac{\phi}{e}$. The slope is h/e , the same for all metals.

8 Answer: (C) Photon flux **Medium** ⌚ 1 min **Calc**

P17-08

SOLUTION $n = \frac{P\lambda}{hc} = \frac{10 \times 662 \times 10^{-9}}{6.62 \times 10^{-34} \times 3 \times 10^8} = \frac{6.62 \times 10^{-6}}{1.986 \times 10^{-25}} \approx 3.3 \times 10^{19} \text{ s}^{-1}$.

SHORTCUT Photon energy $\approx 1240/662 \approx 1.87 \text{ eV} \approx 3 \times 10^{-19} \text{ J}$; $10/(3 \times 10^{-19}) \approx 3.3 \times 10^{19}$.

COMMON TRAP Powers-of-ten slips.

P17 · CHAPTER TEST

1 Answer: (D) Time lag **Easy** ⌚ 0.5 min **Concept**

P17-T1

SOLUTION Each photon transfers its energy to one electron at once.

2 Answer: (A) Photon momentum **Easy** ⌚ 0.5 min **Speed**

P17-T2

SOLUTION $p = h/\lambda = h\nu/c = E/c$.

COMMON TRAP Using the massive-particle formula (D).

3 Answer: 100 Accelerating voltage from wavelength **Easy** ⌚ 0.5 min **NV**

P17-T3

SOLUTION $0.1227 = 1.227/\sqrt{V} \Rightarrow \sqrt{V} = 10 \Rightarrow V = 100 \text{ V}$.

4 Answer: (C) Electron vs photon, same λ **Medium** ⌚ 1 min **Concept**

P17-T4

SOLUTION The momenta are equal (h/λ). Photon: $E = pc \approx 12.4 \text{ keV}$. Electron: $K = p^2/2m \approx 150 \text{ eV}$. The photon's energy is much larger.

SHORTCUT Same $\lambda \Rightarrow$ same p ; $pc \gg p^2/2m$ when $v \ll c$.

COMMON TRAP Choosing (D).

REVISION Revision tracker · Dual Nature

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART II · PHYSICS · UNIT 18

Atoms & Nuclei

18

1-2 per shift

TYPICAL QUESTIONS

Easy

DIFFICULTY

Must-do (easy marks)

PRIORITY

Class 12 · Ch 12, 13

NCERT

~6 hours

STUDY TIME

Concept summary

- **Rutherford's α -scattering:** most α -particles pass straight through gold foil, and a few bounce back. So an atom is mostly empty space, with a tiny, dense, positive nucleus.
- **Bohr's model** (for H-like atoms) quantises angular momentum, $mvr = n\hbar/2\pi$. It gives discrete orbits and energies, and explains the hydrogen spectrum.
- A **nucleus** of Z protons and N neutrons weighs **less** than its parts. The **mass defect** is the binding energy, Δmc^2 .
- **Binding energy per nucleon** peaks near $A \approx 56$. Heavy nuclei release energy by **fission** and light ones by **fusion**.

⚠ WARNING Syllabus note

The 2024–2026 syllabus text for Unit 18 lists α -scattering, Rutherford and Bohr models, the hydrogen spectrum, nuclear composition and size, mass–energy relation, mass defect, BE/A, fission and fusion. **Radioactivity (decay law, half-life, $\alpha/\beta/\gamma$ properties) is not listed** [NTA-SYL]. Give it a short read only.

Formulas: Atoms

Σ FORMULA Bohr model (hydrogen-like ion, atomic number Z)

$$r_n = 0.529 \frac{n^2}{Z} \text{ \AA} \quad v_n = 2.18 \times 10^6 \frac{Z}{n} \text{ m s}^{-1} \quad E_n = -13.6 \frac{Z^2}{n^2} \text{ eV}$$

$$K = -E \quad U = 2E \quad mvr = \frac{nh}{2\pi} \quad 2\pi r_n = n\lambda_{dB}$$

Scaling: $r \propto n^2/Z$, $v \propto Z/n$, $E \propto Z^2/n^2$, time period $\propto n^3/Z^2$.

Rutherford closest approach of an α -particle with kinetic energy K : $d = \frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{K}$.

Σ FORMULA Hydrogen spectrum

$$\frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad R = 1.097 \times 10^7 \text{ m}^{-1} \quad \Delta E = h\nu$$

Series	n_1	Region
Lyman	1	ultraviolet
Balmer	2	visible
Paschen	3	infrared
Brackett	4	infrared
Pfund	5	infrared

- Lines from level n down to ground: $\frac{n(n-1)}{2}$.
- Ionisation energy of H: 13.6 eV. Excitation energies: 10.2 eV ($1 \rightarrow 2$), 12.09 eV ($1 \rightarrow 3$).
- Lyman series limit ≈ 91 nm; Balmer limit ≈ 365 nm; first Balmer line (H- α) ≈ 656 nm.
- **Bohr's limitations:** works only for single-electron atoms; can't explain line intensities or fine structure.

Formulas: Nuclei

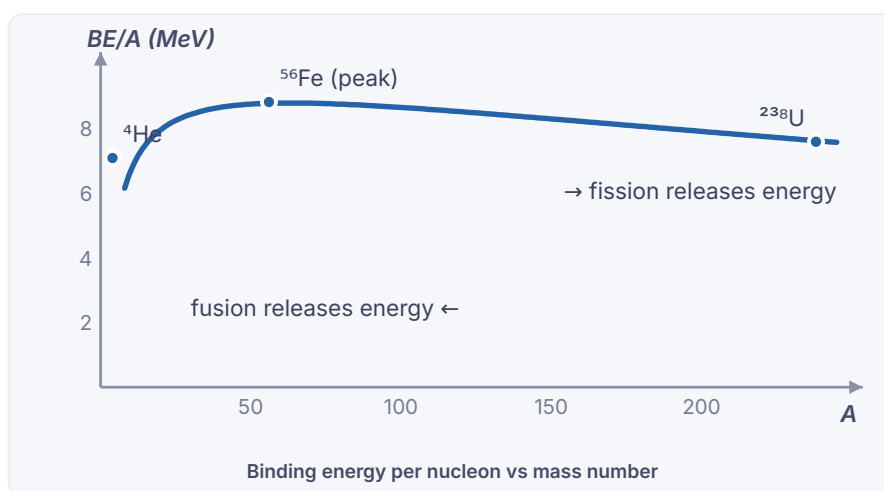
Σ FORMULA Size, mass defect, binding energy

$$R = R_0 A^{1/3}, R_0 \approx 1.2 \text{ fm} \quad \rho_{\text{nucleus}} \approx 2.3 \times 10^{17} \text{ kg m}^{-3} \text{ (independent of } A\text{)}$$

$$\Delta m = [Zm_p + (A - Z)m_n] - M_{\text{nucleus}} \quad BE = \Delta m c^2 \quad 1 \text{ u} = 931.5 \text{ MeV}/c^2$$

$$Q\text{-value} = \left(\sum m_{\text{reactants}} - \sum m_{\text{products}} \right) c^2 \quad Q > 0 \Rightarrow \text{energy released}$$

- **Nuclear force:** strongest force, short range (a few fm), charge independent, saturates.
- **Fission:** $^{235}\text{U} + n \rightarrow$ two medium nuclei + 2–3 neutrons + ~ 200 MeV. **Fusion:** light nuclei combine ($4 \text{ } ^1\text{H} \rightarrow \text{}^4\text{He}$ releases ~ 26.7 MeV in stars). Fusion needs very high temperatures to overcome Coulomb repulsion.



Frequently tested concepts & question patterns

▮ PYQ PATTERN Recurring structures

1. **Bohr scaling** for H, He^+ , Li^{2+} : radius, speed, energy, ratios between orbits.
2. **Spectral wavelengths:** ratio of first lines of two series, series limits, number of lines.
3. **Energy-level diagrams:** which transition emits the longest/shortest wavelength.
4. **Binding energy/nucleon** from given masses; Q-values of reactions.
5. **Nuclear radius/density** scaling with A .
6. **Closest approach** in α -scattering.

Shortcuts

⚡ SHORTCUT Wavelength ratios without R

$$\lambda \propto \frac{1}{\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)}. \text{ For the first lines: Lyman } \propto 4/3, \text{ Balmer } \propto 36/5, \text{ Paschen } \propto 144/7. \text{ Take ratios directly.}$$

⚡ SHORTCUT BE/A curve reading

Anything that moves nucleons **towards the peak** at $A \approx 56$ releases energy. Energy released = $\sum (BE)_{\text{products}} - \sum (BE)_{\text{reactants}}$.

Question classification

Type	Difficulty	Target time
Bohr formulas	E	45 s
Spectral lines / ratios	E	1 min
BE per nucleon	E	1 min
Q-value	E-M	1.5 min
Closest approach	M	1.5 min

Common mistakes

MISTAKE ALERT Mistake alerts

- Using Z instead of Z^2 in energy scaling.
- The **longest** wavelength in a series is its **first** line (smallest ΔE).
- Mass defect uses the **nucleus** mass. If atomic masses are given, electron masses cancel only when you use hydrogen-atom masses for the protons.
- Converting u to MeV with 931 **before** multiplying by the mass defect in u.

Practice questions

P18 · Practice

1 Easy 0.75 min Basic P18-01

The radius of the first Bohr orbit of hydrogen is $0.53 \text{ \AA}$. The radius of the $n = 3$ orbit of Li^{2+} is:

- (A) $0.53 \text{ \AA}$ (B) $1.59 \text{ \AA}$ (C) $4.77 \text{ \AA}$ (D) $0.18 \text{ \AA}$

2 Easy 0.5 min Speed P18-02

The energy of the electron in the $n = 2$ state of He^+ is:

- (A) -3.4 eV (B) -54.4 eV (C) -13.6 eV (D) -6.8 eV

3 Medium 1 min Concept P18-03

The ratio of the wavelength of the first line of the Balmer series to that of the first line of the Lyman series in hydrogen is:

- (A) $9 : 4$ (B) $5 : 27$ (C) $4 : 1$ (D) $27 : 5$

4 Easy 0.5 min NV P18-04

Electrons in a sample of hydrogen are excited to $n = 5$. Find the maximum number of different spectral lines emitted as they return to the ground state.

Your answer (numerical value):

5 Easy 0.5 min Speed P18-05

The ratio of the nuclear radii of nuclei with $A = 216$ and $A = 27$ is:

- (A) 8 (B) 4 (C) 2 (D) $2^{1/3}$

6 Easy 1 min Calc

P18-06

The mass defect of ${}^4_2\text{He}$ is 0.0304 u. Its binding energy per nucleon is about:

- (A) 7.1 MeV (B) 28.3 MeV (C) 14.2 MeV (D) 3.5 MeV

7 Medium 1.5 min Calc

P18-07

A 5 MeV α -particle is aimed head-on at a gold nucleus ($Z = 79$). Its distance of closest approach is about:

- (A) 4.5×10^{-15} m (B) 4.5×10^{-14} m (C) 2.3×10^{-14} m (D) 9.1×10^{-14} m

8 Easy 0.5 min NV

P18-08

Find the ionisation energy (in eV, nearest integer) of Li^{2+} in its ground state.

Your answer (numerical value):

9 Easy 0.75 min Concept

P18-09

Fission of a heavy nucleus releases energy because:

- (A) the products have a larger binding energy per nucleon
 (B) the products have a smaller binding energy per nucleon
 (C) the number of nucleons decreases
 (D) neutrons are destroyed

P18 · Chapter Test

1 Easy 0.5 min Speed

P18-T1

The Balmer series of hydrogen lies in the:

- (A) ultraviolet (B) X-ray region (C) infrared (D) visible

2 Easy 0.5 min Concept

P18-T2

Nuclear density:

- (A) increases with A (B) decreases with A
 (C) is maximum for $A \approx 56$ (D) is nearly the same for all nuclei

3 Easy 0.5 min Speed

P18-T3

The energy equivalent of 1 atomic mass unit is about:

- (A) 931.5 MeV (B) 9.31 MeV (C) 1.6 MeV (D) 0.511 MeV

4 Easy 0.75 min NV

P18-T4

Using $R = 1.097 \times 10^7 \text{ m}^{-1}$, find the series-limit wavelength of the Lyman series (in nm, nearest integer).

Your answer (numerical value):

5 Easy 0.5 min Concept

P18-T5

In the Bohr model, as n increases:

- (A) KE increases and total energy decreases
 (B) both decrease
 (C) both increase
 (D) KE decreases and total energy increases (becomes less negative)

Answers & Solutions

P18 · Practice

1 (B) 2 (C) 3 (D) 4 10 5 (C) 6 (A) 7 (B) 8 122 9 (A)

P18 · Chapter Test

1 (D) 2 (D) 3 (A) 4 91 5 (D)

P18 · PRACTICE

1 Answer: (B) Bohr radius scaling Easy 0.75 min Basic

P18-01

SOLUTION $r = 0.53 \times \frac{n^2}{Z} = 0.53 \times \frac{9}{3} = 1.59 \text{ Å}.$

COMMON TRAP Using n instead of n^2 .

2 Answer: (C) Energy in He^+ Easy 0.5 min Speed

P18-02

SOLUTION $E = -13.6 \times \frac{Z^2}{n^2} = -13.6 \times \frac{4}{4} = -13.6 \text{ eV}.$

COMMON TRAP Using Z instead of Z^2 (-6.8).

3 Answer: (D) Ratio of first lines Medium 1 min Concept

P18-03

SOLUTION $\frac{\lambda_B}{\lambda_L} = \frac{1 - \frac{1}{4}}{\frac{1}{4} - \frac{1}{9}} = \frac{3/4}{5/36} = \frac{27}{5}.$

COMMON TRAP Inverting the ratio.

4 Answer: 10 Number of spectral lines Easy 0.5 min NV

P18-04

SOLUTION $\frac{n(n-1)}{2} = \frac{5 \times 4}{2} = 10.$

COMMON TRAP Counting only direct jumps to $n = 1$ (4 lines).

5 Answer: (C) Nuclear radius scaling Easy 0.5 min Speed

P18-05

SOLUTION $R \propto A^{1/3}: (216/27)^{1/3} = 8^{1/3} = 2.$

6 Answer: (A) Binding energy per nucleon Easy 1 min Calc

P18-06

SOLUTION $BE = 0.0304 \times 931.5 \approx 28.3 \text{ MeV}.$ Per nucleon: $28.3/4 \approx 7.1 \text{ MeV}.$

COMMON TRAP Stopping at the total BE.

7 Answer: (B) Distance of closest approach Medium 1.5 min Calc

P18-07

SOLUTION $d = \frac{k(2e)(79e)}{K} = \frac{9 \times 10^9 \times 158 \times (1.6 \times 10^{-19})^2}{5 \times 1.6 \times 10^{-13}} \approx 4.55 \times 10^{-14} \text{ m}.$

SHORTCUT In MeV-fm units: $ke^2 = 1.44 \text{ MeV fm}$, so $d = 1.44 \times 158/5 \approx 45.5 \text{ fm}.$

COMMON TRAP Forgetting the α -particle's charge $2e$ gives (C).

8 Answer: 122 Ionisation energy of Li^{2+} Easy 0.5 min NV

P18-08

SOLUTION $13.6 \times Z^2 = 13.6 \times 9 = 122.4 \text{ eV}.$

9 Answer: (A) Why fission releases energy Easy 0.75 min Concept

P18-09

SOLUTION Middle-mass nuclei are more tightly bound (higher BE/A). The increase in total binding energy is released.

P18 · CHAPTER TEST

1 Answer: (D) *Balmer region* **Easy** ⌚ 0.5 min **Speed** P18-T1

SOLUTION Transitions to $n = 2$ emit 656 nm to 365 nm, mostly visible.

2 Answer: (D) *Nuclear density* **Easy** ⌚ 0.5 min **Concept** P18-T2

SOLUTION $R^3 \propto A$ and mass $\propto A$, so the density is constant.

COMMON TRAP Confusing it with BE/A (C).

3 Answer: (A) *Energy equivalent of 1 u* **Easy** ⌚ 0.5 min **Speed** P18-T3

SOLUTION $1 \text{ u} \times c^2 \approx 931.5 \text{ MeV}$. (0.511 MeV is the electron's rest energy.)

4 Answer: 91 *Lyman series limit* **Easy** ⌚ 0.75 min **NV** P18-T4

SOLUTION $\lambda = 1/R = 9.116 \times 10^{-8} \text{ m} \approx 91 \text{ nm}$.

5 Answer: (D) *Trends with n* **Easy** ⌚ 0.5 min **Concept** P18-T5

SOLUTION $K \propto 1/n^2$ decreases; $E = -K$ rises towards 0.

 REVISION Revision tracker · Atoms & Nuclei

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

Electronic Devices

19

1–2 per shift

TYPICAL QUESTIONS

Easy

DIFFICULTY

Must-do (easy marks)

PRIORITY

Class 12 · Ch 14

NCERT

~5 hours

STUDY TIME

Concept summary

- In **semiconductors** (Si, Ge) a small band gap lets thermal energy create electron–hole pairs. **Doping** controls conductivity: pentavalent donors give **n-type**, trivalent acceptors give **p-type**.
- A **p–n junction** forms a depletion layer with a barrier potential. It conducts easily in **forward bias** and hardly at all in **reverse bias**, which makes it a rectifier.
- Special diodes**: Zener (voltage regulator in reverse breakdown), LED (light from recombination, forward bias), photodiode (light-controlled reverse current), solar cell (generates emf with no bias).
- Logic gates** process binary signals. NAND and NOR are **universal** gates.

⚠ WARNING Syllabus note

Transistors (BJT action, amplifiers, oscillators) are **not** in the 2024–2026 syllabus text. Unit 19 covers semiconductors, diode I–V characteristics, rectifiers, LED, photodiode, solar cell, Zener diode and logic gates (OR, AND, NOT, NAND, NOR) [NTA-SYL].

Formulas & facts

Σ FORMULA Semiconductors

Type	Dopant (group)	Majority carriers	Examples
Intrinsic	none	$n_e = n_h = n_i$	pure Si, Ge
n-type	pentavalent (15)	electrons	P, As, Sb in Si
p-type	trivalent (13)	holes	B, Al, Ga, In in Si

Mass-action law: $n_e n_h = n_i^2$. A doped semiconductor is still electrically **neutral**.

Band gaps: conductor ≈ 0 (bands overlap), Si ≈ 1.1 eV, Ge ≈ 0.7 eV, insulator > 3 eV. At 0 K an intrinsic semiconductor behaves like an insulator. Conductivity **rises** with temperature (resistance falls).

Σ FORMULA p–n junction & diodes

Barrier potential: Si ≈ 0.7 V, Ge ≈ 0.3 V. **Forward bias** narrows the depletion layer. **Reverse bias** widens it, leaving a tiny reverse saturation current (minority carriers) until breakdown.

Device	Bias	Principle / use
Rectifier diode	forward conducts	half-wave: output ripple at f ; full-wave: at $2f$
Zener diode	reverse (breakdown)	constant V_Z across the load $\rightarrow$ voltage regulator
LED	forward	recombination emits photons, $\lambda = \frac{hc}{E_g}$
Photodiode	reverse	light raises the reverse current (light detector)
Solar cell	none (self-generating)	I–V curve in the 4th quadrant; V_{oc} , I_{sc}

Zener regulator: $I_R = \frac{V_{in} - V_Z}{R_s}$, $I_L = \frac{V_Z}{R_L}$, $I_Z = I_R - I_L$.

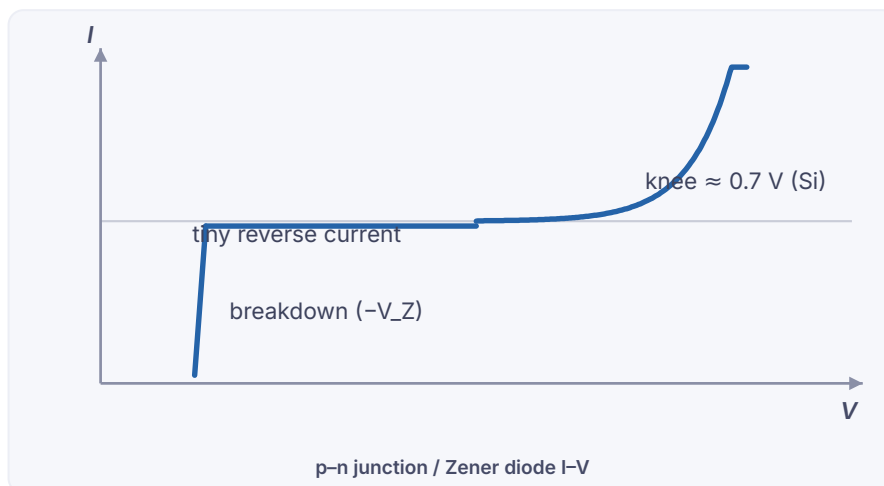
Diode in a circuit: ideal = switch; real Si = 0.7 V drop when conducting.

Σ FORMULA Logic gates

A	B	AND	OR	NAND	NOR	XOR
0	0	0	0	1	1	0
0	1	0	1	1	0	1
1	0	0	1	1	0	1
1	1	1	1	0	0	0

NOT: $\bar{A}$. **De Morgan:** $\overline{A+B} = \bar{A} \cdot \bar{B}$ and $\overline{A \cdot B} = \bar{A} + \bar{B}$.

With NAND gates: NOT = 1 gate (inputs tied), AND = 2 gates, OR = 3 gates. With NOR gates: NOT = 1, OR = 2, AND = 3.



Frequently tested concepts & question patterns

▮ PYQ PATTERN Recurring structures

- Gate combinations:** identify the output or the equivalent gate of a 2–3 gate circuit (very frequent).
- Zener regulator currents.**
- Diode circuits:** which diode conducts; current with ideal or 0.7 V diodes.
- Rectifier** output frequency and waveform.
- Carrier concentration** with $n_e n_h = n_i^2$.
- LED colour ↔ band gap;** photodiode and solar-cell bias facts.

Shortcuts

⚡ SHORTCUT Boolean algebra over truth tables

For gate puzzles, write the Boolean expression and simplify with De Morgan. It's usually faster than a 4-row table and less error-prone.

Example: NOR followed by NOT is $\overline{\overline{A+B}} = A+B$ (OR).

Question classification

Type	Difficulty	Target time
Gate identification	E	45 s
Zener currents	E	1 min
Diode circuit current	E–M	1.5 min
Doping / carrier concentration	E	45 s

Type	Difficulty	Target time
Special diode facts	E	30 s

Common mistakes

MISTAKE ALERT Mistake alerts

- The Zener is used in **reverse** bias, the LED in **forward** bias, the photodiode in **reverse** bias.
- n-type doesn't mean negatively charged. The material stays neutral.
- Full-wave rectifier ripple frequency is **2f** (100 Hz for 50 Hz mains).
- Forgetting the 0.7 V drop when the question says "silicon diode".

Practice questions

P19 · Practice

1 Easy 0.5 min Speed P19-01

Silicon doped with arsenic is:

- (A) p-type, with holes as majority carriers
 (B) an insulator
 (C) intrinsic
 (D) n-type, with electrons as majority carriers

2 Medium 1 min Calc P19-02

An intrinsic semiconductor has $n_i = 1.5 \times 10^{16} \text{ m}^{-3}$. After doping, the electron concentration is $4.5 \times 10^{22} \text{ m}^{-3}$. The hole concentration is:

- (A) $1.5 \times 10^{16} \text{ m}^{-3}$ (B) $3 \times 10^6 \text{ m}^{-3}$ (C) $5 \times 10^9 \text{ m}^{-3}$ (D) $4.5 \times 10^{22} \text{ m}^{-3}$

3 Easy 0.5 min Speed P19-03

The ripple frequency at the output of a full-wave rectifier fed from 50 Hz mains is:

- (A) 100 Hz (B) 50 Hz (C) 25 Hz (D) 200 Hz

4 Medium 1.5 min NV P19-04

A Zener diode with $V_Z = 5 \text{ V}$ is connected, through a 500Ω series resistor, to a 15 V supply. The load across the Zener is $1 \text{ k}\Omega$. Find the Zener current (in mA).

Your answer (numerical value):

5 Easy 0.75 min Basic P19-05

An LED has a band gap of 2.0 eV. The wavelength of the light it emits is about:

- (A) 400 nm (B) 620 nm (C) 500 nm (D) 900 nm

6 Easy 0.5 min Concept P19-06

A NAND gate with both inputs joined together acts as:

- (A) AND (B) OR (C) NOT (D) NOR

7 Easy 0.75 min Concept

P19-07

The output of a NOR gate is fed into a NOT gate. The combination is equivalent to:

- (A) AND (B) XOR (C) NAND (D) OR

8 Easy 1 min Basic

P19-08

A silicon diode (0.7 V drop when conducting) is forward biased in series with a 1 k Ω resistor across a 5 V battery. The current is:

- (A) 5.0 mA (B) 4.3 mA (C) 5.7 mA (D) 0.7 mA

9 Easy 0.5 min Concept

P19-09

A photodiode is normally operated in:

- (A) reverse bias (B) forward bias (C) zero bias only (D) breakdown

P19 · Chapter Test

1 Easy 0.5 min Speed

P19-T1

The correct order of band gaps is:

- (A) conductor > semiconductor > insulator (B) semiconductor > insulator > conductor
(C) insulator > semiconductor > conductor (D) all equal

2 Easy 0.5 min Speed

P19-T2

In reverse bias, the depletion layer of a p-n junction:

- (A) widens (B) narrows (C) disappears (D) is unchanged

3 Easy 0.5 min Concept

P19-T3

A device whose working I-V characteristic lies in the fourth quadrant is a:

- (A) solar cell (B) LED (C) Zener diode (D) rectifier diode

4 Easy 0.5 min NV

P19-T4

What is the minimum number of NAND gates needed to realise an OR gate?

Your answer (numerical value):

5 Easy 0.5 min Speed

P19-T5

At 0 K, a pure semiconductor behaves as:

- (A) a perfect conductor (B) an n-type semiconductor
(C) a superconductor (D) an insulator

Answers & Solutions

P19 · Practice

1 (D) 2 (C) 3 (A) 4 15 5 (B) 6 (C) 7 (D) 8 (B) 9 (A)

P19 · Chapter Test

1 (C) 2 (A) 3 (A) 4 3 5 (D)

P19 · PRACTICE

- 1 Answer: (D) Type of doping** Easy ⌚ 0.5 min Speed P19-01
SOLUTION As is pentavalent (group 15), so it donates electrons.
- 2 Answer: (C) Mass-action law** Medium ⌚ 1 min Calc P19-02
SOLUTION $n_h = \frac{n_i^2}{n_e} = \frac{2.25 \times 10^{32}}{4.5 \times 10^{22}} = 5 \times 10^9 \text{ m}^{-3}$.
COMMON TRAP Assuming $n_h = n_i$.
- 3 Answer: (A) Rectifier ripple** Easy ⌚ 0.5 min Speed P19-03
SOLUTION Both half-cycles are rectified, so the output repeats at $2f$.
COMMON TRAP Using the half-wave value.
- 4 Answer: 15 Zener regulator** Medium ⌚ 1.5 min NV P19-04
SOLUTION $I_R = (15 - 5)/500 = 20 \text{ mA}$; $I_L = 5/1000 = 5 \text{ mA}$; $I_Z = 15 \text{ mA}$.
COMMON TRAP Forgetting to subtract the load current.
- 5 Answer: (B) LED wavelength from band gap** Easy ⌚ 0.75 min Basic P19-05
SOLUTION $\lambda = 1240/2.0 = 620 \text{ nm}$ (red-orange).
- 6 Answer: (C) NAND as NOT** Easy ⌚ 0.5 min Concept P19-06
SOLUTION $\overline{A \cdot A} = \overline{A}$.
- 7 Answer: (D) Gate combination** Easy ⌚ 0.75 min Concept P19-07
SOLUTION $\overline{\overline{A + B}} = A + B$.
- 8 Answer: (B) Silicon diode current** Easy ⌚ 1 min Basic P19-08
SOLUTION $I = (5 - 0.7)/1000 = 4.3 \text{ mA}$.
COMMON TRAP Ignoring the diode drop.
- 9 Answer: (A) Photodiode bias** Easy ⌚ 0.5 min Concept P19-09
SOLUTION In reverse bias the small current is very sensitive to light-generated carriers, so the change is easy to measure.
COMMON TRAP Confusing it with the LED.

P19 · CHAPTER TEST

- 1 Answer: (C) Band gaps** Easy ⌚ 0.5 min Speed P19-T1
SOLUTION Insulator ($> 3 \text{ eV}$) > semiconductor ($\sim 1 \text{ eV}$) > conductor (overlapping bands, ~ 0).
- 2 Answer: (A) Depletion width** Easy ⌚ 0.5 min Speed P19-T2
SOLUTION The applied field adds to the barrier field, pulling carriers away from the junction.
- 3 Answer: (A) Fourth-quadrant device** Easy ⌚ 0.5 min Concept P19-T3
SOLUTION A solar cell delivers power: current flows opposite to the voltage it develops across itself.
- 4 Answer: 3 Universal gates** Easy ⌚ 0.5 min NV P19-T4
SOLUTION $A + B = \overline{\overline{A} \cdot \overline{B}}$: two NANDs as inverters for $\overline{A}$ and $\overline{B}$, then one NAND.
- 5 Answer: (D) Semiconductor at 0 K** Easy ⌚ 0.5 min Speed P19-T5
SOLUTION No thermal energy is available to excite electrons across the gap.



REVISION Revision tracker · Electronic Devices

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART II · PHYSICS · UNIT 20

Experimental Skills

20

0–2 direct per shift
TYPICAL QUESTIONS

Easy
DIFFICULTY

High (predictable)
PRIORITY

Class 11 & 12
Physics Lab
Manuals
NCERT

~6 hours
STUDY TIME

Concept summary

The official syllabus lists **18 experiments** [NTA-SYL]. Questions test three things: **(1)** reading an instrument correctly (least count, zero error), **(2)** the formula and graph behind an experiment, and **(3)** which measurement contributes the most error. Learn each experiment as a one-line card: *aim* → *formula* → *graph* → *main error source*.

Measuring instruments

Σ FORMULA Vernier callipers & screw gauge

Vernier: $LC = 1 \text{ MSD} - 1 \text{ VSD}$. If $N \text{ VSD} = (N - 1) \text{ MSD}$, then $LC = \frac{\text{MSD}}{N}$.

Reading = MSR + (coinciding VSD) × LC. Zero error: vernier zero to the **right** of the main zero is positive; to the **left** is negative. **Corrected = observed – zero error**.

Parts: outer jaws (external diameter), inner jaws (internal diameter), depth strip (depth).

Screw gauge: $LC = \frac{\text{pitch}}{\text{no. of circular divisions}}$. Reading = MSR + CSR × LC. **Backlash error** comes from play in the screw. Avoid it by always turning in one direction.

The 18 syllabus experiments on one page

#	Experiment	Key formula / relation	Graph	Main error source / tip
1	Vernier callipers: internal/external diameter, depth	LC, zero error	—	zero-error sign
2	Screw gauge: thickness/diameter of sheet/wire	LC = pitch/divisions	—	backlash, zero error
3	Simple pendulum: dissipation of energy	$A^2 \propto e^{-bt/m}$; $T = \frac{2\pi\sqrt{L/g}}$	A^2 vs t : exponential decay; L vs T^2 : straight line, slope $g/4\pi^2$	amplitude reading, air drag
4	Metre scale: mass by principle of moments	$m_1 l_1 = m_2 l_2$	—	knife-edge position
5	Young's modulus of a wire (Searle's)	$Y = \frac{MgL}{\pi r^2 \Delta L}$	load vs extension: line through origin	radius (r^2) and ΔL (smallest quantities)
6	Surface tension by capillary rise; effect of detergent	$T = \frac{r h \rho g}{2 \cos \theta}$	—	radius of bore; detergent lowers h
7	Viscosity by terminal velocity	$\eta = \frac{2r^2(\rho - \sigma)g}{9v_t}$	—	timing only after terminal speed is reached; wall effects
8	Speed of sound, resonance tube	$v = 2f(L_2 - L_1)$; $e = 0.3d$	—	end correction cancels in $L_2 - L_1$
9	Specific heat of a solid/liquid (mixtures)	$m_s c_s (T_s - T) = (m_w c_w + W)(T - T_w)$	—	heat loss to the surroundings

#	Experiment	Key formula / relation	Graph	Main error source / tip
10	Resistivity using a metre bridge	$\frac{R}{S} = \frac{l}{100-l}, \rho = \frac{\pi r^2 R}{L}$	—	end errors (interchange R and S and average); balance near 50 cm
11	Resistance by Ohm's law	$R = V/I$	$V-I$ straight line, slope R	heating of the wire
12	Galvanometer resistance & figure of merit (half deflection)	$G = \frac{RS}{R-S}; k = \frac{E}{(R+G)\theta}$	—	use $R \gg G$
13	Focal length by parallax: convex mirror, concave mirror, convex lens	$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$ (mirror), $\frac{1}{v} - \frac{1}{u}$ (lens)	$u-v$ hyperbola; $1/u$ vs $1/v$ line with intercepts $1/f$	parallax removal; for a lens, $u = v = 2f$ point
14	$i-\delta$ curve for a prism	$n = \frac{\sin \frac{A+\delta_m}{2}}{\sin \frac{A}{2}}$	U-shaped, minimum at δ_m	pin alignment
15	Refractive index of a slab with a travelling microscope	$n = \frac{\text{real}}{\text{apparent}} = \frac{R_3 - R_1}{R_3 - R_2}$	—	focusing
16	p-n junction diode, forward & reverse characteristics	knee ≈ 0.7 V (Si)	exponential forward curve; tiny reverse current	use mA in forward, μ A in reverse
17	Zener diode characteristics, breakdown voltage	V_Z at the sharp rise in reverse current	vertical reverse breakdown	series resistor to limit current
18	Identify diode, LED, resistor, capacitor (multimeter)	—	—	see the table below

☆ IMPORTANT Component identification (Experiment 18)

Component	Multimeter behaviour
Resistor	same resistance in both directions
Diode	low resistance one way, very high the other way
LED	conducts one way and glows
Capacitor	momentary deflection (charging), then very high resistance

📊 PYQ PATTERN Recurring structures

1. Vernier/screw-gauge readings with zero error (often combined with the error in a derived quantity).
2. "Which measurement has the largest percentage error?" (usually the one raised to a power or the smallest reading).
3. Metre-bridge null points, interchange; galvanometer half-deflection.
4. Resonance tube: v and end correction from two resonance lengths.
5. Pendulum graphs ($L-T^2$ slope; A^2-t decay).
6. Travelling microscope readings $\rightarrow$ refractive index.

⚠️ MISTAKE ALERT Mistake alerts

- Screw gauge: "the reference line is below/above zero" statements are orientation-dependent. Convert them into a **signed reading with the jaws closed** first.
- Metre bridge: l is measured from the end connected to the **unknown** in its gap, or whichever the question says. Read carefully.
- End correction of the resonance tube is $0.3 \times \text{diameter}$ ($0.6 \times \text{radius}$).

Practice questions

P20 · Practice

1 **Easy** ⌚ 0.5 min **Speed** P20-01

In a Vernier calliper, 20 VSD coincide with 19 MSD, and 1 MSD = 1 mm. The least count is:

- (A) 0.1 mm (B) 0.05 mm (C) 0.02 mm (D) 0.5 mm

2 **Medium** ⌚ 1.5 min **JEE** P20-02

A screw gauge has pitch 1 mm and 100 circular divisions. With the studs closed it reads -0.03 mm. Measuring a wire, the main-scale reading is 4 mm and the circular-scale reading is 52. The corrected diameter is:

- (A) 4.49 mm (B) 4.52 mm (C) 4.55 mm (D) 4.82 mm

3 **Medium** ⌚ 1.5 min **Calc** P20-03

In the half-deflection method, the galvanometer shows deflection θ with $5000\ \Omega$ in series. Connecting a $50\ \Omega$ shunt across the galvanometer reduces the deflection to $\theta/2$. The galvanometer resistance is about:

- (A) $50\ \Omega$ (B) $49.5\ \Omega$ (C) $50.5\ \Omega$ (D) $100\ \Omega$

4 **Easy** ⌚ 0.5 min **Concept** P20-04

In the simple-pendulum energy-dissipation experiment, the graph of $(\text{amplitude})^2$ against time is:

- (A) a straight line with positive slope (B) constant
(C) a parabola (D) an exponentially decaying curve

5 **Easy** ⌚ 1 min **Calc** P20-05

Water rises 3 cm in a capillary of radius 0.5 mm (contact angle 0° , $\rho = 1000$, $g = 10$). The surface tension is:

- (A) $0.075\ \text{N/m}$ (B) $0.15\ \text{N/m}$ (C) $0.0375\ \text{N/m}$ (D) $0.75\ \text{N/m}$

6 **Medium** ⌚ 1 min **Basic** P20-06

Travelling-microscope readings: mark on paper (no slab) 10.20 mm, mark seen through the slab 13.20 mm, top surface of the slab 19.20 mm. The refractive index of the slab is:

- (A) 1.50 (B) 1.33 (C) 1.60 (D) 2.00

7 **Medium** ⌚ 1.5 min **Calc** P20-07

A 0.1 kg metal piece at 100°C is dropped into 0.2 kg of water at 20°C (negligible calorimeter heat capacity), and the final temperature is 24°C . With $c_w = 4200\ \text{J kg}^{-1}\text{K}^{-1}$, the metal's specific heat is about:

- (A) $221\ \text{J/kg K}$ (B) $336\ \text{J/kg K}$ (C) $884\ \text{J/kg K}$ (D) $442\ \text{J/kg K}$

8 **Easy** ⌚ 0.5 min **Speed** P20-08

The end correction of a resonance tube of internal diameter 5 cm is about:

- (A) 0.6 cm (B) 3.0 cm (C) 1.5 cm (D) 5.0 cm

9 Easy 0.5 min Concept

P20-09

Tested with a multimeter, a component shows a momentary deflection that then falls to (almost) zero current. The component is a:

- (A) resistor (B) diode (C) capacitor (D) LED

10 Easy 0.75 min NV

P20-10

In the $u-v$ experiment with a convex lens, the object and image distances are equal at 40 cm. Find the focal length (in cm).

Your answer (numerical value):

P20 · Chapter Test

1 Easy 0.75 min Concept

P20-T1

In Searle's experiment for Young's modulus, the largest contribution to the percentage error usually comes from measuring:

- (A) the length L of the wire (B) the radius r of the wire
(C) the load M (D) g

2 Easy 0.5 min Speed

P20-T2

A screw gauge has pitch 0.5 mm and 50 circular divisions. Its least count is:

- (A) 0.01 mm (B) 0.1 mm (C) 0.001 mm (D) 0.05 mm

3 Easy 0.5 min Speed

P20-T3

In the viscosity experiment, the radius of the ball is doubled (same material). Its terminal velocity becomes:

- (A) 2 times (B) half (C) 8 times (D) 4 times

4 Easy 0.5 min NV

P20-T4

Find the least count (in μm) of a screw gauge with pitch 1 mm and 200 circular divisions.

Your answer (numerical value):

Answers & Solutions

P20 · Practice

1 (B) 2 (C) 3 (C) 4 (D) 5 (A) 6 (A) 7 (D) 8 (C) 9 (C) 10 20

P20 · Chapter Test

1 (B) 2 (A) 3 (D) 4 5

P20 · PRACTICE

1 Answer: (B) Vernier least count Easy 0.5 min Speed

P20-01

SOLUTION $LC = MSD/N = 1/20 = 0.05 \text{ mm}$.

2 Answer: (C) Screw gauge, negative zero error Medium 1.5 min JEE

P20-02

SOLUTION $LC = 0.01 \text{ mm}$; observed = $4 + 0.52 = 4.52 \text{ mm}$; corrected = $4.52 - (-0.03) = 4.55 \text{ mm}$.

SHORTCUT Subtracting a negative error means adding.

COMMON TRAP Subtracting 0.03 gives 4.49.

3 Answer: (C) Half-deflection method **Medium** ⌚ 1.5 min **Calc**

P20-03

SOLUTION $G = \frac{RS}{R-S} = \frac{5000 \times 50}{4950} \approx 50.5 \Omega.$

SHORTCUT When $R \gg G$, $G \approx S$.

4 Answer: (D) Pendulum energy dissipation graph **Easy** ⌚ 0.5 min **Concept**

P20-04

SOLUTION Damping makes the energy ($\propto A^2$) decay exponentially: $A^2 \propto e^{-bt/m}$.

5 Answer: (A) Surface tension from capillary rise **Easy** ⌚ 1 min **Calc**

P20-05

SOLUTION $T = \frac{r h \rho g}{2} = \frac{5 \times 10^{-4} \times 0.03 \times 1000 \times 10}{2} = 0.075 \text{ N/m}.$

COMMON TRAP Forgetting the 2.

6 Answer: (A) Travelling microscope **Medium** ⌚ 1 min **Basic**

P20-06

SOLUTION Real thickness = $19.20 - 10.20 = 9.00 \text{ mm}$. Apparent = $19.20 - 13.20 = 6.00 \text{ mm}$. $n = 9/6 = 1.5$.

COMMON TRAP Using $13.20 - 10.20$ as the apparent thickness (that's the shift).

7 Answer: (D) Method of mixtures **Medium** ⌚ 1.5 min **Calc**

P20-07

SOLUTION $0.1c(76) = 0.2(4200)(4) = 3360 \Rightarrow c = 3360/7.6 \approx 442 \text{ J/kg K}.$

COMMON TRAP Using $100 - 20$ instead of $100 - 24$.

8 Answer: (C) End correction **Easy** ⌚ 0.5 min **Speed**

P20-08

SOLUTION $e = 0.3d = 1.5 \text{ cm}.$

COMMON TRAP Using $0.6d$.

9 Answer: (C) Identifying a capacitor **Easy** ⌚ 0.5 min **Concept**

P20-09

SOLUTION The capacitor charges briefly, then blocks DC.

10 Answer: 20 $u = v$ point of a convex lens **Easy** ⌚ 0.75 min **NV**

P20-10

SOLUTION $|u| = |v| = 2f \Rightarrow f = 20 \text{ cm}.$

P20 · CHAPTER TEST

1 Answer: (B) Largest error in Searle's experiment **Easy** ⌚ 0.75 min **Concept**

P20-T1

SOLUTION r is small and appears squared in $Y = MgL/(\pi r^2 \Delta L)$, so its percentage error is doubled. (The extension ΔL is also critical, which is why a micrometer is used.)

SHORTCUT Small quantity $\times$ high power $\Rightarrow$ largest error.

COMMON TRAP Choosing L (large, easy to measure precisely).

2 Answer: (A) Screw gauge least count **Easy** ⌚ 0.5 min **Speed**

P20-T2

SOLUTION $0.5/50 = 0.01 \text{ mm}.$

3 Answer: (D) Terminal velocity scaling **Easy** ⌚ 0.5 min **Speed**

P20-T3

SOLUTION $v_t \propto r^2.$

4 Answer: 5 Least count in micrometres **Easy** ⌚ 0.5 min **NV**

P20-T4

SOLUTION $1 \text{ mm}/200 = 0.005 \text{ mm} = 5 \mu\text{m}.$

REVISION Revision tracker · Experimental Skills

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

Chemistry

Chemistry strategy and the NCERT method, then Physical, Inorganic and Organic modules with reaction maps, reagent tables and NCERT fact sheets.



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PART III · CHEMISTRY

Chemistry: The Complete Strategy

Why Chemistry decides 99 percentile

Chemistry is usually the **fastest and most accurate** section for well-prepared students. In 2026 analyses it was repeatedly described as the section where NCERT-strong students gained their edge, with several questions taken almost directly from NCERT text and lab manuals [||| THIRD-PARTY ANALYSIS](#) [T-ANALYSIS26]. A realistic 99+ plan treats Chemistry as the **mark bank**: 80–90/100 in about 45 minutes, leaving more time for Mathematics.

VERIFIED FACT Syllabus structure (official)

The official Chemistry syllabus has **20 units**: 8 Physical (Units 1–8), 4 Inorganic (Units 9–12) and 8 Organic (Units 13–20, including Unit 20, Principles Related to Practical Chemistry). The 2024 revision removed States of Matter, Surface Chemistry, Hydrogen, s-Block, Metallurgy, Environmental Chemistry, Polymers and Chemistry in Everyday Life [NTA-SYL], [T-DELETED].

Unit-wise weight and priority

"Typical questions per shift" is compiled from published 2024–2026 analyses [||| THIRD-PARTY ANALYSIS](#) [T-WT-VED], [T-WT-CD]. Priority is this book's recommendation [@ RECOMMENDATION](#).

#	Unit	Branch	Typical Q / shift	Priority	Hours
1	Some Basic Concepts	Physical	~1	High	5
2	Atomic Structure	Physical	1–2	Must-do	7
3	Chemical Bonding	Physical/Inorganic	1–2	Must-do	12
4	Chemical Thermodynamics	Physical	1–2	Must-do	9
5	Solutions	Physical	~1	High	6
6	Equilibrium (chemical + ionic)	Physical	1–2	Must-do	12
7	Redox & Electrochemistry	Physical	1–2	Must-do	9
8	Chemical Kinetics	Physical	1–2	Must-do	7
9	Periodicity	Inorganic	~1	Must-do	5
10	p-Block (Groups 13–18)	Inorganic	1–2	Must-do	12
11	d- & f-Block	Inorganic	1–2	Must-do	8
12	Coordination Compounds	Inorganic	2–3	Must-do	10
13	Purification & Characterisation	Organic	0–1	High	4
14	Basic Principles (GOC)	Organic	1–2	Must-do	12
15	Hydrocarbons	Organic	~1	High	8
16	Halogen compounds	Organic	~1	High	6
17	Oxygen compounds	Organic	2–3	Must-do	14
18	Nitrogen compounds	Organic	~1	High	6
19	Biomolecules	Organic	~1	Must-do (fast)	4
20	Practical Chemistry	Organic/Inorganic	~1	High	5

**PYQ PATTERN Published 2026 counts (all Paper-1 shifts, both sessions)**

Chapter counts for JEE Main 2026 reported by Collegedunia from the ExamSIDE PYQ archive

||| **THIRD-PARTY ANALYSIS** [T-WT-CD]: **Coordination Compounds 36** questions (7.58%) · **Aldehydes, Ketones & Carboxylic Acids 35** (7.37%) · **GOC 27** (5.68%) · **Chemical Kinetics 27** (5.68%) · **Thermodynamics 22** (4.63%) · **Electrochemistry 21** (4.42%) · **Salt analysis 5** (1.05%). On about 19 shifts, that's roughly **2 questions per shift** each for Coordination and Carbonyls.

How to maximise marks in Chemistry

Strong student (≥ 80)

- Target 23–24 attempts at 95% accuracy in ≤ 40 minutes.
- Remaining losses usually come from NCERT exceptions and unusual organic conversions. Keep an **exceptions notebook** and read it weekly.
- Solve 2024–2026 PYQs **line by line against NCERT**, and highlight the NCERT sentence each one came from.

Average student (55–80)

- Physical Chemistry formulas + Coordination + Biomolecules + Practical Chemistry = the quickest 20 marks to add.
- Organic: learn the reaction maps (not isolated reactions) for Units 15–18.
- Inorganic: one NCERT reading **every 10 days** for Units 9–12 (spaced repetition).

Weak student (< 55)

- Start with **NCERT reading**, not question banks. Chemistry is where reading converts to marks most directly.
- Order: Biomolecules → Practical Chemistry → Coordination → Periodicity → Atomic Structure → Mole concept → Kinetics → Solutions → Electrochemistry → GOC → carbonyl compounds.
- Attempt only questions you recognise within 20 seconds. Chemistry questions reward recall, and hesitation usually signals a gap.

PART III · CHEMISTRY

The NCERT Strategy

Why NCERT, and how exactly



VERIFIED FACT Alignment

The 2024 syllabus revision aligned JEE Main with the **rationalised NCERT** textbooks [NTA-SYL]. Inorganic Chemistry, Biomolecules and Practical Chemistry questions are frequently answerable from NCERT text, tables and in-text examples [THIRD-PARTY ANALYSIS](#) [T-ANALYSIS26].

The 4-read method RECOMMENDATION

- Read 1 · Understand** (with the module): read the whole chapter, including boxes, tables and in-text questions.
- Read 2 · Mark** (Day 7): highlight every fact that could be an MCQ: exceptions, colours, shapes, orders, named reactions, uses.
- Read 3 · Extract** (Day 14): copy the highlights into this book's NCERT checklists or your own fact sheet.
- Read 4+ · Recall** (every 10–14 days): cover the page and recall the table, then check. Only the fact sheet is re-read after this point.

NCERT chapter checklist (map to JEE units)

JEE unit	NCERT source (rationalised editions)	Read 1	Read 2	Read 3	Recall
1 Basic Concepts	Class 11 · Ch 1 Some Basic Concepts of Chemistry	☐	☐	☐	☐
2 Atomic Structure	Class 11 · Ch 2 Structure of Atom	☐	☐	☐	☐
3 Bonding	Class 11 · Ch 4 Chemical Bonding & Molecular Structure	☐	☐	☐	☐
4 Thermodynamics	Class 11 · Ch 5 Thermodynamics	☐	☐	☐	☐
5 Solutions	Class 12 · Ch 1 Solutions	☐	☐	☐	☐
6 Equilibrium	Class 11 · Ch 6 Equilibrium	☐	☐	☐	☐
7 Redox & Electrochemistry	Class 11 · Ch 7 Redox Reactions; Class 12 · Ch 2 Electrochemistry	☐	☐	☐	☐
8 Kinetics	Class 12 · Ch 3 Chemical Kinetics	☐	☐	☐	☐
9 Periodicity	Class 11 · Ch 3 Classification of Elements	☐	☐	☐	☐
10 p-Block (Groups 13–18)	Not in the rationalised books. Use the p-Block chapters of the pre-2023 NCERT editions (Class 11 Groups 13–14; Class 12 Groups 15–18) for trends	☐	☐	☐	☐
11 d- & f-Block	Class 12 · Ch 4 The d- and f-Block Elements	☐	☐	☐	☐
12 Coordination	Class 12 · Ch 5 Coordination Compounds	☐	☐	☐	☐
13 Purification & Characterisation	Class 11 · Ch 8 (sections on purification, qualitative & quantitative analysis)	☐	☐	☐	☐
14 GOC	Class 11 · Ch 8 Organic Chemistry: Some Basic Principles & Techniques	☐	☐	☐	☐
15 Hydrocarbons	Class 11 · Ch 9 Hydrocarbons	☐	☐	☐	☐
16 Halogen compounds	Class 12 · Ch 6 Haloalkanes & Haloarenes	☐	☐	☐	☐

JEE unit	NCERT source (rationalised editions)	Read 1	Read 2	Read 3	Recall
17 Oxygen compounds	Class 12 · Ch 7 Alcohols, Phenols & Ethers; Ch 8 Aldehydes, Ketones & Carboxylic Acids	☐	☐	☐	☐
18 Nitrogen compounds	Class 12 · Ch 9 Amines	☐	☐	☐	☐
19 Biomolecules	Class 12 · Ch 10 Biomolecules	☐	☐	☐	☐
20 Practical Chemistry	NCERT Laboratory Manual Chemistry, Class 11 & 12 (salt analysis, titrations, preparations, functional-group tests)	☐	☐	☐	☐

WARNING p-Block needs extra care

The syllabus lists **Groups 13 to 18: general trends and the unique behaviour of the first element**. The rationalised NCERT books no longer contain these chapters. Use the p-Block module in this book (page [213](#)) together with the older NCERT chapters, and **stay at the level of trends and anomalies**. Detailed compound preparations were not the focus of the revised syllabus text.

Important factual areas (what to extract from NCERT)

Tables & trends to memorise

- Periodic trends: atomic/ionic radii, IE, EGE, EN, with every exception
- Shapes & hybridisation (VSEPR table)
- MO diagrams for B_2 to Ne_2 : bond order, magnetism
- d-block: electronic configurations (Cr, Cu anomalies), oxidation states, colours, magnetic moments
- Lanthanoid contraction and its consequences
- Spectrochemical series; crystal field splitting (Δ_o , vs pairing)
- Acid/base strengths: carboxylic acids, phenols, amines (gas vs water)
- Vitamins and deficiency diseases; bases in DNA vs RNA

Reactions, exceptions & lines

- Preparation of $K_2Cr_2O_7$ and $KMnO_4$ (steps and equations)
- All named reactions in Class 12 organic chapters (list on page [254](#))
- Tests: Lucas, iodoform, Tollens', Fehling's, Hinsberg, carbylamine, $FeCl_3$ (phenol)
- Reimer-Tiemann, Kolbe, Cannizzaro, aldol, Gabriel phthalimide, Hofmann bromamide, Sandmeyer, diazo coupling
- Glucose: reactions showing its structure (HI, NH_2OH , Br_2 water, HNO_3 , acetylation)
- Anomalous behaviour of the first element of each p-block group
- Uses of coordination compounds (EDTA, cisplatin, Wilkinson's catalyst, estimation of hardness)
- Every "Why?" statement in the in-text questions

Chemistry revision method

1. **Physical:** formula sheet + 10 numerical questions per chapter per week. Formula recall on a blank sheet every 5 days.
2. **Inorganic:** fact sheets read on a **spaced schedule** (Day 1, 3, 7, 14, 30). 15-minute "flash review" every night in the final 45 days.
3. **Organic:** redraw each **reaction map** from memory once a week; solve 20 conversion questions per week.
4. **Mistake Book:** every Chemistry error tagged *fact* / *concept* / *exception* / *calculation*. Exceptions become flash cards.

PART III • CHEMISTRY • UNIT 01

Some Basic Concepts in Chemistry

01

~1 per shift (+
used in every
numerical)

TYPICAL QUESTIONS

Easy
DIFFICULTY

High
PRIORITY

Class 11 • Ch 1
NCERT

~5 hours
STUDY TIME

Important concepts

- **Laws of chemical combination:** conservation of mass; definite proportions (a compound always has the same composition); multiple proportions (masses of B combining with a fixed mass of A are in small whole-number ratios, as in CO/CO₂); Gay-Lussac's law of gaseous volumes; Avogadro's law (equal volumes of gases at the same T , P contain equal numbers of molecules).
- **Dalton's atomic theory:** matter is made of indivisible atoms. Atoms of an element are identical, and compounds form in fixed whole-number ratios.
- **Atomic mass unit:** $1 \text{ u} = \frac{1}{12}$ of the mass of one ^{12}C atom $= 1.66 \times 10^{-24} \text{ g}$.
- **Mole:** 6.022×10^{23} entities (N_A). Molar mass (g/mol) is numerically equal to the atomic or molecular mass in u.

Formula sheet

Σ FORMULA Mole relations

$$n = \frac{m}{M} = \frac{N}{N_A} = \frac{V_{gas}}{V_m} \quad V_m = 22.4 \text{ L (273 K, 1 atm)} \quad \text{or} \quad 22.7 \text{ L (273.15 K, 1 bar, NCERT STP)}$$

$$\text{Mass \% of element} = \frac{(\text{atoms}) \times (\text{atomic mass})}{\text{molar mass}} \times 100 \quad \text{Molecular formula} = n \times \text{empirical formula}, \quad n = \frac{M}{M_{\text{empirical}}}$$

Σ FORMULA Concentration terms

Term	Definition	Temperature dependent?
Mass % (w/w)	$\frac{m_{\text{solute}}}{m_{\text{solution}}} \times 100$	No
Molarity M	mol solute / L solution	Yes (volume changes)
Molality m	mol solute / kg solvent	No
Mole fraction x	$n_i / \sum n$	No
ppm	$\frac{m_{\text{solute}}}{m_{\text{solution}}} \times 10^6$	No

Conversions: $m = \frac{1000M}{1000d - MM_B}$ (d in g/mL, M_B = molar mass of solute). Dilution: $M_1V_1 = M_2V_2$. Mixing:

$$M_{\text{mix}} = \frac{M_1V_1 + M_2V_2}{V_1 + V_2}$$

Mole fraction of solute in a 1 molal aqueous solution: $\frac{1}{1 + 55.5} \approx 0.018$.

Numerical-solving method: stoichiometry in 4 steps

1. Write and **balance** the equation.
2. Convert every given quantity to **moles**.
3. Find the **limiting reagent**: divide moles by the stoichiometric coefficient; the smallest value is limiting.
4. Use mole ratios from the limiting reagent, then convert to the unit asked.

⚡ SHORTCUT Empirical formula in 30 seconds

Divide each mass % by its atomic mass, then divide by the smallest result. If you get x.5, double everything; x.33 or x.67, triple; x.25, quadruple.

Example: C 40, H 6.67, O 53.3 $\rightarrow$ 3.33 : 6.67 : 3.33 $\rightarrow$ 1 : 2 : 1, so CH_2O .

Common traps**⚠ MISTAKE ALERT Mistake alerts**

- Molarity uses the volume of **solution**; molality uses the mass of **solvent** (not solution).
- Counting atoms vs molecules: 1 mol of NH_3 contains 4 mol of atoms.
- Using 22.4 L for gases **not** at STP. Use $PV = nRT$ instead.
- Forgetting to check the limiting reagent when both reactant amounts are given.
- Mass % of an element: multiply by the **number of atoms** of that element in the formula.

Question patterns**▮ PYQ PATTERN Recurring structures**

1. Moles/atoms/molecules counting; mass of one molecule.
2. Empirical and molecular formula from % composition or combustion data.
3. Limiting reagent and yield.
4. Concentration interconversion (molarity $\leftrightarrow$ molality $\leftrightarrow$ mole fraction) with density.
5. Volume of gases in reactions (usually as a Section-B numerical).

Practice questions**✎ C01 · Practice****1** Easy 1 min Basic

C01-01

The total number of atoms in 4.25 g of NH_3 is about:

- (A) 1.5×10^{23} (B) 3.0×10^{23} (C) 2.4×10^{24} (D) 6.0×10^{23}

2 Easy 1 min Basic

C01-02

A compound contains C 40.0%, H 6.67% and O 53.3% by mass. Its empirical formula is:

- (A) CH_2O (B) CHO (C) $\text{C}_2\text{H}_4\text{O}$ (D) CH_2O_2

3 Medium 1.5 min NV

C01-03

3 g of H_2 reacts with 32 g of O_2 to form water. Find the mass of water formed (in g).

Your answer (numerical value):

4 Easy 0.5 min Speed

C01-04

5.85 g of NaCl is dissolved in water to make 500 mL of solution. The molarity is:

- (A) 0.1 M (B) 0.2 M (C) 0.01 M (D) 2 M

5

Medium

⌚ 1.5 min

Calc

C01-05

A 2 M NaOH solution has density 1.1 g/mL. Its molality is about:

- (A) 1.82 m (B) 2.00 m (C) 1.96 m (D) 2.20 m

6

Easy

⌚ 1 min

Basic

C01-06

The mole fraction of solute in a 1 molal aqueous solution is about:

- (A) 0.1 (B) 0.018 (C) 0.5 (D) 0.001

7

Easy

⌚ 0.5 min

Concept

C01-07

In CO and CO₂, the masses of oxygen that combine with a fixed mass of carbon are in the ratio:

- (A) 1 : 2 (B) 1 : 1 (C) 2 : 1 (D) 3 : 8

8

Easy

⌚ 0.5 min

Speed

C01-08

A compound with empirical formula CH₂O has molar mass 180 g/mol. Its molecular formula is:

- (A) C₆H₁₂O₆ (B) C₃H₆O₃ (C) C₁₂H₂₂O₁₁ (D) C₂H₄O₂

9

Easy

⌚ 1 min

Basic

C01-09

The volume of oxygen at STP (take 22.4 L/mol) needed to burn 1 mol of methane completely is:

- (A) 22.4 L (B) 44.8 L (C) 67.2 L (D) 11.2 L

C01 · Chapter Test

1

Easy

⌚ 0.5 min

Speed

C01-T1

The mass of one ¹²C atom is about:

- (A) 12 g (B) 1.66×10^{-24} g (C) 1.99×10^{-23} g (D) 6.02×10^{-23} g

2

Easy

⌚ 0.5 min

Speed

C01-T2

The percentage of nitrogen in urea, NH₂CONH₂, is about:

- (A) 23.3% (B) 46.7% (C) 28.0% (D) 60.0%

3

Easy

⌚ 0.5 min

NV

C01-T3

Find the number of moles of CO₂ formed by the complete combustion of 2 mol of propane (C₃H₈).

Your answer (numerical value):

4

Easy

⌚ 0.5 min

Speed

C01-T4

100 mL of 0.5 M HCl is diluted to 250 mL. The new molarity is:

- (A) 0.25 M (B) 0.5 M (C) 1.25 M (D) 0.2 M

5

Easy

⌚ 0.75 min

Concept

C01-T5

Which of the following contains the largest number of atoms?

- (A) 1 g He (B) 1 g H₂ (C) 1 g O₂ (D) 1 g N₂

Answers & Solutions

C01 · Practice

1 (D) 2 (A) 3 27 4 (B) 5 (C) 6 (B) 7 (A) 8 (A) 9 (B)

C01 · Chapter Test

1 (C) 2 (B) 3 6 4 (D) 5 (B)

C01 · PRACTICE

1 **Answer: (D)** *Counting atoms* Easy 1 min Basic C01-01

SOLUTION $n = 4.25/17 = 0.25$ mol of molecules; atoms $= 0.25 \times 4 \times N_A = N_A \approx 6.0 \times 10^{23}$.

SHORTCUT $0.25 \text{ mol} \times 4 \text{ atoms} = 1 \text{ mol of atoms}$.

COMMON TRAP Counting molecules only (1.5×10^{23}).

2 **Answer: (A)** *Empirical formula* Easy 1 min Basic C01-02

SOLUTION Moles: C $40/12 = 3.33$, H $6.67/1 = 6.67$, O $53.3/16 = 3.33$. Ratio 1 : 2 : 1.

3 **Answer: 27** *Limiting reagent* Medium 1.5 min NV C01-03

SOLUTION $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$. H_2 : 1.5 mol ($\div 2 = 0.75$); O_2 : 1 mol ($\div 1 = 1$). H_2 is limiting, so it gives 1.5 mol $\text{H}_2\text{O} = 27 \text{ g}$.

SHORTCUT Divide moles by coefficients; the smaller value is limiting.

COMMON TRAP Using O_2 as limiting gives 36 g.

4 **Answer: (B)** *Molarity* Easy 0.5 min Speed C01-04

SOLUTION $n = 5.85/58.5 = 0.1$ mol; $M = 0.1/0.5 = 0.2 \text{ M}$.

COMMON TRAP Dividing by 500 mL as if it were 1 L.

5 **Answer: (C)** *Molarity $\rightarrow$ molality* Medium 1.5 min Calc C01-05

SOLUTION Per litre: solution mass 1100 g, solute $2 \times 40 = 80 \text{ g}$, solvent 1020 g. $m = 2/1.020 \approx 1.96 \text{ m}$.

SHORTCUT $m = \frac{1000M}{1000d - MM_B} = \frac{2000}{1100 - 80}$.

COMMON TRAP Using the solution mass (1.1 kg) gives 1.82.

6 **Answer: (B)** *Mole fraction from molality* Easy 1 min Basic C01-06

SOLUTION 1 mol solute per 1000 g water (55.5 mol): $x = 1/56.5 \approx 0.0177$.

7 **Answer: (A)** *Law of multiple proportions* Easy 0.5 min Concept C01-07

SOLUTION 12 g C combines with 16 g O (CO) and 32 g O (CO_2), a ratio of 1 : 2.

COMMON TRAP Using mass percentages (3 : 8 is the C : O mass ratio in CO_2).

8 **Answer: (A)** *Molecular formula* Easy 0.5 min Speed C01-08

SOLUTION $n = 180/30 = 6$.

9 **Answer: (B)** *Gas volume in combustion* Easy 1 min Basic C01-09

SOLUTION $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$: 2 mol $\text{O}_2 = 44.8 \text{ L}$.

COMMON TRAP Not balancing the equation.

C01 · CHAPTER TEST

1 **Answer: (C)** *Mass of one atom* Easy 0.5 min Speed C01-T1

SOLUTION $12/(6.022 \times 10^{23}) \approx 1.99 \times 10^{-23} \text{ g}$.

COMMON TRAP Choosing (B), which is the value of 1 u.

2 **Answer: (B)** *Percentage composition* Easy 0.5 min Speed C01-T2

SOLUTION $28/60 \times 100 \approx 46.7\%$.

COMMON TRAP Counting only one N (23.3%).

3 Answer: 6 Combustion stoichiometry **Easy** ⌚ 0.5 min **NV**

C01-T3

SOLUTION $\text{C}_3\text{H}_8 + 5\text{O}_2 \longrightarrow 3\text{CO}_2 + 4\text{H}_2\text{O}$; $2 \times 3 = 6$ mol.

SHORTCUT Carbon balance: moles of CO_2 = moles of C atoms.

4 Answer: (D) Dilution **Easy** ⌚ 0.5 min **Speed**

C01-T4

SOLUTION $M_2 = 0.5 \times 100/250 = 0.2$ M.

5 Answer: (B) Maximum number of atoms **Easy** ⌚ 0.75 min **Concept**

C01-T5

SOLUTION Moles of atoms: H_2 : $0.5 \times 2 = 1$; He: 0.25; O_2 : $2/32 = 0.0625$; N_2 : $2/28 \approx 0.071$.

SHORTCUT Lowest mass per atom wins.

 REVISION Revision tracker · Basic Concepts

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III • CHEMISTRY • UNIT 02

Atomic Structure

02

1-2 per shift

TYPICAL QUESTIONS

Easy-Medium

DIFFICULTY

Must-do

PRIORITY

Class 11 • Ch 2

NCERT

~7 hours

STUDY TIME

Important concepts

- **Electromagnetic radiation:** $c = \nu\lambda$, wavenumber $\bar{\nu} = 1/\lambda$. Planck's quantum: $E = h\nu$. The photoelectric effect supports the particle nature of light.
- **Bohr model** (one-electron species): quantised orbits. It explains the hydrogen spectrum but fails for multi-electron atoms, the Zeeman/Stark effects and the uncertainty principle.
- **Dual nature:** $\lambda = h/mv$ (de Broglie). **Heisenberg:** $\Delta x \cdot \Delta p \geq h/4\pi$, so exact orbits are impossible. Hence orbitals.
- **Quantum-mechanical model:** ψ is a one-electron wave function, and ψ^2 is the probability density. Orbitals are described by four quantum numbers.
- **Filling rules:** Aufbau ($n + l$ rule), Pauli exclusion, Hund's rule of maximum multiplicity. Half-filled and fully filled subshells give extra stability.

Formula sheet

Σ FORMULA Spectra & Bohr model

$$E_n = -2.18 \times 10^{-18} \frac{Z^2}{n^2} \text{ J} = -13.6 \frac{Z^2}{n^2} \text{ eV} \quad r_n = 52.9 \frac{n^2}{Z} \text{ pm} \quad v_n = 2.18 \times 10^6 \frac{Z}{n} \text{ m s}^{-1}$$

$$\bar{\nu} = \frac{1}{\lambda} = R_H Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad R_H = 109\,677 \text{ cm}^{-1} \quad \text{lines} = \frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$$

Series: Lyman ($n_1 = 1$, UV), Balmer (2, visible), Paschen (3), Brackett (4), Pfund (5) (IR).

Σ FORMULA Quantum-mechanical results

$$\lambda = \frac{h}{p} \quad \Delta x \Delta p \geq \frac{h}{4\pi} \quad \text{orbital angular momentum} = \sqrt{l(l+1)} \frac{h}{2\pi} \quad \mu_{spin} = \sqrt{n(n+2)} \text{ BM}$$

Radial nodes = $n - l - 1$; angular nodes = l ; total nodes = $n - 1$.

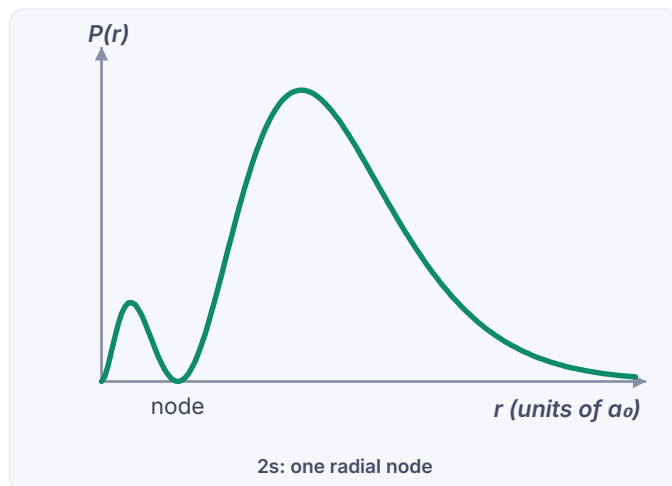
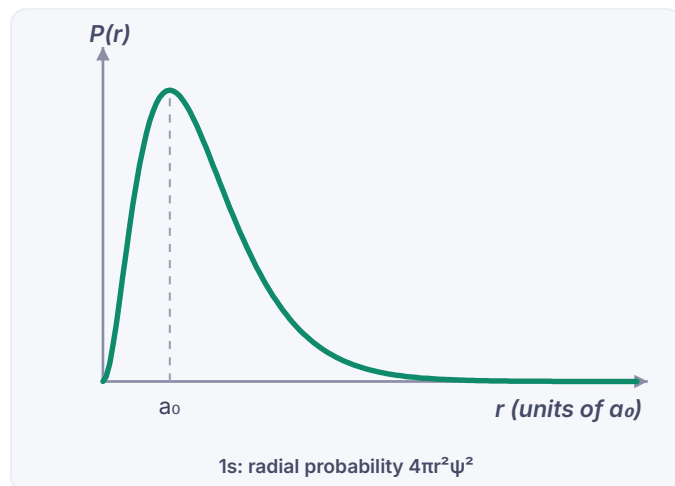
Maximum electrons: shell $2n^2$; subshell $2(2l + 1)$; orbital 2. Orbitals in a shell = n^2 .

Σ FORMULA Quantum numbers

Symbol	Name	Values	Determines
n	principal	1, 2, 3, ...	size, energy (for H)
l	azimuthal	0 ... ($n - 1$)	shape (s, p, d, f)
m_l	magnetic	$-l \dots +l$	orientation
m_s	spin	$\pm \frac{1}{2}$	spin

Electronic configuration: rules and exceptions

- **Order ($n + l$ rule):** 1s 2s 2p 3s 3p 4s 3d 4p 5s 4d 5p 6s 4f 5d 6p ... For equal $n + l$, the lower n fills first.
- **Anomalies to memorise:** Cr $[\text{Ar}]3d^5 4s^1$, Cu $[\text{Ar}]3d^{10} 4s^1$. Similar: Mo $4d^5 5s^1$, Ag $4d^{10} 5s^1$, Au $5d^{10} 6s^1$, Pd $4d^{10} 5s^0$.
- **Ions of d-block elements lose 4s electrons first:** Fe^{2+} is $[\text{Ar}]3d^6$ and Fe^{3+} is $[\text{Ar}]3d^5$.



☆ IMPORTANT ψ^2 plots (NCERT)

For 1s, ψ^2 is maximum at the nucleus and decays. For 2s, ψ^2 has **one radial node**. The radial probability $4\pi r^2 \psi^2$ for 1s peaks at $a_0 = 52.9$ pm. The d_{z^2} orbital has two lobes and a ring (torus). The other four d orbitals have four lobes each.

Common traps

⚠ MISTAKE ALERT Mistake alerts

- l can't equal n : "3f" and "2d" don't exist.
- Using the **atomic** configuration for ions: remove 4s electrons before 3d.
- Unpaired electrons in Cr: **6** ($3d^5 4s^1$), not 4.
- The Heisenberg product uses $\hbar/4\pi$ (NCERT). Some books use $\hbar/2$, which is the same value.
- Wavenumber in cm^{-1} vs m^{-1} : $R_H = 1.097 \times 10^7 \text{ m}^{-1}$.

Question patterns

📊 PYQ PATTERN Recurring structures

1. Valid/invalid quantum-number sets; number of electrons with given n, l, m_l, m_s .
2. Nodes of orbitals (radial/angular/total).
3. Bohr energies, radii, wavelengths for H, He^+ , Li^{2+} ; number of spectral lines.
4. de Broglie / Heisenberg numericals.
5. Configurations and magnetic moments of ions (feeds into d-block and coordination questions).

Practice questions

🔗 C02 · Practice

1 Easy 0.5 min Speed

C02-01

Which set of quantum numbers is **not** allowed?

- (A) $n = 3, l = 3, m_l = 1, m_s = -\frac{1}{2}$ (B) $n = 2, l = 1, m_l = 0, m_s = +\frac{1}{2}$
 (C) $n = 4, l = 2, m_l = -2, m_s = +\frac{1}{2}$ (D) $n = 1, l = 0, m_l = 0, m_s = -\frac{1}{2}$

2 Easy 0.5 min Speed

C02-02

The maximum number of electrons that can have $n = 3$ and $l = 2$ is:

- (A) 2 (B) 6 (C) 18 (D) 10

3 Medium 1 min Calc

C02-03

The wavelength emitted when the electron in a hydrogen atom falls from $n = 3$ to $n = 2$ is about:

- (A) 486 nm (B) 656 nm (C) 122 nm (D) 1875 nm

4 Medium 1.5 min Calc

C02-04

An electron's position is known to within $1 \text{ \AA}$. The minimum uncertainty in its velocity is about ($m_e = 9.1 \times 10^{-31} \text{ kg}$, $h = 6.626 \times 10^{-34} \text{ J s}$):

- (A) $1.2 \times 10^6 \text{ m/s}$ (B) $5.8 \times 10^3 \text{ m/s}$ (C) $5.8 \times 10^5 \text{ m/s}$ (D) $5.8 \times 10^7 \text{ m/s}$

5 Easy 0.5 min NV

C02-05

Find the total number of nodes in a 4d orbital.

Your answer (numerical value):

6 Easy 0.5 min Speed

C02-06

The ground-state electronic configuration of copper ($Z = 29$) is:

- (A) $[\text{Ar}]3d^9 4s^2$ (B) $[\text{Ar}]3d^8 4s^2 4p^1$ (C) $[\text{Ar}]3d^{10} 4s^2$ (D) $[\text{Ar}]3d^{10} 4s^1$

7 Easy 0.75 min Basic

C02-07

The orbital angular momentum of an electron in a d orbital is:

- (A) $\sqrt{2} \frac{h}{2\pi}$ (B) $2 \frac{h}{2\pi}$ (C) $\sqrt{6} \frac{h}{2\pi}$ (D) $\sqrt{12} \frac{h}{2\pi}$

8 Easy 0.75 min Basic

C02-08

The spin-only magnetic moment of Fe^{2+} is about:

- (A) 2.83 BM (B) 4.90 BM (C) 3.87 BM (D) 5.92 BM

9 Easy 0.5 min NV

C02-09

How many unpaired electrons are there in a gaseous Mn^{2+} ion ($Z = 25$)?

Your answer (numerical value):

C02 · Chapter Test

1 Easy 0.5 min Speed

C02-T1

Which orbital is filled first?

- (A) 4s (B) 3d
(C) 4p (D) they fill together

2 Easy 0.5 min Speed

C02-T2

The energy of an electron in the second Bohr orbit of hydrogen is:

- (A) -3.4 eV (B) -13.6 eV (C) -1.51 eV (D) -6.8 eV

3 Easy 0.5 min NV

C02-T3

Find the maximum number of electrons that can be accommodated in the shell with $n = 4$.Your answer (numerical value): **4** Easy 0.5 min Concept

C02-T4

Which orbital is spherically symmetric (non-directional)?

- (A) p_x (B) d_{z^2} (C) s (D) d_{xy}

5 Easy 0.5 min Concept

C02-T5

According to Hund's rule, the three 2p electrons of nitrogen are arranged as:

- (A) all three unpaired, with parallel spins (B) two paired, one unpaired
(C) all three in one orbital (D) all three unpaired, with alternating spins

Answers & Solutions

C02 · Practice

1 (A) 2 (D) 3 (B) 4 (C) 5 3 6 (D) 7 (C) 8 (B) 9 5

C02 · Chapter Test

1 (A) 2 (A) 3 32 4 (C) 5 (A)

C02 · PRACTICE

1 Answer: (A) Allowed quantum numbers Easy 0.5 min Speed

C02-01

SOLUTION l must be $\leq n - 1$. With $n = 3$, $l = 3$ is impossible.**2** Answer: (D) Capacity of a subshell Easy 0.5 min Speed

C02-02

SOLUTION $l = 2$ is a d subshell: $2(2l + 1) = 10$.

COMMON TRAP Answering 18 (the whole shell).

3 Answer: (B) Wavelength of H- α Medium 1 min Calc

C02-03

SOLUTION $\Delta E = 13.6 \left(\frac{1}{4} - \frac{1}{9} \right) = 13.6 \times \frac{5}{36} \approx 1.89 \text{ eV}$; $\lambda = 1240/1.89 \approx 656 \text{ nm}$.

SHORTCUT The first Balmer line is red, at 656 nm.

COMMON TRAP Using $n_1 = 1$ (Lyman).**4** Answer: (C) Uncertainty principle Medium 1.5 min Calc

C02-04

SOLUTION $\Delta v = \frac{h}{4\pi m \Delta x} = \frac{6.626 \times 10^{-34}}{4\pi \times 9.1 \times 10^{-31} \times 10^{-10}} \approx 5.8 \times 10^5 \text{ m/s}$.SHORTCUT $4\pi \times 9.1 \approx 114$.COMMON TRAP Using $h/2\pi$ gives about 1.2×10^6 .**5** Answer: 3 Total nodes Easy 0.5 min NV

C02-05

SOLUTION Total = $n - 1 = 3$ (radial $4 - 2 - 1 = 1$, angular 2).

COMMON TRAP Counting only radial nodes (1).

6 Answer: (D) Anomalous configuration Easy 0.5 min Speed

C02-06

SOLUTION A completely filled 3d subshell gives extra stability.

COMMON TRAP Applying Aufbau blindly (A).

7 Answer: (C) Orbital angular momentum Easy 0.75 min Basic

C02-07

SOLUTION $\sqrt{l(l+1)} \hbar$ with $l = 2$: $\sqrt{6} \hbar$.COMMON TRAP Using Bohr's $n\hbar/2\pi$.

8 Answer: (B) Spin-only magnetic moment **Easy** ⌚ 0.75 min **Basic**

C02-08

SOLUTION Fe^{2+} : $3d^6$ with 4 unpaired electrons (free ion). $\sqrt{4 \times 6} = \sqrt{24} \approx 4.90$ BM.

SHORTCUT Memorise $\sqrt{n(n+2)}$: 1.73, 2.83, 3.87, 4.90, 5.92 for $n = 1-5$.

COMMON TRAP Removing 3d before 4s electrons.

9 Answer: 5 Unpaired electrons in an ion **Easy** ⌚ 0.5 min **NV**

C02-09

SOLUTION Mn is $[\text{Ar}]3d^5 4s^2$, so Mn^{2+} is $3d^5$: 5 unpaired electrons.

C02 · CHAPTER TEST

1 Answer: (A) $(n + l)$ rule **Easy** ⌚ 0.5 min **Speed**

C02-T1

SOLUTION $n + l$: 4s = 4, 3d = 5, 4p = 5. The lowest value fills first.

2 Answer: (A) Energy of the second orbit **Easy** ⌚ 0.5 min **Speed**

C02-T2

SOLUTION $-13.6/4 = -3.4$ eV.

3 Answer: 32 Shell capacity **Easy** ⌚ 0.5 min **NV**

C02-T3

SOLUTION $2n^2 = 32$.

4 Answer: (C) Spherical orbital **Easy** ⌚ 0.5 min **Concept**

C02-T4

SOLUTION s orbitals have $l = 0$, so there are no angular nodes and no directional dependence.

5 Answer: (A) Hund's rule **Easy** ⌚ 0.5 min **Concept**

C02-T5

SOLUTION Degenerate orbitals are singly occupied, with parallel spins, before any pairing.

REVISION Revision tracker · Atomic Structure

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III · CHEMISTRY · UNIT 03

Chemical Bonding & Molecular Structure

03

1–2 per shift
TYPICAL QUESTIONSMedium
DIFFICULTYMust-do
PRIORITYClass 11 · Ch 4
NCERT~12 hours
STUDY TIME

Important concepts

- **Kossel–Lewis approach:** atoms attain noble-gas configurations by transferring electrons (ionic bonds) or sharing them (covalent bonds). **Octet exceptions:** incomplete (BeCl_2 , BF_3), expanded (PCl_5 , SF_6), odd-electron (NO , NO_2).
- **Formal charge** = $V - N_{\text{alone}} - \frac{1}{2}N_{\text{bonding}}$ picks the best Lewis structure.
- **Ionic bonding:** favoured by low IE, high electron-gain enthalpy and high **lattice enthalpy**. Lattice enthalpy rises with ionic charge and falls with ionic size.
- **Fajans' rules:** covalent character of an ionic bond rises with a **small, highly charged cation**, a **large anion**, and a cation with a pseudo-noble-gas (18-electron) configuration.
- **VSEPR:** electron pairs repel. Order of repulsion: $\text{lp-lp} > \text{lp-bp} > \text{bp-bp}$.
- **Valence bond theory & hybridisation** explain geometry. **MO theory** explains bond order, magnetism and species like O_2 (paramagnetic).
- **Hydrogen bonding** (with F, O, N) raises boiling points. Intramolecular H-bonding lowers them compared with the isomer that has intermolecular H-bonding.

Formula sheet

Σ FORMULA Useful counts

$$\text{Steric number (hybridisation)} = \frac{1}{2}(V + M - C + A) \quad \text{Bond order} = \frac{1}{2}(N_b - N_a) \quad \mu = q \times d$$

V = valence electrons of the central atom, M = monovalent atoms attached, C = cation charge, A = anion charge. $1 \text{ D} = 3.336 \times 10^{-30} \text{ C m}$.

$$\text{Percentage ionic character} \approx \frac{\mu_{\text{observed}}}{\mu_{\text{ionic}}} \times 100.$$

VSEPR shape table

Steric no.	Bond pairs + lone pairs	Shape	Examples
2 (sp)	2 + 0	linear	BeCl_2 , CO_2
3 (sp^2)	3 + 0	trigonal planar	BF_3 , NO_3^-
3	2 + 1	bent	SO_2 , O_3
4 (sp^3)	4 + 0	tetrahedral	CH_4 , NH_4^+
4	3 + 1	trigonal pyramidal	NH_3 , PCl_3
4	2 + 2	bent	H_2O
5 (sp^3d)	5 + 0	trigonal bipyramidal	PCl_5
5	4 + 1	see-saw	SF_4
5	3 + 2	T-shaped	ClF_3
5	2 + 3	linear	XeF_2 , I_3^-
6 (sp^3d^2)	6 + 0	octahedral	SF_6

Steric no.	Bond pairs + lone pairs	Shape	Examples
6	5 + 1	square pyramidal	BrF ₅ , XeOF ₄
6	4 + 2	square planar	XeF ₄ , ICl ₄ ⁻
7 (sp ³ d ³)	7 + 0	pentagonal bipyramidal	IF ₇
7	6 + 1	distorted octahedral	XeF ₆

In sp³d the **lone pairs go equatorial**. In sp³d², two lone pairs sit **trans** (opposite).

☆ IMPORTANT Bond angles & dipole moments to remember

- CH₄ 109.5° > NH₃ 107° > H₂O 104.5°. NH₃ (107°) > NF₃ (~102°). PH₃ ~93.5°, H₂S ~92° (little hybridisation).
- μ : NH₃ (1.47 D) $\gg$ NF₃ (0.23 D). In NH₃ the lone-pair moment adds to the bond moments; in NF₃ it opposes them.
- $\mu = 0$: CO₂, BF₃, CCl₄, PCl₅, SF₆, XeF₄, *trans*-dichloroethene, *p*-dichlorobenzene.
- CH₃Cl > CH₂Cl₂ > CHCl₃ > CCl₄ (= 0).

Molecular orbital theory

Σ FORMULA MO energy order

Up to N₂ (14 electrons or fewer): $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < (\pi 2p_x = \pi 2p_y) < \sigma 2p_z < (\pi^*) < \sigma^* 2p_z$

O₂ and beyond: $\sigma 2s < \sigma^* 2s < \sigma 2p_z < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$

Species	Bond order	Magnetic
H ₂	1	dia
He ₂	0 (doesn't exist)	—
B ₂	1	para (2 unpaired π)
C ₂	2	dia
N ₂	3	dia
N ₂ ⁺ , N ₂ ⁻	2.5	para
O ₂ ⁺	2.5	para
O ₂	2	para (2 unpaired π^*)
O ₂ ⁻ (superoxide)	1.5	para
O ₂ ²⁻ (peroxide)	1	dia
F ₂	1	dia

Higher bond order → shorter, stronger bond. Isoelectronic species (e.g. N₂, CO, CN⁻, NO⁺) have the same bond order (3).

Memory tricks

⚡ SHORTCUT Bond order by total electron count (2nd-period diatomics)

Total electrons	10	11	12	13	14	15	16	17	18	19	20
Bond order	1	1.5	2	2.5	3	2.5	2	1.5	1	0.5	0

The bond order rises by 0.5 per electron up to 14 (N₂, CO), then falls by 0.5 per electron. Examples: B₂ (10) = 1, O₂ (16) = 2, O₂⁻ (17) = 1.5, F₂ (18) = 1. **Time saved:** no MO diagram needed.

⚡ SHORTCUT Is it paramagnetic?

Odd total electrons → always paramagnetic. Even electrons → paramagnetic only for B_2 and O_2 in the common set.

Common traps**⚠ MISTAKE ALERT Mistake alerts**

- Lone pairs occupy positions but are **not** included in the "shape" name: NH_3 is pyramidal, not tetrahedral.
- Using the N_2 -type MO order for O_2 gives the wrong magnetism for B_2/C_2 if you mix them up.
- The most covalent halide of a metal is its **iodide**. The most ionic is its **fluoride**.
- Intramolecular H-bonding (o-nitrophenol) → **lower** boiling point, steam-volatile. Intermolecular (p-nitrophenol) → higher boiling point.
- Resonance structures are not real. The actual structure is the hybrid, with equal bond lengths (e.g. all C–O bonds in CO_3^{2-} are equal).

Question patterns**📊 PYQ PATTERN Recurring structures**

1. Shape/hybridisation of a list of species; "which pair has the same shape?"
2. Bond order/magnetism of O_2 , N_2 family ions; ordering bond lengths.
3. Dipole-moment ordering and zero-dipole identification.
4. Fajans' rules / lattice-enthalpy comparisons.
5. Number of lone pairs, σ and π bonds in a molecule.
6. Hydrogen bonding and its effect on properties.
7. Formal charge on atoms in O_3 , CO , NO_3^- .

Practice questions**🔗 C03 · Practice****1**

Easy

⌚ 0.5 min

Speed

C03-01

The shape of XeF_4 is:

- (A) tetrahedral (B) see-saw (C) square planar (D) octahedral

2

Easy

⌚ 0.75 min

Basic

C03-02

The bond order and magnetic nature of O_2^- are:

- (A) 2, paramagnetic (B) 1.5, paramagnetic
(C) 1.5, diamagnetic (D) 2.5, paramagnetic

3

Easy

⌚ 0.75 min

Basic

C03-03

The hybridisation and shape of SF_4 are:

- (A) sp^3 , tetrahedral (B) dsp^2 , square planar
(C) sp^3d^2 , square planar (D) sp^3d , see-saw

4 Easy 0.75 min Concept

C03-04

Which compound has the highest lattice enthalpy?

- (A) MgO (B) NaCl (C) KCl (D) CaO

5 Easy 0.75 min Concept

C03-05

Which compound has the most covalent character?

- (A) LiI (B) LiF (C) KI (D) NaCl

6 Medium 1 min Concept

C03-06

 NH_3 has a much larger dipole moment than NF_3 because:

- (A) N-F bonds are non-polar
 (B) in NH_3 the lone-pair moment reinforces the bond moments, while in NF_3 it opposes them
 (C) NF_3 is planar
 (D) F is less electronegative than H

7 Easy 0.5 min NV

C03-07

Find the number of lone pairs on the Xe atom in XeF_2 .Your answer (numerical value):

8 Easy 0.5 min Speed

C03-08

The correct order of bond angles is:

- (A) all equal (B) $\text{CH}_4 < \text{NH}_3 < \text{H}_2\text{O}$ (C) $\text{NH}_3 < \text{H}_2\text{O} < \text{CH}_4$ (D) $\text{H}_2\text{O} < \text{NH}_3 < \text{CH}_4$

9 Easy 0.5 min Speed

C03-09

Which molecule is paramagnetic?

- (A) N_2 (B) F_2 (C) O_2 (D) C_2

10 Medium 1 min Concept

C03-10

o-Nitrophenol is steam-volatile but p-nitrophenol is not, because:

- (A) o-nitrophenol has intermolecular H-bonding (B) o-nitrophenol has intramolecular H-bonding
 (C) p-nitrophenol is more acidic (D) p-nitrophenol has lower molar mass

11 Medium 1 min Concept

C03-11

In the Lewis structure of O_3 , the formal charge on the central oxygen is:

- (A) 0 (B) +2 (C) -1 (D) +1

C03 · Chapter Test

1 Easy 0.5 min Speed

C03-T1

The hybridisation of carbon in CO_2 is:

- (A) sp^2 (B) sp (C) sp^3 (D) sp^3d

2 Easy 0.75 min NV

C03-T2

Find the number of σ bonds in ethene (C_2H_4).Your answer (numerical value): **3** Easy 0.5 min Speed

C03-T3

Which species is isoelectronic with N_2 and has bond order 3?

- (A) O_2^+ (B) NO (C) CO (D) C_2^{2+}

4 Easy 0.5 min Speed

C03-T4

Which molecule has zero dipole moment?

- (A) CCl_4 (B) CH_2Cl_2 (C) $CHCl_3$ (D) CH_3Cl

5 Easy 0.5 min Speed

C03-T5

The shape of ClF_3 is:

- (A) trigonal planar (B) pyramidal (C) T-shaped (D) see-saw

Answers & Solutions

C03 · Practice

- 1 (C) 2 (B) 3 (D) 4 (A) 5 (A) 6 (B) 7 3 8 (D) 9 (C) 10 (B)
11 (D)

C03 · Chapter Test

- 1 (B) 2 5 3 (C) 4 (A) 5 (C)

C03 · PRACTICE

1 Answer: (C) Shape from VSEPR Easy 0.5 min Speed

C03-01

SOLUTION Xe has 8 valence electrons: 4 bond pairs + 2 lone pairs, steric number 6 (sp^3d^2). The lone pairs sit trans, leaving a square planar shape.

COMMON TRAP Naming the electron geometry (octahedral).

2 Answer: (B) Bond order of superoxide Easy 0.75 min Basic

C03-02

SOLUTION O_2^- has 17 electrons: one more π^* electron than O_2 . Bond order = $\frac{1}{2}(10 - 7) = 1.5$, with one unpaired electron.

SHORTCUT Odd electron count $\rightarrow$ paramagnetic.

3 Answer: (D) Hybridisation of SF_4 Easy 0.75 min Basic

C03-03

SOLUTION S: 6 valence electrons; 4 bonds + 1 lone pair, steric number 5, so sp^3d . The lone pair sits equatorial, giving a see-saw.

SHORTCUT $H = \frac{1}{2}(6 + 4) = 5$.

4 Answer: (A) Lattice enthalpy Easy 0.75 min Concept

C03-04

SOLUTION Lattice enthalpy $\propto \frac{q^+q^-}{r^+ + r^-}$. MgO and CaO have $2+/2-$ ions; Mg^{2+} is smaller than Ca^{2+} .

SHORTCUT Charge first, then size.

5 Answer: (A) Fajans' rules Easy 0.75 min Concept

C03-05

SOLUTION The small Li^+ polarises the large, easily polarisable I^- strongly.

SHORTCUT Small cation + large anion.

6 Answer: (B) Dipole of NH_3 vs NF_3 **Medium** ⌚ 1 min **Concept**

C03-06

SOLUTION In NH_3 the N-H bond moments point towards N, the same direction as the lone pair's moment. In NF_3 they point towards F, away from the lone pair.

7 Answer: 3 Lone pairs on Xe **Easy** ⌚ 0.5 min **NV**

C03-07

SOLUTION 8 valence electrons – 2 used in bonds = 6 electrons, i.e. 3 lone pairs (linear shape).

8 Answer: (D) Bond-angle order **Easy** ⌚ 0.5 min **Speed**

C03-08

SOLUTION More lone pairs → more compression: $104.5^\circ < 107^\circ < 109.5^\circ$.

9 Answer: (C) Paramagnetic molecule **Easy** ⌚ 0.5 min **Speed**

C03-09

SOLUTION O_2 has two unpaired electrons in degenerate π^* orbitals.

10 Answer: (B) Intramolecular hydrogen bonding **Medium** ⌚ 1 min **Concept**

C03-10

SOLUTION Intramolecular H-bonding in the ortho isomer prevents association between molecules, so it is more volatile. The para isomer associates through intermolecular H-bonds.

11 Answer: (D) Formal charge in ozone **Medium** ⌚ 1 min **Concept**

C03-11

SOLUTION Central O: $V = 6$, one lone pair (2 electrons), 3 bonds (6 bonding electrons). $\text{FC} = 6 - 2 - 3 = +1$.

C03 · CHAPTER TEST

1 Answer: (B) Hybridisation of C in CO_2 **Easy** ⌚ 0.5 min **Speed**

C03-T1

SOLUTION Two σ bonds, no lone pairs: linear, sp.

2 Answer: 5 Counting σ bonds **Easy** ⌚ 0.75 min **NV**

C03-T2

SOLUTION 4 C-H σ + 1 C-C σ = 5 (plus 1 π).

COMMON TRAP Counting the double bond as two σ bonds.

3 Answer: (C) Isoelectronic with N_2 **Easy** ⌚ 0.5 min **Speed**

C03-T3

SOLUTION CO has 14 electrons, the same as N_2 , so its bond order is 3.

4 Answer: (A) Zero dipole moment **Easy** ⌚ 0.5 min **Speed**

C03-T4

SOLUTION A symmetric tetrahedron with four identical substituents.

5 Answer: (C) Shape of ClF_3 **Easy** ⌚ 0.5 min **Speed**

C03-T5

SOLUTION 3 bond pairs + 2 lone pairs (sp^3d). Both lone pairs are equatorial, leaving a T shape.

REVISION Revision tracker · Chemical Bonding

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III · CHEMISTRY · UNIT 04

Chemical Thermodynamics

04

1-2 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

Must-do

PRIORITY

Class 11 · Ch 5

NCERT

~9 hours

STUDY TIME

Important concepts

- **System & surroundings:** open, closed and isolated systems. **Extensive** properties depend on amount (V, H, S, G, U). **Intensive** properties don't (T, P , density, molar quantities).
- **State functions:** U, H, S, G, P, V, T . Heat and work are path functions.
- **First law:** $\Delta U = q + w$. **Chemistry (IUPAC) sign convention:** w is work done **on** the system; $w = -P_{ext}\Delta V$.
- **Enthalpy** $H = U + PV$; $\Delta H = \Delta U + \Delta n_g RT$ for reactions involving gases.
- **Hess's law:** ΔH is path-independent, so thermochemical equations can be added and subtracted.
- **Second law:** a spontaneous process increases the entropy of the universe. At constant T, P : $\Delta G < 0$ means spontaneous.

⚠ WARNING Sign convention differs from Physics

In Chemistry (NCERT Class 11), $\Delta U = q + w$ with w = work **done on** the system. In Physics, $\Delta U = Q - W$ with W = work **done by** the system. The physics is the same; only the sign of w differs. Stay consistent within each subject.

Formula sheet

Σ FORMULA Work, heat, enthalpy

$$w_{irrev} = -P_{ext} \Delta V \quad w_{rev, isothermal} = -2.303 nRT \log \frac{V_2}{V_1} \quad q_V = \Delta U \quad q_P = \Delta H$$

$$\Delta H = \Delta U + \Delta n_g RT \quad C_p - C_v = R \quad q = nC\Delta T$$

Free expansion (into vacuum): $w = 0$. Isothermal ideal gas: $\Delta U = \Delta H = 0$.

Σ FORMULA Thermochemistry

$$\Delta_r H^\circ = \sum \Delta_f H^\circ_{products} - \sum \Delta_f H^\circ_{reactants} \quad \Delta_r H = \sum BE_{reactants} - \sum BE_{products}$$

$\Delta_f H^\circ$ of an element in its standard state = 0 (e.g. $O_2(g)$, $C(\text{graphite})$, $Br_2(l)$).

Enthalpies to know by definition: formation, combustion (always negative), atomisation, bond dissociation, sublimation ($= \Delta_{fus}H + \Delta_{vap}H$), phase transition, hydration, ionisation, solution.

Strong acid + strong base neutralisation: $\Delta H \approx -57.1$ kJ per mol of water. It is less exothermic for a weak acid or base, because some energy goes into ionisation.

Σ FORMULA Entropy & Gibbs energy

$$\Delta S = \frac{q_{rev}}{T} \quad \Delta S_{fusion/vap} = \frac{\Delta H_{trans}}{T_{trans}} \quad \Delta S_{total} = \Delta S_{sys} + \Delta S_{surr} > 0 \text{ (spontaneous)}$$

$$\Delta G = \Delta H - T\Delta S \quad \Delta G^\circ = -RT \ln K = -2.303RT \log K \quad \Delta G = \Delta G^\circ + RT \ln Q$$

Isothermal reversible expansion of an ideal gas: $\Delta S = nR \ln \frac{V_2}{V_1}$.

Spontaneity table

ΔH	ΔS	$\Delta G = \Delta H - T\Delta S$	Spontaneous?
–	+	always –	at all temperatures
+	–	always +	never
–	–	– at low T	below $T = \Delta H / \Delta S$
+	+	– at high T	above $T = \Delta H / \Delta S$

⚡ SHORTCUT Δn_g in 5 seconds

Δn_g = (moles of gaseous products) – (moles of gaseous reactants). **Ignore liquids and solids.** Example:
 $C(s) + O_2(g) \longrightarrow CO_2(g)$ has $\Delta n_g = 0$, so $\Delta H = \Delta U$.

Common traps**⚠ MISTAKE ALERT Mistake alerts**

- Mixing J and kJ in $\Delta G = \Delta H - T\Delta S$ (ΔS is usually given in J/K).
- Using log vs ln incorrectly in $\Delta G^\circ = -2.303RT \log K$.
- Bond-energy method: **reactants minus products** (the reverse of the formation-enthalpy method).
- $\Delta_f H^\circ$ of O_3 , diamond or $H_2O(g)$ is **not** zero.
- The entropy of a pure perfect crystal at 0 K is zero (third law). Entropy increases: solid < liquid < gas; dissolving a solid usually increases S .

Question patterns**▮ PYQ PATTERN Recurring structures**

1. ΔH vs ΔU with Δn_g (frequent Section-B numerical).
2. Hess's law manipulation; enthalpy of formation or combustion from given data.
3. Bond-enthalpy calculations.
4. Spontaneity temperature; sign analysis.
5. $\Delta G^\circ \leftrightarrow K$ conversions.
6. Work in isothermal reversible vs irreversible expansion.
7. Intensive/extensive and state/path classification.

Practice questions

C04 · Practice

1 Medium ⌚ 1.5 min NV C04-01

For $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \longrightarrow 2\text{NH}_3(\text{g})$ at 300 K, find $\Delta H - \Delta U$ in kJ/mol. Take $R = 8.3 \text{ J K}^{-1}\text{mol}^{-1}$ and give the answer to the nearest integer (with sign).

Your answer (numerical value):

2 Medium ⌚ 1.5 min JEE C04-02

Given: $\text{C}(\text{s}) + \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g})$, $\Delta H = -393.5 \text{ kJ}$; $\text{CO}(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g})$, $\Delta H = -283.0 \text{ kJ}$. The enthalpy of formation of $\text{CO}(\text{g})$ is:

- (A) -221.0 kJ/mol (B) -676.5 kJ/mol (C) $+110.5 \text{ kJ/mol}$ (D) -110.5 kJ/mol

3 Medium ⌚ 1.5 min Calc C04-03

Using bond enthalpies (kJ/mol) H-H 436, Cl-Cl 243, H-Cl 431, $\Delta_r H$ for $\text{H}_2 + \text{Cl}_2 \longrightarrow 2\text{HCl}$ is:

- (A) -248 kJ (B) $+183 \text{ kJ}$ (C) -183 kJ (D) -431 kJ

4 Medium ⌚ 1 min Concept C04-04

For a reaction, $\Delta H = +30 \text{ kJ/mol}$ and $\Delta S = +100 \text{ J K}^{-1}\text{mol}^{-1}$. It is spontaneous:

- (A) below 300 K (B) above 300 K
(C) at all temperatures (D) at no temperature

5 Medium ⌚ 1 min Calc C04-05

For a reaction at 300 K with $K = 10$, ΔG° is about ($R = 8.314$):

- (A) -5.74 kJ/mol (B) $+5.74 \text{ kJ/mol}$ (C) -2.49 kJ/mol (D) -57.4 kJ/mol

6 Easy ⌚ 0.75 min Basic C04-06

A gas expands against a constant external pressure of 2 bar from 1 L to 6 L. The work done on the gas is (1 L bar = 100 J):

- (A) $+1000 \text{ J}$ (B) -1000 J (C) -1200 J (D) -500 J

7 Easy ⌚ 0.5 min Speed C04-07

Which is an intensive property?

- (A) enthalpy (B) volume
(C) molar heat capacity (D) internal energy

8 Easy ⌚ 1 min NV C04-08

The enthalpy of vaporisation of a liquid is 30 kJ/mol at its boiling point of 300 K. Find $\Delta_{\text{vap}} S$ (in $\text{J K}^{-1}\text{mol}^{-1}$).

Your answer (numerical value):

9 Easy ⌚ 0.75 min Concept C04-09

For which species is $\Delta_f H^\circ = 0$?

- (A) $\text{O}_3(\text{g})$ (B) $\text{C}(\text{diamond})$ (C) $\text{H}_2\text{O}(\text{l})$ (D) $\text{Br}_2(\text{l})$

C04 · Chapter Test

1

Easy

⌚ 0.5 min

Speed

C04-T1

At constant T and P , a process is spontaneous when:

- (A) $\Delta G < 0$ (B) $\Delta G > 0$ (C) $\Delta H > 0$ (D) $\Delta S < 0$

2

Easy

⌚ 0.5 min

Concept

C04-T2

The enthalpy of neutralisation of HCl by NaOH is -57.1 kJ/mol. For CH_3COOH with NaOH it is:

- (A) exactly -57.1 kJ/mol (B) less negative than -57.1 kJ/mol
(C) more negative than -57.1 kJ/mol (D) zero

3

Easy

⌚ 0.5 min

Speed

C04-T3

For the free expansion of an ideal gas into a vacuum:

- (A) $w = 0, q = 0, \Delta U = 0$ (B) $w < 0, q > 0$
(C) $\Delta U > 0$ (D) $\Delta S = 0$

4

Medium

⌚ 1 min

NV

C04-T4

2 mol of an ideal gas expands isothermally and reversibly at 300 K from 1 L to 10 L. Find the magnitude of the work done (in kJ, nearest integer). ($R = 8.314$)

Your answer (numerical value):

5

Easy

⌚ 0.5 min

Speed

C04-T5

Which process involves a decrease in entropy?

- (A) melting of ice (B) evaporation of water
(C) dissolving NaCl in water (D) freezing of water

Answers & Solutions

C04 · Practice

1 -5 2 (D) 3 (C) 4 (B) 5 (A) 6 (B) 7 (C) 8 100 9 (D)

C04 · Chapter Test

1 (A) 2 (B) 3 (A) 4 11 5 (D)

C04 · PRACTICE

1

Answer: -5 $\Delta H - \Delta U$ relation

Medium

⌚ 1.5 min

NV

C04-01

SOLUTION $\Delta n_g = 2 - 4 = -2$. $\Delta H - \Delta U = \Delta n_g RT = -2 \times 8.3 \times 300 = -4980 \text{ J} \approx -5 \text{ kJ/mol}$.

COMMON TRAP Counting Δn_g as +2.

2

Answer: (D) Hess's law

Medium

⌚ 1.5 min

JEE

C04-02

SOLUTION Subtract the second equation from the first: $\text{C} + \frac{1}{2} \text{O}_2 \longrightarrow \text{CO}$, $\Delta H = -393.5 + 283.0 = -110.5 \text{ kJ/mol}$.

COMMON TRAP Adding the two equations.

3

Answer: (C) Bond-enthalpy calculation

Medium

⌚ 1.5 min

Calc

C04-03

SOLUTION $\Delta H = (436 + 243) - 2(431) = 679 - 862 = -183 \text{ kJ}$.

SHORTCUT Bonds broken - bonds formed.

COMMON TRAP Doing products - reactants (+183).

4 Answer: (B) Spontaneity temperature **Medium** ⌚ 1 min **Concept**

C04-04

SOLUTION Both positive, so entropy-driven: spontaneous when $T > \Delta H/\Delta S = 30\,000/100 = 300\text{ K}$.

COMMON TRAP Unit mismatch (0.3 K).

5 Answer: (A) ΔG° from K **Medium** ⌚ 1 min **Calc**

C04-05

SOLUTION $\Delta G^\circ = -2.303 \times 8.314 \times 300 \times \log 10 \approx -5744\text{ J/mol}$.

SHORTCUT $2.303RT \approx 5.74\text{ kJ}$ at 300 K: memorise it.

COMMON TRAP Using $\ln 10 = 1$.

6 Answer: (B) Irreversible work **Easy** ⌚ 0.75 min **Basic**

C04-06

SOLUTION $w = -P_{\text{ext}}\Delta V = -2 \times 5 = -10\text{ L bar} = -1000\text{ J}$.

SHORTCUT Expansion $\rightarrow w < 0$ in the chemistry convention.

COMMON TRAP Positive sign.

7 Answer: (C) Intensive properties **Easy** ⌚ 0.5 min **Speed**

C04-07

SOLUTION Molar (per-mole) quantities are intensive.

SHORTCUT Any extensive quantity $\div$ amount $\rightarrow$ intensive.

8 Answer: 100 Entropy of vaporisation **Easy** ⌚ 1 min **NV**

C04-08

SOLUTION $\Delta S = \Delta H/T = 30\,000/300 = 100\text{ J K}^{-1}\text{ mol}^{-1}$.

9 Answer: (D) Standard enthalpy of formation **Easy** ⌚ 0.75 min **Concept**

C04-09

SOLUTION Bromine's standard state is liquid, so $\text{Br}_2(\text{l})$ is an element in its standard state.

COMMON TRAP Diamond (graphite is carbon's standard state).

C04 · CHAPTER TEST

1 Answer: (A) Condition for spontaneity **Easy** ⌚ 0.5 min **Speed**

C04-T1

SOLUTION At constant T and P , the criterion for a spontaneous process is $\Delta G = \Delta H - T\Delta S < 0$. Neither ΔH nor ΔS_{sys} alone decides it.

2 Answer: (B) Heat of neutralisation **Easy** ⌚ 0.5 min **Concept**

C04-T2

SOLUTION Some heat is used to ionise the weak acid.

3 Answer: (A) Free expansion **Easy** ⌚ 0.5 min **Speed**

C04-T3

SOLUTION No external pressure $\rightarrow$ no work. Adiabatic/ideal $\rightarrow \Delta U = 0$. (But $\Delta S > 0$, since the process is spontaneous.)

COMMON TRAP Choosing (D).

4 Answer: 11 Isothermal reversible work **Medium** ⌚ 1 min **NV**

C04-T4

SOLUTION $|w| = 2.303nRT \log(V_2/V_1) = 2.303 \times 2 \times 8.314 \times 300 \times 1 \approx 11\,488\text{ J} \approx 11\text{ kJ}$.

5 Answer: (D) Entropy increase **Easy** ⌚ 0.5 min **Speed**

C04-T5

SOLUTION Liquid $\rightarrow$ solid is more ordered.

REVISION Revision tracker · Chemical Thermodynamics

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III · CHEMISTRY · UNIT 05

Solutions

05

~1 per shift

TYPICAL QUESTIONS

Easy-Medium

DIFFICULTY

High

PRIORITY

Class 12 · Ch 1

NCERT

~6 hours

STUDY TIME

Important concepts

- Concentration terms:** molarity, molality, mole fraction, mass %, volume %, ppm. For definitions and conversions see Unit 1 (page 165).
- Henry's law:** the solubility of a gas is proportional to its partial pressure: $p = K_H x$. A larger K_H means lower solubility. Solubility of gases falls as temperature rises.
- Raoult's law:** for volatile components, $p_i = p_i^\circ x_i$. For a non-volatile solute, the solution's vapour pressure is $p = p_1^\circ x_1$.
- Ideal solutions** obey Raoult's law exactly: $\Delta_{mix}H = 0$, $\Delta_{mix}V = 0$ (benzene-toluene, n-hexane-n-heptane).
- Colligative properties** depend on the **number** of solute particles, not their nature: RLVP, ΔT_b , ΔT_f , π .
- van't Hoff factor** i corrects for dissociation ($i > 1$) or association ($i < 1$).

Formula sheet

Σ FORMULA Colligative properties

$$\frac{p^\circ - p}{p^\circ} = x_B \quad \Delta T_b = i K_b m \quad \Delta T_f = i K_f m \quad \pi = i CRT$$

$$M_B = \frac{K_b w_B \times 1000}{\Delta T_b w_A} \text{ (same form for } K_f) \quad M_B = \frac{w_B R T}{\pi V}$$

Water: $K_b = 0.52 \text{ K kg mol}^{-1}$, $K_f = 1.86 \text{ K kg mol}^{-1}$. $R = 0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}$ (or $0.083 \text{ L bar K}^{-1} \text{ mol}^{-1}$).

Σ FORMULA van't Hoff factor

$$i = \frac{\text{observed colligative property}}{\text{calculated (normal) value}} = \frac{\text{normal molar mass}}{\text{observed molar mass}}$$

Dissociation into n ions with degree α : $i = 1 + (n - 1)\alpha$. Association of n molecules into one: $i = 1 - \alpha \left(1 - \frac{1}{n}\right)$.

Complete dissociation: NaCl 2, CaCl_2 3, K_2SO_4 3, $\text{K}_4[\text{Fe}(\text{CN})_6]$ 5, $\text{Al}_2(\text{SO}_4)_3$ 5. Benzoic/acetic acid fully dimerised in benzene: $i = 0.5$.

Σ FORMULA Vapour pressure of binary liquid mixtures

$$P_{\text{total}} = p_A^\circ x_A + p_B^\circ x_B \quad y_A = \frac{p_A^\circ x_A}{P_{\text{total}}} \text{ (vapour-phase mole fraction)}$$

The vapour is always richer in the **more volatile** component.

Deviations from Raoult's law

	Positive deviation	Negative deviation
A-B interactions	weaker than A-A, B-B	stronger than A-A, B-B
$\Delta_{mix}H$, $\Delta_{mix}V$	> 0 , > 0	< 0 , < 0

	Positive deviation	Negative deviation
Vapour pressure	higher than ideal	lower than ideal
Azeotrope	minimum-boiling (e.g. ethanol–water, ~95% ethanol)	maximum-boiling (e.g. HNO_3 –water, ~68% acid)
Examples	ethanol + acetone, CS_2 + acetone, ethanol + water	chloroform + acetone (H-bond), HNO_3 + water, phenol + aniline



Solid line: total vapour pressure. Dashed: ideal (Raoult) behaviour for comparison.

Numerical-solving method

1. Identify the colligative property and write its formula **with** i .
2. Molality needs moles of solute $\div$ **kg of solvent**. Convert grams to kg.
3. For electrolytes, decide on complete dissociation vs a given α .
4. For osmotic pressure use C in mol/L and $R = 0.0821$ L atm (answer in atm).

⚡ SHORTCUT Comparing colligative effects

For equal molal concentrations, compare $i \times m$ only. Largest $im \rightarrow$ largest ΔT_b , ΔT_f , π and lowest vapour pressure/freezing point.

Common traps

⚠ MISTAKE ALERT Mistake alerts

- Forgetting i for electrolytes.
- **Higher $K_H \rightarrow$ lower solubility.** It's easy to reverse this.
- Using the mass of the **solution** in molality.
- Osmotic pressure units: with $R = 0.0821$ the answer is in atm. With $R = 8.314$ you need SI units throughout.
- Freezing point depression is **positive** ΔT_f ; the new freezing point is $T_f^\circ - \Delta T_f$.

Question patterns

📊 PYQ PATTERN Recurring structures

1. Molar mass from ΔT_b , ΔT_f , π (common Section-B numerical).
2. i and degree of dissociation/association.
3. Ordering boiling/freezing points of several solutions.
4. Raoult's law with two volatile liquids; vapour composition.
5. Positive/negative deviation identification; azeotropes.
6. Henry's law applications (scuba diving, soft drinks).

Practice questions

C05 · Practice

1 Easy ⌚ 0.75 min Basic C05-01

Pure water has a vapour pressure of 100 mm Hg at some temperature. A non-volatile solute is added so that its mole fraction is 0.02. The solution's vapour pressure is:

- (A) 102 mm (B) 80 mm (C) 2 mm (D) 98 mm

2 Easy ⌚ 1 min Basic C05-02

The freezing-point depression of a 0.1 molal aqueous NaCl solution, assuming complete dissociation ($K_f = 1.86$), is:

- (A) 0.186 K (B) 0.372 K (C) 0.558 K (D) 0.093 K

3 Easy ⌚ 1 min Calc C05-03

The osmotic pressure of 0.1 M glucose at 300 K is about ($R = 0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}$):

- (A) 2.46 atm (B) 24.6 atm (C) 0.246 atm (D) 4.92 atm

4 Medium ⌚ 1.5 min NV C05-04

1.8 g of a non-volatile non-electrolyte dissolved in 100 g of water raises its boiling point by 0.052 K ($K_b = 0.52$). Find the molar mass of the solute (in g/mol).

Your answer (numerical value):

5 Easy ⌚ 0.5 min NV C05-05

Find the van't Hoff factor of $\text{K}_4[\text{Fe}(\text{CN})_6]$ in dilute solution, assuming complete dissociation.

Your answer (numerical value):

6 Easy ⌚ 0.75 min Concept C05-06

Acetic acid in benzene is completely dimerised. Its van't Hoff factor is:

- (A) 2 (B) 1 (C) 0.5 (D) 0.25

7 Easy ⌚ 0.5 min Concept C05-07

Which mixture shows a **positive** deviation from Raoult's law?

- (A) chloroform + acetone (B) nitric acid + water
(C) ethanol + acetone (D) benzene + toluene

8 Easy ⌚ 0.5 min Concept C05-08

At the same temperature, gas X has a larger Henry's-law constant K_H than gas Y in water. Therefore:

- (A) X is more soluble than Y (B) solubility doesn't depend on K_H
(C) both are equally soluble (D) X is less soluble than Y

9 Easy ⌚ 1 min Basic C05-09

Which NaCl concentration (complete dissociation) is isotonic with 0.1 M glucose?

- (A) 0.1 M (B) 0.05 M (C) 0.2 M (D) 0.025 M

C05 · Chapter Test

1 Easy 0.5 min Speed

C05-T1

Colligative properties depend on:

- (A) the number of solute particles (B) the nature of the solute
(C) the molar mass of the solvent only (D) the chemical reactivity of the solute

2 Easy 0.75 min Concept

C05-T2

Among 0.1 m aqueous solutions of glucose, NaCl, CaCl₂ and Al₂(SO₄)₃ (complete dissociation), the lowest freezing point belongs to:

- (A) glucose (B) NaCl (C) CaCl₂ (D) Al₂(SO₄)₃

3 Easy 0.75 min NV

C05-T3

The elevation of boiling point of 0.5 m aqueous urea ($K_b = 0.52$) is $x \times 0.01$ K. Find x .

Your answer (numerical value):

4 Easy 0.5 min Concept

C05-T4

Reverse osmosis requires:

- (A) heating the solution
(B) a pressure greater than the osmotic pressure applied on the solution side
(C) a pressure smaller than the osmotic pressure
(D) no semipermeable membrane

5 Easy 0.5 min Concept

C05-T5

A solution showing large positive deviation from Raoult's law forms:

- (A) a minimum-boiling azeotrope (B) a maximum-boiling azeotrope
(C) no azeotrope (D) an ideal solution

Answers & Solutions

C05 · Practice

1 (D) 2 (B) 3 (A) 4 180 5 5 6 (C) 7 (C) 8 (D) 9 (B)

C05 · Chapter Test

1 (A) 2 (D) 3 26 4 (B) 5 (A)

C05 · PRACTICE

1 Answer: (D) Relative lowering of vapour pressure Easy 0.75 min Basic

C05-01

SOLUTION $p = p^\circ x_{\text{solvent}} = 100 \times 0.98 = 98$ mm Hg.

COMMON TRAP Answering the lowering (2 mm).

2 Answer: (B) Freezing-point depression Easy 1 min Basic

C05-02

SOLUTION $\Delta T_f = iK_f m = 2 \times 1.86 \times 0.1 = 0.372$ K.

COMMON TRAP Forgetting $i = 2$.

3 Answer: (A) Osmotic pressure Easy 1 min Calc

C05-03

SOLUTION $\pi = CRT = 0.1 \times 0.0821 \times 300 \approx 2.46$ atm.

4 Answer: 180 Molar mass from elevation of boiling point **Medium** ⌚ 1.5 min **NV** C05-04

SOLUTION $m = \Delta T_b / K_b = 0.1 \text{ mol/kg}$, so moles of solute = $0.1 \times 0.1 = 0.01$. $M = 1.8 / 0.01 = 180 \text{ g/mol}$.

SHORTCUT $M = \frac{K_b w_B \times 1000}{\Delta T_b w_A} = \frac{0.52 \times 1.8 \times 1000}{0.052 \times 100}$.

COMMON TRAP Using 100 g as 1 kg.

5 Answer: 5 van't Hoff factor of a complex salt **Easy** ⌚ 0.5 min **NV** C05-05

SOLUTION $\text{K}_4[\text{Fe}(\text{CN})_6] \longrightarrow 4\text{K}^+ + [\text{Fe}(\text{CN})_6]^{4-}$: 5 ions. The complex ion doesn't break up.

COMMON TRAP Counting the CN^- ligands (11).

6 Answer: (C) Association in benzene **Easy** ⌚ 0.75 min **Concept** C05-06

SOLUTION Two molecules form one particle, so $i = 1/2$. The observed molar mass is double.

7 Answer: (C) Positive deviation example **Easy** ⌚ 0.5 min **Concept** C05-07

SOLUTION Acetone breaks up ethanol's hydrogen bonding and forms weaker A-B interactions, so the vapour pressure is higher than ideal.

SHORTCUT Chloroform-acetone forms new H-bonds, which gives a negative deviation.

8 Answer: (D) Henry's constant **Easy** ⌚ 0.5 min **Concept** C05-08

SOLUTION $x = p/K_H$: at a given p , a larger K_H means a smaller mole fraction dissolved.

9 Answer: (B) Isotonic solutions **Easy** ⌚ 1 min **Basic** C05-09

SOLUTION Isotonic means equal iC : $1 \times 0.1 = 2 \times C \Rightarrow C = 0.05 \text{ M}$.

C05 · CHAPTER TEST

1 Answer: (A) What colligative properties depend on **Easy** ⌚ 0.5 min **Speed** C05-T1

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

2 Answer: (D) Lowest freezing point **Easy** ⌚ 0.75 min **Concept** C05-T2

SOLUTION $i = 1, 2, 3, 5$. The largest i gives the largest depression.

3 Answer: 26 Boiling-point elevation **Easy** ⌚ 0.75 min **NV** C05-T3

SOLUTION $\Delta T_b = 0.52 \times 0.5 = 0.26 \text{ K}$, so $x = 26$.

4 Answer: (B) Reverse osmosis **Easy** ⌚ 0.5 min **Concept** C05-T4

SOLUTION This forces the solvent out of the solution through the membrane, as in desalination.

5 Answer: (A) Azeotrope type **Easy** ⌚ 0.5 min **Concept** C05-T5

SOLUTION Higher vapour pressure $\rightarrow$ lower boiling point at the azeotropic composition.

REVISION Revision tracker · Solutions

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III · CHEMISTRY · UNIT 06

Equilibrium: Chemical & Ionic

06

1-2 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

Must-do

PRIORITY

Class 11 · Ch 6

NCERT

~12 hours

STUDY TIME

Important concepts

- **Dynamic equilibrium:** forward and backward rates are equal, and concentrations stay constant. It can be reached from either side.
- **Law of mass action:** for $aA + bB \rightleftharpoons cC + dD$, $K_c = \frac{[C]^c[D]^d}{[A]^a[B]^b}$. Pure solids and liquids are omitted.
- **Reaction quotient Q** has the same form at any moment. $Q < K \rightarrow$ forward; $Q > K \rightarrow$ backward; $Q = K \rightarrow$ at equilibrium.
- **Le Chatelier's principle:** a system at equilibrium shifts to partly counteract any imposed change. Only **temperature** changes the value of K .
- **Ionic equilibria:** acids/bases (Arrhenius, Brønsted–Lowry, Lewis), K_w , pH, weak electrolytes, common-ion effect, salt hydrolysis, buffers, solubility product.

Formula sheet: Chemical equilibrium

Σ FORMULA K relations

$$K_p = K_c(RT)^{\Delta n_g} \quad \Delta G^\circ = -RT \ln K \quad \log \frac{K_2}{K_1} = \frac{\Delta H^\circ}{2.303R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

Operation on the equation	New K
reverse	$1/K$
multiply by n	K^n
add two equations	$K_1 K_2$

Degree of dissociation from vapour density ($A \rightarrow nB$): $\alpha = \frac{D-d}{(n-1)d}$, where D = initial and d = equilibrium vapour density.

Σ FORMULA Le Chatelier summary

Change	Effect
add reactant / remove product	shifts forward (K unchanged)
increase pressure (decrease V)	shifts towards fewer gas moles
increase temperature	shifts in the endothermic direction; K rises for endothermic reactions
catalyst	no shift; equilibrium reached faster
inert gas at constant volume	no effect
inert gas at constant pressure	shifts towards more gas moles

Formula sheet: Ionic equilibrium

Σ FORMULA pH and weak electrolytes

$$K_w = [\text{H}^+][\text{OH}^-] = 10^{-14} \text{ (298 K)} \quad \text{pH} = -\log[\text{H}^+] \quad \text{pH} + \text{pOH} = 14$$

$$\text{Weak acid: } \alpha = \sqrt{\frac{K_a}{C}}, [\text{H}^+] = \sqrt{K_a C}, \text{pH} = \frac{1}{2}(\text{p}K_a - \log C) \quad K_a K_b = K_w \text{ (conjugate pair)}$$

Ostwald's dilution law: α increases on dilution. K_w increases with temperature (neutral pH < 7 above 25 °C).
Very dilute strong acid (< 10^{-6} M): include water's H^+ . 10^{-8} M HCl has pH ≈ 6.98 , not 8.

Σ FORMULA Buffers & hydrolysis

$$\text{Acidic buffer: } \text{pH} = \text{p}K_a + \log \frac{[\text{salt}]}{[\text{acid}]} \quad \text{Basic buffer: } \text{pOH} = \text{p}K_b + \log \frac{[\text{salt}]}{[\text{base}]}$$

Salt of	Nature	pH
strong acid + strong base (NaCl)	neutral	7
weak acid + strong base (CH_3COONa)	basic	$7 + \frac{1}{2}\text{p}K_a + \frac{1}{2}\log C$
strong acid + weak base (NH_4Cl)	acidic	$7 - \frac{1}{2}\text{p}K_b - \frac{1}{2}\log C$
weak acid + weak base ($\text{CH}_3\text{COONH}_4$)	depends	$7 + \frac{1}{2}\text{p}K_a - \frac{1}{2}\text{p}K_b$
Maximum buffer capacity at $[\text{salt}] = [\text{acid}]$, i.e. $\text{pH} = \text{p}K_a$.		

Σ FORMULA Solubility product



Type	K_{sp}	Example
AB	s^2	AgCl, BaSO_4
AB_2 / A_2B	$4s^3$	CaF_2 , Ag_2CrO_4
AB_3	$27s^4$	$\text{Fe}(\text{OH})_3$
A_2B_3	$108s^5$	Bi_2S_3

Precipitation occurs when the ionic product > K_{sp} . **Common ion** lowers solubility: AgCl in c M NaCl has $s \approx K_{sp}/c$.

Numerical-solving method (ICE table)

1. Write Initial, Change ($-x$, $+x$ with stoichiometric coefficients) and Equilibrium rows.
2. Substitute into K . If K is small relative to C , approximate $C - x \approx C$.
3. Check: if $x/C > 5\%$, solve the quadratic.

⚡ SHORTCUT pH of a buffer in your head

If $[\text{salt}] = [\text{acid}]$, $\text{pH} = \text{p}K_a$. Every $10\times$ increase in $[\text{salt}]/[\text{acid}]$ adds 1 pH unit.
Acetic acid $\text{p}K_a = 4.74$; NH_3 $\text{p}K_b = 4.74$ (NH_4^+ $\text{p}K_a = 9.26$).

Common traps

MISTAKE ALERT Mistake alerts

- Including solids/liquids in K expressions.
- Δn_g for K_p/K_c : gaseous moles **products – reactants**.
- Catalysts don't change K or the equilibrium composition.
- Diprotic bases: $\text{Ba}(\text{OH})_2$ gives $[\text{OH}^-] = 2C$.
- K_{sp} of Ag_2CrO_4 is $4s^3$ (not s^2): $[\text{Ag}^+] = 2s$.
- Applying the Henderson equation to a solution with no conjugate pair.

Question patterns

PYQ PATTERN Recurring structures

1. K_p – K_c relation; K for combined/reversed reactions.
2. Le Chatelier predictions (pressure, temperature, inert gas).
3. pH of weak acids/bases, very dilute strong acids, mixtures of acid and base (after neutralisation).
4. Buffer pH (Henderson); buffer made by partial neutralisation.
5. $K_{sp} \leftrightarrow$ solubility; common-ion effect; precipitation condition.
6. Salt hydrolysis pH.
7. Degree of dissociation from vapour density or pressure data.

Practice questions

C06 · Practice

1 Easy ⌚ 0.75 min Basic C06-01

For $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$:

- (A) $K_p = K_c$ (B) $K_p = K_c(RT)^{-2}$ (C) $K_p = K_c(RT)^2$ (D) $K_p = K_c(RT)^{-1}$

2 Easy ⌚ 0.5 min Concept C06-02

At some instant $Q < K$ for a reaction. The reaction will:

- (A) proceed backward (B) be at equilibrium
(C) proceed forward (D) stop

3 Easy ⌚ 0.5 min Speed C06-03

For $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$, increasing the total pressure (by compression) at constant temperature:

- (A) shifts it towards N_2O_4 (B) shifts the equilibrium towards NO_2
(C) has no effect (D) changes K_p

4 Medium ⌚ 1 min Calc C06-04

The pH of 0.1 M acetic acid ($K_a = 1.8 \times 10^{-5}$) is about:

- (A) 3.87 (B) 1.00 (C) 4.74 (D) 2.87

5 Medium ⌚ 1 min Basic C06-05

A buffer contains 0.2 M CH_3COONa and 0.02 M CH_3COOH ($\text{p}K_a = 4.74$). Its pH is:

- (A) 3.74 (B) 5.74 (C) 4.74 (D) 6.74

6 Medium 1 min Calc

C06-06

The solubility of CaF_2 is 2×10^{-4} mol/L. Its K_{sp} is:

- (A) 4×10^{-8} (B) 3.2×10^{-11} (C) 8×10^{-12} (D) 1.6×10^{-11}

7 Medium 1 min Concept

C06-07

The solubility of AgCl ($K_{sp} = 1.8 \times 10^{-10}$) in 0.1 M NaCl is about:

- (A) 1.34×10^{-5} M (B) 0.1 M (C) 1.8×10^{-11} M (D) 1.8×10^{-9} M

8 Hard 2 min NV

C06-08

Find the pH of 10^{-8} M HCl at 25°C , to the nearest integer.

Your answer (numerical value):

9 Medium 1.5 min JEE

C06-09

The pH of 0.1 M CH_3COONa ($\text{p}K_a$ of acetic acid = 4.74) is about:

- (A) 7.00 (B) 9.37 (C) 8.87 (D) 5.13

10 Easy 0.5 min Speed

C06-10

The conjugate base of HSO_4^- is:

- (A) SO_4^{2-} (B) H_2SO_4 (C) HSO_3^- (D) SO_3^{2-}

11 Easy 0.75 min Basic

C06-11

For $\text{A} \rightleftharpoons \text{B}$, $K = 4$. For $2\text{B} \rightleftharpoons 2\text{A}$, K is:

- (A) $1/8$ (B) 16 (C) $1/16$ (D) -8

C06 · Chapter Test

1 Easy 0.5 min Speed

C06-T1

A catalyst added to a reaction at equilibrium:

- (A) doesn't change the equilibrium composition, but equilibrium is reached faster
(B) shifts the equilibrium forward
(C) increases K
(D) decreases K

2 Easy 0.5 min Speed

C06-T2

Which is a Lewis acid?

- (A) NH_3 (B) Cl^- (C) H_2O (D) BF_3

3 Easy 1 min NV

C06-T3

Find the pH of 0.001 M $\text{Ba}(\text{OH})_2$ (complete dissociation) to the nearest integer.

Your answer (numerical value):

4 Easy 0.5 min Speed

C06-T4

For Ag_2CrO_4 with molar solubility s , K_{sp} is:

- (A) s^2 (B) $2s^3$ (C) $27s^4$ (D) $4s^3$

5 Easy 0.5 min Concept

C06-T5

On diluting a weak acid, its degree of dissociation:

- (A) decreases (B) increases (C) stays constant (D) becomes zero

Answers & Solutions

C06 · Practice

- 1 (B) 2 (C) 3 (A) 4 (D) 5 (B) 6 (B) 7 (D) 8 7 9 (C) 10 (A)
11 (C)

C06 · Chapter Test

- 1 (A) 2 (D) 3 11 4 (D) 5 (B)

C06 · PRACTICE

1 Answer: (B) K_p and K_c Easy 0.75 min Basic

C06-01

SOLUTION $\Delta n_g = 2 - 4 = -2$.COMMON TRAP Reversing the sign of Δn_g .

2 Answer: (C) Reaction quotient Easy 0.5 min Concept

C06-02

SOLUTION Too few products relative to equilibrium, so the net reaction goes forward until $Q = K$.

3 Answer: (A) Effect of pressure Easy 0.5 min Speed

C06-03

SOLUTION The shift is towards fewer gas moles: 1 mol N_2O_4 vs 2 mol NO_2 . K_p is unchanged.

4 Answer: (D) pH of a weak acid Medium 1 min Calc

C06-04

SOLUTION $[\text{H}^+] = \sqrt{1.8 \times 10^{-6}} \approx 1.34 \times 10^{-3}$; $\text{pH} \approx 2.87$.SHORTCUT $\text{pH} = \frac{1}{2}(4.74 + 1) = 2.87$.

COMMON TRAP Treating it as a strong acid (pH 1).

5 Answer: (B) Buffer pH Medium 1 min Basic

C06-05

SOLUTION $\text{pH} = 4.74 + \log(0.2/0.02) = 4.74 + 1 = 5.74$.

COMMON TRAP Inverting the ratio.

6 Answer: (B) K_{sp} from solubility Medium 1 min Calc

C06-06

SOLUTION $K_{sp} = (s)(2s)^2 = 4s^3 = 4 \times 8 \times 10^{-12} = 3.2 \times 10^{-11}$.COMMON TRAP Using s^2 .

7 Answer: (D) Common-ion effect Medium 1 min Concept

C06-07

SOLUTION $s(0.1 + s) = 1.8 \times 10^{-10} \Rightarrow s \approx 1.8 \times 10^{-9} \text{ M}$.SHORTCUT $s \approx K_{sp}/[\text{common ion}]$.

COMMON TRAP Answering the solubility in pure water (A).

8 Answer: 7 pH of very dilute HCl Hard 2 min NV

C06-08

SOLUTION $[\text{H}^+] = 10^{-8} + x$ with $x(10^{-8} + x) = 10^{-14}$. So $x \approx 9.5 \times 10^{-8}$ and $[\text{H}^+] \approx 1.05 \times 10^{-7}$, giving $\text{pH} \approx 6.98 \rightarrow 7$.SHORTCUT An acid solution can't have $\text{pH} > 7$. At this dilution, water's ions dominate.

COMMON TRAP Answering 8.

9 Answer: (C) *Hydrolysis of sodium acetate* **Medium** ⌚ 1.5 min JEE

C06-09

SOLUTION $\text{pH} = 7 + \frac{1}{2}(4.74) + \frac{1}{2}\log 0.1 = 7 + 2.37 - 0.5 = 8.87$.

SHORTCUT Salt of a weak acid and strong base is basic.

COMMON TRAP Sign of the $\log C$ term.

10 Answer: (A) *Conjugate base* **Easy** ⌚ 0.5 min Speed

C06-10

SOLUTION Remove one H^+ .

COMMON TRAP Giving the conjugate acid.

11 Answer: (C) *K for reversed and doubled reaction* **Easy** ⌚ 0.75 min Basic

C06-11

SOLUTION Reverse: $1/4$. Double: $(1/4)^2 = 1/16$.

C06 · CHAPTER TEST

1 Answer: (A) *Role of a catalyst* **Easy** ⌚ 0.5 min Speed

C06-T1

SOLUTION It speeds up both directions equally.

2 Answer: (D) *Lewis acid* **Easy** ⌚ 0.5 min Speed

C06-T2

SOLUTION B has an incomplete octet and accepts an electron pair.

3 Answer: 11 *pH of a diacidic base* **Easy** ⌚ 1 min NV

C06-T3

SOLUTION $[\text{OH}^-] = 0.002 \text{ M}$, so $\text{pOH} = 2.70$ and $\text{pH} = 11.30 \approx 11$.

COMMON TRAP Forgetting the factor 2 (still pH 11, but the exact value is 11.0 rather than 11.3).

4 Answer: (D) *Ksp expression* **Easy** ⌚ 0.5 min Speed

C06-T4

SOLUTION $[\text{Ag}^+] = 2s$, $[\text{CrO}_4^{2-}] = s$.

5 Answer: (B) *Ostwald's dilution law* **Easy** ⌚ 0.5 min Concept

C06-T5

SOLUTION $\alpha = \sqrt{K_a/C}$ rises as C falls.

REVISION Revision tracker · Equilibrium

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III · CHEMISTRY · UNIT 07

Redox Reactions & Electrochemistry

07

1–2 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

Must-do

PRIORITY

Class 11 · Ch 7;

Class 12 · Ch 2

NCERT

~9 hours

STUDY TIME

Important concepts

- **Oxidation** = loss of electrons, or increase in oxidation number. **Reduction** = gain of electrons, or decrease in oxidation number. The oxidising agent is itself reduced.
- **Types of redox reactions:** combination, decomposition, displacement, and **disproportionation** (the same element is both oxidised and reduced).
- **Galvanic cell:** a spontaneous redox reaction produces electricity. **Anode = oxidation (–), cathode = reduction (+)**. In electrolytic cells the signs reverse, but the anode is still where oxidation happens.
- **Nernst equation** relates cell emf to concentrations. $\Delta G = -nFE$ links electrochemistry and thermodynamics.
- **Conductance:** conductivity κ falls on dilution, while molar conductivity Λ_m rises. **Kohlrausch's law** gives Λ_m° of weak electrolytes.

Formula sheet

Σ FORMULA Oxidation-number rules (in order of priority)

F is always –1. O is –2, except in peroxides (–1), superoxides (–½), OF₂ (+2). H is +1 (–1 in metal hydrides). Alkali metals +1, alkaline-earth metals +2. The sum equals the charge on the species. Watch for **peroxide linkages**: S is +6 in both H₂SO₅ and H₂S₂O₈. Cr is +6 in CrO₅ (butterfly structure).

Σ FORMULA Titration & equivalents

n-factor = electrons exchanged per formula unit: KMnO₄ 5 (acidic), 3 (neutral), 1 (strongly basic); K₂Cr₂O₇ 6 (acidic); oxalic acid 2 (redox) or 2 (acid–base); Fe²⁺ → Fe³⁺ 1. At the equivalence point: $n_1M_1V_1 = n_2M_2V_2$.

Σ FORMULA Cells

$$E_{\text{cell}}^\circ = E_{\text{cathode}}^\circ - E_{\text{anode}}^\circ \quad E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{0.0591}{n} \log Q \quad (298 \text{ K})$$

$$\Delta G = -nFE \quad \Delta G^\circ = -nFE^\circ = -2.303RT \log K \quad \log K = \frac{nE^\circ}{0.0591}$$

$F = 96\,500 \text{ C mol}^{-1}$. Cell notation: anode | anode solution || cathode solution | cathode.

Concentration cell M | Mⁿ⁺(C₁) || Mⁿ⁺(C₂) | M: $E = \frac{0.0591}{n} \log \frac{C_2}{C_1}$ (spontaneous when C₂ > C₁).

Hydrogen electrode: $E = -0.0591 \text{ pH}$ (at 1 bar H₂).

Σ FORMULA Electrolysis (Faraday's laws)

$$m = ZIt = \frac{M}{nF}It \quad \frac{m_1}{m_2} = \frac{E_1}{E_2} \text{ (same charge; } E = M/n)$$

Electrolyte (inert electrodes)	Cathode	Anode
molten NaCl	Na	Cl ₂
aqueous NaCl (concentrated)	H ₂	Cl ₂ (overpotential of O ₂)
aqueous CuSO ₄	Cu	O ₂
aqueous CuSO ₄ , Cu electrodes	Cu	Cu dissolves (purification)
dilute H ₂ SO ₄	H ₂	O ₂

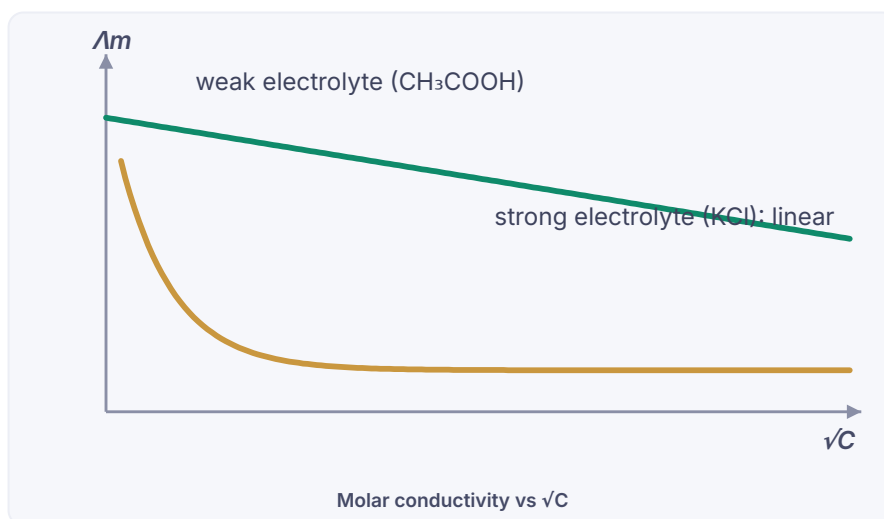
Σ FORMULA Conductance

$$G = \frac{1}{R} \quad \kappa = G \cdot \frac{l}{A} = \frac{G^*}{R} \quad \Lambda_m = \frac{\kappa \times 1000}{M} \text{ (S cm}^2\text{mol}^{-1}, \kappa \text{ in S cm}^{-1}\text{)}$$

Strong electrolyte: $\Lambda_m = \Lambda_m^\circ - A\sqrt{C}$ (linear in $\sqrt{C}$). **Weak electrolyte:** steep rise at low C ; use Kohlrausch.

Kohlrausch: $\Lambda_m^\circ = \nu_+ \lambda_+^\circ + \nu_- \lambda_-^\circ$; e.g. $\Lambda^\circ(\text{CH}_3\text{COOH}) = \Lambda^\circ(\text{CH}_3\text{COONa}) + \Lambda^\circ(\text{HCl}) - \Lambda^\circ(\text{NaCl})$.

Weak electrolyte: $\alpha = \Lambda_m / \Lambda_m^\circ$; $K_a = \frac{C\alpha^2}{1 - \alpha}$.

**Batteries & fuel cells (NCERT reactions)**

Cell	Anode	Cathode	Notes
Dry (Leclanché)	$\text{Zn} \longrightarrow \text{Zn}^{2+} + 2\text{e}^-$	$\text{MnO}_2 + \text{NH}_4^+ + \text{e}^- \longrightarrow \text{MnO}(\text{OH}) + \text{NH}_3$	~1.5 V; primary
Mercury	$\text{Zn}(\text{Hg}) + 2\text{OH}^- \longrightarrow \text{ZnO} + \text{H}_2\text{O} + 2\text{e}^-$	$\text{HgO} + \text{H}_2\text{O} + 2\text{e}^- \longrightarrow \text{Hg} + 2\text{OH}^-$	1.35 V, constant (no ions change in overall reaction)
Lead storage	$\text{Pb} + \text{SO}_4^{2-} \longrightarrow \text{PbSO}_4 + 2\text{e}^-$	$\text{PbO}_2 + \text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^- \longrightarrow \text{PbSO}_4 + 2\text{H}_2\text{O}$	secondary; H ₂ SO ₄ is consumed on discharge (density falls)
H ₂ -O ₂ fuel cell	$2\text{H}_2 + 4\text{OH}^- \longrightarrow 4\text{H}_2\text{O} + 4\text{e}^-$	$\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \longrightarrow 4\text{OH}^-$	~70% efficient; used in Apollo missions; product is water

Common traps

MISTAKE ALERT Mistake alerts

- E° is **intensive**. Don't multiply it by coefficients when balancing (but ΔG° does scale).
- Nernst Q uses products over reactants; solids and pure liquids are omitted.
- For KMnO_4 , the n -factor depends on the medium (5/3/1).
- Molar conductivity units: with κ in S cm^{-1} , use $\times 1000/M$ to get $\text{S cm}^2 \text{ mol}^{-1}$.
- The reducing agent is the one with the **more negative** reduction potential.

Question patterns

PYQ PATTERN Recurring structures

1. Oxidation numbers (including peroxide linkages); balancing and n -factor; titration stoichiometry.
2. E°_{cell} , spontaneity, Nernst calculations (a very common NV type).
3. ΔG° and K from E° .
4. Faraday's laws: mass deposited, time, charge; comparative deposition.
5. Conductivity $\leftrightarrow$ molar conductivity; Kohlrausch; degree of dissociation.
6. Battery and fuel-cell reactions (NCERT text).

Practice questions

C07 · Practice

1 Medium 1 min Concept

C07-01

The oxidation number of sulphur in $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulphuric acid) is:

- (A) +7 (B) +5 (C) +8 (D) +6

2 Easy 0.75 min Concept

C07-02

Which reaction is a disproportionation?

- (A) $\text{Zn} + \text{CuSO}_4 \longrightarrow \text{ZnSO}_4 + \text{Cu}$ (B) $\text{CaCO}_3 \longrightarrow \text{CaO} + \text{CO}_2$
 (C) $2\text{H}_2 + \text{O}_2 \longrightarrow 2\text{H}_2\text{O}$ (D) $\text{Cl}_2 + 2\text{OH}^- \longrightarrow \text{Cl}^- + \text{ClO}^- + \text{H}_2\text{O}$

3 Easy 0.5 min Speed

C07-03

$E^\circ(\text{Zn}^{2+}/\text{Zn}) = -0.76 \text{ V}$ and $E^\circ(\text{Cu}^{2+}/\text{Cu}) = +0.34 \text{ V}$. The standard emf of the Daniell cell is:

- (A) 0.42 V (B) 1.10 V (C) -1.10 V (D) 0.34 V

4 Medium 1.5 min Calc

C07-04

For $\text{Zn} | \text{Zn}^{2+} (0.01 \text{ M}) || \text{Cu}^{2+} (1 \text{ M}) | \text{Cu}$ at 298 K ($E^\circ = 1.10 \text{ V}$), the cell emf is about:

- (A) 1.159 V (B) 1.100 V (C) 1.041 V (D) 1.218 V

5 Easy 1 min Calc

C07-05

For a cell with $n = 2$ and $E^\circ = 1.10 \text{ V}$, ΔG° is about:

- (A) +212 kJ/mol (B) -106 kJ/mol (C) -212 kJ/mol (D) -21.2 kJ/mol

6 Medium 1 min Calc

C07-06

The mass of copper deposited by a current of 2 A passed for 965 s through CuSO_4 solution is ($\text{Cu} = 63.5$):

- (A) 1.27 g (B) 0.635 g (C) 0.3175 g (D) 6.35 g

7 Easy 1 min NV

C07-07

The conductivity of 0.1 M KCl is $1.29 \times 10^{-2} \text{ S cm}^{-1}$. Find its molar conductivity (in $\text{S cm}^2 \text{ mol}^{-1}$).

Your answer (numerical value):

8 Medium 1 min Basic

C07-08

Λ_m° ($\text{S cm}^2 \text{ mol}^{-1}$): CH_3COONa 91, HCl 426, NaCl 126. Λ_m° of CH_3COOH is:

- (A) 391 (B) 643 (C) 461 (D) 209

9 Easy 0.75 min Basic

C07-09

Acetic acid at some concentration has $\Lambda_m = 39.1 \text{ S cm}^2 \text{ mol}^{-1}$, and $\Lambda_m^\circ = 391$. Its degree of dissociation is:

- (A) 0.01 (B) 0.5 (C) 0.1 (D) 10

10 Easy 1 min NV

C07-10

For a cell reaction with $n = 2$ and $E^\circ = 0.0591 \text{ V}$ at 298 K, find the equilibrium constant.

Your answer (numerical value):

11 Easy 0.5 min Concept

C07-11

Electrolysis of concentrated aqueous NaCl with inert electrodes gives:

- (A) Na at the cathode, Cl_2 at the anode (B) H_2 at the cathode, O_2 at the anode
(C) H_2 at the cathode, Cl_2 at the anode (D) Na at the cathode, O_2 at the anode

12 Easy 0.75 min Concept

C07-12

When a lead storage battery discharges:

- (A) H_2SO_4 is produced
(B) Pb is deposited on the cathode
(C) PbSO_4 forms at both electrodes and the acid's density falls
(D) PbO_2 forms at the anode

C07 · Chapter Test

1 Easy 0.5 min Speed

C07-T1

Which is the strongest reducing agent?

- (A) Cu (B) Na (C) Zn (D) Li

2 Easy 0.5 min Speed

C07-T2

The n-factor of KMnO_4 in acidic medium is:

- (A) 1 (B) 5 (C) 3 (D) 7

3 Easy 0.5 min NV

C07-T3

How many moles of electrons are needed to reduce 1 mol of $\text{Cr}_2\text{O}_7^{2-}$ to Cr^{3+} ?Your answer (numerical value): **4** Easy 0.5 min Concept

C07-T4

For a strong electrolyte, a plot of Λ_m against $\sqrt{C}$ is:

- (A) a straight line with negative slope (B) a straight line with positive slope
(C) a curve rising steeply at low C (D) horizontal

5 Easy 0.5 min Speed

C07-T5

The product of an $\text{H}_2\text{-O}_2$ fuel cell is:

- (A) H_2O_2 (B) O_3 (C) H_2O (D) OH^- only

Answers & Solutions

C07 · Practice

1 (D) 2 (D) 3 (B) 4 (A) 5 (C) 6 (B) 7 129 8 (A) 9 (C) 10 100
11 (C) 12 (C)

C07 · Chapter Test

1 (D) 2 (B) 3 6 4 (A) 5 (C)

C07 · PRACTICE

1 Answer: (D) Peroxide linkage Medium 1 min Concept

C07-01

SOLUTION The molecule has one O-O (peroxide) bond. Six O are -2 and two are -1: $2(+1) + 2x + 6(-2) + 2(-1) = 0 \Rightarrow x = +6$.

SHORTCUT S can't exceed +6 (group 16).

COMMON TRAP Assuming all O are -2 gives +7.

2 Answer: (D) Disproportionation Easy 0.75 min Concept

C07-02

SOLUTION Cl goes from 0 to -1 (reduced) and to +1 (oxidised).

3 Answer: (B) Standard cell emf Easy 0.5 min Speed

C07-03

SOLUTION $E^\circ = 0.34 - (-0.76) = 1.10 \text{ V}$.

4 Answer: (A) Nernst equation Medium 1.5 min Calc

C07-04

SOLUTION $Q = [\text{Zn}^{2+}]/[\text{Cu}^{2+}] = 0.01$, $E = 1.10 - \frac{0.0591}{2} \log 0.01 = 1.10 + 0.059 = 1.159 \text{ V}$.

SHORTCUT A smaller Q (fewer products) raises E.

COMMON TRAP Inverting Q gives 1.041 V.

5 Answer: (C) ΔG° from E° Easy 1 min Calc

C07-05

SOLUTION $\Delta G^\circ = -2 \times 96\,500 \times 1.10 \approx -212\,300 \text{ J/mol}$.

6 Answer: (B) Faraday's first law Medium 1 min Calc

C07-06

SOLUTION $Q = 1930 \text{ C} = 0.02 \text{ F}$. Cu^{2+} needs 2 F per mol: $0.01 \text{ mol} = 0.635 \text{ g}$.

COMMON TRAP Using $n = 1$ (1.27 g).

7 Answer: 129 Molar conductivity Easy 1 min NV

C07-07

SOLUTION $\Lambda_m = \frac{\kappa \times 1000}{M} = \frac{1.29 \times 10^{-2} \times 1000}{0.1} = 129$.

- 8 Answer: (A)** Kohlrausch's law **Medium** ⌚ 1 min Basic C07-08
SOLUTION $91 + 426 - 126 = 391$.
SHORTCUT Add the two salts that contain the ions you want; subtract the one containing the unwanted ions.
- 9 Answer: (C)** Degree of dissociation **Easy** ⌚ 0.75 min Basic C07-09
SOLUTION $\alpha = 39.1/391 = 0.1$.
- 10 Answer: 100** Equilibrium constant from E° **Easy** ⌚ 1 min NV C07-10
SOLUTION $\log K = \frac{nE^\circ}{0.0591} = 2 \Rightarrow K = 100$.
- 11 Answer: (C)** Electrolysis of brine **Easy** ⌚ 0.5 min Concept C07-11
SOLUTION Water is reduced more easily than Na^+ . Cl^- is oxidised in preference to water because of oxygen's overpotential.
COMMON TRAP Using the molten- NaCl result (A).
- 12 Answer: (C)** Lead storage battery **Easy** ⌚ 0.75 min Concept C07-12
SOLUTION Both electrodes turn to PbSO_4 , and H_2SO_4 is consumed, producing water.

C07 · CHAPTER TEST

- 1 Answer: (D)** Strongest reducing agent **Easy** ⌚ 0.5 min Speed C07-T1
SOLUTION Li has the most negative E° (-3.05 V), because of its high hydration enthalpy.
COMMON TRAP Choosing Cs/Na by atomic-size reasoning alone.
- 2 Answer: (B)** n -factor of permanganate **Easy** ⌚ 0.5 min Speed C07-T2
SOLUTION $\text{MnO}_4^- (+7) \rightarrow \text{Mn}^{2+} (+2)$.
- 3 Answer: 6** Electrons for dichromate reduction **Easy** ⌚ 0.5 min NV C07-T3
SOLUTION Each Cr goes from +6 to +3 (3 electrons); $\times 2 = 6$.
- 4 Answer: (A)** Λ_m vs $\sqrt{C}$ **Easy** ⌚ 0.5 min Concept C07-T4
SOLUTION Debye-Hückel-Onsager: $\Lambda_m = \Lambda_m^\circ - A\sqrt{C}$.
COMMON TRAP Choosing (C), which describes a weak electrolyte.
- 5 Answer: (C)** Fuel cell **Easy** ⌚ 0.5 min Speed C07-T5
SOLUTION Overall: $2\text{H}_2 + \text{O}_2 \longrightarrow 2\text{H}_2\text{O}$.

 REVISION Revision tracker · Redox & Electrochemistry

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III • CHEMISTRY • UNIT 08

Chemical Kinetics

08

1-2 per shift

TYPICAL QUESTIONS

Easy-Medium

DIFFICULTY

Must-do

PRIORITY

Class 12 • Ch 3

NCERT

~7 hours

STUDY TIME

Important concepts

- **Rate** of $aA + bB \rightarrow cC$: $-\frac{1}{a} \frac{d[A]}{dt} = -\frac{1}{b} \frac{d[B]}{dt} = \frac{1}{c} \frac{d[C]}{dt}$.
- **Rate law** $r = k[A]^x[B]^y$ is **experimental**. The order $x + y$ can be zero or fractional.
- **Molecularity** is the number of species colliding in an **elementary** step (1, 2 or 3, never 0 or fractional). For complex reactions, the slowest step determines the rate law.
- **Temperature** raises the rate sharply, roughly doubling it per 10 °C rise for many reactions. Arrhenius: $k = Ae^{-E_a/RT}$.
- **Catalysts** lower E_a by providing an alternative path. They don't change ΔH , ΔG or K .
- **Collision theory**: rate = $PZ_{AB}e^{-E_a/RT}$. Only collisions with enough energy **and** the proper orientation (steric factor P) react.

Formula sheet

Σ FORMULA Integrated rate laws

Order	Integrated law	Half-life	Linear plot	Units of k
0	$[A] = [A]_0 - kt$	$\frac{[A]_0}{2k}$	$[A]$ vs t	$\text{mol L}^{-1} \text{s}^{-1}$
1	$k = \frac{2.303}{t} \log \frac{[A]_0}{[A]}$	$\frac{0.693}{k}$	$\ln[A]$ vs t (slope $-k$)	s^{-1}
n (general)	—	$\propto [A]_0^{1-n}$	—	$(\text{mol L}^{-1})^{1-n} \text{s}^{-1}$

First order: time for 99.9% completion $\approx 10 t_{1/2}$; 99% $\approx 6.64 t_{1/2}$; 75% $= 2t_{1/2}$; 87.5% $= 3t_{1/2}$.

Gas phase $A(g) \rightarrow B(g) + C(g)$ (first order, total pressure p_t): $k = \frac{2.303}{t} \log \frac{p_0}{2p_0 - p_t}$.

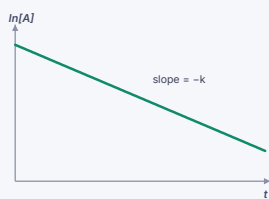
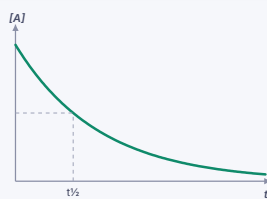
Pseudo-first-order: acid hydrolysis of an ester in excess water; inversion of cane sugar.

Σ FORMULA Temperature dependence

$$k = Ae^{-E_a/RT} \quad \ln k = \ln A - \frac{E_a}{RT} \quad \log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

Plot of $\ln k$ vs $1/T$: slope = $-E_a/R$. Fraction of molecules with energy $\geq E_a$ is $e^{-E_a/RT}$.

Energy profile: $E_a(\text{backward}) = E_a(\text{forward}) - \Delta H$. For an exothermic reaction, the backward barrier is larger.

Zero order: $[A]$ vs t First order: $\ln[A]$ vs t First order: $[A]$ vs t Arrhenius: $\ln k$ vs $1/T$

Numerical-solving method

1. From data tables, find the order by comparing two runs where only one concentration changes: $\frac{r_2}{r_1} = \left(\frac{C_2}{C_1} \right)^x$.

- For first-order time questions, convert "x% completed" into the fraction remaining, then use half-lives if the number is a power of $\frac{1}{2}$.
- For Arrhenius, use $2.303R = 19.15 \text{ J K}^{-1} \text{ mol}^{-1}$ and keep $1/T$ in K^{-1} .

SHORTCUT The 10-degree rule

If the rate doubles for every 10°C rise, a rise of ΔT multiplies the rate by $2^{\Delta T/10}$. Also memorise: a rate that doubles between 300 K and 310 K corresponds to $E_a \approx 53.6 \text{ kJ/mol}$.

Common traps

MISTAKE ALERT Mistake alerts

- Order $\neq$ stoichiometric coefficient (except in elementary steps).
- The half-life of a zero-order reaction **decreases** as $[A]_0$ decreases ($\propto [A]_0$). For first order it is independent of $[A]_0$.
- Mixing up log and ln: $t_{1/2} = 0.693/k$ uses $\ln 2$.
- A catalyst changes k (and E_a), but **not** K_{eq} .
- Rate-of-appearance questions: divide by the stoichiometric coefficients.

Question patterns

PYQ PATTERN Recurring structures

- Order from initial-rate tables (Section-B favourite).
- First-order time/half-life calculations; % completion.
- Arrhenius: E_a from two temperatures; effect of a catalyst on E_a and rate.
- Units of $k \leftrightarrow$ order.
- Graph identification (which plot is linear for which order).
- Relations between rates of disappearance/appearance.
- Kinetic study of the iodide- H_2O_2 reaction (practical chemistry).

Practice questions

C08 · Practice

1

Easy

 0.5 min

Speed

C08-01

The units of the rate constant of a second-order reaction are:

- (A) $\text{L mol}^{-1} \text{s}^{-1}$ (B) $\text{mol L}^{-1} \text{s}^{-1}$ (C) s^{-1} (D) $\text{L}^2 \text{mol}^{-2} \text{s}^{-1}$

2

Medium

 1 min

Calc

C08-02

A first-order reaction has $k = 0.0693 \text{ min}^{-1}$. The time for 99% completion is about:

- (A) 10 min (B) 46 min (C) 66.5 min (D) 100 min

3

Easy

 1 min

NV

C08-03

A zero-order reaction has $[A]_0 = 0.4 \text{ M}$ and $k = 0.01 \text{ M min}^{-1}$. Find its half-life (in minutes).

Your answer (numerical value):

4

Medium

⌚ 1.5 min

Calc

C08-04

The rate constant of a reaction doubles when the temperature rises from 300 K to 310 K. Its activation energy is about ($R = 8.314$, $\log 2 = 0.301$):

- (A) 53.6 kJ/mol (B) 26.8 kJ/mol (C) 107 kJ/mol (D) 5.36 kJ/mol

5

Easy

⌚ 0.5 min

NV

C08-05

When the concentration of a reactant is doubled, the rate becomes 8 times larger. Find the order with respect to that reactant.

Your answer (numerical value):

6

Medium

⌚ 1 min

Concept

C08-06

For $2\text{N}_2\text{O}_5 \rightarrow 4\text{NO}_2 + \text{O}_2$, the rate of formation of O_2 is $1 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$. The rate of formation of NO_2 is:

- (A) 1×10^{-3} (B) 2×10^{-3} (C) 0.25×10^{-3} (D) 4×10^{-3}

7

Easy

⌚ 0.5 min

Concept

C08-07

A catalyst increases the rate of a reaction by:

- (A) increasing ΔH (B) increasing the equilibrium constant
(C) lowering the activation energy (D) increasing the number of collisions only

8

Easy

⌚ 0.5 min

Concept

C08-08

The acid-catalysed hydrolysis of ethyl acetate in a large excess of water is:

- (A) zero order
(B) pseudo-first order
(C) truly second order in the observed rate law
(D) third order

9

Easy

⌚ 0.5 min

Concept

C08-09

A plot of $\ln k$ against $1/T$ is a straight line with slope:

- (A) E_a/R (B) $-E_a/R$ (C) $-E_a/2.303R$ (D) $\ln A$

10

Easy

⌚ 0.5 min

Concept

C08-10

Which statement is correct?

- (A) Order can be zero or fractional. (B) Molecularity can be fractional.
(C) Order is always equal to molecularity. (D) Molecularity is determined experimentally.



C08 · Chapter Test

1

Easy

⌚ 0.5 min

Speed

C08-T1

The half-life of a first-order reaction:

- (A) is proportional to the initial concentration
(B) depends on the square of the initial concentration
(C) is inversely proportional to the initial concentration
(D) is independent of the initial concentration

2 Easy 0.5 min NV

C08-T2

A first-order reaction is 75% complete in 30 min. Find its half-life (in minutes).

Your answer (numerical value):

3 Easy 0.5 min Speed

C08-T3

The units of k for a zero-order reaction are:

- (A) s^{-1} (B) $\text{mol L}^{-1} \text{s}^{-1}$ (C) $\text{L mol}^{-1} \text{s}^{-1}$ (D) dimensionless

4 Easy 0.75 min Basic

C08-T4

If a reaction's rate doubles for every 10°C rise, raising the temperature from 20°C to 50°C multiplies the rate by:

- (A) 3 (B) 6 (C) 8 (D) 9

5 Medium 0.75 min Concept

C08-T5

For an exothermic reaction, the activation energy of the backward reaction is:

- (A) smaller than that of the forward reaction (B) equal to that of the forward reaction
(C) larger than that of the forward reaction (D) zero

Answers & Solutions

C08 · Practice

1 (A) 2 (C) 3 20 4 (A) 5 3 6 (D) 7 (C) 8 (B) 9 (B) 10 (A)

C08 · Chapter Test

1 (D) 2 15 3 (B) 4 (C) 5 (C)

C08 · PRACTICE

1 Answer: (A) Units of k Easy 0.5 min Speed

C08-01

SOLUTION $(\text{mol L}^{-1})^{1-n} \text{s}^{-1}$ with $n = 2$.

2 Answer: (C) 99% completion time Medium 1 min Calc

C08-02

SOLUTION $t = \frac{2.303}{0.0693} \log 100 = 33.23 \times 2 \approx 66.5 \text{ min.}$

SHORTCUT $t_{99\%} \approx 6.64 t_{1/2} = 6.64 \times 10$.

3 Answer: 20 Zero-order half-life Easy 1 min NV

C08-03

SOLUTION $t_{1/2} = [A]_0 / 2k = 0.4 / 0.02 = 20 \text{ min.}$

COMMON TRAP Using $0.693/k$.

4 Answer: (A) Arrhenius two-temperature Medium 1.5 min Calc

C08-04

SOLUTION $0.301 = \frac{E_a}{2.303 \times 8.314} \left(\frac{10}{300 \times 310} \right) \Rightarrow E_a = \frac{0.301 \times 19.15 \times 93\,000}{10} \approx 53\,600 \text{ J/mol.}$

SHORTCUT Standard result: doubling between 300 K and 310 K means $E_a \approx 53.6 \text{ kJ/mol.}$

5 Answer: 3 Order from rate change Easy 0.5 min NV

C08-05

SOLUTION $2^x = 8 \Rightarrow x = 3$.

6 Answer: (D) Relating rates Medium 1 min Concept

C08-06

SOLUTION $\frac{1}{4} \frac{d[\text{NO}_2]}{dt} = \frac{d[\text{O}_2]}{dt} \Rightarrow \frac{d[\text{NO}_2]}{dt} = 4 \times 10^{-3}$.

COMMON TRAP Dividing instead of multiplying.

7 Answer: (C) *Effect of catalyst* Easy 0.5 min Concept

C08-07

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

8 Answer: (B) *Pseudo-first-order reaction* Easy 0.5 min Concept

C08-08

SOLUTION Water's concentration is effectively constant, so the rate law reduces to $r = k'[\text{ester}]$.

9 Answer: (B) *Arrhenius plot* Easy 0.5 min Concept

C08-09

SOLUTION $\ln k = \ln A - \frac{E_a}{R} \cdot \frac{1}{T}$. (For $\log k$ the slope is $-E_a/2.303R$.)

COMMON TRAP Mixing up $\ln$ and $\log$ plots.

10 Answer: (A) *Molecularity vs order* Easy 0.5 min Concept

C08-10

SOLUTION Order is experimental and can take any value. Molecularity is a theoretical positive integer for an elementary step.

C08 · CHAPTER TEST

1 Answer: (D) *First-order half-life* Easy 0.5 min Speed

C08-T1

SOLUTION $t_{1/2} = 0.693/k$.

2 Answer: 15 *Half-life from 75% completion* Easy 0.5 min NV

C08-T2

SOLUTION 75% complete means two half-lives have passed, so $t_{1/2} = 15$ min.

3 Answer: (B) *Units of zero-order k* Easy 0.5 min Speed

C08-T3

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

4 Answer: (C) *Temperature-coefficient rule* Easy 0.75 min Basic

C08-T4

SOLUTION $2^{30/10} = 8$.

COMMON TRAP Multiplying 2×3 .

5 Answer: (C) *Barrier heights* Medium 0.75 min Concept

C08-T5

SOLUTION $E_a(b) = E_a(f) - \Delta H$, and $\Delta H < 0$.

REVISION Revision tracker · Chemical Kinetics

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III · CHEMISTRY · UNIT 09

Classification of Elements & Periodicity

09

~1 per shift

TYPICAL QUESTIONS

Easy

DIFFICULTY

Must-do

PRIORITY

Class 11 · Ch 3

NCERT

~5 hours

STUDY TIME

Important concepts

- **Modern periodic law (Moseley):** properties are periodic functions of **atomic number**. The long form has 18 groups and 7 periods, divided into s, p, d and f blocks by the subshell being filled.
- **Elements with $Z > 100$** get IUPAC temporary names from digit roots: 0 nil, 1 un, 2 bi, 3 tri, 4 quad, 5 pent, 6 hex, 7 sept, 8 oct, 9 enn, plus "-ium". So 119 = ununennium (Uue), 120 = unbinilium (Ubn).
- **Periodic trends** come from effective nuclear charge and shell number: radius, ionisation enthalpy, electron gain enthalpy, electronegativity, valence, metallic character, nature of oxides.

Trends at a glance

Property	Across a period (→)	Down a group (↓)	Key exceptions
Atomic radius	decreases	increases	noble gases (van der Waals radii) look large; Ga < Al (d-contraction)
Ionisation enthalpy	increases	decreases	Be > B, N > O (also Mg > Al, P > S); group 13 order B > Tl > Ga > Al > In
Electron gain enthalpy (more negative)	increases	decreases	Cl more negative than F ; S more negative than O; noble gases, Be, Mg, N ≈ positive
Electronegativity	increases	decreases	F highest (4.0 Pauling)
Metallic character	decreases	increases	—
Oxide nature	basic → amphoteric → acidic	more basic	amphoteric: BeO, Al ₂ O ₃ , ZnO, Ga ₂ O ₃ , SnO ₂ , PbO ₂ ...

☆ IMPORTANT Ionic radii rules

- Cation < parent atom < anion.
- **Isoelectronic species:** radius falls as Z rises. $\text{N}^{3-} > \text{O}^{2-} > \text{F}^- > \text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$ (all 10 electrons).
- Down a group, ions of the same charge get larger.

☆ IMPORTANT Ionisation enthalpy essentials

- $IE_1 < IE_2 < IE_3 < \dots$ A **huge jump** means you've reached a noble-gas core, which tells you the group (e.g. a big jump after IE_2 means group 2).
- Half-filled (N: $2p^3$) and fully filled (Be: $2s^2$) subshells are extra stable, which causes the period-2 exceptions.
- Group 13 anomaly: Ga's poor 3d shielding and Tl's poor 4f/5d shielding raise their IE values.

Σ FORMULA Oxides across period 3

Na_2O (strongly basic) · MgO (basic) · Al_2O_3 (amphoteric) · SiO_2 (weakly acidic) · P_4O_{10} (acidic) · SO_3 (strongly acidic) · Cl_2O_7 (most acidic).

For the same element, a higher oxidation state gives a more acidic oxide (Mn_2O_7 acidic, MnO basic).

Memory tricks

SHORTCUT Exceptions sentence

"Before Boron, Nitrogen beats Oxygen, and Clever chlorine grabs electrons better than fluorine." That covers the three most-asked exceptions: IE (Be > B, N > O) and $\Delta_{eg}H$ (Cl > F).

Common traps

MISTAKE ALERT Mistake alerts

- Electron gain enthalpy is **negative** when energy is released. "Highest electron affinity" means most negative $\Delta_{eg}H$: Cl, not F.
- The smallest atom of period 2 is F, but **He is the smallest atom overall**.
- Radii of noble gases can't be compared with covalent radii.
- IE_2 of Na is exceptionally high (it would break the neon core).

Question patterns

PYQ PATTERN Recurring structures

- Ordering IE / EGE / radius / EN for 3–4 elements (with the exceptions above).
- Isoelectronic ion radii.
- Identifying group/period/block from Z or configuration.
- IUPAC names for $Z > 100$.
- Acidic/basic/amphoteric oxides.
- Successive IE data $\rightarrow$ group identification.

Practice questions

C09 · Practice

1

Easy

 0.5 min

Speed

C09-01

The IUPAC temporary name of the element with $Z = 119$ is:

- (A) unbiennium (B) unennium (C) ununoctium (D) ununennium

2

Easy

 0.5 min

Speed

C09-02

The correct order of ionic radii is:

- (A) $Na^+ > Mg^{2+} > F^- > O^{2-} > N^{3-}$ (B) $Mg^{2+} > Na^+ > F^- > O^{2-} > N^{3-}$
 (C) $N^{3-} > O^{2-} > F^- > Na^+ > Mg^{2+}$ (D) all equal (isoelectronic)

3

Easy

 0.5 min

Speed

C09-03

Which element has the most negative electron gain enthalpy?

- (A) F (B) Cl (C) O (D) Br

4

Medium

 1 min

Concept

C09-04

The correct order of first ionisation enthalpies is:

- (A) $Na < Al < Mg < Si$ (B) $Na < Mg < Al < Si$
 (C) $Al < Na < Mg < Si$ (D) $Si < Al < Mg < Na$

5

Easy

⌚ 0.5 min

Speed

C09-05

Which oxide is amphoteric?

- (A) Na_2O (B) SO_3 (C) Al_2O_3 (D) P_4O_{10}

6

Medium

⌚ 1 min

Concept

C09-06

The atomic radius of Ga (135 pm) is slightly smaller than that of Al (143 pm) because:

- (A) Ga has more shells
 (B) the 3d electrons in Ga shield the nucleus poorly, raising the effective nuclear charge
 (C) Ga is a metalloid
 (D) Al has a completely filled d subshell

7

Easy

⌚ 0.5 min

Speed

C09-07

The most electronegative element is:

- (A) F (B) Cl (C) O (D) N

8

Easy

⌚ 0.75 min

NV

C09-08

Element $Z = 34$ belongs to which group of the periodic table?Your answer (numerical value):

9

Medium

⌚ 1 min

Concept

C09-09

Which element has the highest second ionisation enthalpy?

- (A) Al (B) Mg (C) Na (D) Si

10

Easy

⌚ 0.75 min

Basic

C09-10

The correct order of acidic strength is:

- (A) all equally acidic (B) $\text{SiO}_2 > \text{P}_4\text{O}_{10} > \text{SO}_3 > \text{Cl}_2\text{O}_7$
 (C) $\text{SO}_3 > \text{Cl}_2\text{O}_7 > \text{SiO}_2 > \text{P}_4\text{O}_{10}$ (D) $\text{Cl}_2\text{O}_7 > \text{SO}_3 > \text{P}_4\text{O}_{10} > \text{SiO}_2$



C09 · Chapter Test

1

Easy

⌚ 0.5 min

Speed

C09-T1

An element with configuration $[\text{Ar}]3d^{10}4s^24p^1$ belongs to the:

- (A) p-block (B) s-block (C) d-block (D) f-block

2

Easy

⌚ 0.5 min

Speed

C09-T2

The correct order of size is:

- (A) $\text{F}^- < \text{O}^{2-} < \text{Mg}^{2+} < \text{Na}^+$ (B) $\text{O}^{2-} < \text{F}^- < \text{Na}^+ < \text{Mg}^{2+}$
 (C) $\text{Na}^+ < \text{Mg}^{2+} < \text{O}^{2-} < \text{F}^-$ (D) $\text{Mg}^{2+} < \text{Na}^+ < \text{F}^- < \text{O}^{2-}$

3

Easy

⌚ 0.5 min

Speed

C09-T3

The electron gain enthalpies of the noble gases are:

- (A) large and negative (B) the most negative in their period
 (C) zero exactly (D) positive

4 Easy 0.5 min NV

C09-T4

How many elements are there in the 4th period of the periodic table?

Your answer (numerical value):

5 Easy 0.5 min Speed

C09-T5

Which element is the most metallic?

- (A) Li (B) Na (C) Cs (D) K

Answers & Solutions

C09 · Practice

1 (D) 2 (C) 3 (B) 4 (A) 5 (C) 6 (B) 7 (A) 8 16 9 (C) 10 (D)

C09 · Chapter Test

1 (A) 2 (D) 3 (D) 4 18 5 (C)

C09 · PRACTICE

1 Answer: (D) IUPAC name for $Z = 119$ Easy 0.5 min Speed

C09-01

SOLUTION 1-1-9 → un-un-enn + ium, which is ununennium (Uue).

2 Answer: (C) Isoelectronic radii Easy 0.5 min Speed

C09-02

SOLUTION With the same 10 electrons, a larger nuclear charge pulls them in more tightly.

COMMON TRAP Choosing (D).

3 Answer: (B) Most negative electron gain enthalpy Easy 0.5 min Speed

C09-03

SOLUTION F's small 2p shell has high electron–electron repulsion, which makes its $\Delta_{eg}H$ less negative than Cl's.

COMMON TRAP Choosing F.

4 Answer: (A) First ionisation enthalpy order Medium 1 min Concept

C09-04

SOLUTION Mg ($3s^2$, filled) > Al ($3s^23p^1$), where the 3p electron is easier to remove.

COMMON TRAP Ignoring the Mg > Al exception.

5 Answer: (C) Amphoteric oxide Easy 0.5 min Speed

C09-05

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

6 Answer: (B) Gallium vs aluminium Medium 1 min Concept

C09-06

SOLUTION The ten 3d electrons added between Al and Ga are poor shielders, so the valence electrons feel a higher Z_{eff} .

7 Answer: (A) Highest electronegativity Easy 0.5 min Speed

C09-07

SOLUTION F: 4.0 on the Pauling scale.

8 Answer: 16 Group from atomic number Easy 0.75 min NV

C09-08

SOLUTION $[\text{Ar}]3d^{10}4s^24p^4$: 6 valence (s + p) electrons in the p-block, so group $10 + 6 = 16$ (Se).

SHORTCUT p-block group number = 10 + (valence s + p electrons).

COMMON TRAP Answering 6 (old group notation VI).

9 Answer: (C) Second ionisation enthalpy Medium 1 min Concept

C09-09

SOLUTION Na^+ has a neon configuration, so removing a second electron breaks a noble-gas core.

COMMON TRAP Choosing Si by the general trend.

10 Answer: (D) Acidic strength of oxides Easy 0.75 min Basic

C09-10

SOLUTION Across period 3, non-metallic character and the oxidation state of the central atom increase, so the oxides become more acidic.

C09 · CHAPTER TEST

1 Answer: (A) Block from configuration **Easy** 0.5 min Speed

C09-T1

SOLUTION The last electron enters a p subshell (it is Ga).

2 Answer: (D) Size order **Easy** 0.5 min Speed

C09-T2

SOLUTION Isoelectronic (10 electrons): smaller Z gives a larger ion.

3 Answer: (D) Noble-gas EGE **Easy** 0.5 min Speed

C09-T3

SOLUTION The added electron must enter the next shell, which requires energy.

4 Answer: 18 Period length **Easy** 0.5 min NV

C09-T4

SOLUTION $4s(2) + 3d(10) + 4p(6) = 18$.

5 Answer: (C) Most metallic element **Easy** 0.5 min Speed

C09-T5

SOLUTION Metallic character increases down a group, because the IE falls.

 REVISION Revision tracker · Periodicity

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III • CHEMISTRY • UNIT 10

p-Block Elements (Groups 13–18)

10

1–2 per shift

TYPICAL QUESTIONS

Easy–Medium
(fact-based)

DIFFICULTY

Must-do

PRIORITY

Pre-2023
editions: Class 11
p-Block (Gr 13–
14); Class 12 p-
Block (Gr 15–18)

NCERT

~12 hours

STUDY TIME

 VERIFIED FACT What the syllabus asks

Unit 10: "Group-13 to Group-18 elements: electronic configuration and general trends in physical and chemical properties of elements across the periods and down the groups; unique behaviour of the first element in each group" [NTA-SYL]. This module stays at that level: **trends, anomalies and a few landmark compounds** that questions use to test the trends.

Cross-group principles

- **Inert pair effect:** down Groups 13–15, the ns^2 pair becomes reluctant to take part in bonding, so the **lower** oxidation state becomes more stable: $Tl^+ > Tl^{3+}$, $Pb^{2+} > Pb^{4+}$, $Bi^{3+} > Bi^{5+}$. So Pb^{4+} and Bi^{5+} are strong oxidising agents, while Sn^{2+} is a reducing agent.
- **First element anomalies** (B, C, N, O, F): small size, high electronegativity, **no d orbitals** (maximum covalency 4), and the ability to form $p\pi-p\pi$ multiple bonds.
- **Hydrides:** thermal stability **decreases** down a group. Reducing character **increases**. H-bonding makes NH_3 , H_2O and HF boil anomalously high.

Group-by-group trend cards**Group 13 • B Al Ga In Tl • ns^2np^1**

- Oxidation states +3 and +1 (+1 more stable for Tl).
- B is a non-metal (a metalloid); the rest are metals.
- Radius: **Ga < Al** (poor 3d shielding). IE: $B > Tl > Ga > Al > In$.
- B is **electron-deficient**: BX_3 are Lewis acids. Acidity $BF_3 < BCl_3 < BBr_3 < BI_3$ because $p\pi-p\pi$ back-bonding is strongest in BF_3 .
- **Diborane** B_2H_6 : 4 terminal B–H bonds + **2 three-centre two-electron** (banana) B–H–B bridges.
- **Boric acid** $B(OH)_3$ is a weak **monobasic Lewis acid**: $B(OH)_3 + H_2O \longrightarrow [B(OH)_4]^- + H^+$.
- $AlCl_3$ dimerises to Al_2Cl_6 . Al_2O_3 is amphoteric.
- B anomaly: max covalency 4 (e.g. $[BF_4]^-$), vs $[AlF_6]^{3-}$ for Al.

Group 14 • C Si Ge Sn Pb • ns^2np^2

- Oxidation states +4 and +2 (+2 more stable for Pb). $SnCl_2$ is reducing; PbO_2 is oxidising.
- **Catenation:** $C \gg Si > Ge \approx Sn$ (C–C bond strength). Pb shows virtually none.
- C anomaly: multiple bonds (CO_2 is molecular while SiO_2 is a network), max covalency 4, allotropes.
- **Allotropes of C:** diamond (sp^3 , hard, insulator), graphite (sp^2 , layered, conducts), fullerene C_{60} (sp^2 , 20 six- and 12 five-membered rings).
- CO is neutral and toxic (binds haemoglobin); CO_2 is acidic. SiO_2 is a 3D network.
- CCl_4 is not hydrolysed (no d orbitals on C), while $SiCl_4$ hydrolyses readily.
- Silicones $(R_2SiO)_n$ are water-repellent. Silicates are built from SiO_4^{4-} tetrahedra.

Group 15 · N P As Sb Bi · ns^2np^3

- Oxidation states –3 to +5. +5 stability falls down the group (Bi^{5+} is a strong oxidant).
- Hydrides** EH_3 : basicity $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3 > \text{BiH}_3$; thermal stability falls; reducing power rises. Bond angle NH_3 $107.8^\circ > \text{PH}_3$ $93.6^\circ > \dots > \text{BiH}_3$ $\sim 90^\circ$.
- Boiling points: $\text{BiH}_3 > \text{SbH}_3 > \text{NH}_3 > \text{AsH}_3 > \text{PH}_3$ (NH_3 is raised by H-bonding).
- N anomaly: $\text{N}\equiv\text{N}$ triple bond (very inert N_2); no NCl_5 (no d orbitals), whereas PCl_5 exists.
- Oxides of N: N_2O , NO neutral; N_2O_3 , NO_2 , N_2O_5 acidic. NO_2 is brown and paramagnetic, and dimerises to N_2O_4 .
- Oxoacids of P: H_3PO_2 (monobasic, 2 P–H, strong reducer), H_3PO_3 (dibasic, 1 P–H, reducer), H_3PO_4 (tribasic). **Basicity = number of P–OH groups.**
- White P (P_4 , reactive, glows) vs red P (polymeric, stable). PCl_5 is trigonal bipyramidal; in the solid it is $[\text{PCl}_4]^+[\text{PCl}_6]^-$.

Group 16 · O S Se Te Po · ns^2np^4

- O shows –2 (–1 in peroxides, +2 in OF_2). S, Se, Te: +2, +4, +6.
- Hydrides** H_2E : acidity $\text{H}_2\text{O} < \text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te}$; thermal stability falls; boiling point $\text{H}_2\text{O} \gg \text{H}_2\text{Te} > \text{H}_2\text{Se} > \text{H}_2\text{S}$.
- O anomaly: H-bonding, O_2 paramagnetic, max covalency 4. SF_6 exists but " OF_6 " doesn't.
- Ozone** O_3 : bent, diamagnetic, strong oxidant; formed by silent electric discharge.
- Sulphur allotropes**: rhombic (α , stable below 369 K) and monoclinic (β); S_8 puckered crown ring.
- SF_6 is exceptionally inert (steric protection). H_2SO_4 is dibasic, dehydrating and oxidising.

Group 17 · F Cl Br I At · ns^2np^5

- F shows only –1. The others show –1, +1, +3, +5, +7.
- Oxidising power: $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$.
- Bond dissociation enthalpy**: $\text{Cl}_2 > \text{Br}_2 > \text{F}_2 > \text{I}_2$. F–F is anomalously weak (lone-pair repulsion).
- Acidity of HX (aq)**: $\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$. HF has the highest boiling point (H-bonding).
- Colours: F_2 pale yellow, Cl_2 greenish yellow, Br_2 red-brown, I_2 violet (vapour).
- Oxoacid strength: $\text{HOCl} < \text{HClO}_2 < \text{HClO}_3 < \text{HClO}_4$. Oxidising power runs the other way.
- Interhalogens**: XX' (linear), XX'_3 (T-shaped, e.g. ClF_3), XX'_5 (square pyramidal, BrF_5), XX'_7 (pentagonal bipyramidal, IF_7). They are more reactive than the halogens (except F_2).
- F anomaly: most electronegative element, no d orbitals, only one oxoacid (HOF).

Group 18 · He Ne Ar Kr Xe Rn · ns^2np^6

- Very high IE (decreasing down the group); positive electron gain enthalpy; monatomic gases.
- First noble-gas compound**: $\text{Xe}^+[\text{PtF}_6]^-$ (Bartlett, 1962). The idea came from $\text{O}_2^+[\text{PtF}_6]^-$, because the IE of $\text{O}_2 \approx$ IE of Xe.
- Xenon fluorides: XeF_2 linear, XeF_4 square planar, XeF_6 distorted octahedral. Oxides: XeO_3 pyramidal, XeOF_4 square pyramidal.
- Hydrolysis of XeF_6 gives XeO_3 (complete).
- Uses: He (balloons, diving gas mixtures, low-temperature work), Ar (inert atmosphere, bulbs), Ne (signs).
- He has the lowest boiling point of any substance.

Unique behaviour of the first element: summary

Group	First element	Key anomalies
13	B	non-metal; covalent compounds; max covalency 4; electron-deficient hydrides (diborane)
14	C	catenation; $p\pi-p\pi$ bonds; allotropes; max covalency 4
15	N	diatomic gas with a triple bond; no pentahalides; H-bonding in NH_3
16	O	diatomic paramagnetic gas; H-bonding; mainly –2 state
17	F	only –1; weak F–F bond; HF weak acid with H-bonds; strongest oxidant

Common traps

MISTAKE ALERT Mistake alerts

- "Stronger Lewis acid = more electronegative halogen" is **wrong** for BX_3 . Back-bonding reverses the order.
- Basicity of oxoacids of P = number of **P–OH**, not the total number of H atoms.
- Bond energy of F_2 is **lower** than Cl_2 and Br_2 .
- Acidity of hydrides increases down Groups 16 and 17 (bond strength decreases), even though electronegativity decreases.

Question patterns

PYQ PATTERN Recurring structures

1. Ordering properties (acidity, basicity, bond energy, boiling point, stability) of hydrides/halides/oxoacids down a group.
2. Inert-pair-effect consequences (oxidising/reducing behaviour).
3. Shapes and bonding of landmark compounds (B_2H_6 , PCl_5 , xenon fluorides, interhalogens).
4. Basicity of phosphorus oxoacids.
5. Anomalous behaviour of the first element.

Practice questions

C10 · Practice

1 **Easy** ⌚ 0.5 min **Concept** C10-01

Which ion is the most stable?

- (A) Tl^{3+} (B) Al^{+} (C) Tl^{+} (D) B^{+}

2 **Medium** ⌚ 1 min **Concept** C10-02

The correct order of Lewis acid strength is:

- (A) $BF_3 > BCl_3 > BBr_3$ (B) all equal
(C) $BCl_3 > BF_3 > BBr_3$ (D) $BBr_3 > BCl_3 > BF_3$

3 **Easy** ⌚ 0.5 min **Speed** C10-03

The most basic hydride is:

- (A) PH_3 (B) NH_3 (C) AsH_3 (D) SbH_3

4 **Medium** ⌚ 1 min **Concept** C10-04

The correct order of bond dissociation enthalpy is:

- (A) $F_2 > Cl_2 > Br_2 > I_2$ (B) $I_2 > Br_2 > Cl_2 > F_2$
(C) $Cl_2 > Br_2 > F_2 > I_2$ (D) $Cl_2 > F_2 > Br_2 > I_2$

5 **Easy** ⌚ 0.5 min **Speed** C10-05

Which is the strongest acid in water?

- (A) HI (B) HCl (C) HBr (D) HF

6

Easy

⌚ 0.5 min

NV

C10-06

Find the basicity of phosphorous acid, H_3PO_3 .

Your answer (numerical value):

7

Easy

⌚ 0.5 min

Speed

C10-07

The shape of XeO_3 is:

- (A) trigonal planar (B) tetrahedral (C) T-shaped (D) pyramidal

8

Easy

⌚ 0.75 min

Basic

C10-08

The correct order of boiling points is:

- (A) $\text{H}_2\text{O} > \text{H}_2\text{S} > \text{H}_2\text{Se} > \text{H}_2\text{Te}$ (B) $\text{H}_2\text{Te} > \text{H}_2\text{Se} > \text{H}_2\text{S} > \text{H}_2\text{O}$
(C) $\text{H}_2\text{S} > \text{H}_2\text{Se} > \text{H}_2\text{Te} > \text{H}_2\text{O}$ (D) $\text{H}_2\text{O} > \text{H}_2\text{Te} > \text{H}_2\text{Se} > \text{H}_2\text{S}$

9

Easy

⌚ 0.5 min

Concept

C10-09

Nitrogen can't form NCl_5 while phosphorus forms PCl_5 , because:

- (A) N is less electronegative
(B) N has no vacant d orbitals in its valence shell
(C) N is larger
(D) Cl_2 doesn't react with N

10

Easy

⌚ 0.75 min

NV

C10-10

How many three-centre two-electron ($3c-2e$) bonds are there in B_2H_6 ?

Your answer (numerical value):

11

Easy

⌚ 0.5 min

Speed

C10-11

The strongest acid is:

- (A) HClO_4 (B) HClO_2 (C) HClO_3 (D) HOCl

**C10 · Chapter Test**

1

Easy

⌚ 0.5 min

Speed

C10-T1

The first noble-gas compound prepared (Bartlett, 1962) was:

- (A) $\text{Xe}^+[\text{PtF}_6]^-$ (B) XeF_2 (C) XeO_3 (D) KrF_2

2

Easy

⌚ 0.5 min

Speed

C10-T2

Which element shows the greatest tendency for catenation?

- (A) Si (B) Sn (C) Ge (D) C

3

Easy

⌚ 0.5 min

Speed

C10-T3

The maximum covalency of nitrogen is:

- (A) 3 (B) 4 (C) 5 (D) 6

4 Easy 0.5 min NV

C10-T4

How many P–OH groups are there in hypophosphorous acid, H_3PO_2 ?

Your answer (numerical value):

5 Easy 0.5 min Speed

C10-T5

Iodine vapour is:

- (A) yellow (B) green (C) violet (D) red-brown

Answers & Solutions

C10 · Practice

- 1 (C) 2 (D) 3 (B) 4 (C) 5 (A) 6 2 7 (D) 8 (D) 9 (B) 10 2
11 (A)

C10 · Chapter Test

- 1 (A) 2 (D) 3 (B) 4 1 5 (C)

C10 · PRACTICE

1 **Answer: (C)** *Inert pair effect* Easy 0.5 min Concept C10-01

SOLUTION The inert pair effect makes +1 the more stable state for thallium.

2 **Answer: (D)** *Lewis acidity of boron halides* Medium 1 min Concept C10-02

SOLUTION $p\pi-p\pi$ back-donation from F to B (similar-sized 2p orbitals) is strongest in BF_3 , which reduces boron's electron deficiency the most.

COMMON TRAP Using electronegativity alone.

3 **Answer: (B)** *Basicity of group 15 hydrides* Easy 0.5 min Speed C10-03

SOLUTION N's lone pair is in a small, compact orbital with high electron density, so it is the most available.

4 **Answer: (C)** *Halogen bond energies* Medium 1 min Concept C10-04

SOLUTION F–F is weakened by repulsion between the lone pairs on the two small F atoms.

COMMON TRAP Assuming a simple trend.

5 **Answer: (A)** *Strongest hydrohalic acid* Easy 0.5 min Speed C10-05

SOLUTION H–I is the weakest H–X bond.

6 **Answer: 2** *Basicity of phosphorous acid* Easy 0.5 min NV C10-06

SOLUTION Structure HP(O)(OH)_2 : two ionisable P–OH groups. The P–H hydrogen isn't acidic.

COMMON TRAP Answering 3.

7 **Answer: (D)** *Shape of XeO_3* Easy 0.5 min Speed C10-07

SOLUTION 3 bond pairs + 1 lone pair on Xe.

8 **Answer: (D)** *Boiling points of group 16 hydrides* Easy 0.75 min Basic C10-08

SOLUTION H_2O is raised by H-bonding. For the rest, van der Waals forces increase with molar mass.

9 **Answer: (B)** *Pentahalides* Easy 0.5 min Concept C10-09

SOLUTION N's maximum covalency is 4.

10 **Answer: 2** *Bonds in diborane* Easy 0.75 min NV C10-10

SOLUTION Two B–H–B bridges, each a 3c–2e bond.

COMMON TRAP Counting the 4 terminal bonds.

- 11 Answer: (A)** Strongest oxoacid of chlorine **Easy** ⌚ 0.5 min **Speed** C10-11
SOLUTION More O atoms spread the negative charge of the conjugate base better.

C10 · CHAPTER TEST

- 1 Answer: (A)** First noble-gas compound **Easy** ⌚ 0.5 min **Speed** C10-T1
SOLUTION Direct recall: see the concept summary and formula box of this chapter.

- 2 Answer: (D)** Catenation **Easy** ⌚ 0.5 min **Speed** C10-T2
SOLUTION The C–C bond is strong, and carbon is small.

- 3 Answer: (B)** Maximum covalency of N **Easy** ⌚ 0.5 min **Speed** C10-T3
SOLUTION Only 4 valence orbitals (one 2s + three 2p), e.g. in NH_4^+ .

- 4 Answer: 1** Hypophosphorous acid **Easy** ⌚ 0.5 min **NV** C10-T4
SOLUTION $\text{H}_2\text{P}(\text{O})(\text{OH})$: one P–OH and two P–H, so it is monobasic.

- 5 Answer: (C)** Colour of iodine vapour **Easy** ⌚ 0.5 min **Speed** C10-T5
SOLUTION Direct recall: see the concept summary and formula box of this chapter.

**REVISION Revision tracker · p-Block**

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III • CHEMISTRY • UNIT 11

d- and f-Block Elements

11

1-2 per shift

TYPICAL QUESTIONS

Easy-Medium
(NCERT-heavy)

DIFFICULTY

Must-do
PRIORITYClass 12 • Ch 4
NCERT~8 hours
STUDY TIME

Important concepts

- **Transition elements:** partly filled d subshell in the atom or in a common ion. General configuration $(n-1)d^{1-10}ns^{1-2}$. **Zn, Cd, Hg** (d^{10} in both the atom and the common ion) are not regarded as transition elements.
- **Characteristic properties:** metallic, high melting points (maximum near the middle of each series; W is highest; Zn, Cd, Hg are low, and Hg is liquid), variable oxidation states, coloured ions, paramagnetism, catalytic activity, complex formation, interstitial compounds, alloys.
- **Lanthanoid contraction:** the steady decrease in size from La to Lu, caused by the poor shielding of 4f electrons.
- **Actinoids:** all radioactive; wider range of oxidation states because 5f, 6d and 7s have comparable energies.

Trends & NCERT facts (first transition series)

Σ FORMULA Oxidation states & stability

- Sc shows only +3; Zn only +2. **Mn shows the maximum (+7)** in the first series. Os/Ru reach +8 (OsO_4).
- Stable half-filled d^5 : Mn^{2+} and Fe^{3+} . Mn^{3+} (d^4) is a strong oxidant; Cr^{2+} (d^4) is a strong reductant.
- $E^\circ(\text{M}^{2+}/\text{M})$: **Cu is the only first-series metal with a positive value** (+0.34 V), so it doesn't liberate H_2 from dilute acids.
- Selected $E^\circ(\text{M}^{3+}/\text{M}^{2+})$: Co +1.97, Mn +1.57, Fe +0.77, Cr -0.41 V.
- Higher oxidation states are stabilised by F and O (e.g. MnO_4^- , CrO_4^{2-} , VF_5).

Σ FORMULA Colour & magnetism

$\mu_{\text{spin-only}} = \sqrt{n(n+2)}$ BM. Colour arises from **d-d transitions**, so d^0 and d^{10} ions are colourless (Sc^{3+} , Ti^{4+} , Zn^{2+} , Cu^+).

Aqueous ion	Colour	Aqueous ion	Colour
Ti^{3+} (d^1)	purple	Fe^{2+} (d^6)	pale green
V^{3+} (d^2)	green	Fe^{3+} (d^5)	yellow
Cr^{3+} (d^3)	violet	Co^{2+} (d^7)	pink
Mn^{2+} (d^5)	pale pink	Ni^{2+} (d^8)	green
Cu^{2+} (d^9)	blue	Zn^{2+} (d^{10})	colourless

The intense colours of MnO_4^- (purple) and $\text{Cr}_2\text{O}_7^{2-}$ (orange) come from **charge transfer**, not d-d transitions. Both have d^0 metal centres.

☆ IMPORTANT Catalysts, interstitials, alloys (NCERT)

- Catalysts: V_2O_5 (contact process, $\text{SO}_2 \rightarrow \text{SO}_3$), Fe (Haber process), Ni (hydrogenation), $\text{TiCl}_4 + \text{Al}(\text{CH}_3)_3$ (Ziegler-Natta), MnO_2 (KClO_3 decomposition). They work through variable oxidation states and adsorption on surfaces.
- Interstitial compounds (TiC , Mn_4N , Fe_3H , TiH_2): hard, very high melting points, conducting, chemically inert.
- Alloys form easily because the atomic radii are similar: brass (Cu-Zn), bronze (Cu-Sn), stainless steel.

Potassium dichromate and potassium permanganate (NCERT)

Σ FORMULA $K_2Cr_2O_7$

Preparation from chromite:



pH-dependent equilibrium: $2CrO_4^{2-}$ (yellow) + $2H^+ \rightleftharpoons Cr_2O_7^{2-}$ (orange) + H_2O .

Oxidising action (acid): $Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O$ ($E^\circ = 1.33$ V). It oxidises $I^- \rightarrow I_2$, $Fe^{2+} \rightarrow Fe^{3+}$, $H_2S \rightarrow S$, $Sn^{2+} \rightarrow Sn^{4+}$.

Structure: two CrO_4 tetrahedra sharing one O (Cr–O–Cr angle $\sim 126^\circ$).

Σ FORMULA $KMnO_4$

Preparation: $2MnO_2 + 4KOH + O_2 \longrightarrow 2K_2MnO_4$ (green) + $2H_2O$; then

$3MnO_4^{2-} + 4H^+ \longrightarrow 2MnO_4^- + MnO_2 + 2H_2O$ (disproportionation). Commercially, manganate is oxidised electrolytically.

Oxidising action:

Medium	Half-reaction	n-factor
acidic	$MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$ ($E^\circ 1.52$ V)	5
neutral / faintly alkaline	$MnO_4^- + 2H_2O + 3e^- \longrightarrow MnO_2 + 4OH^-$	3

Acidic: $I^- \rightarrow I_2$; $Fe^{2+} \rightarrow Fe^{3+}$; $C_2O_4^{2-} \rightarrow CO_2$ (at 333 K); $S^{2-} \rightarrow S$; $SO_3^{2-} \rightarrow SO_4^{2-}$; $NO_2^- \rightarrow NO_3^-$.

Neutral/alkaline: $I^- \rightarrow IO_3^-$; thiosulphate $\rightarrow$ sulphate; $Mn^{2+} \rightarrow MnO_2$.

$KMnO_4$ is dark purple and diamagnetic (d^0), with weak temperature-dependent paramagnetism. Titrations with it use **dilute** H_2SO_4 , not HCl (which it would oxidise to Cl_2).

Inner transition elements

Lanthanoids (Ce–Lu)

- Configuration $[Xe]4f^{1-14}5d^{0-1}6s^2$. The common oxidation state is **+3**.
- Other stable states track f^0 , f^7 , f^{14} : Ce^{4+} (f^0 , oxidant), Eu^{2+} (f^7 , reductant), Yb^{2+} (f^{14}), Tb^{4+} (f^7).
- Lanthanoid contraction consequences:** $Zr \approx Hf$ in size (they occur together and are hard to separate); lanthanoids are hard to separate from each other; basicity of $Ln(OH)_3$ decreases from La to Lu; the 5d series has higher IE and density.
- Misch metal ($\sim 95\%$ lanthanoids + Fe) is used in lighter flints.

Actinoids (Th–Lr)

- Configuration $[Rn]5f^{1-14}6d^{0-1}7s^2$. All are radioactive.
- Wider range of oxidation states** (+3 to +7; e.g. Np, Pu up to +7), because 5f, 6d and 7s have comparable energies.
- Actinoid contraction is **greater** than lanthanoid contraction (5f shields even more poorly).
- Early actinoids resemble transition metals more than the lanthanoids do.

Common traps

MISTAKE ALERT Mistake alerts

- Configurations of ions: remove 4s electrons first ($\text{Fe}^{3+} = 3d^5$).
- Cr and Cu atomic anomalies: $3d^5 4s^1$ and $3d^{10} 4s^1$.
- Colour of $\text{KMnO}_4/\text{K}_2\text{Cr}_2\text{O}_7$ is **charge transfer**, not d-d.
- KMnO_4 titrations: never acidify with HCl.
- Zn, Cd, Hg: d-block, but not transition elements.

Question patterns

PYQ PATTERN Recurring structures

- Spin-only magnetic moment of ions; colour/colourless identification.
- Oxidation-state trends; stability (Mn^{2+} vs Mn^{3+} , Cr^{2+} reducing); E° trends.
- $\text{K}_2\text{Cr}_2\text{O}_7/\text{KMnO}_4$ preparation steps, structures, products in each medium, titration stoichiometry.
- Lanthanoid contraction consequences; stable oxidation states of Ce, Eu, Yb.
- Catalysts and interstitial compounds (NCERT lines).

Practice questions

C11 · Practice

1 Easy 0.5 min Speed C11-01

Which first-row transition element shows the highest oxidation state?

- (A) Cr (B) V (C) Fe (D) Mn

2 Easy 0.5 min Speed C11-02

Which aqueous ion is colourless?

- (A) Ti^{3+} (B) Cr^{3+} (C) Zn^{2+} (D) Cu^{2+}

3 Easy 0.5 min Speed C11-03

The spin-only magnetic moment of Cr^{3+} is about:

- (A) 3.87 BM (B) 2.83 BM (C) 1.73 BM (D) 4.90 BM

4 Medium 1.5 min NV C11-04

Find the volume (in mL) of 0.1 M $\text{K}_2\text{Cr}_2\text{O}_7$ (acidified) needed to oxidise 60 mL of 0.1 M Fe^{2+} completely.

Your answer (numerical value):

5 Easy 0.5 min Concept C11-05

Adding acid to a yellow solution of K_2CrO_4 :

- (A) turns it green as Cr^{3+} forms (B) turns it orange as $\text{Cr}_2\text{O}_7^{2-}$ forms
(C) causes no change (D) turns it colourless

6

Easy

⌚ 0.5 min

Speed

C11-06

In neutral or faintly alkaline medium, MnO_4^- is reduced to:

- (A) Mn^{2+} (B) MnO_2 (C) MnO_4^{2-} (D) Mn

7

Easy

⌚ 0.5 min

Concept

C11-07

A consequence of the lanthanoid contraction is that:

- (A) Zr and Hf have nearly the same atomic radius
 (B) lanthanoid hydroxides become more basic from La to Lu
 (C) lanthanoids are easy to separate
 (D) the 5d elements have lower IE than the 4d elements

8

Easy

⌚ 0.75 min

Concept

C11-08

Copper doesn't liberate H_2 from dilute HCl because:

- (A) $E^\circ(\text{Cu}^{2+}/\text{Cu})$ is positive (B) Cu is a transition metal
 (C) Cu forms a protective oxide layer (D) HCl is a weak acid

9

Easy

⌚ 0.5 min

Concept

C11-09

Actinoids show a wider range of oxidation states than lanthanoids because:

- (A) they are radioactive
 (B) they have fewer electrons
 (C) they are larger
 (D) the 5f, 6d and 7s orbitals are close in energy

10

Easy

⌚ 0.5 min

NV

C11-10

Find the number of unpaired electrons in a gaseous Co^{2+} ion ($Z = 27$).

Your answer (numerical value):



C11 · Chapter Test

1

Easy

⌚ 0.5 min

Speed

C11-T1

Zinc is not regarded as a transition element because:

- (A) it is too reactive
 (B) it has no d electrons
 (C) it has a completely filled d subshell in the atom and in Zn^{2+}
 (D) it forms no complexes

2

Easy

⌚ 0.5 min

Speed

C11-T2

The catalyst used in the contact process for H_2SO_4 is:

- (A) Fe (B) Ni (C) Pt only (D) V_2O_5

3 Easy 0.5 min Concept

C11-T3

The intense purple colour of KMnO_4 is due to:

- (A) d-d transitions in Mn^{7+} (B) the K^+ ion
(C) unpaired electrons (D) ligand-to-metal charge transfer

4 Easy 0.5 min NV

C11-T4

Find the oxidation state of Mn in K_2MnO_4 .

Your answer (numerical value):

5 Easy 0.5 min Speed

C11-T5

The most common oxidation state of the lanthanoids is:

- (A) +2 (B) +5 (C) +4 (D) +3

Answers & Solutions

C11 · Practice

1 (D) 2 (C) 3 (A) 4 10 5 (B) 6 (B) 7 (A) 8 (A) 9 (D) 10 3

C11 · Chapter Test

1 (C) 2 (D) 3 (D) 4 6 5 (D)

C11 · PRACTICE

1 Answer: (D) Highest oxidation state, 3d series Easy 0.5 min Speed

C11-01

SOLUTION Mn reaches +7 (in MnO_4^-), using all of its 3d and 4s electrons.

2 Answer: (C) Colourless ion Easy 0.5 min Speed

C11-02

SOLUTION Zn^{2+} is d^{10} , so no d-d transition is possible.

SHORTCUT d^0 or $d^{10} \rightarrow$ colourless.

3 Answer: (A) Magnetic moment of Cr^{3+} Easy 0.5 min Speed

C11-03

SOLUTION d^3 : 3 unpaired, so $\sqrt{15} \approx 3.87$ BM.

4 Answer: 10 Dichromate titration Medium 1.5 min NV

C11-04

SOLUTION $\text{Cr}_2\text{O}_7^{2-} + 6\text{Fe}^{2+} + 14\text{H}^+ \rightarrow 2\text{Cr}^{3+} + 6\text{Fe}^{3+} + 7\text{H}_2\text{O}$. Moles of $\text{Fe}^{2+} = 0.006$, so dichromate = $0.001 \text{ mol} = 10 \text{ mL}$ of 0.1 M.

SHORTCUT Equivalents: $6 \times 0.1 \times V = 1 \times 0.1 \times 60$.

COMMON TRAP Using a 1 : 1 ratio (60 mL).

5 Answer: (B) Chromate-dichromate equilibrium Easy 0.5 min Concept

C11-05

SOLUTION $2\text{CrO}_4^{2-} + 2\text{H}^+ \rightleftharpoons \text{Cr}_2\text{O}_7^{2-} + \text{H}_2\text{O}$ shifts right. This is not a redox change: Cr stays +6.

COMMON TRAP Choosing (A), which would need a reducing agent.

6 Answer: (B) KMnO_4 in neutral medium Easy 0.5 min Speed

C11-06

SOLUTION $\text{MnO}_4^- + 2\text{H}_2\text{O} + 3\text{e}^- \rightarrow \text{MnO}_2 + 4\text{OH}^-$.

7 Answer: (A) Lanthanoid contraction Easy 0.5 min Concept

C11-07

SOLUTION The 4f contraction cancels the size increase expected from the extra shell.

COMMON TRAP Choosing (B): basicity actually **decreases**.

8 Answer: (A) Copper and dilute acid Easy 0.75 min Concept

C11-08

SOLUTION +0.34 V: Cu is a weaker reducing agent than H_2 .

9 Answer: (D) Actinoid oxidation states **Easy** ⌚ 0.5 min **Concept**

C11-09

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

10 Answer: 3 Unpaired electrons in Co^{2+} **Easy** ⌚ 0.5 min **NV**

C11-10

SOLUTION Co^{2+} : $3d^7$, which is 3 unpaired electrons (free ion, Hund's rule).

COMMON TRAP Using the atomic configuration.

C11 · CHAPTER TEST

1 Answer: (C) Zinc's status **Easy** ⌚ 0.5 min **Speed**

C11-T1

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

2 Answer: (D) Contact-process catalyst **Easy** ⌚ 0.5 min **Speed**

C11-T2

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

3 Answer: (D) Origin of KMnO_4 colour **Easy** ⌚ 0.5 min **Concept**

C11-T3

SOLUTION Mn(VII) is d^0 , so there are no d-d transitions. The colour comes from $\text{O} \rightarrow \text{Mn}$ charge transfer.

4 Answer: 6 Oxidation state in manganate **Easy** ⌚ 0.5 min **NV**

C11-T4

SOLUTION $2(+1) + x + 4(-2) = 0 \Rightarrow x = +6$.

5 Answer: (D) Common lanthanoid state **Easy** ⌚ 0.5 min **Speed**

C11-T5

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

REVISION Revision tracker · d- & f-Block

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III · CHEMISTRY · UNIT 12

Coordination Compounds

12

2–3 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

Must-do (top-weight chapter)

PRIORITY

Class 12 · Ch 5

NCERT

~10 hours

STUDY TIME



PYQ PATTERN Weight evidence

36 questions across the 2026 Paper-1 shifts ($\approx 7.6\%$ of Chemistry, the highest single chapter in that dataset)

THIRD-PARTY ANALYSIS [T-WT-CD]. Coordination compounds are also the base for colour, magnetism and isomerism questions in the d-block unit.

Important concepts

- **Werner's theory:** metals have **primary valency** (ionisable, equal to the oxidation state, satisfied by anions) and **secondary valency** (non-ionisable, equal to the coordination number, directed in space).
- **Ligands:** unidentate (Cl^- , NH_3 , H_2O), bidentate (en, $\text{C}_2\text{O}_4^{2-}$), polydentate (EDTA^{4-} , hexadentate). **Ambidentate** ligands bind through two different atoms: $\text{NO}_2^-/\text{ONO}^-$, $\text{SCN}^-/\text{NCS}^-$, CN^-/NC^- . **Chelates** (ring-forming) are extra stable (the chelate effect).
- **Homoleptic** complexes have one kind of ligand ($[\text{Co}(\text{NH}_3)_6]^{3+}$); **heteroleptic** complexes have more than one ($[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$).
- **Bonding:** VBT gives hybridisation and geometry. CFT explains colour, magnetism and the spectrochemical series.

IUPAC nomenclature: the rules

1. Name the **cation first**, then the anion (as for any salt).
2. Inside the complex: **ligands in alphabetical order** (ignoring multiplying prefixes), then the metal.
3. Anionic ligands end in **-o**: chlorido, cyanido, hydroxido, oxido, oxalato, nitrito-N (NO_2), nitrito-O (ONO), thiocyanato-S/-N. Neutral: **aqua**, **ammine**, **carbonyl**, **nitrosyl**, en (ethane-1,2-diamine).
4. Prefixes di-, tri-, tetra-; for ligands whose names already contain a number, bis-, tris-, tetrakis-.
5. Oxidation state of the metal in Roman numerals. In an **anionic complex** the metal name ends in **-ate**: ferrate (Fe), cuprate (Cu), argentate (Ag), aurate (Au), plumbate (Pb), stannate (Sn), cobaltate, nickelate, zincate.

Formula	IUPAC name
$[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$	pentaamminechloridocobalt(III) chloride
$\text{K}_3[\text{Fe}(\text{CN})_6]$	potassium hexacyanidoferrate(III)
$\text{K}_4[\text{Fe}(\text{CN})_6]$	potassium hexacyanidoferrate(II)
$[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$	diamminedichloridoplatinum(II)
$\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$	potassium trioxalatoferrate(III)
$[\text{Co}(\text{en})_3]^{3+}$	tris(ethane-1,2-diamine)cobalt(III) ion
$[\text{Ni}(\text{CO})_4]$	tetracarbonylnickel(0)

Isomerism

Σ FORMULA Types

Structural: ionisation ($[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$ / $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$), linkage ($-\text{NO}_2$ / $-\text{ONO}$), coordination ($[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$ / $[\text{Cr}(\text{NH}_3)_6][\text{Co}(\text{CN})_6]$), solvate/hydrate ($[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ violet / $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$ grey-green / $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$ dark green).

Stereo: geometrical (cis/trans, fac/mer) and optical (non-superimposable mirror images).

Complex type	Geometrical isomers	Optical?
square planar MA_2B_2	2 (cis, trans)	no
square planar MABCD	3	no
tetrahedral (any)	0	only if 4 different groups (rare)
octahedral MA_4B_2	2 (cis, trans)	no
octahedral MA_3B_3	2 (fac, mer)	no
octahedral $\text{M}(\text{AA})_2\text{B}_2$	2	cis form is chiral (1 pair)
octahedral $\text{M}(\text{AA})_3$	—	yes (Δ and Λ)
octahedral $\text{MA}_2\text{B}_2\text{C}_2$	5	one of them is chiral (6 stereoisomers in total)

Valence bond theory: standard examples

Complex	Metal d count	Hybridisation	Shape	Unpaired e^-	Magnetic
$[\text{Ni}(\text{CN})_4]^{2-}$	d^8	dsp^2	square planar	0	dia
$[\text{NiCl}_4]^{2-}$	d^8	sp^3	tetrahedral	2	para (2.83 BM)
$[\text{Ni}(\text{CO})_4]$	d^{10} (Ni^0)	sp^3	tetrahedral	0	dia
$[\text{Fe}(\text{CN})_6]^{3-}$	d^5	$d^2\text{sp}^3$ (inner)	octahedral	1	para (1.73 BM)
$[\text{FeF}_6]^{3-}$	d^5	sp^3d^2 (outer)	octahedral	5	para (5.92 BM)
$[\text{Co}(\text{NH}_3)_6]^{3+}$	d^6	$d^2\text{sp}^3$	octahedral	0	dia
$[\text{CoF}_6]^{3-}$	d^6	sp^3d^2	octahedral	4	para (4.90 BM)
$[\text{Fe}(\text{CN})_6]^{4-}$	d^6	$d^2\text{sp}^3$	octahedral	0	dia

Crystal field theory

Σ FORMULA Splitting & CFSE

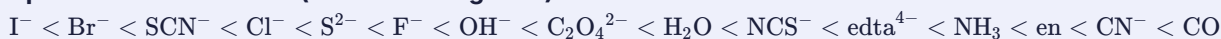
Octahedral: t_{2g} (3 orbitals, $-0.4\Delta_o$ each electron) below e_g (2 orbitals, $+0.6\Delta_o$).

Tetrahedral: reversed (e below t_2), $\Delta_t = \frac{4}{9}\Delta_o$, so tetrahedral complexes are almost always **high spin**.

$$\text{CFSE} = (-0.4 n_{t_{2g}} + 0.6 n_{e_g})\Delta_o \text{ (+ pairing energy terms)}$$

If $\Delta_o > P$ (pairing energy), electrons pair up in t_{2g} (low spin). If $\Delta_o < P$, they occupy e_g first (high spin).

Spectrochemical series (weak → strong field):



☆ IMPORTANT Colour

The colour seen is **complementary** to the light absorbed in the d-d transition. $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ (t_{2g}^1) absorbs in the blue-green region ($\sim 498 \text{ nm}$) and looks violet. Anhydrous CuSO_4 is white; $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is blue (water ligands produce the splitting). Stronger-field ligands give a larger Δ_o , so absorption shifts to shorter wavelength.

☆ IMPORTANT Metal carbonyls

Bonding is **synergic**: CO donates a lone pair to the metal (σ), and the metal back-donates d electrons into CO's π^* orbital (π back-bonding). Result: the M–C bond gains double-bond character, while the C–O bond weakens and lengthens. Examples: $[\text{Ni}(\text{CO})_4]$ (tetrahedral), $[\text{Fe}(\text{CO})_5]$ (trigonal bipyramidal), $[\text{Cr}(\text{CO})_6]$ (octahedral). The metal is in oxidation state 0.

Importance of coordination compounds (NCERT list)

Area	Example
Qualitative analysis	$\text{Ni}^{2+} + \text{dimethylglyoxime} \rightarrow \text{red precipitate}$; $\text{Cu}^{2+} + \text{NH}_3 \rightarrow \text{deep blue } [\text{Cu}(\text{NH}_3)_4]^{2+}$
Hardness of water	EDTA titration of $\text{Ca}^{2+}/\text{Mg}^{2+}$
Metallurgy	extraction of Ag and Au as cyanido complexes, e.g. $[\text{Au}(\text{CN})_2]^-$; purification of Ni via $[\text{Ni}(\text{CO})_4]$
Biology	chlorophyll (Mg), haemoglobin (Fe), vitamin B ₁₂ (Co)
Medicine	cisplatin (anticancer); EDTA for lead poisoning
Catalysis	Wilkinson's catalyst $[(\text{Ph}_3\text{P})_3\text{RhCl}]$ for hydrogenation of alkenes
Photography	$[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-}$ formed when fixing with hypo

Common traps

⚠ MISTAKE ALERT Mistake alerts

- Oxidation state vs coordination number: in $[\text{Co}(\text{en})_3]^{3+}$ the CN is **6** (3 bidentate ligands), not 3.
- Only ions **outside** the square brackets precipitate with AgNO_3 .
- A strong-field ligand does **not** always mean inner-orbital hybridisation. Check the d count: $[\text{Ni}(\text{NH}_3)_6]^{2+}$ (d^8) is sp^3d^2 whatever the ligand.
- Tetrahedral complexes: essentially always high spin, and no geometrical isomers.
- Alphabetical order of ligands ignores prefixes: "tetraammine" is ordered under **a**.

Question patterns

📊 PYQ PATTERN Recurring structures

1. IUPAC names $\leftrightarrow$ formulae (including the -ate endings).
2. Isomer counts (geometrical/optical) for octahedral and square-planar complexes.
3. Hybridisation, geometry, magnetic moment (VBT) of standard complexes.
4. CFT: CFSE, high/low spin, colour, spectrochemical series ordering.
5. Werner-type questions (moles of AgCl precipitated, conductivity/number of ions).
6. Carbonyl bonding; applications (NCERT list).

Practice questions

📝 C12 · Practice

1 Easy ⌚ 0.75 min Basic

C12-01

The IUPAC name of $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ is:

- (A) potassium ferrioxalate(III) (B) potassium trioxalatoiron(III)
 (C) tripotassium oxalatoferrate(II) (D) potassium trioxalatoferrate(III)

2 Easy 0.75 min NV

C12-02

How many moles of AgCl precipitate when excess AgNO_3 is added to a solution containing 1 mol of $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$?

Your answer (numerical value):

3 Easy 0.5 min Speed

C12-03

The number of geometrical isomers of $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ is:

- (A) 1 (B) 4 (C) 3 (D) 2

4 Medium 1 min Concept

C12-04

Which complex is optically active?

- (A) $\text{cis}-[\text{Co}(\text{en})_2\text{Cl}_2]^+$ (B) $\text{trans}-[\text{Co}(\text{en})_2\text{Cl}_2]^+$
(C) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (D) $\text{trans}-[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$

5 Easy 0.75 min Basic

C12-05

$[\text{Ni}(\text{CN})_4]^{2-}$ is:

- (A) sp^3 , tetrahedral, paramagnetic (B) sp^3 , tetrahedral, diamagnetic
(C) dsp^2 , square planar, diamagnetic (D) dsp^2 , square planar, paramagnetic

6 Easy 0.75 min Basic

C12-06

The spin-only magnetic moment of $[\text{NiCl}_4]^{2-}$ is about:

- (A) 0 BM (B) 2.83 BM (C) 1.73 BM (D) 3.87 BM

7 Easy 0.5 min Speed

C12-07

Which ligand produces the largest crystal-field splitting?

- (A) H_2O (B) NH_3 (C) CO (D) F^-

8 Medium 1 min Calc

C12-08

Ignoring pairing energy, the CFSE of a low-spin octahedral d^6 complex is:

- (A) $-2.4\Delta_o$ (B) $-0.4\Delta_o$ (C) $-1.2\Delta_o$ (D) 0

9 Easy 0.5 min Concept

C12-09

Linkage isomerism is possible with the ligand:

- (A) NH_3 (B) H_2O (C) NO_2^- (D) en

10 Easy 0.5 min NV

C12-10

Find the denticity of the EDTA^{4-} ligand.

Your answer (numerical value):

11 Easy 0.75 min Concept

C12-11

In $[\text{Fe}(\text{CO})_5]$, the oxidation state and coordination number of Fe are:

- (A) +2 and 5 (B) 0 and 5 (C) 0 and 10 (D) +5 and 5

12 Medium 1 min Concept

C12-12

The violet colour of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ is due to:

- (A) charge transfer from water
 (B) a d-d transition $t_{2g}^1 \rightarrow e_g^1$
 (C) an unpaired electron alone
 (D) the Ti^{3+} ion's high charge

C12 · Chapter Test

1 Easy 0.5 min Speed

C12-T1

Which ligand can form a chelate?

- (A) Cl^-
 (B) NH_3
 (C) H_2O
 (D) ethane-1,2-diamine (en)

2 Easy 0.75 min Concept

C12-T2

$[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$ and $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$ are:

- (A) ionisation isomers
 (B) linkage isomers
 (C) coordination isomers
 (D) geometrical isomers

3 Easy 0.5 min NV

C12-T3

Find the coordination number of Co in $[\text{Co}(\text{en})_3]^{3+}$.

Your answer (numerical value):

4 Easy 0.5 min Speed

C12-T4

Cisplatin, $\text{cis-}[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$, is used:

- (A) in the treatment of cancer
 (B) as a hydrogenation catalyst
 (C) to treat lead poisoning
 (D) in photography

5 Medium 0.75 min Concept

C12-T5

In metal carbonyls, π back-bonding from the metal to CO:

- (A) strengthens the C-O bond
 (B) weakens the C-O bond and strengthens the M-C bond
 (C) has no effect on bond strengths
 (D) breaks the M-C σ bond

Answers & Solutions

C12 · Practice

- 1 (D) 2 2 3 (D) 4 (A) 5 (C) 6 (B) 7 (C) 8 (A) 9 (C) 10 6
 11 (B) 12 (B)

C12 · Chapter Test

- 1 (D) 2 (A) 3 6 4 (A) 5 (B)

C12 · PRACTICE

1 Answer: (D) IUPAC name **Easy** ⌚ 0.75 min **Basic** C12-01

SOLUTION Anionic complex, so "ferrate". Fe is +3 because $3(+1) + x + 3(-2) = 0$.

COMMON TRAP Using "iron" in an anionic complex.

2 Answer: 2 Werner-type precipitation **Easy** ⌚ 0.75 min **NV** C12-02

SOLUTION Only the two chloride ions outside the coordination sphere are ionisable.

COMMON TRAP Counting 3.

3 Answer: (D) Geometrical isomers **Easy** ⌚ 0.5 min **Speed** C12-03

SOLUTION Octahedral MA_4B_2 : cis and trans.

4 Answer: (A) Optical activity **Medium** ⌚ 1 min **Concept** C12-04

SOLUTION The cis isomer has no plane of symmetry, so its mirror image is non-superimposable. The trans isomer has a mirror plane.

5 Answer: (C) VBT of $[Ni(CN)_4]^{2-}$ **Easy** ⌚ 0.75 min **Basic** C12-05

SOLUTION Ni^{2+} is d^8 . Strong-field CN^- pairs the electrons, freeing one 3d orbital for dsp^2 hybridisation.

COMMON TRAP Mixing it up with $[NiCl_4]^{2-}$.

6 Answer: (B) Magnetic moment of $[NiCl_4]^{2-}$ **Easy** ⌚ 0.75 min **Basic** C12-06

SOLUTION Tetrahedral, weak field: d^8 with 2 unpaired electrons, giving $\sqrt{8} \approx 2.83$ BM.

7 Answer: (C) Strongest field ligand **Easy** ⌚ 0.5 min **Speed** C12-07

SOLUTION CO is at the strong end of the spectrochemical series (π -acceptor).

8 Answer: (A) CFSE of low-spin d^6 **Medium** ⌚ 1 min **Calc** C12-08

SOLUTION $t_{2g}^6 e_g^0$: $6 \times (-0.4) = -2.4\Delta_o$.

COMMON TRAP Using the high-spin value ($t_{2g}^4 e_g^2 = -0.4\Delta_o$).

9 Answer: (C) Linkage isomerism **Easy** ⌚ 0.5 min **Concept** C12-09

SOLUTION NO_2^- is ambidentate: it can bind through N (nitrito-N) or through O (nitrito-O).

10 Answer: 6 Denticity of EDTA **Easy** ⌚ 0.5 min **NV** C12-10

SOLUTION 2 N donors + 4 carboxylate O donors.

11 Answer: (B) Metal carbonyls **Easy** ⌚ 0.75 min **Concept** C12-11

SOLUTION CO is neutral, so Fe is 0. Five CO ligands give CN 5 (trigonal bipyramidal).

12 Answer: (B) Origin of colour **Medium** ⌚ 1 min **Concept** C12-12

SOLUTION The single d electron absorbs light around 498 nm, and the complementary colour (violet) is seen.

C12 · CHAPTER TEST

1 Answer: (D) Chelating ligand **Easy** ⌚ 0.5 min **Speed** C12-T1

SOLUTION A bidentate ligand forms a 5-membered ring with the metal.

2 Answer: (A) Ionisation isomers **Easy** ⌚ 0.75 min **Concept** C12-T2

SOLUTION They swap an ion inside the coordination sphere with one outside it, so they give different ions in solution.

3 Answer: 6 Coordination number with bidentates **Easy** ⌚ 0.5 min **NV** C12-T3

SOLUTION 3 bidentate ligands $\times$ 2 donor atoms = 6.

COMMON TRAP Answering 3.

4 Answer: (A) Cisplatin **Easy** ⌚ 0.5 min **Speed** C12-T4

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

COMMON TRAP Confusing it with EDTA (lead poisoning) or Wilkinson's catalyst (hydrogenation).

5 Answer: (B) *Back-bonding effect* **Medium** ⌚ 0.75 min **Concept**

C12-T5

SOLUTION Electron density enters CO's π^* (antibonding) orbital, which lowers the C–O bond order.

 **REVISION** Revision tracker · Coordination Compounds

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III · CHEMISTRY · UNIT 13

Purification & Characterisation of Organic Compounds

13

0–1 per shift

TYPICAL QUESTIONS

Easy

DIFFICULTY

High (fast marks)

PRIORITY

Class 11 · Ch 8
(purification,
qualitative &
quantitative
analysis)

NCERT

~4 hours

STUDY TIME

Purification methods

Method	Principle	Classic examples
Crystallisation	different solubilities at high/low temperature	sugar, benzoic acid; <i>fractional</i> crystallisation for mixtures
Sublimation	solid → vapour directly	camphor, naphthalene, anthracene, benzoic acid
Simple distillation	boiling points differ widely	chloroform (334 K) + aniline (457 K)
Fractional distillation	boiling points close	crude oil fractions
Distillation under reduced pressure	lowers the boiling point; avoids decomposition	glycerol from spent lye (soap industry)
Steam distillation	steam-volatile, water-immiscible compounds; the mixture boils when $p_{\text{water}} + p_{\text{organic}} = p_{\text{atm}}$ (below 373 K)	aniline , o-nitrophenol, essential oils
Differential extraction	distribution between two immiscible solvents	organic compound from water using ether (separating funnel)
Chromatography	differential adsorption/partition between stationary and mobile phases	column, TLC (adsorption); paper (partition)

Retardation factor: $R_f = \frac{\text{distance moved by compound}}{\text{distance moved by solvent front}}$ (always < 1). Colourless spots are made visible with UV light, iodine vapour or reagent sprays (ninhydrin for amino acids).

Qualitative analysis: Lassaigne's (sodium fusion) test

Σ FORMULA Reactions in the sodium extract



Element	Test	Observation
N	extract + FeSO_4 , boil, acidify (H_2SO_4)	Prussian blue , $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$
S	+ lead acetate / acetic acid	black PbS
S	+ sodium nitroprusside	violet colour
N and S together	+ FeCl_3	blood red , $[\text{Fe}(\text{SCN})]^{2+}$
Halogens	boil with HNO_3 (removes $\text{CN}^-/\text{S}^{2-}$), then AgNO_3	AgCl white (soluble in NH_3), AgBr pale yellow (sparingly soluble), AgI yellow (insoluble)
P	oxidise with Na_2O_2 , then HNO_3 + ammonium molybdate	yellow ammonium phosphomolybdate

Quantitative analysis (basic principles)

Σ FORMULA Percentage formulas (m = mass of compound)

Element	Method	Formula
C	Liebig combustion $\rightarrow \text{CO}_2$ (absorbed by KOH)	$\%C = \frac{12}{44} \times \frac{m_{\text{CO}_2}}{m} \times 100$
H	Liebig $\rightarrow \text{H}_2\text{O}$ (absorbed by anhydrous CaCl_2)	$\%H = \frac{2}{18} \times \frac{m_{\text{H}_2\text{O}}}{m} \times 100$
N	Dumas $\rightarrow \text{N}_2$ volume at STP	$\%N = \frac{28}{22400} \times \frac{V_{\text{N}_2} (\text{mL, STP})}{m} \times 100$
N	Kjeldahl $\rightarrow \text{NH}_3$ absorbed in acid	$\%N = \frac{1.4 \times N_{\text{acid}} \times V_{\text{acid used}} (\text{mL})}{m}$
Halogen X	Carius $\rightarrow \text{AgX}$	$\%X = \frac{A_X}{M_{\text{AgX}}} \times \frac{m_{\text{AgX}}}{m} \times 100$
S	Carius $\rightarrow \text{BaSO}_4$	$\%S = \frac{32}{233} \times \frac{m_{\text{BaSO}_4}}{m} \times 100$
P	$\rightarrow \text{Mg}_2\text{P}_2\text{O}_7$	$\%P = \frac{62}{222} \times \frac{m_{\text{Mg}_2\text{P}_2\text{O}_7}}{m} \times 100$
O	by difference	$100 - (\text{sum of the others})$

Kjeldahl's method fails for nitro and azo compounds and for N in a ring (pyridine), because that N isn't converted to ammonium sulphate.

Common traps

⚠ MISTAKE ALERT Mistake alerts

- In the Kjeldahl formula, N_{acid} is the **normality**: $1 \text{ M H}_2\text{SO}_4 = 2 \text{ N}$.
- Blood-red colour (N and S together) arises only if there is **less** sodium than needed for separate NaCN and Na_2S .
- Boil with HNO_3 before the halogen test only when N or S is present.
- Steam distillation works for compounds that are **immiscible** with water and volatile in steam.

Question patterns

📊 PYQ PATTERN Recurring structures

- Matching method $\leftrightarrow$ mixture (glycerol, aniline, camphor, crude oil).
- Lassaigne colours and the species responsible.
- %N (Kjeldahl/Dumas), %X (Carius), %C/%H numericals: frequent Section-B items.
- R_f calculation and interpretation.
- Limitations of Kjeldahl's method.

Practice questions

✍ C13 · Practice

1 Easy 0.5 min Speed

C13-01

Glycerol is recovered from spent lye in the soap industry by:

- (A) steam distillation (B) distillation under reduced pressure
(C) sublimation (D) crystallisation

2 Easy 0.5 min Speed

C13-02

Aniline is purified by:

- (A) sublimation (B) steam distillation
(C) chromatography only (D) simple crystallisation

3 Easy 0.5 min Speed

C13-03

The Prussian blue colour in Lassaigne's test for nitrogen is due to:

- (A) $\text{Na}_4[\text{Fe}(\text{CN})_6]$ (B) $\text{Fe}(\text{SCN})_3$ (C) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$ (D) $\text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}]$

4 Easy 0.5 min Concept

C13-04

A blood-red colour with FeCl_3 in the sodium extract shows the presence of:

- (A) both N and S (B) S only (C) N only (D) halogen

5 Medium 1.5 min NV

C13-05

In a Kjeldahl estimation, the ammonia from 0.5 g of an organic compound neutralises 12.5 mL of 0.5 M H_2SO_4 . Find the percentage of nitrogen.

Your answer (numerical value):

6 Medium 1.5 min NV

C13-06

0.1 g of an organic compound gives 0.287 g of AgCl in the Carius method. Find the percentage of chlorine (nearest integer).

Your answer (numerical value):

7 Easy 0.5 min Speed

C13-07

In a TLC run, a spot moves 3 cm while the solvent front moves 5 cm. The R_f value is:

- (A) 2 (B) 1.67 (C) 0.4 (D) 0.6

8 Easy 0.5 min Concept

C13-08

Kjeldahl's method can't be used to estimate nitrogen in:

- (A) nitrobenzene (B) acetamide (C) urea (D) aniline

9 Easy 1 min Basic

C13-09

0.2 g of an organic compound gives 0.44 g of CO_2 on combustion. The percentage of carbon is:

- (A) 30% (B) 44% (C) 60% (D) 12%



C13 · Chapter Test

1 Easy 0.5 min Speed

C13-T1

Which substance is purified by sublimation?

- (A) sugar (B) aniline (C) glycerol (D) naphthalene

2 Easy 0.5 min Speed

C13-T2

Silver bromide precipitate is:

- (A) pale yellow (B) white
(C) yellow, insoluble in ammonia (D) black

3 Medium 1.5 min NV

C13-T3

0.3 g of an organic compound gives 50 mL of N_2 at STP in the Dumas method. Find the percentage of nitrogen (nearest integer).

Your answer (numerical value):

4 Easy 0.5 min Concept

C13-T4

In steam distillation the mixture boils:

- (A) below 373 K, when the sum of the vapour pressures equals atmospheric pressure
(B) at the boiling point of the organic compound
(C) above 373 K
(D) only under reduced pressure

5 Easy 0.5 min Speed

C13-T5

Phosphorus in an organic compound is detected as a yellow precipitate with:

- (A) lead acetate
(B) silver nitrate
(C) ammonium molybdate (after oxidation to phosphate)
(D) sodium nitroprusside

Answers & Solutions

C13 · Practice

1 (B) 2 (B) 3 (C) 4 (A) 5 35 6 71 7 (D) 8 (A) 9 (C)

C13 · Chapter Test

1 (D) 2 (A) 3 21 4 (A) 5 (C)

C13 · PRACTICE

1 Answer: (B) Glycerol from spent lye Easy 0.5 min Speed

C13-01

SOLUTION Glycerol decomposes at its normal boiling point, so it is distilled at lower pressure (lower boiling point).

2 Answer: (B) Purifying aniline Easy 0.5 min Speed

C13-02

SOLUTION Aniline is steam-volatile and immiscible with water.

3 Answer: (C) Lassaigne test for N Easy 0.5 min Speed

C13-03

SOLUTION Ferric ferrocyanide.

4 Answer: (A) Blood-red colour Easy 0.5 min Concept

C13-04

SOLUTION $NaSCN$ forms, which gives $[Fe(SCN)]^{2+}$ with Fe^{3+} .

5 Answer: 35 Kjeldahl calculation Medium 1.5 min NV

C13-05

SOLUTION $0.5 \text{ M H}_2\text{SO}_4 = 1 \text{ N}$. $\%N = \frac{1.4 \times 1 \times 12.5}{0.5} = 35\%$.

COMMON TRAP Using molarity (0.5) as normality gives 17.5%.

6 Answer: 71 *Carius method* **Medium** ⌚ 1.5 min **NV**

C13-06

SOLUTION $\%Cl = \frac{35.5}{143.5} \times \frac{0.287}{0.1} \times 100 = 0.2474 \times 287 \approx 71\%$.

7 Answer: (D) *R_f value* **Easy** ⌚ 0.5 min **Speed**

C13-07

SOLUTION $R_f = 3/5$.

8 Answer: (A) *Kjeldahl limitation* **Easy** ⌚ 0.5 min **Concept**

C13-08

SOLUTION The N of nitro groups (and of azo compounds and pyridine) isn't converted into ammonium sulphate.

9 Answer: (C) *Percentage of carbon* **Easy** ⌚ 1 min **Basic**

C13-09

SOLUTION $\frac{12}{44} \times \frac{0.44}{0.2} \times 100 = 60\%$.

C13 · CHAPTER TEST

1 Answer: (D) *Sublimation* **Easy** ⌚ 0.5 min **Speed**

C13-T1

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

2 Answer: (A) *Silver halide colours* **Easy** ⌚ 0.5 min **Speed**

C13-T2

SOLUTION AgCl is white, AgBr pale yellow, AgI yellow.

3 Answer: 21 *Dumas method* **Medium** ⌚ 1.5 min **NV**

C13-T3

SOLUTION $\%N = \frac{28 \times 50}{22400 \times 0.3} \times 100 \approx 20.8 \approx 21\%$.

4 Answer: (A) *Steam distillation principle* **Easy** ⌚ 0.5 min **Concept**

C13-T4

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

5 Answer: (C) *Detection of phosphorus* **Easy** ⌚ 0.5 min **Speed**

C13-T5

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

REVISION Revision tracker · Purification & Characterisation

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III · CHEMISTRY · UNIT 14

Basic Principles of Organic Chemistry (GOC)

14

1–2 per shift (+
underlies all
organic)
TYPICAL QUESTIONS

Medium
DIFFICULTY

Must-do
PRIORITY

Class 11 · Ch 8
NCERT

~12 hours
STUDY TIME

PYQ PATTERN Weight evidence

27 GOC questions across the 2026 Paper-1 shifts ($\approx 5.7\%$ of Chemistry)  THIRD-PARTY ANALYSIS [T-WT-CD].
GOC logic (stability, acidity, basicity, electronic effects) also decides most reaction-mechanism questions in Units 15–18.

Important concepts

- **Tetravalency & hybridisation:** sp^3 (25% s, tetrahedral, 109.5°), sp^2 (33% s, trigonal, 120°), sp (50% s, linear, 180°). More s-character makes carbon more electronegative, so sp C–H bonds are more acidic.
- **Classification & homologous series:** functional groups; successive members differ by $-\text{CH}_2-$.
- **Isomerism:** structural (chain, position, functional, metamerism, tautomerism) and stereoisomerism (geometrical, optical).
- **Bond fission:** homolytic $\rightarrow$ free radicals; heterolytic $\rightarrow$ carbocations/carbanions.
- **Electronic effects:** inductive (permanent, through σ bonds, falls off with distance), electromeric (temporary, on demand), resonance/mesomeric (π delocalisation), hyperconjugation (σ – π delocalisation).
- **Reaction types:** substitution, addition, elimination, rearrangement.

IUPAC nomenclature essentials

1. **Principal chain:** the longest chain containing the principal functional group (and as many multiple bonds as possible).
2. **Lowest locants:** principal group first, then multiple bonds, then substituents (lowest-set rule).
3. **Suffix priority** (highest first): carboxylic acid > sulphonic acid > anhydride > ester > acyl halide > amide > nitrile > aldehyde > ketone > alcohol > amine. Halo, nitro and alkoxy groups are always prefixes.
4. Substituent prefixes in **alphabetical** order (di-, tri- ignored).

Examples: $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{COOH}$ = 3-hydroxybutanoic acid; $\text{CH}_2=\text{CHCHO}$ = prop-2-enal; $\text{CH}_3\text{COCH}_2\text{CH}_2\text{Cl}$ = 4-chlorobutan-2-one.

Stability orders (the core of GOC)

FORMULA Reactive intermediates

Intermediate	Hybridisation / shape	Stability order	Stabilised by
Carbocation	sp^2 , planar	benzyl $\approx$ allyl (resonance) > 3° > 2° > 1° > methyl > vinyl, aryl	+I, hyperconjugation, +R
Free radical	sp^2 , planar (nearly)	benzyl $\approx$ allyl > 3° > 2° > 1° > methyl	hyperconjugation, resonance
Carbanion	sp^3 , pyramidal	methyl > 1° > 2° > 3° (reverse)	–I, –R (e.g. next to C=O, NO_2)

Triphenylmethyl cation/radical/anion are exceptionally stable (extensive resonance).

Σ FORMULA Electronic-effect rankings

–I: $-\text{NO}_2 > -\text{CN} > -\text{COOH} > -\text{F} > -\text{Cl} > -\text{Br} > -\text{I} > -\text{OH} > -\text{OR} > -\text{C}_6\text{H}_5$

+I: $-\text{C}(\text{CH}_3)_3 > -\text{CH}(\text{CH}_3)_2 > -\text{CH}_2\text{CH}_3 > -\text{CH}_3$

+R (electron-donating by resonance): $-\text{O}- > -\text{NH}_2 > -\text{OH} > -\text{OR} > -\text{NHCOR}$, halogens (weak +R, strong –I)

–R (electron-withdrawing): $-\text{NO}_2$, $-\text{CN}$, $-\text{CHO}$, $-\text{COR}$, $-\text{COOH}$, $-\text{SO}_3\text{H}$

Hyperconjugation: the number of α -hydrogens decides it: $\text{CH}_3^+ \ 0 < \text{CH}_3\text{CH}_2^+ \ 3 < (\text{CH}_3)_2\text{CH}^+ \ 6 < (\text{CH}_3)_3\text{C}^+ \ 9$.

☆ IMPORTANT Electrophiles & nucleophiles

Electrophiles (electron-seeking): H^+ , NO_2^+ , Cl^+ , carbocations, BF_3 , AlCl_3 , carbonyl carbon.

Nucleophiles (nucleus-seeking): OH^- , RO^- , CN^- , Cl^- , NH_3 , H_2O , ROH , RNH_2 , carbanions.

Isomerism quick reference

Type	Example
Chain	butane / 2-methylpropane
Position	propan-1-ol / propan-2-ol
Functional	ethanol / methoxymethane; propanal / propanone
Metamerism	diethyl ether / methyl propyl ether
Tautomerism	keto–enol: propanone $\rightleftharpoons$ prop-1-en-2-ol (needs an α -H)
Geometrical	<i>cis/trans</i> -but-2-ene (needs different groups on each double-bond carbon)
Optical	butan-2-ol (one chiral C); lactic acid

Counting stereoisomers: up to 2^n for n chiral centres, fewer when **meso** forms exist (2,3-dibromobutane and tartaric acid have 3: a d/l pair + meso). Meso compounds have an internal mirror plane and are optically inactive. A racemic mixture (50 : 50) is optically inactive by external compensation.

Common traps**⚠ MISTAKE ALERT Mistake alerts**

- Halogens: –I dominates +R, so they deactivate rings, but they still direct ortho/para.
- Vinyl and aryl carbocations are **very unstable** (sp carbon, no resonance help).
- Carbanion stability order is the **reverse** of the carbocation order.
- Geometrical isomerism: but-1-ene has none (two H on C1).
- A molecule with chiral centres may still be achiral (meso).

Question patterns**📊 PYQ PATTERN Recurring structures**

1. Ranking carbocation/radical/carbanion stability.
2. Ranking acidity/basicity using I, R and hyperconjugation effects.
3. IUPAC names of polyfunctional compounds.
4. Counting isomers (structural, stereo); identifying chiral/meso compounds.
5. Classifying reagents (electrophile/nucleophile) and reaction types.
6. Hyperconjugation structures/ α -H counting; resonance structures and their relative stability.

Practice questions

C14 · Practice

1 Medium ⌚ 1 min Concept C14-01

In allene, $\text{CH}_2=\text{C}=\text{CH}_2$, the hybridisations of C1, C2 and C3 are:

- (A) $\text{sp}^2, \text{sp}^2, \text{sp}^2$ (B) $\text{sp}^3, \text{sp}, \text{sp}^3$ (C) $\text{sp}, \text{sp}, \text{sp}$ (D) $\text{sp}^2, \text{sp}, \text{sp}^2$

2 Easy ⌚ 0.75 min Basic C14-02

The IUPAC name of $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{COOH}$ is:

- (A) 3-hydroxybutanoic acid (B) 2-hydroxybutanoic acid
(C) 3-carboxybutan-2-ol (D) β -hydroxybutyric acid

3 Easy ⌚ 0.75 min Concept C14-03

The most stable carbocation is:

- (A) CH_3CH_2^+ (B) $(\text{CH}_3)_2\text{CH}^+$ (C) $\text{C}_6\text{H}_5\text{CH}_2^+$ (D) $\text{CH}_2=\text{CH}^+$

4 Easy ⌚ 0.5 min Speed C14-04

The correct order of $-\text{I}$ effect is:

- (A) $-\text{F} > -\text{NO}_2 > -\text{CN} > -\text{Cl}$ (B) $-\text{NO}_2 > -\text{CN} > -\text{F} > -\text{Cl}$
(C) $-\text{Cl} > -\text{F} > -\text{CN} > -\text{NO}_2$ (D) $-\text{CN} > -\text{NO}_2 > -\text{Cl} > -\text{F}$

5 Medium ⌚ 1 min NV C14-05

Find the total number of stereoisomers of 2,3-dibromobutane.

Your answer (numerical value):

6 Easy ⌚ 0.5 min NV C14-06

How many α -hydrogens are available for hyperconjugation in the tert-butyl cation, $(\text{CH}_3)_3\text{C}^+$?

Your answer (numerical value):

7 Easy ⌚ 0.5 min Speed C14-07

Which compound shows geometrical isomerism?

- (A) but-2-ene (B) but-1-ene (C) 2-methylpropene (D) ethene

8 Easy ⌚ 0.5 min Speed C14-08

Which species is an electrophile?

- (A) NH_3 (B) OH^- (C) NO_2^+ (D) CN^-

9 Easy ⌚ 0.75 min Concept C14-09

Among these alkyl carbanions, the most stable is:

- (A) $(\text{CH}_3)_2\text{CH}^-$ (B) CH_3CH_2^- (C) CH_3^- (D) $(\text{CH}_3)_3\text{C}^-$

10 Easy 0.5 min Concept

C14-10

Which compound shows keto-enol tautomerism?

- (A) CH_3COCH_3 (B) $(\text{CH}_3)_3\text{C}-\text{CHO}$ (C) $\text{C}_6\text{H}_5\text{CHO}$ (D) HCHO

11 Easy 0.5 min Speed

C14-11

Which molecule is chiral?

- (A) propan-2-ol (B) pentan-3-ol
(C) 2-methylpropan-2-ol (D) butan-2-ol

C14 · Chapter Test

1 Easy 0.5 min Speed

C14-T1

Diethyl ether and methyl propyl ether are:

- (A) chain isomers (B) position isomers
(C) functional isomers (D) metamers

2 Easy 0.5 min Speed

C14-T2

Homolytic cleavage of a covalent bond produces:

- (A) carbocations (B) free radicals (C) carbanions (D) ion pairs

3 Easy 0.5 min NV

C14-T3

How many structural isomers does C_5H_{12} have?

Your answer (numerical value):

4 Easy 0.5 min Speed

C14-T4

Which functional group gets the highest priority as the suffix in IUPAC naming?

- (A) $-\text{OH}$ (B) $-\text{CHO}$ (C) $>\text{C}=\text{O}$ (D) $-\text{COOH}$

5 Easy 0.5 min Speed

C14-T5

Which group shows a +R (electron-donating resonance) effect?

- (A) $-\text{NO}_2$ (B) $-\text{OH}$ (C) $-\text{CHO}$ (D) $-\text{CN}$

Answers & Solutions

C14 · Practice

- 1 (D) 2 (A) 3 (C) 4 (B) 5 3 6 9 7 (A) 8 (C) 9 (C) 10 (A)
11 (D)

C14 · Chapter Test

- 1 (D) 2 (B) 3 3 4 (D) 5 (B)

C14 · PRACTICE

1 Answer: (D) Hybridisation in allene Medium 1 min Concept

C14-01

SOLUTION The central carbon forms two π bonds (sp). The terminal carbons form one π bond each (sp^2).

2 Answer: (A) IUPAC naming **Easy** ⌚ 0.75 min **Basic** C14-02

SOLUTION -COOH is the principal group, so it is C1. The OH is on C3.

COMMON TRAP Choosing (D), which is the common name.

3 Answer: (C) Carbocation stability **Easy** ⌚ 0.75 min **Concept** C14-03

SOLUTION The benzyl cation is resonance-stabilised over the ring. Vinyl is the least stable.

4 Answer: (B) $-I$ effect order **Easy** ⌚ 0.5 min **Speed** C14-04

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

5 Answer: 3 Stereoisomers with a meso form **Medium** ⌚ 1 min **NV** C14-05

SOLUTION Two chiral centres, but the molecule is symmetric: a pair of enantiomers + one meso form = 3.

COMMON TRAP Answering $2^2 = 4$.

6 Answer: 9 α -Hydrogens **Easy** ⌚ 0.5 min **NV** C14-06

SOLUTION Three methyl groups $\times$ 3 H.

7 Answer: (A) Geometrical isomerism **Easy** ⌚ 0.5 min **Speed** C14-07

SOLUTION Each double-bond carbon carries two different groups (H and CH_3).

8 Answer: (C) Electrophile **Easy** ⌚ 0.5 min **Speed** C14-08

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

9 Answer: (C) Carbanion stability **Easy** ⌚ 0.75 min **Concept** C14-09

SOLUTION Alkyl groups (+I) destabilise a negative charge.

SHORTCUT Carbanion order is the reverse of the carbocation order.

10 Answer: (A) Keto-enol tautomerism **Easy** ⌚ 0.5 min **Concept** C14-10

SOLUTION Tautomerism needs an α -hydrogen next to the carbonyl group. Acetone has six. The others have none.

11 Answer: (D) Chirality **Easy** ⌚ 0.5 min **Speed** C14-11

SOLUTION C2 of butan-2-ol carries H, OH, CH_3 and C_2H_5 : four different groups.

C14 · CHAPTER TEST

1 Answer: (D) Metamers **Easy** ⌚ 0.5 min **Speed** C14-T1

SOLUTION Same functional group, different alkyl groups on either side of it.

2 Answer: (B) Homolytic fission **Easy** ⌚ 0.5 min **Speed** C14-T2

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

3 Answer: 3 Chain isomers of pentane **Easy** ⌚ 0.5 min **NV** C14-T3

SOLUTION n-pentane, isopentane (2-methylbutane), neopentane (2,2-dimethylpropane).

4 Answer: (D) Principal group priority **Easy** ⌚ 0.5 min **Speed** C14-T4

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

5 Answer: (B) +R group **Easy** ⌚ 0.5 min **Speed** C14-T5

SOLUTION O's lone pair delocalises into the π system.

REVISION Revision tracker · GOC

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III · CHEMISTRY · UNIT 15

Hydrocarbons

15

~1 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

High

PRIORITY

Class 11 · Ch 9

NCERT

~8 hours

STUDY TIME

Alkanes

Σ FORMULA Preparation & reactions

Preparation: hydrogenation of alkenes/alkynes (Ni, Pt or Pd); alkyl halide + Zn/HCl; **Wurtz** $2\text{RX} + 2\text{Na} \xrightarrow{\text{dry ether}} \text{R-R} + 2\text{NaX}$ (poor for unsymmetrical alkanes: gives a mixture); **decarboxylation** $\text{RCOONa} + \text{NaOH} \xrightarrow{\text{CaO}, \Delta} \text{RH} + \text{Na}_2\text{CO}_3$ (one carbon fewer); **Kolbe electrolysis** $2\text{RCOONa} + 2\text{H}_2\text{O} \longrightarrow \text{R-R} + 2\text{CO}_2 + \text{H}_2 + 2\text{NaOH}$ (R-R at the anode).**Reactions:** free-radical **halogenation** (hv). Reactivity $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$; H abstraction $3^\circ > 2^\circ > 1^\circ$.Combustion; controlled oxidation; isomerisation (AlCl_3/HCl); **aromatisation** (n-hexane → benzene over Cr_2O_3 , V_2O_5 or Mo_2O_3 at 773 K); pyrolysis (cracking).

☆ IMPORTANT Free-radical chlorination of methane: mechanism

Initiation: $\text{Cl}_2 \xrightarrow{h\nu} 2\text{Cl}\cdot$ **Propagation:** $\text{Cl}\cdot + \text{CH}_4 \longrightarrow \text{HCl} + \text{CH}_3\cdot$; $\text{CH}_3\cdot + \text{Cl}_2 \longrightarrow \text{CH}_3\text{Cl} + \text{Cl}\cdot$ **Termination:** radicals combine ($\text{Cl}\cdot + \text{Cl}\cdot$, $\text{CH}_3\cdot + \text{CH}_3\cdot \rightarrow$ ethane, which explains traces of ethane in the product).**Conformations of ethane:** staggered (most stable) and eclipsed (least stable), differing by about **12.5 kJ/mol** (torsional strain). They are drawn as **Sawhorse** and **Newman** projections. Free rotation at room temperature means the conformers can't be isolated.

Alkenes

Σ FORMULA Preparation

- Alkynes: **Lindlar's catalyst** (Pd/ CaCO_3 poisoned with S or quinoline) → **cis**-alkene. **Na in liquid NH_3** → **trans**-alkene.
- Dehydrohalogenation:** alkyl halide + alcoholic KOH → alkene (Saytzeff: more substituted alkene is major).
- Dehalogenation** of vicinal dihalides with Zn.
- Acidic dehydration** of alcohols: conc. H_2SO_4 , 443 K ($3^\circ > 2^\circ > 1^\circ$ ease).

🧪 REACTION MAP Reaction map · propene $\text{CH}_3\text{CH}=\text{CH}_2$

H_2 / Ni	→ propane
Br_2 in CCl_4 (anti addition; decolourises)	→ 1,2-dibromopropane
HBr (no peroxide) · Markovnikov	→ 2-bromopropane
HBr + peroxide · anti-Markovnikov (only HBr)	→ 1-bromopropane
H_2O / H^+ · Markovnikov	→ propan-2-ol
cold dil. alkaline KMnO_4 (Baeyer's test)	→ propane-1,2-diol (purple → brown)
hot acidic KMnO_4	→ ethanoic acid + CO_2
O_3 then Zn/ H_2O (ozonolysis)	→ ethanal + methanal
polymerisation	→ polypropene

☆ IMPORTANT Markovnikov & peroxide effect: mechanisms

- **Markovnikov:** H^+ adds first to give the **more stable carbocation** (2° here), and then Br^- attacks it.
- **Peroxide (Kharasch) effect:** $\text{HBr} + \text{ROOR}$ gives a free-radical chain. $\text{Br}^\cdot$ adds first to give the more stable **radical**, so Br ends up on the less substituted carbon. It doesn't occur with HCl or HI (their propagation steps are not both exothermic).

Alkynes

Σ FORMULA Key facts

Preparation: $\text{CaC}_2 + 2\text{H}_2\text{O} \longrightarrow \text{C}_2\text{H}_2 + \text{Ca(OH)}_2$; vicinal dihalide + alc. KOH , then NaNH_2 .

Acidity: terminal alkynes are weakly acidic (sp C-H): $\text{HC}\equiv\text{CH} + \text{Na} \longrightarrow \text{HC}\equiv\text{CNa} + \frac{1}{2}\text{H}_2$. Acidity order: ethyne > ethene > ethane.

Additions: H_2 , X_2 , HX (Markovnikov, twice $\rightarrow$ gem-dihalide).

Hydration: $\text{HgSO}_4/\text{dil. H}_2\text{SO}_4$, 333 K. Ethyne $\rightarrow$ **ethanal**; propyne $\rightarrow$ **propanone** (via the enol).

Cyclic polymerisation: $3\text{C}_2\text{H}_2 \rightarrow$ benzene (red-hot iron tube, 873 K).

Aromatic hydrocarbons

Σ FORMULA Benzene & aromaticity

All C–C bonds are equal (139 pm) because of resonance. **Hückel's rule:** planar, cyclic, fully conjugated, with $(4n + 2)\pi$ electrons $\rightarrow$ aromatic. Examples: benzene (6), naphthalene (10), cyclopentadienyl **anion** (6), tropylium **cation** (6), pyridine. The cyclopentadienyl cation (4) is antiaromatic; cyclooctatetraene is non-planar (tub-shaped), so non-aromatic.

Preparation: cyclic polymerisation of ethyne; decarboxylation of sodium benzoate (soda lime); reduction of phenol with Zn dust.

Σ FORMULA Electrophilic aromatic substitution (EAS)

Mechanism: (1) generate E^+ $\rightarrow$ (2) attack forms the **arenium (σ) complex** (slow, rate-determining) $\rightarrow$ (3) loss of H^+ restores aromaticity.

Reaction	Reagents	Electrophile
Nitration	conc. HNO_3 + conc. H_2SO_4	NO_2^+
Halogenation	Cl_2/Br_2 + $\text{FeCl}_3/\text{AlCl}_3$ (dark, cold)	Cl^+ (Br^+)
Sulphonation	fuming H_2SO_4	SO_3
Friedel–Crafts alkylation	RX + anhydrous AlCl_3	R^+ (rearranges ; polyalkylation possible)
Friedel–Crafts acylation	RCOCl + anhydrous AlCl_3	$\text{RC}\equiv\text{O}^+$ (acylium; no rearrangement)

Addition to benzene: H_2/Ni (high T , P) $\rightarrow$ cyclohexane; $\text{Cl}_2 + \text{UV} \rightarrow$ benzene hexachloride (BHC, $\text{C}_6\text{H}_6\text{Cl}_6$).

Σ FORMULA Directive influence of substituents

Group type	Examples	Directs to	Ring activity
Strongly activating	$-\text{O}-$, $-\text{NH}_2$, $-\text{NHR}$, $-\text{OH}$, $-\text{OR}$	ortho/para	activating
Mildly activating	$-\text{NHCOCH}_3$, $-\text{CH}_3$, alkyl, $-\text{C}_6\text{H}_5$	ortho/para	activating
Halogens	$-\text{F}$, $-\text{Cl}$, $-\text{Br}$, $-\text{I}$	ortho/para	deactivating ($-\text{I} > +\text{R}$)
Meta directors	$-\text{NO}_2$, $-\text{CN}$, $-\text{CHO}$, $-\text{COR}$, $-\text{COOH}$, $-\text{SO}_3\text{H}$, $-\text{NR}_3^+$	meta	deactivating

Friedel–Crafts reactions **fail** on strongly deactivated rings (e.g. nitrobenzene) and on aniline ($-\text{NH}_2$ complexes with AlCl_3).

Polynuclear hydrocarbons such as 3,4-benzpyrene are **carcinogenic** (NCERT).

Common traps

⚠ MISTAKE ALERT Mistake alerts

- The peroxide effect works **only with HBr**.
- Friedel–Crafts alkylation with n-propyl chloride gives mainly **isopropylbenzene** (the 1° carbocation rearranges to 2°).
- Lindlar gives cis; $\text{Na}/\text{liquid NH}_3$ gives trans.
- Hydration of ethyne gives an **aldehyde** (ethanal). Every other alkyne gives ketones.
- Wurtz with two different halides gives **three** alkanes.

Question patterns

📊 PYQ PATTERN Recurring structures

- Products of addition to unsymmetrical alkenes (Markovnikov/anti-Markovnikov), ozonolysis → identifying the parent alkene.
- Stereochemistry of reductions (Lindlar vs Na/NH_3).
- Wurtz/Kolbe/decarboxylation products.
- EAS reagents, electrophiles, and directive effects in disubstituted benzenes.
- Aromaticity identification.
- Conformations (Newman/Sawhorse) and their relative energies.

Practice questions

✍ C15 · Practice

1 Medium ⏱ 1 min NV C15-01
A mixture of CH_3Br and $\text{C}_2\text{H}_5\text{Br}$ is treated with sodium in dry ether. Find the number of different alkanes formed.
Your answer (numerical value):

2 Easy ⏱ 0.5 min Speed C15-02
Propene reacts with HBr in the presence of benzoyl peroxide to give mainly:
(A) 2-bromopropane (B) 1-bromopropane
(C) 1,2-dibromopropane (D) propan-2-ol

3 Easy 0.75 min Basic

C15-03

Ozonolysis of but-2-ene (with Zn/H₂O work-up) gives:

- (A) two molecules of ethanal (B) propanal + methanal
(C) propanone (D) ethanoic acid

4 Easy 0.75 min Basic

C15-04

Propyne with HgSO₄/dil. H₂SO₄ gives:

- (A) propanal (B) propan-1-ol (C) propanone (D) propanoic acid

5 Easy 0.5 min Speed

C15-05

The most acidic hydrocarbon is:

- (A) ethyne (B) ethene (C) ethane (D) methane

6 Medium 1 min Concept

C15-06

Which species is aromatic?

- (A) cyclopentadienyl cation (B) cyclobutadiene
(C) cyclooctatetraene (D) cyclopentadienyl anion

7 Easy 0.5 min Speed

C15-07

Which substituent directs an incoming electrophile to the meta position?

- (A) -OH (B) -CH₃ (C) -Cl (D) -NO₂

8 Easy 0.5 min Concept

C15-08

In ethane, the staggered conformation is more stable than the eclipsed one by about:

- (A) 1.25 kJ/mol (B) 12.5 kJ/mol
(C) 125 kJ/mol (D) 0 (they are equally stable)

9 Easy 0.75 min Basic

C15-09

Kolbe electrolysis of aqueous sodium propanoate gives at the anode:

- (A) ethane (B) butane (C) propane (D) hexane

10 Easy 0.5 min Speed

C15-10

In the chlorination of methane, the chain-initiation step is:

- (A) $\text{Cl}\cdot + \text{CH}_4 \longrightarrow \text{CH}_3\cdot + \text{HCl}$ (B) $\text{CH}_3\cdot + \text{Cl}_2 \longrightarrow \text{CH}_3\text{Cl} + \text{Cl}\cdot$
(C) $\text{Cl}_2 \xrightarrow{h\nu} 2\text{Cl}\cdot$ (D) $\text{CH}_3\cdot + \text{CH}_3\cdot \longrightarrow \text{C}_2\text{H}_6$

11 Medium 1 min Tricky

C15-11

Benzene reacts with 1-chloropropane and anhydrous AlCl₃. The major product is:

- (A) isopropylbenzene (cumene) (B) n-propylbenzene
(C) propiophenone (D) no reaction

C15 · Chapter Test

1 Easy 0.5 min Speed C15-T1

Baeyer's reagent, used to test for unsaturation, is:

- (A) bromine water (B) ammoniacal AgNO_3
(C) acidic $\text{K}_2\text{Cr}_2\text{O}_7$ (D) cold, dilute, alkaline KMnO_4

2 Easy 0.5 min Speed C15-T2

Propene + HCl gives mainly:

- (A) 1-chloropropane (B) 1,2-dichloropropane
(C) 2-chloropropane (D) propan-2-ol

3 Easy 0.5 min NV C15-T3

Find the number of π electrons in naphthalene.

Your answer (numerical value):

4 Easy 0.5 min Speed C15-T4

But-2-yne is converted into *trans*-but-2-ene by:

- (A) H_2 , Lindlar catalyst (B) Na in liquid NH_3
(C) H_2/Ni , excess (D) $\text{HgSO}_4/\text{H}_2\text{SO}_4$

5 Easy 0.5 min Speed C15-T5

Which group deactivates the ring but directs ortho/para?

- (A) $-\text{CH}_3$ (B) $-\text{Cl}$ (C) $-\text{NO}_2$ (D) $-\text{NH}_2$

Answers & Solutions

C15 · Practice

- 1 3 2 (B) 3 (A) 4 (C) 5 (A) 6 (D) 7 (D) 8 (B) 9 (B) 10 (C)
11 (A)

C15 · Chapter Test

- 1 (D) 2 (C) 3 10 4 (B) 5 (B)

C15 · PRACTICE

1 Answer: 3 Wurtz with mixed halides Medium 1 min NV C15-01

SOLUTION Ethane (CH_3-CH_3), propane ($\text{CH}_3-\text{C}_2\text{H}_5$) and butane ($\text{C}_2\text{H}_5-\text{C}_2\text{H}_5$).

COMMON TRAP Answering 1 (only the cross product).

2 Answer: (B) Peroxide effect Easy 0.5 min Speed C15-02

SOLUTION Free-radical addition: $\text{Br}\cdot$ adds to the terminal carbon, giving the more stable 2° radical.

3 Answer: (A) Ozonolysis Easy 0.75 min Basic C15-03

SOLUTION Cleaving $\text{CH}_3\text{CH}=\text{CHCH}_3$ at the double bond gives 2 CH_3CHO .

4 Answer: (C) Hydration of propyne Easy 0.75 min Basic C15-04

SOLUTION Markovnikov hydration gives an enol that tautomerises to the ketone.

COMMON TRAP Choosing propanal.

5 Answer: (A) *Acidity of hydrocarbons* **Easy** ⌚ 0.5 min **Speed** C15-05

SOLUTION An sp carbon (50% s-character) holds the negative charge of the conjugate base best.

6 Answer: (D) *Aromaticity* **Medium** ⌚ 1 min **Concept** C15-06

SOLUTION Planar, cyclic, conjugated, with 6 π electrons ($4n + 2$ with $n = 1$). The cation (4π) and cyclobutadiene (4π) are antiaromatic. COT is non-planar.

7 Answer: (D) *Meta director* **Easy** ⌚ 0.5 min **Speed** C15-07

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

8 Answer: (B) *Ethane conformations* **Easy** ⌚ 0.5 min **Concept** C15-08

SOLUTION Torsional strain in the eclipsed form is about 12.5 kJ/mol (NCERT).

9 Answer: (B) *Kolbe electrolysis* **Easy** ⌚ 0.75 min **Basic** C15-09

SOLUTION $2 \text{CH}_3\text{CH}_2\text{COO}^- \longrightarrow \text{CH}_3\text{CH}_2-\text{CH}_2\text{CH}_3 + 2 \text{CO}_2 + 2 \text{e}^-$.

SHORTCUT Product = 2R, where R = the alkyl group of the acid.

COMMON TRAP Losing the carbon as in decarboxylation (ethane).

10 Answer: (C) *Chain initiation* **Easy** ⌚ 0.5 min **Speed** C15-10

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

11 Answer: (A) *Friedel-Crafts rearrangement* **Medium** ⌚ 1 min **Tricky** C15-11

SOLUTION The 1° carbocation rearranges by a hydride shift to the more stable 2° isopropyl cation.

COMMON TRAP Answering n-propylbenzene.

C15 · CHAPTER TEST

1 Answer: (D) *Baeyer's reagent* **Easy** ⌚ 0.5 min **Speed** C15-T1

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

2 Answer: (C) *Markovnikov product* **Easy** ⌚ 0.5 min **Speed** C15-T2

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

3 Answer: 10 *π electrons in naphthalene* **Easy** ⌚ 0.5 min **NV** C15-T3

SOLUTION Five C=C bonds in the resonance structure, $4n + 2$ with $n = 2$.

4 Answer: (B) *Trans-alkene* **Easy** ⌚ 0.5 min **Speed** C15-T4

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

5 Answer: (B) *Deactivating o/p director* **Easy** ⌚ 0.5 min **Speed** C15-T5

SOLUTION Halogens: -I (deactivating) outweighs +R, but +R still directs ortho/para.

REVISION Revision tracker · Hydrocarbons

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III • CHEMISTRY • UNIT 16

Organic Compounds Containing Halogens

16

~1 per shift
TYPICAL QUESTIONSMedium
DIFFICULTYHigh
PRIORITYClass 12 • Ch 6
NCERT~6 hours
STUDY TIME

Important concepts

- **C–X bond:** polar ($C^{\delta+}-X^{\delta-}$). Bond length $C-F < C-Cl < C-Br < C-I$; bond strength runs the other way, so reactivity is $R-I > R-Br > R-Cl > R-F$.
- **Nucleophilic substitution** (S_N1/S_N2) and **β -elimination** compete. The substrate, nucleophile/base, solvent and temperature decide which wins.
- **Haloarenes** are far less reactive towards nucleophiles: resonance gives C–X partial double-bond character, the carbon is sp^2 , the phenyl cation is unstable, and electron-rich π clouds repel the nucleophile.

Preparation

 Σ FORMULA From alcohols, alkenes and halogen exchange

Route	Reagent	Notes
$ROH \rightarrow RCl$	$HCl + \text{anhydrous } ZnCl_2; PCl_3; PCl_5; SOCl_2$	$SOCl_2$ is best: the byproducts SO_2 and HCl are gases
$ROH \rightarrow RBr / RI$	PBr_3 (red P + Br_2); red P + I_2	—
Alkene $\rightarrow RX$	HX (Markovnikov)	peroxide effect with HBr only
Finkelstein	$RCI/RBr + NaI$ in dry acetone $\rightarrow RI$	$NaCl/NaBr$ precipitate, which drives the reaction
Swarts	$RCI/RBr + AgF, Hg_2F_2, CoF_2$ or $SbF_3 \rightarrow RF$	route to alkyl fluorides
Haloarenes	$ArH + Cl_2$ ($FeCl_3$)	EAS
Sandmeyer	$ArN_2^+ + CuCl/HCl, CuBr/HBr, CuCN/KCN$	$\rightarrow ArCl, ArBr, ArCN$
Gattermann	$ArN_2^+ + Cu$ powder/ HX	$\rightarrow ArX$
Aryl iodide	$ArN_2^+ + KI$	no $Cu(I)$ needed

Physical properties: boiling point rises with halogen mass ($RI > RBr > RCl > RF$) and chain length, and **falls with branching**. The *para*-dihalobenzene has a higher melting point than the ortho or meta isomer (symmetric packing).

Nucleophilic substitution: S_N1 vs S_N2

	S_N2	S_N1
Steps / rate law	one step; $r = k[RX][Nu^-]$	two steps via a carbocation; $r = k[RX]$
Substrate reactivity	$CH_3X > 1^\circ > 2^\circ > 3^\circ$ (steric hindrance)	$3^\circ > 2^\circ > 1^\circ > CH_3X$ (carbocation stability)
Stereochemistry	inversion (Walden)	racemisation (with some inversion)
Favoured by	strong nucleophile, polar aprotic solvent	weak nucleophile, polar protic solvent
Special	—	allylic and benzylic halides react fast (resonance-stabilised cations)

**REACTION MAP** Reaction map • haloalkane R-X (1°, e.g. C₂H₅Br)

aq. KOH / NaOH	→ ROH (alcohol)
alc. KOH, heat	→ alkene (β-elimination, Saytzeff)
NaOR' (Williamson)	→ R-O-R' (ether)
KCN (ionic; attacks via C)	→ R-C≡N (nitrile), major
AgCN (covalent; attacks via N)	→ R-N≡C (isocyanide), major
KNO ₂	→ R-O-N=O (alkyl nitrite)
AgNO ₂	→ R-NO ₂ (nitroalkane)
NH ₃ (excess)	→ RNH ₂ (+ higher amines)
LiAlH ₄	→ RH
Mg, dry ether	→ RMgX (Grignard reagent)
Na, dry ether (Wurtz)	→ R-R

☆ IMPORTANT Grignard reagents

RMgX is made in **dry ether**. It reacts with any compound that has an acidic H (water, alcohols, amines, terminal alkynes) to give the alkane: $\text{CH}_3\text{MgBr} + \text{H}_2\text{O} \rightarrow \text{CH}_4 + \text{Mg}(\text{OH})\text{Br}$. That's why every trace of moisture must be excluded. Reactions with carbonyls are covered in Unit 17.

Σ FORMULA Wurtz family

Wurtz: $2\text{RX} + 2\text{Na} \rightarrow \text{R-R}$. **Wurtz-Fittig:** $\text{ArX} + \text{RX} + 2\text{Na} \rightarrow \text{Ar-R}$ (e.g. toluene from chlorobenzene + chloromethane). **Fittig:** $2\text{ArX} + 2\text{Na} \rightarrow \text{Ar-Ar}$ (biphenyl).

Haloarenes: substitution only when activated

- Chlorobenzene → phenol needs **623 K, 300 atm** NaOH (Dow process).
- NO₂ groups at **ortho/para** positions activate the ring: *p*-nitrochlorobenzene reacts with NaOH at 443 K; the 2,4-dinitro compound at 368 K; the 2,4,6-trinitro compound with warm water.
- EAS** on haloarenes: ortho/para-directing but deactivating (halogenation, nitration, sulphonation, Friedel-Crafts).

Polyhalogen compounds: environmental effects (syllabus)

Compound	Uses	Environmental / health notes (NCERT)
CH ₂ Cl ₂	solvent, paint remover	affects the central nervous system
CHCl ₃	solvent (formerly an anaesthetic)	oxidised by air and light to poisonous phosgene (COCl ₂), so it's stored in dark bottles filled to the brim
CHI ₃ (iodoform)	antiseptic (releases I ₂)	objectionable smell; replaced by other antiseptics
CCl ₄	solvent, refrigerant feedstock	liver damage; depletes the ozone layer
Freons (CFCs)	refrigerants, aerosol propellants	reach the stratosphere, release Cl radicals, deplete ozone
DDT	insecticide (first chlorinated organic insecticide)	not biodegradable; accumulates in fatty tissue; banned in many countries

Common traps

MISTAKE ALERT Mistake alerts

- $\text{KCN} \rightarrow$ nitrile but $\text{AgCN} \rightarrow$ **isocyanide**. $\text{KNO}_2 \rightarrow$ nitrite but $\text{AgNO}_2 \rightarrow$ **nitro** compound.
- Aqueous KOH substitutes; **alcoholic** KOH eliminates.
- $\text{S}_{\text{N}}2$ on a chiral centre inverts the configuration. The R/S label may or may not change, depending on priorities.
- Vinyl and aryl halides don't undergo $\text{S}_{\text{N}}1$ or $\text{S}_{\text{N}}2$ under normal conditions.

Question patterns

PYQ PATTERN Recurring structures

1. $\text{S}_{\text{N}}1$ vs $\text{S}_{\text{N}}2$ reactivity orders; stereochemical outcome.
2. Ambident nucleophiles (CN^- , NO_2^-) and their products.
3. Substitution vs elimination (reagent/conditions); Saytzeff product.
4. Named reactions: Finkelstein, Swarts, Sandmeyer, Gattermann, Wurtz, Wurtz-Fittig, Fittig.
5. Haloarene activation by nitro groups.
6. Environmental facts about CHCl_3 , CCl_4 , freons, DDT.

Practice questions

C16 · Practice

1 Easy 0.5 min Speed C16-01

The best reagent for converting ethanol into pure chloroethane is:

- (A) HCl alone (B) SOCl_2 (C) Cl_2 (D) NaCl

2 Easy 0.5 min Concept C16-02

The Finkelstein reaction converts an alkyl chloride or bromide into an alkyl iodide using:

- (A) NaI in dry acetone (B) KI in water
(C) $\text{I}_2/\text{red P}$ (D) HI

3 Easy 0.75 min Concept C16-03

The correct order of $\text{S}_{\text{N}}2$ reactivity is:

- (A) $(\text{CH}_3)_3\text{CBr} > (\text{CH}_3)_2\text{CHBr} > \text{CH}_3\text{CH}_2\text{Br} > \text{CH}_3\text{Br}$ (B) all equal
(C) $\text{CH}_3\text{CH}_2\text{Br} > \text{CH}_3\text{Br} > (\text{CH}_3)_3\text{CBr} > (\text{CH}_3)_2\text{CHBr}$ (D) $\text{CH}_3\text{Br} > \text{CH}_3\text{CH}_2\text{Br} > (\text{CH}_3)_2\text{CHBr} > (\text{CH}_3)_3\text{CBr}$

4 Easy 0.5 min Speed C16-04

Which halide undergoes $\text{S}_{\text{N}}1$ fastest?

- (A) CH_3Cl (B) $\text{CH}_3\text{CH}_2\text{Cl}$ (C) $\text{C}_6\text{H}_5\text{CH}_2\text{Cl}$ (D) $\text{C}_6\text{H}_5\text{Cl}$

5 Easy 0.5 min Speed C16-05

Ethyl bromide reacts with AgCN to give mainly:

- (A) ethyl cyanide (B) ethyl isocyanide (C) ethylamine (D) ethanol

6

Easy

⌚ 0.5 min

Speed

C16-06

CH_3MgBr reacts with water to give:

- (A) methane (B) methanol (C) ethane (D) formaldehyde

7

Easy

⌚ 0.75 min

Basic

C16-07

2-Bromobutane heated with alcoholic KOH gives mainly:

- (A) but-2-ene (B) but-1-ene (C) butan-2-ol (D) butane

8

Easy

⌚ 0.5 min

Concept

C16-08

Chloroform is stored in dark-coloured bottles filled to the brim because:

- (A) it is highly volatile
(B) it absorbs moisture
(C) it reacts with glass
(D) in air and light it is oxidised to poisonous phosgene

9

Medium

⌚ 1 min

Concept

C16-09

Which compound undergoes nucleophilic substitution with aqueous NaOH most easily?

- (A) chlorobenzene (B) 4-nitrochlorobenzene
(C) 2,4,6-trinitrochlorobenzene (D) 2,4-dinitrochlorobenzene

10

Medium

⌚ 1 min

Concept

C16-10

(*R*)-2-Bromobutane reacts with OH^- by a pure $\text{S}_{\text{N}}2$ mechanism. The product is:

- (A) (*R*)-butan-2-ol (B) a racemic mixture
(C) (*S*)-butan-2-ol (D) butan-1-ol

11

Medium

⌚ 1.5 min

NV

C16-11

Find the number of **structurally** distinct monochloro products from the free-radical chlorination of 2-methylbutane.

Your answer (numerical value):

C16 · Chapter Test

1

Easy

⌚ 0.5 min

Speed

C16-T1

The correct order of boiling points is:

- (A) all equal (B) $\text{CH}_3\text{Cl} > \text{CH}_3\text{Br} > \text{CH}_3\text{I}$
(C) $\text{CH}_3\text{Br} > \text{CH}_3\text{I} > \text{CH}_3\text{Cl}$ (D) $\text{CH}_3\text{I} > \text{CH}_3\text{Br} > \text{CH}_3\text{Cl}$

2

Easy

⌚ 0.5 min

Speed

C16-T2

Chlorobenzene + chloromethane + Na (dry ether) gives:

- (A) biphenyl (B) benzene (C) ethane only (D) toluene

3 Easy 0.5 min NV

C16-T3

How many stereoisomers does 2-chlorobutane have?

Your answer (numerical value):

4 Easy 0.5 min Concept

C16-T4

The main environmental concern with DDT is that it:

- (A) is a greenhouse gas
 (B) is not biodegradable and accumulates in fatty tissues
 (C) depletes ozone
 (D) forms phosgene

5 Medium 0.75 min Concept

C16-T5

Chlorobenzene is less reactive than chloroethane towards nucleophilic substitution mainly because:

- (A) Cl is larger in chlorobenzene
 (B) chlorobenzene is non-polar
 (C) resonance gives the C-Cl bond partial double-bond character
 (D) chlorobenzene forms a stable carbocation

Answers & Solutions

C16 · Practice

- 1 (B) 2 (A) 3 (D) 4 (C) 5 (B) 6 (A) 7 (A) 8 (D) 9 (C) 10 (C)
 11 4

C16 · Chapter Test

- 1 (D) 2 (D) 3 2 4 (B) 5 (C)

C16 · PRACTICE

1 Answer: (B) Best chlorinating agent Easy 0.5 min Speed

C16-01

SOLUTION The byproducts SO_2 and HCl escape as gases, leaving pure product.

2 Answer: (A) Finkelstein reaction Easy 0.5 min Concept

C16-02

SOLUTION NaCl/NaBr are insoluble in acetone and precipitate, pushing the equilibrium forward.3 Answer: (D) $\text{S}_{\text{N}}2$ reactivity order Easy 0.75 min Concept

C16-03

SOLUTION Backside attack is increasingly hindered as alkyl groups are added.

COMMON TRAP Using the $\text{S}_{\text{N}}1$ order.4 Answer: (C) $\text{S}_{\text{N}}1$ reactivity Easy 0.5 min Speed

C16-04

SOLUTION The benzyl carbocation is resonance-stabilised. Chlorobenzene (aryl) doesn't react.

5 Answer: (B) AgCN product Easy 0.5 min Speed

C16-05

SOLUTION AgCN is covalent. The nitrogen lone pair attacks, forming $\text{C}_2\text{H}_5\text{NC}$.

6 Answer: (A) Grignard + water Easy 0.5 min Speed

C16-06

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

7 Answer: (A) Saytzeff elimination Easy 0.75 min Basic

C16-07

SOLUTION Saytzeff: the more substituted alkene is the major product.

COMMON TRAP Choosing the substitution product (C), which needs aqueous KOH .

8 Answer: (D) Storing chloroform Easy 0.5 min Concept

C16-08

SOLUTION $2\text{CHCl}_3 + \text{O}_2 \xrightarrow{\text{light}} 2\text{COCl}_2 + 2\text{HCl}$.

9 Answer: (C) Activated haloarene Medium 1 min Concept

C16-09

SOLUTION Three $-\text{NO}_2$ groups at ortho/para positions stabilise the intermediate anion most. It reacts with warm water.

10 Answer: (C) Stereochemistry of S_N2 Medium 1 min Concept

C16-10

SOLUTION Backside attack inverts the configuration. OH takes the priority position of Br, so the label changes from R to S.

COMMON TRAP Answering racemic (S_N1 thinking).

11 Answer: 4 Monochlorination isomers Medium 1.5 min NV

C16-11

SOLUTION $\text{CH}_3-\text{CH}(\text{CH}_3)-\text{CH}_2-\text{CH}_3$ has 4 types of H: the two equivalent terminal methyls on C2, the C2-H, the CH_2 , and the far CH_3 .

SHORTCUT Count the distinct hydrogen environments.

COMMON TRAP Counting 5 by treating the two methyls on C2 as different.

C16 · CHAPTER TEST

1 Answer: (D) Boiling points Easy 0.5 min Speed

C16-T1

SOLUTION van der Waals forces increase with the size of the halogen.

2 Answer: (D) Wurtz-Fittig Easy 0.5 min Speed

C16-T2

SOLUTION Cross-coupling gives $\text{C}_6\text{H}_5-\text{CH}_3$ (biphenyl and ethane form as side products).

3 Answer: 2 Stereoisomers of 2-chlorobutane Easy 0.5 min NV

C16-T3

SOLUTION One chiral centre: a pair of enantiomers.

4 Answer: (B) DDT Easy 0.5 min Concept

C16-T4

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

5 Answer: (C) Unreactive aryl halides Medium 0.75 min Concept

C16-T5

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

REVISION Revision tracker · Haloalkanes & Haloarenes

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III · CHEMISTRY · UNIT 17

Organic Compounds Containing Oxygen

17

2–3 per shift
TYPICAL QUESTIONSMedium
DIFFICULTYMust-do (top-
weight organic
chapter)
PRIORITYClass 12 · Ch 7, 8
NCERT~14 hours
STUDY TIME PYQ PATTERN Weight evidence

Aldehydes, ketones & carboxylic acids alone: 35 questions across the 2026 Paper-1 shifts ($\approx 7.4\%$ of Chemistry)  **THIRD-PARTY ANALYSIS** [T-WT-CD]. Add alcohols, phenols and ethers, and this unit is the single largest block of organic marks.

Part A · Alcohols

 FORMULA Preparation

- **Alkenes:** acid-catalysed hydration (Markovnikov); **hydroboration–oxidation** (B_2H_6 , then $\text{H}_2\text{O}_2/\text{OH}^-$) gives **anti-Markovnikov** alcohols (propene $\rightarrow$ propan-1-ol).
- **Reduction of carbonyls:** H_2/Pd , NaBH_4 or LiAlH_4 . Aldehydes $\rightarrow$ 1° , ketones $\rightarrow$ 2° . Acids and esters need LiAlH_4 ($\rightarrow$ 1°).
- **Grignard:** $\text{HCHO} \rightarrow 1^\circ$; other aldehydes $\rightarrow 2^\circ$; ketones $\rightarrow 3^\circ$ (after hydrolysis).

 REACTION MAP Reaction map · ethanol $\text{CH}_3\text{CH}_2\text{OH}$

Na	$\rightarrow \text{CH}_3\text{CH}_2\text{ONa} + \frac{1}{2} \text{H}_2$ (acidic character)
RCOOH , conc. H_2SO_4 (Fischer)	$\rightarrow$ ester RCOOC_2H_5
$\text{HCl}/\text{ZnCl}_2 \cdot \text{PCl}_5 \cdot \text{SOCl}_2$	$\rightarrow \text{CH}_3\text{CH}_2\text{Cl}$
conc. H_2SO_4 , 443 K	$\rightarrow$ ethene (dehydration)
conc. H_2SO_4 , 413 K (excess alcohol)	$\rightarrow$ diethyl ether
PCC	$\rightarrow$ ethanal (stops at aldehyde)
acidified $\text{K}_2\text{Cr}_2\text{O}_7 / \text{KMnO}_4$	$\rightarrow$ ethanoic acid
Cu, 573 K	$\rightarrow$ ethanal (dehydrogenation)
I_2/NaOH	$\rightarrow$ iodoform (yellow): ethanol has the $\text{CH}_3\text{CH}(\text{OH})^-$ unit

 IMPORTANT Dehydration mechanism (syllabus-listed)

(1) Protonation of $-\text{OH} \rightarrow \text{R}-\text{OH}_2^+$; (2) loss of H_2O gives a **carbocation** (slow step); (3) loss of $\beta\text{-H}$ gives the alkene (Saytzeff). Ease: $3^\circ > 2^\circ > 1^\circ$, following carbocation stability. Rearrangements are possible.

Σ FORMULA Distinguishing 1°, 2° and 3° alcohols

Test	1°	2°	3°
Lucas (conc. HCl + anhydrous ZnCl ₂)	no turbidity at room temperature	turbidity in ~5 min	immediate turbidity
Oxidation (K ₂ Cr ₂ O ₇ /H ⁺)	aldehyde → acid	ketone	resistant
Cu at 573 K	aldehyde	ketone	alkene

Acidity (in solution): 1° > 2° > 3°. All are weaker acids than water, and much weaker than phenol.

Part B · Phenols

Σ FORMULA Preparation & acidity

Preparation: chlorobenzene (Dow: NaOH, 623 K, 300 atm); benzenesulphonic acid (fuse with NaOH, then acidify); **diazonium salt + warm water**; **cumene process** (cumene + O₂ → cumene hydroperoxide → dil. acid → phenol + **acetone**).

Acidity: phenoxide is resonance-stabilised, so phenol (pK_a 10.0) is far more acidic than ethanol (15.9). It reacts with NaOH, but **not with** NaHCO₃ (picric acid is the exception).

Phenol	pK _a (NCERT)
<i>p</i> -nitrophenol	7.1
<i>o</i> -nitrophenol	7.2
<i>m</i> -nitrophenol	8.3
phenol	10.0
cresols (<i>o</i> , <i>m</i> , <i>p</i>)	~10.1–10.2
2,4,6-trinitrophenol (picric acid)	~0.4 (strongly acidic)

Electron-withdrawing groups at **ortho/para** raise acidity (–R and –I). Electron-donating groups lower it.

REACTION MAP Reaction map · phenol C₆H₅OH

dil. HNO ₃ , 298 K	→ <i>o</i>- + <i>p</i>-nitrophenol (separated by steam distillation: <i>o</i> is volatile)
conc. HNO ₃	→ 2,4,6-trinitrophenol (picric acid)
Br ₂ in CS ₂ , 273 K	→ <i>o</i>- + <i>p</i>-bromophenol
bromine water	→ 2,4,6-tribromophenol (white precipitate)
Kolbe: NaOH, then CO ₂ (400 K, 4–7 atm), then H ⁺	→ salicylic acid (<i>o</i>-hydroxybenzoic acid)
Reimer-Tiemann: CHCl ₃ + aq. NaOH, then H ⁺	→ salicylaldehyde (via dichlorocarbene : CCl₂)
Zn dust, heat	→ benzene
Na ₂ Cr ₂ O ₇ /H ₂ SO ₄	→ benzoquinone
neutral FeCl ₃	→ violet colour (test)

Part C · Ethers

Σ FORMULA Preparation & reactions

Dehydration of alcohols (conc. H_2SO_4 , 413 K): works for 1° alcohols only ($2^\circ/3^\circ$ give alkenes).

Williamson synthesis: $\text{R-X} + \text{R}'\text{-ONa} \longrightarrow \text{R-O-R}' + \text{NaX}$ ($\text{S}_{\text{N}}2$). Use a **1° alkyl halide** and the bulky alkoxide: *tert*-butyl methyl ether = $(\text{CH}_3)_3\text{CONa} + \text{CH}_3\text{Br}$. The reverse combination $((\text{CH}_3)_3\text{CBr} + \text{CH}_3\text{ONa})$ gives isobutylene by elimination.

Structure: C–O–C angle $\approx 111.7^\circ$ (slightly above tetrahedral; bulky groups repel).

Cleavage by HI: $\text{R-O-R}' + \text{HI} \longrightarrow \text{R-I} + \text{R}'\text{-OH}$. With a methyl alkyl ether (1°), I^- attacks the **methyl** carbon ($\text{S}_{\text{N}}2$), giving $\text{CH}_3\text{I} + \text{ROH}$. With a *tert*-alkyl group, the tertiary halide forms ($\text{S}_{\text{N}}1$). **Anisole + HI $\rightarrow$ phenol + CH_3I :** the O–aryl bond doesn't break.

Anisole EAS: the $-\text{OCH}_3$ group is activating and o/p-directing (halogenation, nitration, Friedel–Crafts).

Part D · Aldehydes & Ketones

Σ FORMULA Preparation (named)

Reaction	Transformation
Rosenmund reduction	$\text{RCOCl} + \text{H}_2 \xrightarrow{\text{Pd/BaSO}_4} \text{RCHO}$
Stephen reduction	$\text{RCN} \xrightarrow{\text{SnCl}_2/\text{HCl}} \text{RCH=NH} \xrightarrow{\text{H}_3\text{O}^+} \text{RCHO}$
DIBAL-H	nitriles/esters $\rightarrow$ aldehydes
Etard reaction	toluene + CrO_2Cl_2 (chromyl chloride), then $\text{H}_3\text{O}^+ \rightarrow$ benzaldehyde
Side-chain chlorination	toluene $\rightarrow \text{C}_6\text{H}_5\text{CHCl}_2 \rightarrow$ hydrolysis $\rightarrow$ benzaldehyde
Gattermann–Koch	benzene + CO + HCl, anhydrous $\text{AlCl}_3/\text{CuCl} \rightarrow$ benzaldehyde
Ketones from nitriles	$\text{RCN} + \text{R}'\text{MgX}$, then $\text{H}_3\text{O}^+ \rightarrow \text{RCOR}'$
Ketones from acyl chlorides	$2 \text{RCOCl} + \text{R}'_2\text{Cd} \longrightarrow 2 \text{RCOR}'$
Friedel–Crafts acylation	benzene + $\text{RCOCl}/\text{AlCl}_3 \rightarrow$ aryl ketone
Ozonolysis; hydration of alkynes; oxidation of alcohols	see Units 15 and 17A

☆ IMPORTANT Reactivity of the carbonyl group

The carbonyl carbon is sp^2 , planar and electrophilic. **Nucleophilic addition** reactivity: $\text{HCHO} > \text{CH}_3\text{CHO} > \text{CH}_3\text{COCH}_3$. Alkyl groups donate electrons (+I) and add steric bulk. Aromatic carbonyls are less reactive (resonance with the ring).

REACTION MAP Reaction map · ethanal CH_3CHO (and ketones)

HCN (base)	→ cyanohydrin $\text{CH}_3\text{CH}(\text{OH})\text{CN}$
NaHSO_3	→ bisulphite adduct (crystalline; used for separation)
$\text{R}'\text{MgX}$, then H_3O^+	→ 2° alcohol (ketones → 3°)
ROH , dry HCl	→ hemiacetal → acetal (ketones + ethylene glycol → cyclic ketal)
NH_2OH	→ oxime
NH_2NH_2	→ hydrazone
2,4-DNP (Brady's reagent)	→ orange/yellow 2,4-DNP derivative (test for $\text{C}=\text{O}$)
semicarbazide	→ semicarbazone
NaBH_4 / LiAlH_4	→ ethanol
Clemmensen (Zn-Hg / conc. HCl)	→ ethane ($>\text{C}=\text{O} \rightarrow >\text{CH}_2$)
Wolff-Kishner (NH_2NH_2 , then KOH /ethylene glycol, heat)	→ ethane
Tollens' reagent	→ silver mirror (aldehydes only)
Fehling's solution	→ red Cu_2O (aliphatic aldehydes only)
I_2/NaOH (haloform)	→ iodoform + formate (all methyl carbonyls)
dil. NaOH (aldol)	→ 3-hydroxybutanal → (heat) but-2-enal

FORMULA Condensation & disproportionation

Aldol: aldehydes/ketones with $\alpha\text{-H}$ + dilute alkali → β -hydroxy carbonyl (aldol/ketol) → on heating, α,β -unsaturated carbonyl. **Cross-aldol** between two different carbonyls gives 4 products (fewer if one partner lacks $\alpha\text{-H}$).

Cannizzaro: aldehydes without $\alpha\text{-H}$ (HCHO , $\text{C}_6\text{H}_5\text{CHO}$, $(\text{CH}_3)_3\text{CCHO}$) + conc. NaOH → alcohol + carboxylate salt (one molecule reduced, one oxidised). Cross-Cannizzaro with HCHO : HCHO is oxidised to formate.

Haloform: compounds with CH_3CO^- or $\text{CH}_3\text{CH}(\text{OH})^-$ (ethanal, methyl ketones, ethanol, propan-2-ol) + X_2/NaOH → CHX_3 . Iodoform is a **yellow** precipitate.

EAS on benzaldehyde / acetophenone: $-\text{CHO}$ and $-\text{COR}$ are **meta** directors.

Part E · Carboxylic Acids

FORMULA Preparation

Oxidation of 1° alcohols and aldehydes; **alkylbenzenes + alkaline KMnO_4 → benzoic acid** (whatever the chain length, provided there is a benzylic H); hydrolysis of nitriles and amides; **Grignard + CO_2 (dry ice)**, then H_3O^+ → RCOOH ; hydrolysis of acyl halides, anhydrides and esters.

FORMULA Acidity

The carboxylate ion has two **equivalent** resonance structures, so carboxylic acids are much more acidic than phenols and alcohols. They liberate CO_2 from NaHCO_3 (this is the test).

- EWG raise acidity: $\text{CF}_3\text{COOH} > \text{CCl}_3\text{COOH} > \text{CHCl}_2\text{COOH} > \text{ClCH}_2\text{COOH} > \text{HCOOH} > \text{CH}_3\text{COOH}$.
- α -substituent: $\text{F} > \text{Cl} > \text{Br} > \text{I}$. The effect falls off with distance: 2-chlorobutanoic > 3-chloro > 4-chloro.
- pK_a values: HCOOH 3.75 · $\text{C}_6\text{H}_5\text{COOH}$ 4.19 · CH_3COOH 4.76. *p*-Nitrobenzoic acid is stronger than benzoic acid; *p*-methoxybenzoic acid is weaker.

**REACTION MAP** Reaction map · ethanoic acid CH_3COOH

NaHCO_3	→ $\text{CH}_3\text{COONa} + \text{CO}_2$ (brisk effervescence)
$\text{PCl}_5 / \text{PCl}_3 / \text{SOCl}_2$	→ CH_3COCl (acetyl chloride)
P_2O_5 or heat	→ acetic anhydride
ROH / H^+	→ ester
NH_3 , then heat	→ ammonium acetate → acetamide
LiAlH_4 (not NaBH_4)	→ ethanol
$\text{NaOH} + \text{CaO}$ (soda lime), heat	→ methane (decarboxylation)
HVZ: Cl_2 / red P, then H_2O	→ chloroacetic acid (α -halogenation)

Named-reaction table (Unit 17)

Named reaction	Substrate → product	Key reagent
Reimer–Tiemann	phenol → salicylaldehyde	CHCl_3 , aq. NaOH
Kolbe (phenol)	sodium phenoxide → salicylic acid	CO_2 , 400 K, 4–7 atm
Cumene process	cumene → phenol + acetone	O_2 , then H_3O^+
Williamson	$\text{RX} + \text{R}'\text{ONa} \rightarrow \text{ether}$	1°RX
Hydroboration–oxidation	alkene → anti-Markovnikov alcohol	B_2H_6 ; $\text{H}_2\text{O}_2/\text{OH}^-$
Rosenmund	acyl chloride → aldehyde	H_2 , Pd/BaSO_4
Stephen	nitrile → aldehyde	SnCl_2/HCl , H_3O^+
Etard	toluene → benzaldehyde	CrO_2Cl_2
Gattermann–Koch	benzene → benzaldehyde	CO , HCl , $\text{AlCl}_3/\text{CuCl}$
Clemmensen	$\text{C}=\text{O} \rightarrow \text{CH}_2$	Zn-Hg , conc. HCl
Wolff–Kishner	$\text{C}=\text{O} \rightarrow \text{CH}_2$	NH_2NH_2 ; KOH , glycol, heat
Aldol	carbonyl with $\alpha\text{-H} \rightarrow \beta\text{-hydroxy carbonyl}$	dil. NaOH
Cannizzaro	aldehyde without $\alpha\text{-H} \rightarrow \text{alcohol} + \text{salt}$	conc. NaOH
Haloform	methyl ketone → $\text{CHX}_3 + \text{acid salt}$	X_2 , NaOH
HVZ	acid → $\alpha\text{-halo acid}$	X_2 , red P
Fischer esterification	acid + alcohol $\rightleftharpoons$ ester	conc. H_2SO_4

Distinguishing tests (very high frequency)

Pair	Test that separates them
aldehyde vs ketone	Tollens' (silver mirror), Fehling's (red precipitate, aliphatic aldehydes only)
benzaldehyde vs ethanal	Fehling's (ethanal +) and iodoform (ethanal +)
methyl ketone vs other ketones (pentan-2-one vs pentan-3-one)	iodoform
phenol vs alcohol	neutral FeCl_3 (violet); bromine water (white precipitate); NaOH solubility
phenol vs carboxylic acid	NaHCO_3 (acid effervesces)
$1^\circ/2^\circ/3^\circ$ alcohols	Lucas test
ethanol vs methanol	iodoform (ethanol +)

Common traps

MISTAKE ALERT Mistake alerts

- **Fehling's** is negative for aromatic aldehydes; Tollens' is positive for all aldehydes.
- Iodoform is positive for **ethanol** and **propan-2-ol** (they are oxidised to the methyl carbonyl in situ), but negative for methanol and propan-1-ol.
- Cannizzaro needs **no α -H** and **concentrated** alkali. Aldol needs α -H and **dilute** alkali.
- NaBH_4 doesn't reduce $-\text{COOH}$ or esters.
- Phenol does **not** react with NaHCO_3 , but picric acid does.
- In Williamson synthesis, never use a 3° alkyl halide.

Question patterns

PYQ PATTERN Recurring structures

1. Multi-step conversions ($A \rightarrow B \rightarrow C$) with named reagents; identify an intermediate or the final product.
2. Distinguishing tests (the table above).
3. Acidity orders (substituted phenols and carboxylic acids).
4. Aldol/Cannizzaro/haloform product prediction, including cross reactions.
5. Grignard products and the number of moles consumed.
6. Ether cleavage with HI; Williamson reagent choice.
7. Mechanism steps (alcohol dehydration, nucleophilic addition).

Practice questions

C17 · Practice

1 Easy 0.5 min Speed

C17-01

Which alcohol gives immediate turbidity with Lucas reagent at room temperature?

- (A) butan-1-ol (B) butan-2-ol
(C) methanol (D) 2-methylpropan-2-ol

2 Easy 0.5 min Speed

C17-02

Acetone treated with CH_3MgBr , then hydrolysed, gives:

- (A) propan-2-ol (B) butan-2-ol
(C) 2-methylpropan-2-ol (D) ethanol

3 Easy 0.75 min Concept

C17-03

The correct order of acidity is:

- (A) *p*-nitrophenol > phenol > *p*-cresol (B) *p*-cresol > phenol > *p*-nitrophenol
(C) phenol > *p*-nitrophenol > *p*-cresol (D) all equal

4 Easy 0.5 min Speed

C17-04

Phenol + chloroform + aqueous NaOH (then acidification) gives:

- (A) salicylic acid (B) salicylaldehyde (C) benzaldehyde (D) chlorobenzene

5 Medium 1 min Concept

C17-05

The best way to prepare *tert*-butyl methyl ether is:

- (A) $(\text{CH}_3)_3\text{CBr} + \text{CH}_3\text{ONa}$
(B) $(\text{CH}_3)_3\text{COH} + \text{CH}_3\text{OH}$ with conc. H_2SO_4 at 443 K
(C) $(\text{CH}_3)_3\text{CONa} + \text{CH}_3\text{Br}$
(D) $(\text{CH}_3)_3\text{CCl} + \text{CH}_3\text{OH}$

6 Easy 0.75 min Basic

C17-06

Anisole heated with HI gives:

- (A) iodobenzene + methanol
(B) iodobenzene + iodomethane
(C) benzene + methanol
(D) phenol + iodomethane

7 Easy 0.5 min Speed

C17-07

Benzaldehyde with concentrated NaOH gives:

- (A) benzoic acid only
(B) no reaction
(C) benzoin
(D) benzyl alcohol + sodium benzoate

8 Easy 0.5 min Speed

C17-08

Two molecules of ethanal in dilute NaOH give:

- (A) butanal
(B) 3-hydroxybutanal
(C) ethyl ethanoate
(D) 2-hydroxybutanal

9 Medium 1 min Concept

C17-09

Which compound does **not** give the iodoform test?

- (A) ethanol
(B) propan-2-ol
(C) pentan-3-one
(D) acetone

10 Easy 0.5 min Speed

C17-10

Acetophenone with Zn-Hg and conc. HCl gives:

- (A) ethylbenzene
(B) 1-phenylethanol
(C) styrene
(D) benzoic acid

11 Medium 1 min Concept

C17-11

The correct order of acid strength is:

- (A) $\text{Cl}_2\text{CHCOOH} > \text{ClCH}_2\text{COOH} > \text{CH}_3\text{COOH}$
(B) $\text{CH}_3\text{COOH} > \text{ClCH}_2\text{COOH} > \text{Cl}_2\text{CHCOOH}$
(C) $\text{ClCH}_2\text{COOH} > \text{Cl}_2\text{CHCOOH} > \text{CH}_3\text{COOH}$
(D) all equal

12 Easy 0.75 min Basic

C17-12

Propanoic acid with Cl_2 and red phosphorus gives:

- (A) 3-chloropropanoic acid
(B) 2-chloropropanoic acid
(C) propanoyl chloride
(D) chloroethane

13 Easy 0.75 min Concept

C17-13

Benzaldehyde:

- (A) gives both Tollens' and Fehling's tests
(B) gives Fehling's but not Tollens' test
(C) gives Tollens' but not Fehling's test
(D) gives neither

14 Medium 1 min NV

C17-14

How many structurally isomeric **secondary** alcohols have the formula $C_5H_{12}O$?

Your answer (numerical value):

15 Easy 0.5 min Speed

C17-15

In the Stephen reduction, a nitrile is converted into an aldehyde using:

- (A) $SnCl_2/HCl$, then hydrolysis (B) $LiAlH_4$
(C) $Pd/BaSO_4$ (D) $Zn-Hg/HCl$

C17 · Chapter Test

1 Easy 0.5 min Speed

C17-T1

Sodium phenoxide + CO_2 (400 K, 4–7 atm), then acidification, gives:

- (A) salicylic acid (B) salicylaldehyde (C) benzoic acid (D) phenyl acetate

2 Easy 0.5 min Speed

C17-T2

The catalyst used in Fischer esterification is:

- (A) anhydrous $AlCl_3$ (B) conc. H_2SO_4 (C) $NaOH$ (D) Pd

3 Medium 1 min NV

C17-T3

How many moles of CH_3MgBr does 1 mol of ethyl acetate consume in forming a tertiary alcohol (after hydrolysis)?

Your answer (numerical value):

4 Easy 0.5 min Speed

C17-T4

A carboxylic acid is distinguished from phenol by:

- (A) $FeCl_3$ (B) $NaHCO_3$ (C) $NaOH$ (D) bromine water

5 Easy 0.5 min Speed

C17-T5

Ethylbenzene heated with alkaline $KMnO_4$ and then acidified gives:

- (A) phenylacetic acid (B) benzoic acid
(C) acetophenone (D) benzaldehyde

6 Easy 0.5 min Speed

C17-T6

The cumene process produces:

- (A) phenol and formaldehyde (B) phenol and acetone
(C) benzoic acid and ethanol (D) benzene and propene

7 Easy 0.5 min Speed

C17-T7

Which compound is the most reactive towards nucleophilic addition?

- (A) $C_6H_5COCH_3$ (B) CH_3CHO (C) CH_3COCH_3 (D) $HCHO$

Answers & Solutions

C17 · Practice

1	(D)	2	(C)	3	(A)	4	(B)	5	(C)	6	(D)	7	(D)	8	(B)	9	(C)	10	(A)
11	(A)	12	(B)	13	(C)	14	3	15	(A)										

C17 · Chapter Test

1	(A)	2	(B)	3	2	4	(B)	5	(B)	6	(B)	7	(D)
---	-----	---	-----	---	---	---	-----	---	-----	---	-----	---	-----

C17 · PRACTICE

1 Answer: (D) Lucas test **Easy** ⌚ 0.5 min **Speed** C17-01

SOLUTION A 3° alcohol forms a stable carbocation at once, giving an insoluble chloride.

2 Answer: (C) Grignard + ketone **Easy** ⌚ 0.5 min **Speed** C17-02

SOLUTION Ketone + Grignard gives a 3° alcohol: $(\text{CH}_3)_3\text{COH}$.

3 Answer: (A) Acidity of phenols **Easy** ⌚ 0.75 min **Concept** C17-03

SOLUTION $-\text{NO}_2$ ($-\text{R}$, $-\text{I}$) stabilises phenoxide; $-\text{CH}_3$ ($+\text{I}$, hyperconjugation) destabilises it. $\text{p}K_a$ $7.1 < 10.0 < 10.2$.

4 Answer: (B) Reimer-Tiemann **Easy** ⌚ 0.5 min **Speed** C17-04

SOLUTION The electrophile is dichlorocarbene. It gives the *ortho*-CHO product.

COMMON TRAP Confusing it with the Kolbe product (salicylic acid).

5 Answer: (C) Williamson reagent choice **Medium** ⌚ 1 min **Concept** C17-05

SOLUTION $\text{S}_\text{N}2$ needs an unhindered halide (CH_3Br). In (A), the 3° halide undergoes elimination with the strong base.

SHORTCUT Put the halogen on the smaller group.

COMMON TRAP Choosing (A).

6 Answer: (D) Cleavage of anisole **Easy** ⌚ 0.75 min **Basic** C17-06

SOLUTION The O-aryl bond has partial double-bond character. I^- attacks the methyl carbon by $\text{S}_\text{N}2$.

7 Answer: (D) Cannizzaro **Easy** ⌚ 0.5 min **Speed** C17-07

SOLUTION There is no $\alpha\text{-H}$, so disproportionation occurs.

8 Answer: (B) Aldol product **Easy** ⌚ 0.5 min **Speed** C17-08

SOLUTION The enolate attacks a second ethanal: $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CHO}$.

9 Answer: (C) Iodoform test **Medium** ⌚ 1 min **Concept** C17-09

SOLUTION Pentan-3-one ($\text{C}_2\text{H}_5\text{COC}_2\text{H}_5$) has no CH_3CO^- group.

10 Answer: (A) Clemmensen reduction **Easy** ⌚ 0.5 min **Speed** C17-10

SOLUTION $\text{C}=\text{O}$ becomes CH_2 .

COMMON TRAP Choosing the alcohol (NaBH_4 product).

11 Answer: (A) Acidity of chloro acids **Medium** ⌚ 1 min **Concept** C17-11

SOLUTION More $-\text{I}$ groups stabilise the carboxylate more.

12 Answer: (B) HVZ reaction **Easy** ⌚ 0.75 min **Basic** C17-12

SOLUTION HVZ halogenates the α -carbon.

COMMON TRAP Choosing propanoyl chloride ($\text{PCl}_5/\text{SOCl}_2$ product).

13 Answer: (C) Tollens' vs Fehling's **Easy** ⌚ 0.75 min **Concept** C17-13

SOLUTION Aromatic aldehydes don't reduce Fehling's solution.

14 Answer: 3 Counting 2° alcohols **Medium** ⌚ 1 min **NV**

C17-14

SOLUTION Pentan-2-ol, pentan-3-ol, 3-methylbutan-2-ol.

COMMON TRAP Including 2-methylbutan-2-ol (which is 3°).

15 Answer: (A) Stephen reduction **Easy** ⌚ 0.5 min **Speed**

C17-15

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

COMMON TRAP Confusing it with Rosenmund (Pd/BaSO₄, which works on acyl chlorides).

C17 · CHAPTER TEST

1 Answer: (A) Kolbe's reaction **Easy** ⌚ 0.5 min **Speed**

C17-T1

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

2 Answer: (B) Fischer esterification catalyst **Easy** ⌚ 0.5 min **Speed**

C17-T2

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

3 Answer: 2 Grignard with an ester **Medium** ⌚ 1 min **NV**

C17-T3

SOLUTION The first equivalent substitutes OC₂H₅ (giving acetone); the second adds to the ketone, giving 2-methylpropan-2-ol.

4 Answer: (B) Test for carboxylic acids **Easy** ⌚ 0.5 min **Speed**

C17-T4

SOLUTION Only the acid liberates CO₂ from bicarbonate.

5 Answer: (B) Side-chain oxidation **Easy** ⌚ 0.5 min **Speed**

C17-T5

SOLUTION Any alkyl chain with a benzylic H oxidises down to -COOH.

6 Answer: (B) Cumene process **Easy** ⌚ 0.5 min **Speed**

C17-T6

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

7 Answer: (D) Most reactive carbonyl **Easy** ⌚ 0.5 min **Speed**

C17-T7

SOLUTION It has the least steric hindrance and no electron-donating groups.

REVISION Revision tracker · Oxygen Compounds

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III · CHEMISTRY · UNIT 18

Organic Compounds Containing Nitrogen

18

~1 per shift
TYPICAL QUESTIONSMedium
DIFFICULTYHigh
PRIORITYClass 12 · Ch 9
NCERT~6 hours
STUDY TIME

Amines: important concepts

- **Classification:** 1° RNH_2 , 2° R_2NH , 3° R_3N (and quaternary salts $\text{R}_4\text{N}^+\text{X}^-$). N is sp^3 and pyramidal ($\text{C-N-C} \approx 108^\circ$), with a lone pair that makes amines **bases and nucleophiles**.
- **Boiling points:** $1^\circ > 2^\circ > 3^\circ$ for isomers (N-H hydrogen bonding). Amines boil higher than alkanes but lower than alcohols of similar mass.
- **Basicity** is decided by three effects together: +I effect, **solvation** of the ammonium ion (H-bonding with water), and **steric** hindrance. Aromatic amines are much weaker bases (the lone pair is delocalised into the ring).

Preparation

Σ FORMULA Routes to amines

Method	Transformation	Note
Reduction of nitro compounds	$\text{ArNO}_2 \xrightarrow{\text{Fe/HCl or Sn/HCl or H}_2/\text{Pd}} \text{ArNH}_2$	Fe/HCl preferred (the FeCl_2 formed regenerates HCl)
Ammonolysis of alkyl halides	$\text{RX} + \text{NH}_3 \longrightarrow \text{RNH}_2$ (+ 2° , 3° , quaternary)	a mixture; excess NH_3 favours 1°
Reduction of nitriles	$\text{RCN} \xrightarrow{\text{LiAlH}_4 \text{ or } \text{H}_2/\text{Ni}} \text{RCH}_2\text{NH}_2$	adds one carbon (ascending series)
Reduction of amides	$\text{RCONH}_2 \xrightarrow{\text{LiAlH}_4} \text{RCH}_2\text{NH}_2$	same number of C
Gabriel phthalimide	phthalimide + KOH $\rightarrow$ + RX $\rightarrow$ hydrolysis $\rightarrow$ RNH_2	pure 1° aliphatic amines only (not aniline)
Hofmann bromamide degradation	$\text{RCONH}_2 + \text{Br}_2 + 4\text{NaOH} \longrightarrow \text{RNH}_2 + \text{Na}_2\text{CO}_3 + 2\text{NaBr} + 2\text{H}_2\text{O}$	amine has one carbon fewer

Basicity (NCERT $\text{p}K_b$ values; lower $\text{p}K_b$ = stronger base)

Amine	$\text{p}K_b$	Amine	$\text{p}K_b$
methanamine CH_3NH_2	3.38	ethanamine $\text{C}_2\text{H}_5\text{NH}_2$	3.29
N-methylmethanamine $(\text{CH}_3)_2\text{NH}$	3.27	N-ethylethanamine $(\text{C}_2\text{H}_5)_2\text{NH}$	3.00
N,N-dimethylmethanamine $(\text{CH}_3)_3\text{N}$	4.22	N,N-diethylethanamine $(\text{C}_2\text{H}_5)_3\text{N}$	3.25
ammonia	4.75	benzenamine (aniline)	9.38

☆ IMPORTANT Basicity orders to memorise

- **Methyl series (water):** $(\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_3\text{N} > \text{NH}_3$.
- **Ethyl series (water):** $(\text{C}_2\text{H}_5)_2\text{NH} > (\text{C}_2\text{H}_5)_3\text{N} > \text{C}_2\text{H}_5\text{NH}_2 > \text{NH}_3$.
- **Gas phase** (inductive effect only): $3^\circ > 2^\circ > 1^\circ > \text{NH}_3$.
- **Aromatic amines:** electron-donating groups raise basicity (*p*-toluidine > aniline), electron-withdrawing groups lower it (aniline > *p*-nitroaniline). Aniline $\ll$ NH_3 .

Reactions of amines

REACTION MAP Reaction map · aniline $C_6H_5NH_2$

HCl	→ anilinium chloride (salt)
CH_3COCl or $(CH_3CO)_2O$ / pyridine	→ acetanilide ($-NH_2$ protected; activation reduced)
C_6H_5COCl (benzoylation)	→ benzanilide
$CHCl_3$ + alc. KOH (carbylamine)	→ phenyl isocyanide (foul smell): 1° amines only
$NaNO_2$ + HCl, 273–278 K	→ benzenediazonium chloride
$C_6H_5SO_2Cl$ (Hinsberg)	→ N-phenylbenzenesulphonamide (soluble in alkali, as for all 1° amines)
bromine water	→ 2,4,6-tribromoaniline (white precipitate)
conc. HNO_3 + H_2SO_4 (unprotected)	→ tarry oxidation + <i>p</i> - (51%), <i>m</i> - (47%), <i>o</i> - (2%) nitroaniline
acetylate → nitrate → hydrolyse	→ mainly <i>p</i> -nitroaniline
conc. H_2SO_4 , 453–473 K	→ sulphanilic acid (exists as a zwitterion)
$AlCl_3$ (Friedel–Crafts)	→ no reaction: aniline complexes the Lewis acid

Σ FORMULA Hinsberg test (benzenesulphonyl chloride)

Amine	Product	Solubility in KOH
1°	$C_6H_5SO_2NHR$	soluble (acidic N–H)
2°	$C_6H_5SO_2NR_2$	insoluble
3°	no reaction	—

☆ IMPORTANT Nitrous acid

1° **aliphatic** amines + HNO_2 give an unstable diazonium salt, which decomposes to an alcohol + N_2 (quantitative N_2 evolution). 1° **aromatic** amines give **stable arenediazonium salts** at 273–278 K (stabilised by resonance with the ring).

Diazonium salts: importance in synthesis

Σ FORMULA Replacement of N_2 ($Ar-N_2^+X^- \rightarrow$)

Reagent	Product	Name
$CuCl/HCl$	$ArCl$	Sandmeyer
$CuBr/HBr$	$ArBr$	Sandmeyer
$CuCN/KCN$	$ArCN$	Sandmeyer
Cu powder + HX	ArX	Gattermann
KI	ArI	—
HBf_4 , then heat	ArF	Balz–Schiemann
H_3PO_2 or CH_3CH_2OH	ArH	(reductive removal of $-NH_2$)
H_2O , warm	$ArOH$	—
$NaNO_2/Cu$ (via fluoroborate)	$ArNO_2$	—

Σ FORMULA Coupling (N₂ retained): azo dyes

- Phenol in **alkaline** medium + $\text{C}_6\text{H}_5\text{N}_2^+ \rightarrow p\text{-hydroxyazobenzene}$ (**orange dye**).
- Aniline in **mildly acidic** medium + $\text{C}_6\text{H}_5\text{N}_2^+ \rightarrow p\text{-aminoazobenzene}$ (**aniline yellow**; also listed in Practical Chemistry).
- Coupling is an **electrophilic** substitution: the diazonium ion is a weak electrophile, so it needs a strongly activated ring.

⚡ SHORTCUT Synthesis logic with diazonium salts

To put a group on a benzene ring where direct substitution fails or gives the wrong orientation, go **via the amine**: nitrate $\rightarrow$ reduce $\rightarrow$ (use $-\text{NH}_2$ to direct/activate) $\rightarrow$ diazotise $\rightarrow$ replace. Classic: **1,3,5-tribromobenzene** from aniline (brominate to 2,4,6-tribromoaniline, then diazotise and remove N_2 with H_3PO_2).

Common traps**⚠ MISTAKE ALERT Mistake alerts**

- Gabriel synthesis **cannot** make aniline (aryl halides don't undergo $\text{S}_{\text{N}}2$).
- Hofmann bromamide **loses one carbon**; nitrile reduction **gains one carbon**.
- The carbylamine test works for 1° amines (aliphatic and aromatic) only.
- Direct nitration of aniline gives a lot of the **meta** product, because the anilinium ion ($-\text{NH}_3^+$) forms in acid.
- Order of basicity in water isn't simply $3^\circ > 2^\circ > 1^\circ$. Solvation and sterics matter.

Question patterns**📊 PYQ PATTERN Recurring structures**

- Basicity orders (aliphatic in water/gas; substituted anilines).
- Conversions using diazonium salts (Sandmeyer, Balz–Schiemann, H_3PO_2 , coupling).
- Distinguishing 1°/2°/3° amines (Hinsberg, carbylamine, HNO_2).
- Hofmann bromamide / Gabriel / nitrile reduction: product and carbon count.
- EAS of aniline: bromination, nitration with and without protection, sulphanilic acid.

Practice questions**C18 · Practice****1 Medium** ⌚ 1 min **Concept**

C18-01

The correct order of basic strength in aqueous solution is:

- (A) $(\text{CH}_3)_3\text{N} > (\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2 > \text{NH}_3$ (B) $\text{NH}_3 > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_2\text{NH} > (\text{CH}_3)_3\text{N}$
 (C) $(\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_3\text{N} > \text{NH}_3$ (D) $\text{CH}_3\text{NH}_2 > (\text{CH}_3)_2\text{NH} > (\text{CH}_3)_3\text{N} > \text{NH}_3$

2 Easy ⌚ 0.5 min **Concept**

C18-02

The Gabriel phthalimide synthesis can't be used to prepare:

- (A) aniline (B) ethylamine (C) methylamine (D) benzylamine

3 Easy ⌚ 0.5 min **Speed**

C18-03

Acetamide with Br_2 and aqueous NaOH gives:

- (A) ethylamine (B) acetic acid (C) acetonitrile (D) methylamine

4 Easy 0.5 min Concept

C18-04

Which amine gives the carbylamine (isocyanide) test?

- (A) N-methylaniline (B) aniline
(C) N,N-dimethylaniline (D) triethylamine

5 Easy 0.75 min Concept

C18-05

In the Hinsberg test, a 2° amine gives:

- (A) a sulphonamide insoluble in KOH (B) a sulphonamide soluble in KOH
(C) no reaction (D) an isocyanide

6 Easy 0.5 min Speed

C18-06

Aniline with bromine water at room temperature gives:

- (A) *p*-bromoaniline (B) *o*-bromoaniline
(C) *m*-bromoaniline (D) 2,4,6-tribromoaniline

7 Medium 1 min Concept

C18-07

Direct nitration of aniline gives a large amount of *m*-nitroaniline because:

- (A) $-\text{NH}_2$ is a meta director
(B) in strongly acidic medium aniline is protonated to anilinium ion, whose $-\text{NH}_3^+$ group is meta-directing
(C) the para position is blocked
(D) nitration proceeds by a free-radical mechanism

8 Easy 0.5 min Speed

C18-08

Benzenediazonium chloride treated with CuCN/KCN gives:

- (A) phenyl isocyanide (B) benzonitrile
(C) benzoic acid (D) aniline

9 Easy 0.5 min Speed

C18-09

Benzenediazonium chloride couples with phenol in alkaline medium to give:

- (A) phenol + N_2 (B) *p*-aminoazobenzene (yellow)
(C) *p*-hydroxyazobenzene (orange) (D) benzene

10 Easy 0.75 min Concept

C18-10

Which is the weakest base?

- (A) NH_3 (B) aniline (C) *p*-toluidine (D) *p*-nitroaniline

11 Medium 1 min NV

C18-11

How many structurally isomeric **primary** amines have the formula $\text{C}_4\text{H}_{11}\text{N}$?

Your answer (numerical value):

12 Easy 0.5 min Speed

C18-12

Ethanenitrile (CH_3CN) reduced with LiAlH_4 gives:

- (A) methylamine (B) ethanal (C) ethanamide (D) ethylamine

C18 · Chapter Test

1 Easy 0.5 min Speed C18-T1

Arenediazonium salts are prepared at:

(A) 323 K (B) 373 K (C) 273–278 K (D) 200 K

2 Easy 0.5 min Concept C18-T2

Aniline doesn't undergo Friedel–Crafts reactions because:

(A) its $-\text{NH}_2$ group forms a salt/complex with AlCl_3
 (B) it is too reactive
 (C) it is meta-directing
 (D) it is insoluble

3 Easy 0.5 min NV C18-T3

The NCERT value of $\text{p}K_b$ for aniline is 9.38. Give it to the nearest integer.

Your answer (numerical value):

4 Easy 0.5 min Speed C18-T4

Fluorobenzene is prepared from benzenediazonium chloride by:

(A) CuF/HF (B) HBF_4 , then heating
 (C) KF (D) F_2 gas

5 Easy 0.5 min Concept C18-T5

Aniline is acetylated before nitration to:

(A) reduce the activating effect and protect $-\text{NH}_2$ from oxidation, giving mainly the para product
 (B) increase the activating effect of $-\text{NH}_2$
 (C) make it meta-directing
 (D) increase its solubility

Answers & Solutions

C18 · Practice

1 (C) 2 (A) 3 (D) 4 (B) 5 (A) 6 (D) 7 (B) 8 (B) 9 (C) 10 (D)
 11 4 12 (D)

C18 · Chapter Test

1 (C) 2 (A) 3 9 4 (B) 5 (A)

C18 · PRACTICE

1 Answer: (C) Basicity in water Medium 1 min Concept C18-01

SOLUTION +I effect, solvation and steric factors combine. The 3° amine's cation is poorly solvated (NCERT $\text{p}K_b$: 3.27, 3.38, 4.22, 4.75).

COMMON TRAP Using the gas-phase order (A).

2 Answer: (A) Gabriel synthesis limitation Easy 0.5 min Concept C18-02

SOLUTION Aryl halides don't undergo nucleophilic substitution with phthalimide anion. Benzyl halides do.

- 3 Answer: (D)** *Hofmann bromamide* **Easy** ⌚ 0.5 min **Speed** C18-03
SOLUTION One carbon is lost as carbonate: $\text{CH}_3\text{CONH}_2 \longrightarrow \text{CH}_3\text{NH}_2$.
- 4 Answer: (B)** *Carbylamine test* **Easy** ⌚ 0.5 min **Concept** C18-04
SOLUTION Only primary amines react.
- 5 Answer: (A)** *Hinsberg test* **Easy** ⌚ 0.75 min **Concept** C18-05
SOLUTION The product $\text{C}_6\text{H}_5\text{SO}_2\text{NR}_2$ has no acidic N-H left.
- 6 Answer: (D)** *Bromination of aniline* **Easy** ⌚ 0.5 min **Speed** C18-06
SOLUTION $-\text{NH}_2$ strongly activates all the ortho and para positions.
- 7 Answer: (B)** *Meta product in aniline nitration* **Medium** ⌚ 1 min **Concept** C18-07
SOLUTION Direct recall: see the concept summary and formula box of this chapter.
- 8 Answer: (B)** *Sandmeyer with CuCN* **Easy** ⌚ 0.5 min **Speed** C18-08
SOLUTION Direct recall: see the concept summary and formula box of this chapter.
- 9 Answer: (C)** *Azo coupling* **Easy** ⌚ 0.5 min **Speed** C18-09
SOLUTION Direct recall: see the concept summary and formula box of this chapter.
COMMON TRAP Confusing it with coupling to aniline (B).
- 10 Answer: (D)** *Weakest base* **Easy** ⌚ 0.75 min **Concept** C18-10
SOLUTION $-\text{NO}_2$ pulls the lone pair further into the ring by resonance.
- 11 Answer: 4** *Primary amine isomers* **Medium** ⌚ 1 min **NV** C18-11
SOLUTION Butan-1-amine, butan-2-amine, 2-methylpropan-1-amine, 2-methylpropan-2-amine.
COMMON TRAP Including 2° and 3° amines (8 in total).
- 12 Answer: (D)** *Reduction of a nitrile* **Easy** ⌚ 0.5 min **Speed** C18-12
SOLUTION $\text{CH}_3\text{CN} \longrightarrow \text{CH}_3\text{CH}_2\text{NH}_2$ (the nitrile C becomes CH_2).

C18 · CHAPTER TEST

- 1 Answer: (C)** *Diazotisation temperature* **Easy** ⌚ 0.5 min **Speed** C18-T1
SOLUTION They decompose at higher temperatures.
- 2 Answer: (A)** *Friedel–Crafts and aniline* **Easy** ⌚ 0.5 min **Concept** C18-T2
SOLUTION The resulting positive N strongly deactivates the ring.
- 3 Answer: 9** *pK_b of aniline* **Easy** ⌚ 0.5 min **NV** C18-T3
SOLUTION 9.38 rounds to 9 (compared with NH_3 4.75: aniline is a much weaker base).
- 4 Answer: (B)** *Balz–Schiemann* **Easy** ⌚ 0.5 min **Speed** C18-T4
SOLUTION Direct recall: see the concept summary and formula box of this chapter.
- 5 Answer: (A)** *Why acetylate aniline?* **Easy** ⌚ 0.5 min **Concept** C18-T5
SOLUTION Direct recall: see the concept summary and formula box of this chapter.

 REVISION Revision tracker · Nitrogen Compounds

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III · CHEMISTRY · UNIT 19

Biomolecules

19

~1 per shift
TYPICAL QUESTIONSEasy (pure NCERT
recall)
DIFFICULTYMust-do (fast
marks)
PRIORITYClass 12 · Ch 10
NCERT~4 hours
STUDY TIME

This is an NCERT-fact chapter. Learn the tables below until you can reproduce them on a blank sheet.

Carbohydrates

Σ FORMULA Classification

Class	Examples	Reducing?
Monosaccharides	glucose (aldohexose), fructose (ketohexose), ribose	yes
Disaccharides	sucrose (glucose + fructose)	no
	maltose (2 α -D-glucose), lactose (galactose + glucose)	yes
Polysaccharides	starch, cellulose, glycogen	no

Reducing sugars have a free aldehyde/ketone group (or a free hemiacetal) and reduce Tollens'/Fehling's reagents.

☆ IMPORTANT Structure of glucose: the NCERT evidence

Reaction	Result	Shows
Prolonged heating with HI	n-hexane	six carbons in a straight chain
NH ₂ OH / HCN	oxime / cyanohydrin	a carbonyl group
Bromine water	gluconic acid	the carbonyl is an aldehyde
Acetic anhydride	glucose pentaacetate	five –OH groups
Nitric acid	saccharic (glucaric) acid	one 1° alcohol (–CH ₂ OH)

The open chain fails to explain why glucose gives no Schiff's test, forms no NaHSO₃ adduct, and why its pentaacetate doesn't react with NH₂OH. Hence the **cyclic hemiacetal** (pyranose, C1–C5 ring) with α and β anomers at C1. **Mutarotation** is the interconversion of the anomers in solution.

Σ FORMULA Disaccharides & polysaccharides

Sugar	Units & linkage	Notes
Sucrose	α -D-glucose C1 – β -D-fructose C2	non-reducing; dextrorotatory; hydrolysis gives laevorotatory invert sugar
Maltose	α -D-glucose C1 – C4 α -D-glucose	reducing
Lactose	β -D-galactose C1 – C4 β -D-glucose	reducing; milk sugar
Starch: amylose (15–20%)	α -D-glucose, C1–C4	water-soluble, unbranched
Starch: amylopectin (80–85%)	α -D-glucose, C1–C4 + C1–C6 branches	insoluble, branched
Cellulose	β -D-glucose, C1–C4	straight chains; plant cell walls; not digested by humans
Glycogen	α -D-glucose (like amylopectin, more branched)	"animal starch": liver, muscles, brain

Fructose forms a five-membered (**furanose**) ring (C2–C5) and is laevorotatory: D-(–)-fructose.

Proteins

Σ FORMULA Amino acids & structure

- **α -Amino acids** exist as **zwitterions** ($\text{H}_3\text{N}^+ - \text{CHR} - \text{COO}^-$). They are amphoteric, with an isoelectric point. **Glycine** is the only common achiral amino acid. Naturally occurring amino acids have the L-configuration.
- **Essential amino acids** (must be in the diet): valine, leucine, isoleucine, arginine, lysine, threonine, methionine, phenylalanine, tryptophan, histidine.
- **Peptide bond**: $-\text{CO}-\text{NH}-$ (amide). An n -peptide has $(n - 1)$ peptide bonds.

Level	Description	Held by
Primary	amino-acid sequence	covalent (peptide) bonds
Secondary	α -helix, β -pleated sheet	hydrogen bonds between C=O and N–H
Tertiary	overall 3D folding	H-bonds, disulphide bonds, van der Waals, electrostatic forces
Quaternary	arrangement of subunits (e.g. haemoglobin)	same as tertiary

Fibrous proteins (keratin, myosin) vs **globular** proteins (insulin, albumins).

Denaturation (heat, pH change) destroys secondary and tertiary structure; the **primary structure stays intact**.

Examples: coagulation of egg white, curdling of milk.

Enzymes: mostly globular proteins, highly specific biocatalysts (e.g. maltase, invertase, pepsin).

Vitamins

Vitamin	Solubility	Deficiency disease
A	fat	xerophthalmia (hardening of the cornea), night blindness
B ₁ (thiamine)	water	beriberi (loss of appetite, retarded growth)
B ₂ (riboflavin)	water	cheilosis (fissuring at the corners of the mouth), digestive disorders, burning sensation of the skin
B ₆ (pyridoxine)	water	convulsions
B ₁₂ (cyanocobalamin)	water	pernicious anaemia
C (ascorbic acid)	water	scurvy (bleeding gums)

Vitamin	Solubility	Deficiency disease
D	fat	rickets (children), osteomalacia (adults)
E	fat	increased fragility of RBCs, muscular weakness
K	fat	increased blood-clotting time

Fat-soluble vitamins (A, D, E, K) are stored in the liver and adipose tissue. Water-soluble ones (B group, C) must be taken regularly, **except B₁₂**, which is stored in the liver.

Nucleic acids

	DNA	RNA
Sugar	2-deoxy-D-ribose	D-ribose
Bases	A, G, C, T	A, G, C, U
Structure	double helix	single strand
Types	—	mRNA, rRNA, tRNA
Function	heredity (carries genetic information)	protein synthesis

Nucleoside = base + sugar. **Nucleotide** = nucleoside + phosphate. Nucleotides are joined by **phosphodiester** bonds (5'–3'). Base pairing: **A=T (2 H-bonds)**, **G≡C (3 H-bonds)**.

Hormones (general introduction): peptide hormones (insulin, glucagon), amino-acid derivatives (thyroxine, adrenaline), and steroids (oestrogens, progesterone, testosterone). They are chemical messengers.

Common traps

MISTAKE ALERT Mistake alerts

- Sucrose is **non-reducing**; maltose and lactose are reducing.
- Cellulose has **β**-links; starch and glycogen have **α**-links.
- B₁₂ is water-soluble **but** stored in the liver.
- Denaturation leaves the primary structure intact.
- Thymine is found only in DNA; uracil only in RNA.

Question patterns

PYQ PATTERN Recurring structures

1. Glucose structure evidence (which reaction proves what).
2. Glycosidic linkages and reducing nature of disaccharides.
3. Vitamin ↔ deficiency matching; fat/water solubility.
4. DNA vs RNA (sugar, bases, pairing, H-bonds).
5. Protein structure levels and denaturation; essential amino acids.

Practice questions

C19 · Practice

1

Easy

 0.5 min

Speed

C19-01

On prolonged heating with HI, glucose gives n-hexane. This shows that glucose has:

- (A) an aldehyde group (B) six carbon atoms in a straight chain
(C) five –OH groups (D) a ring structure

2

Easy

⌚ 0.5 min

Speed

C19-02

Which sugar is non-reducing?

- (A) maltose (B) lactose (C) glucose (D) sucrose

3

Easy

⌚ 0.5 min

Speed

C19-03

Lactose is made of:

- (A) α -D-glucose + α -D-glucose (C1-C4) (B) α -D-glucose + β -D-fructose (C1-C2)
(C) β -D-galactose + β -D-glucose (C1-C4) (D) β -D-glucose + β -D-glucose (C1-C6)

4

Easy

⌚ 0.5 min

Speed

C19-04

Deficiency of vitamin B₁₂ causes:

- (A) pernicious anaemia (B) scurvy
(C) beriberi (D) rickets

5

Easy

⌚ 0.5 min

Speed

C19-05

Which vitamin is fat-soluble?

- (A) C (B) B₁ (C) B₆ (D) K

6

Easy

⌚ 0.5 min

Speed

C19-06

Which base is present in RNA but not in DNA?

- (A) adenine (B) uracil (C) thymine (D) guanine

7

Easy

⌚ 0.5 min

Concept

C19-07

When a protein is denatured:

- (A) its primary structure is destroyed
(B) it becomes an amino acid
(C) all peptide bonds are hydrolysed
(D) its secondary and tertiary structures are destroyed but the primary structure remains

8

Easy

⌚ 0.5 min

NV

C19-08

Glucose forms a pentaacetate with acetic anhydride. How many –OH groups does glucose have?

Your answer (numerical value):

9

Easy

⌚ 0.5 min

Concept

C19-09

Hydrolysis of sucrose gives an equimolar mixture called invert sugar because:

- (A) the mixture is dextrorotatory
(B) it doesn't rotate plane-polarised light
(C) the sign of optical rotation changes from (+) to (–)
(D) it reverses the reducing nature

10

Easy

⌚ 0.5 min

Speed

C19-10

Amylopectin differs from amylose in having:

- (A) branching through C1-C6 linkages
(B) β -glycosidic links
(C) fructose units
(D) solubility in cold water

11

Easy

⌚ 0.5 min

NV

C19-11

How many peptide bonds are present in a pentapeptide?

Your answer (numerical value):

12

Easy

⌚ 0.5 min

Speed

C19-12

Which amino acid is essential?

- (A) glycine
(B) lysine
(C) alanine
(D) glutamic acid



C19 · Chapter Test

1

Easy

⌚ 0.5 min

Speed

C19-T1

Cellulose is a polymer of:

- (A) α -D-glucose
(B) β -D-glucose
(C) fructose
(D) galactose

2

Easy

⌚ 0.5 min

Speed

C19-T2

In aqueous solution, amino acids exist mainly as:

- (A) cations only
(B) zwitterions (dipolar ions)
(C) anions only
(D) neutral molecules without charge

3

Easy

⌚ 0.5 min

NV

C19-T3

How many hydrogen bonds form between guanine and cytosine in DNA?

Your answer (numerical value):

4

Easy

⌚ 0.5 min

Speed

C19-T4

Deficiency of vitamin D causes:

- (A) night blindness
(B) scurvy
(C) rickets
(D) beriberi

5

Easy

⌚ 0.5 min

Concept

C19-T5

Which observation is evidence that glucose exists in a cyclic form?

- (A) it gives an oxime
(B) it forms a pentaacetate
(C) it gives gluconic acid with bromine water
(D) it doesn't give Schiff's test

Answers & Solutions

C19 · Practice

- 1 (B) 2 (D) 3 (C) 4 (A) 5 (D) 6 (B) 7 (D) 8 5 9 (C) 10 (A)
11 4 12 (B)

C19 · Chapter Test

- 1 (B) 2 (B) 3 3 4 (C) 5 (D)

C19 · PRACTICE

1 Answer: (B) *Glucose + HI* **Easy** ⌚ 0.5 min **Speed** C19-01

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

2 Answer: (D) *Non-reducing sugar* **Easy** ⌚ 0.5 min **Speed** C19-02

SOLUTION Both anomeric carbons (glucose C1 and fructose C2) are tied up in the glycosidic bond.

3 Answer: (C) *Lactose linkage* **Easy** ⌚ 0.5 min **Speed** C19-03

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

COMMON TRAP Choosing (A), which is maltose.

4 Answer: (A) *Vitamin B₁₂ deficiency* **Easy** ⌚ 0.5 min **Speed** C19-04

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

5 Answer: (D) *Fat-soluble vitamin* **Easy** ⌚ 0.5 min **Speed** C19-05

SOLUTION A, D, E and K are fat-soluble.

6 Answer: (B) *RNA-only base* **Easy** ⌚ 0.5 min **Speed** C19-06

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

7 Answer: (D) *Denaturation* **Easy** ⌚ 0.5 min **Concept** C19-07

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

8 Answer: 5 *Hydroxyl groups in glucose* **Easy** ⌚ 0.5 min **NV** C19-08

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

9 Answer: (C) *Invert sugar* **Easy** ⌚ 0.5 min **Concept** C19-09

SOLUTION Fructose's laevorotation (−92.4°) outweighs glucose's dextrorotation (+52.5°).

10 Answer: (A) *Amylopectin* **Easy** ⌚ 0.5 min **Speed** C19-10

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

11 Answer: 4 *Peptide bonds* **Easy** ⌚ 0.5 min **NV** C19-11

SOLUTION $n - 1 = 4$.

COMMON TRAP Answering 5.

12 Answer: (B) *Essential amino acid* **Easy** ⌚ 0.5 min **Speed** C19-12

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

C19 · CHAPTER TEST

1 Answer: (B) *Cellulose* **Easy** ⌚ 0.5 min **Speed** C19-T1

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

2 Answer: (B) *Zwitterions* **Easy** ⌚ 0.5 min **Speed** C19-T2

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

3 Answer: 3 *G–C hydrogen bonds* **Easy** ⌚ 0.5 min **NV** C19-T3

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

COMMON TRAP Answering 2 (that's A–T).

4 Answer: (C) *Vitamin D deficiency* **Easy** ⌚ 0.5 min **Speed** C19-T4

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

5 Answer: (D) *Evidence for ring structure* **Easy** ⌚ 0.5 min **Concept** C19-T5

SOLUTION The open-chain aldehyde would restore Schiff's reagent's colour. The absence of that colour points to the hemiacetal ring.

**REVISION** Revision tracker · Biomolecules

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III · CHEMISTRY · UNIT 20

Principles Related to Practical Chemistry

20

~1 per shift

TYPICAL QUESTIONS

Easy (Lab-manual recall)

DIFFICULTY

High

PRIORITY

Laboratory Manual
Chemistry, Class
11 & 12

NCERT

~5 hours

STUDY TIME

VERIFIED FACT Syllabus scope (Unit 20)

Detection of N, S and halogens; functional-group tests (alcoholic and phenolic –OH, aldehyde/ketone, –COOH, –NH₂); preparation of Mohr's salt, potash alum, acetanilide, *p*-nitroacetanilide, aniline yellow and iodoform; acid–base titrations with indicators, oxalic acid vs KMnO₄ and Mohr's salt vs KMnO₄; salt analysis for the listed cations and anions (insoluble salts excluded); and four physical-chemistry experiments [NTA-SYL]. 2026 analyses noted questions taken **directly from NCERT lab manuals** [|| THIRD-PARTY ANALYSIS](#) [T-ANALYSIS26].

Functional-group tests

Group	Test	Positive result
Alcoholic –OH	ceric ammonium nitrate	red colour
	sodium metal	effervescence (H ₂)
Phenolic –OH	neutral FeCl ₃	violet/blue/green colour
	bromine water	white precipitate (2,4,6-tribromophenol)
Carbonyl (any)	2,4-DNP (Brady's)	orange-yellow precipitate
Aldehyde	Tollens' reagent	silver mirror
	Fehling's solution	red precipitate (Cu ₂ O); not for aromatic aldehydes
	Schiff's reagent	pink/magenta colour restored
Ketone	sodium nitroprusside + NaOH	red/wine-red colour (methyl ketones)
–COOH	NaHCO ₃	brisk effervescence (CO ₂)
	ethanol + conc. H ₂ SO ₄	fruity smell (ester)
–NH ₂ (1°)	carbylamine (CHCl ₃ + KOH)	offensive smell (isocyanide)
Aromatic –NH ₂ (1°)	azo-dye test: NaNO ₂ /HCl (0–5 °C), then alkaline β-naphthol	orange-red dye

Preparations

Compound	How (principle)	Appearance
Mohr's salt FeSO ₄ · (NH ₄) ₂ SO ₄ · 6 H ₂ O	equimolar FeSO ₄ + (NH ₄) ₂ SO ₄ in water with dil. H ₂ SO ₄ (prevents hydrolysis of Fe ²⁺); crystallise	pale green crystals
Potash alum K ₂ SO ₄ · Al ₂ (SO ₄) ₃ · 24 H ₂ O	equimolar K ₂ SO ₄ + Al ₂ (SO ₄) ₃ with dil. H ₂ SO ₄ ; crystallise	colourless octahedral crystals
Acetanilide	aniline + acetic anhydride (glacial acetic acid)	white crystals
<i>p</i>-Nitroacetanilide	acetanilide + conc. HNO ₃ /conc. H ₂ SO ₄ , kept cold	pale yellow crystals

Compound	How (principle)	Appearance
Aniline yellow (<i>p</i> -aminoazobenzene)	diazotise aniline (0–5 °C), couple with aniline	yellow/orange solid
Iodoform	ethanol or acetone + I ₂ + alkali (e.g. KOH/Na ₂ CO ₃), warm	pale yellow crystals, characteristic smell

Titrimetric exercises

Σ FORMULA Indicators & redox titrations

Titration	Suitable indicator
strong acid vs strong base	phenolphthalein or methyl orange
weak acid vs strong base (e.g. CH ₃ COOH/NaOH)	phenolphthalein (pH 8.3–10; colourless → pink)
strong acid vs weak base (e.g. HCl/NH ₃)	methyl orange (pH 3.1–4.4; red → yellow)

Oxalic acid vs KMnO₄: $2 \text{MnO}_4^- + 5 \text{C}_2\text{O}_4^{2-} + 16 \text{H}^+ \longrightarrow 2 \text{Mn}^{2+} + 10 \text{CO}_2 + 8 \text{H}_2\text{O}$. Heat to **60–70 °C**; acidify with dil. H₂SO₄. KMnO₄ is a **self-indicator**: the end point is a permanent light pink. Mn²⁺ formed catalyses the reaction (autocatalysis).

Mohr's salt vs KMnO₄: $\text{MnO}_4^- + 5 \text{Fe}^{2+} + 8 \text{H}^+ \longrightarrow \text{Mn}^{2+} + 5 \text{Fe}^{3+} + 4 \text{H}_2\text{O}$. **No heating** (air would oxidise Fe²⁺); dil. H₂SO₄ is added.

Mole ratios: KMnO₄ : oxalate = 2 : 5; KMnO₄ : Fe²⁺ = 1 : 5.

Qualitative salt analysis

Σ FORMULA Cation group scheme (syllabus cations)

Group	Group reagent	Cations (precipitate)	Confirmatory tests
0	NaOH, heat	NH ₄ ⁺ (NH ₃ gas)	Nessler's reagent → brown precipitate
I	dil. HCl	Pb ²⁺ (PbCl ₂ white, soluble in hot water)	KI → yellow PbI ₂ ; K ₂ CrO ₄ → yellow PbCrO ₄
II	H ₂ S in dil. HCl	Cu ²⁺ (CuS black); Pb ²⁺ (PbS black)	Cu ²⁺ : NH ₄ OH → deep blue ; K ₄ [Fe(CN) ₆] → chocolate-brown
III	NH ₄ Cl + NH ₄ OH	Al ³⁺ (white gelatinous Al(OH) ₃), Fe ³⁺ (reddish-brown Fe(OH) ₃)	Fe ³⁺ : K ₄ [Fe(CN) ₆] → Prussian blue ; KSCN → blood red . Al ³⁺ : blue lake test
IV	H ₂ S in ammoniacal medium	Zn ²⁺ (ZnS white), Ni ²⁺ (NiS black)	Ni ²⁺ : dimethylglyoxime → rosy-red ; Zn ²⁺ : K ₄ [Fe(CN) ₆] → bluish-white
V	(NH ₄) ₂ CO ₃ + NH ₄ Cl/NH ₄ OH	Ba ²⁺ , Ca ²⁺ (white carbonates)	Ba ²⁺ : K ₂ CrO ₄ → yellow; flame apple green . Ca ²⁺ : ammonium oxalate → white; flame brick red
VI	Na ₂ HPO ₄ + NH ₄ OH	Mg ²⁺ (white crystalline MgNH ₄ PO ₄)	—

Why NH₄Cl in Group III? The common-ion effect suppresses NH₄OH ionisation, keeping [OH[−]] low enough that only the Group III hydroxides (very low *K_{sp}*) precipitate.

Σ FORMULA Anion tests (syllabus anions)

Anion	Preliminary test	Confirmatory test
CO_3^{2-}	dil. $\text{H}_2\text{SO}_4 \rightarrow$ effervescence	CO_2 turns lime water milky
S^{2-}	dil. $\text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}$ (rotten eggs)	lead acetate paper turns black; sodium nitroprusside $\rightarrow$ violet
NO_2^-	dil. $\text{H}_2\text{SO}_4 \rightarrow$ brown fumes	$\text{KI} + \text{starch} \rightarrow$ blue
Cl^-	conc. $\text{H}_2\text{SO}_4 \rightarrow \text{HCl}$ fumes	$\text{AgNO}_3 \rightarrow$ white precipitate, soluble in NH_4OH ; chromyl chloride test (orange-red CrO_2Cl_2 vapours $\rightarrow$ yellow PbCrO_4)
Br^-	conc. $\text{H}_2\text{SO}_4 \rightarrow$ brown Br_2 vapours	$\text{AgNO}_3 \rightarrow$ pale-yellow precipitate (partly soluble in NH_4OH); layer test $\rightarrow$ orange-brown
I^-	conc. $\text{H}_2\text{SO}_4 \rightarrow$ violet I_2 vapours	$\text{AgNO}_3 \rightarrow$ yellow precipitate (insoluble); layer test $\rightarrow$ violet
NO_3^-	conc. $\text{H}_2\text{SO}_4 + \text{Cu} \rightarrow$ brown fumes	brown ring test: $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]^{2+}$
SO_4^{2-}	(no gas)	$\text{BaCl}_2 \rightarrow$ white BaSO_4 , insoluble in HCl

Physical-chemistry experiments (syllabus)

- Enthalpy of solution of CuSO_4 :** measure the temperature change in a calorimeter; $\Delta H = -\frac{(mc + W)\Delta T}{n}$. Anhydrous CuSO_4 dissolves exothermically.
- Enthalpy of neutralisation** (strong acid + strong base): about -57.1 kJ per mol of water.
- Lyophilic sols** (starch, gum, gelatin: warm with water directly) vs **lyophobic sols** ($\text{Fe}(\text{OH})_3$ by hydrolysis of FeCl_3 in boiling water; As_2S_3 by double decomposition).
- Kinetics of $\text{I}^- + \text{H}_2\text{O}_2$:** $\text{H}_2\text{O}_2 + 2\text{I}^- + 2\text{H}^+ \rightarrow \text{I}_2 + 2\text{H}_2\text{O}$. The time for the blue starch-iodine colour to appear (thiosulphate clock) measures the rate at different concentrations.

Common traps**⚠ MISTAKE ALERT Mistake alerts**

- Don't heat in the Mohr's salt- KMnO_4 titration. Do heat ($60-70^\circ\text{C}$) for oxalic acid.
- HCl is never used** to acidify KMnO_4 titrations (it's oxidised to Cl_2).
- The chromyl chloride test is given by **chlorides only**.
- Nessler's reagent (alkaline K_2HgI_4) $\rightarrow$ **brown** precipitate with $\text{NH}_4^+/\text{NH}_3$.
- DMG gives a rosy-red precipitate with Ni^{2+} , not with Zn^{2+} .

Question patterns**📊 PYQ PATTERN Recurring structures**

- Group reagent / confirmatory test / colour matching.
- Indicator choice; conditions for KMnO_4 titrations; titration stoichiometry (NV).
- Reagents and appearance in the listed preparations.
- Functional-group tests for a given compound.
- Principles of the four physical-chemistry experiments.

Practice questions

C20 · Practice

1 Easy 0.5 min Speed

C20-01

A test that distinguishes an aldehyde from a ketone is:

- (A) 2,4-DNP test (B) Tollens' test
(C) sodium bicarbonate test (D) Lucas test

2 Easy 0.5 min Speed

C20-02

The best indicator for titrating acetic acid with NaOH is:

- (A) methyl orange (B) starch (C) litmus only (D) phenolphthalein

3 Easy 0.5 min Concept

C20-03

In the titration of oxalic acid with KMnO_4 :

- (A) no indicator is needed, and the acid solution is warmed to 60–70 °C
(B) phenolphthalein is used at room temperature
(C) HCl is used to acidify
(D) starch is the indicator

4 Easy 0.5 min Concept

C20-04

Dilute H_2SO_4 is added while preparing Mohr's salt to:

- (A) increase its solubility (B) oxidise Fe^{2+}
(C) prevent hydrolysis of Fe^{2+} (D) act as an indicator

5 Easy 0.5 min Speed

C20-05

The brown ring in the test for nitrate is due to:

- (A) $\text{Fe}_2(\text{SO}_4)_3$ (B) NO_2
(C) $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]^{2+}$ (D) $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$

6 Easy 0.5 min Speed

C20-06

The group reagent for Group II cations is:

- (A) H_2S in the presence of dil. HCl (B) dil. HCl
(C) $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$ (D) $(\text{NH}_4)_2\text{CO}_3$

7 Easy 0.5 min Speed

C20-07

Ni^{2+} is confirmed by:

- (A) potassium ferrocyanide
(B) the flame test
(C) Nessler's reagent
(D) dimethylglyoxime in ammoniacal solution, giving a rosy-red precipitate

8 Easy 0.5 min Concept

C20-08

The chromyl chloride test is used to confirm:

- (A) bromide (B) iodide (C) sulphate (D) chloride

9 Easy 0.5 min Speed

C20-09

Barium salts impart which flame colour?

- (A) brick red (B) apple (grassy) green
(C) crimson red (D) golden yellow

10 Medium 1.5 min NV

C20-10

Find the volume (in mL) of 0.02 M KMnO_4 needed to titrate 20 mL of 0.1 M oxalic acid in acidic medium.

Your answer (numerical value):

11 Easy 0.5 min Concept

C20-11

The azo-dye test (diazotisation followed by alkaline β -naphthol) is characteristic of:

- (A) 1° aliphatic amines (B) phenols
(C) 2° amines (D) 1° aromatic amines

12 Easy 0.5 min Speed

C20-12

A lyophobic sol of ferric hydroxide is prepared by:

- (A) dissolving $\text{Fe}(\text{OH})_3$ in cold water
(B) adding NH_4OH to FeCl_3 at room temperature
(C) adding FeCl_3 solution to boiling water (hydrolysis)
(D) dissolving iron in HCl

C20 · Chapter Test

1 Easy 0.5 min Speed

C20-T1

The cations precipitated by NH_4OH in the presence of NH_4Cl (from the syllabus list) are:

- (A) Fe^{3+} , Al^{3+} (B) Zn^{2+} , Ni^{2+} (C) Ca^{2+} , Ba^{2+} (D) Cu^{2+} , Pb^{2+}

2 Easy 0.5 min Speed

C20-T2

Nessler's reagent gives a brown precipitate with:

- (A) Ni^{2+} (B) NH_4^+ (ammonia) (C) Cl^- (D) Fe^{3+}

3 Easy 1 min NV

C20-T3

Find the volume (in mL) of 0.1 M KMnO_4 needed to oxidise 50 mL of 0.1 M Fe^{2+} (Mohr's salt) in acidic medium.

Your answer (numerical value):

4 Easy 0.5 min Speed

C20-T4

Lead chloride precipitated in Group I:

- (A) is insoluble in hot water (B) dissolves in hot water
(C) is black (D) doesn't form

5 Easy 0.5 min Speed

C20-T5

Acetanilide is prepared from aniline using:

- (A) acetic anhydride (or acetyl chloride) (B) acetaldehyde
(C) ethanol (D) CHCl_3/KOH

Answers & Solutions

C20 · Practice

- 1 (B) 2 (D) 3 (A) 4 (C) 5 (C) 6 (A) 7 (D) 8 (D) 9 (B) 10 40
11 (D) 12 (C)

C20 · Chapter Test

- 1 (A) 2 (B) 3 10 4 (B) 5 (A)

C20 · PRACTICE

1 Answer: (B) Aldehyde vs ketone Easy 0.5 min Speed

C20-01

SOLUTION 2,4-DNP is positive for both.

COMMON TRAP Choosing 2,4-DNP.

2 Answer: (D) Indicator for weak acid–strong base Easy 0.5 min Speed

C20-02

SOLUTION The equivalence point is basic ($\text{pH} \approx 8-9$), which falls in phenolphthalein's range.

3 Answer: (A) Oxalic acid titration Easy 0.5 min Concept

C20-03

SOLUTION KMnO_4 is a self-indicator. Warming speeds up the slow initial reaction.

4 Answer: (C) Mohr's salt preparation Easy 0.5 min Concept

C20-04

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

5 Answer: (C) Brown ring Easy 0.5 min Speed

C20-05

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

6 Answer: (A) Group II reagent Easy 0.5 min Speed

C20-06

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

7 Answer: (D) Confirming nickel Easy 0.5 min Speed

C20-07

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

8 Answer: (D) Chromyl chloride test Easy 0.5 min Concept

C20-08

SOLUTION Only chlorides give volatile orange-red CrO_2Cl_2 .

9 Answer: (B) Flame test for barium Easy 0.5 min Speed

C20-09

SOLUTION Ca is brick red, Sr crimson, Na golden yellow.

10 Answer: 40 Permanganate–oxalate titration Medium 1.5 min NV

C20-10

SOLUTION Moles of oxalic acid $= 2 \times 10^{-3}$. KMnO_4 needed $= \frac{2}{5} \times 2 \times 10^{-3} = 8 \times 10^{-4}$ mol. $V = 8 \times 10^{-4} / 0.02 = 0.040 \text{ L} = 40 \text{ mL}$.

SHORTCUT Equivalents: $5 \times 0.02 \times V = 2 \times 0.1 \times 20$.

COMMON TRAP Using a 1 : 1 ratio (100 mL).

11 Answer: (D) Azo-dye test Easy 0.5 min Concept

C20-11

SOLUTION Only aromatic diazonium salts are stable enough to couple.

12 Answer: (C) *Lyophobic sol* **Easy** ⌚ 0.5 min **Speed**

C20-12

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

COMMON TRAP (B) gives a precipitate, not a sol.

C20 · CHAPTER TEST

1 Answer: (A) *Group III* **Easy** ⌚ 0.5 min **Speed**

C20-T1

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

2 Answer: (B) *Nessler's reagent* **Easy** ⌚ 0.5 min **Speed**

C20-T2

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

3 Answer: 10 *Mohr's salt titration* **Easy** ⌚ 1 min **NV**

C20-T3

SOLUTION $\text{MnO}_4^- : \text{Fe}^{2+} = 1 : 5$, so $\text{KMnO}_4 = 5 \times 10^{-3} / 5 = 10^{-3} \text{ mol} = 10 \text{ mL}$.

4 Answer: (B) *Group I* **Easy** ⌚ 0.5 min **Speed**

C20-T4

SOLUTION That's why some Pb^{2+} also appears in Group II.

5 Answer: (A) *Acetanilide* **Easy** ⌚ 0.5 min **Speed**

C20-T5

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

REVISION Revision tracker · Practical Chemistry

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

Mathematics

IV

Mathematics strategy, then 14 unit modules: formulas, identities, standard results, question models, fast methods, practice sets and chapter tests.

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PART IV · MATHEMATICS

Mathematics: The Complete Strategy

What Mathematics in JEE Main actually tests

In 2026 analyses, Mathematics was repeatedly described as the **longliest section and the rank decider**, with multi-step calculations and several concepts combined in single questions. Coaching commentary suggested that students who spent much more than 60–70 minutes on Maths often struggled overall [THIRD-PARTY ANALYSIS](#) [T-ANALYSIS26]. The implication is clear: in Maths, **selection and calculation fitness** matter as much as knowing the chapters.

VERIFIED FACT Syllabus structure (official)

The official Mathematics syllabus has **14 units** [NTA-SYL]. The 2024 revision removed Mathematical Induction and Mathematical Reasoning entirely, and trimmed topics inside units. For example, the 3D Geometry unit text now reads "...Equations of a line", with planes no longer listed; the vector unit lists scalar and vector products (triple products are not listed); Binomial lists "general term and middle term"; Sequences lists AP, GP and AM–GM. Each module below flags what is and isn't listed.

WARNING "Removed" doesn't always mean "never asked"

Third-party PYQ archives list post-2024 3D questions that involve **planes** [THIRD-PARTY ANALYSIS](#) [T-EXAMSIDE]. This book's recommendation [RECOMMENDATION](#) : learn the **basic plane formulas** (normal form, distance of a point from a plane, angle between a line and a plane) as a cheap 2-hour insurance topic, and don't go deeper than that.

Unit-wise weight and priority

Ranges compiled from published 2024–2026 analyses [THIRD-PARTY ANALYSIS](#) [T-WT-VED], [T-WT-CD]. For 2026, published counts (all Paper-1 shifts) include **Sequences & Series 33, Matrices & Determinants 33, 3D Geometry 29, Vector Algebra 27** and **Sets & Relations 13** questions [T-WT-CD]. Priority is this book's recommendation

[RECOMMENDATION](#)

#	Unit	Typical Q / shift	Difficulty	Priority	Hours
1	Sets, Relations & Functions	1–2	Easy–Med	High	8
2	Complex Numbers & Quadratic Equations	2–3	Medium	Must-do	14
3	Matrices & Determinants	2–3	Easy–Med	Must-do	10
4	Permutations & Combinations	1–2	Medium	Must-do	10
5	Binomial Theorem	1–2	Easy–Med	Must-do	6
6	Sequences & Series	1–2	Medium	Must-do	8
7	Limits, Continuity & Differentiability (+ applications of derivatives)	2–3	Medium	Must-do	16
8	Integral Calculus (indefinite, definite, area)	2–3	Med–Hard	Must-do	18
9	Differential Equations	1–2	Medium	Must-do	7
10	Coordinate Geometry (lines, circles, conics)	3–4	Medium	Must-do	20
11	Three-Dimensional Geometry	1–2	Easy–Med	Must-do	7
12	Vector Algebra	1–2	Easy	Must-do	5
13	Statistics & Probability	2–3	Medium	Must-do	12
14	Trigonometry (identities, inverse trig)	~1	Easy–Med	High	6

☆ **IMPORTANT** The "Must-do models" (fastest marks in Maths)

Vectors & 3D (dot/cross product, line equations, shortest distance) · Matrices & determinants (adjoint, inverse, systems of equations, properties) · Binomial general term · AP/GP sums · P&C arrangements · Definite-integral properties (King's rule, even/odd) · Linear differential equations · Probability with Bayes' theorem · Standard conics (focus, directrix, latus rectum, eccentricity) · Complex modulus/argument/locus. **These typically give 12–15 questions a shift** (third-party consensus of unit ranges) and are the backbone of the Weak-Maths strategy (page 16).

How to study a Maths chapter 📌 RECOMMENDATION

1. **Formula sheet first.** Write the chapter's formulas and standard results (from this module) in your own notebook.
2. **Solve the solved examples** in NCERT, then this module's practice set with **full working**.
3. **Model drills:** for each "question model" in the module, solve 8–10 questions in a row. Repetition builds recognition.
4. **Timed PYQs:** chapter-wise PYQs from 2024–2026 at 2.5 minutes per question.
5. **Calculation fitness:** 15 minutes a day of hand arithmetic and algebra (Speed Manual, page 389).
6. **Mixed practice** once a chapter is done: interleave it with earlier chapters. Mixing problem types improves the ability to *choose* a method [LS-ROH].

Mathematics: how to maximise marks

Strong student (≥ 75)

- Attempt 20–22, but go through the paper **twice**: Round 1 takes only questions whose method you see in 20 seconds.
- Most of your lost marks are calculation errors. Use a verification habit on every NV answer (substitute back, check the limiting case).
- Build speed in integration and coordinate geometry. They carry the most marks and are the most lengthy.

Average student (45–75)

- Master the Must-do models above before anything else.
- Target 16–18 attempts at 85%+ accuracy.
- The main loss is spending 8 minutes on one question. Enforce the **3-minute rule** in every mock.
- Learn standard results (area formulas, the $\int_0^{\pi/2}$ results, P&C formulas) cold.

Weak student (< 45)

- Work on the Must-do models only for the first 4–6 weeks.
- Attempt 12–14 questions, all from the models, at 90% accuracy: about 45–50 marks.
- Daily: 20 easy questions from two models, plus 15 minutes of arithmetic drill. See also "Fear of Mathematics" (page 455).

Maths exam-hall rules 📌 RECOMMENDATION

- **Read all 25 questions' first lines within 4 minutes** of starting Maths and tick those you recognise instantly.
- **Section B (NV) Maths** often includes direct questions (a determinant value, a coefficient, a count). Scan for them early.
- **Never do algebra in your head.** Write intermediate steps on the rough sheet. Most Maths errors are transcription slips.
- **Use the options:** substitution, special values, limiting cases and dimension/degree checks eliminate options fast (Answer-Choice Strategy, page 396).
- **Stop rule:** if a question needs a second approach after 3 minutes, mark it and leave it.

PART IV · MATHEMATICS · UNIT 01

Sets, Relations & Functions

01

1-2 per shift

TYPICAL QUESTIONS

Easy-Medium

DIFFICULTY

High

PRIORITY

Class 11 · Ch 1, 2;

Class 12 · Ch 1

NCERT

~8 hours

STUDY TIME

Core concepts

- **Sets:** union, intersection, complement, difference; De Morgan's laws; power set; Cartesian product.
- **Relations** from A to B are subsets of $A \times B$. On a single set they can be reflexive, symmetric, transitive, or all three (**equivalence**).
- **Functions** assign exactly one output to each input. They can be one-one (injective), onto (surjective) or both (bijective). Only bijections have inverses.
- Composition $(f \circ g)(x) = f(g(x))$ is associative but generally **not commutative**.

Important formulas & standard results

Σ FORMULA Sets

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$n(A \cup B \cup C) = \sum n(A) - \sum n(A \cap B) + n(A \cap B \cap C)$$

$$(A \cup B)' = A' \cap B', (A \cap B)' = A' \cup B' \text{ (De Morgan)}. |P(A)| = 2^n. n(A \times B) = n(A)n(B).$$

Σ FORMULA Counting relations and functions ($|A| = m, |B| = n$; relations on a set of size n)

Object	Number
Relations from A to B	2^{mn}
Relations on a set	2^{n^2}
Reflexive relations	2^{n^2-n}
Symmetric relations	$2^{n(n+1)/2}$
Reflexive and symmetric	$2^{n(n-1)/2}$
Functions $A \rightarrow B$	n^m
One-one functions ($m \leq n$)	${}^n P_m = \frac{n!}{(n-m)!}$
Onto functions ($A \rightarrow B$)	$\sum_{k=0}^n (-1)^k \binom{n}{k} (n-k)^m$; for $n = 2: 2^m - 2$
Bijections ($m = n$)	$n!$

Σ FORMULA Domain & range rules

- $\sqrt{g(x)}$ needs $g(x) \geq 0$; $\log g(x)$ needs $g(x) > 0$; $\frac{1}{g(x)}$ needs $g(x) \neq 0$; $\sin^{-1} g(x)$, $\cos^{-1} g(x)$ need $-1 \leq g(x) \leq 1$. With $\log_{g(x)}$ also $g(x) \neq 1$.
- **Range:** solve $y = f(x)$ for x and find the y values that give a valid x ; or use monotonicity/AM-GM.
- $\frac{x^2}{1+x^2} \in [0, 1)$; $x + \frac{1}{x} \geq 2$ for $x > 0$; $a \sin x + b \cos x \in [-\sqrt{a^2+b^2}, \sqrt{a^2+b^2}]$.

Σ FORMULA Special functions

Function	Key facts
$ x $	even; $\sqrt{x^2} = x $
Greatest integer $[x]$	$[x] \leq x < [x] + 1$; $[x + n] = [x] + n$ for integer n ; $[-2.3] = -3$
Fractional part $\{x\} = x - [x]$	$0 \leq \{x\} < 1$; period 1
Signum $\operatorname{sgn} x$	1, 0, -1 for $x > 0, = 0, < 0$
Even / odd	$f(-x) = f(x)$ / $f(-x) = -f(x)$. Every f = even part + odd part
Periods	$\sin kx, \cos kx$: $2\pi/ k $; $\tan kx$: $\pi/ k $; $ \sin x $: π ; $ \sin x + \cos x $: $\pi/2$; $\{x\}$: 1

Common question models

▮ PYQ PATTERN Recurring structures

1. **Classify a relation** (reflexive/symmetric/transitive/equivalence) defined on a small set or by a rule (e.g. $a - b$ divisible by 3).
2. **Count** relations/functions with a property (onto, one-one, reflexive, symmetric).
3. **Domain of a composite expression** with roots, logs and inverse trig.
4. **Range** of rational or trigonometric functions.
5. **Inverse / composition**: $f^{-1}(x)$, $f \circ g$, solving $f(x) = f^{-1}(x)$.
6. **Venn-diagram counting** with 2–3 sets.

Shortcuts & formula recognition

⚡ SHORTCUT Equivalence test in 30 seconds

For a relation given as a list of pairs, check **reflexive** first (all (a, a) present?), then **symmetric** (is every pair's reverse present?). Only if both pass, check **transitive**, and only on chains of pairs with a shared middle element. Most relations fail at the first two checks, which saves the slowest check.

⚡ SHORTCUT Inverse of a Möbius function

If $f(x) = \frac{ax+b}{cx+d}$, then $f^{-1}(x) = \frac{dx-b}{-cx+a}$: swap a and d , negate b and c .

When NOT to use: when the function isn't of this form, or when the domain restricts the inverse.

Common mistakes

⚠ MISTAKE ALERT Mistake alerts

- $\sqrt{x^2} = |x|$, not x . So $g(f(x))$ with $f = x^2$, $g = \sqrt{}$ gives $|x|$.
- Onto-ness depends on the **codomain**. The same rule can be onto or not.
- $[x]$ for negative numbers: $[-0.5] = -1$.
- Forgetting $g(x) \neq 1$ for a logarithm's base, or $\neq 0$ for a denominator, when finding the domain.

Practice questions

M01 · Practice

1 Easy 0.5 min Speed

M01-01

The number of relations on a set with 3 elements is:

- (A) 9 (B) 64 (C) 512 (D) 8

2 Easy 1 min Concept

M01-02

On $A = \{1, 2, 3\}$, $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}$ is:

- (A) reflexive only (B) an equivalence relation
(C) symmetric only (D) transitive but not symmetric

3 Easy 1 min NV

M01-03

Find the number of onto functions from a 4-element set to a 2-element set.

Your answer (numerical value):

4 Medium 1.5 min JEE

M01-04

The domain of $f(x) = \sqrt{x-1} + \frac{1}{\log_e(2-x)}$ is:

- (A) $[1, 2)$ (B) $(-\infty, 2)$ (C) $[1, 2]$ (D) $(1, 2)$

5 Easy 0.5 min Speed

M01-05

If $n(A) = 40$, $n(B) = 30$ and $n(A \cap B) = 10$, then $n(A \cup B)$ is:

- (A) 60 (B) 70 (C) 80 (D) 50

6 Medium 1 min Basic

M01-06

If $f(x) = \frac{2x+3}{x-1}$ ($x \neq 1$), then $f^{-1}(x)$ is:

- (A) $\frac{2x-3}{x+1}$ (B) $\frac{x-1}{2x+3}$ (C) $\frac{x-3}{x+2}$ (D) $\frac{x+3}{x-2}$

7 Medium 1 min Concept

M01-07

The fundamental period of $f(x) = |\sin x| + |\cos x|$ is:

- (A) $\pi/2$ (B) π (C) 2π (D) $\pi/4$

8 Easy 0.75 min NV

M01-08

Find the number of reflexive relations on a set with 3 elements.

Your answer (numerical value):

9 Medium 1 min Concept

M01-09

The range of $f(x) = \frac{x^2}{1+x^2}$, $x \in \mathbb{R}$, is:

- (A) $[0, 1]$ (B) $[0, 1)$ (C) $(0, 1)$ (D) $[0, \infty)$

10 Medium 1 min Tricky

M01-10

If $f(x) = x^2$ and $g(x) = \sqrt{x}$ ($x \geq 0$), then $(g \circ f)(x)$ for real x is:

- (A) x (B) x^2
(C) $|x|$ (D) not defined for $x < 0$

M01 · Chapter Test**1** Easy 0.5 min Speed

M01-T1

A set has 4 elements. Its power set has:

- (A) 16 elements (B) 8 elements (C) 4 elements (D) 24 elements

2 Easy 0.5 min Speed

M01-T2

$f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 3x + 5$ is:

- (A) one-one but not onto (B) bijective
(C) onto but not one-one (D) neither

3 Easy 0.5 min NV

M01-T3

Find the number of one-one functions from $\{a, b\}$ to $\{1, 2, 3, 4\}$.

Your answer (numerical value):

4 Easy 0.75 min Concept

M01-T4

Which function is even?

- (A) $x + \sin x$ (B) $x \cos x$ (C) $x \sin x$ (D) e^x

5 Easy 0.5 min Speed

M01-T5

$[-2.3]$ (greatest integer function) equals:

- (A) -2 (B) -2.3 (C) 2 (D) -3

Answers & Solutions**M01 · Practice**

1 (C) 2 (B) 3 14 4 (D) 5 (A) 6 (D) 7 (A) 8 64 9 (B) 10 (C)

M01 · Chapter Test

1 (A) 2 (B) 3 12 4 (C) 5 (D)

M01 · PRACTICE**1** Answer: (C) Counting relations Easy 0.5 min Speed

M01-01

SOLUTION $2^{n^2} = 2^9 = 512$.

COMMON TRAP Using 2^n .

2 Answer: (B) Equivalence relation check Easy 1 min Concept

M01-02

SOLUTION Reflexive (all (a, a) present), symmetric ($((1, 2)$ and $(2, 1))$), transitive (the only chains give $(1, 1)$ and $(2, 2)$, both present).

SHORTCUT The classes are $\{1, 2\}$ and $\{3\}$: a partition, so it's an equivalence.

3 Answer: 14 *Onto functions onto a 2-set* **Easy** ⌚ 1 min **NV**

M01-03

SOLUTION Total functions $2^4 = 16$, minus the 2 constant functions: 14.

SHORTCUT Onto a 2-set: $2^m - 2$.

COMMON TRAP Forgetting to subtract.

4 Answer: (D) *Domain with root and log* **Medium** ⌚ 1.5 min **JEE**

M01-04

SOLUTION Need $x \geq 1$, $2 - x > 0$ and $\log(2 - x) \neq 0$, i.e. $x \neq 1$. Together: $(1, 2)$.

SHORTCUT Check the endpoint $x = 1$: it makes the log zero.

COMMON TRAP Missing the denominator condition gives $[1, 2)$.

5 Answer: (A) *Union of two sets* **Easy** ⌚ 0.5 min **Speed**

M01-05

SOLUTION $40 + 30 - 10 = 60$.

6 Answer: (D) *Inverse of a rational function* **Medium** ⌚ 1 min **Basic**

M01-06

SOLUTION $y(x - 1) = 2x + 3 \Rightarrow x(y - 2) = y + 3 \Rightarrow x = \frac{y + 3}{y - 2}$.

SHORTCUT Möbius rule: swap $a = 2$ and $d = -1$, negate b and c : $\frac{-x - 3}{-x + 2}$, which equals (D).

COMMON TRAP Taking the reciprocal (B).

7 Answer: (A) *Period of a sum* **Medium** ⌚ 1 min **Concept**

M01-07

SOLUTION $f(x + \pi/2) = |\cos x| + |\sin x| = f(x)$, and no smaller positive period works.

COMMON TRAP Taking the LCM of the individual periods (π) without checking a smaller value.

8 Answer: 64 *Counting reflexive relations* **Easy** ⌚ 0.75 min **NV**

M01-08

SOLUTION The 3 diagonal pairs must be included; each of the remaining 6 pairs is optional: $2^6 = 64$.

SHORTCUT $2^{n^2 - n}$.

9 Answer: (B) *Range of a rational function* **Medium** ⌚ 1 min **Concept**

M01-09

SOLUTION $f = 1 - \frac{1}{1 + x^2}$. At $x = 0$, $f = 0$; as $|x| \rightarrow \infty$, $f \rightarrow 1$ but never reaches it.

COMMON TRAP Including 1.

10 Answer: (C) *Composition with square root* **Medium** ⌚ 1 min **Tricky**

M01-10

SOLUTION $g(f(x)) = \sqrt{x^2} = |x|$, which is defined for all real x because $x^2 \geq 0$.

COMMON TRAP Writing x .

M01 · CHAPTER TEST

1 Answer: (A) *Power set* **Easy** ⌚ 0.5 min **Speed**

M01-T1

SOLUTION $2^4 = 16$.

2 Answer: (B) *Bijective linear function* **Easy** ⌚ 0.5 min **Speed**

M01-T2

SOLUTION A non-constant linear function on $\mathbb{R}$ is strictly monotonic and takes every real value.

3 Answer: 12 *Counting one-one functions* **Easy** ⌚ 0.5 min **NV**

M01-T3

SOLUTION ${}^4P_2 = 4 \times 3 = 12$.

4 Answer: (C) *Even function* **Easy** ⌚ 0.75 min **Concept**

M01-T4

SOLUTION odd $\times$ odd = even. $x \cos x$ and $x + \sin x$ are odd; e^x is neither.

5 Answer: (D) *Greatest integer function* **Easy** ⌚ 0.5 min **Speed**

M01-T5

SOLUTION The greatest integer ≤ -2.3 is -3 .

COMMON TRAP Truncating to -2 .

REVISION Revision tracker · Sets, Relations & Functions

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART IV · MATHEMATICS · UNIT 02

Complex Numbers & Quadratic Equations

02

2–3 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

Must-do

PRIORITY

Class 11 · Ch 4
(Complex
Numbers &
Quadratic
Equations)

NCERT

~14 hours

STUDY TIME

Core concepts

- A complex number $z = a + ib$ is an ordered pair (a, b) , plotted as a point in the **Argand plane**. $|z|$ is its distance from the origin; $\arg z$ is its angle, with principal value in $(-\pi, \pi]$.
- Polar form** $z = r(\cos \theta + i \sin \theta) = re^{i\theta}$ turns multiplication into "multiply moduli, add arguments".
- Geometry**: $|z - z_1|$ is the distance between points, so modulus equations describe **loci** (circles, lines, ellipses).
- Quadratics**: roots, discriminant, Vieta's relations, symmetric functions of roots, and conditions on where the roots lie.

NOTE Syllabus note

The official text lists: complex numbers as ordered pairs, $a + ib$ form, Argand diagram, algebra, modulus, argument; quadratics in real and complex systems, roots and coefficients, nature of roots, forming equations [NTA-SYL]. **Cube roots of unity** and **De Moivre's theorem** are not named; they're included here as time-saving tools only.

Important formulas & identities

Σ FORMULA Algebra of complex numbers

$$\bar{z} = a - ib \quad z\bar{z} = |z|^2 \quad \frac{1}{z} = \frac{\bar{z}}{|z|^2} \quad \operatorname{Re} z = \frac{z + \bar{z}}{2} \quad \operatorname{Im} z = \frac{z - \bar{z}}{2i}$$

$$|z_1 z_2| = |z_1| |z_2| \quad \left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|} \quad \arg(z_1 z_2) = \arg z_1 + \arg z_2 \pmod{2\pi}$$

$$|z_1 \pm z_2|^2 = |z_1|^2 + |z_2|^2 \pm 2\operatorname{Re}(z_1 \bar{z}_2) \quad ||z_1| - |z_2|| \leq |z_1 + z_2| \leq |z_1| + |z_2|$$

Powers of i cycle with period 4: $i, -1, -i, 1$. $(1 + i)^2 = 2i$, $(1 - i)^2 = -2i$.

Principal argument of $a + ib$: find $\alpha = \tan^{-1}|b/a|$, then place it in the right quadrant: I: α ; II: $\pi - \alpha$; III: $-(\pi - \alpha)$; IV: $-\alpha$.

Σ FORMULA Standard loci

Condition	Locus
$ z - z_0 = r$	circle, centre z_0 , radius r
$ z - z_1 = z - z_2 $	perpendicular bisector of $z_1 z_2$
$ z - z_1 + z - z_2 = 2a, 2a > z_1 - z_2 $	ellipse with foci z_1, z_2
$ z - z_1 = k z - z_2 , k \neq 1$	circle (Apollonius)
$\arg \frac{z - z_1}{z - z_2} = \pm \frac{\pi}{2}$	circle with $z_1 z_2$ as diameter (minus the endpoints)
$\operatorname{Re} z = c / \operatorname{Im} z = c$	vertical / horizontal line

Σ FORMULA Tools (not named in the syllabus text)

Cube roots of unity: $1, \omega, \omega^2$ with $\omega = \frac{-1 + i\sqrt{3}}{2}$, $1 + \omega + \omega^2 = 0$, $\omega^3 = 1$. Reduce powers mod 3.

De Moivre: $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$.

Σ FORMULA Quadratic equations $ax^2 + bx + c = 0$

$$x = \frac{-b \pm \sqrt{D}}{2a}, D = b^2 - 4ac \quad \alpha + \beta = -\frac{b}{a} \quad \alpha\beta = \frac{c}{a} \quad |\alpha - \beta| = \frac{\sqrt{D}}{|a|}$$

$$\alpha^2 + \beta^2 = S^2 - 2P \quad \alpha^3 + \beta^3 = S^3 - 3PS \quad \frac{1}{\alpha} + \frac{1}{\beta} = \frac{S}{P} \quad \text{equation with roots } \alpha, \beta : x^2 - Sx + P = 0$$

Nature (real coefficients): $D > 0$ real distinct; $D = 0$ equal; $D < 0$ complex conjugates. Rational coefficients with D a perfect square give rational roots. Irrational/complex roots come in conjugate pairs.

Common root of $a_1x^2 + b_1x + c_1 = 0$ and $a_2x^2 + b_2x + c_2 = 0$: $(c_1a_2 - c_2a_1)^2 = (a_1b_2 - a_2b_1)(b_1c_2 - b_2c_1)$. Often faster: subtract the equations.

Sign of $f(x) = ax^2 + bx + c$: if $D < 0$, f has the sign of a for all x . Extreme value $-\frac{D}{4a}$ at $x = -\frac{b}{2a}$.

Location of roots (both roots greater than k , $a > 0$): $D \geq 0$, $f(k) > 0$, $-\frac{b}{2a} > k$. k lies between the roots iff $a f(k) < 0$.

Common question models**▮ PYQ PATTERN** Recurring structures

1. **Modulus/argument** of a quotient or power; principal argument in the correct quadrant.
2. **Loci** from modulus/argument conditions; maximum/minimum $|z|$ on a circle (distance from the origin to the centre $\pm$ radius).
3. **Equations in z** ($z^2 + |z| = 0$, $\bar{z} = iz^2$): substitute $z = x + iy$ and compare real and imaginary parts.
4. **Vieta / symmetric functions**, forming new equations with transformed roots.
5. **Nature and location of roots**, common roots, parameter ranges.
6. **Powers of i or ω** in sums.

Shortcuts & fast methods**⚡ SHORTCUT** Min/max of $|z|$ on a circle

If $|z - z_0| = r$, then $|z|$ ranges over $[|z_0| - r, |z_0| + r]$. More generally, $|z - w|$ ranges over the distance from w to the centre $\pm r$.

Time saved: a whole parametrisation.

⚡ SHORTCUT Transformed roots

Equation whose roots are $k\alpha, k\beta$: replace x by x/k . Roots $\alpha + h$: replace x by $x - h$. Roots $1/\alpha$: reverse the coefficients.

Common mistakes**⚠ MISTAKE ALERT Mistake alerts**

- Principal argument of $-1 - i$ is $-3\pi/4$, not $\pi/4$ ($\tan^{-1}(b/a)$ loses the quadrant).
- $\sqrt{a}\sqrt{b} = \sqrt{ab}$ fails when both are negative: $\sqrt{-1}\sqrt{-1} = -1$.
- For "real roots", check $D \geq 0$, and also $a \neq 0$ if the problem says "quadratic".
- Conjugate-pair rules need **real** coefficients.

Practice questions**M02 · Practice****1** Easy ⌚ 0.75 min Basic

M02-01

For $z = \frac{1+i}{1-i}$, $|z|$ and $\arg z$ are:

- (A) $1, -\pi/2$ (B) $\sqrt{2}, \pi/4$ (C) $1, \pi/4$ (D) $1, \pi/2$

2 Medium ⌚ 1.5 min Concept

M02-02

The locus of z satisfying $|z - 2| = |z + 2i|$ is the line:

- (A) $x + y = 0$ (B) $x - y = 0$ (C) $x = 2$ (D) $y = -2$

3 Medium ⌚ 1 min Basic

M02-03

If the roots of $x^2 - px + q = 0$ differ by 1, then:

- (A) $p^2 = 4q - 1$ (B) $p^2 = 4q + 1$ (C) $q^2 = 4p + 1$ (D) $p^2 + 4q = 1$

4 Medium ⌚ 1 min Calc

M02-04

If α, β are the roots of $2x^2 - 3x - 5 = 0$, then $\alpha^2 + \beta^2$ equals:

- (A) $9/4$ (B) $19/4$ (C) $29/4$ (D) $-11/4$

5 Easy ⌚ 1 min NV

M02-05

Find the largest integer k for which $kx^2 + 4x + 1 = 0$ has real roots.

Your answer (numerical value):

6 Medium ⌚ 1.5 min JEE

M02-06

If $x^2 + ax + 1 = 0$ and $x^2 + x + a = 0$ ($a \neq 1$) have a common root, then a equals:

- (A) 2 (B) -2 (C) 1 (D) 0

7 Easy ⌚ 0.75 min NV

M02-07

Find the value of $(1 + i)^8$.

Your answer (numerical value):

8 Medium 1 min Concept

M02-08

For non-zero z_1, z_2 , $|z_1 + z_2| = |z_1| + |z_2|$ implies:

- (A) $z_1 = \bar{z}_2$ (B) $\arg z_1 = -\arg z_2$ (C) $\arg z_1 = \arg z_2$ (D) $|z_1| = |z_2|$

9 Medium 1 min Concept

M02-09

If ω is a non-real cube root of unity, then $1 + \omega^{100} + \omega^{200}$ equals:

- (A) 0 (B) 1 (C) 3 (D) ω

10 Medium 1.5 min JEE

M02-10

Both roots of $x^2 - 2kx + k^2 - 1 = 0$ lie strictly between -2 and 4 . Then k lies in:

- (A) $(0, 2)$ (B) $(-2, 4)$ (C) $(-3, 5)$ (D) $(-1, 3)$

M02 · Chapter Test

1 Easy 0.5 min Speed

M02-T1

 i^{2027} equals:

- (A) 1 (B) i (C) $-i$ (D) -1

2 Easy 0.75 min Basic

M02-T2

The conjugate of $\frac{1}{2-i}$ is:

- (A) $\frac{-2+i}{5}$ (B) $\frac{2+i}{5}$ (C) $2+i$ (D) $\frac{2-i}{5}$

3 Easy 1 min NV

M02-T3

If α, β are the roots of $x^2 - 5x + 6 = 0$, find $\alpha^3 + \beta^3$.Your answer (numerical value):

4 Easy 0.5 min Speed

M02-T4

The quadratic with roots $2 \pm \sqrt{3}$ is:

- (A) $x^2 - 4x + 1 = 0$ (B) $x^2 + 4x + 1 = 0$ (C) $x^2 - 4x - 1 = 0$ (D) $x^2 - 2x + 3 = 0$

5 Easy 0.5 min Speed

M02-T5

The locus $|z - 1| = 3$ is a circle with:

- (A) centre $(1, 0)$, radius 9 (B) centre $(0, 1)$, radius 3
(C) centre $(-1, 0)$, radius 3 (D) centre $(1, 0)$, radius 3

Answers & Solutions

M02 · Practice

1 (D) 2 (A) 3 (B) 4 (C) 5 4 6 (B) 7 16 8 (C) 9 (A) 10 (D)

M02 · Chapter Test

1 (C) 2 (D) 3 35 4 (A) 5 (D)

M02 · PRACTICE

1 Answer: (D) Modulus and argument of a quotient **Easy** ⌚ 0.75 min **Basic** M02-01

SOLUTION $\frac{(1+i)^2}{2} = \frac{2i}{2} = i.$

2 Answer: (A) Perpendicular-bisector locus **Medium** ⌚ 1.5 min **Concept** M02-02

SOLUTION $(x-2)^2 + y^2 = x^2 + (y+2)^2 \Rightarrow -4x = 4y \Rightarrow y = -x.$

SHORTCUT The perpendicular bisector of (2, 0) and (0, -2) passes through their midpoint (1, -1) with slope -1.

3 Answer: (B) Roots differing by 1 **Medium** ⌚ 1 min **Basic** M02-03

SOLUTION $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta = p^2 - 4q = 1.$

4 Answer: (C) Symmetric function of roots **Medium** ⌚ 1 min **Calc** M02-04

SOLUTION $S = 3/2, P = -5/2. S^2 - 2P = 9/4 + 5 = 29/4.$

COMMON TRAP Sign of P .

5 Answer: 4 Real-root condition **Easy** ⌚ 1 min **NV** M02-05

SOLUTION $D = 16 - 4k \geq 0 \Rightarrow k \leq 4.$ At $k = 4$ the root is repeated, which is still real (and $k \neq 0$ keeps it quadratic).

COMMON TRAP Using $D > 0$ strictly.

6 Answer: (B) Common root **Medium** ⌚ 1.5 min **JEE** M02-06

SOLUTION Subtracting: $(a-1)x - (a-1) = 0 \Rightarrow x = 1.$ Substituting: $1 + a + 1 = 0 \Rightarrow a = -2.$

SHORTCUT Subtract to find the common root first.

7 Answer: 16 Power of $(1+i)$ **Easy** ⌚ 0.75 min **NV** M02-07

SOLUTION $(1+i)^2 = 2i$, so $(1+i)^8 = (2i)^4 = 16i^4 = 16.$

8 Answer: (C) Triangle-inequality equality **Medium** ⌚ 1 min **Concept** M02-08

SOLUTION Equality holds only when the vectors point in the same direction.

9 Answer: (A) Powers of ω **Medium** ⌚ 1 min **Concept** M02-09

SOLUTION $\omega^{100} = \omega^{99}\omega = \omega; \omega^{200} = \omega^{198}\omega^2 = \omega^2.$ Sum = $1 + \omega + \omega^2 = 0.$

SHORTCUT Reduce the exponents mod 3.

10 Answer: (D) Location of roots **Medium** ⌚ 1.5 min **JEE** M02-10

SOLUTION The roots are $k \pm 1$. Need $k-1 > -2$ and $k+1 < 4$, so $-1 < k < 3.$

SHORTCUT Factor first: $(x-k)^2 = 1.$

COMMON TRAP Using the general location conditions and making an algebra slip.

M02 · CHAPTER TEST

1 Answer: (C) Power of i **Easy** ⌚ 0.5 min **Speed** M02-T1

SOLUTION $2027 = 4 \times 506 + 3$, so $i^3 = -i.$

2 Answer: (D) Conjugate of a reciprocal **Easy** ⌚ 0.75 min **Basic** M02-T2

SOLUTION $\frac{1}{2-i} = \frac{2+i}{5}$, whose conjugate is $\frac{2-i}{5}.$

COMMON TRAP Stopping at $\frac{2+i}{5}.$

3 Answer: 35 Sum of cubes of roots **Easy** ⌚ 1 min **NV** M02-T3

SOLUTION $S^3 - 3PS = 125 - 90 = 35$ (check: $8 + 27$).

4 Answer: (A) Forming a quadratic **Easy** ⌚ 0.5 min **Speed** M02-T4

SOLUTION $S = 4, P = 4 - 3 = 1.$

5 Answer: (D) Circle locus **Easy** ⌚ 0.5 min **Speed** M02-T5

SOLUTION Direct recall: see the concept summary and formula box of this chapter.



REVISION Revision tracker · Complex & Quadratics

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART IV · MATHEMATICS · UNIT 03

Matrices & Determinants

03

2–3 per shift

TYPICAL QUESTIONS

Easy–Medium

DIFFICULTY

Must-do

PRIORITY

Class 12 · Ch 3, 4

NCERT

~10 hours

STUDY TIME



PYQ PATTERN Weight evidence

33 questions across the 2026 Paper-1 shifts ($\approx 7\%$ of Maths, tied for the highest in that dataset)

THIRD-PARTY ANALYSIS

[T-WT-CD]. This is the most reliable scoring unit in Maths.

Core concepts

- A **matrix** is a rectangular array. Matrix multiplication is associative and distributive, but **not commutative** ($AB \neq BA$ in general).
- Determinant** of a square matrix: a scalar that is zero exactly when the matrix is singular (non-invertible).
- Adjoint & inverse**: $A \operatorname{adj} A = |A|I$, so $A^{-1} = \operatorname{adj} A / |A|$ when $|A| \neq 0$.
- Systems of linear equations** are solved as $AX = B$. Consistency depends on $|A|$ and $(\operatorname{adj} A)B$.

Important formulas & standard results



FORMULA Types of matrices

Symmetric: $A^T = A$. Skew-symmetric: $A^T = -A$ (diagonal entries 0). Orthogonal: $AA^T = I$ ($|A| = \pm 1$).Idempotent: $A^2 = A$. Involutory: $A^2 = I$. Nilpotent: $A^k = O$.Every square matrix $= \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$ (symmetric + skew-symmetric). $(AB)^T = B^T A^T$; $(AB)^{-1} = B^{-1} A^{-1}$; $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.FORMULA Determinant results (A is $n \times n$)

$$|AB| = |A||B| \quad |kA| = k^n |A| \quad |A^T| = |A| \quad |A^{-1}| = \frac{1}{|A|}$$

- Swapping two rows changes the sign. Two identical or proportional rows give 0. $R_i \rightarrow R_i + kR_j$ doesn't change the value.
- A skew-symmetric matrix of **odd** order has determinant 0.
- Vandermonde: $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$.
- Area of a triangle: $\Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$. Zero area means the points are collinear.



FORMULA Adjoint & inverse

$$A(\operatorname{adj} A) = (\operatorname{adj} A)A = |A|I \quad |\operatorname{adj} A| = |A|^{n-1} \quad \operatorname{adj}(\operatorname{adj} A) = |A|^{n-2}A$$

$$|\operatorname{adj}(\operatorname{adj} A)| = |A|^{(n-1)^2} \quad \operatorname{adj}(AB) = \operatorname{adj} B \cdot \operatorname{adj} A \quad \operatorname{adj}(kA) = k^{n-1} \operatorname{adj} A$$

For 2×2 : $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (swap the diagonal, negate the off-diagonal).

2×2 **characteristic relation (Cayley–Hamilton)**: $A^2 - (\operatorname{tr} A)A + |A|I = O$.

Σ FORMULA Systems of equations $AX = B$

Condition	Result
$ A \neq 0$	unique solution $X = A^{-1}B$
$ A = 0, (\text{adj } A)B \neq O$	no solution (inconsistent)
$ A = 0, (\text{adj } A)B = O$	infinitely many solutions or none (check the equations directly)
Homogeneous ($B = O$), $ A \neq 0$	only the trivial solution
Homogeneous, $ A = 0$	infinitely many (non-trivial) solutions

Cramer: $x = \Delta_x / \Delta$, etc. If $\Delta = 0$ and some $\Delta_i \neq 0$, there is no solution.

① NOTE Syllabus note

The official text lists matrix algebra and types, evaluation of 2×2 and 3×3 determinants, area of triangles, adjoint and inverse using determinants, and consistency/solution of linear systems [NTA-SYL]. Determinant properties aren't named separately, but you still need them to evaluate determinants quickly.

Common question models**▮ PYQ PATTERN** Recurring structures

1. $|\text{adj } A|, |kA|, |A^{-1}|$ chains with given $|A|$ (very frequent, fast).
2. **Consistency of a 3×3 system** with parameters λ, μ : unique / none / infinitely many.
3. **Matrix powers** via patterns, idempotent/nilpotent properties or Cayley–Hamilton.
4. **Evaluating determinants** with factorisable structure (Vandermonde, cyclic).
5. **Symmetric/skew-symmetric** decomposition and properties; trace questions.
6. **Non-trivial solutions** of homogeneous systems.

Shortcuts & fast methods**⚡ SHORTCUT** Power chains in one line

For $n \times n$: $|\text{adj}(kA)| = (k^{n-1})^n |A|^{n-1}$. Always reduce to powers of $|A|$ and k **before** computing any number.

Common mistake: using n instead of $n - 1$.

⚡ SHORTCUT Parameter systems (λ, μ)

Find λ from $\Delta = 0$ first. Then substitute that λ and check the other determinant (or eliminate) to find μ for consistency. That's two small steps instead of solving the full system.

Common mistakes**⚠ MISTAKE ALERT** Mistake alerts

- $|kA| = k^n |A|$, not $k|A|$.
- $(AB)^{-1} = B^{-1}A^{-1}$. The order reverses.
- $\Delta = 0$ with all $\Delta_i = 0$ doesn't **guarantee** infinitely many solutions for 3 variables. Verify.
- $A^2 = O$ doesn't imply $A = O$. $AB = O$ doesn't imply $A = O$ or $B = O$.

Practice questions

M03 · Practice

- 1** **Easy** ⌚ 0.75 min **Speed** M03-01
If A is 3×3 with $|A| = 5$, then $|\text{adj}(\text{adj } A)|$ is:
(A) 25 (B) 125 (C) 625 (D) 3125
- 2** **Easy** ⌚ 0.5 min **NV** M03-02
If A is 3×3 with $|A| = 3$, find $|2A|$.
Your answer (numerical value):
- 3** **Medium** ⌚ 2 min **JEE** M03-03
The system $x + y + z = 6$, $x + 2y + 3z = 14$, $x + 2y + \lambda z = \mu$ has infinitely many solutions when:
(A) $\lambda = 3, \mu = 14$ (B) $\lambda = 3, \mu \neq 14$ (C) $\lambda \neq 3$ (D) $\lambda = 2, \mu = 10$
- 4** **Easy** ⌚ 0.5 min **Concept** M03-04
The determinant of any 3×3 skew-symmetric matrix is:
(A) 1 (B) 0 (C) -1 (D) the product of its diagonal entries squared
- 5** **Easy** ⌚ 0.75 min **Basic** M03-05
If $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, then A^{-1} is:
(A) $\begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix}$ (B) $\frac{1}{2} \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix}$
(C) $-\frac{1}{2} \begin{pmatrix} 1 & -2 \\ -3 & 4 \end{pmatrix}$ (D) $-\frac{1}{2} \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix}$
- 6** **Easy** ⌚ 1 min **NV** M03-06
Find the area of the triangle with vertices $(1, 1)$, $(4, 1)$ and $(1, 5)$.
Your answer (numerical value):
- 7** **Medium** ⌚ 1.5 min **Concept** M03-07
If $A^2 = A$, then $(I + A)^3$ equals:
(A) $I + 3A$ (B) $I + 8A$ (C) $I + 7A$ (D) $I + A$
- 8** **Medium** ⌚ 1.5 min **JEE** M03-08
The system $kx + y + z = 0$, $x + ky + z = 0$, $x + y + kz = 0$ has non-trivial solutions for:
(A) $k = 1$ only (B) $k = -2$ only (C) no real k (D) $k = 1$ or $k = -2$

9

Medium

⌚ 1 min

Basic

M03-09

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} \text{ equals:}$$

- (A) $(a+b)(b+c)(c+a)$ (B) $(a-b)(b-c)(c-a)$
 (C) abc (D) $a^2 + b^2 + c^2$

10

Medium

⌚ 1 min

Concept

M03-10

A 2×2 matrix has trace 5 and determinant 6. Then A^2 equals:

- (A) $-5A + 6I$ (B) $5A + 6I$ (C) $6A - 5I$ (D) $5A - 6I$



M03 · Chapter Test

1

Easy

⌚ 0.5 min

Speed

M03-T1

$(AB)^T$ equals:

- (A) $B^T A^T$ (B) $A^T B^T$ (C) AB (D) BA

2

Easy

⌚ 0.5 min

Speed

M03-T2

The symmetric part of a square matrix A is:

- (A) $\frac{1}{2}(A - A^T)$ (B) AA^T (C) $\frac{1}{2}(A + A^T)$ (D) A^T

3

Easy

⌚ 0.5 min

NV

M03-T3

Find $\begin{vmatrix} 2 & 1 \\ 3 & 5 \end{vmatrix}$.

Your answer (numerical value):

4

Easy

⌚ 0.5 min

Speed

M03-T4

If A is orthogonal, then $|A|$ is:

- (A) 0 (B) ± 1 (C) 2 (D) any real number

5

Easy

⌚ 0.5 min

Concept

M03-T5

For a 3×3 system, $\Delta = 0$ and $\Delta_x \neq 0$. The system has:

- (A) no solution (B) infinitely many solutions
 (C) a unique solution (D) exactly two solutions

Answers & Solutions

M03 · Practice

1 (C) 2 24 3 (A) 4 (B) 5 (D) 6 6 7 (C) 8 (D) 9 (B) 10 (D)

M03 · Chapter Test

1 (A) 2 (C) 3 7 4 (B) 5 (A)

M03 · PRACTICE

1

Answer: (C) *adj of adj*

Easy

⌚ 0.75 min

Speed

M03-01

SOLUTION $|A|^{(n-1)^2} = 5^4 = 625$.

COMMON TRAP Using $(n-1)$ instead of $(n-1)^2$ gives 25.

2 Answer: 24 Determinant of kA **Easy** ⌚ 0.5 min **NV**

M03-02

SOLUTION $2^3 \times 3 = 24$.

COMMON TRAP $2 \times 3 = 6$.

3 Answer: (A) Infinitely many solutions **Medium** ⌚ 2 min **JEE**

M03-03

SOLUTION Subtracting the second equation from the third: $(\lambda - 3)z = \mu - 14$. $\lambda \neq 3$ gives a unique solution. $\lambda = 3, \mu = 14$ gives infinitely many. $\lambda = 3, \mu \neq 14$ gives none.

SHORTCUT Subtract equations with matching coefficients.

4 Answer: (B) Odd-order skew-symmetric **Easy** ⌚ 0.5 min **Concept**

M03-04

SOLUTION $|A| = |A^T| = |-A| = (-1)^3|A|$, so $|A| = 0$.

5 Answer: (D) 2×2 inverse **Easy** ⌚ 0.75 min **Basic**

M03-05

SOLUTION $|A| = 4 - 6 = -2$. Swap the diagonal and negate the off-diagonal, then divide by -2 .

COMMON TRAP Sign of the determinant.

6 Answer: 6 Area by determinant **Easy** ⌚ 1 min **NV**

M03-06

SOLUTION A right triangle with legs 3 and 4: area = 6. (The determinant gives $\frac{1}{2}|12|$.)

7 Answer: (C) Idempotent matrix power **Medium** ⌚ 1.5 min **Concept**

M03-07

SOLUTION $(I + A)^2 = I + 2A + A^2 = I + 3A$. $(I + A)^3 = (I + 3A)(I + A) = I + 4A + 3A^2 = I + 7A$.

SHORTCUT With $A^n = A$: $(I + A)^n = I + (2^n - 1)A$.

8 Answer: (D) Non-trivial solutions **Medium** ⌚ 1.5 min **JEE**

M03-08

SOLUTION $\Delta = k^3 - 3k + 2 = (k - 1)^2(k + 2) = 0$.

SHORTCUT For the pattern "k on the diagonal, 1 elsewhere": $\Delta = (k - 1)^2(k + 2)$.

9 Answer: (B) Vandermonde determinant **Medium** ⌚ 1 min **Basic**

M03-09

SOLUTION Standard result: it vanishes when any two variables are equal, which fixes the factors. Compare the coefficient of bc^2 to fix the sign.

SHORTCUT Check with $a = 0, b = 1, c = 2$: the determinant is $1 \cdot 4 - 1 \cdot 2 = 2$; (B) gives $(-1)(-1)(2) = 2 \checkmark$.

10 Answer: (D) Cayley–Hamilton for 2×2 **Medium** ⌚ 1 min **Concept**

M03-10

SOLUTION $A^2 - (\text{tr } A)A + |A|I = O$.

M03 · CHAPTER TEST

1 Answer: (A) Transpose of a product **Easy** ⌚ 0.5 min **Speed**

M03-T1

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

2 Answer: (C) Symmetric part **Easy** ⌚ 0.5 min **Speed**

M03-T2

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

3 Answer: 7 2×2 determinant **Easy** ⌚ 0.5 min **NV**

M03-T3

SOLUTION $10 - 3 = 7$.

4 Answer: (B) Orthogonal matrix **Easy** ⌚ 0.5 min **Speed**

M03-T4

SOLUTION $|AA^T| = |A|^2 = 1$.

5 Answer: (A) Cramer's rule **Easy** ⌚ 0.5 min **Concept**

M03-T5

SOLUTION $x = \Delta_x / \Delta$ would need division by zero, so the system is inconsistent.

REVISION Revision tracker · Matrices & Determinants

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART IV · MATHEMATICS · UNIT 04

Permutations & Combinations

04

1-2 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

Must-do

PRIORITY

Class 11 · Ch 6

NCERT

~10 hours

STUDY TIME

Core concepts

- **Fundamental principle of counting:** independent stages multiply (AND). Mutually exclusive alternatives add (OR).
- **Permutation** = arrangement (order matters): nP_r . **Combination** = selection (order doesn't matter): nC_r .
- Most JEE questions are **counting with constraints**: identical objects, restrictions (together/apart), circular arrangements, distributions, digits.

Important formulas & standard results

Σ FORMULA Core formulas

$${}^nP_r = \frac{n!}{(n-r)!} \quad {}^nC_r = \frac{n!}{r!(n-r)!} \quad {}^nC_r = {}^nC_{n-r} \quad {}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$$

$${}^nC_x = {}^nC_y \Rightarrow x = y \text{ or } x + y = n \quad \frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r} \quad r {}^nC_r = n {}^{n-1}C_{r-1}$$

Σ FORMULA Arrangements

Situation	Count
n distinct objects in a row	$n!$
n objects with p alike, q alike, ...	$\frac{n!}{p!q!\dots}$
r from n with repetition allowed	n^r
Circular arrangement of n distinct objects	$(n-1)!$
Circular, clockwise = anticlockwise (necklace, garland)	$\frac{(n-1)!}{2}$
k particular objects together	$(n-k+1)!k!$
No two of k particular objects together (gap method)	$(n-k)! \cdot {}^{n-k+1}P_k$
Derangements of n objects (useful tool)	$D_n = n! \sum_{k=0}^n \frac{(-1)^k}{k!}$; $D_2 = 1$, $D_3 = 2$, $D_4 = 9$, $D_5 = 44$

Σ FORMULA Selections & distributions

- At least one from n distinct objects: $2^n - 1$. From p alike + q alike + r alike: $(p+1)(q+1)(r+1) - 1$.
- Divisors of $N = p^a q^b r^c$: $(a+1)(b+1)(c+1)$ (including 1 and N).
- n **identical** objects into r distinct boxes: ${}^{n+r-1}C_{r-1}$ (empty allowed); ${}^{n-1}C_{r-1}$ (none empty). The same counts give non-negative/positive integer solutions of $x_1 + \dots + x_r = n$.
- Dividing $m+n$ distinct objects into groups of m and n ($m \neq n$): $\frac{(m+n)!}{m!n!}$. Into two **equal** unlabelled groups of n : $\frac{(2n)!}{2!(n!)^2}$.
- Geometry: lines through n points (no 3 collinear): nC_2 ; triangles: nC_3 ; diagonals of an n -gon: ${}^nC_2 - n$.

Common question models



PYQ PATTERN Recurring structures

1. **Word arrangements** (with repeated letters, vowels together/apart, rank of a word in dictionary order).
2. **Digit problems:** numbers formed under conditions (even, divisible by 3/5, greater than a value, digits not repeated).
3. **Selections with constraints:** committees containing at least/at most k of a type.
4. **Distributions:** identical balls into boxes; integer solutions.
5. **Circular arrangements** with restrictions.
6. **Geometry counting:** lines, triangles, diagonals, rectangles in a grid.

Shortcuts & fast methods



SHORTCUT "At least" by complement

Count = total – (count with none). For "at least 2", subtract the cases with 0 and with 1. This almost always has fewer cases than direct counting.

When NOT to use: when the complement is harder (e.g. "exactly 3").



SHORTCUT Divisibility by 3 in digit problems

A number is divisible by 3 iff its digit sum is. Group the digits into residue classes mod 3, pick combinations whose sum is $\equiv 0$, then arrange. Handle the "0 can't lead" rule last, by subtraction.



SHORTCUT Rank of a word

Go letter by letter. For each position, count the letters **smaller** than the current letter still unused, and multiply by the arrangements of the remaining letters (divide for repeats). Add 1 at the end.

Common mistakes



MISTAKE ALERT Mistake alerts

- Treating selections as arrangements: a committee is a combination; a line-up is a permutation.
- Leading zero in digit problems.
- Circular arrangements of **identical-looking** necklace beads: divide by 2.
- "Groups" vs "boxes": unlabelled groups of equal size need an extra division by $k!$.
- Adding when you should multiply. Ask whether it's "and then" ($\times$) or "either/or" ($+$).

Practice questions



M04 · Practice

1

Easy

⌚ 1 min

Basic

M04-01

The number of arrangements of the letters of TRIANGLE in which the three vowels (I, A, E) are together is:

- (A) 720 (B) 4320 (C) 5040 (D) 1440

2

Medium

⌚ 1.5 min

Concept

M04-02

The number of ways to arrange 5 boys and 3 girls in a row so that no two girls are together is:

- (A) $5! \times {}^6P_3$ (B) $5! \times {}^6C_3$ (C) $8! - 3!6!$ (D) $5! \times 3!$

3 Medium 1.5 min NV

M04-03

From 6 men and 4 women, a committee of 3 is formed. Find the number of committees with **at least one** woman.

Your answer (numerical value):

4 Medium 1.5 min JEE

M04-04

The number of non-negative integer solutions of $x_1 + x_2 + x_3 = 10$ is:

- (A) 36 (B) 66 (C) 45 (D) 120

5 Easy 0.75 min Speed

M04-05

The number of ways 6 people can sit around a round table is:

- (A) 720 (B) 360 (C) 60 (D) 120

6 Easy 1 min NV

M04-06

Find the number of diagonals of a decagon (10-sided polygon).

Your answer (numerical value):

7 Medium 1.5 min JEE

M04-07

Using the digits 0, 1, 2, 3, 4 without repetition, how many 3-digit even numbers can be formed?

- (A) 36 (B) 24 (C) 30 (D) 48

8 Easy 0.75 min Speed

M04-08

If ${}^nC_8 = {}^nC_6$, then nC_2 equals:

- (A) 91 (B) 14 (C) 182 (D) 28

9 Medium 1 min NV

M04-09

Find the number of positive divisors of 360.

Your answer (numerical value):

10 Medium 1.5 min Tricky

M04-10

There are 10 points in a plane, of which exactly 4 are collinear (no other three collinear). The number of triangles they form is:

- (A) 120 (B) 110 (C) 116 (D) 104

M04 · Chapter Test

1 Easy 0.5 min Speed

M04-T1

7P_3 equals:

- (A) 35 (B) 5040 (C) 343 (D) 210

2 Easy 0.75 min Basic

M04-T2

The number of distinct arrangements of the letters of ARRANGE is:

- (A) 5040 (B) 2520 (C) 1260 (D) 630

3 Easy 0.75 min NV

M04-T3

In a party of 12 people, each shakes hands with every other person exactly once. Find the number of handshakes.

Your answer (numerical value):

4 Medium 1 min Concept

M04-T4

The number of different garlands that can be made with 6 distinct flowers is:

- (A) 120 (B) 60 (C) 720 (D) 360

5 Medium 1 min Concept

M04-T5

The number of ways to choose at least one fruit from 3 identical apples and 4 identical oranges is:

- (A) 11 (B) 20 (C) 12 (D) 19

Answers & Solutions

M04 · Practice

1 (B) 2 (A) 3 100 4 (B) 5 (D) 6 35 7 (C) 8 (A) 9 24 10 (C)

M04 · Chapter Test

1 (D) 2 (C) 3 66 4 (B) 5 (D)

M04 · PRACTICE

1 Answer: (B) Vowels together Easy 1 min Basic

M04-01

SOLUTION Treat the vowels as one block: 6 units, so $6!$ arrangements, times $3!$ inside the block: $720 \times 6 = 4320$.

COMMON TRAP Forgetting the internal $3!$.

2 Answer: (A) Gap method Medium 1.5 min Concept

M04-02

SOLUTION Arrange the boys ($5!$), which creates 6 gaps. Place the girls in 3 of those gaps in order: 6P_3 . Total = $120 \times 120 = 14400$.

COMMON TRAP Using 6C_3 (forgetting that the girls are distinct).

3 Answer: 100 Committee with at least one woman Medium 1.5 min NV

M04-03

SOLUTION Total ${}^{10}C_3 = 120$; all-men ${}^6C_3 = 20$; so $120 - 20 = 100$.

SHORTCUT Complement.

COMMON TRAP Computing ${}^4C_1 \times {}^9C_2$, which overcounts.

4 Answer: (B) Integer solutions Medium 1.5 min JEE

M04-04

SOLUTION ${}^{10+3-1}C_{3-1} = {}^{12}C_2 = 66$.

SHORTCUT Stars and bars: ${}^{n+r-1}C_{r-1}$.

COMMON TRAP Using the positive-solution formula: ${}^9C_2 = 36$.

5 Answer: (D) Circular arrangement Easy 0.75 min Speed

M04-05

SOLUTION $(6-1)! = 120$.

6 Answer: 35 Diagonals of a polygon Easy 1 min NV

M04-06

SOLUTION ${}^{10}C_2 - 10 = 45 - 10 = 35$.

SHORTCUT $\frac{n(n-3)}{2}$.

7 Answer: (C) Even numbers from digits **Medium** ⌚ 1.5 min JEE

M04-07

SOLUTION Last digit 0: $4 \times 3 = 12$. Last digit 2 or 4 (2 ways): first digit can't be 0 or the last digit (3 choices), middle 3 choices: $2 \times 3 \times 3 = 18$. Total 30.

SHORTCUT Split on whether the units digit is 0.

COMMON TRAP Allowing a leading 0 gives 36.

8 Answer: (A) Symmetry of nCr **Easy** ⌚ 0.75 min Speed

M04-08

SOLUTION $8 + 6 = n = 14$, so ${}^{14}C_2 = 91$.

9 Answer: 24 Number of divisors **Medium** ⌚ 1 min NV

M04-09

SOLUTION $360 = 2^3 \cdot 3^2 \cdot 5$: $(3+1)(2+1)(1+1) = 24$.

10 Answer: (C) Triangles from collinear points **Medium** ⌚ 1.5 min Tricky

M04-10

SOLUTION ${}^{10}C_3 - {}^4C_3 = 120 - 4 = 116$.

SHORTCUT Subtract the degenerate "triangles".

COMMON TRAP Forgetting to subtract.

M04 · CHAPTER TEST

1 Answer: (D) Permutation value **Easy** ⌚ 0.5 min Speed

M04-T1

SOLUTION $7 \times 6 \times 5 = 210$.

COMMON TRAP Giving ${}^7C_3 = 35$.

2 Answer: (C) Arrangements with repetition of letters **Easy** ⌚ 0.75 min Basic

M04-T2

SOLUTION 7 letters with A $\times$ 2 and R $\times$ 2: $\frac{7!}{2!2!} = 1260$.

3 Answer: 66 Handshakes **Easy** ⌚ 0.75 min NV

M04-T3

SOLUTION ${}^{12}C_2 = 66$.

4 Answer: (B) Garland **Medium** ⌚ 1 min Concept

M04-T4

SOLUTION $\frac{(6-1)!}{2} = 60$, because clockwise and anticlockwise arrangements look the same.

5 Answer: (D) Selections with at least one **Medium** ⌚ 1 min Concept

M04-T5

SOLUTION $(3+1)(4+1) - 1 = 19$.

COMMON TRAP Using $2^7 - 1$ (that's for distinct objects).

REVISION Revision tracker · Permutations & Combinations

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART IV · MATHEMATICS · UNIT 05

Binomial Theorem

05

1-2 per shift

TYPICAL QUESTIONS

Easy-Medium

DIFFICULTY

Must-do

PRIORITY

Class 11 · Ch 7

NCERT

~6 hours

STUDY TIME

Core concepts

- For a positive integer n : $(x + y)^n = \sum_{r=0}^n \binom{n}{r} x^{n-r} y^r$, which has $n + 1$ terms.
- Almost every question uses **the general term** T_{r+1} : find a specific coefficient, the term independent of x , or the middle term.
- Applications**: remainders and divisibility (write the base as a multiple ± 1), approximations, sums of coefficients.

NOTE Syllabus note

The official text reads: "Binomial theorem for a positive integral index, general term and middle term, and simple applications" [NTA-SYL]. Detailed identities of binomial coefficients are not listed. The basic sums below are included because they are one-line applications.

Important formulas & standard results

Σ FORMULA General & middle term

$$T_{r+1} = \binom{n}{r} x^{n-r} y^r \quad \text{middle term: } T_{\frac{n}{2}+1} \text{ (} n \text{ even); } T_{\frac{n+1}{2}} \text{ \& } T_{\frac{n+3}{2}} \text{ (} n \text{ odd)}$$

Term independent of x in $\left(ax^p + \frac{b}{x^q}\right)^n$: set the power of x in T_{r+1} to 0: $p(n-r) - qr = 0 \Rightarrow r = \frac{np}{p+q}$ (must be an integer).

Greatest binomial coefficient: $\binom{n}{n/2}$ (n even); $\binom{n}{(n-1)/2} = \binom{n}{(n+1)/2}$ (n odd).

Σ FORMULA Sums & applications

$$\sum_{r=0}^n \binom{n}{r} = 2^n \quad \binom{n}{0} + \binom{n}{2} + \dots = \binom{n}{1} + \binom{n}{3} + \dots = 2^{n-1}$$

- Sum of coefficients** of a polynomial $p(x)$: $p(1)$. Sum of coefficients of the even powers: $\frac{p(1) + p(-1)}{2}$.
- Remainders**: write $a^n = (km \pm 1)^N$ (leftover) and expand. All terms except the last are multiples of m .
- Approximation**: $(1+x)^n \approx 1 + nx + \frac{n(n-1)}{2}x^2$ for small x .
- Divisibility**: $(1+a)^n - 1 - na$ is divisible by a^2 (e.g. $8^n - 7n - 1$ by 49).

Common question models

PYQ PATTERN Recurring structures

- Coefficient of x^k** in a binomial, or in a product such as $(1+x)^m(1-x)^n$ or $(1+x+x^2)(1-x)^n$.
- Term independent of x** (and its value).
- Middle term(s)** and the greatest coefficient.
- Equal coefficients** of two terms $\rightarrow$ find n or r using $\binom{n}{x} = \binom{n}{y} \Rightarrow x = y$ or $x + y = n$.
- Remainder / last digits** of large powers.
- Sums** of coefficients (substituting $x = 1, -1$).

Shortcuts & fast methods

SHORTCUT Power-of- x equation

Write only the exponent of x in T_{r+1} as a linear function of r . Solve for r , then compute the coefficient. **Never expand.** Time: under 1 minute.

Common mistake: a sign error when the second term has a negative sign. Carry $(-1)^r$.

SHORTCUT Remainders mod m

Find a small power of the base close to a multiple of m : $7^2 = 49 = 50 - 1$ (for mod 25 or 50), $2^5 = 32 = 31 + 1$ (mod 31), $3^4 = 81 = 80 + 1$ (mod 80 or 16). Then use $(km \pm 1)^N \equiv (\pm 1)^N$.

Common mistakes

MISTAKE ALERT Mistake alerts

- The r -th term is T_r , not T_{r+1} : indexing is off by one.
- Losing the constant factors $a^{n-r}b^r$ when the terms have coefficients (e.g. $(2x - 3)^n$).
- A negative remainder: convert $-7 \pmod{25}$ to 18.
- "Coefficient" vs "binomial coefficient": the coefficient includes the powers of the constants.

Practice questions

M05 · Practice

1

Easy

 1 min

Basic

M05-01

The coefficient of x^4 in $(2 - x)^7$ is:

- (A) 560 (B) -280 (C) 280 (D) 35

2

Medium

 1 min

NV

M05-02

Find the term independent of x in $\left(x^2 + \frac{1}{x}\right)^9$.

Your answer (numerical value):

3

Medium

 1 min

Basic

M05-03

The middle term of $\left(x - \frac{1}{x}\right)^{10}$ is:

- (A) -252 (B) 252 (C) 210 (D) -210

4

Easy

 0.5 min

Speed

M05-04

The sum of the coefficients in the expansion of $(3x - 2)^5$ is:

- (A) 32 (B) 1 (C) -1 (D) 243

5

Medium

 1.5 min

NV

M05-05

Find the remainder when 7^{103} is divided by 25.

Your answer (numerical value):

6 Medium 1 min NV

M05-06

In $(1+x)^{18}$, the coefficients of the $(2r+4)$ -th and $(r-2)$ -th terms are equal. Find r .

Your answer (numerical value):

7 Easy 0.5 min Speed

M05-07

The greatest binomial coefficient in $(1+x)^9$ is:

- (A) 84 (B) 36 (C) 252 (D) 126

8 Medium 1 min Calc

M05-08

$(1.01)^{10}$, correct to 3 decimal places, is:

- (A) 1.100 (B) 1.010 (C) 1.110 (D) 1.105

9 Medium 1 min Concept

M05-09

The coefficient of x in $(1+x)(1-x)^5$ is:

- (A) -5 (B) -4 (C) -6 (D) 4

10 Easy 0.75 min NV

M05-10

Find $\binom{8}{0} + \binom{8}{2} + \binom{8}{4} + \binom{8}{6} + \binom{8}{8}$.

Your answer (numerical value):

M05 · Chapter Test

1 Easy 0.5 min Speed

M05-T1

The general term of $(a+b)^n$ is:

- (A) $\binom{n}{r} a^r b^{n-r}$ with index T_r (B) $T_{r+1} = \binom{n}{r+1} a^{n-r} b^r$
 (C) $T_{r+1} = \binom{n}{r} a^{n-r} b^r$ (D) $T_r = \binom{n}{r} a^{n-r} b^r$

2 Easy 0.5 min Speed

M05-T2

The number of terms in the expansion of $(a+b)^{15}$ is:

- (A) 16 (B) 15 (C) 14 (D) 30

3 Easy 0.75 min NV

M05-T3

Find the coefficient of x^2 in $(1+2x)^6$.

Your answer (numerical value):

4 Medium 1 min Concept

M05-T4

For every positive integer n , $8^n - 7n - 1$ is divisible by:

- (A) 7 (B) 14 (C) 64 (D) 49

5 Easy 0.5 min Speed

M05-T5

The expansion of $(x+y)^{11}$ has:

- (A) two middle terms, T_6 and T_7 (B) one middle term
 (C) no middle term (D) three middle terms

Answers & Solutions

M05 · Practice

1 (C) 2 84 3 (A) 4 (B) 5 18 6 6 7 (D) 8 (D) 9 (B) 10 128

M05 · Chapter Test

1 (C) 2 (A) 3 60 4 (D) 5 (A)

M05 · PRACTICE

1 **Answer: (C)** Specific coefficient Easy ⌚ 1 min Basic

M05-01

SOLUTION $T_{r+1} = \binom{7}{r} 2^{7-r} (-x)^r$. With $r = 4$: $35 \times 8 \times (+1) = 280$.

SHORTCUT $(-1)^4 = +1$.

COMMON TRAP Dropping 2^{7-r} gives 35.

2 **Answer: 84** Term independent of x Medium ⌚ 1 min NV

M05-02

SOLUTION Exponent of x : $2(9-r) - r = 18 - 3r = 0 \Rightarrow r = 6$. Term = $\binom{9}{6} = 84$.

3 **Answer: (A)** Middle term Medium ⌚ 1 min Basic

M05-03

SOLUTION $n = 10$, so the middle term is $T_6 = \binom{10}{5} x^5 \left(-\frac{1}{x}\right)^5 = -252$.

COMMON TRAP Sign.

4 **Answer: (B)** Sum of coefficients Easy ⌚ 0.5 min Speed

M05-04

SOLUTION Put $x = 1$: $(3-2)^5 = 1$.

5 **Answer: 18** Remainder of a large power Medium ⌚ 1.5 min NV

M05-05

SOLUTION $7^{103} = 7 \cdot (7^2)^{51} = 7(50-1)^{51} \equiv 7(-1)^{51} = -7 \equiv 18 \pmod{25}$.

COMMON TRAP Leaving the answer as -7 .

6 **Answer: 6** Equal coefficients Medium ⌚ 1 min NV

M05-06

SOLUTION $\binom{18}{2r+3} = \binom{18}{r-3}$. Since $2r+3 \neq r-3$, we need $(2r+3) + (r-3) = 18 \Rightarrow r = 6$.

COMMON TRAP Using the term numbers themselves $(2r+4, r-2)$ instead of r -values one less.

7 **Answer: (D)** Greatest coefficient Easy ⌚ 0.5 min Speed

M05-07

SOLUTION $\binom{9}{4} = \binom{9}{5} = 126$.

8 **Answer: (D)** Binomial approximation Medium ⌚ 1 min Calc

M05-08

SOLUTION $1 + 10(0.01) + 45(0.0001) + 120(10^{-6}) + \dots = 1 + 0.1 + 0.0045 + 0.00012 \approx 1.1046 \rightarrow 1.105$.

COMMON TRAP Stopping after the linear term (1.100).

9 **Answer: (B)** Coefficient in a product Medium ⌚ 1 min Concept

M05-09

SOLUTION $(1-x)^5 = 1 - 5x + \dots$. Coefficient of x : $1 \cdot (-5) + 1 \cdot 1 = -4$.

SHORTCUT Collect the ways of making x^1 : $1 \times (-5x)$ and $x \times 1$.

10 **Answer: 128** Sum of even-index coefficients Easy ⌚ 0.75 min NV

M05-10

SOLUTION $2^{n-1} = 2^7 = 128$.

M05 · CHAPTER TEST

1 **Answer: (C)** General term Easy ⌚ 0.5 min Speed

M05-T1

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

2 **Answer: (A)** Number of terms Easy ⌚ 0.5 min Speed

M05-T2

SOLUTION $n+1$.

3 **Answer: 60** Coefficient of x^2 Easy ⌚ 0.75 min NV

M05-T3

SOLUTION $\binom{6}{2} \cdot 2^2 = 15 \times 4 = 60$.

4 Answer: (D) Divisibility via binomial Medium ⌚ 1 min Concept

M05-T4

SOLUTION $8^n = (1+7)^n = 1 + 7n + \binom{n}{2}7^2 + \dots$, so $8^n - 7n - 1$ is a multiple of 49.

SHORTCUT Check $n = 2$: $64 - 15 = 49$ ✓.

COMMON TRAP Choosing 7 (true, but not the strongest divisor offered).

5 Answer: (A) Middle terms for odd n Easy ⌚ 0.5 min Speed

M05-T5

SOLUTION n odd: $T_{(n+1)/2} = T_6$ and $T_{(n+3)/2} = T_7$.



REVISION Revision tracker · Binomial Theorem

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART IV · MATHEMATICS · UNIT 06

Sequences & Series

06

1-2 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

Must-do

PRIORITY

Class 11 · Ch 8

NCERT

~8 hours

STUDY TIME



PYQ PATTERN Weight evidence

33 questions across the 2026 Paper-1 shifts ($\approx 7\%$ of Maths, tied for the top in that dataset)

THIRD-PARTY ANALYSIS [T-WT-CD].

Core concepts

- **Arithmetic progression (AP):** constant difference d . **Geometric progression (GP):** constant ratio r .
- **Means:** inserting n arithmetic or geometric means between two numbers.
- **AM $\geq$ GM** (equality iff all numbers are equal) is the main tool for *minimum/maximum* questions.



NOTE Syllabus note

The official text lists AP, GP, insertion of AMs and GMs, and the AM-GM relation [NTA-SYL]. Sums of special series ($\sum n^2$, $\sum n^3$) and arithmetico-geometric series are **not listed**. They appear below as tools because they often shorten calculations.

Important formulas & standard results



FORMULA AP

$$a_n = a + (n-1)d \quad S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a+l) \quad a_n = S_n - S_{n-1}$$

- If S_n is quadratic in n ($S_n = pn^2 + qn$), the sequence is an AP with $d = 2p$.
- Three terms in AP: $a-d, a, a+d$. Four terms: $a-3d, a-d, a+d, a+3d$.
- a, b, c in AP $\Leftrightarrow 2b = a+c$.
- n AMs between a and b : $d = \frac{b-a}{n+1}$; the sum of the AMs is $n \cdot \frac{a+b}{2}$.



FORMULA GP

$$a_n = ar^{n-1} \quad S_n = \frac{a(r^n - 1)}{r - 1} \quad (r \neq 1) \quad S_\infty = \frac{a}{1-r} \quad (|r| < 1)$$

- Three terms in GP: $\frac{a}{r}, a, ar$ (their product is a^3).
- a, b, c in GP $\Leftrightarrow b^2 = ac$.
- n GMs between a and b : $r = (b/a)^{1/(n+1)}$; the product of the GMs is $(\sqrt[n]{ab})^n$.
- Recurring decimals: $0.\bar{3} = \frac{3}{9} = \frac{1}{3}$; $0.\overline{12} = \frac{12}{99}$.



FORMULA Means & inequalities

$$\text{AM} = \frac{a+b}{2} \geq \text{GM} = \sqrt{ab} \geq \text{HM} = \frac{2ab}{a+b} \quad \text{GM}^2 = \text{AM} \times \text{HM} \text{ (two numbers)}$$

For positive a_i : $\frac{a_1 + \dots + a_n}{n} \geq (a_1 \dots a_n)^{1/n}$. **Minimum of a sum when the product is fixed**, and **maximum of a product when the sum is fixed**, both occur at equality.

Σ FORMULA Tools (not in the syllabus text)

$$\sum_{k=1}^n k = \frac{n(n+1)}{2} \quad \sum k^2 = \frac{n(n+1)(2n+1)}{6} \quad \sum k^3 = \left[\frac{n(n+1)}{2} \right]^2$$

Sum of the first n odd numbers $= n^2$.

Common question models**▮ PYQ PATTERN Recurring structures**

1. **Find terms/sums** from two given conditions (a_p, a_q, S_n).
2. S_n **given as a function of** $n \rightarrow a_n, d$ or r .
3. **Three numbers in AP/GP** with given sum and product.
4. **Inserting means**; relations between AM, GM (and HM).
5. **Minimum/maximum** via AM–GM ($x + k/x, a^x + a^{c-x}$).
6. **Infinite GP** sums and recurring decimals; mixed AP–GP conditions.

Shortcuts & fast methods**⚡ SHORTCUT Symmetric representation**

Choose terms symmetric about the middle ($a - d, a, a + d$, or $a/r, a, ar$). The given sum (for an AP) or product (for a GP) then fixes a immediately.

Time saved: removes one unknown in one step.

⚡ SHORTCUT AM–GM for exponentials

$a^x + a^{c-x} \geq 2\sqrt{a^c}$, with equality at $x = c/2$. Example: $4^x + 4^{1-x} \geq 2\sqrt{4} = 4$.

Common mistakes**⚠ MISTAKE ALERT Mistake alerts**

- Using S_∞ when $|r| \geq 1$.
- $a_n = S_n - S_{n-1}$ is valid for $n \geq 2$. Check $a_1 = S_1$ separately.
- Counting terms: from 5 to 101 in steps of 4 there are $\frac{101-5}{4} + 1 = 25$ terms (don't forget the +1).
- AM–GM needs **positive** quantities.

Practice questions**✍ M06 · Practice****1**

Easy

⌚ 1 min

NV

M06-01

In an AP, the 3rd term is 7 and the 7th term is 19. Find the 20th term.

Your answer (numerical value):

2

Easy

⌚ 0.5 min

Speed

M06-02

$1 + 2 + 4 + \dots + 2^9$ equals:

(A) 511

(B) 1023

(C) 1024

(D) 2047

3 Easy 0.75 min Basic

M06-03

 $1 - \frac{1}{3} + \frac{1}{9} - \frac{1}{27} + \dots$ equals:

- (A) $\frac{3}{2}$ (B) $\frac{2}{3}$ (C) $\frac{3}{4}$ (D) $\frac{1}{2}$

4 Easy 0.75 min Basic

M06-04

Three arithmetic means inserted between 2 and 18 are:

- (A) 6, 10, 14 (B) 4, 8, 12 (C) 5, 10, 15 (D) 6, 12, 16

5 Easy 0.5 min Speed

M06-05

The geometric mean of 4 and 16 is:

- (A) 10 (B) 12 (C) 6.4 (D) 8

6 Easy 0.75 min NV

M06-06

Find the minimum value of $x + \frac{4}{x}$ for $x > 0$.Your answer (numerical value): **7** Medium 1 min Concept

M06-07

If $S_n = 3n^2 + 2n$, the sequence is an AP with common difference:

- (A) 3 (B) 5 (C) 2 (D) 6

8 Medium 1.5 min JEE

M06-08

Three numbers in GP have sum 14 and product 64. Their common ratio (taking $r > 1$) is:

- (A) 4 (B) 3 (C) 2 (D) 1.5

9 Medium 1 min Tricky

M06-09

The minimum value of $4^x + 4^{1-x}$, $x \in \mathbb{R}$, is:

- (A) 2 (B) 4 (C) 5 (D) 1

10 Easy 0.5 min NV

M06-10

Find $\sum_{k=1}^{10} k^2$.Your answer (numerical value): **M06 · Chapter Test****1** Easy 0.5 min Speed

M06-T1

The sum of the first n odd natural numbers is:

- (A) n^2 (B) $n(n+1)$ (C) $2n^2$ (D) $n^2 + 1$

2 Easy 0.75 min Basic

M06-T2

Three numbers in AP have sum 15 and product 105. The numbers are:

- (A) 3, 5, 7 (B) 1, 5, 9 (C) 2, 5, 8 (D) 4, 5, 6

3 Easy 0.5 min NV

M06-T3

How many terms are in the AP 5, 9, 13, ..., 101?

Your answer (numerical value): **4** Easy 0.5 min Concept

M06-T4

For positive a and b , $AM = GM$ exactly when:

- (A) $a = b$ (B) $a = 2b$ (C) $ab = 1$ (D) always

5 Easy 0.5 min Speed

M06-T5

 $0.3 + 0.03 + 0.003 + \dots$ equals:

- (A) 0.33 (B) $1/3$ (C) $3/10$ (D) $10/3$

Answers & Solutions

M06 · Practice

1 **58** 2 **(B)** 3 **(C)** 4 **(A)** 5 **(D)** 6 **4** 7 **(D)** 8 **(C)** 9 **(B)** 10 **385**

M06 · Chapter Test

1 **(A)** 2 **(A)** 3 **25** 4 **(A)** 5 **(B)**

M06 · PRACTICE

1 Answer: **58** AP from two terms Easy 1 min NV

M06-01

SOLUTION $4d = 12 \Rightarrow d = 3$; $a = 7 - 6 = 1$; $a_{20} = 1 + 19(3) = 58$.**2** Answer: **(B)** Finite GP sum Easy 0.5 min Speed

M06-02

SOLUTION $\frac{2^{10} - 1}{2 - 1} = 1023$ (10 terms).

COMMON TRAP Counting 9 terms.

3 Answer: **(C)** Infinite GP Easy 0.75 min Basic

M06-03

SOLUTION $a = 1$, $r = -\frac{1}{3}$: $S_{\infty} = \frac{1}{4/3} = \frac{3}{4}$.COMMON TRAP Using $r = +\frac{1}{3}$ (gives $3/2$).**4** Answer: **(A)** Inserting AMs Easy 0.75 min Basic

M06-04

SOLUTION $d = \frac{18 - 2}{4} = 4$.**5** Answer: **(D)** Geometric mean Easy 0.5 min Speed

M06-05

SOLUTION $\sqrt{64} = 8$.**6** Answer: **4** Minimum by AM-GM Easy 0.75 min NV

M06-06

SOLUTION $x + 4/x \geq 2\sqrt{4} = 4$, with equality at $x = 2$.**7** Answer: **(D)** S_n given Medium 1 min Concept

M06-07

SOLUTION $a_n = S_n - S_{n-1} = 6n - 1$, so $d = 6$.SHORTCUT $d = 2 \times$ (coefficient of n^2).**8** Answer: **(C)** Three numbers in GP Medium 1.5 min JEE

M06-08

SOLUTION The middle term b satisfies $b^3 = 64 \Rightarrow b = 4$. Then $a + c = 10$ and $ac = 16$, so the terms are 2, 4, 8 and $r = 2$.

SHORTCUT Symmetric representation.

9 Answer: (B) AM–GM with exponentials **Medium** ⌚ 1 min **Tricky**

M06-09

SOLUTION $\geq 2\sqrt{4^x \cdot 4^{1-x}} = 2\sqrt{4} = 4$, attained at $x = \frac{1}{2}$.

COMMON TRAP Assuming the minimum is at $x = 0$ (gives 5).

10 Answer: 385 Sum of squares **Easy** ⌚ 0.5 min **NV**

M06-10

SOLUTION $\frac{10 \cdot 11 \cdot 21}{6} = 385$.

M06 · CHAPTER TEST

1 Answer: (A) Sum of odd numbers **Easy** ⌚ 0.5 min **Speed**

M06-T1

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

2 Answer: (A) Three numbers in AP **Easy** ⌚ 0.75 min **Basic**

M06-T2

SOLUTION The middle term is 5. $(5-d)(5+d) = 21 \Rightarrow d = 2$.

3 Answer: 25 Counting terms **Easy** ⌚ 0.5 min **NV**

M06-T3

SOLUTION $\frac{101-5}{4} + 1 = 25$.

COMMON TRAP Forgetting the +1.

4 Answer: (A) AM–GM equality **Easy** ⌚ 0.5 min **Concept**

M06-T4

SOLUTION AM – GM = $\frac{1}{2}(\sqrt{a} - \sqrt{b})^2 \geq 0$, which is zero iff $a = b$.

5 Answer: (B) Recurring decimal **Easy** ⌚ 0.5 min **Speed**

M06-T5

SOLUTION $\frac{0.3}{1-0.1} = \frac{1}{3}$.

REVISION Revision tracker · Sequences & Series

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART IV · MATHEMATICS · UNIT 07

Limits, Continuity & Differentiability (with Applications of Derivatives)

07

2–3 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

Must-do

PRIORITY

Class 11 · Ch 12;
Class 12 · Ch 5, 6

NCERT

~16 hours

STUDY TIME

Core concepts

- **Limit:** the value a function approaches. It exists iff LHL = RHL (finite). **Continuity at a :** $\lim_{x \rightarrow a} f(x) = f(a)$. **Differentiability at a :** LHD = RHD (finite).
- Differentiable $\Rightarrow$ continuous, but **not** conversely ($|x|$ at 0).
- **Derivative rules:** sum, product, quotient, chain; implicit, parametric and logarithmic differentiation; second derivatives.
- **Applications (syllabus):** rate of change, increasing/decreasing functions, maxima and minima of functions of one variable.

NOTE Syllabus note

Unit 7 lists real functions and their graphs, limits, continuity, differentiability, differentiation techniques up to second order, and AOD limited to **rate of change, monotonicity and maxima/minima** [NTA-SYL]. Tangents/normals, mean value theorems and approximations are not listed.

Important formulas & standard results

Σ FORMULA Standard limits ($x \rightarrow 0$ unless stated)

$$\frac{\sin x}{x}, \frac{\tan x}{x}, \frac{\sin^{-1} x}{x}, \frac{\tan^{-1} x}{x} \rightarrow 1 \quad \frac{1 - \cos x}{x^2} \rightarrow \frac{1}{2} \quad \frac{e^x - 1}{x} \rightarrow 1 \quad \frac{a^x - 1}{x} \rightarrow \ln a$$

$$\frac{\ln(1+x)}{x} \rightarrow 1 \quad (1+x)^{1/x} \rightarrow e \quad \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \quad \lim_{x \rightarrow \infty} \left(1 + \frac{a}{x}\right)^{bx} = e^{ab}$$

1^∞ form: $\lim f^g = e^{\lim g(f-1)}$ when $f \rightarrow 1, g \rightarrow \infty$.

Series (fast for $0/0$): $\sin x = x - \frac{x^3}{6} + \dots$, $\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \dots$, $\tan x = x + \frac{x^3}{3} + \dots$, $e^x = 1 + x + \frac{x^2}{2} + \dots$,

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$$

L'Hôpital's rule (a tool, not named in the syllabus): for $0/0$ or ∞/∞ , $\lim \frac{f}{g} = \lim \frac{f'}{g'}$.

Σ FORMULA Derivatives

$f(x)$	$f'(x)$	$f(x)$	$f'(x)$
x^n	nx^{n-1}	$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$
e^x, a^x	$e^x, a^x \ln a$	$\cos^{-1} x$	$-\frac{1}{\sqrt{1-x^2}}$
$\ln x, \log_a x$	$\frac{1}{x}, \frac{1}{x \ln a}$	$\tan^{-1} x$	$\frac{1}{1+x^2}$
$\sin x, \cos x$	$\cos x, -\sin x$	$\cot^{-1} x$	$-\frac{1}{1+x^2}$
$\tan x, \cot x$	$\sec^2 x, -\csc^2 x$	$\sec^{-1} x$	$\frac{1}{ x \sqrt{x^2-1}}$
$\sec x, \csc x$	$\sec x \tan x, -\csc x \cot x$	x^x	$x^x(1+\ln x)$

Parametric: $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}, \frac{d^2y}{dx^2} = \frac{d}{dt} \left(\frac{dy}{dx} \right) / \frac{dx}{dt}$. **Implicit:** differentiate both sides with respect to x , treating y as $y(x)$.

Substitutions for inverse-trig derivatives: $\tan^{-1} \frac{2x}{1-x^2} = 2 \tan^{-1} x$ ($|x| < 1$); $\sin^{-1} \frac{2x}{1+x^2} = 2 \tan^{-1} x$ ($|x| \leq 1$); $\cos^{-1} \frac{1-x^2}{1+x^2} = 2 \tan^{-1} x$ ($x \geq 0$).

Σ FORMULA Continuity & differentiability facts

- Polynomials, $\sin$, $\cos$, e^x are continuous everywhere; rational functions except where the denominator is zero; $\ln x$ for $x > 0$.
- $[x]$ is discontinuous at every integer. $\{x\}$ likewise.
- $|f(x)|$ is non-differentiable at the **simple roots** of f (where f changes sign) where $f' \neq 0$.
- $|x - a_1| + |x - a_2| + \dots$ is non-differentiable exactly at $a_1, a_2, \dots$

Σ FORMULA Applications of derivatives

Rate of change: $\frac{dA}{dt} = \frac{dA}{dr} \cdot \frac{dr}{dt}$ (circle: $2\pi r \dot{r}$; sphere volume: $4\pi r^2 \dot{r}$).

Monotonicity: $f' > 0$ on an interval $\rightarrow$ strictly increasing; $f' < 0 \rightarrow$ strictly decreasing.

Extrema: critical points where $f' = 0$ or f' doesn't exist. First-derivative test: f' changes from $+$ to $-$ means a local max. Second-derivative test: $f'(c) = 0$, $f''(c) < 0$ means a max; > 0 means a min.

Absolute extrema on $[a, b]$: compare f at the critical points **and** the endpoints.

Classic optima: fixed perimeter $\rightarrow$ the square has maximum area; the largest rectangle in a circle is a square; $a \sin x + b \cos x$ has maximum $\sqrt{a^2 + b^2}$.

Common question models**▮ PYQ PATTERN Recurring structures**

- Evaluate limits:** standard forms, rationalisation, series expansion, 1^∞ , limits at infinity with polynomials.
- Find constants for continuity/differentiability** of piecewise functions.
- Count points of non-differentiability** (modulus, greatest-integer and max/min compositions).
- Derivatives:** composite, parametric, implicit, logarithmic, inverse-trig simplification, second derivatives.
- Monotonicity intervals;** local/absolute extrema; optimisation word problems.
- Rates of change** (ladders, balloons, circles).

Shortcuts & fast methods

⚡ SHORTCUT Series expansion beats L'Hôpital

For $\frac{\sin x - x}{x^3}$ -type limits, substitute the first two terms of the series. It's one line, versus three rounds of differentiation.

When NOT to use: when the limit isn't at 0 (substitute $x = a + h$ first).

⚡ SHORTCUT Leading terms at infinity

$\lim_{x \rightarrow \infty} \frac{a_n x^n + \dots}{b_m x^m + \dots}$: if $n = m$ the answer is a_n/b_m ; if $n < m$ it's 0; if $n > m$ it's $\pm\infty$.

Common mistakes

⚠ MISTAKE ALERT Mistake alerts

- Using $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ in degrees. JEE uses **radians**.
- A limit existing (LHL = RHL) doesn't make the function continuous. $f(a)$ must also equal it.
- In optimisation, forgetting the endpoints of a closed interval.
- $\frac{d}{dx}(x^x) \neq x \cdot x^{x-1}$.

Practice questions

🔗 M07 · Practice

1 Easy ⌚ 0.75 min Basic M07-01

$\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{\sin 3x}$ equals:

(A) 1 (B) $3/2$ (C) $2/3$ (D) 6

2 Medium ⌚ 1 min Concept M07-02

$\lim_{x \rightarrow \infty} \left(1 + \frac{2}{x}\right)^{3x}$ equals:

(A) 1 (B) $e^{2/3}$ (C) $e^{3/2}$ (D) e^6

3 Medium ⌚ 1 min Concept M07-03

$\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3}$ equals:

(A) $-1/6$ (B) $1/6$ (C) 0 (D) $-1/3$

4 Easy ⌚ 0.75 min NV M07-04

$f(x) = \frac{x^2 - 4}{x - 2}$ for $x \neq 2$ and $f(2) = k$. Find k if f is continuous at $x = 2$.

Your answer (numerical value):

5 Easy ⌚ 0.75 min M07-05

Find the number of points where $f(x) = |x - 1| + |x - 2|$ is not differentiable.

Your answer (numerical value):

6

Easy

⌚ 0.75 min

Basic

M07-06

$\frac{d}{dx}(x^x)$ for $x > 0$ is:

- (A) $x \cdot x^{x-1}$ (B) $x^x \ln x$ (C) $x^x(1 + \ln x)$ (D) $x^x(1 - \ln x)$

7

Easy

⌚ 0.75 min

Basic

M07-07

If $x = a \cos t$ and $y = a \sin t$, then $\frac{dy}{dx}$ is:

- (A) $\cot t$ (B) $-\cot t$ (C) $\tan t$ (D) $-\tan t$

8

Medium

⌚ 1 min

Basic

M07-08

$f(x) = x^3 - 3x$ is strictly increasing on:

- (A) $(-1, 1)$ (B) $\mathbb{R}$ (C) $(0, \infty)$ (D) $(-\infty, -1) \cup (1, \infty)$

9

Easy

⌚ 1 min

NV

M07-09

A rectangle has perimeter 20. Find its maximum possible area.

Your answer (numerical value):

10

Easy

⌚ 1 min

Basic

M07-10

The radius of a circle grows at 0.5 cm/s. When $r = 4$ cm, its area grows at:

- (A) $4\pi \text{ cm}^2/\text{s}$ (B) $2\pi \text{ cm}^2/\text{s}$ (C) $8\pi \text{ cm}^2/\text{s}$ (D) $16\pi \text{ cm}^2/\text{s}$

11

Easy

⌚ 0.75 min

Basic

M07-11

For $x^2 + y^2 = 25$, $\frac{dy}{dx}$ at $(3, 4)$ is:

- (A) $-3/4$ (B) $3/4$ (C) $-4/3$ (D) $4/3$

12

Medium

⌚ 1 min

Concept

M07-12

For $|x| < 1$, $\frac{d}{dx} \tan^{-1} \left(\frac{2x}{1-x^2} \right)$ equals:

- (A) $\frac{1}{1+x^2}$ (B) $\frac{2}{1+x^2}$ (C) $\frac{2}{1-x^2}$ (D) $\frac{-2}{1+x^2}$

M07 · Chapter Test

1

Easy

⌚ 0.5 min

Speed

M07-T1

$\lim_{x \rightarrow 0} \frac{\tan 2x}{x}$ equals:

- (A) 1 (B) $1/2$ (C) 2 (D) 0

2

Easy

⌚ 0.5 min

x

M07-T2

At $x = 0$, $f(x) = |x|$ is:

- (A) continuous but not differentiable (B) discontinuous
(C) differentiable (D) neither defined nor continuous

3 Medium ⌚ 1 min NV

M07-T3

Find the local maximum value of $f(x) = x^3 - 6x^2 + 9x + 1$.Your answer (numerical value): **4** Easy ⌚ 0.5 min Speed

M07-T4

 $\frac{d}{dx} \sin^{-1} x$ equals:

(A) $\frac{1}{1+x^2}$

(B) $-\frac{1}{\sqrt{1-x^2}}$

(C) $\frac{1}{\sqrt{1-x^2}}$

(D) $\frac{1}{x\sqrt{x^2-1}}$

5 Easy ⌚ 0.5 min Concept

M07-T5

If $f'(c) = 0$ and $f''(c) < 0$, then at $x = c$, f has:

(A) a local minimum

(B) a local maximum

(C) a point of inflection

(D) no conclusion possible

Answers & Solutions

M07 · Practice

1 (C) 2 (D) 3 (A) 4 4 5 2 6 (C) 7 (B) 8 (D) 9 25 10 (A)
 11 (A) 12 (B)

M07 · Chapter Test

1 (C) 2 (A) 3 5 4 (C) 5 (B)

M07 · PRACTICE

1 Answer: (C) Standard limits Easy ⌚ 0.75 min Basic

M07-01

SOLUTION $\frac{e^{2x}-1}{2x} \cdot \frac{3x}{\sin 3x} \cdot \frac{2}{3} \rightarrow \frac{2}{3}$.

SHORTCUT Replace each by its linear approximation: $2x/3x$.**2** Answer: (D) 1^∞ form Medium ⌚ 1 min Concept

M07-02

SOLUTION $e^{\lim 3x \cdot (2/x)} = e^6$.

SHORTCUT e^{ab} .**COMMON TRAP** Answering 1 (treating 1^∞ as 1).**3** Answer: (A) Series expansion limit Medium ⌚ 1 min Concept

M07-03

SOLUTION $\sin x - x = -\frac{x^3}{6} + O(x^5)$.

COMMON TRAP Sign.**4** Answer: 4 Removable discontinuity Easy ⌚ 0.75 min NV

M07-04

SOLUTION $\lim_{x \rightarrow 2} (x+2) = 4$.

5 Answer: 2 Non-differentiability of Easy ⌚ 0.75 min

M07-05

SOLUTION The corners are at $x = 1$ and $x = 2$.**6** Answer: (C) Logarithmic differentiation Easy ⌚ 0.75 min Basic

M07-06

SOLUTION $y = x^x \Rightarrow \ln y = x \ln x \Rightarrow \frac{y'}{y} = 1 + \ln x$.

COMMON TRAP Option (A) uses the power rule wrongly.**7** Answer: (B) Parametric derivative Easy ⌚ 0.75 min Basic

M07-07

SOLUTION $\frac{a \cos t}{-a \sin t} = -\cot t$.

8 Answer: (D) Monotonicity Medium 1 min Basic

M07-08

SOLUTION $f' = 3(x^2 - 1) > 0 \iff |x| > 1$.**9 Answer: 25** Optimisation Easy 1 min NV

M07-09

SOLUTION $A = x(10 - x)$, which is maximised at $x = 5$: $A = 25$.**SHORTCUT** Fixed perimeter $\rightarrow$ square.**10 Answer: (A)** Rate of change Easy 1 min Basic

M07-10

SOLUTION $\dot{A} = 2\pi r\dot{r} = 2\pi(4)(0.5) = 4\pi$.**11 Answer: (A)** Implicit differentiation Easy 0.75 min Basic

M07-11

SOLUTION $2x + 2yy' = 0 \Rightarrow y' = -x/y = -3/4$.**12 Answer: (B)** Inverse-trig simplification Medium 1 min Concept

M07-12

SOLUTION With $x = \tan \theta$ the expression is $\tan^{-1}(\tan 2\theta) = 2\theta = 2\tan^{-1} x$ for $|x| < 1$.**COMMON TRAP** Differentiating directly (slow, error-prone).**M07 · CHAPTER TEST****1 Answer: (C)** Basic limit Easy 0.5 min Speed

M07-T1

SOLUTION Direct recall: see the concept summary and formula box of this chapter.**2 Answer: (A)** Easy 0.5 min x

M07-T2

SOLUTION LHD = -1, RHD = +1.**3 Answer: 5** Local maximum value Medium 1 min NV

M07-T3

SOLUTION $f' = 3(x - 1)(x - 3)$. $f'' = 6x - 12 < 0$ at $x = 1$, so there's a local max: $f(1) = 5$.**COMMON TRAP** Giving the local minimum $f(3) = 1$.**4 Answer: (C)** Derivative of arcsin Easy 0.5 min Speed

M07-T4

SOLUTION Direct recall: see the concept summary and formula box of this chapter.**5 Answer: (B)** Second-derivative test Easy 0.5 min Concept

M07-T5

SOLUTION Direct recall: see the concept summary and formula box of this chapter.**REVISION Revision tracker · Limits & Differentiation**☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART IV · MATHEMATICS · UNIT 08

Integral Calculus

08

2–3 per shift

TYPICAL QUESTIONS

Medium–Hard

DIFFICULTY

Must-do

PRIORITY

Class 12 · Ch 7, 8

NCERT

~18 hours

STUDY TIME

Core concepts

- **Indefinite integral:** an antiderivative, $\int f dx = F + C$ with $F' = f$. Methods: standard forms, **substitution**, **by parts**, **partial fractions**, trigonometric identities.
- **Definite integral:** $\int_a^b f dx = F(b) - F(a)$ (fundamental theorem). Its **properties** often make evaluation unnecessary.
- **Area** between curves: integrate (upper – lower) with respect to x , or (right – left) with respect to y .

Important formulas & standard results

Σ FORMULA Standard integrals (the syllabus list and the basics)

$\int$	Result
$x^n dx$ ($n \neq -1$)	$\frac{x^{n+1}}{n+1}$
$\frac{1}{x} dx$	$\ln x $
$e^{ax} dx$	$\frac{e^{ax}}{a}$
$\sin x, \cos x$	$-\cos x, \sin x$
$\tan x, \cot x$	$\ln \sec x , \ln \sin x $
$\sec x, \csc x$	$\ln \sec x + \tan x , \ln \csc x - \cot x $
$\frac{dx}{x^2 + a^2}$	$\frac{1}{a} \tan^{-1} \frac{x}{a}$
$\frac{dx}{x^2 - a^2}$	$\frac{1}{2a} \ln \left \frac{x-a}{x+a} \right $
$\frac{dx}{a^2 - x^2}$	$\frac{1}{2a} \ln \left \frac{a+x}{a-x} \right $
$\frac{dx}{\sqrt{a^2 - x^2}}$	$\sin^{-1} \frac{x}{a}$
$\frac{dx}{\sqrt{x^2 \pm a^2}}$	$\ln x + \sqrt{x^2 \pm a^2} $
$\sqrt{a^2 - x^2} dx$	$\frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a}$
$\sqrt{x^2 \pm a^2} dx$	$\frac{x}{2} \sqrt{x^2 \pm a^2} \pm \frac{a^2}{2} \ln x + \sqrt{x^2 \pm a^2} $
$\int \frac{dx}{ax^2 + bx + c}$ and $\int \frac{dx}{\sqrt{ax^2 + bx + c}}$: complete the square. $\int \frac{(px+q) dx}{ax^2 + bx + c}$: write $px+q = A \frac{d}{dx}(ax^2 + bx + c) + B$. (+C omitted throughout.)	

Σ FORMULA Methods

By parts: $\int u v dx = u \int v dx - \int (u' \int v dx) dx$. Choose u by **ILATE** (Inverse trig, Log, Algebraic, Trig, Exponential).

Special form: $\int e^x [f(x) + f'(x)] dx = e^x f(x)$.

Partial fractions: $\frac{1}{(x-a)(x-b)} = \frac{1}{a-b} \left(\frac{1}{x-a} - \frac{1}{x-b} \right)$; repeated factor $(x-a)^2 \rightarrow \frac{A}{x-a} + \frac{B}{(x-a)^2}$; quadratic factor $\rightarrow \frac{Bx+C}{x^2+px+q}$.

Trig identities: $\sin^2 x = \frac{1-\cos 2x}{2}$, $\cos^2 x = \frac{1+\cos 2x}{2}$; products $\rightarrow$ sums.

Σ FORMULA Properties of definite integrals

$$\int_a^b f(x) dx = \int_a^b f(a+b-x) dx \quad (\text{King's rule}) \quad \int_0^a f(x) dx = \int_0^a f(a-x) dx$$

$$\int_{-a}^a f = \begin{cases} 2 \int_0^a f & f \text{ even} \\ 0 & f \text{ odd} \end{cases} \quad \int_0^{2a} f = \begin{cases} 2 \int_0^a f & f(2a-x) = f(x) \\ 0 & f(2a-x) = -f(x) \end{cases} \quad \int_0^{nT} f = n \int_0^T f \text{ (period } T)$$

Leibniz rule (from the fundamental theorem): $\frac{d}{dx} \int_{g(x)}^{h(x)} f(t) dt = f(h(x)) h'(x) - f(g(x)) g'(x)$.

Useful values: $\int_0^{\pi/2} \sin^2 x = \int_0^{\pi/2} \cos^2 x = \frac{\pi}{4}$; $\int_0^{\pi/2} \frac{\sin^n x}{\sin^n x + \cos^n x} dx = \frac{\pi}{4}$; $\int_0^{\pi} |\cos x| dx = 2$; $\int_0^{\pi/2} \ln \sin x dx = -\frac{\pi}{2} \ln 2$.

Σ FORMULA Standard areas

Circle πr^2 · ellipse πab · parabola $y^2 = 4ax$ with its latus rectum: $\frac{8}{3}a^2$ · between $y^2 = 4ax$ and $x^2 = 4ay$: $\frac{16a^2}{3}$ · between $y = x^2$ and $y = x$: $\frac{1}{6}$.

Common question models**▮ PYQ PATTERN Recurring structures**

1. **Standard-form integrals** after completing the square, or linear-over-quadratic splitting.
2. **By parts** / $e^x(f + f')$ recognition.
3. **Definite integrals via properties:** King's rule pairs $\left(\frac{f(x)}{f(x)+f(a+b-x)} \right)$ gives $\frac{b-a}{2}$, even/odd, periodicity, modulus and greatest-integer integrands (split at the break points).
4. **Leibniz rule:** limits of integrals, derivatives of integral functions.
5. **Area between curves** (parabola–line, circle–parabola, modulus graphs).
6. **Functional equations with integrals:** $f(x) = x + \int_0^1 f(t) dt$ -type (the integral is a constant).

Shortcuts & fast methods**⚡ SHORTCUT King's-rule pairing**

If $f(x) + f(a+b-x) = c$ (a constant), then $\int_a^b f(x) dx = \frac{c(b-a)}{2}$. Classic case: $\int_a^b \frac{g(x)}{g(x) + g(a+b-x)} dx = \frac{b-a}{2}$.

Time saved: 2–3 minutes. **When NOT to use:** when the substituted integrand doesn't simplify when added back to the original.

⚡ SHORTCUT Differentiate the options

For an indefinite integral MCQ, differentiate each option (starting with the most likely one) and compare with the integrand. That's often faster than integrating.

Common mistakes

MISTAKE ALERT Mistake alerts

- Forgetting to change the limits when substituting in a definite integral.
- $\int \frac{dx}{x^2 - a^2}$ vs $\int \frac{dx}{a^2 - x^2}$: the log's argument flips.
- Area: take the **absolute** value. Split the region wherever the curves cross, or wherever the curve crosses the axis.
- Greatest-integer integrands: split at every integer inside the interval.

Practice questions

M08 · Practice

1 Medium ⌚ 1.5 min Basic M08-01

$\int \frac{dx}{x^2 + 4x + 13}$ equals:

- (A) $\tan^{-1} \frac{x+2}{3} + C$ (B) $\frac{1}{3} \tan^{-1} \frac{x+2}{3} + C$ (C) $\frac{1}{3} \tan^{-1} \frac{x+2}{9} + C$ (D) $\frac{1}{9} \tan^{-1}(x+2) + C$

2 Easy ⌚ 0.75 min Basic M08-02

$\int xe^x dx$ equals:

- (A) $e^x(x-1) + C$ (B) $e^x(x+1) + C$ (C) $xe^x + C$ (D) $\frac{x^2}{2}e^x + C$

3 Easy ⌚ 0.5 min Speed M08-03

$\int e^x(\sin x + \cos x) dx$ equals:

- (A) $e^x \cos x + C$ (B) $e^x(\sin x - \cos x) + C$ (C) $-e^x \sin x + C$ (D) $e^x \sin x + C$

4 Medium ⌚ 1 min Concept M08-04

$\int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} dx$ equals:

- (A) $\pi/2$ (B) 1 (C) $\pi/4$ (D) π

5 Easy ⌚ 0.5 min Speed M08-05

$\int_{-1}^1 x^3 \cos x dx$ equals:

- (A) 0 (B) 2 (C) $2 \cos 1$ (D) 1

6 Easy ⌚ 1 min NV M08-06

Find $\int_0^2 [x] dx$, where $[\cdot]$ is the greatest integer function.

Your answer (numerical value):

7 Easy ⌚ 1 min Basic M08-07

The area enclosed between $y = x^2$ and $y = x$ is:

- (A) $1/2$ (B) $1/6$ (C) $1/3$ (D) $1/12$

8 Easy 0.5 min Speed

M08-08

The area enclosed by $\frac{x^2}{9} + \frac{y^2}{4} = 1$ is:

- (A) 13π (B) 12π (C) 36π (D) 6π

9 Medium 1 min Concept

M08-09

$\frac{d}{dx} \int_0^{x^2} \sin t \, dt$ equals:

- (A) $\sin x^2$ (B) $2x \sin x^2$ (C) $2x \cos x^2$ (D) $\cos x^2$

10 Easy 0.5 min Speed

M08-10

$\int \frac{dx}{\sqrt{9-x^2}}$ equals:

- (A) $\sin^{-1} 3x + C$ (B) $\frac{1}{3} \sin^{-1} \frac{x}{3} + C$ (C) $\sin^{-1} \frac{x}{3} + C$ (D) $\frac{1}{3} \tan^{-1} \frac{x}{3} + C$

11 Medium 1 min Basic

M08-11

$\int \frac{dx}{x(x+1)}$ equals:

- (A) $\ln \left| \frac{x+1}{x} \right| + C$ (B) $\ln |x(x+1)| + C$ (C) $\ln \left| \frac{x}{x+1} \right| + C$ (D) $\frac{1}{x+1} + C$

12 Easy 1 min NV

M08-12

Find $\int_0^\pi |\cos x| \, dx$.

Your answer (numerical value):

M08 · Chapter Test

1 Easy 0.5 min Speed

M08-T1

$\int \sec^2 x \, dx$ equals:

- (A) $\tan x + C$ (B) $\sec x \tan x + C$ (C) $\cot x + C$ (D) $\sec x + C$

2 Easy 0.5 min Speed

M08-T2

$\int_0^a f(x) \, dx$ equals:

- (A) $\int_0^a f(a-x) \, dx$ (B) $\int_0^a f(x-a) \, dx$ (C) $\int_0^{2a} f(x) \, dx$ (D) $\int_{-a}^0 f(x) \, dx$

3 Easy 0.75 min NV

M08-T3

Find $\int_0^3 (2x+1) \, dx$.

Your answer (numerical value):

4 Easy 0.75 min Basic

M08-T4

 $\int \ln x \, dx$ equals:

- (A) $\frac{1}{x} + C$ (B) $x \ln x - x + C$ (C) $x \ln x + C$ (D) $\frac{(\ln x)^2}{2} + C$

5 Easy 0.75 min Basic

M08-T5

 $\int_0^\pi \sin^2 x \, dx$ equals:

- (A) π (B) 1 (C) $\pi/4$ (D) $\pi/2$

Answers & Solutions

M08 · Practice

- 1 (B) 2 (A) 3 (D) 4 (C) 5 (A) 6 1 7 (B) 8 (D) 9 (B) 10 (C)
11 (C) 12 2

M08 · Chapter Test

- 1 (A) 2 (A) 3 12 4 (B) 5 (D)

M08 · PRACTICE

1 Answer: (B) Completing the square Medium 1.5 min Basic

M08-01

SOLUTION $x^2 + 4x + 13 = (x + 2)^2 + 9$, so $\frac{1}{3} \tan^{-1} \frac{x+2}{3}$.

2 Answer: (A) Integration by parts Easy 0.75 min Basic

M08-02

SOLUTION $u = x, v = e^x: xe^x - \int e^x dx = e^x(x - 1)$.

SHORTCUT Differentiate (A): $e^x(x - 1) + e^x = xe^x \checkmark$.

3 Answer: (D) The $e^x(f + f')$ form Easy 0.5 min Speed

M08-03

SOLUTION $f = \sin x, f' = \cos x$.

4 Answer: (C) King's rule Medium 1 min Concept

M08-04

SOLUTION Call it I . Replacing x by $\frac{\pi}{2} - x$ gives the cosine version. Adding the two: $2I = \int_0^{\pi/2} 1 \, dx = \frac{\pi}{2}$, so $I = \frac{\pi}{4}$.

SHORTCUT $\frac{b-a}{2}$.

5 Answer: (A) Odd integrand Easy 0.5 min Speed

M08-05

SOLUTION odd $\times$ even = odd, and the interval is symmetric.

6 Answer: 1 Greatest-integer integrand Easy 1 min NV

M08-06

SOLUTION $\int_0^1 0 \, dx + \int_1^2 1 \, dx = 1$.

7 Answer: (B) Area between curves Easy 1 min Basic

M08-07

SOLUTION They meet at $x = 0$ and 1 : $\int_0^1 (x - x^2) \, dx = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$.

8 Answer: (D) Area of an ellipse Easy 0.5 min Speed

M08-08

SOLUTION $\pi ab = \pi \cdot 3 \cdot 2$.

9 Answer: (B) Leibniz rule Medium 1 min Concept

M08-09

SOLUTION $\sin(x^2) \cdot \frac{d}{dx}(x^2)$.

COMMON TRAP Forgetting the chain factor $2x$.

10 Answer: (C) *Arcsine form* **Easy** ⌚ 0.5 min **Speed**

M08-10

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

COMMON TRAP Adding a spurious $1/3$ (as in the $\tan^{-1}$ form).

11 Answer: (C) *Partial fractions* **Medium** ⌚ 1 min **Basic**

M08-11

SOLUTION $\frac{1}{x(x+1)} = \frac{1}{x} - \frac{1}{x+1}$.

12 Answer: 2 *Integral of modulus* **Easy** ⌚ 1 min **NV**

M08-12

SOLUTION $\int_0^{\pi/2} \cos x \, dx - \int_{\pi/2}^{\pi} \cos x \, dx = 1 + 1 = 2$.

COMMON TRAP Ignoring the modulus gives 0.

M08 · CHAPTER TEST

1 Answer: (A) *Basic integral* **Easy** ⌚ 0.5 min **Speed**

M08-T1

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

2 Answer: (A) *Property* **Easy** ⌚ 0.5 min **Speed**

M08-T2

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

3 Answer: 12 *Definite integral* **Easy** ⌚ 0.75 min **NV**

M08-T3

SOLUTION $[x^2 + x]_0^3 = 9 + 3 = 12$.

4 Answer: (B) *Integral of $\ln x$* **Easy** ⌚ 0.75 min **Basic**

M08-T4

SOLUTION By parts with $u = \ln x$, $v = 1$.

5 Answer: (D) *$\sin^2$ over $[0, \pi]$* **Easy** ⌚ 0.75 min **Basic**

M08-T5

SOLUTION The average of $\sin^2$ over a period is $\frac{1}{2}$: $\frac{1}{2} \cdot \pi$.

REVISION Revision tracker · Integral Calculus

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART IV · MATHEMATICS · UNIT 09

Differential Equations

09

1-2 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

Must-do

PRIORITY

Class 12 · Ch 9

NCERT

~7 hours

STUDY TIME

Core concepts

- A **differential equation** relates a function and its derivatives. The **order** is the highest derivative. The **degree** is the power of that highest derivative once the equation is a polynomial in its derivatives (undefined otherwise, e.g. $\sin y'$).
- The syllabus solves three types: **variables separable**, **homogeneous** and **linear first-order** ($y' + P(x)y = Q(x)$).
- An initial condition picks out one particular solution from the general family.

NOTE Syllabus note

Formation of differential equations is not listed in the 2024–2026 text. It covers order and degree, separation of variables, homogeneous equations and the linear type $\frac{dy}{dx} + p(x)y = q(x)$ [NTA-SYL].

Important formulas & standard results

Σ FORMULA Solution methods

Type	Form	Method
Separable	$\frac{dy}{dx} = f(x)g(y)$	$\int \frac{dy}{g(y)} = \int f(x) dx + C$
Homogeneous	$\frac{dy}{dx} = F\left(\frac{y}{x}\right)$	put $y = vx$: $v + x \frac{dv}{dx} = F(v)$, then separate
Linear in y	$\frac{dy}{dx} + Py = Q$	IF = $e^{\int P dx}$; $y \cdot \text{IF} = \int Q \cdot \text{IF} dx + C$
Linear in x	$\frac{dx}{dy} + P(y)x = Q(y)$	IF = $e^{\int P dy}$; $x \cdot \text{IF} = \int Q \cdot \text{IF} dy + C$

Σ FORMULA Useful integrating factors

$P = \frac{1}{x} \rightarrow \text{IF} = x$; $P = -\frac{1}{x} \rightarrow \frac{1}{x}$; $P = \frac{n}{x} \rightarrow x^n$; $P = k \rightarrow e^{kx}$; $P = \tan x \rightarrow \sec x$; $P = \cot x \rightarrow \sin x$.

Exact-derivative recognition: $x dy + y dx = d(xy)$; $\frac{x dy - y dx}{x^2} = d\left(\frac{y}{x}\right)$; $\frac{x dy - y dx}{x^2 + y^2} = d\left(\tan^{-1} \frac{y}{x}\right)$.

Σ FORMULA Applications

Growth/decay: $\frac{dN}{dt} = kN \Rightarrow N = N_0 e^{kt}$. Doubling time = $\frac{\ln 2}{k}$; it takes twice the doubling time to quadruple.

Newton's law of cooling: $\frac{dT}{dt} = -k(T - T_0) \Rightarrow T - T_0 = (T_i - T_0)e^{-kt}$.

Common question models

PYQ PATTERN Recurring structures

1. **Order and degree**, including "degree not defined" cases and radicals that must be cleared first.
2. **Linear DE with an initial condition** → find y at a point (a very common NV).
3. **Homogeneous DE** → particular solution or curve family.
4. **Separable equations** with exponentials or trig functions.
5. **Equations that are linear in x** (dx/dy form).
6. **Application problems**: growth, cooling, curves whose tangent slope satisfies a condition.

Shortcuts & fast methods

SHORTCUT Choose the dependent variable

If the equation is linear in x but not in y (for example, y appears as y^2 or e^y while x appears linearly), rewrite it as $\frac{dx}{dy}$ and solve for x .

SHORTCUT Verify by substitution

With options of the form $y = f(x)$, substitute into the DE and the initial condition. Often the initial condition alone eliminates 2–3 options.

Common mistakes

MISTAKE ALERT Mistake alerts

- Writing the DE in standard form **before** reading off P : $x y' - y = x^2$ becomes $y' - \frac{y}{x} = x$, so $P = -\frac{1}{x}$.
- Losing the constant of integration before applying the initial condition.
- Degree: clear radicals and fractional powers of the derivatives first.
- $e^{\int P dx}$ with $P = \frac{1}{x}$ gives IF = x , not $\ln x$.

Practice questions

M09 · Practice

1 Easy 0.5 min Speed

M09-01

The order and degree of $\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^2 + y = 0$ are:

- (A) 3, 2 (B) 2, 3 (C) 2, 2 (D) 1, 3

2 Medium 1 min Concept

M09-02

The degree of $\frac{d^2y}{dx^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$ is:

- (A) 1 (B) 1/2 (C) 2 (D) not defined

3 Easy 1 min Basic

M09-03

The solution of $\frac{dy}{dx} = xy$ with $y(0) = 1$ is:

- (A) $y = e^{x^2/2}$ (B) $y = e^{x^2}$ (C) $y = 1 + \frac{x^2}{2}$ (D) $y = e^x$

4 Medium 1 min Basic

M09-04

The general solution of $\frac{dy}{dx} + y = e^{-x}$ is:

- (A) $y = xe^x + C$ (B) $y = (x + C)e^x$ (C) $y = Ce^{-x}$ (D) $y = (x + C)e^{-x}$

5 Medium 1.5 min Concept

M09-05

The general solution of $\frac{dy}{dx} = \frac{x+y}{x}$ ($x > 0$) is:

- (A) $y = x \ln x + Cx$ (B) $y = \ln x + C$ (C) $y = x^2 + C$ (D) $y = Cx$

6 Easy 0.75 min Basic

M09-06

The integrating factor of $x \frac{dy}{dx} - y = x^2$ ($x > 0$) is:

- (A) x (B) $\ln x$ (C) e^x (D) $1/x$

7 Easy 0.5 min NV

M09-07

If $\frac{dy}{dx} = 2x$ and $y(1) = 3$, find $y(2)$.

Your answer (numerical value):

8 Medium 1 min NV

M09-08

A population grows at a rate proportional to its size and doubles in 10 years. Find the number of years it takes to become 4 times its initial size.

Your answer (numerical value):

9 Medium 1.5 min JEE

M09-09

The solution of $\frac{dy}{dx} + \frac{2}{x}y = x$ ($x > 0$) through $(1, 1)$ is:

- (A) $y = \frac{x^2}{4} + \frac{3}{4x^2}$ (B) $y = \frac{x^2}{4} + \frac{1}{x^2}$ (C) $y = x^2$ (D) $y = \frac{x^2}{2} + \frac{1}{2x^2}$

10 Easy 0.5 min Concept

M09-10

The degree of the differential equation $\frac{dy}{dx} + \sin\left(\frac{dy}{dx}\right) = x$ is:

- (A) 1 (B) 0 (C) not defined (D) 2

M09 · Chapter Test

1 Easy 0.5 min Speed

M09-T1

The order of $y''' + y = 0$ is:

- (A) 1 (B) 3 (C) 2 (D) 0

2 Easy 0.5 min Speed

M09-T2

The general solution of $\frac{dy}{dx} = y$ is:

- (A) $y = Cx$ (B) $y = x + C$ (C) $y = Ce^x$ (D) $y = Ce^{-x}$

3 Easy 0.5 min NV

M09-T3

The integrating factor of $\frac{dy}{dx} + 3y = 5$ is e^{kx} . Find k .

Your answer (numerical value):

4 Easy 0.5 min Speed

M09-T4

A homogeneous DE $\frac{dy}{dx} = F(y/x)$ is solved with the substitution:

- (A) $y = e^v$ (B) $y = x + v$ (C) $x = y^2$ (D) $y = vx$

5 Medium 0.75 min Concept

M09-T5

The equation $\frac{dy}{dx} = \frac{y}{x + y^2}$ is best solved by treating it as:

- (A) linear in y (B) linear in x (use $\frac{dx}{dy}$)
(C) separable (D) homogeneous

Answers & Solutions

M09 · Practice

1 (B) 2 (C) 3 (A) 4 (D) 5 (A) 6 (D) 7 6 8 20 9 (A) 10 (C)

M09 · Chapter Test

1 (B) 2 (C) 3 3 4 (D) 5 (B)

M09 · PRACTICE

1 Answer: (B) Order and degree Easy 0.5 min Speed

M09-01

SOLUTION The highest derivative is y'' (order 2), raised to the power 3 (degree 3).

2 Answer: (C) Degree after clearing a radical Medium 1 min Concept

M09-02

SOLUTION Squaring: $(y'')^2 = 1 + (y')^2$, so the degree is 2.

COMMON TRAP Answering 1 without clearing the radical.

3 Answer: (A) Separable with initial condition Easy 1 min Basic

M09-03

SOLUTION $\ln y = \frac{x^2}{2} + C$, and $C = 0$ from $y(0) = 1$.

4 Answer: (D) Linear DE Medium 1 min Basic

M09-04

SOLUTION IF $= e^x$: $ye^x = \int 1 dx = x + C$.

5 Answer: (A) Homogeneous DE Medium 1.5 min Concept

M09-05

SOLUTION $y' - \frac{y}{x} = 1$ (linear) with IF $= \frac{1}{x}$: $\frac{y}{x} = \ln x + C$. (Or put $y = vx$: $xv' = 1$.)

6 Answer: (D) Integrating factor Easy 0.75 min Basic

M09-06

SOLUTION Standard form: $y' - \frac{1}{x}y = x$, so IF $= e^{-\ln x} = \frac{1}{x}$.

COMMON TRAP Reading P before dividing by x .

7 Answer: 6 Direct integration Easy 0.5 min NV

M09-07

SOLUTION $y = x^2 + 2$, so $y(2) = 6$.

8 Answer: 20 Exponential growth Medium 1 min NV

M09-08

SOLUTION $N = N_0 2^{t/10}$; $2^{t/10} = 4 \Rightarrow t = 20$.

SHORTCUT Two doublings.

COMMON TRAP Assuming linear growth (30).

9 Answer: (A) Linear DE with initial condition Medium 1.5 min JEE

M09-09

SOLUTION IF $= x^2$: $x^2 y = \frac{x^4}{4} + C$. At $(1, 1)$: $C = \frac{3}{4}$.

SHORTCUT Check the options at $x = 1$: (A) gives $\frac{1}{4} + \frac{3}{4} = 1$ ✓; (B) gives 1.25 ✗; (D) gives 1 ✓, so test the DE for (D): it fails.

10 Answer: (C) Degree not defined Easy 0.5 min Concept

M09-10

SOLUTION The equation isn't a polynomial in y' .

M09 · CHAPTER TEST

1 Answer: (B) Order Easy 0.5 min Speed

M09-T1

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

2 Answer: (C) Simplest DE Easy 0.5 min Speed

M09-T2

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

3 Answer: 3 Integrating factor exponent Easy 0.5 min NV

M09-T3

SOLUTION $P = 3$, so IF $= e^{3x}$.

4 Answer: (D) Homogeneous substitution Easy 0.5 min Speed

M09-T4

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

5 Answer: (B) Linear in x Medium 0.75 min Concept

M09-T5

SOLUTION $\frac{dx}{dy} = \frac{x + y^2}{y} \Rightarrow \frac{dx}{dy} - \frac{x}{y} = y$, which is linear in x .

REVISION Revision tracker · Differential Equations

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART IV · MATHEMATICS · UNIT 10

Coordinate Geometry: Lines, Circles & Conics

10

3–4 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

Must-do (largest
Maths unit)

PRIORITY

Class 11 · Ch 9, 10

NCERT

~20 hours

STUDY TIME

Core concepts

- **Points & lines:** distance, section formula, slope, forms of a line, angle between lines, distance from a point, concurrency; the centroid, orthocentre and circumcentre of a triangle.
- **Circle:** standard and general forms, circles on a diameter, intersection with lines.
- **Conics** (parabola, ellipse, hyperbola) **in standard form:** focus, directrix, eccentricity, latus rectum, vertices, axes.

NOTE Syllabus note

The official Unit 10 text covers points and lines (including intercepts, concurrency, triangle centres), circles (standard/general form, diameter form, intersection with a line through the origin-centred case) and conics in standard forms [NTA-SYL]. Tangent/normal equations, family of lines and angle bisectors were removed in 2024. This module stays on the listed core.

Important formulas: points & lines

Σ FORMULA Points

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad \text{section } m : n \text{ internal: } \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

Centroid $G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$. **Euler line:** O (orthocentre), G , C (circumcentre) are collinear with $OG : GC = 2 : 1$.

In a right triangle the circumcentre is the midpoint of the hypotenuse, and the orthocentre is the right-angle vertex.

$$\text{Area of a triangle} = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|.$$

Σ FORMULA Lines

Form	Equation
slope-intercept	$y = mx + c$
point-slope	$y - y_1 = m(x - x_1)$
two-point	$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$
intercept	$\frac{x}{a} + \frac{y}{b} = 1$
normal	$x \cos \alpha + y \sin \alpha = p$
general	$ax + by + c = 0$: slope $-a/b$, intercepts $-c/a$, $-c/b$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \quad \text{parallel: } m_1 = m_2 \quad \text{perpendicular: } m_1 m_2 = -1$$

$$\text{distance of } (x_1, y_1) \text{ from } ax + by + c = 0 : \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} \quad \text{between parallel lines: } \frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}$$

$$\text{Concurrency of } a_i x + b_i y + c_i = 0 \ (i = 1, 2, 3): \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0.$$

$$\text{Foot of perpendicular / image of } (x_1, y_1) \text{ in } ax + by + c = 0: \frac{x - x_1}{a} = \frac{y - y_1}{b} = -\frac{ax_1 + by_1 + c}{a^2 + b^2} \text{ (use } -2(\cdot) \text{ for the image).}$$

Important formulas: circles**Σ FORMULA Circle**

$$(x - h)^2 + (y - k)^2 = r^2 \quad x^2 + y^2 + 2gx + 2fy + c = 0 : \text{centre } (-g, -f), r = \sqrt{g^2 + f^2 - c}$$

Diameter form (ends A, B): $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$.

Line and circle: with d = distance from the centre to the line: $d < r$ secant (2 points), $d = r$ touching, $d > r$ no intersection. Chord length = $2\sqrt{r^2 - d^2}$.

Two circles (d = distance between centres): $d > r_1 + r_2$ separate; $= r_1 + r_2$ touch externally; $|r_1 - r_2| < d < r_1 + r_2$ intersect; $= |r_1 - r_2|$ touch internally.

A point is inside, on or outside the circle according to whether $S_1 = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$ is < 0 , $= 0$ or > 0 .

Important formulas: conics (standard forms)**Σ FORMULA Parabola $y^2 = 4ax$ ($a > 0$)**

Vertex $(0, 0)$ · focus $(a, 0)$ · directrix $x = -a$ · axis $y = 0$ · latus rectum $4a$ (ends $(a, \pm 2a)$) · $e = 1$ · focal distance of (x, y) : $x + a$ · parametric $(at^2, 2at)$.

Other orientations: $y^2 = -4ax$, $x^2 = 4ay$ (focus $(0, a)$), $x^2 = -4ay$.

Σ FORMULA Ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b$)

$b^2 = a^2(1 - e^2)$ · foci $(\pm ae, 0)$ · directrices $x = \pm \frac{a}{e}$ · latus rectum $\frac{2b^2}{a}$ · major axis $2a$, minor axis $2b$ · $e < 1$.

Focal property: $PS + PS' = 2a$. Parametric: $(a \cos \theta, b \sin \theta)$. If $b > a$, the major axis is along y and $a^2 = b^2(1 - e^2)$.

Σ FORMULA Hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$b^2 = a^2(e^2 - 1)$ · foci $(\pm ae, 0)$ · directrices $x = \pm \frac{a}{e}$ · latus rectum $\frac{2b^2}{a}$ · transverse axis $2a$ · $e > 1$ · **asymptotes**
 $y = \pm \frac{b}{a}x$ · $|PS - PS'| = 2a$.

Rectangular hyperbola $a = b$: $e = \sqrt{2}$. **Conjugate hyperbola** $\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$: $\frac{1}{e_1^2} + \frac{1}{e_2^2} = 1$.

Conic	e	Latus rectum	Focus-directrix
Circle	0	—	—
Parabola $y^2 = 4ax$	1	$4a$	focus $(a, 0)$, directrix $x = -a$
Ellipse	$\sqrt{1 - b^2/a^2}$	$2b^2/a$	$(\pm ae, 0)$; $x = \pm a/e$
Hyperbola	$\sqrt{1 + b^2/a^2}$	$2b^2/a$	$(\pm ae, 0)$; $x = \pm a/e$

Common question models

▮ PYQ PATTERN Recurring structures

1. **Triangle centres** from vertices or side equations; reflection/image of a point; foot of the perpendicular.
2. **Distance/angle** between lines; lines at a given distance or angle through a point.
3. **Circle from conditions** (centre on a line, passing through points, touching the axes); chord lengths; relative position of two circles.
4. **Conic parameters**: find e , foci, latus rectum, or the equation from given data; ellipse/hyperbola with the same foci.
5. **Locus problems**: eliminate a parameter to get a line, circle or conic.
6. **Area** of regions defined by lines/conics (overlaps with integration).

Shortcuts & fast methods

⚡ SHORTCUT Reflection formula

The image of (x_1, y_1) in $ax + by + c = 0$ is $(x_1, y_1) - \frac{2(ax_1 + by_1 + c)}{a^2 + b^2}(a, b)$. It's a single vector step, so there's no need to find the foot first.

⚡ SHORTCUT Circles touching both axes

Centre $(\pm r, \pm r)$ and radius r : $(x \mp r)^2 + (y \mp r)^2 = r^2$. A circle touching the x-axis has $r = |k|$; touching the y-axis, $r = |h|$.

⚡ SHORTCUT Eccentricity from latus rectum

For an ellipse or hyperbola, $\ell = \frac{2b^2}{a}$ together with $b^2 = a^2(1 \mp e^2)$ gives e in two lines. Keep "ellipse: minus; hyperbola: plus" in mind.

Common mistakes

⚠ MISTAKE ALERT Mistake alerts

- Assuming $a > b$ for an ellipse without checking. If the y^2 denominator is larger, the major axis is vertical.
- Hyperbola: $b^2 = a^2(e^2 - 1)$, not $(1 - e^2)$.
- Radius of $x^2 + y^2 + 2gx + 2fy + c = 0$ needs the coefficients of x^2 and y^2 to equal 1 first. Divide if they don't.
- Distance formula: the line must be in the form $ax + by + c = 0$ before substituting.

Practice questions

M10 · Practice

- 1** Easy 1 min Basic M10-01
 The centroid of the triangle with vertices $(1, 2)$, $(3, 4)$ and $(5, 0)$ is:
 (A) $(3, 3)$ (B) $(9, 6)$ (C) $(3, 2)$ (D) $(2, 3)$
- 2** Easy 1 min NV M10-02
 Find the distance between the parallel lines $3x + 4y - 5 = 0$ and $6x + 8y + 20 = 0$.
 Your answer (numerical value):
- 3** Medium 1.5 min JEE M10-03
 The image of the point $(1, 2)$ in the line $x + y - 5 = 0$ is:
 (A) $(4, 3)$ (B) $(3, 4)$ (C) $(2, 1)$ (D) $(-1, -2)$
- 4** Easy 0.75 min Basic M10-04
 The centre and radius of $x^2 + y^2 - 4x + 6y - 12 = 0$ are:
 (A) $(2, -3)$, 5 (B) $(-2, 3)$, 5 (C) $(2, -3)$, 25 (D) $(4, -6)$, 5
- 5** Medium 1 min NV M10-05
 Find the length of the chord cut by the line $x = 3$ from the circle $x^2 + y^2 = 25$.
 Your answer (numerical value):
- 6** Easy 0.75 min Basic M10-06
 For the parabola $y^2 = 12x$, the focus and the length of the latus rectum are:
 (A) $(3, 0)$, 12 (B) $(12, 0)$, 3 (C) $(0, 3)$, 12 (D) $(3, 0)$, 6
- 7** Medium 1 min Basic M10-07
 The eccentricity of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ is:
 (A) $5/3$ (B) $4/5$ (C) $3/5$ (D) $3/4$
- 8** Medium 1 min Basic M10-08
 The asymptotes of $\frac{x^2}{16} - \frac{y^2}{9} = 1$ are:
 (A) $y = \pm \frac{4}{3}x$ (B) $y = \pm \frac{3}{4}x$ (C) $y = \pm \frac{9}{16}x$ (D) $x = \pm 4$
- 9** Medium 1.5 min Concept M10-09
 The circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 6x + 5 = 0$:
 (A) touch externally (B) touch internally
 (C) are separate (D) intersect at two points

10 Medium 1.5 min JEE

M10-10

An ellipse centred at the origin with foci on the x-axis has eccentricity $1/2$ and latus rectum 6. Its equation is:

- (A) $\frac{x^2}{16} + \frac{y^2}{12} = 1$ (B) $\frac{x^2}{12} + \frac{y^2}{16} = 1$ (C) $\frac{x^2}{4} + \frac{y^2}{3} = 1$ (D) $\frac{x^2}{36} + \frac{y^2}{27} = 1$

11 Medium 1 min Basic

M10-11

The acute angle between $y = 2x + 1$ and $y = -3x + 4$ is:

- (A) 30° (B) 90° (C) 60° (D) 45°

12 Medium 1.5 min Concept

M10-12

The circle with the segment joining (1, 2) and (5, 6) as diameter is:

- (A) $x^2 + y^2 - 6x - 8y + 17 = 0$ (B) $x^2 + y^2 + 6x + 8y + 17 = 0$
(C) $x^2 + y^2 - 6x - 8y - 17 = 0$ (D) $x^2 + y^2 - 3x - 4y + 17 = 0$

M10 · Chapter Test

1 Easy 0.5 min Speed

M10-T1

A line perpendicular to $2x - 3y + 5 = 0$ has slope:

- (A) $2/3$ (B) $-2/3$ (C) $3/2$ (D) $-3/2$

2 Easy 0.5 min Speed

M10-T2

The directrix of $x^2 = 8y$ is:

- (A) $y = -8$ (B) $y = 2$ (C) $x = -2$ (D) $y = -2$

3 Easy 0.75 min NV

M10-T3

Find the distance of (2, 3) from $5x + 12y + 6 = 0$.

Your answer (numerical value):

4 Easy 0.5 min Speed

M10-T4

The eccentricity of a rectangular hyperbola is:

- (A) 1 (B) 2 (C) $\sqrt{2}$ (D) $\sqrt{3}$

5 Easy 0.75 min Basic

M10-T5

A circle of radius 3 in the first quadrant touches both axes. Its equation is:

- (A) $x^2 + y^2 - 6x - 6y = 0$ (B) $x^2 + y^2 - 6x - 6y + 9 = 0$
(C) $x^2 + y^2 + 6x + 6y + 9 = 0$ (D) $x^2 + y^2 - 3x - 3y + 9 = 0$

Answers & Solutions

M10 · Practice

- 1 (C) 2 3 3 (B) 4 (A) 5 8 6 (A) 7 (C) 8 (B) 9 (D) 10 (A)
11 (D) 12 (A)

M10 · Chapter Test

- 1 (D) 2 (D) 3 4 4 (C) 5 (B)

M10 · PRACTICE

1 Answer: (C) Centroid **Easy** ⌚ 1 min Basic

M10-01

SOLUTION $\left(\frac{9}{3}, \frac{6}{3}\right)$.**2 Answer: 3** Distance between parallel lines **Easy** ⌚ 1 min NV

M10-02

SOLUTION Write the second as $3x + 4y + 10 = 0$. Distance = $\frac{|-5 - 10|}{5} = 3$.**SHORTCUT** Normalise the coefficients first.**COMMON TRAP** Using $c_2 = 20$ directly gives 5.**3 Answer: (B)** Image of a point **Medium** ⌚ 1.5 min JEE

M10-03

SOLUTION $\frac{2(1+2-5)}{2} = -2$, so the image is $(1, 2) - (-2)(1, 1) = (3, 4)$.**COMMON TRAP** Giving the foot $(2, 3)$ instead of the image.**4 Answer: (A)** Circle centre and radius **Easy** ⌚ 0.75 min Basic

M10-04

SOLUTION $g = -2$, $f = 3$: centre $(2, -3)$, $r = \sqrt{4 + 9 + 12} = 5$.**5 Answer: 8** Chord length **Medium** ⌚ 1 min NV

M10-05

SOLUTION $d = 3$, so the chord is $2\sqrt{25 - 9} = 8$.**6 Answer: (A)** Parabola focus and latus rectum **Easy** ⌚ 0.75 min Basic

M10-06

SOLUTION $4a = 12 \Rightarrow a = 3$.**7 Answer: (C)** Ellipse eccentricity **Medium** ⌚ 1 min Basic

M10-07

SOLUTION $e^2 = 1 - \frac{16}{25} = \frac{9}{25}$.**8 Answer: (B)** Hyperbola asymptotes **Medium** ⌚ 1 min Basic

M10-08

SOLUTION $y = \pm \frac{b}{a}x$ with $a = 4$, $b = 3$.**9 Answer: (D)** Relative position of circles **Medium** ⌚ 1.5 min Concept

M10-09

SOLUTION $C_1 = (0, 0)$, $r_1 = 2$. $C_2 = (3, 0)$, $r_2 = \sqrt{9 - 5} = 2$. $d = 3$, and $0 < 3 < 4$, so they intersect.**10 Answer: (A)** Ellipse from foci and latus rectum **Medium** ⌚ 1.5 min JEE

M10-10

SOLUTION $b^2 = a^2(1 - \frac{1}{4}) = \frac{3}{4}a^2$ and $\frac{2b^2}{a} = 6 \Rightarrow \frac{3}{2}a = 6 \Rightarrow a = 4$, $b^2 = 12$.**SHORTCUT** Check the options' latus rectum: (A) $2 \cdot 12/4 = 6$ ✓, and $e^2 = 1 - 12/16 = 1/4$ ✓.**11 Answer: (D)** Angle between lines **Medium** ⌚ 1 min Basic

M10-11

SOLUTION $\tan \theta = \left| \frac{2 - (-3)}{1 + (2)(-3)} \right| = \frac{5}{5} = 1$.**12 Answer: (A)** Circle on a diameter **Medium** ⌚ 1.5 min Concept

M10-12

SOLUTION $(x - 1)(x - 5) + (y - 2)(y - 6) = 0 \Rightarrow x^2 + y^2 - 6x - 8y + 5 + 12 = 0$.**SHORTCUT** Centre $(3, 4)$ fixes $-6x - 8y$. The constant is $x_1x_2 + y_1y_2 = 5 + 12$.

M10 · CHAPTER TEST

1 Answer: (D) Perpendicular slopes **Easy** ⌚ 0.5 min Speed

M10-T1

SOLUTION The given slope is $\frac{2}{3}$; perpendicular means $-\frac{3}{2}$.**2 Answer: (D)** Directrix of a parabola **Easy** ⌚ 0.5 min Speed

M10-T2

SOLUTION $4a = 8$, so $a = 2$: the directrix is $y = -2$.**3 Answer: 4** Distance from a point to a line **Easy** ⌚ 0.75 min NV

M10-T3

SOLUTION $\frac{|10 + 36 + 6|}{\sqrt{25 + 144}} = \frac{52}{13} = 4$.

4 Answer: (C) Rectangular hyperbola

Easy

⌚ 0.5 min

Speed

M10-T4

SOLUTION $a = b \Rightarrow e^2 = 2$.**5 Answer: (B)** Circle touching both axes

Easy

⌚ 0.75 min

Basic

M10-T5

SOLUTION Centre $(3, 3)$, $r = 3$: $x^2 + y^2 - 6x - 6y + 18 - 9 = 0$.**REVISION** Revision tracker · Coordinate Geometry☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART IV · MATHEMATICS · UNIT 11

Three-Dimensional Geometry

11

1-2 per shift

TYPICAL QUESTIONS

Easy-Medium

DIFFICULTY

Must-do

PRIORITY

Class 11 · Ch 11 ·

Class 12 · Ch 11

NCERT

~7 hours

STUDY TIME

PYQ PATTERN Weight evidence

29 questions across the 2026 Paper-1 shifts  THIRD-PARTY ANALYSIS [T-WT-CD]. For the time it takes to learn, this unit gives some of the highest returns in Maths.

Core concepts

- A point in space is (x, y, z) . Distance and section formulas extend directly from 2D.
- Direction cosines** (l, m, n) are the cosines of the angles a line makes with the axes, and $l^2 + m^2 + n^2 = 1$. **Direction ratios** (a, b, c) are any numbers proportional to them.
- A line is fixed by a **point and a direction**. Lines in space are parallel, intersecting or **skew** (neither parallel nor intersecting).
- Shortest distance** between two lines is measured along their common perpendicular.

NOTE Syllabus note

The official Unit 11 text lists coordinates, distance, section formula, direction ratios and cosines, angle between intersecting lines, skew lines and the shortest distance between them, and equations of a line [NTA-SYL]. Planes are not listed. The insurance box at the end of this module covers the four plane results worth two hours

 **RECOMMENDATION** (see the warning in the Maths strategy, page 285).

Important formulas & standard results

FORMULA Points

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Section $m : n$ (internal): $\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}, \frac{mz_2 + nz_1}{m+n}\right)$; external: replace n by $-n$.

Centroid of a triangle: $\left(\frac{\sum x_i}{3}, \frac{\sum y_i}{3}, \frac{\sum z_i}{3}\right)$.

Distance of (x, y, z) from the x -axis: $\sqrt{y^2 + z^2}$ (similarly for the other axes). Distance from the xy -plane: $|z|$.

Ratio in which the yz -plane divides PQ : $-x_1 : x_2$.

FORMULA Direction cosines & ratios

$$l = \cos \alpha, m = \cos \beta, n = \cos \gamma \quad l^2 + m^2 + n^2 = 1 \quad \sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 2$$

From DRs (a, b, c) : $l = \frac{a}{\sqrt{a^2 + b^2 + c^2}}$, etc. For the line through P and Q , the DRs are $(x_2 - x_1, y_2 - y_1, z_2 - z_1)$.

Angle between two lines: $\cos \theta = \frac{|a_1 a_2 + b_1 b_2 + c_1 c_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$.

Perpendicular: $a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$. Parallel: $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.

Σ FORMULA Equations of a line

Form	Vector	Cartesian
point $\vec{a}$, direction $\vec{b}$	$\vec{r} = \vec{a} + \lambda\vec{b}$	$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$
two points	$\vec{r} = \vec{a} + \lambda(\vec{b}-\vec{a})$	$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$

General point on $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$: $(x_1 + a\lambda, y_1 + b\lambda, z_1 + c\lambda)$. Almost every line question starts here.

Σ FORMULA Shortest distance

Skew lines $\vec{r} = \vec{a}_1 + \lambda\vec{b}_1$ and $\vec{r} = \vec{a}_2 + \mu\vec{b}_2$:

$$d = \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$$

Parallel lines ($\vec{b}_1 = \vec{b}_2 = \vec{b}$): $d = \frac{|(\vec{a}_2 - \vec{a}_1) \times \vec{b}|}{|\vec{b}|}$.

Intersecting (coplanar) lines: $(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = 0$, i.e. $\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$ (for non-parallel lines).

Σ FORMULA Foot of the perpendicular & image (point P , line L)

1. Take the general point $Q(\lambda)$ on L .
2. Set $\vec{PQ} \cdot \vec{b} = 0$ and solve for λ . Q is the foot; PQ is the perpendicular distance.
3. Image $P' = 2Q - P$.

Common question models**PyQ Pattern** Recurring structures

1. **Shortest distance between skew lines** (the single most repeated 3D model). Often an NV, with the answer as d^2 or $\sqrt{k}$.
2. **Foot of the perpendicular / image** of a point in a line, then a distance or a sum of coordinates.
3. **Intersecting lines:** find the unknown k or the point of intersection.
4. **Angle between lines;** DCs from two conditions ($l + m + n = 0$ and a quadratic).
5. **Section formula** with a coordinate plane; collinearity.
6. **Line through a point meeting / perpendicular to two given lines** (cross product for the direction).

Shortcuts & fast methods**⚡ SHORTCUT** Distance to a line by the cross product

Distance of P from the line through A with direction $\vec{b}$: $\frac{|\vec{AP} \times \vec{b}|}{|\vec{b}|}$. No need to find the foot when only the distance is asked.

Time saved: ~1 minute.

⚡ SHORTCUT Intersection test on one coordinate

To check whether two lines meet, equate the general points on **two** coordinates, solve for λ and μ , and test the third. It is faster than the determinant when the numbers are small.

⚡ SHORTCUT Squared answers

When an NV asks for d^2 or $6d^2$, keep everything squared: $d^2 = \frac{[(\vec{a}_2 - \vec{a}_1) \cdot \vec{n}]^2}{|\vec{n}|^2}$. This avoids surds entirely.

Common mistakes

⚠ MISTAKE ALERT Mistake alerts

- Using DRs as DCs without normalising. $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ cannot be DCs, because $l^2 + m^2 + n^2 \neq 1$.
- Sign slips in the cross product. Expand the middle ($\hat{j}$) term with a minus sign.
- Forgetting the modulus in the shortest-distance formula.
- Reading the direction from the wrong form: in $\frac{2x-1}{3} = \dots$, the x -DR is $\frac{3}{2}$, not 3. First rewrite as $\frac{x-1/2}{3/2}$.

Insurance topic: planes

☆ IMPORTANT Plane insurance (2 hours, not in the official text) 📌 RECOMMENDATION

Result	Formula
Plane with normal (a, b, c) through (x_1, y_1, z_1)	$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$
Distance of a point from $ax + by + cz + d = 0$	$\frac{ ax_1 + by_1 + cz_1 + d }{\sqrt{a^2 + b^2 + c^2}}$
Angle between a line (DRs $\vec{b}$) and a plane (normal $\vec{n}$)	$\sin \phi = \frac{ \vec{b} \cdot \vec{n} }{ \vec{b} \vec{n} }$
Plane through three points	normal $= \vec{AB} \times \vec{AC}$

Practice questions

📌 M11 · Practice

1 Easy ⌚ 0.5 min NV M11-01

Find the distance between $(1, 2, 3)$ and $(4, 6, 15)$.

Your answer (numerical value):

2 Easy ⌚ 0.5 min Basic M11-02

The direction cosines of a line with direction ratios $2, -1, 2$ are:

(A) $\frac{2}{5}, -\frac{1}{5}, \frac{2}{5}$ (B) $2, -1, 2$ (C) $\frac{2}{3}, -\frac{1}{3}, \frac{2}{3}$ (D) $\frac{1}{3}, -\frac{1}{6}, \frac{1}{3}$

3 Easy ⌚ 0.5 min Speed M11-03

If a line makes angles α, β, γ with the coordinate axes, then $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma$ equals:

(A) 1 (B) 0 (C) 3 (D) 2

4 Easy ⌚ 0.75 min Basic M11-04

The angle between lines with direction ratios $(1, 1, 0)$ and $(0, 1, 1)$ is:

(A) 30° (B) 45° (C) 90° (D) 60°

5 Easy 0.75 min Basic

M11-05

The point dividing the join of $(1, 2, 3)$ and $(4, 5, 6)$ internally in the ratio $1 : 2$ is:

- (A) $(2, 3, 4)$ (B) $(3, 4, 5)$ (C) $(2.5, 3.5, 4.5)$ (D) $(1.5, 2.5, 3.5)$

6 Medium 1 min Tricky

M11-06

The angle between the two lines whose direction cosines satisfy $l + m + n = 0$ and $l^2 + m^2 - n^2 = 0$ is:

- (A) 30° (B) 60° (C) 45° (D) 90°

7 Medium 2 min JEE

M11-07

The shortest distance between $\vec{r} = (\hat{i} + 2\hat{j} + \hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k})$ and $\vec{r} = (2\hat{i} - \hat{j} - \hat{k}) + \mu(2\hat{i} + \hat{j} + 2\hat{k})$ is:

- (A) $\sqrt{2}$ (B) $\frac{9}{\sqrt{2}}$ (C) $\frac{3}{\sqrt{2}}$ (D) $3\sqrt{2}$

8 Medium 1.5 min JEE

M11-08

The distance between the parallel lines $\vec{r} = \hat{i} + \hat{j} + \lambda(2\hat{i} - \hat{j} + \hat{k})$ and $\vec{r} = 2\hat{i} + \hat{j} - \hat{k} + \mu(4\hat{i} - 2\hat{j} + 2\hat{k})$ is:

- (A) $\sqrt{\frac{11}{6}}$ (B) $\sqrt{\frac{6}{11}}$ (C) $\sqrt{11}$ (D) $\frac{11}{6}$

9 Medium 2 min JEE

M11-09

The lines $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{4}$ and $\frac{x-3}{1} = \frac{y-k}{2} = \frac{z}{1}$ intersect. Then k equals:

- (A) $\frac{3}{2}$ (B) $\frac{9}{2}$ (C) $\frac{2}{9}$ (D) 5

10 Medium 1.5 min JEE

M11-10

The foot of the perpendicular from $(0, 2, 3)$ to the line $\frac{x+3}{5} = \frac{y-1}{2} = \frac{z+4}{3}$ is:

- (A) $(-3, 1, -4)$ (B) $(2, 3, -1)$ (C) $(7, 5, 2)$ (D) $(2, 3, 1)$

11 Medium 1.5 min JEE

M11-11

The image of $(1, 6, 3)$ in the line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ is:

- (A) $(-1, 0, 7)$ (B) $(1, 3, 5)$ (C) $(7, 0, 1)$ (D) $(1, 0, 7)$

12 Easy 0.75 min NV

M11-12

Find the distance of $(1, 2, 3)$ from the plane $2x - y + 2z + 3 = 0$.

Your answer (numerical value):

M11 · Chapter Test

1 Easy 0.5 min Concept

M11-T1

Which of the following can be the direction cosines of a line?

- (A) $\frac{1}{2}, \frac{1}{2}, \frac{1}{2}$ (B) $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$ (C) 1, 1, 1 (D) $\frac{1}{2}, \frac{1}{\sqrt{2}}, 1$

2 Easy 0.5 min Speed

M11-T2

The distance of $(3, 4, 5)$ from the x -axis is:

- (A) $5\sqrt{2}$ (B) 3 (C) $\sqrt{41}$ (D) $\sqrt{34}$

3 Easy 0.5 min NV

M11-T3

Find the angle (in degrees) between lines with direction ratios $(1, 2, 3)$ and $(-3, 0, 1)$.

Your answer (numerical value):

4 Easy 0.5 min NV

M11-T4

The centroid of the triangle with vertices $(1, 2, 3)$, $(2, 3, 1)$ and $(3, 1, 2)$ is (α, β, γ) . Find $\alpha + \beta + \gamma$.

Your answer (numerical value):

5 Easy 0.5 min Concept

M11-T5

Two lines in space are called skew when they are:

- (A) neither parallel nor intersecting (B) intersecting at right angles
(C) parallel (D) coincident

Answers & Solutions

M11 · Practice

- 1 13 2 (C) 3 (D) 4 (D) 5 (A) 6 (B) 7 (C) 8 (A) 9 (B) 10 (B)
11 (D) 12 3

M11 · Chapter Test

- 1 (B) 2 (C) 3 90 4 6 5 (A)

M11 · PRACTICE

1 Answer: 13 Distance formula Easy 0.5 min NV

M11-01

SOLUTION $\sqrt{9+16+144} = \sqrt{169} = 13$.

2 Answer: (C) DCs from DRs Easy 0.5 min Basic

M11-02

SOLUTION $\sqrt{4+1+4} = 3$; divide each DR by 3.

3 Answer: (D) Sum of squares of sines Easy 0.5 min Speed

M11-03

SOLUTION $\sum \sin^2 = 3 - \sum \cos^2 = 3 - 1 = 2$.

4 Answer: (D) Angle between lines Easy 0.75 min Basic

M11-04

SOLUTION $\cos \theta = \frac{0+1+0}{\sqrt{2} \cdot \sqrt{2}} = \frac{1}{2}$.

5 Answer: (A) Section formula Easy 0.75 min Basic

M11-05

SOLUTION $\left(\frac{4+2}{3}, \frac{5+4}{3}, \frac{6+6}{3}\right) = (2, 3, 4)$.

SHORTCUT The point is one-third of the way from the first point: $(1, 2, 3) + \frac{1}{3}(3, 3, 3)$.

COMMON TRAP Using 2 : 1 by mistake gives $(3, 4, 5)$.

6 Answer: (B) DCs from two conditions Medium 1 min Tricky

M11-06

SOLUTION $n = -(l+m)$, so $l^2 + m^2 - (l+m)^2 = -2lm = 0$. If $l = 0$, the DRs are $(0, 1, -1)$; if $m = 0$, they are $(1, 0, -1)$. $\cos \theta = \frac{1}{\sqrt{2} \cdot \sqrt{2}} = \frac{1}{2}$.

COMMON TRAP Stopping at $lm = 0$ without building both direction vectors.

7 Answer: (C) Shortest distance (skew lines) Medium 2 min JEE

M11-07

SOLUTION $\vec{a}_2 - \vec{a}_1 = (1, -3, -2)$; $\vec{b}_1 \times \vec{b}_2 = (-3, 0, 3)$ with magnitude $3\sqrt{2}$. Dot product: $-3 + 0 - 6 = -9$. $d = \frac{9}{3\sqrt{2}} = \frac{3}{\sqrt{2}}$.

COMMON TRAP Dropping the minus on the $\hat{j}$ term of the cross product.

8 Answer: (A) Shortest distance (parallel lines) **Medium** ⌚ 1.5 min JEE

M11-08

SOLUTION Take $\vec{b} = (2, -1, 1)$, $|\vec{b}| = \sqrt{6}$. $\vec{a}_2 - \vec{a}_1 = (1, 0, -1)$; $(1, 0, -1) \times (2, -1, 1) = (-1, -3, -1)$ with magnitude $\sqrt{11}$. $d = \sqrt{11}/\sqrt{6}$.

COMMON TRAP Using the skew-line formula: $\vec{b}_1 \times \vec{b}_2 = \vec{0}$ for parallel lines.

9 Answer: (B) Intersecting lines **Medium** ⌚ 2 min JEE

M11-09

SOLUTION General points: $(1 + 2\lambda, -1 + 3\lambda, 1 + 4\lambda)$ and $(3 + \mu, k + 2\mu, \mu)$. From z : $\mu = 1 + 4\lambda$. From x : $1 + 2\lambda = 4 + 4\lambda \Rightarrow \lambda = -\frac{3}{2}$, so $\mu = -5$. From y : $-1 - \frac{9}{2} = k - 10 \Rightarrow k = \frac{9}{2}$.

SHORTCUT Two coordinates fix λ and μ ; the third gives k .

10 Answer: (B) Foot of perpendicular **Medium** ⌚ 1.5 min JEE

M11-10

SOLUTION $Q = (5\lambda - 3, 2\lambda + 1, 3\lambda - 4)$; $\vec{PQ} = (5\lambda - 3, 2\lambda - 1, 3\lambda - 7)$. Dot with $(5, 2, 3)$: $38\lambda - 38 = 0 \Rightarrow \lambda = 1$, giving $Q = (2, 3, -1)$.

SHORTCUT Check the options: (D) is not on the line. (B), (A) and (C) are on it ($\lambda = 1, 0, 2$), but only (B) makes $\vec{PQ} \perp \vec{b}$.

11 Answer: (D) Image in a line **Medium** ⌚ 1.5 min JEE

M11-11

SOLUTION $Q = (\lambda, 1 + 2\lambda, 2 + 3\lambda)$; $\vec{PQ} \cdot (1, 2, 3) = 14\lambda - 14 = 0 \Rightarrow Q = (1, 3, 5)$. Image $= 2Q - P = (1, 0, 7)$.

COMMON TRAP Giving the foot $(1, 3, 5)$ as the image.

12 Answer: 3 Point-plane distance (insurance) **Easy** ⌚ 0.75 min NV

M11-12

SOLUTION $\frac{|2 - 2 + 6 + 3|}{\sqrt{4 + 1 + 4}} = \frac{9}{3} = 3$.

M11 · CHAPTER TEST**1 Answer: (B)** Valid direction cosines **Easy** ⌚ 0.5 min Concept

M11-T1

SOLUTION Only (B) has $l^2 + m^2 + n^2 = 1$.

2 Answer: (C) Distance from an axis **Easy** ⌚ 0.5 min Speed

M11-T2

SOLUTION $\sqrt{y^2 + z^2} = \sqrt{16 + 25}$.

COMMON TRAP Answering 3 (the x -coordinate).

3 Answer: 90 Perpendicular lines **Easy** ⌚ 0.5 min NV

M11-T3

SOLUTION $-3 + 0 + 3 = 0$.

4 Answer: 6 Centroid **Easy** ⌚ 0.5 min NV

M11-T4

SOLUTION The centroid is $(2, 2, 2)$.

SHORTCUT Sum of all nine coordinates divided by 3: $18/3 = 6$.

5 Answer: (A) Skew lines **Easy** ⌚ 0.5 min Concept

M11-T5

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

REVISION Revision tracker · 3D Geometry

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART IV · MATHEMATICS · UNIT 12

Vector Algebra

12

1-2 per shift

TYPICAL QUESTIONS

Easy

DIFFICULTY

Must-do

PRIORITY

Class 12 · Ch 10

NCERT

~5 hours

STUDY TIME



PYQ PATTERN Weight evidence

27 questions across the 2026 Paper-1 shifts  THIRD-PARTY ANALYSIS [T-WT-CD]. Most were direct applications of the dot and cross products.

Core concepts

- A **vector** has magnitude and direction; a **scalar** has only magnitude. In components, $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$.
- Addition**: triangle and parallelogram laws. The position vector of the point dividing AB in $m : n$ is $\frac{m\vec{b} + n\vec{a}}{m + n}$.
- Dot product** (a scalar) measures alignment. **Cross product** (a vector perpendicular to both) measures area.



NOTE Syllabus note

The official Unit 12 text lists vectors and scalars, addition, components in 2D and 3D, and **scalar and vector products** [NTA-SYL]. Triple products are not named. The "tools" box below covers them because they compress coplanarity and shortest-distance work  **RECOMMENDATION**.

Important formulas & standard results



FORMULA Dot product

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos \theta = a_1b_1 + a_2b_2 + a_3b_3$$

- $\vec{a} \perp \vec{b} \iff \vec{a} \cdot \vec{b} = 0$ (for non-zero vectors); $\hat{i} \cdot \hat{i} = 1, \hat{i} \cdot \hat{j} = 0$.
- Projection** of $\vec{a}$ on $\vec{b}$: $\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$. Vector projection: $\left(\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \right) \vec{b}$.
- $|\vec{a} \pm \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 \pm 2\vec{a} \cdot \vec{b}$.
- $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}| \iff \vec{a} \perp \vec{b}$.



FORMULA Cross product

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \quad |\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin \theta$$

- $\hat{i} \times \hat{j} = \hat{k}, \hat{j} \times \hat{k} = \hat{i}, \hat{k} \times \hat{i} = \hat{j}$ (cyclic); reversing the order flips the sign.
- $\vec{a} \times \vec{b} = \vec{0} \iff \vec{a} \parallel \vec{b}$ (non-zero vectors).
- Parallelogram with adjacent sides $\vec{a}, \vec{b}$: area = $|\vec{a} \times \vec{b}|$. Triangle: $\frac{1}{2}|\vec{a} \times \vec{b}|$. Parallelogram with diagonals $\vec{d}_1, \vec{d}_2$: area = $\frac{1}{2}|\vec{d}_1 \times \vec{d}_2|$.
- Unit vector perpendicular to both: $\pm \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$.
- Lagrange's identity**: $|\vec{a} \times \vec{b}|^2 + (\vec{a} \cdot \vec{b})^2 = |\vec{a}|^2 |\vec{b}|^2$.

Σ FORMULA Tools (not named in the syllabus text)

Scalar triple product: $[\vec{a} \vec{b} \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$. This is the volume of the parallelepiped; it is zero iff the vectors are coplanar.

Vector triple product: $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$.

Common question models**▮ PYQ PATTERN Recurring structures**

1. **Magnitudes from conditions:** given $|\vec{a}|$, $|\vec{b}|$ and $|\vec{a} + \vec{b}|$, find $\vec{a} \cdot \vec{b}$ or the angle.
2. $\vec{a} + \vec{b} + \vec{c} = \vec{0}$ problems: square both sides.
3. **Projection** and component questions.
4. **Area** of a triangle or parallelogram from vertices or diagonals.
5. **Unknown parameter** from perpendicularity, parallelism or coplanarity.
6. **Vector equations** such as $\vec{r} \times \vec{a} = \vec{b} \times \vec{a}$ and $\vec{r} \cdot \vec{c} = k$: find $\vec{r}$.

Shortcuts & fast methods**⚡ SHORTCUT Square it**

Almost every magnitude question is solved by squaring: $|\vec{a} + \vec{b} + \vec{c}|^2 = \sum |\vec{a}|^2 + 2 \sum \vec{a} \cdot \vec{b}$.

Time saved: avoids components entirely.

⚡ SHORTCUT $\vec{r} \times \vec{a} = \vec{b} \times \vec{a}$

This gives $(\vec{r} - \vec{b}) \times \vec{a} = \vec{0}$, so $\vec{r} = \vec{b} + t\vec{a}$. Use the second condition to find t .

⚡ SHORTCUT Spot linear combinations

Before expanding a 3×3 determinant for coplanarity, check whether one vector is a sum or multiple of the others. If $\vec{c} = \vec{a} + \vec{b}$, the vectors are coplanar at once.

Common mistakes**⚠ MISTAKE ALERT Mistake alerts**

- The projection formula divides by $|\vec{b}|$ (the vector you project **onto**), not by $|\vec{a}|$.
- The cross product isn't commutative: $\vec{b} \times \vec{a} = -\vec{a} \times \vec{b}$.
- The angle between vectors lies in $[0, \pi]$. A negative dot product means an obtuse angle, not an error.
- Triangle area is **half** of $|\vec{AB} \times \vec{AC}|$.

Practice questions**✎ M12 · Practice**

1 Easy 0.5 min Basic

M12-01

If $|\vec{a}| = 3$, $|\vec{b}| = 4$ and $\vec{a} \cdot \vec{b} = 6$, the angle between $\vec{a}$ and $\vec{b}$ is:

- (A) 30° (B) 60° (C) 45° (D) 90°

2 Easy 0.5 min Speed

M12-02

The unit vector along $2\hat{i} - \hat{j} + 2\hat{k}$ is:

- (A) $\frac{1}{5}(2\hat{i} - \hat{j} + 2\hat{k})$ (B) $\frac{1}{9}(2\hat{i} - \hat{j} + 2\hat{k})$ (C) $2\hat{i} - \hat{j} + 2\hat{k}$ (D) $\frac{1}{3}(2\hat{i} - \hat{j} + 2\hat{k})$

3 Easy 0.5 min NV

M12-03

If $\hat{i} + \lambda\hat{j} + 2\hat{k}$ is perpendicular to $2\hat{i} - \hat{j} + \hat{k}$, find λ .Your answer (numerical value): **4** Easy 0.75 min Basic

M12-04

The projection of $\vec{a} = 2\hat{i} + 3\hat{j} + 2\hat{k}$ on $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$ is:

- (A) $\frac{10}{\sqrt{6}}$ (B) $\frac{10}{\sqrt{17}}$ (C) $\sqrt{6}$ (D) $\frac{5}{\sqrt{6}}$

5 Easy 1 min Basic

M12-05

The area of the parallelogram with adjacent sides $\hat{i} + \hat{j} - \hat{k}$ and $\hat{i} - \hat{j} + \hat{k}$ is:

- (A) 4 (B) $\sqrt{2}$ (C) $2\sqrt{2}$ (D) $2\sqrt{3}$

6 Medium 1 min JEE

M12-06

The area of the triangle with vertices $A(1, 1, 1)$, $B(1, 2, 3)$ and $C(2, 3, 1)$ is:

- (A) $\sqrt{21}$ (B) $\frac{\sqrt{21}}{2}$ (C) $\frac{\sqrt{14}}{2}$ (D) $\frac{21}{2}$

7 Easy 0.75 min NV

M12-07

If $|\vec{a}| = 2$, $|\vec{b}| = 5$ and $|\vec{a} \times \vec{b}| = 8$, find $|\vec{a} \cdot \vec{b}|$.Your answer (numerical value): **8** Easy 0.5 min Concept

M12-08

If $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$ for non-zero vectors, then:

- (A) $\vec{a} \perp \vec{b}$ (B) $\vec{a} \parallel \vec{b}$ (C) $|\vec{a}| = |\vec{b}|$ (D) $\vec{a} = \vec{b}$

9 Medium 1 min NV

M12-09

 $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, with $|\vec{a}| = 3$, $|\vec{b}| = 5$ and $|\vec{c}| = 7$. Find the angle between $\vec{a}$ and $\vec{b}$ in degrees.Your answer (numerical value): **10** Medium 1 min Tricky

M12-10

If $\vec{a} + \vec{b} + \vec{c} = \vec{0}$ and all three are unit vectors, then $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$ equals:

- (A) 0 (B) $\frac{3}{2}$ (C) $-\frac{3}{2}$ (D) -3

11 Medium 1 min Basic

M12-11

A unit vector perpendicular to both $\hat{i} + \hat{j} + \hat{k}$ and $\hat{i} + 2\hat{j} + 3\hat{k}$ is:

- (A) $\frac{\hat{i} - 2\hat{j} + \hat{k}}{\sqrt{6}}$ (B) $\frac{\hat{i} + 2\hat{j} + \hat{k}}{\sqrt{6}}$ (C) $\frac{\hat{i} - \hat{j}}{\sqrt{2}}$ (D) $\frac{2\hat{i} - \hat{j}}{\sqrt{5}}$

12 Medium 1 min NV

M12-12

The vectors $\hat{i} + \hat{j} + \hat{k}$, $\hat{i} + 2\hat{j} + 3\hat{k}$ and $2\hat{i} + \lambda\hat{j} + 4\hat{k}$ are coplanar. Find λ .

Your answer (numerical value):

M12 · Chapter Test

1 Easy 0.5 min Speed

M12-T1

$\hat{i} \cdot (\hat{j} \times \hat{k})$ equals:

- (A) 0 (B) -1 (C) 1 (D) $\hat{k}$

2 Easy 0.5 min Speed

M12-T2

$\hat{j} \times \hat{i}$ equals:

- (A) $\hat{k}$ (B) 1 (C) $\vec{0}$ (D) $-\hat{k}$

3 Easy 0.5 min NV

M12-T3

Find $|3\hat{i} - 4\hat{j} + 12\hat{k}|$.

Your answer (numerical value):

4 Easy 0.5 min Concept

M12-T4

$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|$ (both non-zero) when the angle between them is:

- (A) $\pi/2$ (B) 0 (C) π (D) $\pi/4$

5 Easy 0.5 min NV

M12-T5

If $|\vec{a}| = 1$, $|\vec{b}| = 2$ and $\vec{a} \cdot \vec{b} = 1$, find $|\vec{a} + \vec{b}|^2$.

Your answer (numerical value):

Answers & Solutions

M12 · Practice

- 1 (B) 2 (D) 3 4 4 (A) 5 (C) 6 (B) 7 6 8 (A) 9 60 10 (C)
11 (A) 12 3

M12 · Chapter Test

- 1 (C) 2 (D) 3 13 4 (B) 5 7

M12 · PRACTICE

1 Answer: (B) Angle from dot product Easy 0.5 min Basic

M12-01

SOLUTION $\cos \theta = \frac{6}{12} = \frac{1}{2}$.

2 Answer: (D) Unit vector Easy 0.5 min Speed

M12-02

SOLUTION The magnitude is $\sqrt{4+1+4} = 3$.

3 Answer: 4 Perpendicularity Easy 0.5 min NV

M12-03

SOLUTION $2 - \lambda + 2 = 0$.

4 Answer: (A) Projection **Easy** ⌚ 0.75 min Basic

M12-04

SOLUTION $\vec{a} \cdot \vec{b} = 2 + 6 + 2 = 10$ and $|\vec{b}| = \sqrt{6}$.

COMMON TRAP Dividing by $|\vec{a}| = \sqrt{17}$ gives (B).

5 Answer: (C) Parallelogram area **Easy** ⌚ 1 min Basic

M12-05

SOLUTION The cross product is $(0, -2, -2)$, with magnitude $2\sqrt{2}$.

6 Answer: (B) Triangle area from vertices **Medium** ⌚ 1 min JEE

M12-06

SOLUTION $\vec{AB} = (0, 1, 2)$ and $\vec{AC} = (1, 2, 0)$. $\vec{AB} \times \vec{AC} = (-4, 2, -1)$, with magnitude $\sqrt{21}$. Area = $\frac{1}{2}\sqrt{21}$.

COMMON TRAP Forgetting the $\frac{1}{2}$.

7 Answer: 6 Lagrange's identity **Easy** ⌚ 0.75 min NV

M12-07

SOLUTION $(\vec{a} \cdot \vec{b})^2 = 100 - 64 = 36$.

8 Answer: (A) Equal sum and difference **Easy** ⌚ 0.5 min Concept

M12-08

SOLUTION Squaring gives $4\vec{a} \cdot \vec{b} = 0$.

COMMON TRAP Choosing (C): that is the condition for the diagonals of the parallelogram to be perpendicular.

9 Answer: 60 Sum of three vectors is zero **Medium** ⌚ 1 min NV

M12-09

SOLUTION $\vec{c} = -(\vec{a} + \vec{b})$, so $49 = 9 + 25 + 2\vec{a} \cdot \vec{b} \Rightarrow \vec{a} \cdot \vec{b} = \frac{15}{2}$. $\cos \theta = \frac{15/2}{15} = \frac{1}{2}$.

COMMON TRAP Answering 120° (the angle between the sides of the triangle, not between the vectors placed tail to tail).

10 Answer: (C) Unit vectors summing to zero **Medium** ⌚ 1 min Tricky

M12-10

SOLUTION $0 = |\vec{a} + \vec{b} + \vec{c}|^2 = 3 + 2S \Rightarrow S = -\frac{3}{2}$.

11 Answer: (A) Perpendicular unit vector **Medium** ⌚ 1 min Basic

M12-11

SOLUTION The cross product is $(3 - 2, -(3 - 1), 2 - 1) = (1, -2, 1)$.

SHORTCUT Check the options by dot product with both vectors: (A) gives $1 - 2 + 1 = 0$ and $1 - 4 + 3 = 0$.

12 Answer: 3 Coplanarity (tool) **Medium** ⌚ 1 min NV

M12-12

SOLUTION The determinant is $6 - 2\lambda = 0$.

SHORTCUT With $\lambda = 3$ the third vector is the sum of the first two.

M12 · CHAPTER TEST

1 Answer: (C) Unit-vector triple product **Easy** ⌚ 0.5 min Speed

M12-T1

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

2 Answer: (D) Order of the cross product **Easy** ⌚ 0.5 min Speed

M12-T2

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

3 Answer: 13 Magnitude **Easy** ⌚ 0.5 min NV

M12-T3

SOLUTION $\sqrt{9 + 16 + 144} = 13$.

4 Answer: (B) Maximum of a dot product **Easy** ⌚ 0.5 min Concept

M12-T4

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

5 Answer: 7 Magnitude of a sum **Easy** ⌚ 0.5 min NV

M12-T5

SOLUTION $1 + 4 + 2 = 7$.

REVISION Revision tracker · Vector Algebra

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART IV · MATHEMATICS · UNIT 13

Statistics & Probability

13

2–3 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

Must-do

PRIORITY

Class 11 · Ch 13,
14 · Class 12 · Ch
13

NCERT

~12 hours

STUDY TIME

Core concepts

- **Statistics:** measures of central tendency (mean, median, mode) and dispersion (mean deviation, variance, standard deviation) for grouped and ungrouped data.
- **Probability:** events and their probabilities, the addition and multiplication theorems, conditional probability, **total probability and Bayes' theorem**, and random variables with their distributions (mean and variance).

NOTE Syllabus note

The official Unit 13 text lists measures of dispersion (mean, median, mode, SD, variance, mean deviation), probability of an event, the addition and multiplication theorems, Bayes' theorem and the probability distribution of a random variate [NTA-SYL]. Bernoulli trials and the binomial distribution were removed in the 2024 revision [T-DELETED]. The two binomial formulas below are kept only as a quick tool for random-variable questions **RECOMMENDATION**.

Important formulas: statistics

Σ FORMULA Central tendency

Mean $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$. Assumed-mean method: $\bar{x} = A + \frac{\sum f_i d_i}{N}$ with $d_i = x_i - A$ (or step deviation $\times h$).

Median (grouped): $l + \frac{N/2 - C}{f} h$, where C is the cumulative frequency before the median class.

Mode (grouped): $l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} h$.

Empirical relation (moderately skewed data): Mode ≈ 3 Median $- 2$ Mean.

Σ FORMULA Dispersion

$$\sigma^2 = \frac{\sum f_i (x_i - \bar{x})^2}{N} = \frac{\sum f_i x_i^2}{N} - \bar{x}^2 \quad \text{MD}(a) = \frac{\sum f_i |x_i - a|}{N}$$

- **Change of origin and scale:** if $y = ax + b$, then $\bar{y} = a\bar{x} + b$, $\sigma_y = |a|\sigma_x$ and $\sigma_y^2 = a^2\sigma_x^2$. Adding a constant changes the mean, **not** the variance.
- The mean deviation is least when taken about the **median**.
- First n natural numbers: mean $\frac{n+1}{2}$, variance $\frac{n^2-1}{12}$.
- **Combined mean:** $\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$. **Combined variance:** $\sigma^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}$ with $d_i = \bar{x}_i - \bar{x}$.
- **Correcting a wrong observation:** fix $\sum x$ and $\sum x^2$, then recompute. Never "adjust" the variance directly.

Important formulas: probability

Σ FORMULA Rules

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad P(A') = 1 - P(A) \quad P(A | B) = \frac{P(A \cap B)}{P(B)}$$

- **Independent events:** $P(A \cap B) = P(A)P(B)$. Then A' and B' , A and B' , and A' and B are also independent.
- **Mutually exclusive:** $P(A \cap B) = 0$. (Events with non-zero probability can't be both independent and mutually exclusive.)
- Three events: $P(A \cup B \cup C) = \sum P(A) - \sum P(A \cap B) + P(A \cap B \cap C)$.
- **"At least one":** $1 - P(\text{none})$.

Σ FORMULA Total probability & Bayes

If $E_1, \dots, E_n$ partition the sample space:

$$P(A) = \sum_i P(E_i)P(A | E_i) \quad P(E_k | A) = \frac{P(E_k)P(A | E_k)}{\sum_i P(E_i)P(A | E_i)}$$

Draw a **tree**: branches for E_i , then for A . Bayes' theorem is the chosen path divided by the sum of all paths that end in A .

Σ FORMULA Random variables

$$\sum p_i = 1 \quad E(X) = \mu = \sum x_i p_i \quad \text{Var}(X) = E(X^2) - \mu^2$$

Binomial tool: $P(X = r) = \binom{n}{r} p^r q^{n-r}$, mean np , variance npq .

Common question models

▮ PYQ PATTERN Recurring structures

1. **Mean and variance** with a wrong observation corrected, or observations added/removed.
2. **Unknown observations** from a given mean and variance (e.g. find $a^2 + b^2$ or $|a - b|$).
3. **Effect of transformation** ($y = ax + b$) on the mean and SD.
4. **Mean deviation** about the mean or median.
5. **Bayes' theorem** (bags, machines, a test with false positives). This is a very common model.
6. **Random variable:** find k from $\sum p = 1$, then $E(X)$ or $\text{Var}(X)$.
7. **Dice, coins and cards** with conditional probability.

Shortcuts & fast methods

⚡ SHORTCUT Work with Σx and Σx^2

For any mean/variance manipulation, convert to $\sum x = n\bar{x}$ and $\sum x^2 = n(\sigma^2 + \bar{x}^2)$, change those two totals, then convert back.

Time saved: handles every "wrong observation / new observation" question with one method.

⚡ SHORTCUT Bayes by counting

Imagine 1000 items. With machine shares 50/30/20% and defect rates 2/3/4%, the defectives are 10, 9 and 8.

$$P(\text{machine 1} | \text{defective}) = \frac{10}{27}.$$

Common mistakes

MISTAKE ALERT Mistake alerts

- Variance is in **squared units**; SD is its square root. Read which one is asked.
- Adding a constant doesn't change the SD; multiplying by -2 multiplies the SD by 2 (not -2).
- $P(A | B) \neq P(B | A)$. Identify the given condition carefully.
- "At least one boy": the sample space is $\{BB, BG, GB\}$, not $\{BB, BG\}$.
- Mutually exclusive is not the same as independent.

Practice questions

M13 · Practice

1 Easy ⌚ 0.75 min NV M13-01

Find the variance of 2, 4, 6, 8, 10.

Your answer (numerical value):

2 Easy ⌚ 0.5 min Speed M13-02

If the standard deviation of $x_1, \dots, x_n$ is 3, the standard deviation of $2x_1 + 5, \dots, 2x_n + 5$ is:

- (A) 12 (B) 11 (C) 6 (D) 3

3 Easy ⌚ 0.5 min Speed M13-03

The variance of the first 10 natural numbers is:

- (A) 10 (B) 9.17 (C) 5.5 (D) 8.25

4 Easy ⌚ 0.5 min NV M13-04

Find the mean deviation about the mean of 3, 5, 7, 9.

Your answer (numerical value):

5 Medium ⌚ 1 min NV M13-05

The modal class is 20–30 with frequency 12. The classes before and after it have frequencies 8 and 6. Find the mode.

Your answer (numerical value):

6 Easy ⌚ 0.5 min NV M13-06

One section of 30 students has a mean of 40, and another of 20 students has a mean of 50. Find the combined mean.

Your answer (numerical value):

7 Medium ⌚ 1.5 min JEE M13-07

The mean and variance of 5 observations are 4 and 2. One observation, recorded as 5, was actually 10. The correct variance is:

- (A) 8 (B) 6.4 (C) 2 (D) 10

8 Medium 1 min Concept

M13-08

$P(A) = 0.5$, $P(B) = 0.4$ and $P(A \cap B) = 0.2$. Which statement is true?

- (A) A and B are mutually exclusive
 (B) A and B are independent
 (C) $P(A \cup B) = 0.9$
 (D) $P(A | B) = 0.4$

9 Medium 1.5 min JEE

M13-09

Machines M_1 , M_2 and M_3 make 50%, 30% and 20% of a factory's output, with defect rates 2%, 3% and 4%. A randomly chosen item is defective. The probability that it came from M_1 is:

- (A) $\frac{1}{2}$ (B) $\frac{10}{27}$ (C) $\frac{9}{27}$ (D) $\frac{8}{27}$

10 Medium 1 min Basic

M13-10

X takes the values 0, 1 and 2 with probabilities k , $2k$ and $3k$. Then $E(X)$ equals:

- (A) $\frac{4}{3}$ (B) $\frac{5}{3}$ (C) 1 (D) $\frac{2}{3}$

11 Medium 1.5 min JEE

M13-11

For the random variable in M13-10, $\text{Var}(X)$ equals:

- (A) $\frac{16}{9}$ (B) $\frac{7}{3}$ (C) $\frac{5}{9}$ (D) $\frac{2}{3}$

12 Easy 0.5 min Speed

M13-12

Three fair coins are tossed. The probability of at least one head is:

- (A) $\frac{1}{2}$ (B) $\frac{7}{8}$ (C) $\frac{3}{8}$ (D) $\frac{3}{4}$

13 Medium 1 min Tricky

M13-13

A family has two children, and at least one is a boy. The probability that both are boys is:

- (A) $\frac{2}{3}$ (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) $\frac{1}{3}$

M13 · Chapter Test

1 Easy 0.5 min Concept

M13-T1

If every observation is increased by 5, the variance:

- (A) is unchanged (B) increases by 25
 (C) increases by 5 (D) is multiplied by 5

2 Easy 0.75 min Basic

M13-T2

A and B are independent with $P(A) = \frac{1}{3}$ and $P(B) = \frac{1}{4}$. Then $P(A \cup B)$ equals:

- (A) $\frac{1}{2}$ (B) $\frac{7}{12}$ (C) $\frac{1}{12}$ (D) $\frac{5}{12}$

3 Easy 0.5 min Speed

M13-T3

One card is drawn from a pack of 52. The probability that it is a king or a heart is:

- (A) $\frac{17}{52}$ (B) $\frac{4}{13}$ (C) $\frac{1}{4}$ (D) $\frac{1}{52}$

4 Easy 0.5 min Speed

M13-T4

The mean of 5 observations is 10. If the observation 20 is removed, the new mean is:

- (A) 10 (B) 8 (C) 7.5 (D) 6

5 Easy 0.5 min NV

M13-T5

For 10 observations, $\sum x = 50$ and $\sum x^2 = 300$. Find the variance.

Your answer (numerical value):

Answers & Solutions

M13 · Practice

- 1 8 2 (C) 3 (D) 4 2 5 24 6 44 7 (A) 8 (B) 9 (B) 10 (A)
11 (C) 12 (B) 13 (D)

M13 · Chapter Test

- 1 (A) 2 (A) 3 (B) 4 (C) 5 5

M13 · PRACTICE

1 Answer: 8 Variance Easy 0.75 min NV

M13-01

SOLUTION The mean is 6. The squared deviations are $16 + 4 + 0 + 4 + 16 = 40$, so the variance is $40/5 = 8$.

2 Answer: (C) Change of scale Easy 0.5 min Speed

M13-02

SOLUTION $\sigma_y = |2|\sigma_x$; the +5 has no effect.

3 Answer: (D) Variance of natural numbers Easy 0.5 min Speed

M13-03

SOLUTION $\frac{n^2 - 1}{12} = \frac{99}{12} = 8.25$.

4 Answer: 2 Mean deviation Easy 0.5 min NV

M13-04

SOLUTION The mean is 6. $\frac{3 + 1 + 1 + 3}{4} = 2$.

5 Answer: 24 Grouped mode Medium 1 min NV

M13-05

SOLUTION $20 + \frac{12 - 8}{24 - 8 - 6} \times 10 = 20 + 4 = 24$.

6 Answer: 44 Combined mean Easy 0.5 min NV

M13-06

SOLUTION $\frac{1200 + 1000}{50} = 44$.

COMMON TRAP Averaging the two means (45).

7 Answer: (A) Corrected observation Medium 1.5 min JEE

M13-07

SOLUTION Before: $\sum x = 20$ and $\sum x^2 = 5(2 + 4^2) = 90$. After the correction: $\sum x = 25$ and $\sum x^2 = 90 - 25 + 100 = 165$. New mean = 5; new variance = $\frac{165}{5} - 25 = 8$.

SHORTCUT Work with the two totals, never with the variance directly.

COMMON TRAP Correcting only the mean and keeping the variance at 2.

8 Answer: (B) Independent events Medium 1 min Concept

M13-08

SOLUTION $P(A)P(B) = 0.2 = P(A \cap B)$. Also $P(A \cup B) = 0.7$ and $P(A | B) = 0.5$.

9 Answer: (B) Bayes' theorem Medium 1.5 min JEE

M13-09

SOLUTION $\frac{0.5(0.02)}{0.5(0.02) + 0.3(0.03) + 0.2(0.04)} = \frac{0.010}{0.027}$.

SHORTCUT Per 1000 items: 10, 9, 8 defectives.

10 Answer: (A) Mean of a random variable **Medium** ⌚ 1 min Basic

M13-10

SOLUTION $6k = 1 \Rightarrow k = \frac{1}{6}$. $E(X) = 0 + \frac{2}{6} + \frac{6}{6} = \frac{4}{3}$.

11 Answer: (C) Variance of a random variable **Medium** ⌚ 1.5 min JEE

M13-11

SOLUTION $E(X^2) = \frac{2}{6} + \frac{12}{6} = \frac{7}{3}$. $\text{Var} = \frac{7}{3} - \frac{16}{9} = \frac{5}{9}$.

COMMON TRAP Giving $E(X^2)$ as the variance.

12 Answer: (B) At least one **Easy** ⌚ 0.5 min Speed

M13-12

SOLUTION $1 - \frac{1}{8}$.

13 Answer: (D) Conditional probability trap **Medium** ⌚ 1 min Tricky

M13-13

SOLUTION The reduced sample space is $\{BB, BG, GB\}$.

COMMON TRAP Answering $\frac{1}{2}$ by treating the known boy as a specific child.

M13 · CHAPTER TEST

1 Answer: (A) Change of origin **Easy** ⌚ 0.5 min Concept

M13-T1

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

2 Answer: (A) Union of independent events **Easy** ⌚ 0.75 min Basic

M13-T2

SOLUTION $\frac{1}{3} + \frac{1}{4} - \frac{1}{12} = \frac{6}{12}$.

SHORTCUT $1 - P(A')P(B') = 1 - \frac{2}{3} \cdot \frac{3}{4} = \frac{1}{2}$.

3 Answer: (B) Cards **Easy** ⌚ 0.5 min Speed

M13-T3

SOLUTION $\frac{4 + 13 - 1}{52} = \frac{16}{52}$.

COMMON TRAP Double-counting the king of hearts (17/52).

4 Answer: (C) Removing an observation **Easy** ⌚ 0.5 min Speed

M13-T4

SOLUTION $\frac{50 - 20}{4}$.

5 Answer: 5 Variance from sums **Easy** ⌚ 0.5 min NV

M13-T5

SOLUTION $30 - 25 = 5$.

REVISION Revision tracker · Statistics & Probability

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART IV · MATHEMATICS · UNIT 14

Trigonometry & Inverse Trigonometric Functions

14

~1 per shift

TYPICAL QUESTIONS

Easy–Medium

DIFFICULTY

High (also a tool
for calculus and
complex
numbers)

PRIORITY

Class 11 · Ch 3 ·
Class 12 · Ch 2

NCERT

~6 hours

STUDY TIME

Core concepts

- **Trigonometric functions** of any angle: signs by quadrant, allied-angle reduction, periodicity.
- **Identities:** compound, multiple and sub-multiple angles; sum $\leftrightarrow$ product conversions.
- **Inverse trigonometric functions:** principal-value branches, domains and ranges, and the standard properties.
- Trigonometry appears inside other chapters far more often than as its own question. Calculus, complex numbers (polar form) and conics (parametric form) all rely on it.

NOTE Syllabus note

The official Unit 14 text lists trigonometric identities, trigonometric functions, inverse trigonometric functions and their properties [NTA-SYL]. Heights and distances were removed in 2024. Trigonometric *equations* are not named separately, but general solutions are included below because identity and inverse-trig questions often reduce to solving one  **RECOMMENDATION**.

Important formulas: identities

Σ FORMULA Compound & multiple angles

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B \quad \cos(A \pm B) = \cos A \cos B \mp \sin A \sin B \quad \tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A = 2 \sin A \cos A = \frac{2 \tan A}{1 + \tan^2 A} \quad \cos 2A = 1 - 2 \sin^2 A = 2 \cos^2 A - 1 = \frac{1 - \tan^2 A}{1 + \tan^2 A}$$

$$\sin 3A = 3 \sin A - 4 \sin^3 A \quad \cos 3A = 4 \cos^3 A - 3 \cos A \quad \tan 3A = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A}$$

Σ FORMULA Sum $\leftrightarrow$ product

$$\sin C + \sin D = 2 \sin \frac{C+D}{2} \cos \frac{C-D}{2} \quad \cos C + \cos D = 2 \cos \frac{C+D}{2} \cos \frac{C-D}{2}$$

$$\sin C - \sin D = 2 \cos \frac{C+D}{2} \sin \frac{C-D}{2} \quad \cos C - \cos D = -2 \sin \frac{C+D}{2} \sin \frac{C-D}{2}$$

$$2 \sin A \cos B = \sin(A+B) + \sin(A-B); 2 \cos A \cos B = \cos(A+B) + \cos(A-B); 2 \sin A \sin B = \cos(A-B) - \cos(A+B).$$

Σ FORMULA Standard values & special products

Angle	sin	cos	tan
15°	$\frac{\sqrt{6} - \sqrt{2}}{4}$	$\frac{\sqrt{6} + \sqrt{2}}{4}$	$2 - \sqrt{3}$
18°	$\frac{\sqrt{5} - 1}{4}$	$\frac{\sqrt{10 + 2\sqrt{5}}}{4}$	—
36°	$\frac{\sqrt{10 - 2\sqrt{5}}}{4}$	$\frac{\sqrt{5} + 1}{4}$	—
22.5°	$\frac{\sqrt{2 - \sqrt{2}}}{2}$	$\frac{\sqrt{2 + \sqrt{2}}}{2}$	$\sqrt{2} - 1$

$\sin \theta \sin(60^\circ - \theta) \sin(60^\circ + \theta) = \frac{1}{4} \sin 3\theta$; $\cos \theta \cos(60^\circ - \theta) \cos(60^\circ + \theta) = \frac{1}{4} \cos 3\theta$; $\tan \theta \tan(60^\circ - \theta) \tan(60^\circ + \theta) = \tan 3\theta$.

$$\cos A \cos 2A \cos 4A \cdots \cos 2^{n-1} A = \frac{\sin 2^n A}{2^n \sin A}.$$

Σ FORMULA Range & conditional identities

$$a \sin x + b \cos x \in [-\sqrt{a^2 + b^2}, \sqrt{a^2 + b^2}].$$

If $A + B + C = \pi$: $\tan A + \tan B + \tan C = \tan A \tan B \tan C$ and $\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$.

General solutions: $\sin \theta = \sin \alpha \Rightarrow \theta = n\pi + (-1)^n \alpha$; $\cos \theta = \cos \alpha \Rightarrow \theta = 2n\pi \pm \alpha$; $\tan \theta = \tan \alpha \Rightarrow \theta = n\pi + \alpha$ ($n \in \mathbb{Z}$).

Important formulas: inverse trigonometric functions**Σ FORMULA** Principal branches

Function	Domain	Range (principal)
$\sin^{-1} x$	$[-1, 1]$	$[-\pi/2, \pi/2]$
$\cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1} x$	$\mathbb{R}$	$(-\pi/2, \pi/2)$
$\cot^{-1} x$	$\mathbb{R}$	$(0, \pi)$
$\sec^{-1} x$	$ x \geq 1$	$[0, \pi] - \{\pi/2\}$
$\csc^{-1} x$	$ x \geq 1$	$[-\pi/2, \pi/2] - \{0\}$

Σ FORMULA Properties

$$\sin^{-1} x + \cos^{-1} x = \tan^{-1} x + \cot^{-1} x = \sec^{-1} x + \csc^{-1} x = \frac{\pi}{2}.$$

Negative arguments: $\sin^{-1}(-x) = -\sin^{-1} x$ and $\tan^{-1}(-x) = -\tan^{-1} x$, but $\cos^{-1}(-x) = \pi - \cos^{-1} x$ and $\cot^{-1}(-x) = \pi - \cot^{-1} x$.

$$\tan^{-1} x + \tan^{-1} y = \tan^{-1} \frac{x+y}{1-xy} \quad (xy < 1) \quad \tan^{-1} x - \tan^{-1} y = \tan^{-1} \frac{x-y}{1+xy} \quad (xy > -1)$$

For $x, y > 0$ and $xy > 1$, add π to the sum formula.

$$2 \tan^{-1} x = \sin^{-1} \frac{2x}{1+x^2} \quad (|x| \leq 1) = \cos^{-1} \frac{1-x^2}{1+x^2} \quad (x \geq 0) = \tan^{-1} \frac{2x}{1-x^2} \quad (|x| < 1).$$

Compositions: $\sin^{-1}(\sin x) = x$ only for $x \in [-\pi/2, \pi/2]$. For $x \in [\pi/2, 3\pi/2]$ it equals $\pi - x$. Always reduce into the principal range first.

Common question models

PYQ PATTERN Recurring structures

1. **Evaluate** expressions like $\sin^{-1}(\sin \frac{5\pi}{6})$ or $\cos^{-1}(\cos \frac{7\pi}{6})$ (principal-range traps).
2. **Sums of $\tan^{-1}$ and telescoping series** $\sum \tan^{-1} \frac{1}{1+k+k^2} = \sum [\tan^{-1}(k+1) - \tan^{-1} k]$.
3. **Maximum/minimum** of $a \sin x + b \cos x + c$, or via AM–GM.
4. **Special products** ($\cos 20^\circ \cos 40^\circ \cos 80^\circ$) and standard values.
5. **Number of solutions** of an equation in an interval (graph both sides).
6. **Domain questions** involving inverse trig functions (often inside Sets & Functions).

Shortcuts & fast methods

SHORTCUT Telescoping $\tan^{-1}$

Write the general term as $\tan^{-1} \frac{b-a}{1+ab} = \tan^{-1} b - \tan^{-1} a$. For $\frac{1}{1+k(k+1)}$, take $a = k$ and $b = k+1$. The sum up to n collapses to $\tan^{-1}(n+1) - \tan^{-1} 1$.

Time saved: 3–4 minutes compared with adding terms.

SHORTCUT Substitute a value

For identity-type MCQs, put $\theta = 0, \frac{\pi}{4}$ or $\frac{\pi}{6}$ in both the expression and each option. Two values usually leave one option (Answer-Choice Strategy, page 396).

Common mistakes

MISTAKE ALERT Mistake alerts

- $\cos^{-1}(-\frac{1}{2}) = \frac{2\pi}{3}$, not $-\frac{\pi}{3}$: the range of $\cos^{-1}$ is $[0, \pi]$.
- $\sin^{-1}(\sin x) = x$ only in the principal range.
- Applying the $\tan^{-1}$ sum formula without checking $xy < 1$.
- $\sin^{-1} x \neq \frac{1}{\sin x}$. The notation means the inverse function.
- Squaring an equation introduces extra roots. Check each solution in the original.

Practice questions

M14 · Practice

1

Easy

 0.5 min

NV

M14-01

Find the maximum value of $3 \sin x + 4 \cos x$.

Your answer (numerical value):

2

Easy

 0.5 min

Speed

M14-02

$\sin 15^\circ$ equals:

(A) $\frac{\sqrt{6} + \sqrt{2}}{4}$

(B) $\frac{\sqrt{6} - \sqrt{2}}{4}$

(C) $\frac{\sqrt{3} - 1}{2}$

(D) $\frac{2 - \sqrt{3}}{4}$

3 Easy 0.5 min Speed

M14-03

 $\tan 75^\circ$ equals:

- (A) $2 - \sqrt{3}$ (B) $\sqrt{3} + 1$ (C) $2 + \sqrt{3}$ (D) $\frac{\sqrt{3} + 1}{2}$

4 Easy 0.75 min Tricky

M14-04

 $\sin^{-1}\left(\sin \frac{2\pi}{3}\right)$ equals:

- (A) $\frac{2\pi}{3}$ (B) $\frac{\pi}{6}$ (C) $-\frac{\pi}{3}$ (D) $\frac{\pi}{3}$

5 Easy 0.75 min Basic

M14-05

 $\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3}$ equals:

- (A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{2}$

6 Medium 1 min JEE

M14-06

 $\cos 20^\circ \cos 40^\circ \cos 80^\circ$ equals:

- (A) $\frac{1}{16}$ (B) $\frac{1}{4}$ (C) $\frac{\sqrt{3}}{8}$ (D) $\frac{1}{8}$

7 Easy 0.5 min NV

M14-07

How many solutions does $\sin x = \frac{1}{2}$ have in $[0, 2\pi]$?Your answer (numerical value): **8** Medium 1 min Basic

M14-08

If $\sin \theta + \cos \theta = \frac{1}{5}$, then $\sin 2\theta$ equals:

- (A) $\frac{1}{25}$ (B) $\frac{24}{25}$ (C) $-\frac{12}{25}$ (D) $-\frac{24}{25}$

9 Medium 1 min NV

M14-09

Find the minimum value of $9 \tan^2 \theta + 4 \cot^2 \theta$.Your answer (numerical value): **10** Easy 0.5 min Tricky

M14-10

 $\cos^{-1}\left(-\frac{1}{2}\right)$ equals:

- (A) $-\frac{\pi}{3}$ (B) $\frac{2\pi}{3}$ (C) $\frac{\pi}{3}$ (D) $\frac{4\pi}{3}$

11 Easy 0.5 min Speed

M14-11

If $\sin^{-1} x = \frac{\pi}{5}$, then $\cos^{-1} x$ equals:

- (A) $\frac{\pi}{5}$ (B) $\frac{4\pi}{5}$ (C) $\frac{3\pi}{10}$ (D) $\frac{7\pi}{10}$

12 Medium 1.5 min Tricky

M14-12

The range of $f(x) = \sin^{-1} x + \cos^{-1} x + \tan^{-1} x$ is:

- (A) $\left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$ (B) $[0, \pi]$ (C) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (D) $\left[\frac{\pi}{2}, \pi\right]$

13 Medium 1 min Concept

M14-13

 $2 \tan^{-1} \frac{1}{3}$ equals:

- (A) $\tan^{-1} \frac{4}{3}$ (B) $\tan^{-1} \frac{2}{3}$ (C) $\tan^{-1} \frac{3}{4}$ (D) $\tan^{-1} \frac{1}{9}$

M14 · Chapter Test

1 Easy 0.5 min Speed

M14-T1

 $\cos 2\theta$ equals:

- (A) $\frac{2 \tan \theta}{1 - \tan^2 \theta}$ (B) $\frac{2 \tan \theta}{1 + \tan^2 \theta}$ (C) $\frac{1 + \tan^2 \theta}{1 - \tan^2 \theta}$ (D) $\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$

2 Medium 0.75 min Tricky

M14-T2

 $\tan^{-1} \sqrt{3} - \sec^{-1}(-2)$ equals:

- (A) π (B) $-\frac{\pi}{3}$ (C) $\frac{\pi}{3}$ (D) 0

3 Easy 0.5 min Speed

M14-T3

The maximum value of $\sin x \cos x$ is:

- (A) $\frac{1}{2}$ (B) 1 (C) $\frac{1}{\sqrt{2}}$ (D) 2

4 Easy 0.5 min Speed

M14-T4

 $\sin 18^\circ$ equals:

- (A) $\frac{\sqrt{5} + 1}{4}$ (B) $\frac{\sqrt{5} - 1}{4}$ (C) $\frac{\sqrt{5} - 1}{2}$ (D) $\frac{\sqrt{3} - 1}{4}$

5 Medium 0.75 min NV

M14-T5

Find $\tan 1^\circ \tan 2^\circ \tan 3^\circ \cdots \tan 89^\circ$.

Your answer (numerical value):

Answers & Solutions

M14 · Practice

- 1 5 2 (B) 3 (C) 4 (D) 5 (A) 6 (D) 7 2 8 (D) 9 12 10 (B)
11 (C) 12 (A) 13 (C)

M14 · Chapter Test

- 1 (D) 2 (B) 3 (A) 4 (B) 5 1

M14 · PRACTICE

1 Answer: 5 Range of $a \sin x + b \cos x$ Easy 0.5 min NV

M14-01

SOLUTION $\sqrt{9+16} = 5$.

2 Answer: (B) Standard value **Easy** ⌚ 0.5 min **Speed**

M14-02

SOLUTION $\sin(45^\circ - 30^\circ) = \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \cdot \frac{1}{2}.$

SHORTCUT $\sin 15^\circ \approx 0.26$; only (B) is close (≈ 0.259).

3 Answer: (C) Compound angle **Easy** ⌚ 0.5 min **Speed**

M14-03

SOLUTION $\tan(45^\circ + 30^\circ) = \frac{1 + 1/\sqrt{3}}{1 - 1/\sqrt{3}} = \frac{\sqrt{3} + 1}{\sqrt{3} - 1} = 2 + \sqrt{3}.$

4 Answer: (D) Principal value **Easy** ⌚ 0.75 min **Tricky**

M14-04

SOLUTION $\sin \frac{2\pi}{3} = \sin \frac{\pi}{3}$, and $\frac{\pi}{3}$ lies in $[-\frac{\pi}{2}, \frac{\pi}{2}]$.

COMMON TRAP Cancelling $\sin^{-1}$ and $\sin$ directly.

5 Answer: (A) Sum of inverse tangents **Easy** ⌚ 0.75 min **Basic**

M14-05

SOLUTION $\frac{1/2 + 1/3}{1 - 1/6} = 1.$

6 Answer: (D) Special product **Medium** ⌚ 1 min **JEE**

M14-06

SOLUTION With $\theta = 20^\circ$: $\cos \theta \cos(60^\circ - \theta) \cos(60^\circ + \theta) = \frac{1}{4} \cos 60^\circ = \frac{1}{8}.$

7 Answer: 2 Counting solutions **Easy** ⌚ 0.5 min **NV**

M14-07

SOLUTION $x = \frac{\pi}{6}$ and $\frac{5\pi}{6}.$

8 Answer: (D) Squaring an identity **Medium** ⌚ 1 min **Basic**

M14-08

SOLUTION Squaring: $1 + \sin 2\theta = \frac{1}{25}.$

9 Answer: 12 AM-GM minimum **Medium** ⌚ 1 min **NV**

M14-09

SOLUTION AM-GM: $\geq 2\sqrt{36} = 12$, attained when $\tan^2 \theta = \frac{2}{3}.$

10 Answer: (B) Principal value of $\cos^{-1}$ **Easy** ⌚ 0.5 min **Tricky**

M14-10

SOLUTION $\pi - \cos^{-1} \frac{1}{2} = \pi - \frac{\pi}{3}.$

COMMON TRAP Answering $-\frac{\pi}{3}$, which lies outside $[0, \pi]$.

11 Answer: (C) Complementary inverses **Easy** ⌚ 0.5 min **Speed**

M14-11

SOLUTION $\frac{\pi}{2} - \frac{\pi}{5} = \frac{3\pi}{10}.$

12 Answer: (A) Range of a combination **Medium** ⌚ 1.5 min **Tricky**

M14-12

SOLUTION $f(x) = \frac{\pi}{2} + \tan^{-1} x$ with $x \in [-1, 1]$ (the domain of $\sin^{-1}$), so $\tan^{-1} x \in [-\frac{\pi}{4}, \frac{\pi}{4}]$.

COMMON TRAP Using the full range of $\tan^{-1}$ and ignoring the domain restriction.

13 Answer: (C) Double-angle inverse **Medium** ⌚ 1 min **Concept**

M14-13

SOLUTION $\tan^{-1} \frac{2/3}{1 - 1/9} = \tan^{-1} \frac{2/3}{8/9} = \tan^{-1} \frac{3}{4}.$

M14 · CHAPTER TEST**1 Answer: (D)** Double angle in tan **Easy** ⌚ 0.5 min **Speed**

M14-T1

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

2 Answer: (B) Principal values **Medium** ⌚ 0.75 min **Tricky**

M14-T2

SOLUTION $\frac{\pi}{3} - \frac{2\pi}{3}$ (the range of $\sec^{-1}$ is $[0, \pi] - \{\frac{\pi}{2}\}$).

COMMON TRAP Taking $\sec^{-1}(-2) = -\frac{\pi}{3}.$

3 Answer: (A) Maximum of a product **Easy** ⌚ 0.5 min **Speed**

M14-T3

SOLUTION $\frac{1}{2} \sin 2x \leq \frac{1}{2}.$

4 Answer: (B) *Standard value* **Easy** ⌚ 0.5 min **Speed**

M14-T4

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

SHORTCUT $\sin 18^\circ \approx 0.309$ and $(\sqrt{5} - 1)/4 \approx 0.309$.

COMMON TRAP Confusing it with $\cos 36^\circ = (\sqrt{5} + 1)/4$.

5 Answer: 1 *Product of tangents* **Medium** ⌚ 0.75 min **NV**

M14-T5

SOLUTION Pair $\tan \theta \tan(90^\circ - \theta) = 1$. The middle term is $\tan 45^\circ = 1$.

 REVISION Revision tracker · Trigonometry

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART V



Master Formula Handbook

Every formula, constant, identity, reaction and trend, organised for 1-day, 3-day, 7-day and final 24-hour revision.

• How to Use the Formula Handbook	367
• Physics Formula Sheets	368
• Chemistry Formula & Fact Sheets	375
• Mathematics Formula Sheets	380

• Constants, Units & Frequently Forgotten Formulas	383
• Formula Revision Plans: 1-Day, 3-Day, 7-Day & Final 24 Hours	386

PART V · MASTER FORMULA HANDBOOK

How to Use the Formula Handbook

This part compresses the formula content of all 54 modules into sheets you can revise in minutes. Nothing here is new: every line comes from the chapter modules, where the derivations, conditions and examples are. Use it for revision, not for first learning.

Physics sheets · page [368](#)

All 20 units, with sign conventions and the conditions each formula needs.

Chemistry sheets · page [375](#)

Physical formulas, inorganic trends and exceptions, organic reagents and named reactions.

Maths sheets · page [380](#)

All 14 units: identities, standard results, integrals and conic parameters.

Constants & frequently forgotten formulas · page [383](#)

Physical constants (CODATA values), unit conversions, and the formulas students forget most often.

Formula revision plans · page [386](#)

1-Day, 3-Day and 7-Day cycles and the Final 24-Hour plan, with checkboxes.

The three-pass method RECOMMENDATION

1 Pass 1 · Recall

Cover the right column of a sheet and write each formula from memory on a rough sheet. Retrieval practice is one of the best-supported learning techniques [LS-DUN], [LS-RK].

2 Pass 2 · Check & mark

Uncover and compare. Put a dot next to every formula you got wrong or hesitated on.

3 Pass 3 · Repeat the dots

Re-test only the dotted formulas the next day, then after 3 and 7 days (spaced revision, page [449](#)). A formula with three dots goes into the Frequently Forgotten list.

** IMPORTANT Conditions are part of the formula**

Most formula errors in JEE come from using a correct formula outside its conditions: $v = u + at$ only for constant acceleration; S_∞ only for $|r| < 1$; $\Delta G^\circ = -RT \ln K$ uses the standard-state K . The sheets list the condition next to each formula where it matters. Revise the condition along with the formula.

** TIP Printing**

The sheets are designed for A4 print at 100% scale. For a pocket version, print two pages per sheet (A5 each). Text stays readable down to about 70%.

PART V • MASTER FORMULA HANDBOOK

Physics Formula Sheets

Each sheet links to its full module. The symbols follow NCERT usage.

Σ FORMULA P01 • Units & Measurements • page 36

Result	Formula / rule
Relative error of $Q = \frac{a^p b^q}{c^r}$	$\frac{\Delta Q}{Q} = p \frac{\Delta a}{a} + q \frac{\Delta b}{b} + r \frac{\Delta c}{c}$ (errors always add)
Sum or difference $Q = a \pm b$	$\Delta Q = \Delta a + \Delta b$ (absolute errors add)
Least count, vernier	$LC = 1 \text{ MSD} - 1 \text{ VSD} = \frac{1 \text{ MSD}}{n}$
Least count, screw gauge	$LC = \frac{\text{pitch}}{\text{number of circular divisions}}$
Useful dimensions	$[G] = \text{M}^{-1} \text{L}^3 \text{T}^{-2} \cdot [h] = \text{ML}^2 \text{T}^{-1} \cdot [\epsilon_0] = \text{M}^{-1} \text{L}^{-3} \text{T}^4 \text{A}^2 \cdot [\mu_0] = \text{MLT}^{-2} \text{A}^{-2}$
Significant figures	$\times / \div$: keep the fewest significant figures $\cdot + / -$: keep the fewest decimal places

Σ FORMULA P02 • Kinematics • page 43

Result	Formula	Condition
Equations of motion	$v = u + at \cdot s = ut + \frac{1}{2}at^2 \cdot v^2 = u^2 + 2as$	constant a
Distance in the n th second	$s_n = u + \frac{a}{2}(2n - 1)$	constant a
Projectile	$T = \frac{2u \sin \theta}{g} \cdot H = \frac{u^2 \sin^2 \theta}{2g} \cdot R = \frac{u^2 \sin 2\theta}{g}$	level ground
Trajectory	$y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta} = x \tan \theta \left(1 - \frac{x}{R}\right)$	—
Relative velocity; river	$\vec{v}_{AB} = \vec{v}_A - \vec{v}_B \cdot \text{shortest time } t = d/v_{br} \cdot \text{zero drift if } \sin \theta = v_r/v_{br}$	—
Circular motion	$a_c = \frac{v^2}{r} = \omega^2 r \cdot a_t = \frac{dv}{dt} \cdot a = \sqrt{a_c^2 + a_t^2}$	—

Σ FORMULA P03 • Laws of Motion • page 50

Result	Formula
Newton's second law	$\vec{F} = \frac{d\vec{p}}{dt} = m\vec{a}$ (constant mass) $\cdot \text{impulse} = \Delta p = \int F dt$
Friction	$f_s \leq \mu_s N \cdot f_k = \mu_k N \cdot \text{angle of repose } \tan \theta = \mu_s$
Incline (rough)	block slides if $\tan \theta > \mu_s \cdot a = g(\sin \theta - \mu_k \cos \theta)$
Atwood machine	$a = \frac{(m_1 - m_2)g}{m_1 + m_2} \cdot T = \frac{2m_1 m_2 g}{m_1 + m_2}$
Banking	$\tan \theta = \frac{v^2}{rg}$ (no friction) $\cdot v_{max} = \sqrt{rg \frac{\mu + \tan \theta}{1 - \mu \tan \theta}}$
Flat curve	$v_{max} = \sqrt{\mu rg}$
Vertical circle (string)	$v_{bottom} \geq \sqrt{5gl} \cdot v_{top} \geq \sqrt{gl} \cdot T_{bottom} - T_{top} = 6mg$

Σ FORMULA P04 · Work, Energy & Power · page 57

Result	Formula
Work	$W = \int \vec{F} \cdot d\vec{s}$ · work-energy theorem $W_{net} = \Delta K$
Spring	$U = \frac{1}{2}kx^2$ · springs in series $\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2}$ · parallel $k = k_1 + k_2$
Conservative force	$F = -\frac{dU}{dx}$ · stable equilibrium where $\frac{d^2U}{dx^2} > 0$
Power	$P = \frac{dW}{dt} = \vec{F} \cdot \vec{v}$
Collisions (1D)	$e = \frac{\text{separation speed}}{\text{approach speed}}$ · elastic, equal masses: velocities exchange
Elastic collision, m_2 at rest	$v_1 = \frac{m_1 - m_2}{m_1 + m_2}u_1$ · $v_2 = \frac{2m_1}{m_1 + m_2}u_1$
KE-momentum	$K = \frac{p^2}{2m}$

Σ FORMULA P05 · Rotational Motion & Centre of Mass · page 63

Result	Formula
Centre of mass	$\vec{r}_{cm} = \frac{\sum m_i \vec{r}_i}{\sum m_i}$ · semicircular ring $\frac{2R}{\pi}$ · semicircular disc $\frac{4R}{3\pi}$ · hemisphere (solid) $\frac{3R}{8}$ · cone $\frac{h}{4}$ from base
Moments of inertia	ring MR^2 · disc $\frac{1}{2}MR^2$ · solid sphere $\frac{2}{5}MR^2$ · hollow sphere $\frac{2}{3}MR^2$ · rod (centre) $\frac{1}{12}ML^2$ · rod (end) $\frac{1}{3}ML^2$
Axis theorems	parallel $I = I_{cm} + Md^2$ · perpendicular $I_z = I_x + I_y$ (planar bodies)
Rotation	$\tau = I\alpha$ · $L = I\omega$ · $K = \frac{1}{2}I\omega^2$ · $\tau = \frac{dL}{dt}$ (L conserved if $\tau_{ext} = 0$)
Rolling without slipping	$v = \omega R$ · $K = \frac{1}{2}mv^2 \left(1 + \frac{k^2}{R^2}\right)$ · on an incline $a = \frac{g \sin \theta}{1 + k^2/R^2}$
Radius of gyration	$I = Mk^2$

Σ FORMULA P06 · Gravitation · page 70

Result	Formula
Field & potential	$g = \frac{GM}{R^2}$ · $V = -\frac{GM}{r}$ · $U = -\frac{GMm}{r}$
Variation of g	height $g_h = \frac{g}{(1 + h/R)^2} \approx g \left(1 - \frac{2h}{R}\right)$ · depth $g_d = g \left(1 - \frac{d}{R}\right)$
Orbits	$v_o = \sqrt{\frac{GM}{r}}$ · $T^2 = \frac{4\pi^2}{GM}r^3$ · $E = -\frac{GMm}{2r}$ · $K = -E$, $U = 2E$
Escape speed	$v_e = \sqrt{\frac{2GM}{R}} = \sqrt{2gR} = \sqrt{2}v_o$ (surface orbit) · Earth: ≈ 11.2 km/s
Kepler II	areal velocity $\frac{dA}{dt} = \frac{L}{2m} = \text{constant}$
Inside a uniform sphere	$g = \frac{GMr}{R^3}$ · shell: $g = 0$ inside

Σ FORMULA P07 · Properties of Solids & Liquids · page 76

Result	Formula
Elasticity	$Y = \frac{FL}{A\Delta L}$ · $B = -\frac{\Delta P}{\Delta V/V}$ · energy density = $\frac{1}{2}$ stress $\times$ strain
Fluids	$P = P_0 + \rho gh$ · continuity $A_1v_1 = A_2v_2$ · Bernoulli $P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$ · Torricelli $v = \sqrt{2gh}$
Viscosity	$F = \eta A \frac{dv}{dx}$ · Stokes $F = 6\pi\eta rv$ · terminal $v_t = \frac{2r^2(\rho - \sigma)g}{9\eta}$
Surface tension	excess pressure: drop $\frac{2T}{r}$, bubble $\frac{4T}{r}$ · capillary rise $h = \frac{2T \cos \theta}{r\rho g}$
Thermal expansion	$\Delta L = L\alpha\Delta T$ · $\beta = 2\alpha$ · $\gamma = 3\alpha$ · thermal stress = $Y\alpha\Delta T$
Heat	$Q = mc\Delta T$ · $Q = mL$ · conduction $\frac{dQ}{dt} = \frac{kA\Delta T}{L}$ · $R_{th} = \frac{L}{kA}$
Radiation & cooling	Stefan $P = \sigma eAT^4$ · Wien $\lambda_m T = b$ · Newton $\frac{dT}{dt} = -k(T - T_0)$

Σ FORMULA P08 · Thermodynamics · page 84

Process	Work by gas W	Key relation
First law	—	$\Delta Q = \Delta U + W$ · $\Delta U = nC_V\Delta T$ always (ideal gas)
Isothermal	$nRT \ln \frac{V_2}{V_1}$	$PV = \text{const}$ · $\Delta U = 0$
Adiabatic	$\frac{P_1V_1 - P_2V_2}{\gamma - 1}$	$PV^\gamma = \text{const}$ · $TV^{\gamma-1} = \text{const}$ · $Q = 0$
Isobaric	$P\Delta V = nR\Delta T$	$Q = nC_P\Delta T$
Isochoric	0	$Q = nC_V\Delta T$
Cyclic	area enclosed	$\Delta U = 0$
Carnot efficiency $\eta = 1 - \frac{T_2}{T_1}$ · refrigerator COP = $\frac{T_2}{T_1 - T_2}$ · $C_P - C_V = R$ · slope: adiabatic = $\gamma \times$ isothermal.		

Σ FORMULA P09 · Kinetic Theory of Gases · page 90

Result	Formula
Pressure	$P = \frac{1}{3}\rho v_{rms}^2 = \frac{1}{3}\frac{mN}{V}v_{rms}^2$
Speeds	$v_{rms} = \sqrt{\frac{3RT}{M}}$ · $v_{avg} = \sqrt{\frac{8RT}{\pi M}}$ · $v_{mp} = \sqrt{\frac{2RT}{M}}$ ($v_{rms} > v_{avg} > v_{mp}$)
Energy	average KE per molecule = $\frac{3}{2}k_B T$ · equipartition $\frac{1}{2}k_B T$ per degree of freedom
Specific heats	$C_V = \frac{f}{2}R$ · $\gamma = 1 + \frac{2}{f}$ · monatomic $f = 3$ ($\gamma = \frac{5}{3}$) · diatomic $f = 5$ ($\gamma = \frac{7}{5}$)
Mixture	$C_{V,mix} = \frac{n_1C_{V1} + n_2C_{V2}}{n_1 + n_2}$
Mean free path	$\lambda = \frac{1}{\sqrt{2}\pi d^2 n}$

Σ FORMULA P10 • Oscillations & Waves • page 95

Result	Formula
SHM	$x = A \sin(\omega t + \phi) \cdot v = \omega \sqrt{A^2 - x^2} \cdot a = -\omega^2 x \cdot E = \frac{1}{2} m \omega^2 A^2$
Periods	spring $T = 2\pi \sqrt{\frac{m}{k}}$ • pendulum $T = 2\pi \sqrt{\frac{l}{g}}$ • in a lift $g_{eff} = g \pm a$
Wave speed	string $v = \sqrt{\frac{T}{\mu}}$ • sound $v = \sqrt{\frac{\gamma P}{\rho}}$ • $v = f \lambda$
Strings and pipes	string / open pipe $f_n = \frac{nv}{2L}$ (all harmonics) • closed pipe $f_n = \frac{nv}{4L}$, n odd
Beats & Doppler	$f_{beat} = f_1 - f_2 \cdot f' = f \frac{v \pm v_o}{v \mp v_s}$ (upper signs: approaching)
Travelling wave	$y = A \sin(kx - \omega t) \cdot v = \frac{\omega}{k} \cdot \text{intensity} \propto A^2 \omega^2$

Σ FORMULA P11 • Electrostatics & Capacitors • page 101

Result	Formula
Coulomb, field, potential	$F = \frac{kq_1q_2}{r^2} \cdot E = \frac{kq}{r^2} \cdot V = \frac{kq}{r} \cdot E = -\frac{dV}{dr} \cdot k = \frac{1}{4\pi\epsilon_0}$
Dipole	axial $E = \frac{2kp}{r^3}$ • equatorial $\frac{kp}{r^3}$ • $\tau = \vec{p} \times \vec{E} \cdot U = -\vec{p} \cdot \vec{E}$
Gauss's law	$\oint \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\epsilon_0}$ • line $\frac{\lambda}{2\pi\epsilon_0 r}$ • sheet $\frac{\sigma}{2\epsilon_0}$ • conductor surface $\frac{\sigma}{\epsilon_0}$
Capacitors	$C = \frac{\epsilon_0 A}{d}$ • with dielectric K • slab: $C = \frac{\epsilon_0 A}{d - t + t/K}$ • sphere $4\pi\epsilon_0 R$
Energy	$U = \frac{1}{2} CV^2 = \frac{Q^2}{2C}$ • energy density $\frac{1}{2} \epsilon_0 E^2$
Combinations	series $\frac{1}{C} = \sum \frac{1}{C_i}$ • parallel $C = \sum C_i$ • sharing: common $V = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$

Σ FORMULA P12 • Current Electricity • page 108

Result	Formula
Current & drift	$I = neAv_d \cdot v_d = \frac{eE\tau}{m} \cdot \rho = \frac{m}{ne^2\tau} \cdot R = \frac{\rho l}{A}$
Stretched wire	length $\times n$ at constant volume: $R \rightarrow n^2 R$
Temperature	$R_T = R_0(1 + \alpha \Delta T)$
Cell	$V = \epsilon - Ir \cdot P_{max}$ to load when $R = r$ • cells in series $\epsilon_{eq} = \sum \epsilon, r_{eq} = \sum r$
Cells in parallel	$\epsilon_{eq} = \frac{\sum \epsilon_i / r_i}{\sum 1/r_i}$
Bridges & meters	Wheatstone balance $\frac{P}{Q} = \frac{R}{S}$ • meter bridge $\frac{R}{S} = \frac{l}{100-l}$ • potentiometer $\frac{\epsilon_1}{\epsilon_2} = \frac{l_1}{l_2} \cdot r = R \left(\frac{l_1}{l_2} - 1 \right)$
Power	$P = VI = I^2 R = \frac{V^2}{R}$

Σ FORMULA P13 · Magnetic Effects & Magnetism · page 114

Result	Formula
Fields	long wire $\frac{\mu_0 I}{2\pi r}$ · loop centre $\frac{\mu_0 I}{2R}$ · axis $\frac{\mu_0 I R^2}{2(R^2 + x^2)^{3/2}}$ · solenoid $\mu_0 n I$ · arc $\frac{\mu_0 I \theta}{4\pi R}$
Forces	$\vec{F} = q\vec{v} \times \vec{B}$ · $\vec{F} = I\vec{L} \times \vec{B}$ · parallel wires $\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d}$
Charge in B	$r = \frac{mv}{qB}$ · $T = \frac{2\pi m}{qB}$ (independent of v) · pitch $= v_{\parallel} T$
Loop	$\vec{m} = NI\vec{A}$ · $\tau = \vec{m} \times \vec{B}$ · $U = -\vec{m} \cdot \vec{B}$
Galvanometer conversions	ammeter: shunt $S = \frac{I_g G}{I - I_g}$ · voltmeter: series $R = \frac{V}{I_g} - G$
Magnetic materials	$B = \mu_0(H + M)$ · $\chi = \frac{M}{H}$ · $\mu_r = 1 + \chi$ · Curie $\chi \propto \frac{1}{T}$ (paramagnets)

Σ FORMULA P14 · EMI & Alternating Current · page 121

Result	Formula
Faraday & motional emf	$\varepsilon = -\frac{d\Phi}{dt}$ · rod $\varepsilon = Blv$ · rotating rod $\frac{1}{2}B\omega l^2$ · coil $NBA\omega \sin \omega t$
Inductance	$L = \mu_0 n^2 Al$ · $U = \frac{1}{2}LI^2$ · $M = \mu_0 n_1 n_2 Al$
Transients	LR: $I = I_0(1 - e^{-t/\tau})$, $\tau = \frac{L}{R}$ · RC: $q = CV(1 - e^{-t/RC})$
AC	$V_{rms} = \frac{V_0}{\sqrt{2}}$ · $X_L = \omega L$ · $X_C = \frac{1}{\omega C}$ · $Z = \sqrt{R^2 + (X_L - X_C)^2}$ · $\tan \phi = \frac{X_L - X_C}{R}$
Resonance	$\omega_0 = \frac{1}{\sqrt{LC}}$ · $Q = \frac{\omega_0 L}{R}$ · bandwidth $= \frac{R}{L}$
Power	$P = V_{rms} I_{rms} \cos \phi$ · power factor $\cos \phi = \frac{R}{Z}$
Transformer	$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$ (ideal)

Σ FORMULA P15 · Electromagnetic Waves · page 128

Result	Formula
Speed & ratio	$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \cdot \frac{E_0}{B_0} = c$ · in a medium $v = \frac{c}{\sqrt{\mu_r \varepsilon_r}}$
Displacement current	$I_d = \varepsilon_0 \frac{d\Phi_E}{dt}$
Energy	$u_{avg} = \frac{1}{2} \varepsilon_0 E_0^2$ · $I = u_{avg} c$ · radiation pressure $\frac{I}{c}$ (absorbed), $\frac{2I}{c}$ (reflected)
Spectrum (increasing λ)	γ · X-rays · UV · visible · IR · microwaves · radio

Σ FORMULA P16 · Optics: Ray & Wave · page 132

Result	Formula
Mirrors & lenses	$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ (mirror) · $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ (lens) · $m = -\frac{v}{u}$ (mirror), $\frac{v}{u}$ (lens)
Lens maker	$\frac{1}{f} = (n-1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$ · combination $P = \sum P_i$
Refraction	Snell $n_1 \sin i = n_2 \sin r$ · apparent depth $= \frac{\text{real}}{n}$ · $\sin C = \frac{1}{n}$ · curved surface $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$
Prism	$A = r_1 + r_2$ · $\delta = i + e - A$ · $n = \frac{\sin \frac{A+\delta_m}{2}}{\sin \frac{A}{2}}$ · thin prism $\delta = (n-1)A$
Instruments	microscope $m = \frac{L}{f_o} \frac{D}{f_e}$ · telescope $m = \frac{f_o}{f_e}$ (normal adjustment)
YDSE	$\beta = \frac{\lambda D}{d}$ · $I = 4I_0 \cos^2 \frac{\phi}{2}$ · path difference $\Delta x = \frac{yd}{D}$ · sheet shift $\frac{(\mu-1)tD}{d}$
Diffraction & polarisation	single slit minima $a \sin \theta = n\lambda$ · central width $\frac{2\lambda D}{a}$ · Malus $I = I_0 \cos^2 \theta$ · Brewster $\tan i_B = n$

Σ FORMULA P17 · Dual Nature · page 139

Result	Formula
Photon	$E = h\nu = \frac{hc}{\lambda}$ · $E(\text{eV}) \approx \frac{1240}{\lambda(\text{nm})}$ · $p = \frac{h}{\lambda}$
Photoelectric	$K_{max} = h\nu - \phi = eV_0$ · threshold $\nu_0 = \frac{\phi}{h}$ · the stopping potential depends on frequency, not intensity
de Broglie	$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}}$ · electron: $\lambda(\text{\AA}) \approx \frac{12.27}{\sqrt{V}}$

Σ FORMULA P18 · Atoms & Nuclei · page 144

Result	Formula
Bohr (hydrogen-like)	$r_n = 0.529 \frac{n^2}{Z} \text{\AA}$ · $v_n = 2.18 \times 10^6 \frac{Z}{n} \text{ m/s}$ · $E_n = -13.6 \frac{Z^2}{n^2} \text{ eV}$
Spectrum	$\frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$ · lines from level n : $\frac{n(n-1)}{2}$
Nucleus	$R = R_0 A^{1/3}$ (density independent of A) · $BE = \Delta m c^2$ · $1 \text{ u} = 931.5 \text{ MeV}/c^2$
Decay	$N = N_0 e^{-\lambda t}$ · $T_{1/2} = \frac{\ln 2}{\lambda}$ · mean life $\tau = \frac{1}{\lambda}$ · activity $A = \lambda N$

Σ FORMULA P19 · Electronic Devices · page 150

Result	Formula / fact
Semiconductors	$n_e n_h = n_i^2$ · n-type: pentavalent donor · p-type: trivalent acceptor
Diode	forward bias: barrier falls · reverse: tiny current until breakdown · Zener: regulator in reverse bias
Rectifiers	half-wave output frequency f · full-wave $2f$
Logic gates	NAND and NOR are universal · $\overline{A \cdot B} = \bar{A} + \bar{B}$ · $\overline{A + B} = \bar{A} \cdot \bar{B}$

Σ FORMULA P20 · Experimental Skills · page 155

Experiment	Key relation
Vernier / screw gauge	reading = MSR + (coincidence $\times$ LC) – zero error (with sign)
Simple pendulum	slope of T^2 vs $l = \frac{4\pi^2}{g}$
Young's modulus (Searle)	$Y = \frac{MgL}{\pi r^2 \Delta l}$
Surface tension (capillary)	$T = \frac{r h \rho g}{2 \cos \theta}$
Resonance tube	$\lambda = 2(l_2 - l_1)$ · end correction $e = \frac{l_2 - 3l_1}{2}$
Metre bridge	$X = R \frac{l}{100 - l}$
Focal length (u–v)	$f = \frac{uv}{u + v}$ (magnitudes, real image)

PART V · MASTER FORMULA HANDBOOK

Chemistry Formula & Fact Sheets

Physical Chemistry

Σ FORMULA C01-C02 · Mole concept & atomic structure · pages 165, 170

Result	Formula
Moles	$n = \frac{m}{M} = \frac{N}{N_A} = \frac{V_{\text{gas, STP}}}{22.7 \text{ L}}$ (IUPAC STP, 1 bar) · 22.4 L at 1 atm and 273 K
Concentration	$M = \frac{n_{\text{solute}}}{V(\text{L})} \cdot m = \frac{n_{\text{solute}}}{\text{kg solvent}} \cdot M = \frac{10 x d}{M_{\text{solute}}}$ ($x = \% \text{ w/w}$, d in g/mL)
Equivalents	n -factor method: equivalents = $n \times n\text{-factor} \cdot N = M \times n\text{-factor}$
Photon & Bohr	$E = h\nu \cdot E_n = -13.6 \frac{Z^2}{n^2} \text{ eV} \cdot r_n = 0.529 \frac{n^2}{Z} \text{ Å}$
Uncertainty & de Broglie	$\Delta x \Delta p \geq \frac{h}{4\pi} \cdot \lambda = \frac{h}{p}$
Nodes	radial $n - l - 1$ · angular l · total $n - 1$
Orbital angular momentum	$\sqrt{l(l+1)} \frac{h}{2\pi}$ · spin-only moment $\mu = \sqrt{n(n+2)} \text{ BM}$

Σ FORMULA C03 · Chemical bonding · page 175

Result	Formula / rule
Hybridisation (central atom)	steric number = bond pairs + lone pairs: $2 sp \cdot 3 sp^2 \cdot 4 sp^3 \cdot 5 sp^3d \cdot 6 sp^3d^2$
Bond order (MOT)	$\frac{1}{2}(N_b - N_a)$ · paramagnetic if unpaired MO electrons: $\text{O}_2, \text{B}_2, \text{O}_2^-$
Dipole moment	$\mu = q \times d$ · zero for symmetrical shapes ($\text{CO}_2, \text{BF}_3, \text{CH}_4, \text{SF}_6, \text{XeF}_4$)
Formal charge	$V - L - \frac{1}{2}B$
Bond angle order	$\text{CH}_4 (109.5^\circ) > \text{NH}_3 (107^\circ) > \text{H}_2\text{O} (104.5^\circ)$ — lone pairs compress angles

Σ FORMULA C04 · Thermodynamics · page 181

Result	Formula
First law	$\Delta U = q + w$ (IUPAC sign: w done on the system) · $w = -P_{\text{ext}}\Delta V$
Enthalpy	$\Delta H = \Delta U + \Delta n_g RT$
Reversible isothermal	$w = -2.303 nRT \log \frac{V_2}{V_1}$
Hess & bond enthalpy	$\Delta_r H = \sum \Delta_f H(\text{products}) - \sum \Delta_f H(\text{reactants}) = \sum BE(\text{broken}) - \sum BE(\text{formed})$
Entropy	$\Delta S = \frac{q_{\text{rev}}}{T}$ · phase change $\Delta S = \frac{\Delta H}{T}$
Gibbs	$\Delta G = \Delta H - T\Delta S \cdot \Delta G^\circ = -RT \ln K = -2.303RT \log K \cdot \Delta G^\circ = -nFE^\circ$
Spontaneity	$\Delta H < 0, \Delta S > 0$: always · $\Delta H > 0, \Delta S > 0$: above $T = \frac{\Delta H}{\Delta S}$

Σ FORMULA C05 · Solutions · page 186

Property	Formula
Raoult's law	$p_A = x_A p_A^\circ$ · relative lowering $\frac{p^\circ - p}{p^\circ} = x_{\text{solute}}$
Elevation / depression	$\Delta T_b = i K_b m$ · $\Delta T_f = i K_f m$
Osmotic pressure	$\pi = i C R T$
van 't Hoff factor	association $i = 1 - \alpha \left(1 - \frac{1}{n}\right)$ · dissociation $i = 1 + (n - 1)\alpha$
Henry's law	$p = K_H x$ (higher $K_H \rightarrow$ lower solubility)
Deviations	positive (A-B weaker than A-A, B-B): $\Delta H_{\text{mix}} > 0$, minimum-boiling azeotrope · negative: $\Delta H_{\text{mix}} < 0$, maximum-boiling azeotrope

Σ FORMULA C06 · Equilibrium · page 191

Result	Formula
K_p and K_c	$K_p = K_c (RT)^{\Delta n_g}$
Reaction quotient	$Q < K$: forward · $Q > K$: backward
Weak acid	$[\text{H}^+] = \sqrt{K_a C}$ · $\alpha = \sqrt{K_a / C}$ (Ostwald, small α)
pH	$\text{pH} + \text{pOH} = 14$ (25 °C) · $K_w = 10^{-14}$
Buffers	$\text{pH} = \text{p}K_a + \log \frac{[\text{salt}]}{[\text{acid}]}$ · $\text{pOH} = \text{p}K_b + \log \frac{[\text{salt}]}{[\text{base}]}$
Salt hydrolysis	WA-SB: $\text{pH} = 7 + \frac{1}{2}(\text{p}K_a + \log C)$ · SA-WB: $7 - \frac{1}{2}(\text{p}K_b + \log C)$ · WA-WB: $7 + \frac{1}{2}(\text{p}K_a - \text{p}K_b)$
Solubility	AB: $K_{sp} = s^2$ · AB ₂ : $4s^3$ · A ₂ B ₃ : $108s^5$ · precipitation if $Q_{sp} > K_{sp}$

Σ FORMULA C07 · Electrochemistry · page 197

Result	Formula
Cell emf	$E_{\text{cell}}^\circ = E_{\text{cathode}}^\circ - E_{\text{anode}}^\circ$ (reduction potentials)
Nernst	$E = E^\circ - \frac{0.0591}{n} \log Q$ (298 K) · at equilibrium $\log K = \frac{nE^\circ}{0.0591}$
Conductance	$\kappa = \frac{G^*}{R}$ ($G^* = l/A$) · $\Lambda_m = \frac{1000 \kappa}{M}$ (S cm ² mol ⁻¹)
Kohlrausch	$\Lambda_m = \Lambda_m^\circ - A\sqrt{C}$ (strong) · $\Lambda_m^\circ = \sum \nu_i \lambda_i^\circ$ · $\alpha = \frac{\Lambda_m}{\Lambda_m^\circ}$
Faraday	$m = \frac{MIt}{nF}$ · $F = 96485$ C/mol

Σ FORMULA C08 · Chemical kinetics · page 203

Order	Integrated law	Half-life	Units of k
0	$[A] = [A]_0 - kt$	$\frac{[A]_0}{2k}$	mol L ⁻¹ s ⁻¹
1	$k = \frac{2.303}{t} \log \frac{[A]_0}{[A]}$	$\frac{0.693}{k}$	s ⁻¹
n	—	$\propto [A]_0^{1-n}$	(mol L ⁻¹) ¹⁻ⁿ s ⁻¹

Arrhenius $k = Ae^{-E_a/RT}$ · $\log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$ · first order: 99.9% completion $\approx 10 t_{1/2}$.

Inorganic Chemistry: trends & exceptions

REFERENCE C09 • Periodicity • page 208

Property	Trend	Exceptions to remember
Atomic radius	decreases across a period, increases down a group	Ga $\approx$ Al (d-contraction) • 4d $\approx$ 5d (lanthanoid contraction)
Ionisation enthalpy	increases across a period	Be > B, N > O (and Mg > Al, P > S)
Electron gain enthalpy	more negative across a period	Cl > F (most negative is Cl) • noble gases and N: positive or near zero
Electronegativity	increases across a period	F highest (4.0, Pauling)
Isoelectronic ions	size decreases as Z increases	$\text{N}^{3-} > \text{O}^{2-} > \text{F}^- > \text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$

REFERENCE C10–C12 • p-, d-, f-block & coordination • pages 213, 219, 225

Topic	Key facts
Inert-pair effect	stability of the lower state increases down groups 13–15: Tl^+ , Pb^{2+} , Bi^{3+}
Oxoacid strength	$\text{HClO}_4 > \text{HClO}_3 > \text{HClO}_2 > \text{HClO}$ • hydrogen-halide acidity $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$
Hydrides	thermal stability $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3$ • boiling point $\text{H}_2\text{O} > \text{H}_2\text{Te} > \text{H}_2\text{Se} > \text{H}_2\text{S}$
d-block	magnetic moment $\sqrt{n(n+2)}$ BM • Mn shows +7 • KMnO_4 : acidic $\rightarrow \text{Mn}^{2+}$ ($n = 5$), neutral/weakly basic $\rightarrow \text{MnO}_2$ ($n = 3$)
$\text{K}_2\text{Cr}_2\text{O}_7$	orange; $\text{Cr}_2\text{O}_7^{2-} \rightleftharpoons \text{CrO}_4^{2-}$ (yellow) with pH • acidic $\rightarrow \text{Cr}^{3+}$ ($n = 6$)
Lanthanoids	common state +3 • Ce +4 (oxidant) • Eu, Yb +2 • lanthanoid contraction
Coordination: CFSE (octahedral)	$(-0.4n_{t_{2g}} + 0.6n_{e_g})\Delta_o$ • $\Delta_o \cdot \Delta_t = \frac{4}{9}\Delta_o$
Spectrochemical series	$\text{I}^- < \text{Br}^- < \text{Cl}^- < \text{F}^- < \text{OH}^- < \text{H}_2\text{O} < \text{NH}_3 < \text{en} < \text{CN}^- < \text{CO}$
Isomer counts	$[\text{MA}_4\text{B}_2]$: 2 geometric • $[\text{MA}_3\text{B}_3]$: fac/mer • $[\text{M}(\text{AA})_2\text{B}_2]$: cis is optically active (3 stereoisomers)

Organic Chemistry

REFERENCE C14 • GOC orders • page 237

Quantity	Order
Carbocation stability	$3^\circ > 2^\circ > 1^\circ > \text{CH}_3^+$ • benzylic/allylic stabilised by resonance
Carbanion stability	$\text{CH}_3^- > 1^\circ > 2^\circ > 3^\circ$ (opposite of carbocations)
Free radicals	$3^\circ > 2^\circ > 1^\circ$ • benzyl $\approx$ allyl $> 3^\circ$
Acidity	$\text{RCOOH} > \text{phenol} > \text{H}_2\text{O} > \text{ROH} > \text{alkyne} > \text{alkene} > \text{alkane}$ • EWG increases acidity
Basicity (aqueous)	$(\text{C}_2\text{H}_5)_2\text{NH} > (\text{C}_2\text{H}_5)_3\text{N} > \text{C}_2\text{H}_5\text{NH}_2 > \text{NH}_3 > \text{aniline}$
Effects	$-I$: $\text{NO}_2 > \text{CN} > \text{COOH} > \text{F} > \text{Cl}$ • $+M$: $-\text{NH}_2, -\text{OH}, -\text{OR}$

REFERENCE Reagent table (C15–C18) · pages 242–264

Reagent	Does
H ₂ /Lindlar (Pd–BaSO ₄ , quinoline)	alkyne → <i>cis</i> -alkene
Na/liquid NH ₃	alkyne → <i>trans</i> -alkene
HBr + peroxide	anti-Markovnikov addition (HBr only)
Alkaline KMnO ₄ (cold, dilute)	Baeyer's test: alkene → <i>syn</i> -diol
O ₃ , then Zn/H ₂ O	ozonolysis → aldehydes/ketones
LiAlH ₄	reduces –COOH, esters, aldehydes, ketones, amides and nitriles
NaBH ₄	reduces aldehydes and ketones only
PCC	1° alcohol → aldehyde (stops there)
K ₂ Cr ₂ O ₇ /H ⁺	1° alcohol → acid · 2° → ketone
DIBAL-H	ester / nitrile → aldehyde
SOCl ₂ (pyridine)	ROH → RCl (by-products SO ₂ , HCl are gases)
alcoholic KOH	dehydrohalogenation (elimination)
aqueous KOH	substitution (RX → ROH)
Br ₂ /NaOH	Hofmann bromamide: amide → amine with one C fewer
NaNO ₂ /HCl, 0–5 °C	diazotisation of aromatic 1° amines
Sn/HCl or Fe/HCl	ArNO ₂ → ArNH ₂

REFERENCE Named reactions (C15–C18)

Reaction	Change
Wurtz	2RX + 2Na → R–R (dry ether)
Friedel–Crafts	ArH + RCl/RCOCl + anhydrous AlCl ₃ → alkyl/acyl arene
Kolbe (electrolytic)	RCOONa → R–R
Kolbe (phenol)	phenoxide + CO ₂ → salicylic acid
Reimer–Tiemann	phenol + CHCl ₃ /NaOH → salicylaldehyde
Williamson	RONa + R'X → ROR' (R'X must be 1°)
Aldol	aldehyde/ketone with α-H + dilute base → β-hydroxy carbonyl
Cannizzaro	aldehyde without α-H + conc. base → alcohol + carboxylate
Clemmensen / Wolff–Kishner	C=O → CH ₂ (Zn(Hg)/HCl · NH ₂ NH ₂ /KOH, glycol)
Rosenmund	RCOCl + H ₂ (Pd–BaSO ₄) → RCHO
Stephen	RCN + SnCl ₂ /HCl, then H ₃ O ⁺ → RCHO
HVZ	acid + X ₂ /red P → α-halo acid
Gabriel phthalimide	1° aliphatic amines only
Sandmeyer	ArN ₂ ⁺ + CuCl/CuBr/CuCN → ArCl/ArBr/ArCN
Carbylamine	1° amine + CHCl ₃ + KOH → isocyanide (foul smell)
Etard / Gattermann–Koch	toluene → benzaldehyde (CrO ₂ Cl ₂) · benzene + CO + HCl (AlCl ₃ /CuCl) → benzaldehyde

 REFERENCE Distinguishing tests (C13–C20) · page 277

Test	Positive for
Tollens' (silver mirror)	aldehydes (aliphatic and aromatic), formic acid, reducing sugars
Fehling's (red Cu_2O)	aliphatic aldehydes (not benzaldehyde)
Iodoform (I_2/NaOH)	CH_3CO^- or $\text{CH}_3\text{CH}(\text{OH})^-$: ethanol, acetaldehyde, acetone, 2-propanol
Lucas (ZnCl_2/HCl)	turbidity: 3° at once · 2° in ~5 min · 1° none at room temperature
Neutral FeCl_3	violet colour with phenols
NaHCO_3	effervescence with carboxylic acids (not phenol, except picric acid)
Hinsberg reagent	1° : sulphonamide soluble in alkali · 2° : insoluble · 3° : no reaction
Lassaigne's test	N: Prussian blue · S: violet with nitroprusside · halogen: AgNO_3 precipitate

PART V · MASTER FORMULA HANDBOOK

Mathematics Formula Sheets

Σ FORMULA M01-M02 · Sets, functions, complex numbers & quadratics · pages 287, 292

Result	Formula
Counting	$n(A \cup B) = n(A) + n(B) - n(A \cap B)$ · relations on an n -set: 2^{n^2} · reflexive: 2^{n^2-n} · symmetric: $2^{n(n+1)/2}$
Functions $A \rightarrow B$	total $ B ^{ A }$ · one-one mP_n ($n = A \leq m = B $) · onto from n to 2 elements: $2^n - 2$
Complex numbers	$ z_1 z_2 = z_1 z_2 $ · $\arg(z_1 z_2) = \arg z_1 + \arg z_2$ · $z \bar{z} = z ^2$ · $ z_1 + z_2 \leq z_1 + z_2 $
Cube roots of unity	$1 + \omega + \omega^2 = 0$ · $\omega^3 = 1$ · $\omega = \frac{-1 + i\sqrt{3}}{2}$
Loci	$ z - z_1 = z - z_2 $: perpendicular bisector · $ z - z_0 = r$: circle
Quadratic	$\alpha + \beta = -\frac{b}{a}$ · $\alpha\beta = \frac{c}{a}$ · $D = b^2 - 4ac$ · $ \alpha - \beta = \frac{\sqrt{D}}{ a }$
Root location ($a > 0$)	both roots $> k$: $D \geq 0$, $f(k) > 0$, $-\frac{b}{2a} > k$ · k between the roots: $f(k) < 0$

Σ FORMULA M03 · Matrices & determinants · page 298

Result	Formula
Adjoint & inverse	$A^{-1} = \frac{\text{adj } A}{ A }$ · $A(\text{adj } A) = A I$
Determinant rules ($n \times n$)	$ kA = k^n A $ · $ \text{adj } A = A ^{n-1}$ · $ \text{adj}(\text{adj } A) = A ^{(n-1)^2}$ · $ A^{-1} = \frac{1}{ A }$
Products	$(AB)^{-1} = B^{-1}A^{-1}$ · $(AB)^T = B^T A^T$ · $\text{adj}(AB) = \text{adj } B \text{ adj } A$
Special matrices	symmetric $A^T = A$ · skew $A^T = -A$ (odd-order skew: $ A = 0$) · orthogonal $AA^T = I$
Linear systems $AX = B$	$ A \neq 0$: unique · $ A = 0$ and $(\text{adj } A)B \neq 0$: no solution · $ A = 0$ and $(\text{adj } A)B = 0$: infinitely many or none (check)
Area of a triangle	$\frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$

Σ FORMULA M04–M06 · P&C, binomial, sequences · pages 303, 308, 313

Result	Formula
Permutations & combinations	${}^nP_r = \frac{n!}{(n-r)!} \cdot {}^nC_r = \frac{n!}{r!(n-r)!} \cdot {}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$
Repetition	$\frac{n!}{p!q!r!} \cdot \text{circular } (n-1)! \cdot \text{necklace } \frac{(n-1)!}{2}$
Distribution	n identical into r groups: ${}^{n+r-1}C_{r-1}$ (each ≥ 1 : ${}^{n-1}C_{r-1}$)
Derangements	$D_n = n! \sum_{k=0}^n \frac{(-1)^k}{k!} \cdot D_3 = 2, D_4 = 9, D_5 = 44$
Binomial general term	$T_{r+1} = {}^nC_r x^{n-r} y^r \cdot \text{middle term: } T_{n/2+1} \text{ (} n \text{ even)}$
Coefficient sums	$\sum {}^nC_r = 2^n \cdot \sum_{\text{odd}} = \sum_{\text{even}} = 2^{n-1}$
AP	$a_n = a + (n-1)d \cdot S_n = \frac{n}{2}[2a + (n-1)d]$
GP	$a_n = ar^{n-1} \cdot S_n = \frac{a(r^n - 1)}{r - 1} \cdot S_\infty = \frac{a}{1-r} \text{ (} r < 1 \text{)}$
Means	$AM \geq GM \geq HM \cdot \sum k^2 = \frac{n(n+1)(2n+1)}{6} \cdot \sum k^3 = \left[\frac{n(n+1)}{2} \right]^2$

Σ FORMULA M07 · Limits, continuity, differentiability & applications · page 318

Result	Formula
Standard limits	$\lim \frac{\sin x}{x} = 1 \cdot \lim \frac{e^x - 1}{x} = 1 \cdot \lim \frac{\ln(1+x)}{x} = 1 \cdot \lim \frac{a^x - 1}{x} = \ln a \cdot \lim \frac{x^n - a^n}{x - a} = na^{n-1}$
1^∞ form	$\lim f^g = e^{\lim g(f-1)}$
Expansions	$\sin x \approx x - \frac{x^3}{6} \cdot \cos x \approx 1 - \frac{x^2}{2} \cdot \tan x \approx x + \frac{x^3}{3} \cdot e^x \approx 1 + x + \frac{x^2}{2}$
Derivatives	$(\tan x)' = \sec^2 x \cdot (\sec x)' = \sec x \tan x \cdot (\sin^{-1} x)' = \frac{1}{\sqrt{1-x^2}} \cdot (\tan^{-1} x)' = \frac{1}{1+x^2} \cdot (a^x)' = a^x \ln a$
Rules	product, quotient, chain · parametric $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} \cdot x^x$: take logs
Applications	rate of change · increasing where $f' > 0$ · maximum: $f' = 0, f'' < 0$ (Rolle/LMVT are not in the 2024+ text)

Σ FORMULA M08–M09 · Integration & differential equations · pages 324, 330

Result	Formula
Standard integrals	$\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} \cdot \int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \frac{x}{a} \cdot \int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left \frac{x-a}{x+a} \right $
More	$\int \frac{dx}{\sqrt{x^2 \pm a^2}} = \ln x + \sqrt{x^2 \pm a^2} \cdot \int e^x [f + f'] dx = e^x f \cdot \int \sec x dx = \ln \sec x + \tan x $
By parts	$\int uv dx = u \int v - \int (u' \int v) \text{ (ILATE)}$
Definite properties	$\int_a^b f(x) dx = \int_a^b f(a+b-x) dx \cdot \text{even: } 2 \int_0^a \cdot \text{odd: } 0 \cdot \text{period } T: \int_0^{nT} = n \int_0^T$
Standard results	$\int_0^{\pi/2} \sin^n x dx \text{ (Wallis)} \cdot \int_0^{\pi/2} \ln \sin x dx = -\frac{\pi}{2} \ln 2 \cdot \int_0^{\pi/2} \frac{\sin^n}{\sin^n + \cos^n} = \frac{\pi}{4}$
Leibniz	$\frac{d}{dx} \int_{g(x)}^{h(x)} f(t) dt = f(h)h' - f(g)g'$
Areas	$y^2 = 4ax \text{ and } x^2 = 4by: \frac{16ab}{3} \cdot \text{ellipse } \pi ab \cdot \text{parabola } y^2 = 4ax \text{ with its latus rectum: } \frac{8a^2}{3}$
Linear DE	$y' + Py = Q: \text{IF} = e^{\int P dx} \cdot y \cdot \text{IF} = \int Q \cdot \text{IF} dx + C$

Σ FORMULA M10 · Coordinate geometry · page 335

Result	Formula			
Lines	distance $\frac{ ax_1 + by_1 + c }{\sqrt{a^2 + b^2}} \cdot \tan \theta = \left \frac{m_1 - m_2}{1 + m_1 m_2} \right \cdot \text{image: } \frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{-2(ax_1 + by_1 + c)}{a^2 + b^2}$			
Circle	centre $(-g, -f)$, $r = \sqrt{g^2 + f^2 - c}$ · chord $2\sqrt{r^2 - d^2}$			
Conic	e	Focus	Directrix	Latus rectum
$y^2 = 4ax$	1	$(a, 0)$	$x = -a$	$4a$
$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b$)	$\sqrt{1 - \frac{b^2}{a^2}}$	$(\pm ae, 0)$	$x = \pm \frac{a}{e}$	$\frac{2b^2}{a}$
$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	$\sqrt{1 + \frac{b^2}{a^2}}$	$(\pm ae, 0)$	$x = \pm \frac{a}{e}$	$\frac{2b^2}{a}$

Ellipse: $PS_1 + PS_2 = 2a$ · hyperbola: $|PS_1 - PS_2| = 2a$ · rectangular hyperbola $e = \sqrt{2}$.

Σ FORMULA M11-M12 · 3D geometry & vectors · pages 342, 348

Result	Formula
DCs	$l^2 + m^2 + n^2 = 1 \cdot \cos \theta = \frac{ a_1 a_2 + b_1 b_2 + c_1 c_2 }{\sqrt{\sum a_1^2} \sqrt{\sum a_2^2}}$
Shortest distance	skew $\frac{ (\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) }{ \vec{b}_1 \times \vec{b}_2 }$ · parallel $\frac{ (\vec{a}_2 - \vec{a}_1) \times \vec{b} }{ \vec{b} }$
Foot / image in a line	general point $Q(\lambda)$, then $\vec{PQ} \cdot \vec{b} = 0$ · image $2Q - P$
Products	$\vec{a} \cdot \vec{b} = a b \cos \theta$ · $ \vec{a} \times \vec{b} = a b \sin \theta$ · projection $\frac{\vec{a} \cdot \vec{b}}{ \vec{b} }$
Areas & volume	triangle $\frac{1}{2} \vec{AB} \times \vec{AC} $ · $[\vec{a} \vec{b} \vec{c}] = \text{volume}$ · coplanar iff $[\vec{a} \vec{b} \vec{c}] = 0$
Identities	$ \vec{a} \times \vec{b} ^2 + (\vec{a} \cdot \vec{b})^2 = a ^2 b ^2$ · $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$

Σ FORMULA M13-M14 · Statistics, probability & trigonometry · pages 353, 359

Result	Formula
Variance	$\sigma^2 = \frac{\sum x^2}{n} - \bar{x}^2$ · $y = ax + b$: $\sigma_y = a \sigma_x$ · first n naturals $\frac{n^2 - 1}{12}$
Grouped median / mode	$l + \frac{N/2 - C}{f} h$ · $l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} h$
Probability	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$ · $P(A B) = \frac{P(A \cap B)}{P(B)}$ · independent: $P(A \cap B) = P(A)P(B)$
Bayes	$P(E_k A) = \frac{P(E_k)P(A E_k)}{\sum P(E_i)P(A E_i)}$
Random variable	$E(X) = \sum xp$ · $\text{Var} = E(X^2) - [E(X)]^2$ · binomial: mean np , variance npq
Trig identities	$\sin 2A = \frac{2 \tan A}{1 + \tan^2 A}$ · $\cos 2A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$ · $\sin 3A = 3 \sin A - 4 \sin^3 A$ · $\cos 3A = 4 \cos^3 A - 3 \cos A$
Range	$a \sin x + b \cos x \in [-\sqrt{a^2 + b^2}, \sqrt{a^2 + b^2}]$
Inverse trig	$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ · $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \frac{x+y}{1-xy}$ ($xy < 1$) · $\cos^{-1}(-x) = \pi - \cos^{-1} x$

PART V · MASTER FORMULA HANDBOOK

Constants, Units & Frequently Forgotten Formulas

VERIFIED FACT Physical constants

Exact values are fixed by the 2019 SI redefinition. The others are CODATA recommended values, rounded to the precision JEE uses [CODATA]. Use whatever values a question gives.

Constant	Value	JEE working value
Speed of light c	299 792 458 m/s (exact)	3×10^8 m/s
Planck constant h	$6.626\,070\,15 \times 10^{-34}$ J s (exact)	6.63×10^{-34} J s
Elementary charge e	$1.602\,176\,634 \times 10^{-19}$ C (exact)	1.6×10^{-19} C
Avogadro N_A	$6.022\,140\,76 \times 10^{23}$ mol $^{-1}$ (exact)	6.02×10^{23}
Boltzmann k_B	$1.380\,649 \times 10^{-23}$ J/K (exact)	1.38×10^{-23} J/K
Gas constant R	8.314 J mol $^{-1}$ K $^{-1}$	$8.314 \cdot 0.0821$ L atm · 0.0831 L bar · 1.987 cal
Faraday F	$96\,485$ C/mol	$96\,500$ C/mol
Gravitational G	6.674×10^{-11} N m 2 kg $^{-2}$	6.67×10^{-11}
ϵ_0	8.854×10^{-12} F/m	$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
μ_0	$1.256\,637 \times 10^{-6}$ N/A 2	$4\pi \times 10^{-7}$
Electron mass m_e	9.109×10^{-31} kg	9.1×10^{-31} kg
Proton mass m_p	1.6726×10^{-27} kg	1.67×10^{-27} kg
Atomic mass unit	1.6605×10^{-27} kg	931.5 MeV/ c^2
Rydberg R_∞	1.097×10^7 m $^{-1}$	13.6 eV (hydrogen ground state)
Stefan–Boltzmann σ	5.670×10^{-8} W m $^{-2}$ K $^{-4}$	5.67×10^{-8}
Wien's constant b	2.898×10^{-3} m K	2.9×10^{-3} m K

VERIFIED FACT Conversions & handy numbers

Quantity	Value
1 eV	1.6×10^{-19} J · $hc \approx 1240$ eV nm = 12400 eV Å
1 atm	$101\,325$ Pa = 1.01325 bar = 760 mm Hg
1 cal	4.184 J
$1 \text{ Å} \cdot 1 \text{ fm}$	10^{-10} m · 10^{-15} m
1 light year	9.46×10^{15} m
1 horsepower	746 W
$2.303RT/F$ at 298 K	0.0591 V
$\log 2 \cdot \log 3 \cdot \log 7 \cdot \ln 10$	$0.3010 \cdot 0.4771 \cdot 0.8451 \cdot 2.303$
$\sqrt{2} \cdot \sqrt{3} \cdot \sqrt{5} \cdot \pi^2$	$1.414 \cdot 1.732 \cdot 2.236 \cdot 9.87 (\approx g)$
$e \cdot e^{-1} \cdot \ln 2$	$2.718 \cdot 0.368 \cdot 0.693$

The frequently forgotten list RECOMMENDATION

These are the formulas students most often confuse or forget under time pressure, based on the traps collected in the modules. Revise this list in the last week and again on exam eve.

MISTAKE ALERT Physics

Often confused	Correct version
Lens vs mirror formula	lens $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ · mirror $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$
Drop vs soap bubble	excess pressure $\frac{2T}{r}$ (drop, one surface) vs $\frac{4T}{r}$ (bubble, two surfaces)
Closed vs open pipe	closed: odd harmonics only, $f_1 = \frac{v}{4L}$ · open: all harmonics, $f_1 = \frac{v}{2L}$
g at height vs depth	height: $\left(1 - \frac{2h}{R}\right)$ · depth: $\left(1 - \frac{d}{R}\right)$
Axial vs equatorial dipole field	axial $\frac{2kp}{r^3}$ · equatorial $\frac{kp}{r^3}$ (and opposite to $\vec{p}$)
v_{rms} vs v_{mp}	$\sqrt{3RT/M}$ vs $\sqrt{2RT/M}$
Wire vs loop field	wire $\frac{\mu_0 I}{2\pi r}$ · loop centre $\frac{\mu_0 I}{2R}$
Moment of inertia of a rod	about the centre $\frac{1}{12}ML^2$ · about an end $\frac{1}{3}ML^2$
Photoelectric	K_{max} depends on frequency · current depends on intensity

MISTAKE ALERT Chemistry

Often confused	Correct version
Sign of work	chemistry (IUPAC): $\Delta U = q + w$, with w done on the system · physics: $\Delta Q = \Delta U + W$, with W done by the gas
Nernst sign	$E = E^\circ - \frac{0.0591}{n} \log Q$ (minus, and Q = products over reactants)
Molarity vs molality	per litre of solution vs per kg of solvent
$K_p = K_c(RT)^{\Delta n}$	Δn counts gaseous moles only
First-order half-life	independent of concentration · zero order: $t_{1/2} \propto [A]_0$
Carbocation vs carbanion order	opposite orders
NaBH_4 vs LiAlH_4	NaBH_4 does not reduce acids or esters
Fehling's vs Tollens'	benzaldehyde gives Tollens' but not Fehling's
Electron gain enthalpy	most negative: Cl, not F

 MISTAKE ALERT Mathematics

Often confused	Correct version
Ellipse vs hyperbola	ellipse $b^2 = a^2(1 - e^2)$ · hyperbola $b^2 = a^2(e^2 - 1)$
$ kA $	$k^n A $, not $k A $
Distributing identical objects	${}^{n+r-1}C_{r-1}$ (with zeros allowed), not r^n
T_{r+1}	the term number is $r + 1$, so the 5th term uses $r = 4$
Principal range of $\cos^{-1}$	$[0, \pi]$, so $\cos^{-1}(-x) = \pi - \cos^{-1}x$
Variance under $y = ax + b$	$a^2\sigma^2$ (the constant b has no effect)
$\int_0^{2a} f$ with $f(2a - x) = f(x)$	$2 \int_0^a f$ (if $f(2a - x) = -f(x)$: zero)
IF of $y' - \frac{y}{x} = \dots$	$\frac{1}{x}$, not x

PART V · MASTER FORMULA HANDBOOK

Formula Revision Plans: 1-Day, 3-Day, 7-Day & Final 24 Hours

All four plans use the three-pass method (page 367): recall first, then check, then repeat what you missed. Reading a sheet without first trying to recall it feels productive, but it builds far less memory than testing yourself [LS-DUN], [LS-RK]. The plans are recommendations  **RECOMMENDATION**; adjust them to your weak areas from the Mistake Book.

The 7-Day Formula Cycle

Use this cycle once a month during preparation, and as the formula part of the final week.

Day	Session 1 (60 min)	Session 2 (45 min)	Re-test (15 min)	Done
1	Physics P01–P07	Maths M01–M03	yesterday's dots	☐
2	Physics P08–P12	Maths M04–M06	Day 1 dots	☐
3	Physics P13–P20	Maths M07–M09	Day 2 dots	☐
4	Chemistry C01–C08 (physical)	Maths M10	Day 1 + Day 3 dots	☐
5	Chemistry C09–C12 (inorganic) + NCERT tables	Maths M11–M14	Day 4 dots	☐
6	Chemistry C14–C20 (organic): reagent and named-reaction tables	Constants & frequently forgotten list	Day 2 + Day 5 dots	☐
7	Full self-test: 60 random formulas across all three subjects	Rewrite every missed formula 3 times	all remaining dots	☐

The 3-Day Formula Sprint

For the gap between two mocks, or when you have lost touch with the formulas after a break.

Day 1 · Physics

- ☐ Morning: sheets P01–P10 (recall, then check)
- ☐ Afternoon: sheets P11–P20
- ☐ Evening: frequently forgotten list (Physics); re-test the dots

Day 2 · Chemistry

- ☐ Morning: physical sheets C01–C08
- ☐ Afternoon: inorganic trends, then the reagent and named-reaction tables
- ☐ Evening: distinguishing tests; re-test the Day 1 and Day 2 dots

Day 3 · Maths

- ☐ Morning: M01–M07
- ☐ Afternoon: M08–M14, conic table from memory
- ☐ Evening: constants; the full dot list from all three days

The 1-Day Formula Blitz

For the day before a mock, or one revision day in the final fortnight. About 6 hours, with a 10-minute break every hour.

1

Hour 1 · Physics mechanics

P02–P06 sheets. Write the vertical-circle, banking, rolling and orbit results from memory.

2

Hour 2 · Physics rest

P07–P20. Focus on thermodynamics processes, optics sign conventions and the AC table.

3 Hour 3 · Physical chemistry

C01–C08. Write out every formula with its units.

4 Hour 4 · Inorganic & organic

Trend exceptions, the reagent table, named reactions and tests. Say each reaction aloud.

5 Hour 5 · Maths

M01–M14. Conic table, definite-integral properties, determinant rules, P&C distribution.

6 Hour 6 · Frequently forgotten + dots

All three trap tables, then everything you dotted today. Stop by evening and sleep normally.

The Final 24 Hours**☆ IMPORTANT The last 24 hours are for confidence and sleep, not new material**

Research on sleep and memory consolidation shows that sleep after learning helps you retain it, and sleep deprivation impairs attention and working memory [LS-RB], [LS-AASM]. Do **not** study past your normal bedtime on the eve of the exam.

Time (for a morning shift)	Formula task
Eve, morning	7-Day cycle "Day 7" self-test, but only 30 formulas per subject
Eve, afternoon	Frequently forgotten list; the personal dot list from the last 7 days
Eve, evening (max 60 min)	Read the Last-Minute Revision Book sheets (page 485) once, calmly. No new problems.
Exam morning	Optional 10-minute glance at your personal "top 20 forgotten" card. Nothing else.
In the hall	Before starting, write any 3–5 formulas you tend to forget on the rough sheet (only after the exam officially starts).

REVISION Formula cycle log

Cycle	Dates	Formulas missed	Top 3 to fix	Done
1	_____	_____	_____	☐
2	_____	_____	_____	☐
3	_____	_____	_____	☐
4	_____	_____	_____	☐
5	_____	_____	_____	☐
6	_____	_____	_____	☐

PART VI

Speed & Shortcut Manual

Legitimate speed methods (when to use, when not to, time saved, common mistake) and the answer-choice strategy.

- The JEE Main Speed & Shortcut Manual **389**
- Calculation Fitness: Tables & Daily Drills **394**

- Answer-Choice Strategy

396

PART VI · SPEED & SHORTCUT MANUAL

The JEE Main Speed & Shortcut Manual

In JEE Main, "speed" is not fast writing. It is three separate skills:

1 · Selection

Recognising within about 20 seconds which questions to attempt now, later or never. This is the biggest time saver (Exam-Hall Protocol, page [480](#)).

2 · Recognition

Seeing the question model ("this is a shortest-distance problem") so the method comes to mind at once. It comes from model drills in every chapter.

3 · Execution

Doing the algebra and arithmetic cleanly by hand, with no calculator. This manual and the drills in the next chapter train it.

The sixteen techniques below are **legitimate methods**: each one is either a mathematically valid shortcut or a way to eliminate options with certainty. Each gives when to use it, when **not** to use it, a worked example, the approximate time saved and the most common mistake. The time savings are this book's estimates  **RECOMMENDATION**, not measured data.

#	Technique	Main subjects
1	Dimensional analysis	Physics
2	Limiting cases	Physics, Maths
3	Special-value substitution	Maths
4	Back-substitution of options	Maths, Chemistry
5	Symmetry	Physics, Maths
6	One-significant-figure estimation	Physics, Chemistry
7	Ratio (proportionality) method	Physics, Chemistry
8	Conservation laws first	Physics
9	The standard-result bank	all
10	Graph instead of algebra	Maths, Physics
11	Equivalents and the n -factor	Chemistry
12	Log and pH shortcuts	Chemistry
13	Working backwards in organic chemistry	Chemistry
14	Elimination by impossibility	all
15	Casting out nines	all (arithmetic checks)
16	Cancel before you multiply	all

⚡ SHORTCUT 1 · Dimensional analysis**Use it when**

the options differ in the powers of physical quantities

Don't use it when

the options differ only in pure numbers ($2, \pi, \frac{1}{2}$) or in dimensionless groups

Example. The speed of a transverse wave on a string with tension T and linear mass density μ is (A) $\sqrt{T/\mu}$ (B) $\sqrt{T\mu}$ (C) T/μ (D) $\sqrt{\mu/T}$. $[T] = \text{MLT}^{-2}$ and $[\mu] = \text{ML}^{-1}$, so $[T/\mu] = \text{L}^2\text{T}^{-2}$ and only $\sqrt{T/\mu}$ is a speed.

Time saved: 1–2 minutes (no derivation).

Common mistake: treating angles, strains or e^x arguments as dimensional. Exponents and trig arguments must be dimensionless.

⚡ SHORTCUT 2 · Limiting cases**Use it when**

a formula depends on masses, angles or distances that can be pushed to 0, ∞ or equality

Don't use it when

the limit changes the physical situation (e.g. a block that stops sliding once μ gets large enough)

Example. For an Atwood machine, $a = \frac{(m_1 - m_2)g}{m_1 + m_2}$. Check: $m_2 \rightarrow 0$ gives $a \rightarrow g$ (free fall) ✓; $m_1 = m_2$ gives $a = 0$ ✓. An option such as $\frac{(m_1 - m_2)g}{2m_1}$ passes the second test but fails the first ($m_2 \rightarrow 0$ gives $g/2$). Test two limits.

Time saved: 1–3 minutes on formula-choice MCQs.

Common mistake: using only one limit. Several options often pass the same one.

⚡ SHORTCUT 3 · Special-value substitution**Use it when**

the options are formulas in n, x or θ and the statement is an identity

Don't use it when

the options agree at every easy value, or the question asks for a count of solutions

Example. $\sum_{k=1}^n k \cdot k!$ equals (A) $(n+1)! - 1$ (B) $n! + 1$ (C) $(n+1)!$ (D) $n \cdot n!$. At $n = 1$ the sum is 1: (A) and (D) survive. At $n = 2$ the sum is $1 + 4 = 5$: (A) gives 5 and (D) gives 4. The answer is (A).

Time saved: 2–4 minutes compared with proving the result.

Common mistake: testing only $n = 1$ or $x = 0$, where several options coincide. Always test a second value.

⚡ SHORTCUT 4 · Back-substitution of options**Use it when**

the options are numbers and checking one is faster than solving

Don't use it when

numerical-value questions (there are no options), or options that contain parameters

Example. Solve $\sqrt{x+5} + \sqrt{x} = 5$: options 4, 9, 1, 16. At $x = 4$: $3 + 2 = 5$ ✓. No squaring is needed, so no extraneous roots can appear.

Time saved: 1–2 minutes.

Common mistake: accepting an option that "almost" works. The check must be exact.

⚡ SHORTCUT 5 · Symmetry**Use it when**

a circuit, charge distribution or integral is unchanged by a reflection, rotation or $x \rightarrow a + b - x$

Don't use it when

one element breaks the symmetry (a single different resistor, an asymmetric limit)

Example. $I = \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$. Replacing x by $\frac{\pi}{2} - x$ swaps $\sin$ and $\cos$, so $2I = \int_0^{\pi/2} 1 dx$ and $I = \frac{\pi}{4}$.

Time saved: 3–5 minutes.

Common mistake: assuming a circuit is symmetric without checking every branch.

⚡ SHORTCUT 6 · One-significant-figure estimation**Use it when**

the options differ by a factor of 2 or more

Don't use it when

the options are within about 10% of each other, or you are subtracting nearly equal numbers

Example. Energy of a 500 nm photon: (A) 1.3×10^{-19} J (B) 4×10^{-19} J (C) 4×10^{-18} J (D) 4×10^{-20} J. Estimate: $\frac{(6.6 \times 10^{-34})(3 \times 10^8)}{5 \times 10^{-7}} \approx 4 \times 10^{-19}$ J. (In eV it is $1240/500 \approx 2.5$ eV. An option of 2.5×10^{-19} J would be a unit trap.)

Time saved: about 1 minute per calculation-heavy question.

Common mistake: estimating in mass-defect or small-difference problems, where the difference is the answer.

⚡ SHORTCUT 7 · Ratio (proportionality) method**Use it when**

the question changes one quantity and asks for the effect on another

Don't use it when

quantities combine additively ($R + r$, $u + at$), so no single power law applies

Example. A planet orbits at 4 times Earth's orbital radius. Since $T \propto r^{3/2}$, $T = 4^{3/2} = 8$ years. Chemistry: the time for equal volumes to effuse scales as $\sqrt{M}$, so O_2 takes $\sqrt{32/2} = 4$ times as long as H_2 .

Time saved: 1–2 minutes (no constants needed).

Common mistake: using the wrong power, such as $r^{2/3}$ instead of $r^{3/2}$.

⚡ SHORTCUT 8 · Conservation laws first**Use it when**

the question asks for a speed, height or final velocity and the forces are conservative (or collisions are involved)

Don't use it when

time, force or acceleration at an instant is asked, or friction does unknown work

Example. A bead slides from rest down a frictionless curved wire through a height h . Its speed at the bottom is $\sqrt{2gh}$, whatever the shape of the wire. Force analysis would need the curve's equation.

Time saved: 2–5 minutes.

Common mistake: conserving kinetic energy in an inelastic collision. Only momentum is conserved there.

⚡ SHORTCUT 9 • The standard-result bank**Use it when**

the situation matches a standard case exactly

Don't use it when

a condition differs (a projectile launched from a height, a non-uniform rod)

Example. Spin-only magnetic moments for $n = 1$ to 5 unpaired electrons are 1.73, 2.83, 3.87, 4.90 and 5.92 BM. For Fe^{3+} (high-spin d^5) you know 5.92 BM without computing $\sqrt{35}$. Other entries in the bank: $\int_0^{\pi/2} \ln \sin x \, dx = -\frac{\pi}{2} \ln 2$; the maximum range at 45° ; area between $y^2 = 4ax$ and $x^2 = 4by$ is $\frac{16ab}{3}$.

Time saved: 1–3 minutes each.

Common mistake: applying a standard result outside its conditions.

⚡ SHORTCUT 10 • Graph instead of algebra**Use it when**

the question asks for the number of solutions or the sign of a root

Don't use it when

the exact value of a root is required

Example. Number of real solutions of $e^x = x^2$. Sketch both curves. For $x < 0$ they cross once. For $x > 0$, $e^x - x^2$ is always positive (its minimum there is $2 - 2 \ln 2 > 0$). Answer: 1.

Time saved: 2–4 minutes.

Common mistake: careless sketching near tangency or asymptotes. Check one or two exact points.

⚡ SHORTCUT 11 • Equivalents and the n-factor**Use it when**

titration, redox stoichiometry or electrolysis questions

Don't use it when

disproportionation or an unclear product, where the n -factor is ambiguous. Balance the equation instead

Example. Volume of 0.1 M KMnO_4 (acidic, $n = 5$) needed for 50 mL of 0.1 M FeSO_4 ($n = 1$): milliequivalents of $\text{Fe}^{2+} = 50 \times 0.1 \times 1 = 5$, so $V = \frac{5}{0.1 \times 5} = 10$ mL.

Time saved: 1–2 minutes (no balanced equation needed).

Common mistake: using $n = 5$ for KMnO_4 in neutral or weakly basic medium, where it is 3.

⚡ SHORTCUT 12 • Log and pH shortcuts**Use it when**

weak acids and bases with small α , and buffers

Don't use it when

very dilute solutions (below about 10^{-6} M), where water's own ions matter

Example. 0.01 M acetic acid with $\text{p}K_a = 4.74$: $\text{pH} = \frac{1}{2}(\text{p}K_a - \log C) = \frac{1}{2}(4.74 + 2) = 3.37$. A buffer with $[\text{salt}] = 10[\text{acid}]$ has $\text{pH} = \text{p}K_a + 1$.

Time saved: about 1 minute.

Common mistake: giving pH 8 for 10^{-8} M HCl. An acid solution can't be basic; the answer is just below 7.

⚡ SHORTCUT 13 • Working backwards in organic chemistry**Use it when**

the products are given and the starting compound is asked (ozonolysis, hydrolysis, degradation)

Don't use it when

carbocation rearrangements are possible. Then the forward mechanism is needed

Example. An alkene gives acetone and acetaldehyde on ozonolysis. Join the two carbonyl carbons with a double bond: $(\text{CH}_3)_2\text{C}=\text{CHCH}_3$ (2-methylbut-2-ene).

Time saved: 1–3 minutes.

Common mistake: miscounting carbons. Each $\text{C}=\text{O}$ carbon came from one alkene carbon.

⚡ SHORTCUT 14 · Elimination by impossibility**Use it when**

always, as a first filter: probability > 1 , efficiency > 1 , negative speed or count, wrong sign of a quantity

Don't use it when

you are tempted to eliminate an option that merely *looks* unusual (a negative focal length is valid)

Example. Carnot efficiency between 500 K and 300 K: (A) 0.6 (B) 1.67 (C) 0.4 (D) 0.8. (B) is impossible, and $1 - \frac{300}{500} = 0.4$.

Time saved: varies; turns some four-way guesses into two-way decisions (Answer-Choice Strategy, page 396).

Common mistake: eliminating on "looks" instead of on a physical or mathematical impossibility.

⚡ SHORTCUT 15 · Casting out nines**Use it when**

checking a long multiplication or addition before you commit an NV answer

Don't use it when

as proof of correctness: it cannot detect swapped digits

Example. Is $347 \times 26 = 9022$? Digit sums reduce to $3 + 4 + 7 = 14 \rightarrow 5$ and $2 + 6 = 8$; $5 \times 8 = 40 \rightarrow 4$. $9 + 0 + 2 + 2 = 13 \rightarrow 4 \checkmark$. A mismatch would prove an error.

Time saved: prevents lost marks rather than saving time: about 15 seconds per check.

Common mistake: treating a match as proof. 9202 gives the same check digit.

⚡ SHORTCUT 16 · Cancel before you multiply**Use it when**

always: every fraction with products above and below

Don't use it when

—

Example. $\frac{22}{11} \times \frac{49}{7} \times 3$: cancel 22 with 11 (giving 2) and 49 with 7×7 (giving 1), leaving $2 \times 3 = 6$. No four-digit numbers are ever written.

Time saved: 30–60 seconds per calculation, with fewer arithmetic slips.

Common mistake: rounding intermediate values (e.g. $\pi \approx 3$) and then being "between two options". Keep exact values until the last step.

PART VI · SPEED & SHORTCUT MANUAL

Calculation Fitness: Tables & Daily Drills

JEE Main allows no calculator (page 100). Hand calculation is a trainable skill, like fitness: 15 minutes a day for eight weeks makes a visible difference to both speed and error rate  **RECOMMENDATION**.

Numbers worth knowing by heart

Squares 11–30

n	n^2	n	n^2
11	121	21	441
12	144	22	484
13	169	23	529
14	196	24	576
15	225	25	625
16	256	26	676
17	289	27	729
18	324	28	784
19	361	29	841
20	400	30	900

Cubes & powers

n	n^3	2^n
6	216	64
7	343	128
8	512	256
9	729	512
10	1000	1024
11	1331	2048
12	1728	4096
13	2197	8192
14	2744	16384
15	3375	32768

Reciprocals

$\frac{1}{n}$	value	$\frac{1}{n}$	value
$\frac{1}{7}$	0.1429	$\frac{1}{14}$	0.0714
$\frac{1}{8}$	0.125	$\frac{1}{15}$	0.0667
$\frac{1}{9}$	0.1111	$\frac{1}{16}$	0.0625
$\frac{1}{11}$	0.0909	$\frac{1}{17}$	0.0588
$\frac{1}{12}$	0.0833	$\frac{1}{18}$	0.0556
$\frac{1}{13}$	0.0769	$\frac{1}{19}$	0.0526

Roots

$$\begin{aligned}\sqrt{2} &= 1.414 \cdot \sqrt{3} = 1.732 \cdot \sqrt{5} = 2.236 \cdot \sqrt{6} = 2.449 \cdot \sqrt{7} = 2.646 \cdot \sqrt{10} = 3.162 \\ \cdot \sqrt[3]{2} &= 1.26 \cdot \sqrt[3]{3} = 1.442\end{aligned}$$

Logs & exponentials

$$\begin{aligned}\log 2 &= 0.301 \cdot \log 3 = 0.477 \cdot \log 5 = 0.699 \cdot \log 7 = 0.845 \cdot \ln 2 = 0.693 \cdot \ln 10 = 2.303 \cdot e^2 = 7.39 \cdot e^{-1} = 0.368\end{aligned}$$

Factorials & trig

$$\begin{aligned}6! &= 720 \cdot 7! = 5040 \cdot 8! = 40320 \cdot 10! = 3628800 \cdot \sin 37^\circ \approx 0.6 \cdot \cos 37^\circ \approx 0.8 \text{ (the 3-4-5 triangle)}\end{aligned}$$

Mental-arithmetic methods

SHORTCUT Squares near a base

$(a \pm b)^2 = a^2 \pm 2ab + b^2$: $48^2 = (50 - 2)^2 = 2500 - 200 + 4 = 2304$. Numbers ending in 5: $n5^2 = n(n+1)$ followed by 25, so $65^2 = 4225$.

SHORTCUT Products near 100

97×94 : the deficits are 3 and 6. $97 - 6 = 91$ and $3 \times 6 = 18$, so the product is 9118. This works for any pair near a power of 10.

SHORTCUT Percentages & ratios

$x\%$ of $y = y\%$ of x : 16% of $25 = 25\%$ of $16 = 4$. Successive changes of $a\%$ and $b\%$ combine to $a + b + \frac{ab}{100}$ percent.

SHORTCUT Division by estimation

To divide by an awkward number, scale both parts first: $\frac{3.6}{0.012} = \frac{3600}{12} = 300$. Write powers of ten separately: $\frac{6 \times 10^{-7}}{3 \times 10^{-4}} = 2 \times 10^{-3}$.

The 15-minute daily drill RECOMMENDATION

1 Minutes 0–4 · Tables

Recall one table from this page (squares, cubes, reciprocals, logs) from memory. Check it; redo the misses.

2 Minutes 4–10 · Timed set

12 mixed calculations of the kind that appear in solutions: fraction simplification, powers of ten, a square root, a log. Write every step.

3 Minutes 10–13 · Algebra

3 expansions or factorisations, such as $(a + b + c)^2$ or $x^3 - y^3$, or completing the square.

4 Minutes 13–15 · Log it

Record time and errors in the tracker below. Aim for fewer errors first, then speed.

TIP Rotate the content

Week 1–2: arithmetic only. Week 3–4: add logs, powers and roots. Week 5+: take the calculations from your own mock solutions (the arithmetic you actually got wrong). This makes the drill specific to you.

REVISION Drill log (4 weeks)

Week	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Errors this week
1	☐	☐	☐	☐	☐	☐	☐	_____
2	☐	☐	☐	☐	☐	☐	☐	_____
3	☐	☐	☐	☐	☐	☐	☐	_____
4	☐	☐	☐	☐	☐	☐	☐	_____

PART VI · SPEED & SHORTCUT MANUAL

Answer-Choice Strategy

The options in an MCQ are information. Using them well is a legitimate exam skill. Guessing blindly is not: with -1 for a wrong answer, it adds risk for almost no expected gain.

The mathematics of attempting

Under the verified $+4 / -1$ scheme (page 10), suppose k options remain and you have no further information, so each is equally likely. The expected marks from attempting are:

$$E = \frac{4}{k} - \frac{k-1}{k} = \frac{5-k}{k}$$

Options remaining	Expected marks if you attempt	Probability of -1	Book rule ⓘ RECOMMENDATION
4 (pure guess)	+0.25	75%	Never.
3 (one eliminated)	+0.67	67%	Skip, unless the remaining three are not equally likely and your reasoning favours one
2 (two eliminated with certainty)	+1.50	50%	Attempt.
1 (solved)	+4.00	only if you erred	Attempt, then verify quickly

⚠ WARNING Why the book is stricter than the arithmetic

The table assumes your eliminations are **certainly right**. Under time pressure they often aren't, and every wrong answer also worsens your incorrect-to-correct ratio, which is a tie-breaker (page 10). Hence the rule used throughout this book: **attempt only after eliminating at least two options through real reasoning**.

☆ IMPORTANT Numerical-value questions: never guess

Section B answers are typed in, so there is nothing to eliminate and a guess is almost never right. Enter an NV answer only after solving it. A blank costs 0; a wrong answer costs 1.

Seven legitimate ways to use the options

1 · Impossibility filter

Remove options that are physically or mathematically impossible: probability > 1 , efficiency ≥ 1 , negative counts, wrong sign of ΔG for a spontaneous process (Technique 14).

2 · Dimensions & units

In Physics, remove every option with the wrong dimensions (Technique 1).

3 · Limiting / special cases

Test each surviving option at a special value or limit (Techniques 2 and 3).

4 · Back-substitution

If the options are numbers, substitute them into the condition (Technique 4).

5 · Estimation

If the options are far apart, compute to one significant figure (Technique 6).

6 · Structure of the options

If two options differ only in sign or by a factor of 2, the question is testing that detail. Solve for that detail specifically.

7 · Consistency across parts

In "Statement I / Statement II" and "match the columns" questions, decide the item you are surest of first. It often removes two or three options at once.

Match-the-column and statement questions

These are common in Chemistry and Physics. Don't evaluate every pair in order.

STRATEGY Method

1. Find the **one pairing you are most certain of** and eliminate every option that contradicts it.
2. Repeat with the next most certain pairing. Usually two pairings leave one option.
3. For "Statement I and II" questions, decide each statement separately (true/false) before reading the options. Only then check whether II explains I, if the options ask for it.

What not to do**MISTAKE ALERT Myths that cost marks**

- **"Option C is correct most often."** There is no reliable answer-position pattern to exploit in NTA papers, and this book's own question bank deliberately has balanced keys.
- **"The longest option is usually right."** No evidence. It is a habit from poorly written tests.
- **"If I've left many questions, I should guess a few."** No. Skipped questions cost 0. The only way to raise your score is to solve more, or to eliminate with certainty.
- **Changing an answer on a "feeling"** during the final minutes. Change an answer only if you find a specific error in your working.

Mark-for-review discipline**SHORTCUT Use the review status as a queue, not a verdict**

Use **Marked for Review** only for questions you intend to return to, with a note on your rough sheet of how far you got ("Q14: $\lambda = 2$ found, need d "). In the last 10 minutes, review only questions where you already know the next step (Exam-Hall Protocol, page [480](#)).

NOTE How marked answers are evaluated

In recent NTA instructions, a question that is answered and also "marked for review" is evaluated, while one only marked (with no answer) is not. Confirm this in the instructions on your screen before the exam starts.

THIRD-PARTY ANALYSIS

PYQ Intelligence

Trend analysis through 2026, the PYQ Priority Matrix, master priority lists and a system for working through previous-year papers.

• PYQ Trend Analysis 2024–2026	399
• The PYQ Priority Matrix	402
• Repeated Question Structures	406

• Master Priority Lists	408
• The PYQ Practice System	413

PART VII · PYQ INTELLIGENCE

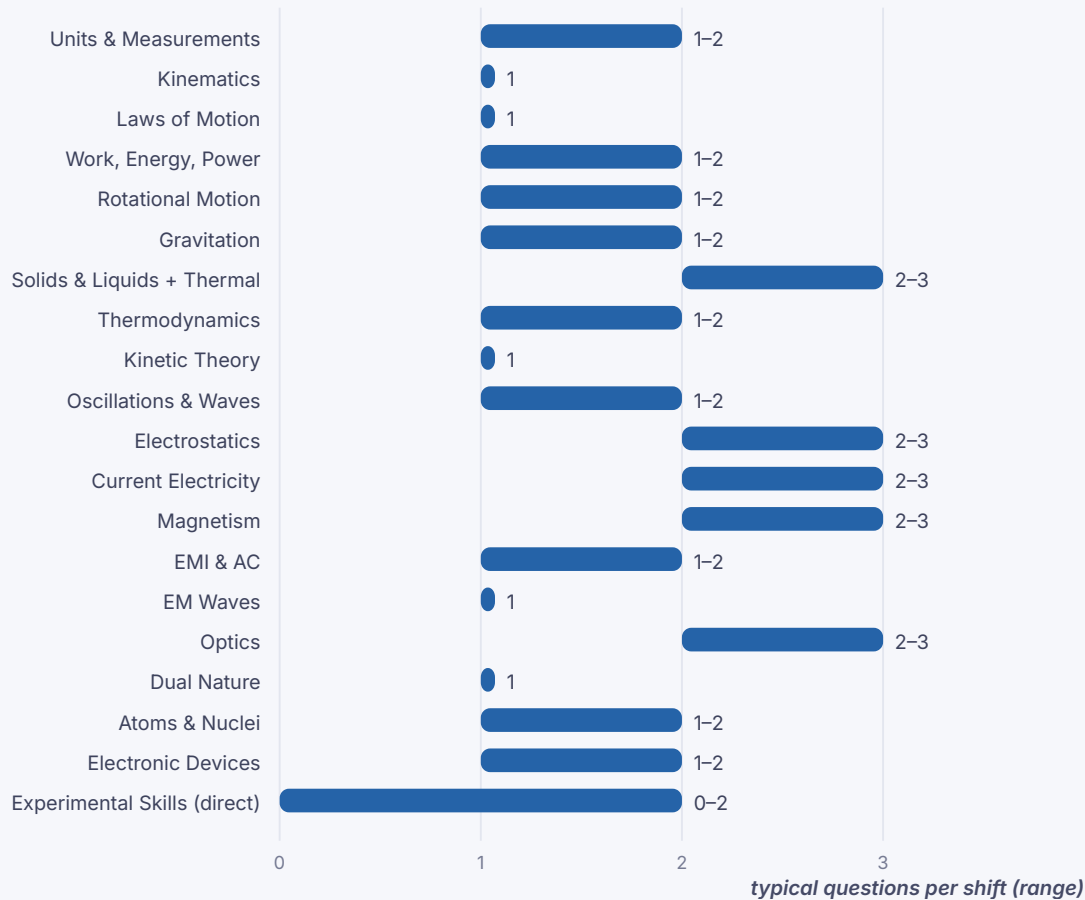
PYQ Trend Analysis 2024–2026

⚠ WARNING How to read this part

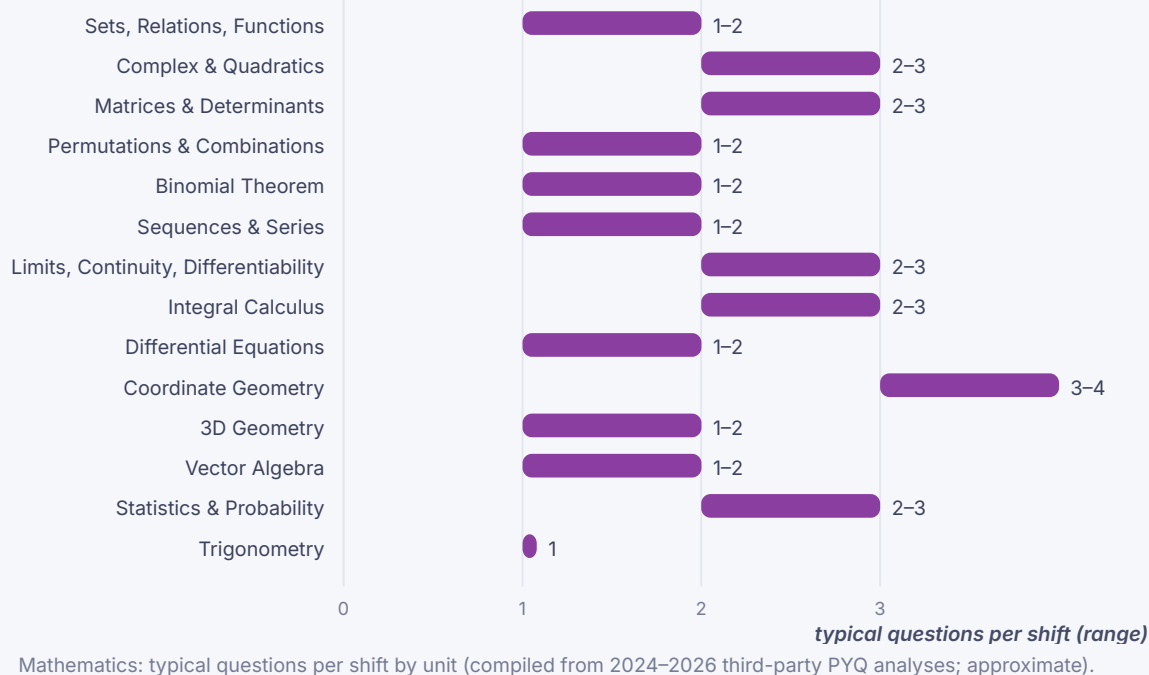
NTA does **not** publish chapter-wise statistics. Every weight in this part comes from **third-party analyses** of the 2024–2026 papers (coaching institutes and PYQ archives), compiled into ranges **THIRD-PARTY ANALYSIS** [T-WT-VED], [T-WT-CD], [T-PHY26], [T-ANALYSIS26]. The ranges are approximate: individual shifts vary, and different analysts group topics differently. The priorities built on them are this book's recommendations

RECOMMENDATION . The 2024 syllabus revision [NTA-SYL] makes 2024–2026 the only fully comparable years, so older papers are used for practice only (page [413](#)).

Unit weights at a glance



Physics: typical questions per shift by unit (compiled from 2024–2026 third-party PYQ analyses; approximate).



What the 2024–2026 papers show

Physics

- Mostly **one principle plus a clean calculation**. A minority of each paper is lengthy.
- Modern Physics, Electronic Devices, EM Waves and Units & Errors form a fast-scoring **easy block** of 5–8 questions (page [33](#)).
- Electrostatics, Magnetism, Current Electricity, Optics and Solids & Liquids carry the most weight.
- Experimental Skills (vernier, screw gauge, meter bridge) appear as direct questions in some shifts.

Chemistry

- Repeatedly described as the section where **NCERT-strong students gain their edge** [T-ANALYSIS26].
- In 2026 counts, **Coordination Compounds (36)** and **Aldehydes, Ketones & Acids (35)** were the largest chapters; GOC and Kinetics had 27 each [T-WT-CD].
- Inorganic questions often come almost directly from NCERT text and tables.
- Physical Chemistry numericals are mostly single-step with simple numbers.

Mathematics

- Repeatedly described as the **lengthiest section and the rank decider** [T-ANALYSIS26].
- Coordinate geometry, calculus and algebra (complex numbers, matrices, sequences) dominate.
- In 2026 counts, **Sequences & Series** and **Matrices & Determinants** had 33 each, 3D 29 and Vectors 27 [T-WT-CD].
- Many questions combine two units (e.g. a conic with a definite integral).

Five patterns that hold across 2024–2026 RECOMMENDATION

1 • Standard models repeat

The same question structures (listed chapter by chapter in the "Recurring structures" boxes and on page [406](#)) return every session with new numbers. Recognising a model is the fastest route to marks.

2 • Shifts differ in difficulty

Some shifts are clearly harder than others. Normalisation (page [10](#)) compensates, so a hard paper is not a disaster. Don't let a difficult start change your plan.

3 • NCERT is the base in Chemistry

Statements, exceptions, colours and tables from NCERT recur. The NCERT strategy (page [163](#)) is non-negotiable.

4 • Maths rewards selection

Because Maths is lengthy, the order of attempting (easy models first) matters more than in the other two subjects.

5 • Numerical questions are compulsory

Since 2025, all five Section B questions in each subject count, with negative marking. Numerical fluency is now worth up to 60 marks.

Topic-level shifts after the 2024 revision

VERIFIED FACT Removed in 2024 (don't study for JEE Main) [NTA-SYL], [T-DELETED]

Physics: Communication Systems; several details inside units, such as some experimental topics. **Chemistry:** States of Matter, Surface Chemistry, Hydrogen, s-Block, Metallurgy, Environmental Chemistry, Polymers, Chemistry in Everyday Life. **Maths:** Mathematical Induction, Mathematical Reasoning, Heights & Distances; several subtopics inside units. Each module's syllabus note lists what is and isn't in the official text.

NOTE Old books and old papers

Pre-2024 question banks still contain questions on removed topics. When you use them, skip those questions rather than spending time learning removed material. The one deliberate exception in this book is the two-hour plane-geometry insurance in 3D (page [342](#)).

PART VII · PYQ INTELLIGENCE

The PYQ Priority Matrix

The matrix ranks **topics** (finer than units) by four factors: how often they appear, how hard they are, how much study time they need, and how quickly their knowledge fades. It is **expert-derived**  **RECOMMENDATION** : built from the unit ranges on page [399](#) and from the question models in each module. It is not official data.

NOTE Key

Frequency: Very high (in almost every shift) · High (most shifts) · Medium (about half the shifts) · Low (occasional).

Priority: P1 Must-do · P2 High · P3 Medium · P4 Low.

Revision frequency: how often to revisit the topic once it has been learned, up to the final month (spaced revision, page [452](#)).

Physics

Topic	Unit	Frequency	Difficulty	Priority	Revise
Errors & dimensions	1	High	Easy	P1	weekly
Projectile & relative motion	2	Medium	Medium	P2	2 weeks
Friction, pulleys, circular dynamics	3	Medium	Medium	P2	2 weeks
Work–energy theorem, collisions	4	High	Medium	P2	2 weeks
Moment of inertia, rolling, angular momentum	5	High	Hard	P1	weekly
Centre of mass	5	Medium	Medium	P2	2 weeks
Orbits, escape speed, variation of g	6	High	Easy–Med	P1	2 weeks
Fluids: Bernoulli, viscosity, surface tension	7	High	Medium	P1	weekly
Elasticity	7	Medium	Easy	P2	monthly
Calorimetry, expansion, conduction, radiation	7	Medium	Medium	P2	2 weeks
Thermodynamic processes & cycles	8	High	Medium	P1	weekly
KTG: speeds, degrees of freedom, γ	9	High	Easy	P1	2 weeks
SHM: springs, pendulums, energy	10	Medium	Medium	P2	2 weeks
Waves: strings, pipes, beats, Doppler	10	Medium	Medium	P2	2 weeks
Coulomb's law, field, Gauss's law	11	High	Medium	P1	weekly
Potential, dipoles	11	Medium	Medium	P2	2 weeks
Capacitors & dielectrics	11	High	Medium	P1	weekly
Kirchhoff's laws, Wheatstone bridge, cells	12	High	Easy–Med	P1	weekly
Potentiometer, meter bridge	12	Medium	Easy	P2	monthly
Biot–Savart & Ampère	13	High	Medium	P1	weekly
Forces on charges & currents; galvanometer conversion	13	High	Medium	P1	2 weeks
Magnetic materials	13	Low	Easy	P3	monthly
Motional emf, inductance	14	Medium	Medium	P2	2 weeks
AC: LCR, resonance, power, transformer	14	Medium	Medium	P2	2 weeks
EM waves: spectrum, E_0/B_0 , energy	15	Medium	Easy	P1	monthly
Ray optics: lenses, mirrors, prism, instruments	16	High	Medium	P1	weekly
Wave optics: YDSE, diffraction, polarisation	16	Medium	Medium	P1	2 weeks

Topic	Unit	Frequency	Difficulty	Priority	Revise
Photoelectric effect & de Broglie	17	High	Easy	P1	2 weeks
Bohr model, spectra	18	Medium	Easy	P1	2 weeks
Nuclei: binding energy, decay	18	Medium	Easy	P1	2 weeks
Semiconductors, diodes, logic gates	19	High	Easy	P1	2 weeks
Experimental skills	20	Low–Medium	Easy	P2	monthly

Chemistry

Topic	Unit	Frequency	Difficulty	Priority	Revise
Mole concept, concentration terms	1	High	Easy	P2	2 weeks
Bohr model, quantum numbers, nodes	2	Medium	Easy	P1	2 weeks
Hybridisation & VSEPR shapes	3	High	Medium	P1	weekly
MOT & bond order	3	Medium	Medium	P1	2 weeks
Thermochemistry, Hess's law	4	Medium	Medium	P1	2 weeks
Entropy & Gibbs energy	4	Medium	Medium	P1	2 weeks
Colligative properties	5	Medium	Easy	P2	2 weeks
Chemical equilibrium (K_p , K_c , Le Chatelier)	6	Medium	Medium	P1	2 weeks
Ionic equilibrium (pH, buffers, K_{sp})	6	Medium	Medium	P1	weekly
Nernst equation, cells	7	Medium	Medium	P1	2 weeks
Conductance, Kohlrausch's law	7	Low–Medium	Easy	P2	monthly
Rate laws, integrated equations, half-life	8	High	Medium	P1	weekly
Arrhenius equation	8	Medium	Easy	P1	2 weeks
Periodic trends & exceptions	9	High	Easy	P1	weekly
Groups 15–17 compounds, oxoacids	10	High	Medium	P1	weekly
Groups 13–14, noble gases	10	Medium	Easy	P2	2 weeks
d-block properties, KMnO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$	11	High	Easy–Med	P1	weekly
Lanthanoids & actinoids	11	Low–Medium	Easy	P2	monthly
Werner theory, nomenclature, isomerism	12	High	Medium	P1	weekly
CFT: colour, magnetism, CFSE	12	High	Medium	P1	weekly
Purification, Lassaigne's test, estimation	13	Low–Medium	Easy	P2	monthly
GOC: effects, stability, acidity, basicity	14	Very high	Medium	P1	weekly
Isomerism (structural & stereo)	14	Medium	Medium	P1	2 weeks
Hydrocarbon reactions, aromaticity, EAS	15	Medium	Medium	P2	2 weeks
Haloalkanes: SN_1 , SN_2 , elimination	16	Medium	Medium	P2	2 weeks
Alcohols, phenols, ethers	17	High	Medium	P1	weekly
Aldehydes, ketones, carboxylic acids	17	Very high	Medium	P1	weekly
Amines & diazonium salts	18	Medium	Medium	P2	2 weeks
Biomolecules	19	Medium	Easy	P1	2 weeks
Practical: salt analysis, titrations, tests	20	Medium	Easy	P2	2 weeks

Mathematics

Topic	Unit	Frequency	Difficulty	Priority	Revise
Sets & relations (types, counting)	1	Medium	Easy	P2	monthly
Functions: domain, range, types	1	Medium	Medium	P2	2 weeks
Complex numbers: modulus, argument, locus	2	High	Medium	P1	weekly
Quadratic equations: roots, location	2	Medium	Medium	P1	2 weeks
Matrices: adjoint, inverse, properties	3	High	Easy	P1	weekly
Determinants & linear systems	3	High	Easy–Med	P1	weekly
P&C: arrangements, selection, distribution	4	Medium	Medium	P1	2 weeks
Binomial: general term, coefficient sums	5	Medium	Easy	P1	2 weeks
AP, GP, AM–GM	6	High	Medium	P1	weekly
Limits (standard forms, 1^∞)	7	Medium	Medium	P1	2 weeks
Continuity & differentiability	7	Medium	Medium	P1	2 weeks
Derivatives & applications (monotonicity, extrema)	7	Medium	Medium	P1	2 weeks
Indefinite integration	8	Medium	Medium	P2	2 weeks
Definite integrals & properties	8	High	Med–Hard	P1	weekly
Area under curves	8	Medium	Medium	P2	2 weeks
Differential equations	9	Medium	Medium	P1	2 weeks
Straight lines	10	Medium	Easy	P2	2 weeks
Circles	10	Medium	Medium	P1	2 weeks
Parabola	10	Medium	Medium	P1	2 weeks
Ellipse & hyperbola	10	Medium	Medium	P1	2 weeks
3D: lines, shortest distance, foot and image	11	High	Easy–Med	P1	weekly
Vectors: dot, cross, projection, area	12	High	Easy	P1	weekly
Statistics: mean, variance, corrections	13	Medium	Easy	P1	2 weeks
Probability: conditional, Bayes, distributions	13	High	Medium	P1	weekly
Trigonometric identities & values	14	Low–Medium	Easy–Med	P2	monthly
Inverse trigonometric functions	14	Low–Medium	Easy	P2	monthly

Priority lists by tier RECOMMENDATION

P1 • Must-do (learn to exam level, revise weekly or fortnightly)

Physics: errors; rotation; gravitation; fluids; thermodynamics; KTG; electrostatics & capacitors; circuits; magnetic effects; ray & wave optics; modern physics (dual nature, atoms, nuclei); semiconductors; EM waves.

Chemistry: atomic structure; bonding; thermodynamics; equilibrium; electrochemistry; kinetics; periodicity; p-block (15–17); d-block; coordination; GOC & isomerism; alcohols to acids; biomolecules.

Maths: complex numbers; quadratics; matrices & determinants; P&C; binomial; sequences; limits and derivatives; definite integrals; DE; circles and conics; 3D; vectors; statistics & probability.

P2 • High (learn fully, revise every 2–4 weeks)

Physics: kinematics; laws of motion; work–energy; centre of mass; SHM & waves; EMI & AC; elasticity; heat transfer; potentiometer; experimental skills.

Chemistry: mole concept; solutions; conductance; groups 13–14 and 18; lanthanoids; purification; hydrocarbons; haloalkanes; amines; practical chemistry.

Maths: sets, relations and functions; indefinite integration; area; straight lines; trigonometry and inverse trigonometry.

P3 • Medium (core ideas only, then PYQs)

Physics: magnetic materials (classification, Curie's law, B – H basics); polarisation beyond Malus and Brewster.

Chemistry: isolated NCERT facts outside the main tables (e.g. uses of individual noble gases).

Maths: mean deviation for grouped data; unusual function types (greatest-integer compositions).

P4 • Low (only after everything above is secure)

Physics: lengthy multi-body rotation problems (solve 2–3 per week for exposure, don't chase them).

Chemistry: detailed actinoid chemistry; rare named reactions not in NCERT.

Maths: planes in 3D beyond the insurance box (page [342](#)); general solutions of complicated trigonometric equations.

PART VII · PYQ INTELLIGENCE

Repeated Question Structures

These are the question structures that recur most visibly in 2024–2026 papers, classified from third-party PYQ archives [||| THIRD-PARTY ANALYSIS](#) [T-EXAMSIDE], [T-MATHONGO]. They are described here in the book's own words, not reproduced. Each has original practice questions in its module. Once you can spot these structures, a large share of every paper is familiar before you start writing.

Physics

#	Structure	What is really tested	Practise at
1	"Find the percentage error in $Q = a^p b^q / c^r$ "	adding relative errors with powers	page 36
2	Dimensions of a constant in a given equation	homogeneity	page 36
3	Rolling body on an incline: acceleration, speed or time ratio	$\frac{k^2}{R^2}$ of the body	page 63
4	Satellite: orbital speed, period, energy or escape-speed ratio	$v \propto r^{-1/2}$, $T \propto r^{3/2}$	page 70
5	Terminal velocity or capillary rise with changed radius	$v_t \propto r^2$, $h \propto 1/r$	page 76
6	P – V cycle: work, heat or efficiency	area of the loop, first law	page 84
7	v_{rms} or γ of a gas mixture	degrees of freedom	page 90
8	Capacitor with a dielectric slab, or charge sharing	series/parallel reduction	page 101
9	Galvanometer converted to an ammeter or voltmeter	shunt / series resistance	page 114
10	Field at the centre of a combination of arcs and straight wires	superposition	page 114
11	LCR at resonance: current, power factor, Q	$X_L = X_C$	page 121
12	YDSE with a changed medium or a thin sheet	$\beta = \lambda D/d$, fringe shift	page 132
13	Photoelectric: stopping potential from two wavelengths	Einstein's equation	page 139
14	Hydrogen spectrum: wavelength ratio or number of lines	Rydberg formula	page 144
15	Logic-gate circuit truth table	Boolean reduction	page 150

Chemistry

#	Structure	What is really tested	Practise at
1	Molarity/molality conversion with density	definitions	page 165
2	Number of radial/angular nodes, or quantum-number validity	n, l rules	page 170
3	Count species with a given shape, hybridisation or zero dipole	VSEPR	page 175
4	Bond order and magnetism of O_2 , N_2 and their ions	MO filling	page 175
5	ΔG , K and E° interconversion	$\Delta G^\circ = -RT \ln K = -nFE^\circ$	pages 181 , 197
6	Colligative property with a van 't Hoff factor	i for dissociation/association	page 186
7	pH of a buffer or salt solution	Henderson equation, hydrolysis	page 191
8	Half-life or order from data	integrated rate laws	page 203
9	Arrange by ionisation enthalpy or electron gain enthalpy	trends and exceptions	page 208
10	Oxoacid structures: P–OH, P–H and P=O counts	NCERT structures	page 213
11	Spin-only magnetic moment of a complex	CFT, strong/weak field	page 225

#	Structure	What is really tested	Practise at
12	Number of geometrical/optical isomers of a complex	isomerism rules	page 225
1 3	Rank acidity/basicity of substituted compounds	$-I, +M, -M$ effects	page 237
1 4	Multi-step conversion: identify A, B, C	reagent table	pages 242–264
15	Test that distinguishes two compounds	Tollens', Fehling's, iodoform, Lucas, Hinsberg	page 277

Mathematics

#	Structure	What is really tested	Practise at
1	Number of relations/functions of a type	counting	page 287
2	Locus of z from a modulus or argument condition	geometry of complex numbers	page 292
3	Conditions on the roots of a quadratic	sum/product, discriminant, location	page 292
4	Determinant of $\text{adj } kA$ or A^{-1}	determinant rules	page 298
5	System of equations: unique, none or infinite solutions	$ A $ and consistency	page 298
6	Coefficient of x^k or the term independent of x	general term	page 308
7	AP/GP with two conditions $\rightarrow$ a term or sum	symmetric representation	page 313
8	Limit of type 1^∞ or $\frac{0}{0}$ with expansions	standard limits	page 318
9	Definite integral using King's rule or even/odd	properties	page 324
10	Linear DE with an initial condition $\rightarrow$ value at a point	integrating factor	page 330
11	Conic parameters: eccentricity, latus rectum, foci	standard forms	page 335
12	Shortest distance between skew lines (often d^2 as NV)	cross-product formula	page 342
13	Vector magnitude from $ \vec{a} + \vec{b} $ conditions	squaring	page 348
14	Mean/variance after correcting or adding observations	$\sum x, \sum x^2$	page 353
15	Bayes' theorem with two or three sources	total probability	page 353

PART VII · PYQ INTELLIGENCE

Master Priority Lists

☆ IMPORTANT Expert-derived lists

Every list in this chapter is **expert-derived**  **RECOMMENDATION**. The rankings combine the third-party unit weights (page 399), the difficulty of each topic, the study time it needs, and how often it overlaps with other chapters. They are a judgement, not official data. Use them to decide the order of work, and to choose what to revise when time is short.

Top 20 topics per subject

Physics

1. Electrostatics & capacitors
2. Current electricity
3. Magnetic effects of current
4. Ray optics
5. Rotational motion
6. Modern physics (photoelectric, Bohr, nuclei)
7. Semiconductors & logic gates
8. Thermodynamics
9. Fluids & surface tension
10. Wave optics (YDSE)
11. Gravitation
12. Kinetic theory
13. Errors & dimensions
14. EMI & AC
15. Work, energy & collisions
16. SHM
17. Waves & sound
18. Heat transfer & calorimetry
19. EM waves
20. Laws of motion & friction

Chemistry

1. Coordination compounds
2. Aldehydes, ketones & acids
3. GOC (effects, stability)
4. Chemical kinetics
5. p-Block (groups 15–17)
6. Chemical bonding
7. Chemical thermodynamics
8. Electrochemistry
9. Equilibrium (ionic + chemical)
10. d- & f-Block
11. Alcohols, phenols & ethers
12. Periodicity
13. Atomic structure
14. Amines & diazonium salts
15. Biomolecules
16. Haloalkanes & haloarenes
17. Hydrocarbons
18. Solutions
19. Practical chemistry
20. Mole concept

Mathematics

1. Definite integrals & area
2. Matrices & determinants
3. Sequences & series
4. Conics (parabola, ellipse, hyperbola)
5. 3D geometry
6. Vectors
7. Probability (incl. Bayes)
8. Complex numbers
9. Differential equations
10. Limits, continuity & differentiability
11. Binomial theorem
12. Permutations & combinations
13. Circles
14. Applications of derivatives
15. Quadratic equations
16. Statistics
17. Functions
18. Straight lines
19. Sets & relations
20. Inverse trigonometry

The 30 most important formulas

#	Formula	Why it ranks
1	$\frac{\Delta Q}{Q} = \sum p_i \frac{\Delta a_i}{a_i}$	error questions in most shifts
2	$K_{max} = h\nu - \phi$	photoelectric effect, every year
3	$E_n = -13.6 Z^2/n^2 \text{ eV}$	Bohr model in Physics and Chemistry
4	$a = \frac{g \sin \theta}{1 + k^2/R^2}$	rolling
5	$v_o = \sqrt{GM/r}, v_e = \sqrt{2GM/R}$	orbits
6	$\Delta Q = \Delta U + W$ with the process table	thermodynamics
7	$v_{rms} = \sqrt{3RT/M}, \gamma = 1 + 2/f$	kinetic theory

#	Formula	Why it ranks
8	$C = \frac{\epsilon_0 A}{d - t + t/K}$	capacitors
9	$B = \frac{\mu_0 I}{2\pi r}, \frac{\mu_0 I}{2R}, \mu_0 nI$	magnetic fields
10	$Z = \sqrt{R^2 + (X_L - X_C)^2}$	AC circuits
11	$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ and the lens-maker formula	ray optics
12	$\beta = \lambda D/d$	YDSE
13	$\Delta G^\circ = -RT \ln K = -nFE^\circ$	links three chapters
14	$E = E^\circ - \frac{0.0591}{n} \log Q$	electrochemistry
15	$t_{1/2} = 0.693/k$ and $k = \frac{2.303}{t} \log \frac{[A]_0}{[A]}$	kinetics
16	$\text{pH} = \text{p}K_a + \log \frac{[\text{salt}]}{[\text{acid}]}$	buffers
17	$\Delta T_f = iK_f m, \pi = iCRT$	colligative properties
18	$\mu = \sqrt{n(n+2)} \text{ BM}$	d-block and coordination
19	$\text{CFSE} = (-0.4n_{t_{2g}} + 0.6n_{e_g})\Delta_o$	coordination
20	$K_p = K_c(RT)^{\Delta n_g}$	equilibrium
21	$T_{r+1} = {}^nC_r x^{n-r} y^r$	binomial
22	$ \text{adj } A = A ^{n-1}, kA = k^n A $	determinants
23	$\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$	definite integrals
24	$\text{IF} = e^{\int P dx}$	differential equations
25	Shortest distance $\frac{ (\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) }{ \vec{b}_1 \times \vec{b}_2 }$	3D
26	e , foci and latus rectum of the standard conics	coordinate geometry
27	S_n of an AP/GP, $S_\infty = \frac{a}{1-r}$	sequences
28	$\sigma^2 = \frac{\sum x^2}{n} - \bar{x}^2$	statistics
29	Bayes' theorem	probability
30	$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ and the 1^∞ rule	limits

The 20 most important reactions

1. **Aldol condensation** (α -H carbonyls, dilute base)
2. **Cannizzaro reaction** (no α -H, concentrated base)
3. **Hofmann bromamide degradation** (one carbon fewer)
4. **Reimer-Tiemann reaction** (salicylaldehyde)
5. **Kolbe's reaction** (salicylic acid)
6. **Sandmeyer reaction** (diazonium $\rightarrow$ Cl, Br, CN)
7. **Diazo coupling** (azo dyes with phenol or aniline)
8. **Gabriel phthalimide synthesis** (1° amines)
9. **HVZ reaction** (α -halo acids)
10. **Friedel-Crafts alkylation & acylation**
11. **Ozonolysis** (locating the double bond)
12. **Markovnikov vs peroxide (anti-Markovnikov) addition of HBr**
13. **Clemmensen & Wolff-Kishner reductions**
14. **Rosenmund & Stephen reductions** (to aldehydes)
15. **Etard & Gattermann-Koch reactions** (benzaldehyde)
16. **Williamson ether synthesis**
17. **Wurtz reaction**
18. **SN1 vs SN2** (substrate, solvent, stereochemistry)
19. **Iodoform reaction** (CH_3CO^- / $\text{CH}_3\text{CH}(\text{OH})^-$)
20. **KMnO₄ and K₂Cr₂O₇ redox** in acidic medium (inorganic)

The 20 most important NCERT facts

#	Fact
1	N ₂ is unreactive because of its very strong triple bond (bond enthalpy about 941 kJ/mol).
2	Solid PCl ₅ exists as [PCl ₄] ⁺ [PCl ₆] ⁻ .
3	H ₃ PO ₃ is dibasic (two P-OH); H ₃ PO ₂ is monobasic and a strong reducing agent.
4	Boric acid is a weak monobasic Lewis acid: it accepts OH ⁻ rather than donating H ⁺ .
5	Diborane has two three-centre two-electron (B-H-B) "banana" bonds.
6	XeF ₂ is linear, XeF ₄ square planar, XeF ₆ distorted octahedral; ClF ₃ is T-shaped.
7	O ₂ is paramagnetic; O ₃ is diamagnetic.
8	Hydride acidity increases down group 16: H ₂ O < H ₂ S < H ₂ Se < H ₂ Te.
9	Mn shows the highest oxidation state (+7) in the 3d series.
10	Cu ⁺ disproportionates in water: $2\text{Cu}^+ \rightarrow \text{Cu}^{2+} + \text{Cu}$.
11	The colour of KMnO ₄ is due to charge transfer, not d-d transitions (Mn(VII) is d ⁰).
12	Lanthanoid contraction makes Zr and Hf almost the same size.
13	Ce ⁴⁺ is an oxidising agent; Eu ²⁺ and Yb ²⁺ are reducing agents.
14	[Ni(CO) ₄]: tetrahedral, diamagnetic • [Ni(CN) ₄] ²⁻ : square planar, diamagnetic • [NiCl ₄] ²⁻ : tetrahedral, paramagnetic.
15	EDTA is a hexadentate ligand.
16	Glucose pentaacetate doesn't react with hydroxylamine, one evidence for the cyclic structure of glucose.
17	Sucrose is non-reducing; maltose and lactose are reducing sugars.
18	Water-soluble vitamins (B, C) must be taken regularly, except B ₁₂ ; fat-soluble ones (A, D, E, K) are stored in the body.
19	Denaturation destroys the secondary and tertiary structure of a protein; the primary structure stays intact.
20	Aniline doesn't undergo Friedel-Crafts reactions (it forms a salt with AlCl ₃).

The 15 most important question types

1. Shortest distance between lines / foot of the perpendicular (Maths)
2. Definite integral by properties (Maths)
3. Determinant and adjoint properties; systems of equations (Maths)
4. Bayes' theorem and conditional probability (Maths)

5. Conic parameters (Maths)
6. Error propagation and dimensions (Physics)
7. Photoelectric effect and de Broglie wavelength (Physics)
8. Capacitor combinations with dielectrics (Physics)
9. Magnetic field of wire and arc combinations (Physics)
10. Thermodynamic cycles and processes (Physics)
11. Spin-only moment and isomer counting for complexes (Chemistry)
12. Multi-step organic conversion (identify A, B, C) (Chemistry)
13. Order of acidity, basicity or stability (Chemistry)
14. Kinetics: half-life or order from data (Chemistry)
15. Statement I / Statement II on NCERT facts (Chemistry)

The 15 most costly mistakes

MISTAKE ALERT Ranked by estimated cost and frequency **RECOMMENDATION**

1. Misreading "incorrect", "not", "false" or "except" in the question.
2. Unit slips: cm vs m, kJ vs J, mL vs L, °C vs K.
3. Sign-convention errors (optics, thermodynamics, work in chemistry vs physics).
4. Answering a different quantity than the one asked (radius instead of diameter, d instead of d^2).
5. Arithmetic or transcription slips on the rough sheet.
6. Using a formula outside its conditions (constant acceleration, small α , $|r| < 1$).
7. Wrong n -factor, or Δn counted with non-gaseous species.
8. Principal-value errors in inverse trigonometry.
9. Counting errors: the +1 in the number of terms, or overlapping cases in P&C.
10. Assuming events are independent, or mutually exclusive, without justification.
11. Forgetting the $\frac{1}{2}$ (triangle area, kinetic energy, dipole energy density).
12. Over-attempting: guessing with fewer than two options eliminated.
13. Spending more than 3 minutes on one question in Round 1.
14. Not converting Section B answers as instructed (rounding, units).
15. Changing a correct answer without finding a specific error.

The Mistake Book (page [449](#)) turns this list into a prevention system.

The 15 highest-return revision topics

For the final month, these give the most marks per hour of revision  **RECOMMENDATION** : they are formula- or fact-direct, frequent, and quick to refresh.

#	Topic	Subject	Why
1	Dual nature, atoms & nuclei	Physics	formula-direct, 2–3 questions
2	Semiconductors & logic gates	Physics	fact- and truth-table-based
3	EM waves	Physics	1 question, 15-minute revision
4	Errors & dimensions	Physics	almost every shift
5	Kinetic theory	Physics	few formulas
6	Biomolecules	Chemistry	pure NCERT
7	Periodicity exceptions	Chemistry	short list
8	Coordination (CFT, isomers)	Chemistry	largest 2026 chapter
9	p-Block NCERT tables and structures	Chemistry	direct recall
10	Practical chemistry tests	Chemistry	direct recall
11	Vectors & 3D	Maths	short formula set, 2–3 questions
12	Matrices & determinants	Maths	rule-based, fast
13	Binomial theorem	Maths	one formula family
14	Statistics	Maths	two formulas cover most questions
15	Probability (Bayes)	Maths	one standard model

PART VII · PYQ INTELLIGENCE

The PYQ Practice System

Previous-year questions are the most realistic practice available. This book doesn't reproduce them. It tells you **where to find them, in what order to use them, and how to extract the most from each one.**

Where to get the papers

Source	What you get	Trust level
NTA (jeemain.nta.nic.in)	Question papers with provisional answer keys during each session's challenge window, then final answer keys	Official. The final key decides correctness [NTA-WEB]
PYQ archives such as ExamSIDE and MathonGo	Shift-wise and chapter-wise PYQs with solutions	Third-party. Excellent for practice, but check disputed answers against the NTA final key THIRD-PARTY ANALYSIS [T-EXAMSIDE], [T-MATHONGO]
This book	Original questions modelled on the repeated structures (page 406)	Original practice, not PYQs



NTA JEE Main (answer keys & notices)

jeemain.nta.nic.in

The three-phase plan 📌 RECOMMENDATION

- 1 **Phase 1 · Chapter-wise (while studying a chapter)**
After each module, solve that chapter's 2024–2026 PYQs, timed at 2 minutes per question. Log every error in the Mistake Book.
- 2 **Phase 2 · Mixed revision (after the syllabus is complete)**
Solve PYQs unit-mixed: 30 questions from 6 chapters per session. Mixing forces you to choose the method, which is what the exam requires [LS-ROH].
- 3 **Phase 3 · Full shifts as mocks (last 45–60 days)**
Solve complete shifts in 3 hours at your exam time. **Keep the 6 most recent shifts untouched** until the final three weeks, so your last mocks are papers you haven't seen.

🌟 IMPORTANT Older papers (2019–2023)

They are still useful for Physics and Maths practice, but skip questions on topics removed in 2024 (page [399](#)). Don't count scores on old papers as mock scores: the pattern and syllabus have changed.

Getting full value from every PYQ

📌 STRATEGY The 4-question review (after each PYQ set)

1. **Did I recognise the structure within 30 seconds?** If not, which clue should have told me?
2. **Was my method the shortest one?** Compare it with the solution. If a shortcut exists, add it to your shortcut list.
3. **If I got it wrong, which mistake type was it?** Classify it with the 8 mistake types (page [449](#)).
4. **Would I get a variant right?** Change one number or condition and re-solve it in your head.

Shift-wise tracker (2024–2026 papers)

Tick a shift once you have attempted it as a timed full paper, and write your score. Dates are as reported for Paper 1 (B.E./B.Tech.) **THIRD-PARTY ANALYSIS** [R-DATES24], [R-DATES25], [R-DATES26]. A dash marks a slot with no Paper-1 shift (8 April 2025 had only the afternoon shift).

2024

Date	S1	Score	S2	Score
27 Jan	☐	_____	☐	_____
29 Jan	☐	_____	☐	_____
30 Jan	☐	_____	☐	_____
31 Jan	☐	_____	☐	_____
1 Feb	☐	_____	☐	_____
4 Apr	☐	_____	☐	_____
5 Apr	☐	_____	☐	_____
6 Apr	☐	_____	☐	_____
8 Apr	☐	_____	☐	_____
9 Apr	☐	_____	☐	_____

2025

Date	S1	Score	S2	Score
22 Jan	☐	_____	☐	_____
23 Jan	☐	_____	☐	_____
24 Jan	☐	_____	☐	_____
28 Jan	☐	_____	☐	_____
29 Jan	☐	_____	☐	_____
2 Apr	☐	_____	☐	_____
3 Apr	☐	_____	☐	_____
4 Apr	☐	_____	☐	_____
7 Apr	☐	_____	☐	_____
8 Apr	—	—	☐	_____

2026

Date	S1	Score	S2	Score
21 Jan	☐	_____	☐	_____
22 Jan	☐	_____	☐	_____
23 Jan	☐	_____	☐	_____
24 Jan	☐	_____	☐	_____
28 Jan	☐	_____	☐	_____
2 Apr	☐	_____	☐	_____
4 Apr	☐	_____	☐	_____
5 Apr	☐	_____	☐	_____
6 Apr	☐	_____	☐	_____
8 Apr	☐	_____	☐	_____



TIP Chapter-wise PYQ tracking

The chapter-by-chapter PYQ tracker (2024, 2025 and 2026 columns for every unit) is in the Trackers part, page **497**.

PART VIII

Question Bank & Mock Tests

Mixed, calculation-heavy, tricky and time-pressure sets, a full-length original mock test, and the mock-test system with analysis sheets.

• The Question Bank	417
• Full-Length Mock Test 1	429

• The Mock-Test System	444
• Mock Analysis Sheet	446

PART VIII · QUESTION BANK & MOCK TESTS

The Question Bank

Sixty original questions in six sets, each built for a different exam skill. All are written for this book and modelled on the repeated structures (page 406). None is copied from a past paper.

Set	Skill trained	Questions	Target time
QB-A	Mixed Physics	10	16 min
QB-B	Mixed Chemistry	10	12 min
QB-C	Mixed Mathematics	10	18 min
QB-D	Calculation-heavy (by hand)	8	24 min
QB-E	Tricky / trap-laden	10	15 min
QB-F	Time pressure (sprint)	12	12 min

🔗 STRATEGY How to use the sets 🔗 RECOMMENDATION

Take each set **timed** and in one sitting, after the relevant chapters are done. Mark your confidence (sure / 50-50) next to each answer before checking. Take QB-E only after QB-A to QB-C: its purpose is to see whether you fall for traps when you aren't warned. Take QB-F again every two weeks, since your time on it is a clean measure of recall speed.

📌 QB-A · Mixed Physics

1 Easy ⌚ 1 min Kinematics

QB-A01

A ball is thrown vertically upward at 20 m/s ($g = 10 \text{ m/s}^2$). Its maximum height and total time of flight are:

- (A) 20 m, 4 s (B) 40 m, 4 s (C) 20 m, 2 s (D) 10 m, 2 s

2 Medium ⌚ 1.5 min Rotation

QB-A02

A solid sphere and a hollow sphere of the same mass and radius roll without slipping down the same incline from rest. The ratio of their speeds at the bottom (solid : hollow) is:

- (A) 1 (B) $\sqrt{21}/5$ (C) $5/\sqrt{21}$ (D) $7/5$

3 Medium ⌚ 1.5 min Gravitation NV

QB-A03

A satellite orbits at a height R above Earth's surface, where $R = 6400 \text{ km}$ is Earth's radius and $g = 10 \text{ m/s}^2$ at the surface. Its orbital speed is $\sqrt{x} \text{ km/s}$. Find x .

Your answer (numerical value):

4 Medium ⌚ 1.5 min Thermodynamics

QB-A04

One mole of an ideal monatomic gas is heated at constant pressure through 100 K ($R = 8.3 \text{ J/mol K}$). The heat supplied is:

- (A) 2905 J (B) 1245 J (C) 830 J (D) 2075 J

5 Easy ⌚ 0.75 min Kinetic theory

QB-A05

The rms speed of oxygen molecules at temperature T is v . At the same temperature, the rms speed of hydrogen molecules is:

- (A) $v/4$ (B) $4v$ (C) $16v$ (D) $2v$

6

Medium

⌚ 1.5 min

Capacitors

NV

QB-A06

An air-filled parallel-plate capacitor has capacitance $10\ \mu\text{F}$. A slab of dielectric constant 4, with thickness half the plate separation, is inserted parallel to the plates and covers their full area. Find the new capacitance in μF .

Your answer (numerical value):

7

Easy

⌚ 1 min

Current electricity

NV

QB-A07

A cell of emf $12\ \text{V}$ and internal resistance $1\ \Omega$ is connected to a variable external resistor. Find the maximum power (in W) that can be delivered to the resistor.

Your answer (numerical value):

8

Medium

⌚ 1.5 min

Magnetism

QB-A08

A proton and an alpha particle with equal kinetic energies move perpendicular to the same uniform magnetic field. The ratio of their radii $r_p : r_\alpha$ is:

(A) 2 : 1

(B) 1 : 2

(C) 1 : 1

(D) $1 : \sqrt{2}$

9

Easy

⌚ 1 min

Optics

QB-A09

A convex lens of focal length $20\ \text{cm}$ forms a real image $60\ \text{cm}$ from the lens. The object distance is:

(A) $30\ \text{cm}$ (B) $15\ \text{cm}$ (C) $40\ \text{cm}$ (D) $60\ \text{cm}$

10

Easy

⌚ 1 min

Modern physics

NV

QB-A10

Light of wavelength $310\ \text{nm}$ falls on a metal with work function $2\ \text{eV}$. Find the stopping potential in volts.

Your answer (numerical value):



QB-B · Mixed Chemistry

1

Easy

⌚ 0.75 min

Mole concept

NV

QB-B01

How many moles of oxygen atoms are present in $49\ \text{g}$ of H_2SO_4 ($M = 98$)?

Your answer (numerical value):

2

Easy

⌚ 0.5 min

Atomic structure

QB-B02

The number of radial nodes in a $3p$ orbital is:

(A) 3

(B) 0

(C) 2

(D) 1

3

Easy

⌚ 0.5 min

Bonding

QB-B03

Which molecule has zero dipole moment?

(A) BF_3 (B) NH_3 (C) H_2O (D) CHCl_3

4

Medium

⌚ 1 min

Thermodynamics

NV

QB-B04

For a reaction, $\Delta H = +30\ \text{kJ/mol}$ and $\Delta S = +100\ \text{J K}^{-1}\text{mol}^{-1}$. Find the temperature (in K) above which it is spontaneous.

Your answer (numerical value):

5 Easy 0.75 min Equilibrium

QB-B05

The pH of a solution that is 0.1 M in CH_3COOH and 0.1 M in CH_3COONa ($\text{p}K_a = 4.74$) is:

- (A) 9.26 (B) 4.74 (C) 7.00 (D) 2.87

6 Easy 0.75 min Kinetics NV

QB-B06

A first-order reaction has a half-life of 10 min. Find the time (in min) for 87.5% completion.

Your answer (numerical value):

7 Medium 1 min Coordination

QB-B07

The spin-only magnetic moment of $[\text{Fe}(\text{CN})_6]^{3-}$ is:

- (A) 5.92 BM (B) 1.73 BM (C) 3.87 BM (D) 0 BM

8 Medium 1 min Periodicity

QB-B08

The correct order of first ionisation enthalpy is:

- (A) $\text{Be} < \text{B} < \text{C} < \text{N}$ (B) $\text{Be} < \text{B} < \text{O} < \text{N}$ (C) $\text{B} < \text{Be} < \text{N} < \text{O}$ (D) $\text{B} < \text{Be} < \text{O} < \text{N}$

9 Easy 0.75 min GOC

QB-B09

The most acidic compound among these is:

- (A) phenol (B) *p*-cresol (C) *p*-nitrophenol (D) ethanol

10 Medium 1 min Carbonyls

QB-B10

A compound $\text{C}_3\text{H}_6\text{O}$ gives a positive iodoform test and a negative Tollens' test. It is:

- (A) prop-2-en-1-ol (B) propanal (C) propanone (D) methoxyethene

QB-C · Mixed Mathematics

1 Medium 1.5 min Complex numbers

QB-C01

The locus of z satisfying $|z - 2| = |z + 2i|$ is:

- (A) $x + y = 0$ (B) $x - y = 0$ (C) $x + y = 2$ (D) a circle

2 Medium 1.5 min Matrices NV

QB-C02

A is a 3×3 matrix with $|A| = 4$. Find $|\text{adj}(2A)|$.

Your answer (numerical value):

3 Easy 1 min Binomial NV

QB-C03

Find the term independent of x in $\left(x^2 + \frac{1}{x}\right)^6$.

Your answer (numerical value):

4 Easy 0.75 min P&C NV

QB-C04

In how many ways can the letters of BANANA be arranged?

Your answer (numerical value):

5 Medium 2 min Sequences NV

QB-C05

An infinite GP has sum 20, and the sum of the squares of its terms is 100. Find its first term.

Your answer (numerical value):

6 Easy 1 min Limits NV

QB-C06

Find $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$.

Your answer (numerical value):

7 Medium 1.5 min Definite integrals

QB-C07

$\int_0^\pi x \sin x \, dx$ equals:

- (A) $\pi/2$ (B) π (C) 2π (D) 0

8 Easy 1 min Differential equations NV

QB-C08

If $\frac{dy}{dx} = \frac{y}{x}$ and $y(1) = 2$, find $y(3)$.

Your answer (numerical value):

9 Easy 1 min Conics

QB-C09

The eccentricity of the hyperbola $9x^2 - 16y^2 = 144$ is:

- (A) $3/4$ (B) $4/5$ (C) $5/3$ (D) $5/4$

10 Easy 1 min Probability

QB-C10

Two cards are drawn without replacement from a pack of 52. The probability that both are aces is:

- (A) $1/221$ (B) $1/169$ (C) $1/13$ (D) $2/221$

QB-D · Calculation-Heavy

1 Medium 2.5 min Laws of motion NV

QB-D01

A 2 kg block is pushed up a rough incline ($\theta = 37^\circ$, $\mu = 0.5$) at constant velocity by a force parallel to the incline. Take $g = 10 \text{ m/s}^2$, $\sin 37^\circ = 0.6$. Find the force in newtons.

Your answer (numerical value):

2 Medium 2.5 min Wave optics NV

QB-D02

In a YDSE, $d = 0.3 \text{ mm}$, $D = 1.5 \text{ m}$ and $\lambda = 600 \text{ nm}$. Find the distance (in mm) between the 2nd bright fringe on one side of the central maximum and the 4th bright fringe on the other side.

Your answer (numerical value):

3 Medium 2.5 min AC NV

QB-D03

A series LCR circuit has $R = 30 \, \Omega$, $X_L = 80 \, \Omega$ and $X_C = 40 \, \Omega$, and is connected to a 200 V (rms) supply. Find the average power in watts.

Your answer (numerical value):

4 Medium 2 min Electrochemistry

QB-D04

For $\text{Zn} | \text{Zn}^{2+} (0.01 \text{ M}) || \text{Cu}^{2+} (1 \text{ M}) | \text{Cu}$ with $E^\circ = 1.10 \text{ V}$ at 298 K , the cell emf is:

- (A) 1.159 V (B) 1.041 V (C) 1.10 V (D) 1.218 V

5 Medium 1.5 min Solutions

QB-D05

The freezing-point depression of 0.1 molal aqueous K_2SO_4 , assuming complete dissociation ($K_f = 1.86 \text{ K kg/mol}$), is:

- (A) 0.186 K (B) 0.558 K (C) 0.372 K (D) 0.744 K

6 Medium 2 min Area

QB-D06

The area bounded by $y = x^2$ and $y = 2x$ is:

- (A) $8/3$ (B) $4/3$ (C) $2/3$ (D) $16/3$

7 Medium 2.5 min Statistics NV

QB-D07

Ten observations have mean 5 and variance 4. Two observations, 1 and 9, are found to be invalid and removed. Find the variance of the remaining eight.

Your answer (numerical value):

8 Hard 3 min 3D NV

QB-D08

If d is the shortest distance between $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$, find $6d^2$.

Your answer (numerical value):

QB-E · Tricky & Trap-Laden

1 Medium 1.5 min Rotation

QB-E01

A uniform rod of length L , pivoted at one end, is released from rest in a horizontal position. Its initial angular acceleration is:

- (A) $2g/L$ (B) $6g/L$ (C) $3g/(2L)$ (D) g/L

2 Easy 0.5 min Concept

QB-E02

A particle moves in a circle at constant speed. Which quantity stays constant?

- (A) velocity (B) acceleration (C) momentum (D) kinetic energy

3 Easy 0.75 min Circuits

QB-E03

Resistors of 4Ω and 12Ω are connected in parallel across an ideal 6 V battery. The current through the 4Ω resistor is:

- (A) 2 A (B) 0.5 A (C) 1.5 A (D) 1 A

4 Medium 1 min Waves NV

QB-E04

A pipe 50 cm long is closed at one end. Taking $v = 340 \text{ m/s}$, find the frequency (in Hz) of its second overtone.

Your answer (numerical value):

5 Easy 0.5 min Biomolecules

QB-E05

Which of the following is **not** a reducing sugar?

- (A) lactose (B) glucose (C) maltose (D) sucrose

6 Medium 1 min Ionic equilibrium

QB-E06

The pH of 10^{-8} M HCl at 25 °C is approximately:

- (A) 6 (B) 8 (C) 7 (D) 6.98

7 Medium 1 min Isomerism NV

QB-E07

How many structural isomers of $C_4H_{10}O$ are alcohols?

Your answer (numerical value):

8 Medium 1 min Quadratics NV

QB-E08

Find the number of real roots of $x^2 - 5|x| + 6 = 0$.

Your answer (numerical value):

9 Easy 1 min Probability

QB-E09

If $P(A) = 0.6$, $P(B) = 0.5$ and $P(A \cup B) = 0.8$, then $P(A | B)$ equals:

- (A) 0.5 (B) 0.3 (C) 0.6 (D) 0.75

10 Medium 1.5 min Functions

QB-E10

The domain of $f(x) = \sqrt{\log_{1/2}(x-1)}$ is:

- (A) $(1, 2]$ (B) $[2, \infty)$ (C) $(1, \infty)$ (D) $[1, 2]$

QB-F · Time-Pressure Sprint (target: 12 minutes)

1 Easy 0.5 min Speed

QB-F01

The dimensional formula of impulse is:

- (A) MLT^{-2} (B) MLT^{-1} (C) ML^2T^{-1} (D) $ML^{-1}T$

2 Easy 0.5 min Speed

QB-F02

The de Broglie wavelength of an electron accelerated through 100 V is about:

- (A) $1.23 \text{ \AA}$ (B) $12.3 \text{ \AA}$ (C) $0.123 \text{ \AA}$ (D) $123 \text{ \AA}$

3 Easy 0.5 min Speed

QB-F03

A nuclide has a half-life of 5 days. The fraction left after 15 days is:

- (A) $1/3$ (B) $1/8$ (C) $1/6$ (D) $1/16$

4 Easy 0.5 min Speed NV

QB-F04

Find the equivalent capacitance (in μF) of three $3 \mu F$ capacitors in series.

Your answer (numerical value):

5 Easy 0.5 min Speed

QB-F05

The hybridisation of Xe in XeF_4 is:

- (A) $sp^3 d^2$ (B) $sp^3 d$ (C) sp^3 (D) dsp^2

6 Easy 0.5 min Speed NV

QB-F06

Find the oxidation state of Cr in $\text{K}_2\text{Cr}_2\text{O}_7$.Your answer (numerical value):

7 Easy 0.5 min Speed NV

QB-F07

How many σ bonds are there in ethyne?Your answer (numerical value):

8 Easy 0.5 min Speed NV

QB-F08

How many unpaired electrons does a ground-state Cr atom ($Z = 24$) have?Your answer (numerical value):

9 Easy 0.5 min Speed

QB-F09

 i^{2027} equals:

- (A) $-i$ (B) i (C) -1 (D) 1

10 Easy 0.5 min Speed NV

QB-F10

Find ${}^{10}C_3$.Your answer (numerical value):

11 Easy 0.5 min Speed NV

QB-F11

Find $\frac{d}{dx}e^{2x}$ at $x = 0$.Your answer (numerical value):

12 Easy 0.5 min Speed

QB-F12

The slope of $3x - 4y + 7 = 0$ is:

- (A) $-3/4$ (B) $3/4$ (C) $4/3$ (D) $-4/3$

Answers & Solutions

QB-A · Mixed Physics

1 (A) 2 (C) 3 32 4 (D) 5 (B) 6 16 7 36 8 (C) 9 (A) 10 2

QB-B · Mixed Chemistry

1 2 2 (D) 3 (A) 4 300 5 (B) 6 30 7 (B) 8 (D) 9 (C) 10 (C)

QB-C · Mixed Mathematics

1 (A) 2 1024 3 15 4 60 5 8 6 2 7 (B) 8 6 9 (D) 10 (A)

QB-D · Calculation-Heavy

1 20 2 18 3 480 4 (A) 5 (B) 6 (B) 7 1 8 1

QB-E · Tricky & Trap-Laden

1	(C)	2	(D)	3	(C)	4	850	5	(D)	6	(D)	7	4	8	4	9	(C)	10	(A)
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QB-F · Time-Pressure Sprint (target: 12 minutes)

1	(B)	2	(A)	3	(B)	4	1	5	(A)	6	6	7	3	8	6	9	(A)	10	120
11	2	12	(B)																

QB-A · MIXED PHYSICS

1 **Answer: (A)** Vertical projection **Easy** ⌚ 1 min Kinematics

QB-A01

SOLUTION $H = \frac{u^2}{2g} = \frac{400}{20} = 20 \text{ m}; T = \frac{2u}{g} = 4 \text{ s}.$

SHORTCUT Time up = $u/g = 2 \text{ s}$, so the total is 4 s.

COMMON TRAP Taking the time to the top (2 s) as the total time.

2 **Answer: (C)** Rolling race **Medium** ⌚ 1.5 min Rotation

QB-A02

SOLUTION $v^2 = \frac{2gh}{1 + k^2/R^2}$: solid $\frac{10gh}{7}$, hollow $\frac{6gh}{5}$. Ratio = $\sqrt{\frac{10/7}{6/5}} = \sqrt{\frac{25}{21}}.$

SHORTCUT The body with the smaller k^2/R^2 is faster, so the ratio must exceed 1. Only (C) and (D) do, and (D) is a ratio of $1 + k^2/R^2$, not of speeds.

COMMON TRAP Choosing 1 because "mass and radius are equal".

3 **Answer: 32** Orbital speed **Medium** ⌚ 1.5 min Gravitation NV

QB-A03

SOLUTION $v = \sqrt{\frac{GM}{2R}} = \sqrt{\frac{gR}{2}} = \sqrt{\frac{10 \times 6.4 \times 10^6}{2}} \text{ m/s} = \sqrt{3.2 \times 10^7} \text{ m/s} = \sqrt{32} \text{ km/s}.$

SHORTCUT $gR = 64 \text{ km}^2/\text{s}^2$, so $v^2 = 32 \text{ km}^2/\text{s}^2$.

COMMON TRAP Using $r = R$ instead of $2R$ (gives 64).

4 **Answer: (D)** Heat at constant pressure **Medium** ⌚ 1.5 min Thermodynamics

QB-A04

SOLUTION $Q = nC_P\Delta T = \frac{5}{2}(8.3)(100) = 2075 \text{ J}.$

SHORTCUT $Q : \Delta U : W = 5 : 3 : 2$ for a monatomic gas at constant pressure.

COMMON TRAP Using C_V (1245 J is ΔU ; 830 J is the work).

5 **Answer: (B)** rms speed ratio **Easy** ⌚ 0.75 min Kinetic theory

QB-A05

SOLUTION $v_{rms} \propto \frac{1}{\sqrt{M}}; \sqrt{32/2} = 4.$

COMMON TRAP Forgetting the square root (16v).

6 **Answer: 16** Partial dielectric slab **Medium** ⌚ 1.5 min Capacitors NV

QB-A06

SOLUTION $C = \frac{\epsilon_0 A}{d - t + t/K}$ with $t = \frac{d}{2}$: the denominator is $\frac{d}{2} + \frac{d}{8} = \frac{5d}{8}$, so $C = \frac{8}{5}C_0 = 16 \mu\text{F}.$

SHORTCUT Two capacitors in series: $2C_0$ (air) and $8C_0$ (dielectric) give $\frac{16}{10}C_0$.

COMMON TRAP Multiplying by K as if the gap were completely filled (40 μF).

7 **Answer: 36** Maximum power transfer **Easy** ⌚ 1 min Current electricity NV

QB-A07

SOLUTION Maximum power occurs at $R = r$: $P = \frac{\epsilon^2}{4r} = \frac{144}{4} = 36 \text{ W}.$

COMMON TRAP Computing $\epsilon^2/r = 144 \text{ W}$ (the power at short circuit, dissipated inside the cell).

8 **Answer: (C)** Radius in a magnetic field **Medium** ⌚ 1.5 min Magnetism

QB-A08

SOLUTION $r = \frac{\sqrt{2mK}}{qB}$. For the alpha particle, $m \rightarrow 4m$ and $q \rightarrow 2e$: $\frac{\sqrt{4}}{2} = 1.$

SHORTCUT $r \propto \frac{\sqrt{m}}{q}$ at fixed K .

COMMON TRAP Using $r \propto m$ alone (equal speeds) gives 1 : 2.

9 Answer: (A) Thin lens Easy 1 min Optics

QB-A09

SOLUTION $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$; $\frac{1}{u} = \frac{1}{60} - \frac{1}{20} = -\frac{1}{30}$, so $u = -30$ cm.

COMMON TRAP Adding the reciprocals ($\frac{1}{20} + \frac{1}{60}$) gives 15 cm.

10 Answer: 2 Stopping potential Easy 1 min Modern physics NV

QB-A10

SOLUTION $E = \frac{1240}{310} = 4$ eV, so $K_{max} = 2$ eV and $V_0 = 2$ V.

QB-B · MIXED CHEMISTRY**1 Answer: 2** Atoms in a sample Easy 0.75 min Mole concept NV

QB-B01

SOLUTION 0.5 mol of H_2SO_4 contains $0.5 \times 4 = 2$ mol of O atoms.

COMMON TRAP Answering 0.5 (moles of molecules).

2 Answer: (D) Nodes Easy 0.5 min Atomic structure

QB-B02

SOLUTION Radial nodes $= n - l - 1 = 3 - 1 - 1 = 1$.

COMMON TRAP Giving the total number of nodes (2).

3 Answer: (A) Dipole moment Easy 0.5 min Bonding

QB-B03

SOLUTION BF_3 is trigonal planar and symmetrical, so the bond dipoles cancel.

4 Answer: 300 Temperature of spontaneity Medium 1 min Thermodynamics NV

QB-B04

SOLUTION $T = \frac{\Delta H}{\Delta S} = \frac{30000}{100} = 300$ K.

COMMON TRAP Mixing kJ and J (gives 0.3).

5 Answer: (B) Buffer pH Easy 0.75 min Equilibrium

QB-B05

SOLUTION Equal concentrations: $pH = pK_a + \log 1 = 4.74$.

COMMON TRAP Answering pK_b (9.26).

6 Answer: 30 Half-lives Easy 0.75 min Kinetics NV

QB-B06

SOLUTION 87.5% complete means $\frac{1}{8}$ remains: 3 half-lives.

7 Answer: (B) Low-spin complex Medium 1 min Coordination

QB-B07

SOLUTION Fe^{3+} is d^5 ; CN^- is strong-field, giving t_{2g}^5 with 1 unpaired electron: $\sqrt{3} = 1.73$ BM.

COMMON TRAP The high-spin value (5.92 BM) applies to weak-field ligands such as F^- .

8 Answer: (D) Ionisation enthalpy order Medium 1 min Periodicity

QB-B08

SOLUTION Two exceptions: $Be > B$ (removal from $2p$ in B) and $N > O$ (half-filled $2p^3$ in N).

COMMON TRAP Using the plain left-to-right trend.

9 Answer: (C) Acidity Easy 0.75 min GOC

QB-B09

SOLUTION The $-NO_2$ group at the para position stabilises the phenoxide ion by $-M$ and $-I$ effects.

10 Answer: (C) Iodoform without Tollens' Medium 1 min Carbonyls

QB-B10

SOLUTION Acetone has CH_3CO^- (iodoform positive) and is a ketone (Tollens' negative).

COMMON TRAP Propanal is an aldehyde (Tollens' positive) and has no CH_3CO^- group.

QB-C · MIXED MATHEMATICS**1 Answer: (A)** Locus in the Argand plane Medium 1.5 min Complex numbers

QB-C01

SOLUTION The points equidistant from (2, 0) and (0, -2) lie on the perpendicular bisector, through (1, -1) with slope -1: $y = -x$.

SHORTCUT Test $z = 0$: $|-2| = |2i|$ ✓, so the locus passes through the origin. Only (A) and (B) do; (B) fails at $z = 1 + i$.

2 Answer: 1024 Determinant of an adjoint Medium 1.5 min Matrices NV

QB-C02

SOLUTION $|\text{adj } B| = |B|^2$ for 3×3 matrices, and $|2A| = 8 \cdot 4 = 32$. So the answer is $32^2 = 1024$.

COMMON TRAP Using $|2A| = 2|A|$.

3 Answer: 15 Term independent of x **Easy** ⌚ 1 min Binomial NV QB-C03

SOLUTION $T_{r+1} = {}^6C_r x^{12-3r}$; $r = 4$ gives ${}^6C_4 = 15$.

4 Answer: 60 Arrangements with repetition **Easy** ⌚ 0.75 min P&C NV QB-C04

SOLUTION $\frac{6!}{3!2!} = 60$.

5 Answer: 8 Infinite GP conditions **Medium** ⌚ 2 min Sequences NV QB-C05

SOLUTION $\frac{a}{1-r} = 20$ and $\frac{a^2}{1-r^2} = 100$. Dividing the second by the square of the first: $\frac{1-r}{1+r} = \frac{1}{4} \Rightarrow r = \frac{3}{5}$, so $a = 20 \cdot \frac{2}{5} = 8$.

6 Answer: 2 Standard limit **Easy** ⌚ 1 min Limits NV QB-C06

SOLUTION $1 - \cos 2x = 2 \sin^2 x$, and $\frac{2 \sin^2 x}{x^2} \rightarrow 2$.

SHORTCUT $1 - \cos kx \approx \frac{k^2 x^2}{2}$.

7 Answer: (B) King's property **Medium** ⌚ 1.5 min Definite integrals QB-C07

SOLUTION By King's rule, $I = \int_0^\pi (\pi - x) \sin x \, dx$, so $2I = \pi \int_0^\pi \sin x \, dx = 2\pi$.

COMMON TRAP Assuming odd symmetry (the interval is not symmetric about 0).

8 Answer: 6 Separable DE **Easy** ⌚ 1 min Differential equations NV QB-C08

SOLUTION $\ln y = \ln x + C$, so $y = kx$ with $k = 2$.

9 Answer: (D) Hyperbola eccentricity **Easy** ⌚ 1 min Conics QB-C09

SOLUTION $\frac{x^2}{16} - \frac{y^2}{9} = 1$, so $e = \sqrt{1 + \frac{9}{16}} = \frac{5}{4}$.

COMMON TRAP Swapping a^2 and b^2 gives $\sqrt{1 + 16/9} = 5/3$.

10 Answer: (A) Without replacement **Easy** ⌚ 1 min Probability QB-C10

SOLUTION $\frac{4}{52} \cdot \frac{3}{51} = \frac{1}{221}$.

COMMON TRAP 1/169 is the answer with replacement.

QB-D · CALCULATION-HEAVY

1 Answer: 20 Pushing up a rough incline **Medium** ⌚ 2.5 min Laws of motion NV QB-D01

SOLUTION $F = mg \sin \theta + \mu mg \cos \theta = 20(0.6) + 0.5 \times 20 \times 0.8 = 12 + 8 = 20 \text{ N}$.

COMMON TRAP Friction acts down the incline when the block moves up. Subtracting it gives 4 N.

2 Answer: 18 Fringe positions **Medium** ⌚ 2.5 min Wave optics NV QB-D02

SOLUTION $\beta = \frac{\lambda D}{d} = \frac{600 \times 10^{-9} \times 1.5}{0.3 \times 10^{-3}} = 3 \text{ mm}$. Distance = $(2 + 4)\beta = 18 \text{ mm}$.

COMMON TRAP Subtracting the positions (2β) as if both fringes were on the same side.

3 Answer: 480 Power in an LCR circuit **Medium** ⌚ 2.5 min AC NV QB-D03

SOLUTION $Z = \sqrt{30^2 + 40^2} = 50 \, \Omega$, so $I_{rms} = 4 \text{ A}$ and $P = I^2 R = 480 \text{ W}$.

SHORTCUT $P = \frac{V^2 R}{Z^2} = \frac{40000 \times 30}{2500}$.

COMMON TRAP Using $VI = 800 \text{ W}$ (it ignores the power factor 0.6).

4 Answer: (A) Nernst equation **Medium** ⌚ 2 min Electrochemistry QB-D04

SOLUTION $E = 1.10 - \frac{0.059}{2} \log \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]} = 1.10 - 0.0295(-2) = 1.159 \text{ V}$.

SHORTCUT A lower product concentration (Zn^{2+}) must raise the emf, so (B) and (C) are out immediately.

COMMON TRAP Inverting Q gives 1.041 V.

5 Answer: (B) Depression with a van 't Hoff factor **Medium** ⌚ 1.5 min Solutions QB-D05

SOLUTION $i = 3$: $\Delta T_f = 3 \times 1.86 \times 0.1 = 0.558 \text{ K}$.

COMMON TRAP Using $i = 2$ (0.372 K).

6 Answer: (B) Area between curves **Medium** ⌚ 2 min Area

QB-D06

SOLUTION The curves meet at $x = 0$ and $x = 2$. $\int_0^2 (2x - x^2) dx = 4 - \frac{8}{3} = \frac{4}{3}$.**SHORTCUT** Area between $y = x^2$ and $y = mx$ is $\frac{m^3}{6}$.**7 Answer: 1** Removing observations **Medium** ⌚ 2.5 min Statistics NV

QB-D07

SOLUTION $\sum x = 50$ and $\sum x^2 = 10(4 + 25) = 290$. After removal: $\sum x = 40$, $\sum x^2 = 290 - 1 - 81 = 208$. Mean = 5, variance = $26 - 25 = 1$.**COMMON TRAP** Assuming the variance is unchanged because the mean is unchanged.**8 Answer: 1** Shortest distance **Hard** ⌚ 3 min 3D NV

QB-D08

SOLUTION $\vec{a}_2 - \vec{a}_1 = (1, 2, 2)$; $\vec{b}_1 \times \vec{b}_2 = (-1, 2, -1)$ with magnitude $\sqrt{6}$. Dot product = $-1 + 4 - 2 = 1$, so $d = \frac{1}{\sqrt{6}}$.**SHORTCUT** Keep it squared: $d^2 = \frac{1}{6}$.

QB-E · TRICKY & TRAP-LADEN

1 Answer: (C) Rod pivoted at one end **Medium** ⌚ 1.5 min Rotation

QB-E01

SOLUTION $\tau = mg\frac{L}{2}$ and $I = \frac{1}{3}mL^2$, so $\alpha = \frac{3g}{2L}$.**COMMON TRAP** Using $I_{cm} = \frac{1}{12}mL^2$ gives $6g/L$. The axis is at the end.**2 Answer: (D)** Uniform circular motion **Easy** ⌚ 0.5 min Concept

QB-E02

SOLUTION The speed is constant, so KE is constant. The directions of velocity, momentum and acceleration change continuously.**COMMON TRAP** Choosing acceleration: its magnitude is constant, but its direction is not.**3 Answer: (C)** Current division **Easy** ⌚ 0.75 min Circuits

QB-E03

SOLUTION Each resistor has the full 6 V across it: $I = \frac{6}{4} = 1.5$ A.**COMMON TRAP** Splitting the total current (2 A) in the ratio of the resistances instead of inversely gives 0.5 A.**4 Answer: 850** Overtones of a closed pipe **Medium** ⌚ 1 min Waves NV

QB-E04

SOLUTION $f_1 = \frac{v}{4L} = 170$ Hz. A closed pipe has only odd harmonics, so the second overtone is the 5th harmonic: 850 Hz.**COMMON TRAP** Answering $3f_1$ (the first overtone) or $2f_1$ (not present in a closed pipe).**5 Answer: (D)** Reducing sugars **Easy** ⌚ 0.5 min Biomolecules

QB-E05

SOLUTION In sucrose, both anomeric carbons are tied up in the glycosidic bond.**COMMON TRAP** Missing the word "not".**6 Answer: (D)** Very dilute acid **Medium** ⌚ 1 min Ionic equilibrium

QB-E06

SOLUTION Include water's ions: $x^2 - 10^{-8}x - 10^{-14} = 0$ gives $[H^+] \approx 1.05 \times 10^{-7}$, so pH ≈ 6.98 .**SHORTCUT** An acid solution can't have pH > 7. (B) and (C) are impossible.**COMMON TRAP** Taking pH = $-\log 10^{-8} = 8$.**7 Answer: 4** Counting isomers **Medium** ⌚ 1 min Isomerism NV

QB-E07

SOLUTION Butan-1-ol, butan-2-ol, 2-methylpropan-1-ol and 2-methylpropan-2-ol.**COMMON TRAP** Including the three ethers (the question asks for alcohols only), or counting the enantiomers of butan-2-ol (not structural isomers).**8 Answer: 4** Modulus equation **Medium** ⌚ 1 min Quadratics NV

QB-E08

SOLUTION $|x| = 2$ or 3 , so $x = \pm 2, \pm 3$.**COMMON TRAP** Answering 2 by ignoring the modulus.**9 Answer: (C)** Conditional probability **Easy** ⌚ 1 min Probability

QB-E09

SOLUTION $P(A \cap B) = 0.6 + 0.5 - 0.8 = 0.3$, so $P(A | B) = \frac{0.3}{0.5} = 0.6$.**COMMON TRAP** Stopping at $P(A \cap B) = 0.3$.

10 Answer: (A) Log with base below 1 **Medium** ⌚ 1.5 min Functions

QB-E10

SOLUTION We need $\log_{1/2}(x-1) \geq 0$. Since the base is below 1, this means $0 < x-1 \leq 1$.

COMMON TRAP Not reversing the inequality for a base below 1 gives $[2, \infty)$.

QB-F · TIME-PRESSURE SPRINT (TARGET: 12 MINUTES)

1 Answer: (B) Dimensions **Easy** ⌚ 0.5 min Speed

QB-F01

SOLUTION Impulse = change in momentum.

COMMON TRAP Confusing it with force (MLT^{-2}).

2 Answer: (A) de Broglie wavelength **Easy** ⌚ 0.5 min Speed

QB-F02

SOLUTION $\lambda = \frac{12.27}{\sqrt{100}} \text{ \AA}$.

3 Answer: (B) Radioactive decay **Easy** ⌚ 0.5 min Speed

QB-F03

SOLUTION Three half-lives: $(1/2)^3$.

4 Answer: 1 Series capacitors **Easy** ⌚ 0.5 min Speed **NV**

QB-F04

SOLUTION $\frac{1}{C} = \frac{1}{3} \times 3$.

5 Answer: (A) Hybridisation **Easy** ⌚ 0.5 min Speed

QB-F05

SOLUTION 4 bond pairs + 2 lone pairs give a steric number of 6 (square planar shape).

COMMON TRAP Deducing hybridisation from the square planar shape (dsp^2) instead of the steric number.

6 Answer: 6 Oxidation state **Easy** ⌚ 0.5 min Speed **NV**

QB-F06

SOLUTION $2 + 2x - 14 = 0$.

7 Answer: 3 Sigma bonds **Easy** ⌚ 0.5 min Speed **NV**

QB-F07

SOLUTION Two C-H bonds and one C-C σ bond (plus two π bonds).

8 Answer: 6 Unpaired electrons **Easy** ⌚ 0.5 min Speed **NV**

QB-F08

SOLUTION $[\text{Ar}] 3d^5 4s^1$.

COMMON TRAP Using $3d^4 4s^2$ (gives 4).

9 Answer: (A) Powers of i **Easy** ⌚ 0.5 min Speed

QB-F09

SOLUTION $2027 = 4 \times 506 + 3$, and $i^3 = -i$.

10 Answer: 120 Combinations **Easy** ⌚ 0.5 min Speed **NV**

QB-F10

SOLUTION $\frac{10 \cdot 9 \cdot 8}{6}$.

11 Answer: 2 Derivative **Easy** ⌚ 0.5 min Speed **NV**

QB-F11

SOLUTION $2e^{2x}$ at $x = 0$.

12 Answer: (B) Slope of a line **Easy** ⌚ 0.5 min Speed

QB-F12

SOLUTION $-\frac{a}{b} = -\frac{3}{-4}$.

COMMON TRAP Sign error with $-a/b$.

PART VIII · QUESTION BANK & MOCK TESTS

Full-Length Mock Test 1

⌚ MOCK TEST Instructions (same pattern as the 2026 paper [NTA-IB26])

- **75 questions, 300 marks, 3 hours.** Each subject has 20 MCQs (Section A) and 5 numerical-value questions (Section B), all compulsory.
- **Marking:** +4 correct, -1 incorrect, 0 unanswered, in **both** sections.
- For numerical answers, enter the integer value.
- No calculator. Use only blank rough sheets.
- Start at your shift time. Before checking answers, fill in Part 1 of the Mock Analysis Sheet (page [446](#)).

📌 NOTE About this paper

All 75 questions are original and written for this book, in the style and difficulty mix of recent shifts: about 40% easy, 45% medium and 15% hard 📌 **RECOMMENDATION**. The answer key and full solutions follow the paper (page [437](#)).

Start time · End time

Physics

📌 Mock 1 · Physics

1

MT1-P01

The percentage errors in measuring the mass and the side of a cube are 1% and 2%. The maximum percentage error in its density is:

- (A) 9% (B) 5% (C) 3% (D) 7%

2

MT1-P02

The position of a particle is $x = 2t^3 - 3t^2 + 4$ (in m, t in s). Its acceleration at $t = 1$ s is:

- (A) 12 m/s^2 (B) 0 (C) 6 m/s^2 (D) -6 m/s^2

3

MT1-P03

Two projectiles are launched with the same speed at 30° and 60° to the horizontal. The ratio of their maximum heights is:

- (A) 1 : 3 (B) 1 : 1 (C) 3 : 1 (D) $1 : \sqrt{3}$

4

MT1-P04

A 5 kg block rests on a horizontal floor ($\mu_s = 0.4$, $\mu_k = 0.3$, $g = 10 \text{ m/s}^2$). A horizontal force of 12 N is applied. The friction force on the block is:

- (A) 20 N (B) 12 N (C) 15 N (D) 0

5

MT1-P05

A 1 kg ball moving at 6 m/s collides head-on elastically with a stationary 2 kg ball. Their velocities after the collision are:

- (A) 0 and 3 m/s (B) 2 m/s and 2 m/s (C) -2 m/s and 4 m/s (D) -4 m/s and 5 m/s

MT1-P06

6

The moment of inertia of a uniform disc (mass M , radius R) about a tangent in its own plane is:

- (A) $\frac{3}{2}MR^2$ (B) $\frac{5}{4}MR^2$ (C) $\frac{1}{4}MR^2$ (D) $\frac{3}{4}MR^2$

MT1-P07

7

A planet has twice Earth's radius and the same mean density. Its escape speed (Earth: 11.2 km/s) is:

- (A) 22.4 km/s (B) 11.2 km/s (C) 15.8 km/s (D) 44.8 km/s

MT1-P08

8

Water flows through a horizontal pipe whose cross-section narrows from 4 cm^2 to 1 cm^2 . The speed in the wide part is 1 m/s. The pressure difference between the wide and narrow parts is:

- (A) 8000 Pa (B) 15000 Pa (C) 7500 Pa (D) 3750 Pa

MT1-P09

9

Two rods of equal length and cross-section, with thermal conductivities K and $2K$, are joined end to end. The equivalent conductivity of the combination is:

- (A) $\frac{2}{3}K$ (B) $\frac{3}{2}K$ (C) $3K$ (D) $\frac{4}{3}K$

MT1-P10

10

A monatomic ideal gas ($\gamma = \frac{5}{3}$) is compressed adiabatically to $\frac{1}{8}$ of its volume. Its absolute temperature becomes:

- (A) 2 times (B) 8 times (C) 4 times (D) 16 times

MT1-P11

11

A mixture contains 1 mol of helium and 1 mol of oxygen (rigid diatomic). Its γ is:

- (A) 1.67 (B) 1.50 (C) 1.40 (D) 1.33

MT1-P12

12

A block on a spring oscillates with period T . The spring is cut into two equal halves and the same block is attached to one half. The new period is:

- (A) $\sqrt{2}T$ (B) $T/\sqrt{2}$ (C) $T/2$ (D) $2T$

MT1-P13

13

A string fixed at both ends vibrates in its 3rd harmonic at 300 Hz. If the tension is made four times as large (same length and string), the 3rd-harmonic frequency becomes:

- (A) 150 Hz (B) 1200 Hz (C) 300 Hz (D) 600 Hz

MT1-P14

14

Charges $+q$ and $-q$ are a distance d apart. At the midpoint:

- (A) $E = \frac{4kq}{d^2}$, $V = \frac{2kq}{d}$ (B) $E = 0$, $V = 0$
 (C) $E = \frac{8kq}{d^2}$, $V = 0$ (D) $E = 0$, $V = \frac{4kq}{d}$

15

MT1-P15

A $4\ \mu\text{F}$ capacitor is charged to 50 V. The energy stored is:

- (A) 5 mJ (B) 10 mJ (C) 0.1 mJ (D) 2.5 mJ

16

MT1-P16

In a meter bridge, an unknown resistance X is in the left gap and $30\ \Omega$ in the right gap. The balance point is 40 cm from the left end. X is:

- (A) $30\ \Omega$ (B) $45\ \Omega$ (C) $12\ \Omega$ (D) $20\ \Omega$

17

MT1-P17

The magnetic field 5 cm from a long straight wire carrying 10 A is:

- (A) $4 \times 10^{-4}\ \text{T}$ (B) $2 \times 10^{-5}\ \text{T}$ (C) $4 \times 10^{-5}\ \text{T}$ (D) $8 \times 10^{-5}\ \text{T}$

18

MT1-P18

A 0.5 m rod moves at 4 m/s perpendicular to a 0.2 T magnetic field, with the rod, velocity and field mutually perpendicular. The induced emf is:

- (A) 0.4 V (B) 0.1 V (C) 4 V (D) 1.6 V

19

MT1-P19

An electromagnetic wave in vacuum has electric-field amplitude 30 V/m. Its magnetic-field amplitude is:

- (A) $3 \times 10^{-7}\ \text{T}$ (B) $9 \times 10^9\ \text{T}$ (C) $10^{-7}\ \text{T}$ (D) $10^{-6}\ \text{T}$

20

MT1-P20

The output of a NAND gate is 0 when:

- (A) both inputs are 0 (B) both inputs are 1
(C) exactly one input is 1 (D) at least one input is 0

Physics · Section B (numerical value)

21

NV

MT1-P21

A thin prism has refracting angle 5° and refractive index 1.6. Find its angle of minimum deviation in degrees.

Your answer (numerical value):

22

NV

MT1-P22

Electrons in a sample of hydrogen atoms are excited to $n = 4$. Find the maximum number of spectral lines emitted as they return to the ground state.

Your answer (numerical value):

23

NV

MT1-P23

The activity of a sample falls from 800 to 50 counts per minute in 20 hours. Find the half-life in hours.

Your answer (numerical value):

24 NV

MT1-P24

A disc spins freely at 30 rad/s about its axis. An identical disc, initially at rest, is gently placed on it coaxially, and the two move together. Find the final angular speed in rad/s.

Your answer (numerical value):

25 NV

MT1-P25

Cells of emf 2 V and 4 V, each with internal resistance $1\ \Omega$, are connected in parallel with like terminals together. Find the emf of the equivalent cell in volts.

Your answer (numerical value):

Chemistry

Mock 1 · Chemistry

1

MT1-C01

4 g of NaOH is dissolved to make 500 mL of solution. The molarity is:

- (A) 0.2 M (B) 0.1 M (C) 0.4 M (D) 2 M

2

MT1-C02

The energy of the electron in the second orbit of He^+ is:

- (A) -3.4 eV (B) -13.6 eV (C) -54.4 eV (D) -6.8 eV

3

MT1-C03

The bond order of O_2^+ is:

- (A) 2 (B) 2.5 (C) 1.5 (D) 3

4

MT1-C04

The shape of SF_4 is:

- (A) trigonal pyramidal (B) tetrahedral
(C) square planar (D) see-saw

5

MT1-C05

For $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \longrightarrow 2\text{NH}_3(\text{g})$ at 298 K, $\Delta H - \Delta U$ is approximately:

- (A) -4.96 kJ (B) +4.96 kJ (C) -2.48 kJ (D) 0

6

MT1-C06

Which 0.1 m aqueous solution has the highest boiling point?

- (A) CaCl_2 (B) glucose (C) NaCl (D) urea

7

MT1-C07

For $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$:

- (A) $K_p = K_c(RT)^2$ (B) $K_p = K_c$ (C) $K_p = K_c/(RT)$ (D) $K_p = K_c RT$

8

MT1-C08

The solubility of AgCl ($K_{sp} = 1.8 \times 10^{-10}$) in 0.1 M NaCl is:

- (A) 1.8×10^{-11} M (B) 1.34×10^{-5} M (C) 1.8×10^{-10} M (D) 1.8×10^{-9} M

9

MT1-C09

The charge needed to deposit 1 mol of Al from Al^{3+} is:

- (A) F/3 (B) 1 F (C) 2 F (D) 3 F

10

MT1-C10

For a zero-order reaction, the half-life is:

- (A) directly proportional to the initial concentration
(B) independent of the initial concentration
(C) inversely proportional to the initial concentration
(D) proportional to the square of the initial concentration

11

MT1-C11

Which ion has the largest radius?

- (A) Na^+ (B) O^{2-} (C) F^- (D) N^{3-}

12

MT1-C12

The basicity of H_3PO_3 is:

- (A) 3 (B) 2 (C) 1 (D) 0

13

MT1-C13

Which xenon compound is square planar?

- (A) XeO_3 (B) XeF_2 (C) XeF_4 (D) XeF_6

14

MT1-C14

Which ion is colourless in aqueous solution?

- (A) Ti^{3+} (B) Sc^{3+} (C) Cu^{2+} (D) Fe^{2+}

15

MT1-C15

The IUPAC name of $\text{K}_3[\text{Fe}(\text{CN})_6]$ is:

- (A) potassium hexacyanidoiron(III) (B) potassium hexacyanidoferrate(II)
(C) potassium hexacyanidoferrate(III) (D) tripotassium hexacyanidoferrate(II)

16

MT1-C16

Which carbocation is the most stable?

- (A) $(\text{CH}_3)_2\text{CH}^+$ (B) CH_3CH_2^+ (C) CH_3^+ (D) $(\text{CH}_3)_3\text{C}^+$

17

MT1-C17

HBr adds to propene in the presence of benzoyl peroxide. The major product is:

- (A) propane (B) 2-bromopropane
(C) 1,2-dibromopropane (D) 1-bromopropane

18

MT1-C18

Which reacts fastest by the S_N2 mechanism?

- (A) $\text{CH}_3\text{CH}_2\text{Br}$ (B) $(\text{CH}_3)_3\text{CBr}$ (C) $(\text{CH}_3)_2\text{CHBr}$ (D) CH_3Br

19

MT1-C19

Which compound undergoes the Cannizzaro reaction?

- (A) HCHO (B) CH_3CHO (C) CH_3COCH_3 (D) $\text{CH}_3\text{CH}_2\text{CHO}$

20

MT1-C20

Which amine gives the carbylamine test?

- (A) $(\text{CH}_3)_2\text{NH}$ (B) CH_3NH_2 (C) $(\text{CH}_3)_3\text{N}$ (D) $\text{C}_6\text{H}_5\text{N}(\text{CH}_3)_2$

Chemistry · Section B (numerical value)

21

NV

MT1-C21

How many unpaired electrons are in $[\text{NiCl}_4]^{2-}$?

Your answer (numerical value):

22

NV

MT1-C22

A first-order reaction is 75% complete in 60 min. Find its half-life in minutes.

Your answer (numerical value):

23

NV

MT1-C23

Find the pH of 0.001 M NaOH at 25 °C.

Your answer (numerical value):

24

NV

MT1-C24

Find the mass of CO_2 (in g) formed by complete combustion of 16 g of methane.

Your answer (numerical value):

25

NV

MT1-C25

How many chiral carbon atoms are there in open-chain D-glucose?

Your answer (numerical value):

Mathematics

Mock 1 · Mathematics

1

MT1-M01

If $f(x) = 2x + 3$ and $g(x) = x^2$, then $(f \circ g)(2)$ is:

- (A) 7 (B) 49 (C) 11 (D) 14

2

MT1-M02

The number of reflexive relations on a set with 3 elements is:

- (A) 8 (B) 512 (C) 64 (D) 27

3

MT1-M03

$|(1+i)^8|$ equals:

- (A) 4 (B) 8 (C) 256 (D) 16

4

MT1-M04

If α and β are the roots of $x^2 - 5x + 6 = 0$, then $\alpha^2 + \beta^2$ is:

- (A) 13 (B) 25 (C) 37 (D) 11

5

MT1-M05

If $A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$, then A^{-1} is:

- (A) $\begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$
 (C) $\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$ (D) $\begin{bmatrix} -1 & 1 \\ 1 & -2 \end{bmatrix}$

6

MT1-M06

The system $x + y + z = 3$, $x + 2y + 3z = 6$, $x + 3y + kz = 9$ has infinitely many solutions for k equal to:

- (A) 3 (B) 5 (C) 4 (D) 6

7

MT1-M07

A committee of 3 is chosen from 5 men and 4 women. The number of committees with at least one woman is:

- (A) 74 (B) 84 (C) 70 (D) 64

8

MT1-M08

The coefficient of x^3 in $(1 + 2x)^5$ is:

- (A) 40 (B) 80 (C) 10 (D) 160

9

MT1-M09

The sum of the first 20 terms of $3, 7, 11, \dots$ is:

- (A) 820 (B) 800 (C) 840 (D) 780

10

MT1-M10

$\lim_{x \rightarrow \infty} \left(1 + \frac{2}{x}\right)^x$ equals:

- (A) e^2 (B) e (C) $2e$ (D) $\sqrt{e}$

11

MT1-M11

$f(x) = \frac{\sin 3x}{x}$ for $x \neq 0$ and $f(0) = k$. f is continuous at 0 when k is:

- (A) 3 (B) 1 (C) $\frac{1}{3}$ (D) 0

12

MT1-M12

$f(x) = x^3 - 3x$ is decreasing on:

- (A) $(-\infty, -1)$ (B) $(1, \infty)$ (C) $(-1, 1)$ (D) $(0, 3)$

13

MT1-M13

$\int x e^x dx$ equals:

- (A) $(x+1)e^x + C$ (B) $(x-1)e^x + C$ (C) $x e^x + C$ (D) $\frac{1}{2} x^2 e^x + C$

14

MT1-M14

$\int_{-1}^1 (x^3 + x \cos x + 1) dx$ equals:

- (A) 1 (B) 0 (C) 2 (D) 4

15

MT1-M15

The integrating factor of $\frac{dy}{dx} + y \tan x = \sec x$ is:

- (A) $\cos x$ (B) $\sec x$ (C) $e^{\tan x}$ (D) $\tan x$

16

MT1-M16

The distance between $3x + 4y = 5$ and $6x + 8y = 20$ is:

- (A) 3 (B) 1 (C) $\frac{3}{2}$ (D) 2

17

MT1-M17

The circle $x^2 + y^2 - 4x + 6y - 12 = 0$ has:

- (A) centre $(-2, 3)$, radius 5 (B) centre $(2, -3)$, radius 5
(C) centre $(2, -3)$, radius $\sqrt{12}$ (D) centre $(4, -6)$, radius 5

18

MT1-M18

The length of the latus rectum of $\frac{x^2}{25} + \frac{y^2}{16} = 1$ is:

- (A) $\frac{16}{5}$ (B) $\frac{25}{2}$ (C) $\frac{32}{5}$ (D) $\frac{8}{5}$

19

MT1-M19

The angle between $\hat{i} + \hat{j} + \hat{k}$ and $\hat{i} - \hat{j} + \hat{k}$ is:

- (A) $\cos^{-1} \frac{1}{3}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{2}$ (D) $\cos^{-1} \frac{2}{3}$

20

MT1-M20

A die is thrown twice. Given that the first throw is even, the probability that the sum is 8 is:

(A) $\frac{1}{12}$

(B) $\frac{5}{36}$

(C) $\frac{1}{6}$

(D) $\frac{1}{3}$

Mathematics · Section B (numerical value)

21

NV

MT1-M21

Find the angle (in degrees) between lines with direction ratios $(1, 1, 2)$ and $(\sqrt{3} - 1, -\sqrt{3} - 1, 4)$.

Your answer (numerical value):

22

NV

MT1-M22

If σ^2 is the variance of $1, 2, \dots, 10$, find $4\sigma^2$.

Your answer (numerical value):

23

NV

MT1-M23

If $|\vec{a}| = 3$, $|\vec{b}| = 4$ and $|\vec{a} + \vec{b}| = 5$, find $|\vec{a} - \vec{b}|^2$.

Your answer (numerical value):

24

NV

MT1-M24

Find the maximum value of $5 \sin x + 12 \cos x + 7$.

Your answer (numerical value):

25

NV

MT1-M25

Find $\int_0^3 [x] dx$, where $[x]$ is the greatest integer $\leq x$.

Your answer (numerical value):

Answer Key & Solutions

Mock 1 · Physics

1	(D)	2	(C)	3	(A)	4	(B)	5	(C)	6	(B)	7	(A)	8	(C)	9	(D)	10	(C)
11	(B)	12	(B)	13	(D)	14	(C)	15	(A)	16	(D)	17	(C)	18	(A)	19	(C)	20	(B)
21	3	22	6	23	5	24	15	25	3										

Mock 1 · Chemistry

1	(A)	2	(B)	3	(B)	4	(D)	5	(A)	6	(A)	7	(D)	8	(D)	9	(D)	10	(A)
11	(D)	12	(B)	13	(C)	14	(B)	15	(C)	16	(D)	17	(D)	18	(D)	19	(A)	20	(B)
21	2	22	30	23	11	24	44	25	4										

Mock 1 · Mathematics

1	(C)	2	(C)	3	(D)	4	(A)	5	(C)	6	(B)	7	(A)	8	(B)	9	(A)	10	(A)
11	(A)	12	(C)	13	(B)	14	(C)	15	(B)	16	(B)	17	(B)	18	(C)	19	(A)	20	(C)
21	60	22	33	23	25	24	20	25	3										

MOCK TEST Score yourself

Physics _____ / 100 · Chemistry _____ / 100 · Maths _____ / 100 · Total _____ / 300

Now complete the Mock Analysis Sheet (page 446) before reading the solutions.

MOCK 1 · PHYSICS

- 1 Answer: (D) Error in density** Easy 1 min Units MT1-P01
SOLUTION $\rho = \frac{m}{L^3}$, so $\frac{\Delta\rho}{\rho} = 1\% + 3(2\%) = 7\%$.
COMMON TRAP Adding without the power (3%).
- 2 Answer: (C) Acceleration from position** Easy 1 min Kinematics MT1-P02
SOLUTION $v = 6t^2 - 6t$ and $a = 12t - 6$, which is 6 m/s^2 at $t = 1$.
COMMON TRAP $v = 0$ at $t = 1 \text{ s}$. Giving the velocity instead of the acceleration.
- 3 Answer: (A) Complementary angles** Easy 1 min Projectile MT1-P03
SOLUTION $H \propto \sin^2 \theta$: $\frac{1}{4} : \frac{3}{4}$.
COMMON TRAP Equal ranges for complementary angles don't mean equal heights.
- 4 Answer: (B) Static friction** Medium 1.5 min Laws of motion MT1-P04
SOLUTION Limiting friction is $0.4 \times 50 = 20 \text{ N} > 12 \text{ N}$, so the block stays at rest and static friction balances the push: 12 N .
COMMON TRAP Using $\mu_s N$ (20 N) or $\mu_k N$ (15 N) when the block isn't sliding.
- 5 Answer: (C) Elastic collision** Medium 1.5 min Collisions MT1-P05
SOLUTION $v_1 = \frac{1-2}{3}(6) = -2 \text{ m/s}$; $v_2 = \frac{2(1)}{3}(6) = 4 \text{ m/s}$.
SHORTCUT Check the options: momentum is 6 in (C), (B), (A) and (D), but only (C) conserves KE ($2 + 16 = 18 \text{ J}$).
COMMON TRAP (B) is the perfectly inelastic case.
- 6 Answer: (B) Axis theorems** Medium 1.5 min Rotation MT1-P06
SOLUTION About a diameter: $\frac{1}{4}MR^2$ (perpendicular-axis theorem). Parallel shift by R : $\frac{1}{4}MR^2 + MR^2$.
COMMON TRAP $\frac{3}{2}MR^2$ is for a tangent **perpendicular** to the plane.
- 7 Answer: (A) Escape speed scaling** Medium 1 min Gravitation MT1-P07
SOLUTION $v_e = \sqrt{\frac{2GM}{R}}$ with $M \propto \rho R^3$, so $v_e \propto R\sqrt{\rho}$.
COMMON TRAP Assuming the same mass (gives $11.2/\sqrt{2}$).
- 8 Answer: (C) Bernoulli in a pipe** Medium 1.5 min Fluids MT1-P08
SOLUTION Continuity: $v_2 = 4 \text{ m/s}$. $\Delta P = \frac{1}{2}\rho(v_2^2 - v_1^2) = 500 \times 15$.
COMMON TRAP Forgetting the $\frac{1}{2}$ (15000 Pa).
- 9 Answer: (D) Rods in series** Medium 1.5 min Heat conduction MT1-P09
SOLUTION Thermal resistances add: $\frac{2L}{K_{eq}A} = \frac{L}{KA} + \frac{L}{2KA}$, so $K_{eq} = \frac{4K}{3}$.
SHORTCUT Harmonic mean of K and $2K$.
COMMON TRAP Arithmetic mean ($\frac{3}{2}K$) is for rods in parallel.
- 10 Answer: (C) Adiabatic compression** Medium 1 min Thermodynamics MT1-P10
SOLUTION $TV^{\gamma-1}$ is constant: $T_2/T_1 = 8^{2/3} = 4$.
COMMON TRAP Using the isothermal or ideal-gas-at-constant- P relation (8).
- 11 Answer: (B) γ of a mixture** Medium 1.5 min Kinetic theory MT1-P11
SOLUTION $C_V = \frac{\frac{3}{2}R + \frac{5}{2}R}{2} = 2R$, $C_P = 3R$, $\gamma = 1.5$.
COMMON TRAP Averaging the two γ values gives 1.53 , which is not correct.

12 Answer: (B) *Cut spring* **Medium** ⌚ 1 min SHM

MT1-P12

SOLUTION Each half has stiffness $2k$, so $T \propto 1/\sqrt{k}$ gives $T/\sqrt{2}$.**COMMON TRAP** Thinking a shorter spring is "weaker".**13 Answer: (D)** *Tension and harmonics* **Easy** ⌚ 1 min Waves

MT1-P13

SOLUTION $v = \sqrt{T/\mu}$ doubles, so every harmonic frequency doubles.**COMMON TRAP** Scaling with T instead of $\sqrt{T}$ (1200 Hz).**14 Answer: (C)** *Field and potential of a pair* **Easy** ⌚ 1 min Electrostatics

MT1-P14

SOLUTION The fields add (both point towards $-q$): $2 \cdot \frac{kq}{(d/2)^2}$. The potentials cancel.**COMMON TRAP** Assuming $V = 0$ means $E = 0$.**15 Answer: (A)** *Energy in a capacitor* **Easy** ⌚ 0.75 min Capacitors

MT1-P15

SOLUTION $\frac{1}{2}CV^2 = \frac{1}{2}(4 \times 10^{-6})(2500) = 5 \times 10^{-3}$ J.**COMMON TRAP** Forgetting the $\frac{1}{2}$ (10 mJ).**16 Answer: (D)** *Meter bridge* **Easy** ⌚ 1 min Current electricity

MT1-P16

SOLUTION $\frac{X}{30} = \frac{40}{60}$.**COMMON TRAP** Using $\frac{60}{40}$ (45 Ω).**17 Answer: (C)** *Field of a long wire* **Easy** ⌚ 0.75 min Magnetism

MT1-P17

SOLUTION $\frac{\mu_0 I}{2\pi r} = \frac{2 \times 10^{-7} \times 10}{0.05}$.**COMMON TRAP** Using r in cm.**18 Answer: (A)** *Motional emf* **Easy** ⌚ 0.75 min EMI

MT1-P18

SOLUTION $Blv = 0.2 \times 0.5 \times 4$.**19 Answer: (C)** *E-B relation* **Easy** ⌚ 0.5 min EM waves

MT1-P19

SOLUTION $B_0 = \frac{E_0}{c} = \frac{30}{3 \times 10^8}$.**COMMON TRAP** Multiplying by c .**20 Answer: (B)** *NAND gate* **Easy** ⌚ 0.5 min Electronic devices

MT1-P20

SOLUTION $\text{NAND} = \overline{A \cdot B}$, which is 0 only when $A = B = 1$.**21 Answer: 3** *Thin prism* **Easy** ⌚ 1 min Optics **NV**

MT1-P21

SOLUTION $\delta = (n - 1)A = 0.6 \times 5$.**22 Answer: 6** *Number of lines* **Easy** ⌚ 0.75 min Atoms **NV**

MT1-P22

SOLUTION $\frac{n(n-1)}{2} = 6$.**COMMON TRAP** Counting only the lines that end on $n = 1$ (3).**23 Answer: 5** *Half-life from activity* **Medium** ⌚ 1 min Nuclei **NV**

MT1-P23

SOLUTION $\frac{50}{800} = \frac{1}{16}$, which is 4 half-lives in 20 h.**24 Answer: 15** *Angular momentum conservation* **Medium** ⌚ 1 min Rotation **NV**

MT1-P24

SOLUTION $I\omega = 2I\omega'$.**COMMON TRAP** Using energy conservation (gives $30/\sqrt{2}$). Energy is lost as the discs slip against each other.**25 Answer: 3** *Cells in parallel* **Medium** ⌚ 1.5 min Current electricity **NV**

MT1-P25

SOLUTION $\varepsilon_{eq} = \frac{\varepsilon_1/r_1 + \varepsilon_2/r_2}{1/r_1 + 1/r_2} = \frac{2+4}{2}$.**SHORTCUT** Equal internal resistances: simple average.**COMMON TRAP** Adding the emfs (6 V) as if in series.

MOCK 1 · CHEMISTRY

- 1 Answer: (A) Molarity** Easy ⌚ 0.75 min Basic concepts MT1-C01
SOLUTION $\frac{4/40}{0.5} = 0.2 \text{ M}$.
COMMON TRAP Forgetting to convert mL to L.
- 2 Answer: (B) Bohr energy for He⁺** Easy ⌚ 0.75 min Atomic structure MT1-C02
SOLUTION $-13.6 \frac{Z^2}{n^2} = -13.6 \cdot \frac{4}{4}$.
COMMON TRAP Using $Z = 1$ (−3.4 eV).
- 3 Answer: (B) Bond order of an ion** Medium ⌚ 1 min Bonding MT1-C03
SOLUTION O₂ has bond order 2. O₂⁺ loses an antibonding π^* electron, which raises the bond order to 2.5.
COMMON TRAP Removing a bonding electron (1.5).
- 4 Answer: (D) VSEPR shape** Easy ⌚ 0.75 min Bonding MT1-C04
SOLUTION 4 bond pairs + 1 lone pair: trigonal bipyramidal arrangement, with the lone pair equatorial.
- 5 Answer: (A) $\Delta H - \Delta U$** Medium ⌚ 1.5 min Thermodynamics MT1-C05
SOLUTION $\Delta n_g = 2 - 4 = -2$, so $\Delta H - \Delta U = -2RT = -2 \times 8.314 \times 298 \text{ J}$.
COMMON TRAP Sign of Δn_g , or taking $\Delta n = -1$.
- 6 Answer: (A) Boiling-point elevation** Easy ⌚ 0.75 min Solutions MT1-C06
SOLUTION i values are 2, 1, 3 and 1, so CaCl₂ has the largest elevation.
- 7 Answer: (D) K_p and K_c** Easy ⌚ 0.75 min Equilibrium MT1-C07
SOLUTION $\Delta n_g = 2 - 1 = 1$.
- 8 Answer: (D) Common-ion effect** Medium ⌚ 1 min Ionic equilibrium MT1-C08
SOLUTION $s = \frac{K_{sp}}{[\text{Cl}^-]} = \frac{1.8 \times 10^{-10}}{0.1}$.
COMMON TRAP $1.34 \times 10^{-5} \text{ M}$ is the solubility in pure water.
- 9 Answer: (D) Faraday's law** Easy ⌚ 0.5 min Electrochemistry MT1-C09
SOLUTION $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$.
- 10 Answer: (A) Zero-order half-life** Easy ⌚ 0.75 min Kinetics MT1-C10
SOLUTION $t_{1/2} = \frac{[A]_0}{2k}$.
COMMON TRAP (B) is true for first order.
- 11 Answer: (D) Isoelectronic radii** Easy ⌚ 0.5 min Periodicity MT1-C11
SOLUTION Isoelectronic ions (10 electrons): the lowest nuclear charge gives the largest ion.
- 12 Answer: (B) Basicity of an oxoacid** Easy ⌚ 0.75 min p-Block MT1-C12
SOLUTION H₃PO₃ has two P–OH bonds and one P–H bond. Only the OH hydrogens are ionisable.
COMMON TRAP Counting all three hydrogens.
- 13 Answer: (C) Xenon fluoride shapes** Easy ⌚ 0.5 min p-Block MT1-C13
SOLUTION XeF₄: 4 bond pairs + 2 lone pairs, with sp^3d^2 hybridisation.
- 14 Answer: (B) Colourless ion** Easy ⌚ 0.75 min d-Block MT1-C14
SOLUTION Sc³⁺ is d^0 , so no d–d transition is possible.
- 15 Answer: (C) IUPAC name** Medium ⌚ 1 min Coordination MT1-C15
SOLUTION Fe is +3 ($3 + x - 6 = 0$), and an anionic complex takes the "-ate" ending (ferrate).
COMMON TRAP Using "iron" in an anionic complex.

16 Answer: (D) Carbocation stability **Easy** ⌚ 0.5 min GOC

MT1-C16

SOLUTION 3° : the most hyperconjugation and $+I$ stabilisation.

17 Answer: (D) Peroxide effect **Easy** ⌚ 0.75 min Hydrocarbons

MT1-C17

SOLUTION Free-radical (anti-Markovnikov) addition. It applies to HBr only.

COMMON TRAP Markovnikov's product (2-bromopropane) forms without peroxide.

18 Answer: (D) SN_2 reactivity **Easy** ⌚ 0.75 min Haloalkanes

MT1-C18

SOLUTION SN_2 is fastest for the least hindered substrate (methyl).

COMMON TRAP Using the SN_1 order.

19 Answer: (A) Cannizzaro reaction **Easy** ⌚ 0.75 min Carbonyls

MT1-C19

SOLUTION Cannizzaro requires an aldehyde without α -hydrogens.

COMMON TRAP Aldehydes with α -H undergo aldol condensation instead.

20 Answer: (B) Carbylamine test **Easy** ⌚ 0.5 min Amines

MT1-C20

SOLUTION Only primary amines form isocyanides.

21 Answer: 2 Tetrahedral nickel complex **Medium** ⌚ 1 min Coordination **NV**

MT1-C21

SOLUTION Ni^{2+} is d^8 . Tetrahedral, weak-field Cl^- : $e^4t_2^4$ has 2 unpaired electrons.

COMMON TRAP Using square planar (0 unpaired), which applies to $[Ni(CN)_4]^{2-}$.

22 Answer: 30 Half-life from completion **Easy** ⌚ 0.75 min Kinetics **NV**

MT1-C22

SOLUTION 75% complete means 2 half-lives.

23 Answer: 11 pH of a base **Easy** ⌚ 0.5 min Ionic equilibrium **NV**

MT1-C23

SOLUTION $pOH = 3$, so $pH = 11$.

COMMON TRAP Answering 3.

24 Answer: 44 Combustion stoichiometry **Easy** ⌚ 0.5 min Basic concepts **NV**

MT1-C24

SOLUTION 1 mol CH_4 gives 1 mol CO_2 .

25 Answer: 4 Chiral centres **Easy** ⌚ 0.75 min Biomolecules **NV**

MT1-C25

SOLUTION C-2, C-3, C-4 and C-5.

COMMON TRAP Counting C-1 (the aldehyde carbon) or C-6.

MOCK 1 · MATHEMATICS

1 Answer: (C) Composition **Easy** ⌚ 0.5 min Functions

MT1-M01

SOLUTION $f(g(2)) = f(4) = 11$.

COMMON TRAP $(g \circ f)(2) = 49$.

2 Answer: (C) Counting relations **Easy** ⌚ 0.75 min Sets & relations

MT1-M02

SOLUTION $2^{n^2-n} = 2^6$.

COMMON TRAP $2^9 = 512$ counts all relations.

3 Answer: (D) Modulus of a power **Easy** ⌚ 0.5 min Complex numbers

MT1-M03

SOLUTION $(\sqrt{2})^8 = 16$.

4 Answer: (A) Symmetric functions of roots **Easy** ⌚ 0.75 min Quadratics

MT1-M04

SOLUTION $(\alpha + \beta)^2 - 2\alpha\beta = 25 - 12$.

5 Answer: (C) Inverse of a 2×2 matrix **Easy** ⌚ 1 min Matrices

MT1-M05

SOLUTION $|A| = 1$; swap the diagonal and negate the off-diagonal entries.

SHORTCUT Check: $A \cdot A^{-1} = I$ for (C).

COMMON TRAP (A) negates the off-diagonal entries but forgets to swap the diagonal.

6 Answer: (B) *Infinitely many solutions* **Medium** ⌚ 2 min **Determinants**

MT1-M06

SOLUTION The determinant is $k - 5$. At $k = 5$, subtracting consecutive equations gives $y + 2z = 3$ twice, so the system is consistent with infinitely many solutions.

COMMON TRAP Stopping at $|A| = 0$ without checking consistency.

7 Answer: (A) *At least one* **Medium** ⌚ 1 min **P&C**

MT1-M07

SOLUTION Total minus all-men: ${}^9C_3 - {}^5C_3 = 84 - 10$.

COMMON TRAP Computing ${}^4C_1 \cdot {}^8C_2 = 112$ counts some committees more than once.

8 Answer: (B) *Coefficient* **Easy** ⌚ 0.75 min **Binomial**

MT1-M08

SOLUTION ${}^5C_3 \cdot 2^3 = 10 \cdot 8$.

COMMON TRAP Forgetting 2^3 .

9 Answer: (A) *AP sum* **Easy** ⌚ 0.75 min **Sequences**

MT1-M09

SOLUTION $S_{20} = 10[6 + 19(4)] = 820$.

10 Answer: (A) *1^∞ form* **Easy** ⌚ 0.75 min **Limits**

MT1-M10

SOLUTION $e^{\lim x \cdot (2/x)} = e^2$.

COMMON TRAP Answering 1 (1^∞ is indeterminate).

11 Answer: (A) *Continuity* **Easy** ⌚ 0.75 min **Continuity**

MT1-M11

SOLUTION $\lim_{x \rightarrow 0} \frac{\sin 3x}{x} = 3$.

12 Answer: (C) *Decreasing interval* **Easy** ⌚ 1 min **Applications of derivatives**

MT1-M12

SOLUTION $f'(x) = 3(x^2 - 1) < 0$ for $|x| < 1$.

13 Answer: (B) *Integration by parts* **Easy** ⌚ 1 min **Integrals**

MT1-M13

SOLUTION $xe^x - \int e^x dx$.

SHORTCUT Differentiate the options: only (B) gives xe^x .

14 Answer: (C) *Odd functions* **Easy** ⌚ 0.75 min **Definite integrals**

MT1-M14

SOLUTION x^3 and $x \cos x$ are odd, so they integrate to 0 on $[-1, 1]$. What remains is $\int_{-1}^1 1 dx = 2$.

COMMON TRAP Declaring the whole integrand odd (0).

15 Answer: (B) *Integrating factor* **Easy** ⌚ 0.75 min **Differential equations**

MT1-M15

SOLUTION $e^{\int \tan x dx} = e^{\ln \sec x}$.

16 Answer: (B) *Distance between parallel lines* **Medium** ⌚ 1 min **Straight lines**

MT1-M16

SOLUTION The second line is $3x + 4y = 10$, so $d = \frac{|10 - 5|}{5} = 1$.

COMMON TRAP Not making the coefficients equal first (gives 3).

17 Answer: (B) *Centre and radius* **Easy** ⌚ 0.75 min **Circles**

MT1-M17

SOLUTION $g = -2$ and $f = 3$, so the centre is $(2, -3)$ and $r = \sqrt{4 + 9 + 12} = 5$.

COMMON TRAP Sign of the centre.

18 Answer: (C) *Latus rectum* **Easy** ⌚ 0.75 min **Ellipse**

MT1-M18

SOLUTION $\frac{2b^2}{a} = \frac{32}{5}$.

COMMON TRAP Using $\frac{b^2}{a}$ (half the length).

19 Answer: (A) *Angle between vectors* **Easy** ⌚ 0.75 min **Vectors**

MT1-M19

SOLUTION $\cos \theta = \frac{1 - 1 + 1}{\sqrt{3}\sqrt{3}}$.

20 Answer: (C) Conditional probability with dice **Medium** ⌚ 1 min Probability MT1-M20

SOLUTION Of 18 outcomes with an even first throw, (2, 6), (4, 4) and (6, 2) give sum 8: $\frac{3}{18}$.

COMMON TRAP $\frac{5}{36}$ is the unconditional probability.

21 Answer: 60 Angle between lines **Medium** ⌚ 1.5 min 3D NV MT1-M21

SOLUTION Dot product: $\sqrt{3} - 1 - \sqrt{3} - 1 + 8 = 6$. Magnitudes $\sqrt{6}$ and $\sqrt{24}$: $\cos \theta = \frac{6}{12} = \frac{1}{2}$.

22 Answer: 33 Variance of naturals **Easy** ⌚ 0.75 min Statistics NV MT1-M22

SOLUTION $\sigma^2 = \frac{n^2 - 1}{12} = \frac{99}{12}$.

23 Answer: 25 Perpendicular vectors **Medium** ⌚ 1 min Vectors NV MT1-M23

SOLUTION $25 = 9 + 16 + 2\vec{a} \cdot \vec{b}$, so $\vec{a} \cdot \vec{b} = 0$ and $|\vec{a} - \vec{b}|^2 = 25$.

24 Answer: 20 Maximum of a trig expression **Easy** ⌚ 0.75 min Trigonometry NV MT1-M24

SOLUTION $\sqrt{25 + 144} + 7$.

25 Answer: 3 Greatest-integer integral **Medium** ⌚ 1 min Definite integrals NV MT1-M25

SOLUTION $0 + 1 + 2 = 3$.

PART VIII · QUESTION BANK & MOCK TESTS

The Mock-Test System

A mock test has two jobs: to **rehearse the exam** and to **show you what to fix next**. Most students do the first and skip the second, which wastes most of the mock's value 📌 RECOMMENDATION.

How many mocks, and when 📌 RECOMMENDATION

Phase	When	Mocks	Type
A · Building	while the syllabus is still in progress	1 every 2 weeks	part-syllabus tests on completed chapters
B · Full syllabus	from the end of the syllabus to 45 days before the exam	1 per week	full-length (original mocks, older PYQ shifts)
C · Intensive	last 45 to 10 days	2–3 per week	full-length, including recent PYQ shifts
D · Final	last 10 days	1 every 2–3 days, the last one 2–3 days before the exam	recent unseen shifts, at your exam time
Exam eve	—	none	formula sheets and rest only

☆ **IMPORTANT** The 1 : 1 rule

Spend at least as long analysing a mock as taking it (3 hours), within 24 hours. If a week has no time for analysis, take fewer mocks, not less analysis.

Mock conditions (non-negotiable)

Replicate the exam

- ☐ Same start time as your shift (9:00 or 15:00)
- ☐ 3 hours, no breaks, no phone in the room
- ☐ Only blank rough sheets and a pen
- ☐ On-screen CBT format whenever possible, with the NTA-style palette (answered / not answered / marked)
- ☐ Water allowed; no food during the paper

Replicate the plan

- ☐ Follow your subject order and round plan (page [480](#))
- ☐ Note the time at each checkpoint (e.g. 0:55, 1:55, 2:40)
- ☐ Write a code next to each skipped question: **S** (skip for now), **N** (never)
- ☐ Stop exactly at 3:00, then fill Part 1 of the analysis sheet **before** checking answers

The six-step analysis

- 1 · Before checking answers**
Record time per subject, the number attempted, and your confidence in each answer (sure / 50-50 / guess). This baseline shows whether your confidence is well calibrated.
- 2 · Score & breakdown**
Marks per subject; correct, wrong and unattempted counts; accuracy = correct ÷ attempted.
- 3 · Question-by-question**
Give every question a status: **C** correct · **W** wrong · **UK** unattempted but you *knew* how · **UN** unattempted and not known. UK questions are free marks lost.
- 4 · Classify every W and UK**
Assign one of the 8 mistake types (page [449](#)) and write one line on what happened.

5 • Time audit

Which questions took more than 3 minutes? Did easy questions get skipped because time ran out?

6 • Three action items

No more than three, each specific ("re-derive rolling on an incline and solve 10 questions", not "revise rotation"). Put them into next week's plan.

📌 STRATEGY The re-attempt

Seven days after a mock, re-attempt every W and UK question without looking at the solution. If you still get it wrong, it goes into the Mistake Book as a **repeat**, the highest-priority category.

Reading your mock trend

📌 STRATEGY What the numbers mean **📌 RECOMMENDATION**

- **Accuracy below 80%** means slow down. You are over-attempting. Cut the attempts from the questions you are least sure of.
- **Accuracy above 92% with few attempts** means you are too cautious, or too slow. Work on speed in Round 1.
- **Many UK questions** is a time-management problem, not a knowledge problem. Fix selection and the 3-minute rule.
- **One subject much lower** decides the next two weeks of your plan. See the scenario tables (page [16](#)).
- **A single bad mock means nothing**. Judge the average of the last three.

PART VIII · QUESTION BANK & MOCK TESTS

Mock Analysis Sheet

Print one copy per mock (this chapter is two pages), or fill it in on screen. The codes are defined on the previous pages; the mistake types are on page [449](#).

MOCK TEST Part 1 · Before checking answers

Mock name / source _____ Date _____ Start time _____

Sleep the night before (hours) _____ Energy at start (1-5) _____

Subject	Order	Time used	Attempted	Sure	50-50
Physics	_____	_____	_____	_____	_____
Chemistry	_____	_____	_____	_____	_____
Maths	_____	_____	_____	_____	_____

MOCK TEST Part 2 · Score

Subject	Correct	Wrong	Unattempted	Marks (/100)	Accuracy %
Physics	_____	_____	_____	_____	_____
Chemistry	_____	_____	_____	_____	_____
Maths	_____	_____	_____	_____	_____
Total	_____	_____	_____	_____	_____

Marks lost to negatives _____ · Free marks lost (UK × 4) _____ · Estimated percentile (page [16](#))

MOCK TEST Part 3 · Question status (C / W / UK / UN, plus the mistake code for W and UK)

Q	Physics	Chemistry	Maths	Q	Physics	Chemistry	Maths
1	_____	_____	_____	14	_____	_____	_____
2	_____	_____	_____	15	_____	_____	_____
3	_____	_____	_____	16	_____	_____	_____
4	_____	_____	_____	17	_____	_____	_____
5	_____	_____	_____	18	_____	_____	_____
6	_____	_____	_____	19	_____	_____	_____
7	_____	_____	_____	20	_____	_____	_____
8	_____	_____	_____	21 NV	_____	_____	_____
9	_____	_____	_____	22 NV	_____	_____	_____
10	_____	_____	_____	23 NV	_____	_____	_____
11	_____	_____	_____	24 NV	_____	_____	_____
12	_____	_____	_____	25 NV	_____	_____	_____
13	_____	_____	_____				

MOCK TEST Part 4 · Mistake summary (count W + UK by type)

Type	Code	Physics	Chemistry	Maths	Total
Concept gap	CG				
Formula / fact recall	FR				
Calculation	CA				
Misreading	MR				
Transcription & clicking	TR				
Time & selection	TS				
Guessing (over-attempt)	GU				
Pressure (panic, fatigue)	PR				

MOCK TEST Part 5 · Time audit

Checkpoints reached (planned → actual): 1st subject done · 2nd subject done

· Round 2 started

Questions that took more than 3 minutes (Q numbers)

Easy questions left unattempted because of time (Q numbers)

MOCK TEST Part 6 · Three action items for next week

1.
2.
3.

Re-attempt of W and UK questions (7 days later): date ☐ done · still wrong:

→ Mistake Book ☐

MOCK TEST Part 7 · Reflection (one line each)

What went well

What I will do differently in the next mock

PART IX

Mistakes, Revision & Mindset

IX

The Mistake Book, the spaced revision system, and practical, evidence-based methods for focus, anxiety and consistency.

- The Mistake Book 449
- The Revision System 452

- Mindset, Focus & Wellbeing 455

PART IX · MISTAKES, REVISION & MINDSET

The Mistake Book

Most marks between 95 and 99+ percentile are not lost to topics you don't know. They are lost to **repeated, preventable mistakes** in topics you do know. The Mistake Book is a system for making each mistake once only.

☆ IMPORTANT The one rule

Every wrong or "knew-it-but-skipped" question from practice, PYQs and mocks gets **one entry**, classified into **one of eight types**. Each type has its own prevention routine. Re-attempt the entry after 7 days. If you get it wrong again, it becomes a **repeat**: the highest priority in your weekly review.

The eight mistake types

1 · CG · Concept gap

You didn't understand the idea, or you applied the wrong principle.

Example: using energy conservation in an inelastic collision.

Prevention: re-learn the concept from the module and NCERT. Explain it aloud in 60 seconds without notes. Then solve 5 **fresh** questions on the same model. Reading the solution is not enough.

2 · FR · Formula or fact recall

You knew the topic but misremembered a formula, condition, exception or NCERT fact.

Example: $\frac{4T}{r}$ used for a drop instead of a bubble.

Prevention: add it to your personal "forgotten" card. Test it with the three-pass method (page **367**) on days 1, 3 and 7. Write the **condition** next to the formula.

3 · CA · Calculation

The method was right; the arithmetic or algebra went wrong.

Example: $\frac{3}{5} \times \frac{10}{9} = \frac{3}{2}$ (wrong; it is $\frac{2}{3}$).

Prevention: write every step in a boxed area of the rough sheet. Cancel before multiplying. Verify NV answers by substitution or a quick estimate. Do the 15-minute daily drill (page **394**).

4 · MR · Misreading

You solved a different question from the one asked: missed "not", "incorrect" or "except"; wrong unit; radius instead of diameter; d instead of d^2 .

Prevention: underline the **asked quantity** and any negative word before solving. Write "Find: ___" at the top of your rough-sheet box. Re-read the last line of the question before you mark the answer.

5 · TR · Transcription & clicking

The right answer in your rough work, but a wrong value copied, a wrong option clicked, or a digit dropped while typing.

Prevention: number your rough-sheet boxes by question. Before saving, compare the screen with the rough sheet: "option letter, value, sign". In the NTA interface, check the palette colour after saving.

6 · TS · Time & selection

You spent too long on one question, or ran out of time with easy questions unread.

Prevention: the 3-minute rule; the round system and checkpoint times (page **480**); a quick first scan of each section. Track the number of UK questions per mock. It should fall week by week.

7 · GU · Guessing (over-attempt)

You answered with fewer than two options eliminated, or entered an NV answer you hadn't finished.

Prevention: apply the two-elimination rule (page [396](#)). In every mock, mark each guess with "?" on the rough sheet and count how many were right. Most students find their guesses score below break-even.

8 · PR · Pressure

Mistakes you would never make in calm practice: blanking, panic, careless errors late in the paper.

Prevention: rehearse under exam conditions at exam time; sleep 8 hours; use the reset routine (page [455](#)). Just before the test, write briefly about your worries (shown to improve exam performance in anxious students [LS-RAM]). Treat the adrenaline as fuel, not as a threat [LS-JAM].

How to keep the book**1 · Log within 24 hours**

Use the entry format below, in a notebook or on the printable pages. One line of diagnosis is enough. Accuracy matters more than neatness.

2 · Tag the type

One of the eight codes. If two apply, choose the **first** thing that went wrong.

3 · Write the fix

A specific action ("revise lens sign convention; solve 5 lens questions"), not a feeling ("be careful").

4 · Re-attempt on day 7

Cover the solution and re-solve. Right: tick it. Wrong: mark **R** (repeat) and schedule it again for day 14.

5 · Weekly review (30 min)

Count entries by type. The most frequent type decides next week's focus (e.g. many CA entries: daily drill plus verification habit).

TIP What a good entry looks like

P-13 · Mock 4 Q17 · MR · Asked for the field at $2R$ *inside* a hollow sphere (answer 0). I used the outside formula.

Fix: draw the region before choosing a formula. Re-attempt on 14 Mar ☐ ✓ / R

Entry pages

PRACTICE Mistake Book · page 1

#	Date	Source (mock/PYQ/set, Q)	Subject · chapter	Type	What went wrong	Fix	Day 7 ✓	R
1							☐	☐
2							☐	☐
3							☐	☐
4							☐	☐
5							☐	☐
6							☐	☐
7							☐	☐
8							☐	☐
9							☐	☐
10							☐	☐
11							☐	☐
12							☐	☐
13							☐	☐
14							☐	☐
15							☐	☐

PRACTICE Weekly type count

PART IX · MISTAKES, REVISION & MINDSET

The Revision System

Forgetting is predictable: without review, much of what you learn today fades within days. Two of the best-supported findings in learning research give the remedy. **Spaced** review, spread over increasing intervals, beats the same time spent in one block [LS-CEP]. **Retrieval** (testing yourself) beats re-reading [LS-RK], [LS-DUN]. This system builds both into a fixed schedule, so revision never depends on how you feel that day.

The six revision points

Point	When	Time per chapter	What you do
R0 · Same day	the evening you study it	15–20 min	Blank-page recall: write the chapter's key formulas and ideas from memory, then check. Solve 5 quick questions.
R1 · Day 2	two days later	15 min	Formula recall (cover-and-write) and re-solve the day's wrong questions.
R2 · Day 7	one week later	30–45 min	A mixed set of 15 questions from the chapter, timed. Log errors in the Mistake Book.
R3 · Day 14	two weeks later	30 min	Chapter-wise PYQs (2024–2026), about 15 questions.
R4 · Day 30 / monthly	once a month	45–60 min	The chapter test from the module, the formula sheet (page 367) and the Mistake Book entries for the chapter.
R5 · Final	last 30 days	per the final plans	Formula Handbook cycles, Last-Minute sheets, repeats from the Mistake Book (page 467).

☆ IMPORTANT Budget ☉ RECOMMENDATION

Revision should take about **20–25% of daily study time** during the learning phase, and most of it in the final phase. If revision is regularly squeezed out, you'll relearn chapters later at a much higher cost.

💡 TIP Interleave once the chapters pile up

After about six chapters, make R2 and R3 sets **mixed**: questions from several chapters in random order. Mixed practice is harder in the moment, but trains you to choose the method, which is what the exam tests [LS-ROH].

A normal day with revision built in

1 Morning block

New learning (the chapter of the day).

2 Afternoon block

Practice on the new chapter, then R1 for the chapter from two days ago.

3 Evening (45–60 min)

R0 for today's chapter, R2 for the chapter from a week ago, Mistake Book re-attempts due today.

4 Sunday

Weekly review: R3/R4 items due, the Mistake Book type count, next week's plan (page 475).

Spaced Revision Calendar

Write each chapter the day you finish it, then fill in its due dates (+2, +7, +14 and +30 days). Tick each review when done. Most students forget reviews, not chapters. This calendar prevents that.

 REVISION Spaced Revision Calendar · page 1

[illegible]

 REVISION Spaced Revision Calendar · page 2

[illegible]

 NOTE Enough rows?

There are 54 modules. Print pages 1–2 three times, or use one row per *group* of short chapters (e.g. "P15 + P17").

PART IX · MISTAKES, REVISION & MINDSET

Mindset, Focus & Wellbeing

Preparation is a long effort, and the problems in this chapter are normal parts of it. Each section names a problem, gives the evidence where there is good evidence, and a concrete routine. No quotes, no "just believe in yourself". Methods only.

Procrastination

STRATEGY Implementation intentions

Instead of "I will study Physics today", decide **when, where and what**: "At 6:30 pm, at my desk, I will open Module P11 and solve Practice Q1–Q5." A meta-analysis of 94 studies found that these if–then plans substantially increase follow-through [LS-GOLL].

Start small

- **The 5-question start:** commit to five questions, not "Rotation". Starting is the hardest part, and most people continue once started.
- **Make the first step obvious:** leave the book open at the right page the night before.
- **Time blocks of 50–90 minutes** with a written target for each block.

Remove friction

- Plan tomorrow in 5 minutes tonight: three blocks, each with a specific output.
- Keep a "parking list" on your desk: any distracting thought ("check the result date") goes on paper, to be handled at the next break.
- Track *completed blocks* in the daily tracker, not hours "sat at the desk".

Phone and screen use

WARNING The mere presence of a phone costs attention

In experiments, participants with their smartphone on the desk (even face-down and silent) performed worse on working-memory and fluid-intelligence tasks than those whose phone was in another room [LS-WARD].

1 Rule 1 · Another room

During study blocks the phone is outside the room, not in a bag on the desk.

2 Rule 2 · Scheduled checks

Check messages only in breaks (e.g. 10 minutes after every second block).

3 Rule 3 · Study apps on another device

If you need videos or PDFs, use a laptop or tablet with notifications off and only the needed app open.

4 Rule 4 · Screen curfew

No recreational screens in the hour before sleep. Screens and late messaging push bedtime later.

5 Rule 5 · Weekly allowance

Decide your leisure screen time for the week in advance (e.g. 1 h/day plus a longer slot on the rest day). Planned leisure is fine; unplanned scrolling is the problem.

Fear of Mathematics

STRATEGY Make the fear smaller and the evidence larger

1. **Expressive writing before tests.** Writing for about 10 minutes about your worries just before an exam improved the scores of anxious students in controlled studies [LS-RAM]. Try it before every mock for three weeks.
2. **Graded exposure.** Start each Maths session with 5 questions you can certainly solve (the Must-do models, page **285**), then move to harder ones. Success first builds momentum.
3. **Separate "I can't" from "I haven't yet".** Log every Maths error by type. Most turn out to be CA or MR (page **449**), which are habits, not ability.
4. **Timed confidence sets.** Once a week, take a 10-question set you are likely to finish. Record your time, and watch it fall over the weeks.

Mock-test anxiety

Reframe the arousal

A racing heart before a test is the body preparing to perform. Students told that arousal can *help* performance scored higher on a standardised maths test than a control group [LS-JAM]. Say it plainly: "This is energy for the paper."

Reframe the mock

A mock is **data, not a verdict**. Its purpose is to find mistakes before the exam does. A low mock score with a good analysis is more valuable than a high one you don't learn from. Judge the average of your last three mocks, never a single one.

TIP The 60-second reset (in any mock or in the exam)

Put the pen down. Breathe in slowly for 4 counts and out for 6 counts, five times. Read the next question's last line only. Then continue with the easiest unattempted question in the section.

Burnout and energy

IMPORTANT Warning signs RECOMMENDATION

Two weeks or more of: constant tiredness despite sleep, dread of opening books, falling mock scores despite more hours, irritability, or loss of interest in things you normally enjoy. These call for a **reset week**, not more hours.

Sleep

For 13–18-year-olds, the American Academy of Sleep Medicine recommends **8–10 hours** a night [LS-AASM]. Sleep also consolidates what you learned during the day [LS-RB]. Every plan in this book protects at least 7.5–8 hours.

Movement

The WHO recommends an average of **60 minutes a day** of moderate-to-vigorous physical activity for adolescents [LS-WHO]. A 20–30-minute walk, cycle or workout still counts. Timetables include it.

Rest

One lighter half-day per week, planned in advance. Rest is part of training. Performance falls when there is none.

STRATEGY The reset week

Keep the daily routine, but cut study to about 70% of normal: revision and easy questions only, no new chapters, no mocks. Sleep fully, exercise daily, and talk to someone you trust. Then resume the plan.

Motivation that lasts

1 Process goals

Aim for what you control ("4 blocks today, 30 questions, Mistake Book updated"), not only outcomes ("99 percentile").

2 Visible progress

Use the trackers: chapters done, questions solved, mock trend. Evidence of progress sustains effort better than slogans.

3 A weekly win

Each Sunday, write one thing that improved this week (a faster time, a fixed mistake type).

4 Your reasons

Write down why this matters to you (a college, a branch, a kind of work) and keep it inside the cover.

When to talk to someone

WARNING Get help early

If low mood, anxiety, sleep problems or hopeless thoughts last for more than two weeks, or if you ever feel unsafe, talk to a parent, teacher or doctor **now**. You don't have to wait until after the exam. In India, **Tele-MANAS** (the government's free mental-health helpline) is available on **14416** or **1-800-891-4416** [TELEMANAS]. Asking for help is part of looking after your preparation, not a sign of weakness.



Study Plans & Timetables

30- to 180-day plans at 6, 8 and 10 hours a day, the final 30/14/7/3-day plans, daily timetables and the weekly system.

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PART X · STUDY PLANS & TIMETABLES

How the Plans Work

Every plan in this part is built from the same parts: the five phases of the roadmap (page 15), the chapter order below, and a **block system** that turns one plan into its 6 h, 8 h and 10 h variants. All plans are recommendations  **RECOMMENDATION**. Adjust them with the Plan Selector (page 30) and your weekly review.

The block system (6 h · 8 h · 10 h)

A **block** is a focused session of about 2 hours (or 1–1.5 hours on school days) with a written target. Every day in every plan lists its blocks in priority order:

Variant	Blocks per day	Which blocks you do
6 h	3 blocks of 2 h (or 4 shorter blocks on school days)	Blocks 1–3 only
8 h	4 blocks	Blocks 1–4
10 h	5 blocks	Blocks 1–5
Hybrid (6 h weekdays, 10 h weekends)	3 on weekdays, 5 on Sat–Sun	Tasks marked W move to the weekend

☆ **IMPORTANT** What the 6 h variant gives up

The 6 h variant covers the **same chapters** in the same weeks, but with fewer practice questions per chapter and fewer extension sets. It relies more on the Must-do models. If your mock scores stall on the 6 h variant, the first fix is **selection and accuracy**, not more hours.

Daily time allocation by phase

Activity	Phase 1 · Build (6 / 8 / 10 h)	Phase 2 · Strengthen	Phase 3 · Mock & Fix	Phase 4 · Peak
New learning	2 / 2.5 / 3 h	0.5 / 1 / 1 h (second pass)	—	—
Practice & PYQs	2.25 / 3 / 4 h	3 / 4 / 5 h	2 / 3 / 4 h (targeted repair)	1 / 1.5 / 2 h
Revision (R0–R4)	1 / 1.5 / 1.75 h	1.5 / 2 / 2.5 h	1.5 / 2 / 2.5 h	2.5 / 3.5 / 4.5 h
Mistake Book	0.25 / 0.5 / 0.5 h	0.25 / 0.25 / 0.5 h	0.5 / 0.5 / 0.5 h	0.5 / 0.5 / 0.5 h
Tests & analysis (weekly average)	0.5 / 0.5 / 0.75 h	0.75 / 0.75 / 1 h	2 / 2 / 2.5 h	2 / 2.5 / 2.5 h

The chapter order

The order interleaves heavy and light chapters, so every week has one demanding chapter per subject at most. It follows the sequence in which ideas build on each other (e.g. GOC before the organic reaction chapters, calculus before area and DEs).

Physics

P01 → P02 → P03 → P04 → P05 →
P06 → P07 → P08 → P09 → P10 → P11
→ P12 (+ P20) → P13 → P14 → P15 →
P16 → P17 → P18 → P19

Chemistry

C01 → C02 → C03 → C09 → C04 →
C05 → C06 → C07 → C08 → C14 →
C13 → C15 → C16 → C10 → C11 → C12
→ C17 → C18 → C19 → C20

Mathematics

M01 → M02 → M03 → M04 → M05 →
M06 → M07 → M08 → M09 → M14 →
M10 → M11 → M12 → M13

 TIP Already studied some chapters at school or coaching?

Don't skip them. Do the module's **chapter test first**. Score $\geq 80\%$: do only the practice set and PYQs, which frees 50–70% of that chapter's time for weaker ones. Score below 60%: study the module in full.

PART X · STUDY PLANS & TIMETABLES

The 180-Day Plan

Phases: 0 · Diagnose (Days 1–3) → 1 · Build (Days 4–85) → 2 · Strengthen (Days 86–130) → 3 · Mock & Fix (Days 131–166) → 4 · Peak (Days 167–180).

Phase 1 · Build (Weeks 1–12)

Week	Physics	Chemistry	Maths	Tests (Sunday)	Done
1	Diagnostic (Days 1–3) · P01, P02	C01, C02	M01	Diagnostic test	☐
2	P03, P04	C03	M02	chapter tests	☐
3	P05	C09, C04	M03	chapter tests	☐
4	P06, P07	C05, C06 (start)	M04, M05	Part test 1 (Weeks 1–4)	☐
5	P08, P09	C06	M06	chapter tests	☐
6	P10	C07, C08	M07	chapter tests	☐
7	P11	C14	M07 (finish), M08 (start)	chapter tests	☐
8	P12, P20	C13, C15	M08	Part test 2 (Weeks 5–8)	☐
9	P13	C16, C10 (start)	M09, M14	chapter tests	☐
10	P14, P15	C10, C11	M10 (lines, circles)	chapter tests	☐
11	P16	C12, C17 (start)	M10 (conics), M11	chapter tests	☐
12	P17, P18, P19	C17, C18, C19, C20	M12, M13	Full Mock 1 + Part test 3	☐

Gate 1 (end of Week 12): every Must-do chapter has its practice set at $\geq 75\%$ and its chapter test taken ☐

Phase 2 · Strengthen (Weeks 13–18)

Week	Focus (all variants)	Extension (8 h / 10 h)	Mock	Done
13	PYQs: P01–P06, C01–C04, M01–M03	second pass on the weakest chapter per subject	Mock 2	☐
14	PYQs: P07–P10, C05–C08, M04–M06	QB-A to QB-C sets (page 417)	Mock 3	☐
15	PYQs: P11–P13, C09–C12 + NCERT tables	calculation drill $\times 2$ daily	Mock 4	☐
16	PYQs: P14–P20, C13–C16, M07–M09	QB-D, QB-E sets	Mock 5	☐
17	PYQs: C17–C20, M10–M14	second pass: two weakest chapters from the Mistake Book	Mock 6	☐
18	Weak-chapter repair; blank-sheet formula test (target $\geq 90\%$)	7-Day Formula Cycle (page 386)	Mock 7	☐

Gate 2: PYQs done for all Must-do + High chapters, formula recall $\geq 90\%$ ☐

Phase 3 · Mock & Fix (Weeks 19–23)

Week	Mocks	Between mocks	Done
19	2 (Tue, Sat)	analysis → three action items → targeted repair	☐
20	2	+ one PYQ shift from 2024	☐
21	3 (Mon, Wed, Sat)	+ Mistake Book repeats only	☐

Week	Mocks	Between mocks	Done
22	3	+ 2025 shifts	☐
23	3	+ formula cycle	☐

Gate 3: last 5 mocks average in your target band, with ≤ 8 negative marks each ☐

Phase 4 · Peak (Days 167–180)

Follow the Final 14 Days plan (page [469](#)).

Session 1 → Session 2 bridge (about 65–70 days)

For students who take Session 1 in January and aim to improve in April (the better NTA score counts, page [10](#)).

Period	What to do	Done
Days 1–2 after Session 1	Rest. Then compare your response sheet with the provisional answer key when NTA releases it	☐
Days 3–5	Full analysis of your Session 1 paper with the Mock Analysis Sheet. Re-take the Plan Selector with your actual score	☐
Weeks 2–5	Repair the three weakest chapters per subject (module + PYQs); solve the other Session 1 2027 shifts subject-wise	☐
Weeks 6–8	Phase 3 again: 2–3 mocks a week, using 2027 Session 1 shifts you haven't seen	☐
Final 14 days	Final 14 Days plan (page 469); board-exam dates permitting	☐



WARNING Class 12 board exams

Board exams usually fall between Session 1 and Session 2. During board weeks, switch to the **6 h variant or lower** and keep only R-revision and one mock per week. Board syllabi overlap strongly with JEE Physics, Chemistry and Maths, so board preparation is not lost time.

PART X · STUDY PLANS & TIMETABLES

The 120-Day Plan

NOTE Personalised for Gayathri

Starting **Monday 28 September 2026** as **Day 5**. The Day 1–4 tasks (profile, diagnostic, set-up) fit into the first weekend, 3–4 October. Day 120 is **21 January 2027**, the eve of the first Session 1 exam day (22 January, NTA tentative calendar [NTA-CAL27]). **If your allotted shift is later, add the extra days to Phase 3.**

Phases: 0 · Diagnose (Days 1–4) → 1 · Build (Days 5–67) → 2 · Strengthen (Days 68–95) → 3 · Mock & Fix (Days 96–106) → 4 · Peak (Days 107–120).

Phase 1 · Build (Weeks 1–9)

Week · dates	Physics	Chemistry	Maths	Sunday	Done
1 · 28 Sep–4 Oct	P01, P02, P03	C01, C02	M01, M02	chapter tests	☐
2 · 5–11 Oct	P04, P05	C03, C09	M03, M04	chapter tests	☐
3 · 12–18 Oct	P06, P07	C04, C05	M05, M06	Part test 1	☐
4 · 19–25 Oct	P08, P09, P10	C06, C07	M07	chapter tests	☐
5 · 26 Oct–1 Nov	P11, P12	C08, C14	M08	Mock 1	☐
6 · 2–8 Nov	P13, P14	C13, C15, C16	M09, M14	Part test 2	☐
7 · 9–15 Nov	P15, P16	C10, C11	M10	chapter tests	☐
8 · 16–22 Nov	P17, P18, P19, P20	C12, C17	M11, M12	Mock 2	☐
9 · 23–29 Nov	backlog + weakest Physics chapter	C18, C19, C20	M13	Part test 3	☐

TIP 6 h variant in Phase 1

The pace above assumes you have already met most Class 11 topics at school. On the 6 h variant, cover the **Must-do models** of each chapter in the listed week, and move the remaining practice to the Phase 2 extension slots.

Phase 2 · Strengthen (Weeks 10–13)

Week · dates	Focus (all variants)	Extension (8 h / 10 h)	Mock	Done
10 · 30 Nov–6 Dec	PYQs: P01–P10, C01–C08	QB-A to QB-C	Mock 3	☐
11 · 7–13 Dec	PYQs: P11–P20, C09–C12, M01–M06	QB-D, QB-E	Mock 4	☐
12 · 14–20 Dec	PYQs: C13–C20, M07–M14	second pass on weakest chapters	Mock 5	☐
13 · 21–27 Dec	Mistake Book repeats; blank-sheet formula test	7-Day Formula Cycle	Mock 6	☐

Phase 3 · Mock & Fix (28 Dec–7 Jan)

Dates	Plan	Done
28 Dec–31 Dec	Mocks on 28 and 30 Dec (exam time); analysis and repair on the other days	☐
1 Jan–7 Jan	Mocks on 2, 4 and 6 Jan; 2025 and 2026 PYQ shifts; formula cycle	☐

Phase 4 · Peak (8–21 January)

Follow the Final 14 Days plan (page **469**). Exam eve (21 January): page **478**.

PART X · STUDY PLANS & TIMETABLES

The 90-Day Plan

Phases: 0 · Diagnose (Days 1–2) → 1 · Build (Days 3–45) → 2 · Strengthen (Days 46–66) → 3 · Mock & Fix (Days 67–76) → 4 · Peak (Days 77–90).

Week	Physics	Chemistry	Maths	Sunday	Done
1	Diagnostic · P01–P03	C01, C02, C03	M01, M02	chapter tests	☐
2	P04, P05, P06	C09, C04, C05	M03, M04, M05	Part test 1	☐
3	P07, P08, P09	C06, C07	M06, M07	chapter tests	☐
4	P10, P11, P12	C08, C14, C13	M08	Mock 1	☐
5	P13, P14, P15	C15, C16, C10	M09, M14, M10 (part)	Part test 2	☐
6	P16, P17, P18, P19, P20	C11, C12, C17, C18, C19, C20	M10, M11, M12, M13	Mock 2	☐
7–9	PYQs of all Must-do chapters (one subject block per day, rotating)	+ second pass on weak chapters (8 h / 10 h)	Mocks 3–5 (one a week)	☐	
10–11 (Days 67–76)	3 mocks, analysis, targeted repair, formula cycle			☐	
12–13 (Days 77–90)	Final 14 Days (page 469)			☐	

WARNING Week 6 is deliberately dense

If a Week 6 chapter is new to you, cover only its Must-do models and PYQ structures (page [406](#)), and move the rest to the Phase 2 extension slots. On the 6 h variant, P3/P4 topics (page [402](#)) wait until Phase 3.

PART X · STUDY PLANS & TIMETABLES

The 60-Day Plan

Phases: 0 · Diagnose (Days 1–2) → 1 · Build P1 topics (Days 3–30) → 2 · Strengthen (Days 31–40) → 3 · Mock & Fix (Days 41–46) → 4 · Peak (Days 47–60).

 IMPORTANT Sixty days means selection

Cover every **P1 (Must-do)** topic in the Priority Matrix (page **402**) first. P2 topics enter only on the 8 h and 10 h variants, or once P1 is done. P3 and P4 topics are skipped.

Days	Physics	Chemistry	Maths	Test	Done
1–2	Diagnostic + plan			Diagnostic	☐
3–9	P01, P05, P06, P08, P09	C02, C03, C04, C09	M02, M03	chapter tests	☐
10–16	P07, P11, P12	C06, C07, C08	M05, M06, M04	Mock 1	☐
17–23	P13, P16	C10, C11, C12	M07, M08	Part test	☐
24–30	P15, P17, P18, P19	C14, C17, C19	M09, M10, M11, M12, M13	Mock 2	☐
31–40	PYQs of all P1 topics; P2 topics on 8 h / 10 h			Mocks 3–4	☐
41–46	3 mocks + repair + formula cycle				☐
47–60	Final 14 Days (page 469)				☐

PART X · STUDY PLANS & TIMETABLES

The 30-Day Plan

 IMPORTANT Thirty days: a revision-and-mock plan

This plan assumes you have studied most of the syllabus at least once. It builds marks through **P1 topics, PYQs, mocks and mistake elimination**, not through new chapters. If a P1 chapter is completely new, learn only its Must-do models.

Days	Focus	Blocks 1–3 (all variants)	Blocks 4–5 (8 h / 10 h)	Done
1	Diagnose	Full mock at exam time + analysis	Plan the 30 days with the Priority Matrix	☐
2–6	Physics P1 + formulas	Formula sheets → PYQs (P1 topics, 40/day)	Mistake Book, QB-A	☐
7	Mock 2	mock + analysis	repair	☐
8–12	Chemistry P1 + NCERT	NCERT tables → PYQs (40/day)	reagent tables, named reactions	☐
13	Mock 3	mock + analysis	repair	☐
14–18	Maths P1	Must-do models → PYQs (30/day)	calculation drill, QB-C, QB-D	☐
19–23	Mixed + mocks	Mocks on Days 19, 21 and 23; repair between	formula cycle	☐
24–30	Final 7 Days (page 470)			☐

PART X · STUDY PLANS & TIMETABLES

The Final Plans: 30, 14, 7 & 3 Days

The last month decides how much of your preparation turns into marks. The principles below hold for every final plan  **RECOMMENDATION** :

Do

- Take mocks **in your exam time slot** (9:00 or 15:00) so your body is alert at that hour.
- Revise from **your own material**: the Formula Handbook, the Mistake Book, the Last-Minute sheets.
- Keep the sleep schedule fixed. Wake at the time you'll wake on exam day.
- Keep exercise and meals normal.

Don't

- Start new chapters or new books in the last 14 days.
- Take more than one mock a day, or a mock on the exam eve.
- Compare mock scores with friends. Different series have different difficulty.
- Study past your bedtime. Sleep consolidates memory [LS-RB], and tiredness costs more marks than one extra hour gains.

Using the tables: blocks are about 2 hours. **6 h** = Blocks 1–3 · **8 h** = Blocks 1–4 · **10 h** = Blocks 1–5 (page **459**). On mock days, the mock and its analysis fill Blocks 1–3 for every variant.

PART X · STUDY PLANS & TIMETABLES

Final 30 Days

Days are counted down: D-30 is 30 days before your exam, D-1 is the eve.

Days	Blocks 1-3 (all variants)	Block 4 (8 h / 10 h)	Block 5 (10 h)	Done
D-30	Mock (exam slot) + analysis	repair action item 1	7-Day Formula Cycle · day 1	☐
D-29	Physics: weakest P1 chapter, module + 30 PYQs	Mistake Book repeats	formula cycle · day 2	☐
D-28	Chemistry: weakest P1 chapter + NCERT tables	organic reagent table	formula cycle · day 3	☐
D-27	Mock + analysis	repair	formula cycle · day 4	☐
D-26	Maths: weakest P1 chapter + 25 PYQs	calculation drill + QB-D	formula cycle · day 5	☐
D-25	Physics second-weakest chapter + PYQs	QB-E	formula cycle · day 6	☐
D-24	Mock + analysis	repair	formula cycle · day 7 (self-test)	☐
D-23	Chemistry: inorganic full NCERT pass (C09-C12)	p-block structures	Last-Minute Chemistry sheet	☐
D-22	Maths second-weakest chapter + PYQs	QB-C	Mistake Book weekly count	☐
D-21	Mock + analysis	repair	Frequently Forgotten list	☐
D-20	Physics: 2025 shift, Physics part only, timed (60 min) + review	repair	formula cycle 2 · day 1	☐
D-19	Chemistry: 2025 shift, Chemistry part only (50 min) + review	organic named reactions	formula cycle 2 · day 2	☐
D-18	Mock + analysis	repair	formula cycle 2 · day 3	☐
D-17	Maths: 2025 shift, Maths part only (70 min) + review	calculation drill	formula cycle 2 · day 4	☐
D-16	Mistake Book: every repeat entry re-solved	QB-F sprint ×2	formula cycle 2 · day 5	☐
D-15	Mock + analysis	repair	formula cycle 2 · day 6	☐
D-14 to D-1	Final 14 Days (next page)			☐

PART X · STUDY PLANS & TIMETABLES

Final 14 Days

Day	Blocks 1–3 (all variants)	Block 4 (8 h / 10 h)	Block 5 (10 h)	Done
D–14	Mock (exam slot, recent unseen shift) + analysis	repair top action item	formula cycle 2 · day 7	☐
D–13	Physics P1 revision: formula sheets P01–P20 + 40 mixed PYQs	Mistake Book (Physics)	Last-Minute Physics sheet	☐
D–12	Mock + analysis	repair	formula self-test (60 random)	☐
D–11	Chemistry: NCERT tables, trends, exceptions + 40 mixed PYQs	reagent and named-reaction tables	Last-Minute Chemistry sheet	☐
D–10	Mock + analysis	repair	Frequently Forgotten list	☐
D–9	Maths Must-do models: 10 timed questions per model	calculation drill + QB-F	Last-Minute Maths sheet	☐
D–8	Mock + analysis	repair	Mistake Book repeats	☐
D–7 to D–1	Final 7 Days (next page)			☐

 TIP If the exam is in the afternoon slot

Move every mock to 15:00–18:00, and shift the other blocks around it. Keep your wake-up time fixed at the time you'd need on exam day.

PART X · STUDY PLANS & TIMETABLES

Final 7 Days

Day	Blocks 1–3 (all variants)	Block 4 (8 h / 10 h)	Block 5 (10 h)	Done
D-7	Mock (exam slot) + analysis	repair	Last-Minute Physics sheet	☐
D-6	Physics: Mistake Book entries + 30 mixed questions (easy–medium)	formula sheets P01–P20	Frequently Forgotten (Physics)	☐
D-5	Mock + analysis	repair	Last-Minute Chemistry sheet	☐
D-4	Chemistry: NCERT tables + reactions + 30 mixed questions	inorganic exceptions list	Last-Minute Maths sheet	☐
D-3	Final mock (exam slot). Afternoon: check only the wrong answers, calmly	Maths: 20 easy Must-do questions	early night	☐
D-2	Light day: three passes of the Formula Handbook; Mistake Book repeats; 15 easy questions per subject	documents and admit card check; route to the centre	rest	☐
D-1	Exam eve (page 478)			☐

☆ IMPORTANT Why the last mock is on D-3

A mock two days before the exam gives you a final rehearsal and still leaves a full recovery day. A bad score on D-3 changes nothing about your preparation: treat it only as a reminder of the traps to watch for.

PART X · STUDY PLANS & TIMETABLES

Final 3 Days

A detailed version of D-3 to D-1 for a **9:00 shift** (shift everything by 6 hours for a 15:00 shift).

Time	D-3	D-2	D-1 (eve)
6:00	wake, water, 15-min walk	wake, walk	wake, walk
7:00	breakfast (exam-day menu)	breakfast	breakfast
8:00–9:00	Last-Minute sheet (one subject)	Formula Handbook pass 1	Last-Minute sheets, one calm read
9:00–12:00	Final mock	15 easy questions per subject, untimed	light: the Command Sheet (page 483), the Frequently Forgotten list
12:00–14:00	lunch, rest	Mistake Book repeats	lunch; pack the exam bag
14:00–16:00	check wrong answers only	Formula Handbook pass 2	walk, rest, no new problems
16:00–18:00	20 easy Maths questions	documents, route, centre visit if possible	family time, light reading
18:00–19:00	exercise	exercise	light stretching
19:00–21:00	dinner, Formula pass (one subject)	dinner, Formula pass 3	dinner by 20:00; 20 min of formulas at most
21:30	screens off	screens off	screens off by 21:00
22:00	sleep	sleep	sleep by 22:00



REVISION Final-days checklist

- ☐ Admit card downloaded and printed (check every detail on it)
 ☐ Photo ID ready
 ☐ Photos as specified on the admit card
 ☐ Centre route checked, travel planned
 ☐ Only items the admit card permits (e.g. a transparent water bottle, if allowed)
 ☐ Exam-day clothes as per the admit-card rules
 ☐ Alarm set (two alarms)
 ☐ Command Sheet read

PART X · STUDY PLANS & TIMETABLES

Daily Timetables

Four model days. Each protects **8 hours of sleep**, includes **exercise** and real breaks, and limits screens. Change the clock times to fit school, coaching and family routines, but keep the structure: hardest work when you're freshest, revision in the evening, screens off an hour before bed  **RECOMMENDATION**.

Fixed rules for every day

- Sleep 22:00–6:00 (or any fixed 8-hour window). Same time on weekends.
- 20–45 minutes of exercise or brisk walking (the WHO recommends an average of 60 minutes of activity a day for adolescents [LS-WHO]).
- Phone outside the study room during blocks (page **455**).
- A 10–15-minute break after every block, away from the desk.

Screen limits

- Study videos only in planned slots, at up to 1.5× speed, with notes.
- Leisure screens: up to about 60 minutes a day, after the last block, and **not in the last hour before sleep**.
- Social media: checked in breaks only, twice a day.
- Rest day: a longer leisure slot is fine, planned in advance.

Timetable A · School day (≈ 6 h of study)

Time	Activity	Type
6:00	Wake, water, 5 minutes of stretching	routine
6:15–7:00	Block 0: formula recall (R1 of the chapter from two days ago)	revision
7:00–7:30	Breakfast, get ready	break
7:45–14:15	School (travel included)	—
14:30–15:15	Lunch, rest (a nap of 20 minutes at most)	break
15:15–16:45	Block 1: new learning (hardest subject of the day)	learning
16:45–17:00	Break: snack, walk around	break
17:00–18:30	Block 2: practice on today's chapter	practice
18:30–19:15	Exercise / outdoors	health
19:15–20:00	Dinner, family	break
20:00–21:30	Block 3: second subject: practice or PYQs	practice
21:30–22:00	R0 recap + Mistake Book (on paper, no screens)	revision
22:00	Sleep	sleep

Timetable B · Full study day (8 h)

Time	Activity	Type
6:00	Wake, water	routine
6:15–6:45	Walk or exercise	health
6:45–7:15	Breakfast	break
7:15–9:15	Block 1: new learning (hardest subject)	learning
9:15–9:30	Break	break
9:30–11:30	Block 2: practice on Block 1's chapter	practice
11:30–11:45	Break	break

Time	Activity	Type
11:45–12:15	Block 3a: revision R1/R2 (30 min)	revision
12:15–13:30	Lunch, rest	break
13:30–15:30	Block 3b: second subject: learning and practice	learning
15:30–16:00	Break: snack, fresh air	break
16:00–17:00	Block 4: third subject: PYQs (1 h)	practice
17:00–18:00	Exercise / hobby	health
18:00–20:00	Free time, dinner, family	break
20:00–20:30	R0 recap + Mistake Book (on paper)	revision
20:30–21:00	Leisure (screens allowed until 21:00)	break
22:00	Sleep	sleep

Timetable C · Holiday or dropper day (10 h)

Time	Activity	Type
6:00	Wake, water	routine
6:10–6:40	Exercise	health
6:40–7:00	Breakfast	break
7:00–9:00	Block 1: new learning (hardest subject)	learning
9:00–9:15	Break	break
9:15–11:15	Block 2: practice	practice
11:15–11:30	Break	break
11:30–12:30	Block 3: revision R1/R2	revision
12:30–13:30	Lunch, rest	break
13:30–15:30	Block 4: second subject	learning / practice
15:30–16:00	Break	break
16:00–18:00	Block 5: third subject, PYQs or a timed set	practice
18:00–19:00	Outdoors / exercise	health
19:00–20:00	Dinner, family	break
20:00–21:00	Block 6: R0 recap + Mistake Book + tomorrow's plan	revision
21:00–22:00	Wind-down: no screens, light reading	break
22:00	Sleep	sleep

Timetable D · Mock day (exam slot 9:00–12:00)

Time	Activity
6:00	Wake (the same time you'll wake on exam day)
6:15–6:40	Walk, stretching
6:40–7:10	Breakfast: the same menu you plan for exam day
7:10–8:30	Light warm-up: one Last-Minute sheet, 5 easy questions. No new material
8:30–9:00	Set up the desk as in the exam: rough sheets, water, timer; phone outside
9:00–12:00	Mock (no breaks)
12:00–12:20	Part 1 of the Mock Analysis Sheet, before checking answers

Time	Activity
12:20–13:30	Lunch, rest
13:30–16:30	Analysis (Parts 2–7) and Mistake Book entries
16:30–17:30	Exercise / outdoors
17:30–19:30	Repair action item 1 (8 h and 10 h variants)
19:30–21:00	Dinner, family, a short formula pass
22:00	Sleep

 NOTE Periods and health

Plan lighter blocks (more revision, less new learning) on days you know you'll feel low in energy, rather than skipping days. If pain or fatigue regularly disrupts study, speak to a doctor. Effective options exist, and planning around the exam dates is easier with advance advice.

PART X · STUDY PLANS & TIMETABLES

The Weekly System & Scorecard

A week is the natural unit of preparation: long enough to finish chapters, short enough to correct course. Each week follows the same rhythm, and ends with a 45-minute review that decides the next week RECOMMENDATION.

The weekly rhythm

Day	Main work	Evening
Monday	New chapters (all three subjects)	R0 + R1
Tuesday	New chapters	R0 + R1
Wednesday	New chapters + one timed mixed set	R0 + R2
Thursday	New chapters	R0 + R1
Friday	New chapters	R0 + R2
Saturday	Chapter tests of the week; PYQs; backlog	R3 items due
Sunday	Morning: test or mock (exam slot) · afternoon: analysis · evening: weekly review	light; early night

TIP The half-day of rest

Take one half-day off each week: Sunday evening after the review, or Wednesday evening. Decide it in advance, so the rest doesn't make you feel guilty.

The 45-minute weekly review

- 1 · Score the week (10 min)**
Fill in the scorecard below honestly.
- 2 · Mistake Book count (10 min)**
Count this week's entries by type (page 449). The largest type gets a specific fix next week.
- 3 · Plan check (10 min)**
Compare the plan's week with what you actually did. Move any backlog into named slots; don't just "catch up later".
- 4 · Next week (15 min)**
Write each day's three blocks with specific outputs. Put the three mock action items into the first half of the week.

Weekly Scorecard

Score each line 0, 1 or 2 against its target. 16 or more is a strong week, 12–15 a normal week. Below 12 for two weeks in a row: re-check the plan's hours (page 30) and the warning signs of burnout (page 455).

REVISION Weekly Scorecard · Week _____ · dates _____

Measure	Target	Actual	Score (0–2)
Study blocks completed	as per plan (e.g. 24 for 8 h)		
Chapters completed (Phase 1) or PYQ sets done (Phase 2+)	as per plan		
Questions solved	250+ (8 h)		
Practice accuracy	$\geq 75\%$ (Phase 1) · $\geq 85\%$ (later)		
Revisions done on their due day	100% of the calendar		
Mistake Book entries logged within 24 h	all		
Mock / test score vs target band	in band		
Average sleep	≥ 7.5 h		
Exercise days	≥ 5		
Leisure screen time within limit	≥ 6 days		
Total	/ 20		

 Biggest win this week

 Fix for next week (one line)
REVISION Weekly score log (write the total each Sunday)

Week	1	2	3	4	5	6	7	8	9	10	11	12	13
Score													
Week	14	15	16	17	18	19	20	21	22	23	24	25	26
Score													

XI

Exam Day

Exam-eve and exam-day plans, the exam-hall protocol, subject-specific scoring strategies and the one-page Command Sheet.

• The Exam Eve	478
• The Exam-Day Plan	479
• The Exam-Hall Protocol	480

• Subject Scoring Strategies	482
• Exam Day Command Sheet	483

PART XI · EXAM DAY

The Exam Eve

The goal of the exam eve is simple: **arrive tomorrow rested, calm and on time**. Nothing you study today will add as many marks as a good night's sleep protects  **RECOMMENDATION**.

☆ IMPORTANT Three rules for the eve

1. **No new problems and no mock.** A hard question tonight can only shake your confidence.
2. **Revise your own sheets only:** the Command Sheet, the Frequently Forgotten list and one pass of the Last-Minute sheets.
3. **Screens off by 21:00, asleep by 22:00** (for a 9:00 shift).

Time	What to do	Done
Morning	Wake at exam-day time. Walk. One calm pass of the Last-Minute sheets (page 485)	☐
Late morning	Re-read your Mistake Book's repeat entries only	☐
Afternoon	Rest, light activity, family time. Pack the exam bag (checklist below)	☐
17:00	Read the Command Sheet (page 483) once, then put it in the bag	☐
18:00	Light exercise or a walk	☐
19:30–20:00	Early, familiar dinner. Nothing new or heavy	☐
20:00–20:30	Optional 20 minutes: the Frequently Forgotten list. Then close the books	☐
21:00	Screens off. Shower, light reading, set two alarms	☐
22:00	Lights out	☐

REVISION Exam bag checklist (check against your admit card, which overrides this list)

- ☐ Admit card (printed, all pages as instructed)
 ☐ Original valid photo ID
 ☐ Passport-size photographs as specified
☐ Any declaration/undertaking the admit card requires
 ☐ Transparent water bottle (if permitted)
 ☐ Simple clothes that meet the dress guidelines, and no metal items
 ☐ Money and a charged phone for the journey (to be left outside/as instructed)
 ☐ Route and travel arranged; leave with a margin

TIP If you can't fall asleep

Don't check the time or your phone. Lie still with your eyes closed and breathe slowly (4 counts in, 6 out). Quiet rest still helps, and one imperfect night rarely decides an exam. Adrenaline will carry you through the morning.

PART XI · EXAM DAY

The Exam-Day Plan

For a **9:00 shift** (shift everything by 6 hours for a 15:00 shift). Reporting and gate-closing times are printed on your admit card. In 2026, gates closed 30 minutes before the exam started [NTA-IB26].

Time	Step	Done
5:45	Wake (your fixed mock-day time). Water, 5 minutes of stretching	☐
6:00–6:30	Familiar light breakfast (e.g. idli, poha or toast with eggs or paneer) and water. Nothing new	☐
6:30	Glance at your personal "top 5 forgotten formulas" card for 10 minutes at most	☐
Leave home	With a margin for traffic, so that you reach the centre well before reporting opens	☐
At the centre	Follow staff instructions; identity check and frisking. Don't discuss topics or predictions with other candidates	☐
Seated	Check your details on screen. Breathe: 4 in, 6 out, five times. Recall the first 5 minutes of the protocol (page 480)	☐
9:00	Start. Execute the plan you have practised in every mock	☐
After the exam	Don't analyse the paper with friends on the way out. Eat, rest, and switch off for the evening	☐

☆ IMPORTANT Between shifts and sessions

If you have taken Session 1 and plan to take Session 2, use the bridge plan (page **461**). Analyse your response sheet only when NTA publishes the provisional key, not from memory on the day.

PART XI · EXAM DAY

The Exam-Hall Protocol

This protocol turns 180 minutes into a routine you have rehearsed in every mock. Every part of it is a recommendation  **RECOMMENDATION**. Test it in at least five mocks, adjust it, and then **don't change it on exam day**.

The first 5 minutes

1 Minute 0–1 · Settle

Breathe slowly: 4 counts in, 6 out, three times. Pen in hand.

2 Minute 1–2 · Rough sheet

Draw three columns (P, C, M) to log question numbers you skip. Write your checkpoint times at the top (see below).

3 Minute 2–3 · Memory dump

Write the 3–5 formulas you tend to forget. They are now out of your head and on paper.

4 Minute 3–5 · Start with your first subject

Open Question 1 of your first subject. Begin Round 1.

NOTE Instructions time

Read the on-screen instructions calmly before the exam begins, especially the NV answer format and the question-palette colours. They should already be familiar from mocks. Follow the invigilator's instructions about when you may write on the rough sheet.

Subject order

Option	Order	Best for
Default	Chemistry → Physics → Maths	most students: Chemistry is fastest, banks marks early and settles nerves
Strongest first	your strongest → middle → weakest	students whose confidence depends on a strong start
Maths first	Maths → Chemistry → Physics	Maths-strong students who want the lengthiest section while fresh

Choose one after testing each in two mocks. Record the choice on the Command Sheet.

The round system

Round 1 · Sure marks

Go through the whole subject once. Attempt every question where you **see the method within about 20 seconds** and can finish in ≤ 2 minutes. Mark everything else **S** on your rough sheet and move on without guilt.

Round 2 · Medium questions

Return to the S questions of that subject. Give each **up to 3 minutes**. If you're still unsure after 3 minutes, mark it for review, note how far you got, and move on.

Round 3 · Final pass (last 20 min)

Across all subjects: finish the questions where you already have partial progress. Attempt an MCQ only after eliminating **at least two** options (page [396](#)). Enter NV answers only when solved.

Time checkpoints (9:00 start, default order)

Clock	Elapsed	Checkpoint
9:45	0:45	Chemistry Rounds 1 + 2 done; move to Physics

Clock	Elapsed	Checkpoint
10:45	1:45	Physics Rounds 1 + 2 done; move to Maths
11:40	2:40	Maths Rounds 1 + 2 done; start Round 3
11:50	2:50	Last 10 minutes: verification only
12:00	3:00	End

⚠ WARNING Behind schedule?

At each checkpoint, move on **even if the subject isn't finished**. Mark the rest S and return in Round 3. Coaching analyses of 2026 papers suggested that spending much more than 60–70 minutes on Maths often hurt overall scores [T-ANALYSIS26]. The checkpoints exist so that no subject eats another's time.

The last 20 minutes

- Stop starting long questions. Work only on those where you already have progress.
- Re-check every **NV answer entered**: value, sign, rounding and units as the question asks.
- Check the question palette: every question you meant to answer should show as answered.

The last 10 minutes

- **No new attempts** unless the answer is certain within a minute.
- Change an answer **only** if you find a specific error in your working, not because of a feeling.
- Make sure your final responses are saved (answered-and-marked questions count as answered; confirm the rule in the on-screen instructions).

Panic control

🕒 STRATEGY When a section feels hard

1. **Remember normalisation.** Your percentile is computed within your shift (page **10**). If the paper is hard for you, it is hard for everyone in that shift.
2. **Reset for 60 seconds.** Pen down; five slow breaths (4 in, 6 out); read the *last line* of the next question only.
3. **Find an easy question.** Scroll to the next question in that subject that looks short. One solved question restores momentum.
4. **Switch subjects early** if the whole section is hard. Return in Round 3.
5. **Never chase a lost question.** It is worth 4 marks. Twenty minutes on it costs the 12–16 marks you could have scored elsewhere.

PART XI · EXAM DAY

Subject Scoring Strategies

Recommended attempt and accuracy targets by level  **RECOMMENDATION** . "Level" is your current mock score in that subject (out of 100). Aim to hit the **accuracy** target first; raise attempts only once accuracy holds for three mocks.

Subject	Level	Attempt	Accuracy target	Where the marks come from
Physics	Strong (≥ 70)	21–23	$\geq 90\%$	everything; watch calculation traps in mechanics
Physics	Average (40–70)	16–19	$\geq 85\%$	the easy block (modern physics, semiconductors, units, EM waves, KTG), current electricity, optics formulas
Physics	Weak (< 40)	12–14	$\geq 90\%$	the easy block only, plus direct formula questions (page 33)
Chemistry	Strong (≥ 75)	23–25	$\geq 95\%$	everything; NCERT exceptions decide the last marks
Chemistry	Average (45–75)	19–21	$\geq 90\%$	inorganic NCERT, physical numericals, biomolecules, GOC orders
Chemistry	Weak (< 45)	15–17	$\geq 90\%$	NCERT inorganic facts, biomolecules, practical chemistry, one-step physical questions (page 161)
Maths	Strong (≥ 75)	20–22	$\geq 90\%$	two passes; verify every NV
Maths	Average (45–75)	16–18	$\geq 85\%$	the Must-do models; the 3-minute rule
Maths	Weak (< 45)	12–14	$\geq 90\%$	Must-do models only (page 285)

 IMPORTANT Section B in every subject

Section B (5 NV questions per subject) has negative marking, and a guess almost never scores. Scan it in Round 1: NV questions are often direct (a count, a determinant, a stopping potential). Solve those early; leave the rest for Round 2.

PART XI · EXAM DAY

Exam Day Command Sheet

BEFORE THE EXAM

- ☐ Admit card, photo ID, photos as the admit card specifies
- ☐ Wake at mock-day time · familiar breakfast · water
- ☐ Leave with a margin; follow the reporting time on the admit card
- ☐ Seated: breathe **4 in · 6 out**, five times

FIRST 5 MINUTES

1. Settle: three slow breaths, pen in hand.
2. Rough sheet: P · C · M skip columns + checkpoint times.
3. Memory dump: my 3–5 forgotten formulas.
4. Open my first subject, start **Round 1**.

MY ORDER & CHECKPOINTS

Order	Subject	Done by (9:00 shift)	(15:00 shift)
1		9:45	15:45
2		10:45	16:45
3		11:40	17:40
R3	all subjects	11:50 → verify	17:50 → verify

THE ROUNDS

R1 · sure marks: the method is visible in about 20 s and it takes ≤ 2 min. Otherwise mark **S** and move on.

R2 · medium: at most **3 minutes** each, then leave it.

R3 · last 20 min: questions with partial progress; MCQ only with **2 options eliminated**.

THE RULES

- Underline the **asked quantity** and any **NOT / INCORRECT / EXCEPT**.
- Read the last line again before marking an answer.
- **NV:** enter only a solved answer. Check value, sign and rounding.
- Never chase a lost question: 4 marks isn't worth 20 minutes.
- Change an answer only for a **specific error** you've found.

LAST 20 MIN → no new long questions · check every NV entry · check the palette.

LAST 10 MIN → verification only · all responses saved.

PANIC RESET (60 s)

Pen down → 5 slow breaths → read only the last line of the next question → do the next *easy* question. **A hard paper is hard for the whole shift.** Normalisation handles it.



MISTAKE ALERT My personal top 3 traps (from the Mistake Book)

1. _____
2. _____
3. _____

PART XII

Last-Minute Revision Book

Compressed one-page sheets for Physics, Chemistry and Maths, reaction and shortcut sheets, and the final checklists.

• How to Use the Last-Minute Book	485
• Physics in One Page	486
• Physical & Inorganic Chemistry in One Page	487
• Organic Reactions in One Page	488

• Maths in One Page	489
• Shortcut Checklist	490
• Mistake Checklist	491
• Strategy Checklist	492

PART XII · LAST-MINUTE REVISION BOOK

How to Use the Last-Minute Book

Seven one-page sheets for the last three days and the exam morning. They hold **only the highest-yield lines**: the formulas, facts and rules that decide the most marks and are easiest to get wrong. The full versions are in the Formula Handbook (page [367](#)).

Sheet	Page	When to read
Physics in one page	486	D-3, D-1
Physical & inorganic chemistry in one page	487	D-3, D-1
Organic reactions in one page	488	D-2, D-1
Maths in one page	489	D-2, D-1
Shortcut checklist	490	D-2
Mistake checklist	491	D-1, exam morning
Strategy checklist	492	D-1, exam morning

 IMPORTANT Read, don't study

On these days, read each sheet once, calmly, and cover-test only the lines you mark as shaky. If a line looks unfamiliar, **don't** open the full chapter. Note it on your "top 5 forgotten" card and move on.

PART XII • LAST-MINUTE REVISION BOOK

Physics in One Page

Mechanics

$$R = \frac{u^2 \sin 2\theta}{g} \cdot H = \frac{u^2 \sin^2 \theta}{2g} \cdot T = \frac{2u \sin \theta}{g}$$

$$\text{Banking: } \tan \theta = \frac{v^2}{rg} \cdot \text{vertical circle: } v_{\text{bottom}} \geq \sqrt{5gl}$$

$$\text{Elastic, } m_2 \text{ at rest: } v_1 = \frac{m_1 - m_2}{m_1 + m_2} u, v_2 = \frac{2m_1}{m_1 + m_2} u$$

$$I: \text{ring } MR^2 \cdot \text{disc } \frac{1}{2}MR^2 \cdot \text{solid sphere } \frac{2}{5}MR^2 \cdot \text{rod } \frac{1}{12}ML^2 / \frac{1}{3}ML^2$$

$$\text{Rolling: } a = \frac{g \sin \theta}{1 + k^2/R^2} \cdot L = I\omega \text{ conserved if } \tau = 0$$

$$v_o = \sqrt{GM/r} \cdot v_e = \sqrt{2gR} \cdot T^2 \propto r^3 \cdot E = -\frac{GMm}{2r}$$

$$g_h \approx g(1 - 2h/R) \cdot g_d = g(1 - d/R)$$

Matter, heat, thermodynamics

$$\text{Bernoulli: } v_t = \frac{2r^2(\rho - \sigma)g}{9\eta} \cdot h = \frac{2T \cos \theta}{r\rho g} \cdot \text{drop } \frac{2T}{r}, \text{ bubble } \frac{4T}{r}$$

$$W_{iso} = nRT \ln \frac{V_2}{V_1} \cdot \text{adiabatic } PV^\gamma \text{ const, } W = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$$

$$\eta = 1 - \frac{T_2}{T_1} \cdot v_{rms} = \sqrt{\frac{3RT}{M}} \cdot \gamma = 1 + \frac{2}{f}$$

$$\text{Stefan } \sigma T^4 \cdot \text{Wien } \lambda_m T = b \cdot \text{Newton's cooling is exponential}$$

Oscillations & waves

$$T = 2\pi\sqrt{m/k} \cdot 2\pi\sqrt{l/g} \cdot v = \sqrt{T/\mu} \cdot \text{closed pipe: odd harmonics, } f_1 = \frac{4L}{v}$$

$$\text{Doppler } f' = f \frac{v \pm v_o}{v \mp v_s} \cdot \text{beats } |f_1 - f_2|$$

Electricity & magnetism

$$\text{Dipole: axial } \frac{2kp}{r^3}, \text{ equatorial } \frac{kp}{r^3} \cdot \text{sheet } \frac{\sigma}{2\epsilon_0} \cdot \text{conductor } \frac{\sigma}{\epsilon_0}$$

$$C = \frac{\epsilon_0 A}{d - t + t/K} \cdot U = \frac{1}{2}CV^2 \cdot \text{sharing: common } V = \frac{\sum \epsilon_0 V}{\sum C}$$

$$R \rightarrow n^2 R \text{ when stretched } n \text{ times} \cdot P_{max} \text{ at } R = r \cdot \text{meter bridge}$$

$$\frac{X}{R} = \frac{l}{100 - l}$$

$$B: \text{wire } \frac{\mu_0 I}{2\pi r} \cdot \text{loop } \frac{\mu_0 I}{2R} \cdot \text{solenoid } \mu_0 n I \cdot \text{arc } \frac{\mu_0 I \theta}{4\pi R}$$

$$r = \frac{mv}{qB} \cdot T = \frac{2\pi m}{qB} \cdot \text{shunt } S = \frac{I_g G}{I - I_g}$$

$$\epsilon = Blv \cdot \tau_{LR} = L/R \cdot Z = \sqrt{R^2 + (X_L - X_C)^2} \cdot \omega_0 = \frac{1}{\sqrt{LC}} \cdot P =$$

$$V_{rms} I_{rms} \cos \phi$$

$$\frac{E_0}{B_0} = c \cdot \text{spectrum: } \gamma, X, \text{ UV, visible, IR, micro, radio}$$

Optics

$$\text{Lens } \frac{1}{v} - \frac{1}{u} = \frac{1}{f} \cdot \text{mirror } \frac{1}{v} + \frac{1}{u} = \frac{1}{f} \cdot P = \sum P_i$$

$$\text{Prism } \delta_{min}: n = \frac{\sin \frac{A+\delta}{2}}{\sin \frac{A}{2}} \cdot \text{thin prism } \delta = (n - 1)A$$

$$\text{YDSE } \beta = \frac{\lambda D}{d} \cdot \text{slit central width } \frac{2\lambda D}{a} \cdot \text{Malus } \cos^2 \theta \cdot \text{Brewster } \tan i_B = n$$

Modern physics & devices

$$E(\text{eV}) = \frac{1240}{\lambda(\text{nm})} \cdot K_{max} = h\nu - \phi = eV_0 \cdot \lambda_e = \frac{12.27}{\sqrt{V}} \text{ \AA}$$

$$E_n = -13.6 \frac{Z^2}{n^2} \text{ eV} \cdot \text{lines } \frac{n(n-1)}{2} \cdot R = R_0 A^{1/3} \cdot T_{1/2} = \frac{0.693}{\lambda}$$

$$\text{NAND, NOR universal} \cdot \overline{AB} = \bar{A} + \bar{B} \cdot \text{full-wave output } 2f$$

PART XII · LAST-MINUTE REVISION BOOK

Physical & Inorganic Chemistry in One Page

Physical chemistry

$$\Delta H = \Delta U + \Delta n_g RT \cdot \Delta G = \Delta H - T\Delta S \cdot \Delta G^\circ = -RT \ln K = -nFE^\circ$$

$$K_p = K_c(RT)^{\Delta n_g} \text{ (gases only)} \cdot Q < K: \text{forward}$$

$$\text{Weak acid } [H^+] = \sqrt{K_a C} \cdot \text{buffer pH} = pK_a + \log \frac{\text{salt}}{\text{acid}}$$

$$K_{sp}: AB \ s^2 \cdot AB_2 \ 4s^3 \cdot A_2B_3 \ 108s^5 \cdot \text{common ion lowers } s$$

$$\Delta T_f = iK_f m \cdot \pi = iCRT \cdot i = 1 + (n-1)\alpha$$

$$\text{Nernst } E = E^\circ - \frac{0.0591}{n} \log Q \cdot \log K = \frac{nE^\circ}{0.0591} \cdot \Lambda_m = \frac{1000\kappa}{M}$$

$$\text{First order } t_{1/2} = \frac{0.693}{k} \cdot \text{zero order } t_{1/2} = \frac{[A]_0}{2k} \cdot 87.5\% = 3 \text{ half-lives}$$

$$\text{Arrhenius } \log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$\text{Nodes: radial } n-l-1, \text{ angular } l \cdot E_n = -13.6 \frac{Z^2}{n^2} \text{ eV}$$

$$\text{Bond order } \frac{1}{2}(N_b - N_a) \cdot \text{paramagnetic: } O_2, B_2, O_2^- \cdot O_2^+ \text{ bond order } 2.5$$

Inorganic chemistry

IE exceptions: Be > B, N > O, Mg > Al, P > S · most negative EGE: Cl

Isoelectronic: $N^{3-} > O^{2-} > F^- > Na^+ > Mg^{2+} > Al^{3+}$

Shapes: XeF_2 linear · XeF_4 square planar · SF_4 see-saw · ClF_3 T-shaped · XeF_6 distorted octahedral

Basicity: H_3PO_4 3 · H_3PO_3 2 · H_3PO_2 1 · H_3BO_3 Lewis acid

Solid $PCl_5 = [PCl_4]^+ [PCl_6]^-$ · B_2H_6 : 3c-2e bonds

$KMnO_4$: acidic $n = 5$, neutral $n = 3$ · $K_2Cr_2O_7$ acidic $n = 6$

Spin-only μ : 1.73 · 2.83 · 3.87 · 4.90 · 5.92 BM ($n = 1$ to 5)

CFSE (oct) = $(-0.4t + 0.6e)\Delta_o$ · $\Delta_t = \frac{4}{9}\Delta_o$

Series: $I^- < Br^- < Cl^- < F^- < OH^- < H_2O < NH_3 < en < CN^- < CO$

$[Ni(CO)_4]$ tetrahedral, dia · $[Ni(CN)_4]^{2-}$ square planar, dia ·

$[NiCl_4]^{2-}$ tetrahedral, para

$[MA_4B_2]$: 2 geometrical · $[MA_3B_3]$: fac/mer · cis- $[M(AA)_2B_2]$: optically active

Colourless: d^0, d^{10} ($Sc^{3+}, Ti^{4+}, Zn^{2+}$) · $KMnO_4$ colour = charge transfer

Lanthanoids: +3 common · Ce^{4+} oxidant · Eu^{2+}, Yb^{2+} reductants · $Zr \approx Hf$

Cu^+ disproportionates in water · Mn: max +7 in the 3d series

PART XII · LAST-MINUTE REVISION BOOK

Organic Reactions in One Page

Reagents

$\text{H}_2/\text{Lindlar} \rightarrow \text{cis-alkene}$ · $\text{Na/liq. NH}_3 \rightarrow \text{trans-alkene}$
 $\text{HBr} + \text{peroxide} \rightarrow \text{anti-Markovnikov (HBr only)}$
 cold dilute alkaline $\text{KMnO}_4 \rightarrow \text{syn-diol (Baeyer)}$ · $\text{O}_3/\text{Zn-H}_2\text{O} \rightarrow \text{carbonyls}$
 LiAlH_4 reduces acids, esters, amides, nitriles · NaBH_4 : aldehydes and ketones only
 PCC : $1^\circ \rightarrow \text{aldehyde}$ · $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}^+$: $1^\circ \rightarrow \text{acid}$, $2^\circ \rightarrow \text{ketone}$
 DIBAL-H : ester/nitrile $\rightarrow$ aldehyde · SOCl_2 : $\text{ROH} \rightarrow \text{RCl}$
 alcoholic KOH : elimination · aqueous KOH : substitution
 Br_2/NaOH on amide: Hofmann (one C fewer) · NaNO_2/HCl , 0–5 $^\circ\text{C}$: diazonium
 Sn/HCl : $\text{ArNO}_2 \rightarrow \text{ArNH}_2$ · Zn(Hg)/HCl or $\text{NH}_2\text{NH}_2/\text{KOH}$: $\text{C=O} \rightarrow \text{CH}_2$

Mechanism rules

SN_2 : methyl $> 1^\circ > 2^\circ > 3^\circ$, inversion · SN_1 : $3^\circ > 2^\circ > 1^\circ$, racemisation
 Carbocation $3^\circ > 2^\circ > 1^\circ$ · carbanion the reverse · EWG raises acidity
 Aqueous basicity: $(\text{C}_2\text{H}_5)_2\text{NH} > (\text{C}_2\text{H}_5)_3\text{N} > \text{C}_2\text{H}_5\text{NH}_2 > \text{NH}_3 > \text{aniline}$

Named reactions

Aldol: $\alpha\text{-H} + \text{dilute base}$ · Cannizzaro: no $\alpha\text{-H} + \text{concentrated base}$
 Reimer–Tiemann: phenol + $\text{CHCl}_3/\text{NaOH} \rightarrow \text{salicylaldehyde}$ · Kolbe: phenoxide + $\text{CO}_2 \rightarrow \text{salicylic acid}$
 Sandmeyer: $\text{ArN}_2^+ + \text{CuX} \rightarrow \text{ArX}$ · coupling with phenol $\rightarrow$ orange azo dye
 Gabriel: 1° aliphatic amines only · HVZ: $\alpha\text{-halo acid}$ · Wurtz: $2\text{RX} + 2\text{Na} \rightarrow \text{R-R}$
 Friedel–Crafts: AlCl_3 (not on aniline or nitrobenzene)
 Rosenmund: $\text{RCOCl} \rightarrow \text{RCHO}$ · Stephen: $\text{RCN} \rightarrow \text{RCHO}$ · Etard / Gattermann–Koch $\rightarrow$ benzaldehyde
 Williamson: $\text{RONa} + \text{R}'\text{X}$ ($\text{R}'\text{X}$ primary)

Tests

Tollens': all aldehydes, HCOOH · Fehling's: aliphatic aldehydes only
 Iodoform: CH_3CO^- or $\text{CH}_3\text{CH}(\text{OH})^-$ (ethanol, acetone, 2-propanol)
 Lucas: 3° immediate, $2^\circ \sim 5 \text{ min}$, 1° none · neutral FeCl_3 : violet with phenols
 NaHCO_3 : CO_2 with acids · Hinsberg: 1° soluble in alkali, 2° insoluble, 3° no reaction
 Carbylamine: 1° amines only · Lassaigne: N gives Prussian blue
 Biomolecules: sucrose non-reducing · denaturation keeps 1° structure · RNA has U instead of T

PART XII • LAST-MINUTE REVISION BOOK

Maths in One Page

Algebra

$\alpha + \beta = -\frac{b}{a}$, $\alpha\beta = \frac{c}{a}$ • $|z_1 z_2| = |z_1| |z_2|$ • $1 + \omega + \omega^2 = 0$
 $|kA| = k^n |A|$ • $|\text{adj } A| = |A|^{n-1}$ • $(AB)^{-1} = B^{-1} A^{-1}$
 $|A| = 0$ and $(\text{adj } A)B = 0 \rightarrow$ check for infinitely many solutions
 ${}^n C_r$, $\frac{n!}{p!q!}$, circular $(n-1)!$ • identical into r groups: ${}^{n+r-1} C_{r-1}$
 $T_{r+1} = {}^n C_r x^{n-r} y^r$ • $\sum {}^n C_r = 2^n$
 AP $S_n = \frac{n}{2}[2a + (n-1)d]$ • GP $S_\infty = \frac{a}{1-r}$ • AM $\geq$ GM
 Relations on n elements: 2^{n^2} • reflexive 2^{n^2-n}

Calculus

$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ • 1^∞ : $e^{\lim g(f-1)}$ • $1 - \cos kx \approx \frac{k^2 x^2}{2}$
 $(\tan^{-1} x)' = \frac{1}{1+x^2}$ • $(\sin^{-1} x)' = \frac{1}{\sqrt{1-x^2}}$
 $\int e^x (f + f') = e^x f$ • $\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \frac{x}{a}$
 King: $\int_a^b f(x) = \int_a^b f(a+b-x)$ • odd function on $[-a, a]$: 0
 Area $y^2 = 4ax$, $x^2 = 4by$: $\frac{16ab}{3}$ • $y = x^2$ and $y = mx$: $\frac{m^3}{6}$
 Linear DE: IF $= e^{\int P}$ • $P = \frac{1}{x} \rightarrow x \cdot P = \tan x \rightarrow \sec x$

Coordinate geometry

$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$ • parallel lines: equalise the coefficients first
 Circle: centre $(-g, -f)$, $r = \sqrt{g^2 + f^2 - c}$ • chord $2\sqrt{r^2 - d^2}$
 Parabola $y^2 = 4ax$: focus $(a, 0)$, latus rectum $4a$
 Ellipse: $b^2 = a^2(1 - e^2)$ • hyperbola: $b^2 = a^2(e^2 - 1)$ • latus rectum $\frac{2b^2}{a}$

3D & vectors

$l^2 + m^2 + n^2 = 1$ • SD $= \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$
 Foot: $Q(\lambda)$ with $\vec{PQ} \cdot \vec{b} = 0$ • image $2Q - P$
 Projection $\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$ • triangle $\frac{1}{2} |\vec{AB} \times \vec{AC}|$ • $|a \times b|^2 + (a \cdot b)^2 = a^2 b^2$

Statistics, probability, trigonometry

$\sigma^2 = \frac{\sum x^2}{n} - \bar{x}^2$ • $y = ax + b$: $\sigma_y = |a| \sigma_x$ • naturals: $\frac{n^2 - 1}{12}$
 Bayes: chosen path $\div$ sum of paths • $E(X) = \sum xp$ • $\text{Var} = E(X^2) - \mu^2$
 $a \sin x + b \cos x \in [-\sqrt{a^2 + b^2}, \sqrt{a^2 + b^2}]$ • $\cos^{-1}(-x) = \pi - \cos^{-1} x$
 $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ • $\sin^{-1}(\sin x) = x$ only on $[-\frac{\pi}{2}, \frac{\pi}{2}]$

PART XII · LAST-MINUTE REVISION BOOK

Shortcut Checklist

Before solving any MCQ the long way, run down this list (Speed Manual, page [389](#)).

- | | |
|--|--|
| <ul style="list-style-type: none"> ☐ Dimensions: do the options differ in dimensions? Eliminate. ☐ Limiting cases: push a mass, angle or distance to 0, ∞ or equality. Test two limits. ☐ Special values: $n = 1, 2$ or $x = 0, \frac{\pi}{4}$ in identity-type options. ☐ Back-substitution: are the options numbers that are quick to check? ☐ Symmetry: can $x \rightarrow a + b - x$, or a symmetric circuit, cut the work? ☐ Estimation: are the options a factor of 2 or more apart? One significant figure. ☐ Ratios: is only one quantity changing? Use the power law. ☐ Conservation first: is a speed or height asked?
Energy/momentum before forces. | <ul style="list-style-type: none"> ☐ Standard results: is this a textbook case (rolling, area, μ values)? ☐ Graph: "number of solutions"? Sketch both sides. ☐ n-factor: titration or redox? Equivalents, not equations. ☐ pH: buffer? $pK_a + \log(\text{ratio})$. Very dilute? pH stays below 7 for an acid. ☐ Organic: products given? Work backwards. ☐ Impossibility: probability > 1? Efficiency ≥ 1? Wrong sign? ☐ Casting out nines on long NV arithmetic. ☐ Cancel before multiplying. Keep exact values until the end. |
|--|--|

PART XII · LAST-MINUTE REVISION BOOK

Mistake Checklist

The eight types: one-line prevention

- ☐ **CG · Concept:** am I sure which principle applies here?
- ☐ **FR · Recall:** does the formula's **condition** hold?
- ☐ **CA · Calculation:** every step written; verified by substitution or estimate?
- ☐ **MR · Misreading:** asked quantity and NOT/EXCEPT underlined?
- ☐ **TR · Transcription:** screen = rough sheet (letter, value, sign)?
- ☐ **TS · Time:** past 3 minutes? Leave it.
- ☐ **GU · Guessing:** two options eliminated? If not, don't attempt.
- ☐ **PR · Pressure:** stuck or panicking? 60-second reset.

The most costly slips

- ☐ $\text{cm} \leftrightarrow \text{m}$, $\text{kJ} \leftrightarrow \text{J}$, $\text{mL} \leftrightarrow \text{L}$, $^{\circ}\text{C} \leftrightarrow \text{K}$
- ☐ Lens vs mirror formula; drop vs bubble
- ☐ Closed pipe = odd harmonics only
- ☐ Δn_g counts gases only; n -factor of KMnO_4 depends on the medium
- ☐ Principal value of $\cos^{-1}$ lies in $[0, \pi]$
- ☐ The +1 when counting terms; the $\frac{1}{2}$ in triangle areas
- ☐ T_{r+1} : the 5th term uses $r = 4$
- ☐ d vs d^2 and radius vs diameter: answer what is asked
- ☐ "Incorrect statement" questions: pick the false one

My own repeats (from the Mistake Book):

 ·

PART XII · LAST-MINUTE REVISION BOOK

Strategy Checklist

Before the exam

- ☐ 8 hours of sleep on each of the last three nights
- ☐ Exam-day wake-up time used for all final mocks
- ☐ Admit card, photo ID and photos packed (admit card rules checked)
- ☐ Route and travel time checked; leave with a margin
- ☐ Familiar breakfast planned; water
- ☐ Command Sheet read (page [483](#))

In the first 5 minutes

- ☐ Three slow breaths
- ☐ Rough sheet: skip columns and checkpoint times
- ☐ Memory dump of the 3–5 forgotten formulas

During the paper

- ☐ Subject order:
- ☐ Round 1: method visible in 20 s, ≤ 2 min, else **S**
- ☐ Round 2: 3 minutes maximum per question
- ☐ Checkpoints: 0:45 · 1:45 · 2:40
- ☐ MCQ only after eliminating two options; NV only if solved
- ☐ A hard section? Normalisation: it's hard for everyone. Next easy question.

The last 20 minutes

- ☐ Only questions with partial progress
- ☐ Every NV entry checked
- ☐ No answer changes without a specific error
- ☐ All responses saved

Trackers & Final Checklist

Printable, fillable trackers for hours, chapters, questions, PYQs, mocks, accuracy, weak chapters, formulas and the Mistake Book.

• Trackers	494
• Mock Test Log	502

• The Final Checklist	504
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PART XIII · TRACKERS & FINAL CHECKLIST

Trackers

Every tracker here is fillable on screen (checkboxes and text fields) and printable. Tick as you go: the trackers show at a glance which parts of the plan are done, and the weekly review (page [475](#)) reads from them.

Tracker	Page
Study hours (monthly)	494
Chapter completion	495
PYQ tracker (chapter-wise)	497
Questions solved & accuracy	499
Weak chapters	500
Formula revision	500
Error log summary	501
Mock test log	502
Final checklist	504

Study Hours Tracker

Write the hours studied each day (blocks × block length). Month · target per day h

 **REVISION** Study hours

Day	1	2	3	4	5	6	7	8	9	10	11
Hours											
Day	12	13	14	15	16	17	18	19	20	21	22
Hours											
Day	23	24	25	26	27	28	29	30	31	Total	
Hours											

Chapter Completion Tracker

Tick each stage; write the practice-set and chapter-test percentages.



REVISION Physics

Unit	Chapter	Module read	Practice %	Test %	R2	R4	Done
P01	Units & Measurements	☐	_____	_____	☐	☐	☐
P02	Kinematics	☐	_____	_____	☐	☐	☐
P03	Laws of Motion	☐	_____	_____	☐	☐	☐
P04	Work, Energy & Power	☐	_____	_____	☐	☐	☐
P05	Rotational Motion & Centre of Mass	☐	_____	_____	☐	☐	☐
P06	Gravitation	☐	_____	_____	☐	☐	☐
P07	Properties of Solids & Liquids	☐	_____	_____	☐	☐	☐
P08	Thermodynamics	☐	_____	_____	☐	☐	☐
P09	Kinetic Theory of Gases	☐	_____	_____	☐	☐	☐
P10	Oscillations & Waves	☐	_____	_____	☐	☐	☐
P11	Electrostatics & Capacitors	☐	_____	_____	☐	☐	☐
P12	Current Electricity	☐	_____	_____	☐	☐	☐
P13	Magnetic Effects of Current & Magnetism	☐	_____	_____	☐	☐	☐
P14	Electromagnetic Induction & Alternating Current	☐	_____	_____	☐	☐	☐
P15	Electromagnetic Waves	☐	_____	_____	☐	☐	☐
P16	Optics: Ray & Wave	☐	_____	_____	☐	☐	☐
P17	Dual Nature of Matter & Radiation	☐	_____	_____	☐	☐	☐
P18	Atoms & Nuclei	☐	_____	_____	☐	☐	☐
P19	Electronic Devices	☐	_____	_____	☐	☐	☐
P20	Experimental Skills	☐	_____	_____	☐	☐	☐

REVISION Chemistry

Unit	Chapter	Module read	Practice %	Test %	R2	R4	Done
C01	Some Basic Concepts in Chemistry	☐	_____	_____	☐	☐	☐
C02	Atomic Structure	☐	_____	_____	☐	☐	☐
C03	Chemical Bonding & Molecular Structure	☐	_____	_____	☐	☐	☐
C04	Chemical Thermodynamics	☐	_____	_____	☐	☐	☐
C05	Solutions	☐	_____	_____	☐	☐	☐
C06	Equilibrium: Chemical & Ionic	☐	_____	_____	☐	☐	☐
C07	Redox Reactions & Electrochemistry	☐	_____	_____	☐	☐	☐
C08	Chemical Kinetics	☐	_____	_____	☐	☐	☐
C09	Classification of Elements & Periodicity	☐	_____	_____	☐	☐	☐
C10	p-Block Elements	☐	_____	_____	☐	☐	☐
C11	d- and f-Block Elements	☐	_____	_____	☐	☐	☐
C12	Coordination Compounds	☐	_____	_____	☐	☐	☐
C13	Purification & Characterisation of Organic Compounds	☐	_____	_____	☐	☐	☐
C14	Basic Principles of Organic Chemistry	☐	_____	_____	☐	☐	☐
C15	Hydrocarbons	☐	_____	_____	☐	☐	☐
C16	Organic Compounds Containing Halogens	☐	_____	_____	☐	☐	☐
C17	Organic Compounds Containing Oxygen	☐	_____	_____	☐	☐	☐
C18	Organic Compounds Containing Nitrogen	☐	_____	_____	☐	☐	☐
C19	Biomolecules	☐	_____	_____	☐	☐	☐
C20	Principles Related to Practical Chemistry	☐	_____	_____	☐	☐	☐

REVISION Mathematics

Unit	Chapter	Module read	Practice %	Test %	R2	R4	Done
M01	Sets, Relations & Functions	☐	_____	_____	☐	☐	☐
M02	Complex Numbers & Quadratic Equations	☐	_____	_____	☐	☐	☐
M03	Matrices & Determinants	☐	_____	_____	☐	☐	☐
M04	Permutations & Combinations	☐	_____	_____	☐	☐	☐
M05	Binomial Theorem	☐	_____	_____	☐	☐	☐
M06	Sequences & Series	☐	_____	_____	☐	☐	☐
M07	Limits, Continuity & Differentiability	☐	_____	_____	☐	☐	☐
M08	Integral Calculus	☐	_____	_____	☐	☐	☐
M09	Differential Equations	☐	_____	_____	☐	☐	☐
M10	Coordinate Geometry: Lines, Circles & Conics	☐	_____	_____	☐	☐	☐
M11	Three-Dimensional Geometry	☐	_____	_____	☐	☐	☐
M12	Vector Algebra	☐	_____	_____	☐	☐	☐
M13	Statistics & Probability	☐	_____	_____	☐	☐	☐
M14	Trigonometry & Inverse Trigonometric Functions	☐	_____	_____	☐	☐	☐

PYQ Tracker (chapter-wise)

Tick each year's chapter-wise PYQs when done (Phase 1 of the PYQ system, page **413**) and write the accuracy. Tick **Redo** once the wrong ones are re-solved.



REVISION Physics PYQs

Unit	Chapter	2024	2025	2026	Accuracy %	Redo
P01	Units & Measurements	☐	☐	☐	_____	☐
P02	Kinematics	☐	☐	☐	_____	☐
P03	Laws of Motion	☐	☐	☐	_____	☐
P04	Work, Energy & Power	☐	☐	☐	_____	☐
P05	Rotational Motion & Centre of Mass	☐	☐	☐	_____	☐
P06	Gravitation	☐	☐	☐	_____	☐
P07	Properties of Solids & Liquids	☐	☐	☐	_____	☐
P08	Thermodynamics	☐	☐	☐	_____	☐
P09	Kinetic Theory of Gases	☐	☐	☐	_____	☐
P10	Oscillations & Waves	☐	☐	☐	_____	☐
P11	Electrostatics & Capacitors	☐	☐	☐	_____	☐
P12	Current Electricity	☐	☐	☐	_____	☐
P13	Magnetic Effects of Current & Magnetism	☐	☐	☐	_____	☐
P14	Electromagnetic Induction & Alternating Current	☐	☐	☐	_____	☐
P15	Electromagnetic Waves	☐	☐	☐	_____	☐
P16	Optics: Ray & Wave	☐	☐	☐	_____	☐
P17	Dual Nature of Matter & Radiation	☐	☐	☐	_____	☐
P18	Atoms & Nuclei	☐	☐	☐	_____	☐
P19	Electronic Devices	☐	☐	☐	_____	☐
P20	Experimental Skills	☐	☐	☐	_____	☐

REVISION Chemistry PYQs

Unit	Chapter	2024	2025	2026	Accuracy %	Redo
C01	Some Basic Concepts in Chemistry	☐	☐	☐	_____	☐
C02	Atomic Structure	☐	☐	☐	_____	☐
C03	Chemical Bonding & Molecular Structure	☐	☐	☐	_____	☐
C04	Chemical Thermodynamics	☐	☐	☐	_____	☐
C05	Solutions	☐	☐	☐	_____	☐
C06	Equilibrium: Chemical & Ionic	☐	☐	☐	_____	☐
C07	Redox Reactions & Electrochemistry	☐	☐	☐	_____	☐
C08	Chemical Kinetics	☐	☐	☐	_____	☐
C09	Classification of Elements & Periodicity	☐	☐	☐	_____	☐
C10	p-Block Elements	☐	☐	☐	_____	☐
C11	d- and f-Block Elements	☐	☐	☐	_____	☐
C12	Coordination Compounds	☐	☐	☐	_____	☐
C13	Purification & Characterisation of Organic Compounds	☐	☐	☐	_____	☐
C14	Basic Principles of Organic Chemistry	☐	☐	☐	_____	☐
C15	Hydrocarbons	☐	☐	☐	_____	☐
C16	Organic Compounds Containing Halogens	☐	☐	☐	_____	☐
C17	Organic Compounds Containing Oxygen	☐	☐	☐	_____	☐
C18	Organic Compounds Containing Nitrogen	☐	☐	☐	_____	☐
C19	Biomolecules	☐	☐	☐	_____	☐
C20	Principles Related to Practical Chemistry	☐	☐	☐	_____	☐

REVISION Mathematics PYQs

Unit	Chapter	2024	2025	2026	Accuracy %	Redo
M01	Sets, Relations & Functions	☐	☐	☐	_____	☐
M02	Complex Numbers & Quadratic Equations	☐	☐	☐	_____	☐
M03	Matrices & Determinants	☐	☐	☐	_____	☐
M04	Permutations & Combinations	☐	☐	☐	_____	☐
M05	Binomial Theorem	☐	☐	☐	_____	☐
M06	Sequences & Series	☐	☐	☐	_____	☐
M07	Limits, Continuity & Differentiability	☐	☐	☐	_____	☐
M08	Integral Calculus	☐	☐	☐	_____	☐
M09	Differential Equations	☐	☐	☐	_____	☐
M10	Coordinate Geometry: Lines, Circles & Conics	☐	☐	☐	_____	☐
M11	Three-Dimensional Geometry	☐	☐	☐	_____	☐
M12	Vector Algebra	☐	☐	☐	_____	☐
M13	Statistics & Probability	☐	☐	☐	_____	☐
M14	Trigonometry & Inverse Trigonometric Functions	☐	☐	☐	_____	☐

Questions Solved & Accuracy

Weekly totals of practice questions (modules, PYQs, question bank; mocks excluded) and their accuracy.



REVISION Weekly questions and accuracy

Week	Physics Q	Physics %	Chemistry Q	Chemistry %	Maths Q	Maths %	Total Q
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							
14							
15							
16							
17							
18							
19							
20							
21							
22							
23							
24							
25							
26							

Weak Chapter Tracker

A chapter is **weak** if its practice or PYQ accuracy is below 70%, or if it appears in three or more Mistake Book entries in a month. Repair it with the module plus 20 fresh questions, then re-test after 7 days.



REVISION Weak chapters

Chapter	Why weak (% , entries)	Repair plan	Repaired on	Re-test %	Fixed
					☐
					☐
					☐
					☐
					☐
					☐
					☐
					☐
					☐
					☐
					☐
					☐
					☐
					☐
					☐
					☐
					☐
					☐
					☐

Formula Revision Tracker

Tick each Formula Handbook sheet every time it passes a blank-sheet self-test (page [386](#)).



REVISION Formula sheets × cycles

Sheet	Cycle 1	Cycle 2	Cycle 3	Cycle 4	Cycle 5	Cycle 6
Physics P01–P06	☐	☐	☐	☐	☐	☐
Physics P07–P10	☐	☐	☐	☐	☐	☐
Physics P11–P14	☐	☐	☐	☐	☐	☐
Physics P15–P20	☐	☐	☐	☐	☐	☐
Chemistry C01–C08 (physical)	☐	☐	☐	☐	☐	☐
Chemistry C09–C12 (inorganic)	☐	☐	☐	☐	☐	☐
Chemistry: reagents & named reactions	☐	☐	☐	☐	☐	☐
Chemistry: distinguishing tests	☐	☐	☐	☐	☐	☐
Maths M01–M06	☐	☐	☐	☐	☐	☐
Maths M07–M09	☐	☐	☐	☐	☐	☐
Maths M10–M12	☐	☐	☐	☐	☐	☐
Maths M13–M14	☐	☐	☐	☐	☐	☐
Constants & frequently forgotten	☐	☐	☐	☐	☐	☐

Error Log Summary

A monthly roll-up of the Mistake Book (page 449). The goal is falling counts in CA, MR, TR and GU, the four types that are pure habit.

REVISION Mistake types by month										
Month	CG	FR	CA	MR	TR	TS	GU	PR	Repeats	Main fix
Oct										
Nov										
Dec										
Jan										
Feb										
Mar										
Apr										

PART XIII · TRACKERS & FINAL CHECKLIST

Mock Test Log

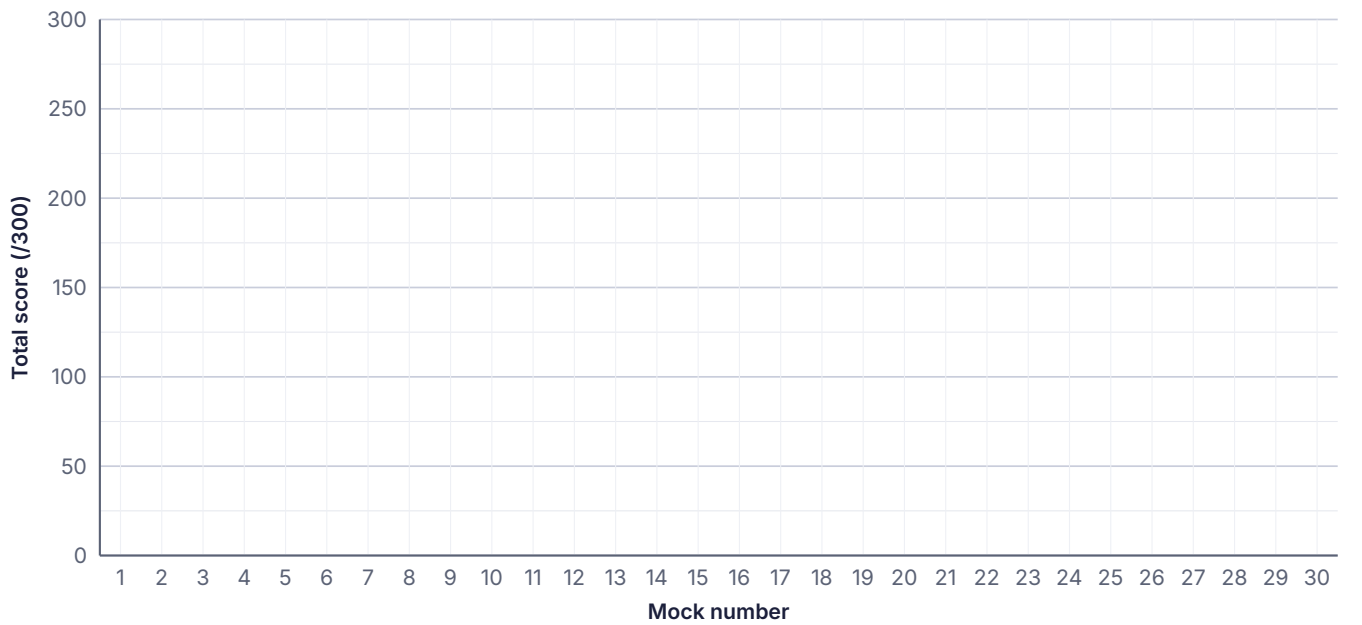
One line per full-length mock. The trend matters more than any single score: judge progress by the average of your last three mocks (page [444](#)).


MOCK TEST Mock log

#	Date	Source / paper	P	C	M	Total	Neg.	Acc. %	Top fix (from the analysis)
1									
2									
3									
4									
5									
6									
7									
8									
9									
10									
11									
12									
13									
14									
15									
16									
17									
18									
19									
20									
21									
22									
23									
24									
25									
26									
27									
28									
29									
30									

Score trend

Plot each mock's total and join the points. Draw your **target band** (page 16) as two horizontal lines in a different colour. Three points in a row inside the band show that the level is stable.



Blank score-trend chart: mock number (1–30) against total score out of 300.

PART XIII · TRACKERS & FINAL CHECKLIST

The Final Checklist

Work through this list in the last 14 days. Every item should be ticked before exam day. Any unticked item goes into the Final 7 Days plan (page [470](#)).

Content readiness

- ☐ Every P1 (Must-do) topic studied and its chapter test taken (page [402](#))
- ☐ Chapter-wise PYQs 2024–2026 done for all P1 and P2 chapters (page [497](#))
- ☐ At least two full 7-Day Formula Cycles completed (page [386](#))
- ☐ Blank-sheet formula test $\geq 90\%$ in each subject
- ☐ NCERT inorganic tables and exceptions read twice (page [163](#))
- ☐ Organic reagent and named-reaction tables recalled without notes
- ☐ Every Mistake Book **repeat** re-solved correctly
- ☐ Last-Minute sheets read once (page [485](#))

Exam readiness

- ☐ At least 15 full mocks analysed with the Mock Analysis Sheet (target: 20, page [15](#))
- ☐ Last 5 mocks: in the target band, with ≤ 8 negative marks
- ☐ Subject order and checkpoints tested in ≥ 5 mocks and fixed
- ☐ Round system and 3-minute rule used without thinking
- ☐ At least 3 mocks taken in the exact exam slot
- ☐ Command Sheet filled in with personal traps (page [483](#))
- ☐ Panic reset practised in at least two mocks
- ☐ Exam-day breakfast and routine rehearsed on mock days

Logistics

- ☐ Admit card downloaded; name, photo, date, shift and centre checked
- ☐ Valid photo ID and photographs as specified on the admit card
- ☐ Any required declaration or undertaking filled in
- ☐ Centre route and travel time known; travel companion arranged
- ☐ Clothing that follows the admit-card guidelines ready
- ☐ Two alarms set; phone plan for the journey

Body and mind

- ☐ Sleep at a fixed time for the last 7 nights (8 hours)
- ☐ Wake-up time matched to the exam slot for 7 days
- ☐ Exercise or a walk every day
- ☐ Screens off an hour before sleep
- ☐ Worries written down before the last mock and on the exam morning (page [455](#))
- ☐ Reminder: a hard paper is hard for the whole shift. Normalisation handles it

☆ IMPORTANT After Session 1

Take two days' rest, then follow the Session 1 → Session 2 bridge (page [461](#)). Only the better of your two NTA scores is used (page [10](#)), so Session 2 is a genuine second chance.

SOURCES & REFERENCES

Sources & References

How to read the labels in this book

✓ VERIFIED

Taken from an **official** document (NTA, JoSAA, NCERT), or from a news report that quotes one. Re-check it against the 2027 bulletin when it is released (page **10**).

||| THIRD-PARTY ANALYSIS

From **third-party** analyses: coaching institutes, education portals and PYQ archives. Useful but approximate; different analysts group topics differently.

📌 RECOMMENDATION

This book's **recommendation**: a strategy, plan, estimate or ranking derived from the sources and from standard preparation practice. It is a judgement, not a fact.

All web sources were accessed in September 2026. Codes in square brackets in the text (e.g. [NTA-IB26]) refer to the entries below. **No question in this book is reproduced from a past paper.** Past papers are referenced only as sources to practise from.

Official sources

Code	Source	Used for
NTA-IB26	National Testing Agency, <i>Information Bulletin, JEE (Main) 2026</i> (released 31 Oct 2025, revised Nov 2025), via jeemain.nta.nic.in/public-notice	pattern, marking, normalisation, exam-day rules
NTA-SYL	NTA, <i>JEE (Main) 2026 Syllabus</i> , jeemain.nta.nic.in/document/syllabus-2026	unit lists and syllabus notes in every module
NTA-CALC	NTA, <i>Clarification on Use of Virtual Calculator in JEE (Main) 2026</i> (Nov 2025), PDF on the NTA site	no calculator
NTA-CAL27	NTA, <i>Tentative examination calendar 2026–27</i> (public notice dated 16 Sep 2026), nta.ac.in ; reported by Business Standard	JEE Main 2027 Session 1 dates
NTA-WEB	NTA JEE (Main) portal, jeemain.nta.nic.in	answer keys, notices, question papers
JOsAA	Joint Seat Allocation Authority, josaa.nic.in	female-only (supernumerary) seat pool, opening and closing ranks
NCERT	National Council of Educational Research and Training, textbooks, ncert.nic.in/textbook.php	rationalised Class 11–12 content, chapter numbers
CODATA	NIST, <i>CODATA Recommended Values of the Fundamental Physical Constants</i> , physics.nist.gov/cuu/Constants	constants table
TELEMANAS	Ministry of Health & Family Welfare, Tele-MANAS helpline, telemanas.mohfw.gov.in	wellbeing chapter

Reported official information (news and education portals)

Code	Source	Used for
R-2026STAT	Shiksha, <i>JEE Main 2026 Result Statistics: 15,38,468 unique candidates appeared</i> , shiksha.com	candidates appeared, rank estimate
R-CUTOFF	Shiksha, <i>JEE Main Cutoff 2026 for General</i> , shiksha.com	JEE Advanced qualifying percentiles
R-CALC	Careers360, <i>JEE Main 2026: use of virtual calculator not allowed, says NTA</i> , news.careers360.com	calculator clarification
R-TIE	Careers360, <i>JEE 2026 Tie Rule Explanation</i> , engineering.careers360.com	tie-breaking order

Code	Source	Used for
R-FEE	Collegedunia, <i>JEE Main 2026 Registration Fee</i> , collegedunia.com	fee for female candidates
R-NORM	Careers360, <i>JEE Main 2026 Normalisation Explained</i> , news.careers360.com	percentile formula
R-REFORM	Shiksha, <i>NTA introduces four-tier question paper checking system</i> , shiksha.com	2026–27 reforms
R-DATES24	CollegeDekho, <i>JEE Main 2024 Session 2 shift-wise exam dates</i> , collegedekho.com ; Careers360, <i>JEE Main 2024 session 1 exam dates</i> , news.careers360.com	PYQ shift tracker
R-DATES25	Careers360, <i>JEE Mains 2025 session 2 exams from April 2 to 9</i> , news.careers360.com	2025 dates and shifts
R-DATES26	Shiksha, <i>JEE Main 2026 Session 2 Exam Date</i> , shiksha.com	2026 dates and shifts

Third-party analyses and PYQ archives

Code	Source	Used for
T-WT-CD	Collegedunia, <i>JEE Main chapter-wise weightage</i> (2026 counts compiled from the ExamSIDE archive), collegedunia.com ; <i>Coordination Compounds weightage in JEE Main 2026</i> , collegedunia.com	2026 chapter counts
T-WT-VED	Vedantu, <i>JEE Main Chapter Wise Weightage 2026</i> , vedantu.com	unit weight ranges
T-PHY26	Collegedunia, <i>JEE Mains 2026 Physics Weightage (April attempt)</i> , collegedunia.com	Physics weights
T-ANALYSIS26	Careers360, <i>JEE Main 2026 Exam Analysis</i> , engineering.careers360.com ; eSaraI, <i>JEE Main 2026 Paper Analysis: All Shifts</i> , esaral.com	section-wise character of the 2026 papers
T-VED-25	Vedantu, <i>Marks vs Percentile JEE Main 2025 (shift-wise)</i> , vedantu.com	2025 marks for 99 percentile
T-VED-26	Vedantu, <i>Shift-wise JEE Main 2026 Marks vs Percentile, January Session 1</i> , vedantu.com	January 2026 range
T-COMP	Shiksha, <i>JEE Main 2026 Marks vs Percentile: shift-wise analysis</i> , shiksha.com	cross-check of 2026 ranges
T-PW-S2	Physics Wallah, <i>Shift-wise JEE Main 2026 Session 2 Marks vs Percentile</i> , pw.live	April 2026 range
T-DELETED	Shiksha, <i>JEE Main Reduced Syllabus: deleted chapters and topics</i> , shiksha.com	topics removed in 2024
T-EXAMSIDE	ExamSIDE, <i>JEE Main previous year questions</i> , questions.examside.com	PYQ archive, repeated structures
T-MATHONGO	MathonGo, <i>JEE Main chapter-wise questions with solutions</i> , mathongo.com	PYQ archive

Learning science and health

Code	Reference	Used for
LS-DUN	Dunlosky, J., Rawson, K. A., Marsh, E. J., Nathan, M. J., & Willingham, D. T. (2013). Improving students' learning with effective learning techniques. <i>Psychological Science in the Public Interest</i> , 14(1), 4–58. doi:10.1177/1529100612453266	practice testing, distributed practice
LS-RK	Roediger, H. L., & Karpicke, J. D. (2006). Test-enhanced learning. <i>Psychological Science</i> , 17(3), 249–255. doi:10.1111/j.1467-9280.2006.01693.x	retrieval practice
LS-CEP	Cepeda, N. J., Pashler, H., Vul, E., Wixted, J. T., & Rohrer, D. (2006). Distributed practice in verbal recall tasks. <i>Psychological Bulletin</i> , 132(3), 354–380. doi:10.1037/0033-2909.132.3.354	spaced revision
LS-R0H	Rohrer, D., & Taylor, K. (2007). The shuffling of mathematics problems improves learning. <i>Instructional Science</i> , 35, 481–498. doi:10.1007/s11251-007-9015-8	interleaved practice
LS-RB	Rasch, B., & Born, J. (2013). About sleep's role in memory. <i>Physiological Reviews</i> , 93(2), 681–766. doi:10.1152/physrev.00032.2012	sleep and memory

Code	Reference	Used for
LS-AASM	Paruthi, S., et al. (2016). Recommended amount of sleep for pediatric populations: a consensus statement of the American Academy of Sleep Medicine. <i>Journal of Clinical Sleep Medicine</i> , 12(6), 785–786. doi:10.5664/jcsm.5866	8–10 h for ages 13–18
LS-WARD	Ward, A. F., Duke, K., Gneezy, A., & Bos, M. W. (2017). Brain drain: the mere presence of one's own smartphone reduces available cognitive capacity. <i>Journal of the Association for Consumer Research</i> , 2(2), 140–154. doi:10.1086/691462	phone rules
LS-GOLL	Gollwitzer, P. M., & Sheeran, P. (2006). Implementation intentions and goal achievement: a meta-analysis of effects and processes. <i>Advances in Experimental Social Psychology</i> , 38, 69–119. doi:10.1016/S0065-2601(06)38002-1	procrastination
LS-JAM	Jamieson, J. P., Mendes, W. B., Blackstock, E., & Schmader, T. (2010). Turning the knots in your stomach into bows: reappraising arousal improves performance on the GRE. <i>Journal of Experimental Social Psychology</i> , 46(1), 208–212. doi:10.1016/j.jesp.2009.08.015	test anxiety
LS-RAM	Ramirez, G., & Beilock, S. L. (2011). Writing about testing worries boosts exam performance in the classroom. <i>Science</i> , 331(6014), 211–213. doi:10.1126/science.1199427	expressive writing
LS-WHO	World Health Organization (2020). <i>WHO guidelines on physical activity and sedentary behaviour</i> . who.int	60 min/day activity for adolescents

**NOTE Corrections**

If you find an error in this book, or a rule that has changed in the 2027 bulletin, correct it by hand where it appears and note it here: _____